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MATHEMATICS FOR ECONOMISTS

B C. MEHTA

M.A., M.Sc., Ph.D.

Professor

*Department of Economics
Sukhadia University, Udaipur*

G.M.K. MADNANI

B.Sc., M.A., Ph.D.

Associate Professor

*Department of Economics
Sukhadia University, Udaipur*

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Show that there are decreasing returns to scale and deduce that the total product is $> a$ times the marginal product is a plus b times the marginal product of b

13. If the production function is given as

$$x = A.a^{1/2}.b^{1/3}.c^{1/6}$$

find expressions for the marginal products of a , b , and c .

Hence show that if these marginal products are all equal, then $x = \frac{A.a}{\sqrt{2}}$

14. What do you understand by 'an expansion path'?

Illustrate your answer by discussing the expansion path of the least cost combination of two factors, labour and capital, in a Cobb Douglas model

Does the behaviour of the expansion path change if assumptions regarding returns to scale are varied?

15. Show that a production function homogeneous of degree one generates a linear production expansion path and also a linear total cost function

16. Explain the concept of elasticity of substitution

Investigate its value for the following production functions

(a) $x = A L^{\alpha} K^{\beta}$

(b) $x^{-\alpha} = A L^{-\alpha} + B K^{-\alpha}$

17. Prove that in the Cobb-Douglas production function

$$P = A L^{\alpha} K^{\beta}$$

α and β are the elasticities of production with respect to labour and capital respectively

18. If $U = \log (X+a)^{\alpha} (X+b)^{\beta}$ is one form of a utility function, find the marginal substitution between the goods X and Y , and deduce the elasticity of substitution is :

$$\pi = 1 + \frac{b\alpha X + a\beta Y}{(\alpha + \beta)XY}$$

19. Explain the concept of a production function. State in what respects the Cobb-Douglas formulation of a production function differs from the more recent formulation of a production function given by Arrow, Chenery, Minhas and Solow

20. Define the concept of factor substitution

Derive an expression for such an elasticity assuming a linear homogeneous production function

21. Prove that the elasticity of substitution of a Cobb-Douglas production function with constant returns to scale, is unity. Write out another production function in which this elasticity is constant but not necessary equal to unity

22 The production function of a monopolist is $Q = AX_1^a X_2^b$ and the demand curve function is $Q = KP^d$ where Q is output and X_1 and X_2 are inputs p is the price of output and p_1 and p_2 are fixed output prices. Derive his equilibrium output, price and profit.

23 What are the properties of a well behaved production function? Examine if $q = Ax_1^\alpha x_2^{1-\alpha}$ is a well behaved production function.

can also be solved with the help of Differential Calculus, Linear Programming or Theory of Games

Consumer's equilibrium and equilibrium of the firm involve decisions at the margin and marginal analysis is nothing but extension of differential calculus. We also know that price change on account of output change depends upon elasticity of demand and supply and 'Elasticity' is, in fact, a mathematical concept

Basic relationships in Micro and Macro-Economics can usually be put in the form of functions and equations. Quadratic, Simultaneous, Differential and Difference. For example, in Growth Economics where the time paths of national income are to be found out (depending on some crucial factors) difference and differential equations are indispensable. In planning models, sectoral targets are fixed only with the help of Input-Output Analysis and Linear Programming. Determinants and Matrix Algebra are of immense use in such techniques.

Thus in almost all fields of economics, mathematics is useful. Economic phenomena are so complex and interdependent that some abstractions — *ceteris paribus* assumptions — have to be made. It is often not possible for a student of Economics to understand the effect of many factors simultaneously on a particular economic phenomenon. Mathematics is especially helpful in these situations. A simplified model is first built wherein relationship between the different variables is established by making assumptions and keeping many factors constant. Thereafter, the *ceteris paribus* assumptions are relaxed one by one and the effects analysed.

The proof of pudding lies in eating. After going through this book the reader will find out that how easy, smooth and systematic it becomes to understand the economic processes and derivation of various results by learning a little of mathematics. For example, the cobweb theorem and the stability conditions can be derived and understood very neatly and briefly in not more than two pages. This cannot be possible without the use of the tools which have been used in this volume.

Dynamisation of Economics Traditional economics explains the situation at a point of time, whereas actual economy witnesses process of change over time. Hence 'time' must be accounted for and, therefore, dynamisation is a step towards reality. Growth economics is an outgrowth of dynamisation of Economics.

Estimation Since economics does not deal with pure theorizing alone but also with the relevance of theory in practice, there must be some methods of testing the correspondence between theory and reality. And here comes a new branch of knowledge, i.e., Econometrics, to our rescue. In Econometrics we use mathematics to specify the relationships as given by economic theory.

and apply tools of statistics and theory of probability to test theorems of economic theory estimate parameters and then predict the results. In the words of J. Tinbergen *Econometrics is mathematical economics working with measured data to give empirical content to economic theory*

Programming Research Problems of maximisation and minimisation are basic to Economics. For example we have to minimise cost of transportation of goods from different godowns to different markets. But against this maximum capacity of each godown is fixed so also the maximum demand in each of the markets. These are termed as constraints or side conditions. Such problems are solved by programming techniques making economics a complete practical science.

All these revolutions thus involve increasing use of mathematics.

A. Chiang in his book *Fundamental Methods of Mathematical Economics* rightly considers the mathematical approach as a quick mode of transportation from a set of postulates to a set of conclusions. For example if a person intends to go to a place five miles away he will definitely prefer driving to walking if he possesses a car unless he has time to kill or wants to exercise his legs. Similarly a theorist who wishes to get to his conclusions more quickly will find it convenient to drive the vehicle of mathematical techniques appropriate for his particular purpose. But all the same he must know the driving and for this driving lessons are essential. In other words, the theorist must accustom himself with mathematical techniques and their applications which are driving lessons to elucidate the problems of economic theory.

One must also be on guard lest the use of mathematics should turn into a disadvantage. Mathematics like fire is a bad master although a good servant. It is also an art and naturally one gets fascinated by its aesthetic values after perfecting its techniques. At times therefore, addiction to mathematics tempts us to make inappropriate economic assumptions and limit ourselves only to the problems that can be solved mathematically. In this connection we would like to reproduce here the conversation between Prof. Samuelson and his student.

Student I am interested in economic theory. I know little mathematics. And when I look at the journals I am greatly troubled. Must I give up hope of being a theorist? Must I learn mathematics?

Samuelson Some of the most distinguished economic theorists, past and present have been innocent of mathematics. Some of the most distinguished theorists have known some mathematics. Obviously you can become a great theorist without knowing mathematics. Yet it is fair to say that you will have to be that much more clever and brilliant. Mathematics is neither necessary nor a sufficient condition for a fruitful career in economic theory. It can help.¹

¹ Paul A. Samuelson, *Economic Theory and Mathematical Analysis*, 4th Edition, *Journal of American Economic Review*, Vol. XLII, No. 2, May 1952, pp. 55-66.

Thus use of mathematics can only better good economics, it can never compensate for bad economics.

The book is specifically meant for use by students and teachers in India who have got a rudimentary knowledge of mathematics and as such it is presumed that the readers are acquainted only with elementary Algebra and Geometry.

In the first part we have introduced the highly useful Algebra of Matrices and extended the discussion to cover Input Output Analysis and Linear Programming. In the second part Differential Calculus — the Calculus of related changes — is given in great detail with several chapters dealing with applications in the various branches of micro-economic theory. In the third part Integral Calculus the technique of deriving the macro from the micro analysis has been dealt with. Differential and Difference equations are covered in the last four chapters with illustrations from macro-economic theory and the theory of economic growth.

Though we have tried to make the book self-contained, we do not claim that we have achieved perfection. It is almost impossible to compress the entire toolbox provided by mathematics in a single volume intended to be studied in a single academic session. The purpose of this work, then, is just to give an introduction to basic mathematics capable of providing tools for use in economic theory and Econometrics.

This is not a book on economic theory. Hence understanding of economic theory is presumed. Only those who have cared to study their economics well will be benefited by this book. It is in no way a substitute for books on economic theory.

We will be glad to receive suggestions for improvement of the volume from those who may use the book as students or as teachers.

JAIPUR
July 25, 1973

MEHTA
MADHAVANI

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Chapter 1

Theory of Sets

1.1 In mathematics the idea of a set is fundamental, yet the theory of sets is of recent origin. George Cantor (1845-1918) is regarded as the founder of this mathematical system. This system has become popular only in the last two decades.

In mathematical analysis and its applications, we often deal with groups, collections, aggregates, populations, families, etc. They consist of concrete or abstract, finite or infinite or similar or dissimilar objects.

A set is a collection of well defined distinct objects thought of as a whole. It is synonymous with the term aggregate.

The objects must be precisely defined. The objects constituting a set are called its 'elements' or 'members'.

(i) A set, consists of well defined objects,

(ii) No two elements of a set are identical.

Ex 1. (i) Books owned by Ramesh

(ii) All girl students of B.A. in Bombay University

(iii) The aggregate of all odd integers (natural numbers)

(iv) A team of football players

(v) Collection of coins with Mahesh

(vi) Family of pets and cattle owned by the sarpanch of my village

These are aggregates of objects. The number of elements may be finite (as in examples i, ii, iv, v and vi) or infinite (as iii) similar or dissimilar (as i, v, vi).

1.2. Notations for a Set. We use capital letters ($A, B, \dots$) to denote sets and small letters for the objects. One way of representing a set is complete enumeration of all the objects or elements (without repeating any of the elements) and putting them in brace brackets $\{ \}$. Thus,

$$A = \{a, b, c, d, e\}$$

Here set A contains five elements a, b, c, d, e . We have listed/enumerated all the elements. This is termed as Roster or Tabulation or Enumeration method. This is also known as defining a set by extension.

Ex. 2. $A = \{1, 2, 3, 4, 5, 6\}$

is a set of six numbers, the elements are the first six natural numbers. Here, we have enumerated all the members.

Ex 3 $A = \{\text{pen pencil rubber sharpener, inkpot}\}$

is a set of 5 members enumerated or specified individually. The set A has been defined by extension.

Many times it is not possible to list all the members of a set as when a set contains very large number of members. In many other cases it is not necessary to list all members as when the members are related to each other and the relation can be put in a phrase or sentence or rule or property.

Ex 4 $A = \{a \mid a \text{ is } P\}$

$A = \{a \mid a \text{ is } P\}$

This is to be read as A is set of all objects (or elements) of the type a such that a has the property P . P here describes the common property.

If the property is described in a phrase or sentence, then the representation is termed as *descriptive phrase method* but if the property is described symbolically by a rule, it is termed as *rule method*. This method of defining a set by a common property is called *definition by intension*.

Many sets can be defined either by extension (Tabulation) or intension (Descriptive phrase or rule method). The set of digits can be defined by enumeration (extension) or by intension. However many sets can be defined only by intension (Descriptive phrase or Rule). This is always the case if the set contains infinite or unknown number of objects. How will we define the set of all fishes in the Indian Ocean at 11 M on July 30, 1991? Even if there are finite number of fishes in the ocean we do not know their number we cannot enumerate them. There may even be no fish in the set. In such a case definition by intension is the only possibility.

Ex 5 $A = \{a \mid a \text{ is a digit}\}$ Descriptive phrase method

$A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ Tabulation method

$A = \{a \mid 0 \leq a \leq 9\}$ Rule method

Ex 6 $A = \{1, 3, 5, 7, 9\}$ Tabulation method

$A = \{a \mid a \text{ odd numbers from 1 to 9}\}$ Descriptive phrase method

$A = \{a \mid a \text{ first five odd numbers}\}$ Descriptive phrase method

$A = \{a \mid a \text{ (odd) } 1 \leq a \leq 9\}$ Rule method

$A = \{a \mid a \leq 10 \text{ and an even number}\}$ Rule method

A set contains objects which are called its elements. The elements belong to the set A . Symbolically

$$a \in A$$

means a belongs to the set A or a is an element of the set A .

1.3 Finite and Infinite Sets A set is finite if it contains finite number of elements.

$$\begin{aligned}\text{Ex 7} \quad A &= \{a/a \text{ students in my class}\} \\ B &= \{a/a \text{ citizens of India}\} \\ C &= \{x/x \ 0 \leq x \leq 9\} \\ D &= \{a/a \text{ coins in India}\}\end{aligned}$$

are all finite sets

But if the number of elements is very large and infinite then the set is *infinite set*

$$\begin{aligned}\text{Ex 8} \quad A &= \{x \mid x \text{ natural number}\} \\ B &= \{x \mid x \text{ is a point on line AB}\} \\ C &= \{a \mid a \leq 50\}\end{aligned}$$

are all infinite sets

1.4 Null Set A set containing no element is called *null set* or *empty set* or *zero set* and is denoted by the symbol ϕ

$$\begin{aligned}\text{Ex 9} \quad \text{The set of Indians on the moon is a null set} \\ A &= \{ \quad \quad \quad \} \\ A &= \{a/a \text{ Indian on moon}\} \\ &= \phi\end{aligned}$$

Ex 10 The set of living female Prime Ministers of India since independence is a null set since all first female prime Ministers have expired

$$\begin{aligned}\text{Therefore } A &= \{x/x \text{ female prime Ministers of India since} \\ &\quad \text{independence}\} \\ &= \phi\end{aligned}$$

1.5 Unit Set or Singleton Set A set containing only one element is termed as unit set or singleton

$$\begin{aligned}\text{Ex 11} \quad A &= \{a\} \text{ is a unit set} \\ A &= \{a/a \text{ present Prime Minister of India}\} \text{ is a} \\ &\quad \text{unit set} \\ A &= \text{Set of minimum possible marks in a test} \\ &= \{0\} \\ A &\text{ is a unit set}\end{aligned}$$

It is to be noted that $\lambda = \{0\}$

contains one element 0, hence it is not a null set.

1.6 Universal Set Consider a given totality of elements from all possible sets A, B, C, D . From these elements the set of totality of elements is called the universal set (Ω) and the set of no elements is the empty set ($\{\}$). Thus if in a discussion sets A, B, C, D are formed then the totality of all possible elements out of which the elements of the sets A, B, C, D are taken is called the universal set with reference to this particular discussion.

$$\begin{aligned}\text{Ex 12} \quad & \text{If } A = \{2, 4, 6, 8\} \\ & B = \{8, 12, 16, 20, \dots\} \\ & C = \{a \mid a = 2n \text{ positive integer}\}\end{aligned}$$

Now, all these sets are formed out of the positive even integers. Thus as far as this discussion is concerned for these and similar sets, the set of all positive even integers is the universal set. Hence

$$\begin{aligned}& = \{2, 4, 6, 8, 10, 12, \dots\} \\ & = \{a \mid a = \text{all positive even numbers}\}\end{aligned}$$

$$\begin{aligned}\text{Ex 13} \quad & \text{If } A = \{\text{All male students in Rajasthan}\} \\ & B = \{\text{Students in university girls college in Rajasthan}\} \\ & C = \{\text{Boys getting scholarship in Rajasthan}\} \\ & D = \{\text{Girls committing suicide in Rajasthan after declaration of their examination results}\}\end{aligned}$$

The parent set for all these sets in this particular discussion consists of all students—males, females, college school—in Rajasthan. Set of all students in Rajasthan is the universal set for such sets A, B, C, D and like.

1.7 Equal (or identical) Sets and Equivalent Sets. If two sets A and B contain the same distinct elements, then A and B are equal or identical sets.

$$\begin{aligned}\text{Ex 14} \quad & A = \{1, 2, 3, 4\} \\ & = \{x \mid x = \text{positive integers} \leq 4\} \\ & B = \{2, 3, 4, 1\} \\ & = \{x \mid x = \text{positive integers} \leq 4\}\end{aligned}$$

Since sets A and B both contain the same 4 distinct objects $\{1, 2, 3, 4\}$ hence they are equal sets. Hence $A = B$.

$$\begin{aligned}\text{Ex 15.} \quad & A = \{\text{price, quantity demanded, income}\} \\ & B = \{\text{income, price, quantity demanded}\} \\ \text{Here} \quad & A = B\end{aligned}$$

Two sets are equivalent if there is one to one correspondence between the elements of the two sets. Equivalent sets have same number of distinct elements but not the same elements.

$$\begin{aligned}\text{Ex 16} \quad & A = \{p, l, a, n\} \\ & B = \{1, 2, 3, 4\}\end{aligned}$$

are equivalent sets, written as

$$\begin{aligned}A & \sim B \text{ or,} \\ A & \leftrightarrow B\end{aligned}$$

Ex 17 Sets formed from the numbers 1, 2, 3, 4, 5, 6 and the alphabets in "massive" are equivalent, since "massive" contains 6 distinct alphabets.

forming a set each element of which can be put in one to one correspondence with the elements of the set $\{1, 2, 3, 4, 5, 6\}$

EXERCISE 1.1

1 Express each of the following sets in descriptive phrase form or rule form

- (i) The set of all odd integers
- (ii) Set of all integers divisible by 4
- (iii) Set of all women Presidents of India
- (iv) The set of squares of integers from 1 to 5
- (v) The set of numbers such that $x = x-1$

2 Describe each of the following sets in words

- (i) $A = \{1, 8, 27, 64\}$
- (ii) $A = \{x : x = 2^n, \text{ is an integer}\}$
- (iii) $A = \{x : 15 \leq x < 60 \text{ years of age}\}$
- (iv) $X = \{x/x^2 - 7x + 12 = 0\}$
- (v) $A = \{x/x^2\}$

3 Which of the following sets is finite or infinite?

- (i) $A = \{x : x \text{ is a multiple of } 3\}$
- (ii) $B = \{x : x \text{ is persons in age group } 15-60 \text{ years in India}\}$
- (iii) $C = \{x : x \text{ is mango trees in India}\}$
- (iv) $D = \{x : x \text{ is ray emanating from the sun at any moment of time}\}$
- (v) Set of Indians getting Nobel prize in Economics
- (vi) Set of Indians in the world

4 Which of the following is correct or incorrect?

- (i) $\{a, b, c\} = \{\alpha, \beta, \gamma\}$
- (ii) $\{a, b, c\} = \{\alpha, \beta, \gamma\}$
- (iii) $\{0, 2, 6, 4\} = \{4, 6, 0, 2\}$
- (iv) $\phi = \{0\}$
- (v) If $A = \{1, 3, 5\}$
 $B = \{7, 9, 11\}$
 $C = \{1, 5, 9, 13, \dots\}$

Then $\Omega \{x : x = \text{all positive integers}\}$

- (i) $\{x/x^2 - 5x + 6 = 0\} = \{2, 3\}$

5 The following sets have been defined by intension which of these can (practically) be defined by extension?

- (i) Set of all Presidents of India since independence
- (ii) The set of all citizens of India alive on 15th August 1991
- (iii) The set of all blue-eyed persons in the world
- (iv) The set of all stars of a particular film
- (v) The set of all M.L.A.s who are 10 feet tall
- (vi) The set of integers between 10 and 50
- (vii) The set of all integers between the smallest 3 digit number and the largest 6 digit number

6 Discuss whether the following sets are equal or equivalent

$$(i) \quad A = \{x/x^2 - 3x + 2 = 0\} \quad B = \{1, 2, 1\} \quad C = \{2, 2, 1\}$$

$$(ii) \quad A = \{4, 5, 7\} \quad B = \{6, 3, 7\}$$

$$(iii) \quad A = \{x/x \text{ is a letter of the word 'follow'}\}$$

$$B = \{x/x \text{ is a letter of the word 'flow'}\}$$

7 Show that the following are null sets

$$(i) \quad A = \{x/x^2 + 9 = 0, x \text{ is real}\}$$

$$(ii) \quad A = \{x/x^2 = 9, x \text{ is odd}\}$$

$$(iii) \quad A = \{\text{Indians aged 200 years}\}$$

1.8 Set Operations

Q

Venn diagram Sets and set operations can be represented by drawing diagrams termed as Venn Diagrams. The universal set Ω is represented by points within a rectangle and set A, B, C etc. are shown by points with circles or ovals drawn inside the rectangle.

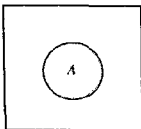


Fig. 1

Fig. 1 represents a set A (consisting of points within the circle) drawn from the universal set Ω .

Subsets A subset of a set A is a set B such that it consists only of some or all of the elements of A —every element of B is also an element of A . This implies that if $A \subset B$ (i.e. element a is contained in the

set B) then $a \in A$. We put it symbolically as $B \subset A$.

It should be noted that we have not said anything about the elements of set A in relation to the set B .

Ex. 18 Let the universal set be the set of all girl students of the college. Let the set A be the set consisting of 5 girls—Roma, Neeta, Nutan, Nalini and Isha. Let B be the set of blue-eyed girls—Roma and Isha.

Thus $\Omega = \{x \mid x \text{ girl students of this college}\}$
 $A = \{\text{Roma, Neeta, Nilam, Nalini, Lila}\}$
 $B = \{\text{Roma, Lila}\}$

Then every element of B (Roma, or Lila) is a member of the set A i.e.,

Roma $\in B$ and Roma $\in A$

Lila $\in B$ and Lila $\in A$

Then B is a subset of A , denoted by $B \in A$

See Fig 2

Proper Subset In fact B is a particular kind of subset of A in the above example. Every element of B belongs to A but every element of A does not belong to B .

Roma and Lila of set B belong to the set A but Neeta, Nilam and Nalini of set A do not belong to B .

thus if we know that every element of B belongs to A but every element of A is not an element of B , then B is said to be a proper subset of A written as $B \subset A$ or $A \supset B$, i.e., $A \neq B$.

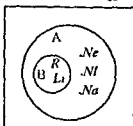


Fig 2

Clearly, no null set ϕ is a proper subset of all the non null sets. If there is no red-eyed girl, then the set of red-eyed girls (C), which is a null set is a subset of A as well as B i.e., $C \subset B$ and $C \subset A$, or $\phi \subset A$ and $\phi \subset B$. But what does the statement $A \subseteq \phi$ mean? Since ϕ contains no element A can be a subset of ϕ only if A itself is an empty set. Hence $A \subseteq \phi$ only if $A = \phi$. Of course, A cannot be a proper subset of ϕ .

Thus a subset B cannot have number of elements more than that in the parent set (called super set) A . Hence if the set A is a finite set B will also be a finite set. However, if the set A is an infinite set, the subset B can be a finite or infinite set.

Ex 19 Let $A = \{x \mid x \text{ is a positive integer}\}$
 $B = \{x \mid x \text{ is an even number}\}$
 $C = \{x \mid 1 \leq x \leq 10 \text{ and } x \text{ is an integer}\}$

A is an infinite set B is also an infinite set. Every element of B is contained in the set A , but every element of A is not contained in B . Hence B is proper subset of A , i.e.,

$$B \subset A$$

Similarly C is a finite set of 10 elements. It is also a proper subset of A , i.e.

$$C \subset A$$

But it is to be noted that elements of C , viz. 1, 3, 5, 7, 9 (all odd numbers) are not contained in the set B [which contains only even numbers 2, 4, 6, 8, 10, 12, ...]. Hence C is not a subset of B i.e.,

$$C \not\subset B$$

Ex 20 The set of females in Rajasthan (X) is a proper subset of set of

females in India (Y) also a proper subset of set of persons in Rajasthan (Z) or in India (W)

$$X \subset Y$$

$$X \subset Z$$

$$X \subset W$$

Clearly all these are finite sets

Equal Sets Again Two sets A and B are equal only if every element of A is contained in B and every element of B is contained in A i.e. A is subset of B and B is subset of A . Hence

$$A \subset B \text{ and } B \subset A$$

or to be more precise

$$\begin{array}{l} A \subseteq B \text{ and } B \subseteq A \text{ then } A = B \\ \text{or if } x \in A \text{ implies } x \in B \quad (A \subseteq B) \end{array}$$

$$\begin{array}{l} \text{that is, } x \in B \Rightarrow x \in A \quad (B \subseteq A) \\ \text{then } A = B \end{array}$$

Ex 21 If A is a set of alphabets in 'class' and B is set of the alphabets in 'lacs', then clearly both A and B have only 4 distinct elements each distinct element of A belongs to B and vice versa, hence

$$A = B$$

Union of sets $A \cup B$ The set C which contains all the elements of the two given sets A and B avoiding duplications, is called the union of the two sets A and B , and is written as

$$\begin{aligned} C &= A \cup B \\ &= A \text{ union } B \end{aligned}$$

Obviously, C is a set such that all its elements belong to either A or B . Hence,

$$C = \{x \mid x \in A \text{ or } x \in B\}$$

Ex 22 If A = set of alphabets in Bombay

B = set of alphabets in Calcutta

Find $A \cup B$

Now $A = \{B, O, M, A, Y\}$

$B = \{C, A, L, U, T\}$

Hence $C = A \cup B$ is formed of all elements in A or B avoiding duplication,

$$C = \{B, O, M, A, Y, C, U, T, L\}$$

The following results should be noted immediately

(i) Union of A with A , $A \cup A = A$

(ii) Union of A and the universal set, $A \cup \Omega = \Omega$ since Ω contains all

elements of A

(iii) Union of a non null set and a null set, is the non null set itself

$$A \cup \phi = A$$

(iv) Union of A with B is same as union of B with A ,

$$A \cup B = B \cup A$$

This is termed as the *Commutative Law of Union*

(v) $A \cup (B \cap C) = (A \cup B) \cap C = A \cup B \cap C$ This is termed as *associative law of union*

(vi) If $B \subset A$ then $A \cup B = A$ The union of a set with its subset is the set itself

Ex 23. If $A = \{1, 2, 3, 4, 5\}$

$$B = \{4, 5, 6, 7\}$$

Then, $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

Here clearly A and B overlap and elements 4, 5 are common See Fig 3 (i)

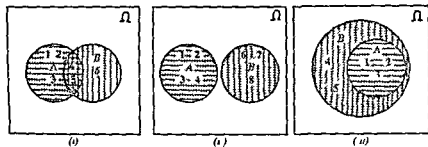


Fig 3

Ex 24 If $A = \{1, 2, 3, 4\}$

$$B = \{6, 7, 8\}$$

Then $A \cup B = \{1, 2, 3, 4, 6, 7, 8\}$

Clearly A and B do not overlap. If two sets have no common elements, they are disjoint sets. See Fig 3 (ii)

Ex 25 If $A = \{1, 2, 3, 4, 5\}$

$$B = \{1, 2, 3\}$$

Then, clearly $B \subset A$, hence $A \cup B = \{1, 2, 3, 4, 5\}$ See Fig 3 (iii)

Intersection of sets $A \cap B$ The set C which contains all the elements which are present in both the sets A and B (contains all the common elements of A and B) is called the intersection of the two sets A and B and is written as

$$C = A \cap B$$

and is read as A intersection B

$A \cap B$ implies that, each element of C is contained in A

$$[C = \{x | x \in A\}]$$

and also in B [$C = \{x | x \in B\}]$ i.e.

$$C = A \cap B = \{x | x \in A \wedge x \in B\}$$

$\wedge$ stands for 'and' or as well as

Clearly $\subseteq$ is a subset of A 'as well as' B

Ex. 26. If $A = \{1, 2, 3, 4, 5\}$

$$B = \{4, 5, 6\}$$

Then $A \cap B = \{4, 5\}$, elements 4, 5 are common to both the sets. See Fig 4 (i). Clearly $C = \{4, 5\}$ is a subset of both A and B

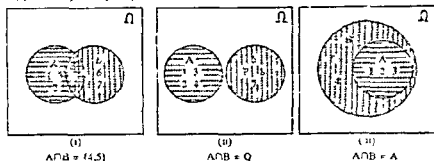


Fig 4

Ex. 27. If $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$

then there is no element common to both sets, hence $A \cap B$ is a null set ϕ . (Fig 4 (ii))

$$A \cap B = \phi$$

Ex. 28. If $A = \{1, 2, 3\}$

$$B = \{1, 2, 3, 4, 5\}$$

Clearly A is a proper subset of B . In such a case the intersection of A and B will consist of elements of the set A [(Fig 4 (iii))], hence

$$A \cap B = \{1, 2, 3\} = A$$

Ex. 29. If $A = \{1, 2, 3, 4\}$

$$B = \phi$$

Then there is no common element, hence $A \cap B = \phi$

Ex. 30. $A \cap \Omega = A$ since A is subset of the universal set Ω

Disjoint sets Disjoint sets do not have any common element

Hence

$$A \cap B = \phi$$

Complement of a Set If Ω is the universal set for the set A then the set A containing all the elements of Ω except those of A is called the complement of set A with reference to Ω . Thus each element of A is contained in Ω but not contained in A i.e.



Fig 5

$$A^c = \{x/x \in \Omega \text{ and } x \notin A\}$$

It should be noted that if the universal set changes then the complement of A will also change

Ex 31 If $A = \{\text{All females in Rajasthan}\}$
and $B = \{\text{All persons in Rajasthan}\}$
then $A^c = \{\text{All non females in Rajasthan}\}$

$$= \{\text{All persons in Rajasthan but not females}\}$$

$$x \in \Omega \quad \text{and} \quad x \notin A$$

But if $\Omega = \{\text{All females in India}\}$

then $A^c = \{\text{All females in India } (x \in \Omega) \text{ except those in Rajasthan } x \notin A\}$
 $= \{\text{females in India in States other than Rajasthan}\}$

Ex 32 If $A = \{1, 2, 3\}$
 $\Omega = \text{All digits} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
 $A^c = \{0, 4, 5, 6, 7, 8, 9\}$
 $= \{0, 4, 5, 6, 7, 8, 9\}$

Clearly A^c contains elements of Ω except those in A , hence A^c is complement of A

Results to be noted

- (i) If A is complement of A then A is complement of A i.e.
 $(A^c)^c = A$
- (ii) Complement of a universal set is a null set and *vice versa*.
 $\Omega^c = \phi$
 $\phi^c = \Omega$
- (iii) The union of A with A^c is the universal set
 $A \cup A^c = \Omega$
- (iv) The intersection of A and A^c is the null set
 $A \cap A^c = \phi$

De Morgan's Rule Suppose we are to find the complements of the Union and Intersection of two sets A and B , i.e. we have to find $(A \cup B)^c$ and $(A \cap B)^c$

Now $A \cup B$ is the set containing all the elements in A or B without duplication in Fig 6 it is the shaded area. Clearly $(A \cup B)^c$ is the non shaded area of Ω

Form the set A^c by deleting the elements of A from Ω which leaves all if B

area outside A . Form the set B' , by taking the area outside B . Clearly the non-shaded area is intersection of the area excluding A and area excluding B i.e., $A' \cap B'$

Hence $(A \cup B)' = A' \cap B'$

Let us make it more clear by taking a concrete example

Ex 33 Let $A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6, 7\}$

and $\Omega = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Then $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

and $(A \cup B)' = \{8, 9, 0\}$

$= \{\text{elements of } \Omega \text{ not in } A \cup B\}$ (1)

Now $A' = \{\text{elements of } \Omega \text{ not in } A\}$

$= \{0, 6, 7, 8, 9\}$

$B' = \{0, 1, 2, 3, 8, 9\}$

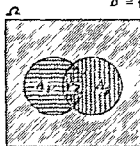


Fig 7

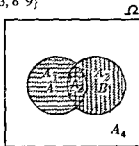


Fig 8

Intersection of A' and B' is $A' \cap B'$ contains all elements common to A' and B' ,

$A' \cap B' = \{0, 8, 9\}$ (2)

Hence from (1) and (2) we have

$(A \cap B)' = A' \cup B'$
 $= \text{Area } A_4$

Similarly, $(A \cap B) = \{4, 5\}$

$(A \cap B)' = \{0, 1, 2, 3, 6, 7, 8, 9\}$ (3)

Again, $A' = \{0, 6, 7, 8, 9\}$

$B' = \{0, 1, 2, 3, 8, 9\}$

Hence, $A' \text{ union } B' = A' \cup B'$

$$\begin{aligned}
 &= \{0, 6, 7, 8, 9, 1, 2, 3\} \\
 &= \{0, 1, 2, 3, 6, 7, 8, 9\}
 \end{aligned} \tag{4}$$

Hence, from (3) and (4) we have

$$\begin{aligned}
 (A \cap B)' &= A' \cup B = \text{complement of area } A_2 \\
 &= \text{area } A_1 + A_3 + A_4
 \end{aligned}$$

These are termed as De Morgan's rules

Difference of Sets The difference of two sets A and B is the set of all elements belonging to A but not to B , i.e.,

$$A - B = \{x/x \in A \cap x \notin B\}$$

If A is the set of all Congressmen, and B is the set of Congressmen who wear khadi, $A - B$ is the set of all Congressmen who do not wear khadi. Every person in this set belongs to the set A (of Congressmen) but not to set B (of khadi wearing Congressmen)

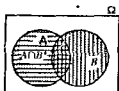


Fig 9

It is possible to define the difference of two sets with the help of the operation of complementation. Now those x such that $x \notin B$ are simply the elements of B' — the complementation being with respect to the universal set Ω of all persons in India. Hence using the definition of set intersection we can rephrase our definition of difference of two sets as follows

$$A - B = A \cap B$$

Notice that if $A \subset B$, then $A \cap B'$ will be ϕ . The set of non-khadi-wearing Congressmen will be empty

Ex. 33 If $A = \{2, 4, 6\}$ and $B = \{1, 3, 7\}$ find $A - B$

Let us define the universal set, this is the set of digits. Hence

$$B' = \{0, 2, 4, 5, 6, 8, 9\}$$

$$\begin{aligned}
 \text{Hence } A - B &= A \cap B' = \{2, 4, 6\} \cap \{0, 2, 4, 5, 6, 8, 9\} \\
 &= \{2, 4, 6\} = A
 \end{aligned}$$

When the sets are disjoint sets, $A - B = A$

Ex. 34 If $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 3, 4\}$
 then $A - B = A \cap B' = \{1, 2, 3, 4, 5\} \cap \{0, 1, 5, 6, 7, 8, 9\}$
 $= \{1, 5\}$,

Clearly $B - A = B \cap A'$
 $= \{2, 3, 4\} \cap \{0, 6, 7, 8, 9\} = \phi$

Hence if $B \subset A$, then $B - A = \phi$, $A - B$ is non empty

Ex. 35 If $A = \{1, 2, 3, 4, 5\}$
 and $B = \{4, 5, 6, 7\}$

$$\begin{aligned}
 \text{then } A-B &= A \cap B \\
 &= \{1, 2, 3, 4, 5\} \cap \{0, 1, 2, 3, 8, 9\} \\
 &= \{1, 2, 3\} \\
 \text{Also } B-A &= B \cap A \\
 &= \{4, 5, 6\} \cap \{0, 6, 7, 8, 9\} \\
 &= \{6, 7\}
 \end{aligned}$$

Clearly if $A \neq B$ and A and B are overlapping both $A-B$ and $B-A$ are non-empty sets

Ex 36 $x = \{1, 2, 3, 4, 5\}$ are non empty subsets of X such that $A \cup B = x$, $A \cap B = \phi$ and $B = \{1\}$ find A

$$\begin{aligned}
 \text{Since } B &= \{1\} \text{ and } A \cup B = x \\
 A \cap B &= \phi \text{ Hence } A \text{ must be } \{2, 3, 4, 5\}
 \end{aligned}$$

Ex 37 A university department restricts the admission to its Ph.D course to those having (i) an M.A. degree in Economics or Politics or Sociology and (ii) two or more years' research experience. Express the set of candidates eligible for admission to the Course in terms of the sets implied in requirement (i) and (ii)

Define the following sets

$$(A) = \{\text{M.A. is Econ}\}$$

$$(B) = \{\text{M.A. is Pol}\}$$

$$(C) = \{\text{M.A. is Sociology}\}$$

$(D) = \{\text{Students with two or more years' research experience}\}$ then the set of eligible candidates is

$$(A \cup B \cup C) \cap D$$

1.9 Something more about Properties of Sets

(i) Reflexive property A set is a subset of itself $A \subseteq A$

(ii) Anti-symmetric Property If $A \subseteq B$ and $B \subseteq A$ then $A = B$

(iii) Transitive Property If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$

A relation which is reflexive, anti-symmetric and transitive is called a partial order relation. Thus $\subseteq$ is a partial order relation defined on sets

Power set The set of all subsets of a set A is called the power set of A and is denoted by $P(A)$

Ex 38 If $A = \{a, b, c\}$

Then all the subsets can be enumerated as follows, ϕ , $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{b, c\}$, $\{a, c\}$, $\{a, b, c\}$

Hence the power set is the set of all these 8 subsets, i.e.,

$$P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$$

Index set From the definition of union set and intersection set we have

$$A_1 \cup A_2 \cup A_3 \quad \cap A_n = \bigcup_{i=1}^n A_i = \bigcup_{i \in I} A_i$$

Note that in ordinary algebra $\sum_{i=1}^n$ is used instead of $\bigcup_{i=1}^n$

And

$$A_1 \cup A_2 \cup A_3 \quad \cup A_n = \bigcup_{i=1}^n A_i = \bigcup_{i \in I} A_i$$

Note that in ordinary algebra $\prod_{i=1}^n$ is used instead of $\bigcap_{i=1}^n$

The set $i \in I$ means that i is an element of set I of positive integers which is termed as I

EXERCISE 1.2

1 Of the sets given below which one are included in others

- (i) A = set of all Prime Ministers of India
- (ii) B = set of all MP's in India
- (iii) C = set of all Adults in India
- (iv) D = set of all women Prime Ministers of India
- (v) E = set of all female adults in India
- (vi) F = set of all female MP's in India

2 Can there be more than one empty set? Why?

3 Which of the following statements is correct?

- (i) set $\{1, 2\}$ is a subset of ϕ
- (ii) $\phi \subset A$ means $A = \phi$
- (iii) Every subset of a finite set is finite
- (iv) Every subset of an infinite set is infinite
- (v) $A \subset \phi$ means $A = \phi$

4 If A = the set of all mothers

B = the set of all fathers

C = the set of all widows

D = the set of all widowers

then describe in words the following sets

- (i) $A \cup B$
- (ii) $A \cap C$
- (iii) $B \cap D$
- (iv) $A \cap B$
- (v) $A \cap D$

5 Draw Venn Diagrams to show the results of exercise 4

6. Interpret the following in ex 4 above

- (i) $A \cap B$ (ii) $A' \cap B'$ (iii) $C \subset A$ (iv) $A \subset C$

7. Show that $B - A = B \cap A'$

8. Show that $(A - B) \cap B = \phi$ [Hint $(A - B) \cap B = A \cap B' \cap B$]

9. Show that $(A - B) = (B - A)$ only if $A = B$

10. Find in solved example 29

$A \cap B$, $A \cup B$, $A' \cup B'$ and $A' \cap B'$

Also prepare Venn diagrams to show the results

11. Find $A \cap B$ and $A \cup B$, when (show by Venn diagrams)

(i) A and B are disjoint

(ii) A and B are overlapping

(iii) $B \subset A$ (iv) $A \subset B$

12. prove the De Morgan's rules for the following sets

A

B

(i) $\{a, b, c, d, e\}$

$\{a, c, e, f, g\}$

(ii) $\{x/x \text{ is odd digits}\}$

$\{x/x \text{ is digit}\}$

(iii) $\{x/x \text{ is } 0 \leq x \leq 5\}$

$\{x/x \text{ is } 3 \leq x \leq 7\}$

13. If $A = \{1, 2, 3, 4, 5\}$

$B = \{3, 4, 5, 6, 7\}$

$C = \{0, 1, 8, 9\}$

Find (i) $A \cup B$

(ii) $A \cap B$

(iii) $A' \cap B$

(iv) $B' \cap B$

(v) $A \cup B \cup C$

(vi) $A \cup (B \cap C)$

(vii) $A - B$

(viii) $(A \cup B) - C$

(ix) $B \cap C$

(x) $(A \cup B) - C'$

14. If $A = \{a, b, c\}$, $B = \{1, 2, c\}$ and $C = \{c, f\}$

Find $A - B$, $B - C$ and $A - C$ and show that $(A \cup B) = A' \cap B'$

15. if $A = \{0, 1, 2\}$, define the power set

16. Show that the roots of the equation $x^2 + 2x - 8 = 0$ form a set which is a proper subset of the set of integers

17. Let $A = [\{a, b\}, \{b, c\}, d, e]$

which of the following statements is correct

(i) $\{a, b\} \subset A$

(ii) $\{a, b\} \in A$

(iii) $\{\{b, c\}\} \in A$

(iv) $C \in A$

(v) $C \subset A$

(vi) $\{\{a, b\}\} \subset A$

18. If $A - B$ denotes $A \cap B'$ for $B \subseteq A$, show that $A - A = A'$, $A - A = \phi$ and

$A - 0 = A$, show that $(A - B) - C = A - (B + C)$ can be written if B and C are disjoint and such that BCA and CCA . Draw Venn diagrams

19. How will you show that A is not a subset of B . . .

20. Let $A = \{a, b, c\}$, $B = \{a, b\}$ and $C = \{a, b, d\}$ which of the following statements are correct.

(i) $B \subseteq A$ (ii) $B \not\subseteq C$ (iii) $a \in A$

(iv) $a \subseteq A$, (v) $\{a\} \in A$

21. Show that

(i) $A \cup A' = \Omega$ (ii) $A \cap \Omega = A$ (iii) $(A \cup B)' = A' \cap B'$

(iv) $A \cap (B - A) = \phi$ (v) $A \cup B = (A - B) \cup B$

22. If $A = \{x \mid x^2 - 6x + 8 = 0\}$ and $B = \{x^2 - 6x + 4 = 0\}$

write $A \cup B$, $A \cap B$, $A - B$

23. Prove that $A \cap (A \cap B) = A \cup (A \cap B) = A$

24. Prove that $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

(Hint $A - B = A \cap B'$ and $B - A = B \cap A'$)

1.10. Three and more Sets : Venn Diagram

Ex. 39. Fig 10 gives a Venn diagram for three sets A, B, C

Ω

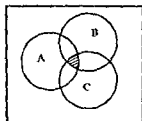


Fig 10

$A \cap B \cap C$



Fig 11

$A \cup B \cup C$

Let $A = \{1, 2, 3, 4\}$

$B = \{2, 4, 5, 6\}$

$C = \{0, 3, 4, 7, 8\}$

Then,

$A \cup B = \{1, 2, 3, 4, 5, 6\}$

$A \cap B = \{2, 4\}$

$A \cup B = A + B - (A \cap B)$

Hence

(i) Now

$(A \cup B) \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 0\}$

$A \cup (B \cup C) = \{1, 2, 3, 4\} \cup \{2, 4, 5, 6, 7, 8, 0\}$
 $= \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

$A \cup (B \cap C) = (A \cup B) \cap C$ states Associative Law for set union (Fig 11)

$$(ii) \quad (A \cap B) \cap C = \{2, 4\} \cap \{0, 3, 4, 7, 8\} \\ = \{4\}$$

also $A \cap (B \cap C) = \{1, 2, 3, 4\} \cap \{4\} \\ = \{4\}$

$A \cap (B \cap C) = (A \cap B) \cap C$ states Associative Law of set intersection (Fig 10)

$$(iii) \quad A \cup (B \cap C) = \{1, 2, 3, 4\} \cap \{4\} \\ = \{1, 2, 3, 4\}$$

also $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4, 5, 6\} \cap \{1, 2, 3, 4, 0, 7, 8\} \\ = \{1, 2, 3, 4\}$

Hence $A \cap (B \cap C) = (A \cup B) \cap (A \cup C)$ states Distributive Law of set union with respect to set intersection (Fig 12)

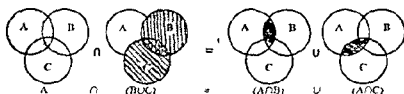


Fig 12

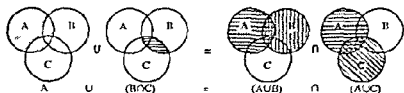


Fig 11

$$(iv) \quad A \cap (B \cup C) = \{1, 2, 3, 4\} \cap \{2, 4, 5, 6, 0, 3, 7, 8\} \\ = \{2, 3, 4\}$$

also $(A \cap B) \cup (A \cap C) = \{2, 4\} \cup \{3, 4\} \\ = \{2, 3, 4\}$

Hence, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Distributive Law for set intersection with respect to set union (Fig 13)

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

$$(A \cup B \cup C) = \{9\}$$

$$A = \{0, 5, 6, 7, 8, 9\}$$

$$B = \{0, 1, 3, 7, 8, 9\}$$

$$C' = \{1, 2, 5, 6, 9\}$$

$$A \cap B \cap C = \{9\}$$

Hence, $(A \cap B \cap C) = A' \cap B \cap C'$

Similarly,

$$(A \cap B \cap C)' = A' \cup B' \cup C'$$

These are De Morgan's Laws

1.11. Summary of Operational Rules in Set Algebra

TABLE 1 Operational Rules for Sets

	Rule	Union ($\cup$)	Intersection ($\cap$)
1	Closure	$A \cup B$ is a set	$A \cap B$ is a set
2	Associative	$A \cup (B \cup C) = (A \cup B) \cup C$	$A \cap (B \cap C) = (A \cap B) \cap C$
3	Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
4	Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
5	Idempotent	$A \cup A = A$	$A \cap A = A$
6	Complements	$A \cup A' = \Omega$	$A \cap A' = \phi$
7	De Morgan's Law	$(A \cup B)' = A' \cap B'$	$(A \cap B)' = A' \cup B'$
8	Involutions	$(A')' = A$	$(A')' = A$
9	Bounds	$A \cap \phi = A$ $A \cap \Omega = A$	$A \cap \phi = \phi$ $A \cap \Omega = A$

1.12. Boolean Algebra

The Algebra of sets is not the same as the algebra of ordinary numbers. It is an example of what is known as *Boolean Algebra* after Boole (1815-64). If the algebraic operations with certain entities satisfy the set of rules set out in Table 1 then the algebra is Boolean. This is, however, not the only Boolean Algebra which can be devised.

We have seen that Boolean Algebra has many merits. But a serious deficiency in Boolean algebra is that unlike those of ordinary algebra it does not define reciprocals (and hence division) and cancellation is not valid. For example,

If $A = \phi$ or $B = \phi$
then, $A \cap B = \phi$ {similar to $A + B = 0$ }

But the converse is not true i.e., if $A \cap B = \phi$ then it does not imply that either $A = \phi$ or $B = \phi$. In fact none may be empty, they may be *disjoint* sets.

Again in ordinary algebra, if $A \times B = 1$ then B is inverse of A . Nothing like this can be defined in set algebra.

Some Illustrative Examples

Ex. 38. Simplify

$$(i) \quad (A \cap B) \cup (B' \cap A)$$

$$(ii) \quad (A \cup B') \cap (A' \cup C) \cap (B \cup C')$$

Now, we know that

$$A \cup B \cap C = (A \cup B) \cap (A \cup C)$$

$$\text{and } A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\text{Hence, } (A \cap B) \cup (A \cap C) = A \cap (B \cup C)$$

$$\text{On substituting } C = B' = (A \cap B) \cup (A \cap C)$$

$$(A \cap B) \cup (B' \cap A) = (A \cap B) \cup (A \cap B')$$

$$= A \cap (B \cup B')$$

$$= A \cap \Omega$$

$$= A$$

From Venn Diagram, we have,

$$\text{Let } A = \{1, 2, 3, 4, 5\}$$

$$B = \{4, 5, 6, 7, 8\}$$

$$\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$A' = \{6, 7, 8, 9\}$$

$$B' = \{1, 2, 3, 9\}$$

$$\text{Now } A \cap B = \text{area } A_2 = \{4, 5\}$$

$$B' \cap A = \text{area } A_1 = \{1, 2, 3\}$$

The union of (A_2) and (A_1) = set A

$$(A \cap B) \cup (B' \cap A) = \{4, 5, 1, 2, 3\} = A$$

$$(ii) \quad (A \cup B') \cap (A' \cup C) \cap (B \cup C')$$

From Venn diagram we have

$$A \cap B' = \text{Area } A_1 + A_2 + A_3 + A_4 \text{ and } A_8 = \{1, 2, 3, 4, 5, 9\}$$

$$\text{Let } C = \{2, 3, 8, 4\}$$

$$\therefore C' = \{1, 9, 5, 6, 7\}$$

$$\text{Then } (A' \cup C) = \{6, 7, 8, 9, 2, 3, 4\}$$

$$= \text{Area } A_3 + A_4 + A_5 = A_6 + A_7 + A_8$$

$$\text{Also } B \cup C = \{4, 5, 6, 7, 8, 1, 9\}$$

$$= \text{Area } A_2 + A_3 + A_5 + A_6 + A_7 + A_8$$

$$\text{Hence } (A \cup B') \cap (A' \cup C) \cap (B \cup C')$$

$$= \{1, 2, 3, 4, 5, 9\}$$

$$\cap \{6, 7, 8, 9, 2, 3, 4\}$$

$$\cap \{4, 5, 6, 7, 8, 1, 9\}$$

$$= \{4, 9\}$$

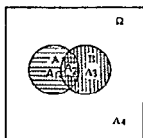


Fig 14

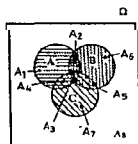


Fig 15

From Venn Diagram

$$\begin{aligned} & (A \cup B) \cap (A' \cup C) \cup (B \cup C') \\ &= \text{Area } (A_1 + A_2 + A_3 + A_4 + A_7 + A_8) \cap (A_3 + A_4 + A_5 + A_7 + A_6 + A_8) \\ & \quad \cap (A_2 + A_3 + A_4 + A_6 + A_1 + A_5) \\ &= \text{Area } A_3 \text{ and } A_8 \end{aligned}$$

That is union of area A_3 and area A_8 .

Now Area A_3 is $A \cap B \cap C$

Area A_8 is complement of $A \cup B \cup C$

$$\begin{aligned} \text{Hence } & (A \cup B) \cap (A' \cup C) \cap (B \cup C') \\ &= (A \cap B \cap C) \cup (A \cup B \cup C)' \\ &= (A \cap B \cap C) \cup (A' \cap B' \cap C') \end{aligned}$$

In many operations we can use the sign $+$ for $\cup$ and $\times$ for $\cap$. Keeping in mind that $A \cup A$ (or $A + A$) is not $2A$ but A in set algebra. Hence in this question

$$\begin{aligned} & (A \cup B) \cap (A' \cup C) \cap (B \cup C') \\ &= (A + B) \times (A' + C) \times (B + C') \\ &= (A \times A' + B' \times A' + A \times C + B' \times C) (B + C') \\ &= A \times A' \times B + B' \times A' \times B + A \times C \times B + B' \times C \times B \\ & \quad + A \times A' \times C' + B' \times A' \times C' + A \times C \times C' + B' \times C \times C' \\ &= A \times A' \times B + B' \times B \times A' + A \times B \times C + B \times B \times C \\ & \quad + A \times A' \times C' + A' \times B \times C' + A \times C \times C' + B' \times C \times C' \\ & \quad (\text{Remembering that } A \cap B = B \cap A) \\ &= \phi + \phi + A \times B \times C + \phi + \phi + A' \times B' \times C' + \phi + \phi \\ & \quad (\text{Remembering that } A \cap A' = \phi) \end{aligned}$$

Reverting back to the notation of set algebra

$$\begin{aligned} &= \phi \cup A \cap B \cap C \cup A' \cap B' \cap C' \\ &= A \cap B \cap C \cup A' \cap B' \cap C' \end{aligned}$$

1.13. Number of elements in a Set

Since we can count a finite set, we can proceed to operate with various counts of finite sets A , B , C

Write $n(A)$, $n(B)$ as the number of elements in the sets A and B respectively

From the Venn diagram we can define $n(A+B)$ —number of elements in the union of A and B

Clearly, $A_1 = A \cap B'$

$$A_2 = A \cap B$$

$$A_3 = B \cap A'$$

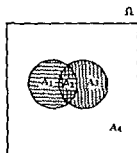


Fig 16

$$A_4 = A \cap B \text{ or } (A \cup B)$$

Hence

$$\begin{aligned} n(A+B) &= \text{Area } (A_1 + A_2 + A_3) \\ &= n(A \cap B) + n(A \cap B) + n(B \cap A) \\ &= n(A \cap B) + n(A \cap B) + n(B \cap A) \\ &\quad + n(A \cap B) - n(A \cap B) \end{aligned}$$

$$n(A+B) = n(A) + n(B) - n(A \cap B)$$

If A and B are disjoint sets then $n(A \cap B) = 0$

$$\text{Hence } n(A+B) = n(A) + n(B)$$

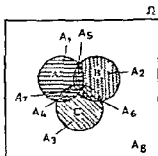


Fig 17

Similarly 3 sets A, B, C will form eight distinct and exhaustive subsets of the totality of elements as shown in the Fig (17)

$$n(A+B+C) = \text{Area } A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8$$

Set $A_7 = ABC$ (or $A \cap B \cap C$)

$A_6 = BCA$ (or $B \cap C \cap A$)

$A_5 = ACB$

$A_4 = A \cap B$

$A_3 = CA \cap B$

$A_2 = BA \cap C$

$A_1 = AB \cap C$

$A_8 = A \cap B \cap C$

$$\text{Now } n(ABC) = n(AB) - n(ABC)$$

$$n(BC \cap A) = n(BC) - n(ABC)$$

$$n(AB \cap C) = n(AB) - n(AB) - n(AC) + n(ABC)$$

$$n(B \cap C) = n(B) - n(AB) - n(BC) + n(ABC)$$

$$n(C \cap A \cap B) = n(C) - n(CA) - n(BC) + n(ABC)$$

Hence

$$\begin{aligned} n(A+B+C) &= n(AB) - n(ABC) + n(BC) - n(ABC) \\ &\quad + n(AC) - n(ABC) + n(A) - n(AB) - n(AC) \\ &\quad + n(ABC) + n(B) - n(BC) - n(AB) + n(ABC) \\ &\quad + n(C) - n(AC) - n(BC) + n(ABC) + n(ABC) \\ &= n(A) + n(B) + n(C) - n(AB) - n(BC) \\ &\quad - n(AC) + n(ABC) \end{aligned}$$

Note It is required that each of the areas A_1, A_2, A_3 should be non-negative i.e., the given numbers are consistent

Ex 39 A research team interviews 100 people and finds that

I	Rich	28
P	Poor	33

W	Willing to work	14
IP	Idle and Poor	12
IW	Idle and willing	9
PW	Poor and willing	6
IPW	Idle, poor and willing	5

(Rajkrishna, Economic and Political Weekly, March 3, 1973)

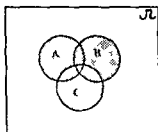


Fig. 18

Are the data consistent ?

Also find the persons who are either idle or poor or willing to work

To test consistency we find the value of all the disjoint sets

$$IPW = 5$$

$$\text{Set } A_4 = n(IP) - n(IPW) = 12 - 5 = 7$$

$$\text{Set } A_5 = n(IW) - n(IPW) = 9 - 5 = 4$$

$$\text{Set } A_6 = n(PW) - n(IPW) = 6 - 5 = 1$$

$$\begin{aligned}\text{Set } A_1 &= n(I) - n(IP) - n(IW) + n(IPW) \\ &= 28 - 12 - 9 + 5 = 12\end{aligned}$$

or more directly,

$$\begin{aligned}&= n(I) - [\text{Set } A_4, A_5, A_6] \\ &= 28 - 7 - 4 - 5 = 12\end{aligned}$$

$$\text{Set } 2 = 33 - 7 - 5 - 1 = 20$$

$$\text{Set } 3 = 14 - 4 - 5 - 1 = 4$$

$$\begin{aligned}\text{Hence } n(I + P + W) &= n(I) + n(W) + n(P) - n(IP) - n(PW) - n(IW) + n(IPW) \\ &= 28 + 33 + 14 - 12 - 9 - 6 + 5 = 53\end{aligned}$$

$$\text{Hence } n(I + P + W)' = 100 - 53 = 47$$

Hence the data are consistent

Ex 40 Sets A and B are such that A has 25 members B has 20 members and $A \cup B$ has 35 members Draw a Venn diagram to represent the above situation and hence find the number of members in the set $A \cap B$

Now we know that

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$35 = 25 + 20 - n(A \cap B)$$

$$n(A \cap B) = 10$$

Ex 41 A is the set of persons in a factory whose monthly incomes are between 100 and 300 B is the set of persons in the factory whose monthly incomes are between Rs 200 and Rs 500 C is the set of persons in the factory whose monthly incomes are between Rs 400 and Rs 600 Find

- | | | | |
|-------|-----------------------|------|---------------------|
| (i) | $A \cap B$ | (ii) | $A \cup B \cup C$, |
| (iii) | $(A \cup B) \cap C$, | (iv) | $A \cap B \cap C$ |

Clearly according to the given information a person cannot be in all the three sets neither can a person be in the 1st and 3rd set simultaneously, hence $A \cap B \cap C$ and $A \cap C$ are null sets

$A \cap B$ = set of persons with income in the brackets 100 to 300 and 200 to 500 i.e. with income between Rs. 200 and 300

$A \cup B \cup C$ = Set of persons with income Rs. 100 to 600—all persons

$(A \cup B) \cap C$ = Set of persons with income either to the 1st and 2nd brackets and also in the third brackets i.e. persons with income between Rs. 400 to 500 which is equal to $B \cap C$

Ex 42 In a class of 25 students of Economics and Politics 12 students have taken Economics. Out of these 8 have taken Economics but not Politics. Find the number of students who have taken Economics and Politics and those who have taken Politics but not Economics.

Total number of students = 25

A = No. of students taking Economics = 12

B = No. of students taking Politics

$A \cap B = 8$

To find $A \cap B$ and $B \cap A$ we have

$$A = (A \cap B) \cup (A \cap B^c)$$

$$n(A) = n(A \cap B) + n(A \cap B^c)$$

$$12 = 8 + n(A \cap B^c)$$

$$n(A \cap B^c) = A \cap B = 4$$

$$\text{Now } n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$25 = 12 + n(B) - 4$$

$$\text{or } n(B) = 17$$

$$\text{again } n(B) = n(A \cap B) + n(A \cap B^c)$$

$$\text{or } 17 = 4 + n(A \cap B)$$

$$\text{or } n(A \cap B) \text{ or } B \cap A = 13$$

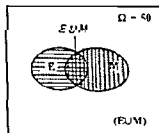


Fig. 19

Ex 43 In a class of 50 students 30 students take Economics, 25 students take Mathematics and 10 take both. Find the number of students taking neither of the two subjects.

Hence number of students taking Economics or Mathematics is

$$L \cup M = E + M - E \cap M$$

Since $E \cap M$ is counted twice, once in E and then in M

$$L \cup M = 30 + 25 - 10 = 45$$

Hence students not taking E or M are (LUM)

which is = nonshaded part of the figure. 19

$$= 5$$

Lx 44 In an examination 53% students passed in mathematics, 61% in Economics and 60% in Statistics. 24% passed in Mathematics as well as Economics, 35% in Maths and Statistics and 27% in Economics and Statistics. 5% passed in none of these subjects. How many students passed in all the 3 subjects? How many students failed in exactly one subject? How many passed in exactly one subject?

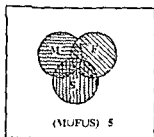


Fig. 20

We have

$$\begin{aligned} M \cup E \cup S &= M + E + S - (M \cap E) - (M \cap S) - (E \cap S) + (M \cap E \cap S) \\ &= 53 + 61 + 60 - 24 - 27 - 35 + (M \cap E \cap S) \end{aligned}$$

Now since number of students failing in all subjects is 5 those who pass either in M or E or S, that is,

$$M \cup E \cup S = 100 - 5 = 95$$

$$95 = 174 - 86 + M \cap E \cap S$$

$$M \cap E \cap S = 7$$

Students passing in exactly one subject are represented by MUEUS minus the intersection areas. Now MUEUS=95 and intersection area is

$$(24-7) + (27-7) + (35-7) = 65$$

Hence students passing exactly in one subject number

$$95 - 65 = 30$$

Now students failing in exactly one subject form the complement set of the students passing in exactly two subjects. Here students passing in exactly 2 subjects number

$$(24-8) + (27-7) + (35-7) = 65$$

Complement of this is $100 - 65 = 35$

Hence 35% students fail exactly in one subject

Ex 45 In a survey of 300 companies the number of companies using different media for advertisements are as follows

$$N=200 \quad R=100 \quad H=80$$

$$N \cap R = 50 \quad R \cap H = 30 \quad N \cap H = 50 \quad N \cap R \cap H = 20$$

Find the number of companies using media other than newspapers (N) radio (R) and handbills (H) for advertisements

Here, $\Omega = 300$

Companies using either N or R or H are $N \cup R \cup H$,

$$\begin{aligned} N \cup R \cup H &= N + R + H - N \cap R - R \cap H - N \cap H + N \cap R \cap H \\ &= 200 + 100 + 80 - 50 - 50 - 50 + 20 \\ &= 270 \end{aligned}$$

Hence number of companies using none of these methods is

$$(N \cup R \cup H)^c = 300 - 270 = 30$$

Ex 46 Prove that

(i) $(A - C) \cap (B - C) = (A \cap B) - C$ and

(ii) $A \cap (B - C) = (A \cap B) - (A \cap C)$

(i) In other words,

$$\begin{aligned} (A - C) \cap (B - C) &\rightarrow \{x \text{ in } A \text{ but not in } C\} \cap \\ &\quad \{x \text{ in } B \text{ but not in } C\} \\ &\rightarrow \{x \text{ in } A \text{ and } B \text{ but not in } C\} \\ &\rightarrow \{x(A \cap B) \text{ but } x \notin C\} \\ &\rightarrow x \notin (A \cap B) - C \end{aligned}$$

$$(A - C) \cap (B - C) \subset (A \cap B) - C \quad (1)$$

Similarly $(A \cap B) - C \rightarrow x \text{ in } A \text{ and } x \text{ in } B \text{ but } x \text{ not in } C$

$\rightarrow x \text{ in } A \text{ but not in } C \text{ and } x \text{ in } B \text{ and not in } C$

$$\rightarrow x \in (A - C) \cap x \in (B - C)$$

$$(A \cap B) - C \subset (A - C) \cap (B - C) \quad (2)$$

Hence from (1) and (2) we have

$$(A - C) \cap (B - C) = (A \cap B) - C$$

(ii) Let $x \in A \cap (B - C) \Rightarrow (x \text{ is } A \text{ and } x \text{ is } B - C)$

$$\Rightarrow x \text{ in } A \text{ and } \{x \text{ is } B \text{ but } x \text{ not in } C\}$$

$$\Rightarrow x \in A \cap B \text{ but } x \notin A \cap C$$

$$A \cap (B - C) \subset (A \cap B) - (A \cap C)$$

Similarly $(A \cap B) - (A \cap C) \subset A \cap (B - C)$

Hence proved

Ex 47 In a survey conducted of 1000 teachers in Udaipur, it was found that 48% preferred coffee (C) 54% tea (T) and 64% were smokers (S). 28% took coffee as well as tea, 30% took tea and also smoked, and 30% were smokers and coffee users. 6% did none of these.

Find the number (i) using tea, coffee and also smoking

(ii) took only coffee and

(iii) smokers taking tea but not coffee

Information given is

$$n(C) = 48 \quad n(T) = 54 \quad n(S) = 64$$

$$n(C \cap T) = 28, \quad n(T \cap S) = 32, \text{ and}$$

$$n(S \cap C) = 30$$

Find (i) $n(C \cap T \cap S)$ (ii) $n(C \cap T^c \cap S^c)$ and $n(S \cap T \cap C^c)$

Percentage of teachers who used atleast one product is given by

$$n(C \cup T \cup S) = n(C) + n(T) + n(S) - n(C \cap T) - n(C \cap S) - n(T \cap S) + n(C \cap T \cap S)$$

$$(100 - 6) = 48 + 54 + 64 - 28 - 32 - 30 - n(C \cap T \cap S)$$

$$\therefore n(C \cap T \cap S) = 18\% \text{ or } 180$$

(ii) Now,

$$n(S \cap T) = n(S \cap T \cap C) + n(S \cap T \cap C^c)$$

$$32 = 18 + n(S \cap T \cap C^c)$$

$$\therefore n(S \cap T \cap C^c) = 32 - 18 = 14\% \text{ or } 140$$

$$\begin{aligned} \text{(iii)} \quad n(C \cap T^c \cap S^c) &= n(C) - n(C \cap T \cap S) - n(C \cap T \cap S^c) - n(C \cap T \cap S) \\ &= 48 - n(C \cap T^c \cap S) - n(C \cap T \cap S^c) = 18\% \text{ or } 180 \end{aligned}$$

$$\begin{aligned} \text{Again, } n(C \cap T^c \cap S) &= n(C \cap S) - n(C \cap S \cap T) \\ &= 30 - 18 = 12\% \end{aligned}$$

$$\begin{aligned} n(C \cap T^c \cap S^c) &= n(C \cap T^c) - n(C \cap S \cap T^c) \\ &= 28 - 18 = 10\% \end{aligned}$$

Hence,

$$n(C \cap T^c \cap S^c) = 48 - 12 - 10 - 18 = 8\% \text{ or } 80$$

EXERCISE 1.3

1. If $A = \{1, 2\}$, $B = \{2, 3\}$

and $\Omega = \{1, 2, 3, 4\}$

Find $(A \cap B)$ and $(A \cup B)$

If C is also subset of the same universal set, $C = \{1, 4\}$

Find (i) $A \cup B \cup C$

(ii) $A \cap B \cap C$

(iii) $A \cup (B \cap C)$

(iv) $A^c \cup (B \cap C)$

(v) $(A \cup B) \cap (B \cup C)$

(vi) $(A \cap B^c) \cup (B^c \cup C)$

2. Show the above sets by Venn diagrams

3. State whether the following statements are correct or incorrect if the universal set is set of all digits

- (i) $\{1, 2, 3, 6, 7\} \cup [\{4, 5, 6, 7\} \cap \{5, 1, 4, 6\}] = \{7\}$
 (ii) $\{4, 5, 0\} \cap \{1, 2, 3\} \cup \{5, 6, 0, 9\} = \emptyset$
 (iii) $[\{5, 1, 4, 6\}' \cap \{4, 5, 6, 7\}] \cap \{1, 2, 3, 6, 7\} = \{7\}$

4. Show the following by Venn diagrams

- (i) ACB and BCC
 (ii) ACB but $B \not\subset C$
 (iii) $A \cup B \cap C'$
 (iv) $A' \cup B' \cap C$

5. If $X = \{1, 2, 4, 5\}$, $Y = \{4, 5, 6, 7\}$ $Z = \{2, 3, 5, 6\}$

- Find (i) $X \cup Y \cup Z$ (ii) $X \cap Y \cap Z$
 (iii) $X \cup (X \cap Z)$ (iv) $X \cup (Z \cap Y)$
 (v) $(X \cup Z) \cap (X \cup Y)$ (vi) $X \cap (Z \cup Y)$
 (vii) $(X \cap Z) \cup (X \cap Y)$

6. In Ex 6 find (if $\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$)

- (i) $(X \cup Y \cup Z)'$ (ii) $X' \cup Y' \cup Z'$
 (iii) $(X \cup Y)' \cap (Y \cap Z)$ (iv) $(X \cup Y)' \cap (Y' \cup Z)$

7. Prove the operational Rules for sets with the help of ex 5, 6 above.

8. If $A = \{a, b, c\}$, $B = \{1, 2, c\}$ and $C = \{c, f\}$, describe the following

sets

$A \cup B$, $A \cup C$, $A \cap B$, $A \cap C$, $(A \cup B) \cap C$, $A \cap (B \cup C)$ and verify the distributive laws

9. If A , B , C , D denote respectively the set of male passengers, the set of female passengers, the set of male children and the set of female children travelling in a bus, express in terms of set notation (i) set of children, (ii) set of adults (iii) set of male adults and, (iv) set of female adults.

10. A research team interviews 100 people, asking each whether he does any or all of the items, the team returns the following numbers of overlapping categories

Category	Number	Category	Number
ABC = used all three	3	A Used college library	42
AB College and own library	7	B Had their own library	17
BC own library and borrowed	8	C Borrowed book	27
AC college library and borrowed	12		

Find number of students using none of the three categories. Are the data consistent. Find the value of all the disjoint distinct groups which can be formed. Are the data consistent if there is no student using none of the three?

11 If

$A = \{x \mid x \text{ non negative integers including zero}\}$

$B = \{x \mid x \text{ is Positive non zero integer}\}$

$C = \{x \mid x \text{ is Positive even number}\}$

$D = \{x \mid x \text{ is Positive prime number}\}$

$E = \{x \mid x \text{ is Positive integer divisible by 3}\}$

$F = \{x \mid x \text{ is non positive integer}\}$

Then prove that

(a) $B \subset A$

(b) $C \subset B$

(c) $D \subset B$

(d) $B \cap F = \phi$

(e) $C \cap F = \phi$

(f) $(B \cup E) \subset (A \cup E)$

(g) $(B \cup E) - B \cap E$

(h) $[B \cup (C \cap D)] = B \cup [C \cap D]$

12 I prove by means of Venn Diagrams or otherwise

(i) If $D \subset A$, $D \subset B$, $D \subset C$ then
 $D \subset A \cap (B \cap C)$

(ii) $A \cap (B \cap C) = (A \cap B) \cap C$

(iii) $A \cup (B \cap C) \neq (A \cup B) \cap C$

(iv) $[A \cup (B \cap C)] = A \cap [B \cup C]$

(v) $\{(A - C) \cap C\} = (B - C) \cup (A - C)$

13 A market research team interviews 100 people asking each whether he smokes any or all of the items A cigarettes B cigars C pipe tobacco. The team returns the following data

Category	No	Category	No
ABC	3	A	42
AB	7	B	17
BC	13	C	27
AC	18		
		Total	100

Are the returns consistent ?

14 An insurance company classifies a set of 50 lives according as they are men (or women) married (or single) and Indian (or foreign) and gives the groups

Men 18

Married Men ?

Married 20

Indian Men 14

Indian 29

Indian Married persons ?

Find the disjoint or non overlapping distinct groups and show that out of 11 Foreign 'lives' 4 are single women

15 If the number of married 'lives' in Ex. 14 is entered as 25 (not 20) check that the data are then inconsistent

16 In a survey of 100 students it was found that 50 used the college library, 40 had their own library and 40 borrowed books, 20 used both college library and their own, 15 borrowed books and used their own books, whereas 10 borrowed books and used college library. Assuming that all students use either college library books or their own or borrowed books find the number of students using all the three sources.

17 If in Ex. 16 the number of students using no book at all is 10 and the number of students using all the three is 20 show that the information is inconsistent

18 Asked if you will vote for congress the following responses are recorded

	Yes	No	Don't know
Adult Male	10	20	5
Adult Female	20	15	5
Youth just to enter adulthood	10	5	10

Write A = set of adults

C = set of women and children

Y = set of 'Yes' answers

N = set of 'No' answers

Find (i) $n(A)$ (ii) $n(A \cap C)$ (iii) $n(Y \cup N)$ (iv) $n[A \cap (Y \cap N)]$ (v) $n(C \cap Y)$

19 State with reasons or examples whether the following statements are true or false

(i) If $A \subset B$ and $B \subset C$ then $A \subset C$

(ii) If $A \subset C$ then $A \cup B = A$

(iii) If $X = \{a, b, c\}$ then $\{a\} \in X$

20 Give an example of three sets A , B and C since that $A \cap B = \phi$, $B \cap C \neq \phi$, $A \cap C \neq \phi$ but $A \cap B \cap C = \phi$

21 In a class of 50 M.A. (Econ.) students, 12 students have taken Econometrics, 5 have taken econometrics but not mathematical economics. Find the number of students who have taken econometrics and mathematical economics and those who have taken mathematical economics but not econometrics

1.14 Ordered Pairs

We have stated that $\{a, b\} = \{b, a\}$ i.e., in defining a set the order of elements is not material. But the sets can have quite different structures

Let X and Y be two sets and write $x \in X$ and $y \in Y$ as representative elements. Thus X is $\{x_1, x_2, x_3\}$ and Y is $\{y_1, y_2, y_3\}$. We form the set of pairs of x and y i.e., (x, y) such that x from X or $x \in X$ is first and y from $(y \in Y)$ is the second element. All the pairs contain x and y in this particular order, hence this set is called set of ordered pairs (x, y) . The significance of the term ordered pairs is that the order in which we take the elements, the first from X and the second from Y , is important. If the order changes then the meaning attached to the ordered pair will change. Thus $\{x, y\} \neq \{y, x\}$ if $x \in Y$ and X and Y are different sets.

This can be illustrated in graphical terms. Let X and Y be sets of real numbers. Mark off the elements of X on the axis OX and element of Y on axis OY . The ordered pair (x, y) can now be shown by a point or dot in the plane oxy . If the ordered pair (x, y) is $(1, 2)$ then the point can be plotted by taking $x=1$ along OX and $y=2$ along OY . But if we take the ordered pair $(y, x) = (2, 1)$ and plot y on OX -axis and x on OY axis the point will be different.

1.15. Cartesian Product

The set of all distinct possible ordered pairs of the form (x, y) where $x \in X$ and $y \in Y$ is known as the cartesian product or Cartesian set of the two sets X and Y and is symbolically represented as $X \times Y$ or $X \cdot Y$ or X and Y or X cross Y .

Hence Cartesian product of sets X and Y is

$$X \cdot Y = \{(x, y) \mid x \in X, y \in Y\}$$

or $X \cdot Y = \{x, y \mid x \in X, y \in Y\}$

Since the order of writing x and y is essential, the Cartesian products $X \cdot Y \neq Y \cdot X$ is general. The number of pairs will be equal to the product of the number of elements in each set.

For three sets, the Cartesian product can be defined similarly

$$XYZ = \{(x, y, z) \mid x \in X, y \in Y, z \in Z\}$$

$$\text{Let } V = \{a, b\}$$

$$^w Y = \{x, y\}$$

Then $X \cdot Y = \{(a, x), (b, x), (a, y), (b, y)\}$

and $Y \cdot X = \{(x, a), (y, a), (x, b), (y, b)\}$

Hence (i) $Y \cdot Y \neq Y \cdot X$

(ii) Number of pairs in each case is 4

Now $X \cdot X = \{(a, a), (b, b), (a, b), (b, a)\}$

Ex. 48 $A = \{1, 2\}$

$$B = \{1, 2, 3\}$$

Form the cartesian products $A \cdot B$, $B \cdot A$, $A \cdot A$ and $B \cdot B$ and show them graphically

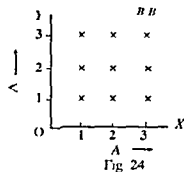
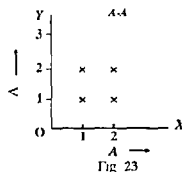
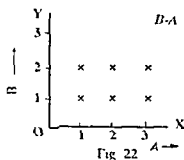
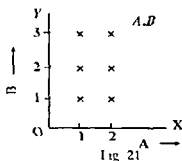
$$A \cdot B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\} \quad [2 \times 3 \text{ pairs}]$$

$$BA = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\} \quad [2 \times 3 \text{ pairs}]$$

$$AA = \{(1, 1), (1, 2), (2, 1), (2, 2)\} \quad [2 \times 2 \text{ pairs}]$$

$$BB = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\} \quad [3 \times 3 \text{ pairs}]$$

AB is represented in Fig 21, BA , AA and BB in Fig 22, 23, 24 respectively



Note (i) The diagrams clearly show that $AB \neq BA$. Thus Cartesian product is non commutative.

$$(ii) \quad n(A \cdot B) = n(A) \cdot n(B)$$

$$= 2 \times 3 = 6 \text{ in the above example.}$$

Ex. 49. Let A be the set of all possible outcomes for the throw of a coin (head or tail). What is the set of all possible outcomes where the coin is tossed thrice?

Now clearly

$$A = \{H, T\} \quad (\text{set of possible outcomes in a single throw})$$

Similarly sets B and C for second and third throws will be

$$B = \{H, T\}$$

$$C = \{H, T\}$$

The set of all possible outcomes in 3 throws will be the Cartesian product of A, B, C , i.e.

$$\begin{aligned} A \times B \times C &= \{(H, H, H), (H, H, T), (H, T, T), (H, T, H), \\ &\quad (T, H, H), (T, T, T), (T, H, T), (T, T, H)\} \\ &= \text{Eight outcomes} \end{aligned}$$

The result can be obtained directly

$$\begin{aligned} n(A \times B \times C) &= n(A) \times n(B) \times n(C) \\ &= 2 \times 2 \times 2 = 8 \end{aligned}$$

Ex. 50 Find $X \cap Y$ if,

$$x = \{x/x=1, 2\}$$

$$y = \{y/y=x-1\}$$

putting the value of $x=1$ and $x=2$ in the equation $y=x-1$, we have $y=0$ and 1

$$x = \{1, 2\}$$

$$y = \{0, 1\}$$

Hence $x \cap y = \{1, 0\}, \{1, 1\}, \{2, 0\}, \{2, 1\}$

Ex. 51. If $X \cap Y = Y \cap X$, show that $y = X$

$$(x, y) \in (X \times Y)$$

$$X \cap Y = \{(x, y)/x \in X \text{ and } y \in Y\}$$

Similarly $Y \cap X = \{(y, x)/y \in Y \text{ and } x \in X\}$

Since $X \cap Y = Y \cap X$ hence each pair (x, y) is same as (y, x) , i.e., $x \in Y$ as well as $x \in Y$. Similarly $y \in Y$ and $x \in X$, which proves that $X=Y$

Lx. 52. Prove that

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

If $A = \{a/a \in A\}$, $B = \{a/a \in B\}$

and $C = \{b/b \in C\}$

then $A \cup B = \{a/a \in A \vee a \in B\}$

and $(A \cup B) \cap C = \{(a, b)/a \in A \vee a \in B \text{ and } b \in C\}$ (1)

$$A \cap C = \{(a, b)/a \in A \text{ and } b \in C\}$$

$$B \cap C = \{(a, b)/a \in B \text{ and } b \in C\}$$

$$(A \cap C) \cup (B \cap C) = \{(a, b)/a \in A \text{ and } b \in C \vee a \in B \text{ and } b \in C\}$$
 (2)

Now (1) implies that $(A \cup B) \cap C$ is the set of ordered pairs (a, b) such that a is from A or B and b is from C , (2) implies that it is union of two sets of ordered pairs, in one (a, b) is such that a is from A and b from C and in the other a is from B and b from C . In taking union the repeated ordered pairs will be eliminated.

For example

$$A = \{1, 1\}$$

$$B = \{0, 2\}$$

$$C = \{1, 2\}$$

Now $A \cup B = \{1, 2, 0\}$

and $(A \cup B) \cap C = \{(1, 1), (1, 2), (2, 1), (2, 2), (0, 1), (0, 2)\}$

Again $A \cap C = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$
 and $B \subset C = \{(0, 1), (0, 2), (2, 1), (2, 2)\}$
 Hence $(A \cap C) \cup B = \{(1, 1), (1, 2), (2, 1), (2, 2), (0, 1), (0, 2)\}$

1.16 Relation

Consider the sets

$$A_1 = \{1, 2, 3, 4\}$$

$$A_2 = \{1, 2, 3\}$$

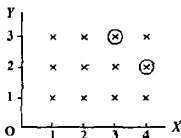


Fig. 25

Then the Cartesian product $A_1 \times A_2$ is given by the set of 12 ordered pairs presented in diagram

Now if a condition (sometimes referred to as a 'sentence') be imposed on pairing of x and y

$$x + y = 6$$

Then only by the pairs $(3, 3)$ and $(4, 2)$ satisfy the condition. The condition or limitation imposed on x and y by equation

$x + y = 6$ defines a relationship between x and y

The set consists of pairs $(3, 3)$, $(4, 2)$ and is clearly a subset of the Cartesian product $A \times B$ (Fig. 25). We can now define a relation

Any non empty proper subset R of the Cartesian product $A \times B$ is called the relation R from the set A to the set B which satisfies the given condition and is written as

$$R = \{(x, y) / x \in A, y \in B, x \in y\} \quad \text{Let}$$

where the statement reads as x is related to y

Since R consists of some chosen elements of $A \times B$ we must specify the rule by which the choice of these elements has been made and xRy is exactly the rule relating $x \in A$ to $y \in B$

The relation may involve equality or an inequality. The rule or statement or condition here relates the variables x and y or the elements of two sets hence these relations are known as binary relations. We shall confine our attention to binary relations.

Domain of a Relation In the Cartesian product $A \times B$ each element $x \in A$ has $y \in B$ to correspond to it. But in xRy many $x \in A$ may not have a $y \in B$ and vice versa. The set of all first elements (x) of ordered pairs which belong to the relation R is called the domain of the relation R .

Range of a Relation The set of all second elements (y) of ordered pairs (x, y) which belong to the relation R is called the range of the relation R .

$$\{(1, 2), (2, 1), (3, 1), (4, 1)\} \rightarrow \{2, 1\}$$

Ex 53 If $A = \{1, 2, 4\}$ and $A \cap B \neq \emptyset$
 $B = \{1, 2, 3, 4\}$ then find xRy if

(i) R is given by $y < x$

(ii) $y = x$

(iii) $x = 2y$

Let us form the Cartesian product $A \times A$

$$A \times A = \{(x, y) \mid x \in A \text{ and } y \in A\}$$

$$\{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

(i) Condition $y < x$ implies that those pairs where $y > x$ are non permissible. Eliminating these we have the relation

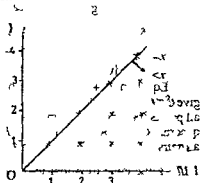
$$R = \{(x, y) \mid x \in A \text{ and } y \in A, y < x\}$$

$$\{(x, y) \mid x, y \in A, y < x\}$$

$$\{(2, 1), (3, 1), (4, 1), (3, 2), (4, 2), (4, 3)\}$$

and the domain which is the set of the first elements of each pair is $\{2, 3, 4\}$ and the range is $\{1, 2, 3\}$

If we plot gives the relation as a set of pairs (marked by cross) below the line $y = x$



(ii) Condition $y = x$ $R = \{(x, y) \in A \times A \mid y = x\}$ is clearly $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ which is given by the dots on the line $y = x$

Both domain and range are $\{1, 2, 3, 4\}$

(iii) Condition $x = 2y$ $R = \{(x, y) \in A \times A \mid x = 2y\}$ only the pairs possible in which the first element is twice the second element $\{(2, 1), (4, 2)\}$ and is given by the crosses on the line $x = 2y$ the domain and range is $\{1, 2\}$

Let us find the domain and range of the relation R defined by $x = 2y$

$$R_1 = \{(x, y) \in A \times A \mid x = 2y\}$$

$$R_2 = \{(x, y) \in A \times A \mid x = 2y\}$$

Let us find the domain and range of the relation R defined by $x = 2y$

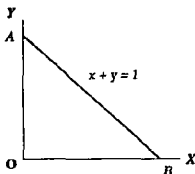


Fig 27

Now both the sets are infinite sets, hence the Cartesian set will be a set of an infinite number of ordered pairs, all of which cannot be shown in the graph

The condition $x+y=1$ implies that the elements of each pair should satisfy this equality condition. Hence the relation R_1 comprises all points on the line AB ($x+y=1$) in the positive quadrant; the domain as well as range are the set of all real numbers lying between 0 and 1.

Similarly R_2 comprises all points on $x+y=1$ and below it, i.e., in the triangle OAB , including its boundaries (Fig 27)

Domain = Range = set of all real numbers from 0 to 1

Ex. 55. Find $R_1 = \{(x, y) / x \in X, y \in Y,$
 $x^2 + y^2 = 1$

$R_2 = \{(x, y) / x \in X, y \in Y, x^2 + y^2 < 1\}$

Equation $x^2 + y^2 = 1$ when plotted will give a circle with radius 1. Hence R_1 comprises all points (pairs) on the segment AB in positive quadrant of the circle with radius 1. R_2 is set of all pairs in the area OAB but not on AB (Fig 28)

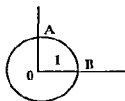


Fig 28

1.17 Functional Relations and Functions

The concept of a relation was defined very generally in the preceding sections. We shall now take some rather special types of relations. Let us consider a relation R and the corresponding statement xRy . The relation R will then be said to be a **functional relation** if, whatever element of x of its domain is given, there is exactly one object y such that the statement xRy is true.

A more formal definition of a functional relation will be as follows:

A functional relation f from a set A ($x \in A$) to a set B ($y \in B$) is a subset of $A \times B$ such that if $(x, y^1) \in f$ and $(x, y^2) \in f$, then $y^1 = y^2$, where $y^1 = y^2$ means that y^1 and y^2 are the same object. A relation is therefore, a functional relation where for each x (or several x) there is only one corresponding y , i.e., for each $x \in A$ there is a unique element $y \in B$. A relation (subset of Cartesian product) is a functional relation if no two distinct ordered pairs of the relation have the same first component. Thus a subset $(1, 2), (1, 3)$ cannot be a function, whereas $(1, 2), (2, 3)$ can be.

Functional relations which occur in mathematics are known as **functions**,

and we say that y is a function of x . There is a special notation for functions. Instead of xRy or yRx we write $y=f(x)$ or $y=g(x)$.

The symbols x and y stand for the domain and range (or converse domain) of R respectively, and are frequently referred to as variables, x being called the independent variable and y the dependent variable. We may also call x the argument and y the value of the function.

The function just defined has a unique y in the range for each specified x in the domain and the function $y=f(x)$ is often called a single valued function.*

Ex 56. If $A=\{1, 2, 3, 4, 5\}$, $x \in A$

and $B=\{-1, -2, 0, 1, 2\}$, $y \in B$

and the conditions is (i) $y = x^2$

(ii) $y = x$

(iii) $x = y^2$

Do these relations define functions?

(i) The subset of $A.B$ which satisfies the condition $y=x^2$ is

x	1
$y=x^2$	1

Range $\{1\}$, domain $\{1\}$, hence a unique y (1) in the range corresponds to the specified x ($=1$) in the domain $\{(x, y)/x, y \in A.B, y=x^2\}$. $y=x^2$ is a functional relation or function, $y=f(x)$. Similarly x is a function of y .

(ii) $y = x$, then the subset of $A.B$ which satisfies the condition $y=x$ consists of the ordered pairs

x	1	2
$y=x$	1	2

Hence, to each x in the domain corresponds a unique y in the range. Hence, y is a function of x , $i.e.$ $\{(x, y)/x, y \in A.B, y=x\}$ is a functional relation.

Similarly, to each y there corresponds a unique x in the range, hence x is a function of y .

(iii) $x = y^2$. The subset of the Cartesian product $A.B$ which satisfies the condition $x = y^2$ consists of the ordered pairs

* The definition given here does not cover what is called a multi valued function. These need to be split into single valued branches.

x	1	4	1	4
$y = \sqrt{x}$ or $x = y^2$	-1	-2	1	2

Two pairs have the first components as 1 and the other two have the first components as 4. That is to one x (=1 say) in the domain, there exists more than one value of y ($\in -1$ and 1) in the range. Hence y is not one function of x . However the relation can be divided into two branches, one has negative values in the range but other positive values in the range. One functional relation will be given by the set of ordered pairs $\{(1, -1), (4, -2)\}$ and the other functional relation by the set of ordered pairs $\{(1, 1), (4, 2)\}$.

However we see that for each value of y there corresponds a unique value of x , i.e., x is a function of y , now the first components will be $\{-1, -2, 1, 2\}$ which defines the domain, no component is repeated.

Hence the relation $\{(x, y)/x \in A, y \in B, x = y^2\}$ defines a functional relation ship but the relation $\{(x, y)/x \in A, y \in B, x = y^2\}$ does not define a functional relationship unless it is split into two single valued functions i.e.

$$\{x, y/x \in A, y \in B, y = +\sqrt{x}\}$$

$$\text{and } \{(x, y)/x \in A, y \in B, y = -\sqrt{x}\}$$

which are functional relations.

Thus if $y=f(x)$, it is not necessarily true that $x=f(y)$.

EXERCISE 1.4

1. Find the Cartesian products $A \times A, A \times B, B \times A$ and $B \times B$

$$A = \{0, 1\}$$

$$\text{and } B = \{-3, -2\}$$

2. Find $A \times A$ and $A \times B$ if

$$A = \{x/x=1, 2\} \text{ and } B = \{y/y=x+2\}$$

3. If $A = \{1, 2\}$, $B = \{3, 4\}$ and $C = \{1, 4, 2\}$

Prove that

$$A \cap (B \cap C) = (A \cap B) \cap (A \cap C)$$

$$\text{and } A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

4. prove the results of Ex 3 generally

5. Deduce that

$$(A \cap B) \cap (C \cup D) = [(A \cap C) \cup (A \cap D)] \cap [(B \cap C) \cup (B \cap D)]$$

6. If $A = \{0, 1\}$

$$B = \{2, 3\}$$

$$C = \{2, 3, 4\}$$

Find (i) $(A \cup B) \cap C$ (ii) $A \cap (B \cap C)$

(iii) $(A \cup B) \cap (B \cap C)$

7 If A, B, C are three sets, show that

(i) $A \times B \supseteq (A \cap B) \cap (B \times A)$

(ii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(iii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

8 Prove the above results if

(i) $A = \{1, 4\}, B = \{2, 3\}, C = \{3, 5\}$

(ii) $A = \{a, b\}, B = \{c, d\}, C = \{c, f\}$

(iii) $A = \{1, 2, 3\}, B = \{2, 3, 4\}, C = \{1, 3, 4\}$

19 $D = \{2, 4, 5\}$ in the above two examples verify that

$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$

10 What is the set of possible outcomes in two throws with a fixed dice. What is the set when a four faced die is thrown twice

11 Find the domain and range of the relation

$A = \{0, 1, 2, 3\}$

$B = \{-3, -2, -1, 0, 1, 2, 3\}$

If the relation from A to B is $R = \{(x, y) : x = y + 1\}$

Graph the relation

12 Let $A = \{0, 1, 2, 3\}$

and $x \in A$ and $y \in A$ then find the relation

(i) $y < x$

(ii) $x = y$

(iii) $x = 2y$

Find the domain and range in each case

Graph the Cartesian product AB in each case

13 Let $X = \{1, 2, 3, 4, 5\}$ and consider the set of ordered pairs (x, y) such that $x \in X$ and $y \in X$

(1) Which of the pairs belong to the relation

(i) $y < x$ (ii) $y = x$ (iii) y is a factor of x

14 Let $A = \{1, 2, 3, 4\}$

or $B = \{-3, -2, -1, 1, 2, 3\}$

Find $R_1 = \{(x, y) : x, y \in A, x = y^2\}$

$R_2 = \{(x, y) : x, y \in A, x = y^2\}$

Are these functional relations? Graph the relations

15 Show that if y is a function of x and x is a function of z then y is a function of z .

16 Determine which of the following relations are functional

- (a) y is weight of x .
- (b) y is square root of x
- (c) y is reciprocal of x
- (d) y is Positive square root of x
- (e) y is a great-grand function of x .

17. Determine which of the following relations are functional relations

- (i) y is son of x
- (ii) $y = (x-3)^2$
- (iii) Consumption is proportional to income

18. Let R be the relation $>$ from $A = \{2, 4, 6, 8\}$ to $B = \{3, 5, 7, 9, 11\}$

Write R as a set

of ordered pairs. Specify the domain and range of R

19. A function is defined by $f(x) = x^3 - 4x^2 + 3$ and domain of the function is $\{-2, 1, 2, 3\}$. Determine $f(-2)$, $f(1)$, $f(2)$, $f(3)$. Find the range of the relation

20. If $A = \{-1, -2, 0, 4, 5\}$

and $B = \{-2, -1, 1, 2, 3\}$

Find $R = \{x, y / (x, y) \in A \times B, y < x\}$

and graph it. Find range and domain of the relation. Is it a functional relation?

21. If R is the relation from $A = \{2, 3, 4\}$ to $B = \{3, 6, 8, 9, 10\}$ defined by " x divides y " write R as a set of pairs

22. Let $A = \{1, 2, 3\}$ and R be the relation defined by $x + 3y < 9$, $x \in A$ and $y \in A$. Find R . Graph it.

23. If X and Y are each set of all real numbers explain why the statements $y = |x| - 1$ and $y = x^2 - 1$ give the same relation.

[Hint: $|x|$ = absolute value of x , i.e., x always taken positive.]

24. If $X = \{1, 2\}$ and $Y = \{1, 2, 3, 4, 5\}$, find the relation

$$R = \{x, y / x \in X, y \in Y, y = x^2\}$$

25. Examine the relationship between sets A and B on one hand and sets C, D, E on the other given below, the universal set of all persons residing in Bombay city

A is the set of unemployed persons

B is the set of graduates

C is the set of unemployed non-graduates

D is the set of employed graduates

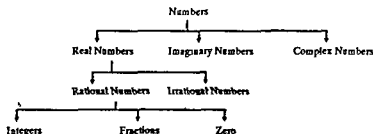
E is the set of unemployed graduates.

Chapter 2

The Number System

2.1 Use of Numbers is Essential in Mathematics.

Number in a two-dimensional field can be classified as follows



Natural Numbers and Integers A natural number is a whole number used in counting for example, a child learns to count the number of his fingers. Thus 1, 2, 3, 10, 150 are all natural numbers.

The problems associated with this process of counting have given rise to many types of relations and operations associated with numbers, e.g., addition, subtraction, multiplication and division.

Zero Zero is an additional number which is not used in counting and is thus not a natural number. It is included in the set of integers which also includes the negatives of natural numbers.

Rational numbers and Fractions. If we take one integer (m) and divide it by another integer (n) other than zero (it should be remembered that division by zero is not permissible) then we obtain a fraction m/n . If any number can be put in the form of fraction of two integers then that number is rational number. For example, $1.5 = 3/2$, is a ratio of two integers 3 and 2, hence 1.5 is a rational number. Similarly $6 = 6/1$ is a rational number. Hence all integers are rational numbers.

Thus rational numbers comprise all the simple numbers dealt with in elementary arithmetic; positive and negative integers and fractions (ratios of integers), together with zero as the number separating the positive from the negative.

The set of rational numbers is denoted by R . Rationals can be represented as fractions m/n where m and n are positive or negative integers, together with 0.

As a set, the remarkable feature of the rationals is that they are closed with respect to operations of addition, subtraction, multiplication and division. Add

or, subtract, multiply or divide any two rationals (except by zero), and we shall get a rational number.

Irrational Numbers But all numbers cannot be put in the form of a fraction of the integers. Such numbers which are not rational are termed as irrational numbers. Consider $\sqrt{2}$. Now,

$$2 = 1.414$$

$\sqrt{2}$ cannot be put in the form of a fraction of the integers. It is irrational. The division goes on.

Ex 1 Are the following rational numbers?

(i) $\sqrt{6.25}$ (ii) $\sqrt{3}$ (iii) $\frac{1}{2}$ (iv) 2.78

(i) $\sqrt{6.25} = 2.5 = 2\frac{5}{10} = \frac{5}{2}$

This is ratio of the integers 5 and 2 hence it is a rational number.

(ii) $\sqrt{3} = 1.732$

This is non-recurring infinite expansion which cannot be put as a fraction hence it is irrational number.

(iii) $\frac{1}{3} = .333 = \bar{3}$

This is a recurring infinite expansion, where 3 repeats itself infinitely.

As is obvious $\bar{3}$ can be put as fraction $\frac{1}{3}$ hence it is a rational number.

All recurring infinite expansions can be put in fraction form, hence are rational numbers. All non-recurring infinite expansions cannot be put in fraction form hence are irrational numbers.

(iv) $2.78 = 2.787878$
 $= 2.\overline{78}$

$$100 \times 2.\overline{78} = 278.\overline{78}$$

$$= 276 + 2.\overline{78}$$

or $100 \times 2.\overline{78} - 2.\overline{78} = 276$

or $2.\overline{78} (100 - 1) = 276$

or $2.\overline{78} = \frac{276}{99}$

Thus, $2.\overline{78}$ has been expressed as a fraction of two integers 276 and 99, hence it is a rational number.

Real Numbers The numbers so far discussed—rational, irrational and zero—constitute the set of real numbers—the real number system.

It will be useful to remember two propositions about real numbers.

(i) There are an infinite number of rational numbers between any two

rational numbers.

Example 1 and 2 are two rational numbers

$$\frac{4}{3}, \frac{6}{4}, \frac{6}{5}, \frac{7}{6}, \frac{7}{4}, \frac{8}{7}, \frac{8}{6}, \frac{8}{5}$$

are infinite numbers of fractions (rational numbers) between them

(ii) There is at least one irrational number between two rational numbers

Example Between 1 and 2 at least one $\sqrt{2}$ is an irrational number

Imaginary Numbers. We know that if $x^2 = 4$

then $x = \pm \sqrt{4} = \pm 2$

But if, $x^2 = \sqrt{-4}$ then

$$x = \pm \sqrt{-4}$$

$$x = \pm 2i$$

In fact, we do not know the squareroot of negative real numbers (or numbers with negative sign). $\sqrt{-4}$ is, therefore, termed as imaginary number. No real number can show the value of $\sqrt{-4}$.

To simplify matter, we always write,

$$\sqrt{-1} = i \text{ (iota)}$$

so that $i^2 = -1$

Then, $\sqrt{-4}$ can be written as

$$x^2 = -4$$

$$= -1 \times 4$$

$$x = \pm \sqrt{-1} \sqrt{4}$$

$$= \pm i2 = \pm 2i$$

In this way we have expressed an imaginary number as product of a real number and an imaginary quantity i .

Example $x = 2 + i3$

Real Imaginary

Ex 2 Give two integers whose quotient is not a rational number.

A rational number is defined as a ratio of two integers (positive or negative)

If p and q are two integers and $q \neq 0$

then $\frac{p}{q}$ is a rational number,

and if $\frac{p}{q}$ is not a rational number then q must be zero, so that

$\frac{p}{q} = \frac{p}{0}$ is not a rational number

Ex. 3. If x and y are real numbers, state whether the following are true or false

(i) if $x < y$ then $x^2 < y^2$, (ii) if $x > 0$ then $x^2 > x$

(i) False e.g., $-3 < 2$ but $(-3)^2 \nless (2)^2$ or $9 \nless 4$

(ii) False

if $x = \frac{1}{2}$ which is > 0

then $x^2 = \frac{1}{4}$ which is $< x$

Ex. 4. If x is a real number and $x^2 > 25$, find x ,

Given $x^2 > 25$

$x > \pm 5$

Ex. 5. Find one rational number and one irrational number between 1 and

$\sqrt{3}$

Rational number is 1.2 (any decimal number between 1 and $\sqrt{3}$)

Irrational number is $\sqrt{2}$.

Ex. 6. Find x such that $x^{2/3}$ is a rational number

Let $x^{2/3} = \frac{p}{q}$

$$\begin{aligned} \therefore x &= \left(\frac{p}{q}\right)^{3/2} \\ &= \sqrt{\left(\frac{p^3}{q^3}\right)} \end{aligned}$$

where p and q are some integers.

Ex. 7. Show that $\frac{4m+5n}{m+n}$ is a rational number between 4 and 5 where m

and n are positive rationals.

Now, $\frac{4m+5n}{m+n} = \frac{4(m+n)+n}{m+n}$

$= 4 + \frac{n}{m+n}$

$= 4 + k \quad \left(\text{where } k = \frac{n}{m+n} > 0\right)$

$\frac{4m+5n}{m+n} > 4$

$$\begin{aligned}\text{Again } \frac{4m+5n}{m+n} &= \frac{5(m+n) - m}{m+n} \\ &= 5 - \frac{m}{m+n} < 5 \text{ Hence proved}\end{aligned}$$

EXERCISE 2.1

- 1 Which of the following are rational numbers ?

$$3\sqrt{16} - \frac{5}{2}, 2\sqrt{3}$$

- 2 Prove that $\sqrt{5}$ is an irrational number
- 3 Insert two rational and irrational numbers between $\sqrt{73}$ and $\sqrt{74}$
- 4 Write the recurring decimal numbers $7.\overline{234}$ and $3.\overline{58}$ as fractions
- 5 Prove that if x is a rational number then $x - \sqrt{3}$ will be an irrational number
- 6 Solve
- $x^2 + 25 = 0$
 - $x^4 - 16 = 0$
- 7 Point out real and imaginary numbers in the number $3 - i - 3\sqrt{2}$
- 8
- Give a rational number between π and $22/7$
 - Give a value of x such that (1) $\sqrt{x}$ is rational (2) $\sqrt{x}$ is irrational
- 9 Give one illustration each of the following
- A rational number which is not an integer
 - An irrational number greater than 1

2.2 Axiomatic Properties of Real Numbers

The real number system is *complete ordered field*.

Field A field is a set of numbers for which the algebraic operations of addition, difference, product, division hold. The numbers obtained from these operations belong to the field.

Following are the operational rules of algebra which are also termed as Axioms of the Field.

Closure. The remarkable feature of the real numbers (and also the rationals) is that they are closed with respect to the operations of addition, multiplication, subtraction and division. The combination of real numbers by means of $+$, $-$, $\times$, $\div$ always gives another real number and never anything outside the set. Thus the set is self-contained, i.e. if a and b are real numbers $a + b$ and $a \times b$, So also $a - b$ and a/b belong to the set.

Association. Order of repeated addition or multiplication is immaterial

that is $(a+b)+c=a+(b+c)$ and

$$a \times (b \times c) = (a \times b) \times c$$

Commutation When two real numbers are combined (by + or $\times$) it is a symmetric operation

$a + b$ is same as $b + a$ and ab is same as ba

Distribution Distribution axiom links the two operations of addition and multiplication

$$(a+b)c = ac+bc$$

Identity The set of real numbers contains a unique member (zero) which does not alter any other member by addition ($a+0=a$) and another unique member (1) which does not alter other members by multiplication ($a \times 1 = a$, or $1 \times a = a$)

Inverse. Every real number has its negative, the two add to zero. The negative of a is $-a$ such that $a+(-a)=0$. Negative of a number is also called its *additive inverse*. Similarly,

$$a-b = a+(-b)$$

where $(-b)$ is additive inverse or negative of b

Similarly, every real number (except zero) has a number which on multiplication with it yields one

$$a \times \frac{1}{a} = 1$$

hence $\frac{1}{a}$ is the multiplicative inverse of a . Obviously $\frac{1}{\frac{1}{a}}$ is multiplicative inverse of $\frac{1}{a}$. Similarly

$$\frac{a}{b} = a \times b^{-1} = 1$$

where b^{-1} is multiplicative inverse of b

Cancellation. If $a+b=b+c$ then $a=c$ which follows from the axiom of additive inverse.

If $ab=bc$ then $a=c$ which follows from the axiom of multiplicative inverse. Of course, b should not be zero otherwise b^{-1} will not be defined.

Ex 8 Show that if $ac=ab$ and $a \neq 0$ then $c=b$

$$(ac)a^{-1} = (ab)a^{-1}$$

$$(ca)a^{-1} = (ba)a^{-1}$$

$$c(aa^{-1}) = b(aa^{-1}) \text{ hence } c=b$$

Ex 9 If $a \times a = a$ then a must be equal to 1.

Here

$$(ax)a^{-1} = aa^{-1}$$

$$a(axa^{-1}) = aa^{-1}$$

$$a=1$$

Ex 10 Show that $\frac{a}{b} \div \frac{c}{d} = \frac{ac}{bd} \times 1$ if $b \neq 0, d \neq 0$

$$\text{Now } (ab^{-1})(cd^{-1}) = (b^{-1}a)(cd^{-1}) \\ = b^{-1}(ac)d^{-1}$$

$$= (ac)d^{-1}b^{-1}$$

$$= (ac)b^{-1}d^{-1}$$

$$= \frac{ac}{bd}$$

$$= \frac{ac}{bd}$$

Ex 11 Show that $(ab)^{-1} = a^{-1}b^{-1}$ if $a \neq 0$, $b \neq 0$

$$(ab)^{-1} = \frac{1}{ab}$$

$$= a^{-1}b^{-1}$$

$$(ab)^{-1} = a^{-1}b^{-1}$$

$$\frac{1}{ab} = b^{-1}a^{-1}$$

$$\frac{1}{ab} = (a^{-1}b^{-1})(ab)$$

$$= a(a^{-1}b^{-1})b$$

$$= aa^{-1}b^{-1}b$$

$$= aa^{-1}bb^{-1}$$

$$= 1 \times 1$$

$$= 1$$

Ex 12 Show that $(x-a)(x-b) = x^2 - ax - bx + ab$

From the axiom of distribution we have

$$(x-a)(x-b) = x(x-b) + (-a)(x-b)$$

$$= x \cdot x + x(-b) + (-a)x + (-a)(-b)$$

$$= x^2 - xb - ax + ab$$

Ex 13 Attempt to make subtraction $a - b$ a basic operation in the set of rationals and check that the operational rules do not hold

$$(a-b)c \neq a(b-c)$$

$$(a-b)-c \neq a-(b-c)$$

hence the associative law does not hold

EXERCISE 2.2

1. Show that if $a+b=0$ then $b=-a$

2. Show that $(a+b) = a+b$

3. If $ab=1$ and $a \neq 0$ then $b = 1/a$

4. If $b \neq 0$ then $x = b/a$ for $a \neq 0$

5. Show $(a+b)^2 = a^2 + 2ab + b^2$
6. Show that there is no real number which is reciprocal of 0
7. Show that if $a \times b = a \times c$ ($a \neq 0$) then $b = c$
8. Show that if $a \times b = 0$, then either $a = 0$ or $b = 0$
9. Attempt to make division a/b a basic operation in the set of rational numbers, check that the operational rules do not all hold
10. Show that the operation 'to the power' [$(a^b) = a$ to the power b] is not associative in the set of rationals

$$\left[1 \text{ e. } a^{b^c} \neq (a^b)^c \right]$$

11. Prove that the cube of a rational number cannot be 2, i.e. $\sqrt[3]{2}$ is irrational
12. If p, q, m and n are integers such that

$$\left(\frac{p}{q} \right)^2 = \frac{m}{n},$$

prove that m and n are both squares of integers. Deduce that the square root of a rational number which is not of the form u^2/v^2 (where u and v are integers) is irrational

13. If a, b, c are real numbers, prove that

$$(a) \quad 0 = -0$$

$$(b) \quad (a-b) = -(b-a)$$

2.3. Order Field

We have seen the set of rationals is a *field*. So also is the set of real numbers.

Rational numbers (so also real numbers) have an equally important property of *being ordered*. The symbol $<$ is used for 'less than', $>$ for 'greater than' and $=$ for 'equal to'.

Let a, b, c be real (or rational) numbers. We discuss the following properties of an *ordered field*.

1. **Trichotomy** Only one and one of $a < b, a = b, b > a$ is true. Obviously if $2 < 3$ then the statements $2 = 3$ and $2 > 3$ cannot be true.

2. **Transitivity**. If $a < b$ and $b < c$ then $a < c$ or if $a > b$ and $b > c$ then $a > c$. Thus, if $2 < 3$ and $3 < 4$ then $2 < 4$, i.e., 2, 3, 4 can be put in a sequence of order.

3. **Density**. If $a < b$, there is member ' c ' of the set of real numbers such that $a < c < b$. That is, there is always a real (or rational) number between two real (or rational) numbers.

4 **Extension** For any $a < b$ if $a < b$ then c exists so that $a < c < b$ and d exists so that $b < d$ the set can be extended at both the ends

5 **Consistency** (a) *Consistency in addition* If $a < b$ then $a + c < b + c$ for any c

(b) *Consistency of multiplication* If $a < b$ then $ax < bx$ for any positive x

2.4 The Property of Completeness

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If a cut is made in the set of numbers and the dividing point is always the member of the same set then nothing new is produced the numbers are said to form a *complete ordered field*

If a cut of the real numbers is made just a real number will be produced no number other than real number will emerge. Hence real numbers form a *complete ordered field*. But if a cut of rational numbers is made either a rational or an irrational number can be produced quite a new number not member of the original set can emerge hence rationals do not form a *complete ordered field*.

The property (or axiom) of completeness can be put in another form. It states that

If a set S has a lower bound then it has a Greatest Lower Bound (GLB) and if S has an upper bound then it has a Lowest Upper Bound (LUB)

The set S has a lower bound a if $a < x$ for each x in S . It has a greatest lower bound if no member of the set is less than a and a is a lower bound of S . Similarly, the set S has an upper bound b if $b > x$ for each x in S . It has a lowest upper bound if no member of the set is greater than b and b is an upper bound of the set.

Sets of numbers need not have GLB or LUB. Set of positive real numbers has no LUB, set of negative real numbers has no GLB, set of integers has no GLB or LUB. What the axiom of completeness states is that if there is any lower bound then there is also GLB and so for the upper bound.

Let us consider two examples: (a) set of rationals x such that $x^2 > \sqrt{3}$ and (b) set of real numbers x such that $x^2 > 3$.

(a) Now if x is a rational number $> \sqrt{3}$ then there is a lower bound or many lower bounds $x=0, x=1$ because for each x in the set defined by $x^2 > 3$, $a=0$ or 1 is less than x . But what about GLB? Certainly GLB is $\sqrt{3}$ which is not a rational number. Hence there is no rational number's GLB. Hence the set of rational numbers x such that $x^2 > \sqrt{3}$ is not a complete ordered field. In other words, a set of rational numbers is not complete.

(b) Evidently $\sqrt{3}$ is the GLB hence set of real numbers x such that $x^2 > \sqrt{3}$ is a complete ordered field.

Ex 14 Show that an inequality is reversed if both sides are multiplied by the same negative number

Let $a < b$ and $c < 0$ then show that $ac > bc$
 we are given, $ac > bc$
 then,

$$a(-c) < b(-c)$$

$$-ac < -bc$$

$$-ac + [ac + bc] < -bc + [ac + bc]$$

$$\text{(adding } ac + bc \text{ to both sides)}$$

or, $bc < ac$

or, $ac > bc$ hence proved

Ex 15 Prove that $x > \sqrt{xy} > y$ only if $x > y > 0$

From that axiom of transitivity since $x > \sqrt{xy} > y$ hence $x > \sqrt{xy}$ $\sqrt{xy} > y$
 and $x > y$

$$x > \sqrt{xy} \quad \text{is same as} \quad \sqrt{x} \sqrt{x} > \sqrt{x} \sqrt{y}$$

Using the axiom of multiplication

Since $x > y$ and $\sqrt{x} \sqrt{x} > \sqrt{x} \sqrt{y}$ hence $\sqrt{x} > 0$ $x > 0$

Similarly from $\sqrt{xy} > y$ $\sqrt{y} \sqrt{y}$ we have

$\sqrt{y} > 0$ or $y > 0$ Proved

Ex 16. Solve $|x - 4| \geq 2$

Now $|x - 4|$ is the absolute value of $x - 4$. Absolute value of 4 and -4 is 4
 So $2 - 4$ and $4 - 2$ is same ($= 2$). Hence absolute value (or modulus) of $(x - 4)$ and $(4 - x)$ is same and written as $|x - 4|$. Conversely $|x - 4|$ can be written as $(x - 4)$ or $(4 - x)$

$$|x - 4| \geq 2 \text{ is same as}$$

$$(x - 4) \geq 2 \quad \text{and} \quad 4 - x \geq 2$$

$$x \geq 2 + 4 \quad \text{and} \quad 4 - 2 \geq x$$

$$x \geq 6 \quad \text{and} \quad 2 \geq x$$

$$2 \leq x \leq 6 \text{ is required solution}$$

EXERCISE 2.3

1. Show that if $a > b$ then $-a < -b$
2. Prove that $ab > 0$ if and only if a, b are both positive or both negative
3. Show that if $x + y + z < x + 2y$ then $y > z$.
4. Which of the following statements is correct?
 - (i) If $x + y > y + 3$ then $x > 3$
 - (ii) If $x - 10 > x - 2y$ then $10 > 2y$
 - (iii) $(-4)(-3) > (-2)(-3)$
 - (iv) The set of positive even numbers is a complete ordered field

(v) The product of two irrational numbers is irrational

(vi) The sum of two irrational numbers is irrational

5 Solve the following inequalities

(i) $|2x-3| \leq 3$

(ii) $|3-5x| > 0$

(iii) $\left| \frac{x-3}{2} \right| > 5$

(iv) $|4-x| \geq 0$

2.5 More on Complex Numbers

A complex number can be written as $a+ib$ where a and b are two real numbers and $i^2 = -1$. A complex number z is thus an ordered pair (a, b) of real numbers. Thus $z=a+ib$. The force of order is that (a, b) is not same as the pair (b, a) if $a \neq b$ because (b, a) implies the complex number $b+ia$ and $a+ib \neq b+ia$ if $a \neq b$.

Two complex numbers are equal only if the respective elements are equal

$$a_1 + ib_1 = a_2 + ib_2 \quad \text{if } a_1 = a_2 \quad \text{and} \quad b_1 = b_2$$

The sum of two complex numbers $a_1 + ib_1$ and $a_2 + ib_2$ or (a_1, b_1) and (a_2, b_2) is complex number $(a_1 + a_2, b_1 + b_2)$ i.e. $a_1 + a_2 + i(b_1 + b_2)$.

It can be verified that the commutative and associative laws hold for complex numbers z_1, z_2, z_3

$$\begin{aligned} \text{i.e.} \quad z_1 + z_2 &= z_2 + z_1 \\ \text{and} \quad z_1 + (z_2 + z_3) &= (z_1 + z_2) + z_3 \end{aligned}$$

The multiplication of the complex numbers yields

$$\begin{aligned} z_1 z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= a_1 a_2 + ib_1 a_2 + ia_1 b_2 + i^2 b_1 b_2 \\ &= (a_1 a_2 - b_1 b_2) + i(b_1 a_2 + a_1 b_2) \end{aligned}$$

Hence $z_1 z_2$ is number $(a_1 a_2 - b_1 b_2, b_1 a_2 + a_1 b_2)$ which is again a complex number.

Obviously the commutative law of multiplication holds, so also the associative law.

There is a complex number $(0, 0)$ denoted by 0. This has the property that

$$z + 0 = z$$

$$\text{or} \quad (a, b) + (0, 0) = (a, b)$$

$$\text{i.e.} \quad (a+ib) + 0 + i0 = a+ib$$

There is a complex number denoted by 1 , which has the property enjoyed by the real number 1 , namely

$$1z = z1 = z$$

1 is complex number $(1, 0)$, hence

$$(a + ib)(1 + i0) = a + ib$$

For any complex number $z = (a, b)$ there is a complex number which can be written as $-z$, such that

$$z - z = 0$$

obviously

$$-z \text{ is } (-a, -b)$$

Finally every complex number z except zero (0) has an inverse complex number z^{-1} such that $zz^{-1} = 1$

$$\text{Let } z = (a, b) \quad \text{and} \quad z^{-1} = (p, q)$$

$$\text{then } zz^{-1} = (a, b)(p, q) = (a + ib)(p + iq)$$

$$= (ap - bq, bp + aq) = 1 = (1, 0)$$

Hence from the property of equality we have,

$$ap - bq = 1$$

$$bp + aq = 0$$

$$\text{Solving for } p \text{ and } q \text{ we have } p = \frac{a}{a^2 + b^2}$$

$$\text{and } q = \frac{-b}{a^2 + b^2}$$

$$\text{Hence the inverse of } (a, b) \text{ is } \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right)$$

$$\text{this is, } z^{-1} = \frac{a}{a^2 + b^2} - i \frac{b}{a^2 + b^2}$$

$\sqrt{a^2 + b^2}$ is modulus of the complex number $z = a + ib$ and is written as $|z|$, hence

$$z^{-1} = \frac{1}{|z|^2} (a - ib)$$

$$z^{-1} = \left(\frac{a}{|z|^2}, \frac{-b}{|z|^2} \right)$$

Hence complex numbers satisfy all the properties (or axioms) of a field

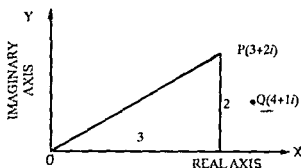


Fig 1

2.6. Graphical representation of complex numbers: The Argand Diagram

Real numbers can be thought of as points on a real line, complex numbers can be represented as points on a plane (two dimensional plane). Since we represent z by (a, b) or alternatively (x, y) (note ordering is important here), we plot the point whose Cartesian coordinates are (x, y) . The point $P(x, y)$ represents the complex number $x + iy$. In Fig. 1 we have plotted the complex number $3 + 2i$.

Axis of x is the real axis and axis of y is the imaginary axis. The modulus of P is OP , because it is $\sqrt{3^2 + 2^2} = \sqrt{x^2 + y^2}$.

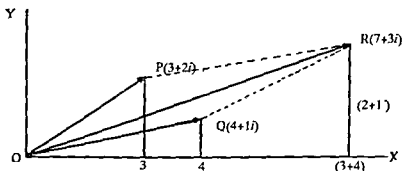


Fig 2

If $z_1 = 3 + 2i$ and $z_2 = (2 + 1i)$ then $z_1 + z_2$ is obviously $(3+2) + i(2+1) = (5+3i)$ which is the point R . R can be obtained geometrically by completing the parallelogram with OP and OQ as two sides (see Fig 2).

There is another representation, $z_1(=3+2i)$ can be represented by a directed line OP (which is termed as a vector OP and $z_2(=4+1i)$ by the vector,

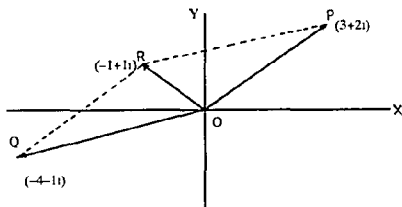


Fig 3

OQ then the sum $z_1 + z_2$ will be the vector OR which is the main diagonal (from origin) of the parallelogram constructed on OP and OQ

If we have to subtract z_2 from z_1 , that is if we find $z_1 - z_2$ or $(x_1 + iy_1) - (x_2 + iy_2)$ then we plot $-z_2$ instead of z_2 , and then construct the parallelogram, (see Fig 3)

Complex numbers are widely used in economics, specially in the explanation of business cycles

Chapter 3

Resume of High School Algebra

3.1. Equations and Identities

A statement of equality between two quantities is called an Equation

For instance, $5x=15$ states the equality between $5x$ and 15 , hence it is an equation. If the equality is true for all values assigned to x then it is called an *identity*. Equalities true for particular value or values are called equations.

Thus, $\frac{x^2+4x+4}{x+2} = x+2$, $(x^2-4) = (x-2)(x+2)$ are identities, and

$(x-3)^2 = 5$, $\frac{x+3}{5} = 8$ are equations

Linear and Quadratic Equations $\frac{x+3}{5} = 8$ is an equation in one unknown

x . Since only first power of the unknown appears in the given equation, the equation is called a *Linear Equation*.

Equations in which largest power of the unknowns is only *one* are called Linear Equations. Such equations when graphed give rise to straight lines.

For example $ax+b=0$, $\frac{x-a}{x+a} + \frac{3b-x}{2b+x} = 0$ and $\frac{a}{bx} - a^2 = b^2 - \frac{b}{ax}$

(a and b are constants) are linear equations, while $(x-3)^2=5$ is not a linear equation.

$(x-3)^2=5$, $\therefore x^2-6x+9=5$, or $x^2-6x+4=0$

The unknown x in the equation possesses the power 2, hence it is not a linear equation, but is called *quadratic equation* or *equation of second degree*.

(i) $x^2-4=0$, (ii) $2x^2+5x=8$, (iii) $3x^2-7x=8$

are the examples of quadratic equations. In all these equations unknown x has power 2.

Solution of Equations By solving an equation we mean the determination of the particular value of the unknown which satisfies the given equation.

• Suppose, $5x+8=0$ $\therefore x=-8/5$

$x=-\frac{8}{5}$ satisfies the equation $5x+8=0$. No other value can satisfy this equation.

Note: In linear equations we get only one value of the unknown

Solution of Linear Equation in one Variable We illustrate the process of solving the linear equations in one unknown by the following few examples

Ex 1 Solve $\frac{3x-7}{5} - \frac{3x+7}{4} = \frac{5x-9}{8} - \frac{3x+9}{6}$

L.C.M. = 120

Given equation becomes

$$24 \frac{(3x-7)}{5} - 30 \frac{(3x+7)}{4} = 15 \frac{(5x-9)}{8} - 20 \frac{(3x+9)}{6}$$

$$24(3x-7) - 30(3x+7) = 15(5x-9) - 20(3x+9)$$

$$\text{or } 33x = 63$$

$$x = \frac{21}{11}$$

Ex 2 Solve $\frac{a}{bx} - a^2 = b^2 - \frac{b}{ax}$ (a and b are constants)

L.C.M. = abx $a^2 - a^3bx = ab^3x - b^2$ or $a^3bx - ab^3x = a^2 - b^2$

$$\text{or } (a^3b - ab^3)x = a^2 - b^2 \text{ i.e. } x = \frac{a^2 + b^2}{ab(a^2 + b^2)} = \frac{1}{ab}$$

Solution of Quadratic Equations In quadratic equations the unknown is of second degree or power. The general form of such equations is given by $ax^2 + bx + c = 0$ (a, b, c are constants)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here a is the coefficient of x^2 , b is the coefficient of x and c is the constant quantity in the equation $ax^2 + bx + c = 0$

Ex 3 Solve $x^2 - 16x + 48 = 0$

Comparing the given equation with that of the general form of quadratic equation $ax^2 + bx + c = 0$ we find

$$a=1, b=-16 \text{ and } c=48$$

$$x = \frac{(-16) \pm \sqrt{(-16)^2 - 4(1)(48)}}{2 \times 1} = \frac{16 \pm \sqrt{64}}{2} = \frac{16 \pm 8}{2}$$

$$x = \frac{16+8}{2} = 12 \text{ or } x = \frac{16-8}{2} = 4$$

EXERCISE 3.1

Solve the equations

1 $4x+3=2x+5$

2 $3x+2-x+6$

3 $13x-4(5x-8)+17=0$

4. $14(x-4)+3(x+5)=6(7-2x)+4$

5. $8x+5(x+7)+9(2x+23)-3(x+6)=0$

6. $(x-3)(x-5)-(x-4)(x-6)=7$

7. $\frac{3}{5}(x-1)+\frac{2x}{7}-\frac{x-7}{14}=\frac{x-1}{5}+13.$

8. $\frac{x}{6}+\frac{x}{8}=\frac{x+1}{12}+\frac{x}{12}+3$

9. $\frac{2x-7}{2x-3}=\frac{x+7}{x+11}$

10. $\frac{5}{7}\left(\frac{3x}{5}-1\right)=\frac{2}{3}\left(\frac{x}{8}-6\right)+3$

11. $\frac{x}{12}-\frac{1}{8}(8-x)-\frac{1}{4}(1+x)+\frac{7}{4}=0$

12. $\frac{5}{7}(x-9)+\frac{7}{9}(x-5)=\frac{9}{5}(x-7)+1\frac{2}{3}$

13. $5-3(4-x)+4(3-2x)=0$

14. $3x-4[9-(2x+7)+3x]=13$

15. $\frac{x}{ab}+\frac{x}{bc}+\frac{x}{ac}=a+b+c$

16. $\frac{1}{3}(75-x)+\frac{1}{5}(47+2x)=(3-\frac{1}{15})x$

17. $x-\frac{2x-3}{7}=\frac{5-x}{35}$

18. $\frac{9x-8}{7}+\frac{x+2}{4}+\frac{2}{9}=0$

19. $\frac{x}{2}-\frac{05x-7.5}{6}=\frac{25x+3.8}{3}$

20. $a\frac{x-b}{a-b}+b\frac{x-a}{b-a}=1$

21. $6x^2-10x+4=0$

22. $7x^2+5x=-3$

23. $(x-3)^2+(x-2)^2=1$

24. $\frac{x-2}{x-1}+\frac{x+2}{x+1}=0$

25. $4x^2-37x-7=0.$

26. $3x^2-5x-2=0$

27. $9x^2-105x+300=0$

28. $a^2-100a+24=0$

29. $\frac{x-3}{x-2}-\frac{x+2}{x}=\frac{3}{2}$

30. $\frac{x}{x-1}-\frac{x-1}{x}=\frac{x^2+2x-1}{x^2-1}$

31. $\frac{1}{x^4}-\frac{11}{5x^2}=-\frac{6}{5}$

32. $\left(x-\frac{1}{x}\right)^2+7\left(x-\frac{1}{x}\right)=\frac{51}{4}$

33. $\sqrt{\frac{x+1}{x-1}}+\sqrt{\frac{x-1}{x+1}}=\frac{5}{2}$

34. $\sqrt{\frac{x}{x-3}}+\sqrt{\frac{x-3}{x}}=\frac{5}{2}$

35. $\sqrt{x}+\frac{8}{\sqrt{x}}=6$

Simultaneous Equations. If an equation contains two unknown quantities, say x and y , then by giving definite value to one of the unknown quantities a corresponding value for another can be obtained

Suppose $2x-y=1$ is the equation. There are two unknowns

If $x = 1$, then $y = 2(1) - 1$ or $y = 1$, $\therefore y = 2x - 1$

if $x = 2$, then $y = 2(2) - 1$ or $y = 3$,

if $x = 3$, then $y = 2(3) - 1$ or $y = 5$, and so on

Thus, for any assigned value of x corresponding value of y can be obtained. In a similar manner if values are assigned to y , corresponding values of x can be obtained

Let us consider another equation $3x + y = 9$. Again, by assigning any value to either x or y , a corresponding value of the other unknown can be obtained

If $x = 1$, then $y = 9 - 3$ or $y = 6$,

if $x = 2$, then $y = 9 - 6$ or $y = 3$,

if $x = 3$, then $y = 9 - 9$ or $y = 0$, and so on

In this way we can have infinite number of values of x and y when both these equations are considered *separately*

But on taking both these equations simultaneously it is found that only one pair of values (of x and y) satisfies both the equations. These values are $x = 2$ and $y = 3$

In equation $2x - y = 1$, when $x = 2$ then $y = 3$

and in equation $3x + y = 9$, when $x = 2$ then $y = 3$

Equations such as $2x - y = 1$, $3x + y = 9$ which are satisfied by the same value of the unknown quantities are called *Simultaneous Equations*

Solution of Simultaneous Equations. To solve simultaneous equations we require as many distinct and independent equations as there are unknowns to be found

If two unknowns have to be determined, two distinct equations are required; if three unknowns, then three equations are needed, and so on

In the solution of simultaneous equations containing two unknowns, we follow the method of elimination

Let us solve a few examples :

Ex. 4. Solve the equations $\cdot \quad 3x - 2y = 5 \quad \dots (1)$

$5x - y = 3 \quad \dots (2)$

Multiply (1) by 5 and (2) by 3. This will make the terms in x the same in both the equations

$\therefore \quad 5(3x - 2y = 5)$ or, $15x - 10y = 25 \quad \dots (i)$

$3(5x - y = 3)$ or, $15x - 3y = 9 \quad \dots (ii)$

on subtracting (ii) from (i)

we get $-7y = 16$

$y = -\frac{16}{7}$

Substitute value of y in (1) $\therefore 3x - 2\left(-\frac{16}{7}\right) = 5$ or, $x = \frac{1}{7}$

Therefore, the required solution of equation is $x = \frac{1}{7}$

$$\text{and } y = -\frac{16}{7}$$

Ex. 5. Solve : $6x - 5y = 4$ (1)

$$8x - 3y = 4 \quad (2)$$

To make the coefficients of x equal in both the equations, we multiply equation (1) by 4 and (2) by 3

$$4(6x - 5y = 4) \text{ or, } 24x - 20y = 16$$

$$3(8x - 3y = 4) \text{ or, } 24x - 9y = 12$$

$$\text{on subtracting } -11y = 4,$$

$$y = -\frac{4}{11}$$

We can now obtain the value of the other unknown x

Ex. 6. Solve $51x - 31y = 51$ (1)

$$31x - 51y = 31 \quad (2)$$

By inspection, we find here that coefficients of the unknowns are interchanged. Such equations are solved by a method which is easier than the usual one of elimination

$$\begin{aligned} \text{Add both equations, we get, } 82x - 82y &= 82 \\ 82(x - y) &= 82 \\ \therefore x - y &= 1 \end{aligned} \quad (3)$$

$$\begin{aligned} \text{Subtract (2) from (1), we get } 20x + 20y &= 20 \\ 20(x + y) &= 20 \\ \therefore x + y &= 1 \end{aligned} \quad (4)$$

Thus we get equations (3) and (4) which can be easily solved by the usual method

$$x - y = 1 \quad \dots (3)$$

$$\begin{array}{r} x - y = 1 \\ x + y = 1 \\ \hline 2x = 2, \text{ on addition} \end{array} \quad \dots (4)$$

that is, $x = 1$ and $y = 0$

$$\text{Ex. 7 Solve, } \frac{5}{x} - \frac{8}{y} = 1 \quad (1)$$

$$\frac{4}{x} - \frac{15}{y} = 2 \quad (2)$$

Both the unknowns are in the denominator. We follow another method to solve such equations

Let $\frac{1}{x} = A$ and $\frac{1}{y} = B$, so that equations become

$$5A - 8B = 1 \quad (3)$$

$$4A - 15B = 2 \quad (4)$$

Now we follow the usual method. Multiply Eqn (3) by 4 and (4) by 5

$$4(5A - 8B = 1), \quad 20A - 32B = 4$$

$$5(4A - 15B = 2), \quad 20A - 75B = 10$$

$$\hline 43B = -6, \text{ on subtracting}$$

$$B = -\frac{6}{43}, \text{ but } B = \frac{1}{y}$$

$$\frac{1}{y} = -\frac{6}{43}$$

or, $y = -\frac{43}{6} = -7\frac{1}{6}$

On substituting value of y in equation (1) or (2), we get the value of

$$x = -43. \text{ Thus, the solution is } x = -43 \text{ and } y = -7\frac{1}{6}$$

Applications In Economics

Ex. 8 The supply and demand curves for a commodity are known to be

$$q_s = p - 1 \text{ and } q_d = \frac{12}{p} \quad (q_s = \text{quantity supplied; } q_d = \text{quantity demanded})$$

Find the equilibrium price. $q_s = q_d$

$$p - 1 = \frac{12}{p}$$

$$\text{or, } p^2 - p - 12 = 0$$

$$\text{or, } (p+3)(p-4) = 0$$

$$p = 4 \text{ or } -3$$

Hence, equilibrium price is 4 units. (Can it be -3 units?)

Ex. 9. A market demand curve is given by $D = 50 - 5p$. Find (i) the maximum price anybody will pay for the commodity, (ii) the amount demanded when commodity is a free good

(i) Given $D = 50 - 5p$

$$5p = 50 - D$$

$$5p = 50, \text{ when } D \text{ is zero}$$

Hence, $p = 10$, which is the maximum price on which there will be no demand ($D = 0$). In other words, it is the maximum price anybody will pay

(ii) $p = 0$ when commodity is a free good

$$D = 50 - 5p$$

$$D = 50,$$

$$\text{when } p = 0$$

Ex. 10. One unit of commodity *A* is produced by using 1 unit of land, 2 units of labour and 3 units of capital. For producing 1 unit of commodity *B* 2 units of land, 3 units of labour and 1 unit of capital are required. For producing 1 unit of commodity *C*, 3 units of land, 1 unit of labour and 4 units of capital are required. If the prices per unit of these commodities are Rs. 14, 11 and 17 respectively, find the rent (*r*), wage (*w*) and the interest (*x*).

The given problem can be put in the form (on the assumption of perfect competition).

$$r+2w+3x=14 \quad (1)$$

$$2r+3w+x=11 \quad (2)$$

$$3r+w+4x=17 \quad (3)$$

Add Lqs. (1) and (2) and subtract from Eq. (3), then $w=2$

Put $w=2$ in any two equations and solve. This gives

$$r=1, w=2 \text{ and } x=3 \text{ units}$$

Ex. 11. If the propensity to save in a country is given by the expression $S=0.2y-50$ and the level of investment is given by $I=0.1y$ where *y* is the income, find the equilibrium level of income.

On the equilibrium level of income $I=S$

That is, $0.1y=0.2y-50$

$$0.1y=50,$$

$y=500$, that is equilibrium level of income is Rs. 500

EXERCISE 3.2

Solve the equations

1. $7x-5y=1$

2. $13x+16y=58$

3. $28x-15y=41$

$$3x+2y=13$$

$$5x-11y=9$$

$$21x+13y=55$$

2. $19x+24y=34$

5. $65x-14y=9$

6. $14x-81y=9$

$$23x+36y=62$$

$$91x-15y=31$$

$$91x-15y=31$$

7. $33x+35y=4$

8. $(x+y) = \frac{1}{3}(x-y)$

9. $\frac{1}{5}(x+y) = \frac{1}{7}(x-y)$

$$55x-55y=-16$$

$$3x+11y=4$$

$$3x+17y=2$$

10. $\frac{4x+5y}{40} = x-y$

11. $\frac{x+y}{2} = \frac{3x-5y}{2} = 2$

$$\frac{2x-y}{3} + 2y = \frac{1}{2}$$

$$\frac{x}{14} + \frac{y}{18} = 1$$

12. $\frac{x}{8} + \frac{y}{3} = 15$

13. $x + \frac{y-11}{3} = 15$

14. $3x+5y = \frac{155}{14}$

$$\frac{x}{4} - \frac{y}{5} = 4$$

$$\frac{x-5}{4} + 2y = 36$$

$$9x-10y = -\frac{30}{7}$$

$$15. \quad 3x + 2\frac{2}{3}y - 4(y-1) = 2(x+y-1)$$

$$16. \quad 5x + 11y = 146$$

$$11x + 5y = 110$$

$$17. \quad 104x - 202y = 104$$

$$202x - 104y = 202$$

$$18. \quad ax - by = a^2$$

$$bx - ay = b^2$$

$$19. \quad \frac{4}{x} + \frac{10}{y} = 2$$

$$20. \quad \frac{2}{x} + \frac{3}{y} = 2$$

$$\frac{3}{x} + \frac{2}{y} = \frac{19}{20}$$

$$\frac{5}{x} + \frac{10}{y} = 5\frac{5}{6}$$

21. Sketch the supply function $S = 20 + 3p$ where S is the quantity supplied and p the price. Find the equilibrium price and quantity when the demand is $D = 160 - 2p$. Write the inverse function for $S = 20 + 3p$.

22. Demand and supply curves of a commodity are given to be $D = 35 - 3p$ and $S = 2p$. What amount of tax or subsidy would have to be imposed to double the equilibrium price?

23. Demand and supply curves of a commodity are given to be $D = 1 - p$ and $S = p$. Obtain the equilibrium quantity and price before and after the imposition of a specific tax of Rs. 1 per unit sold.

24. Demand and supply curves for sugar are estimated as $D = 16 - 0.5p$ and $S = 0.7p + 0.4$. Find the equilibrium price, quantity sold and total amount of tax or subsidy after the imposition of

(i) a tax of one paise

(ii) a tax of five paise and

(iii) a subsidy of three paise per unit sold

25. A unit of commodity A is produced by combining one acre of land and 2 days of labour. A unit of B commodity requires 2 acres of land and 4 days of labour. Given that $p_a = 20$ and $p_b = 40$, show that it is not possible to determine the rent R and the wage W why?

3.2 Ratio, Proportion, and Variation

Ratio A ratio is a comparative relationship of two values expressed in the same unit.

For example, the ratio of two numbers, say m and n , is expressed

(i) as $m : n$ (read as m is to n) or

(ii) as a fraction $\frac{m}{n}$, provided n is not zero

Thus ratio only indicates the division between two quantities, so that, $m : n = \frac{m}{n}$ if and only if $n \neq 0$.

The first term m of the ratio $m : n$ is called the *antecedent* and the second

term n is called *consequent*

When a ratio $m : n$ is compounded with itself, the resulting ratio is $m^2 : n^2$ which is called the *Duplicate ratio* of $m : n$

Similarly $m^3 : n^3$ is called the *Triplicate ratio* and $\sqrt[m]{m} : \sqrt[n]{n}$ is called the *Subtriplicate ratio* of $m : n$

For instance

(i) The duplicate ratio of 4 : 9 is 16 : 81

(ii) The subduplicate ratio of 16 : 81 is 4 : 9 and

(iii) The triplicate ratio of 4 : 9 is 64 : 729

On the basis of above definitions we frame a rule that the value of the ratio does not change if

(i) it is reduced to its lowest term, and

(ii) its terms are multiplied by the same number

For example

$$(i) \quad 144 : 36 = \frac{144}{36} = \frac{12}{3} = \frac{4}{1} = 4 : 1$$

$$(ii) \quad (ab^2 - a^2b) : (a-b)^2 = \frac{ab^2 - a^2b}{(a-b)^2} = \frac{ab(b-a)}{(a-b)^2} = \frac{(-)ab}{a-b} \\ = -(ab) : (a-b)$$

$$(iii) \quad a^2b^3 : 3ab^4 = \frac{a^2b^3}{3ab^4} = \frac{a}{3b} = a : 3b$$

$$(iv) \quad 4 : 5 = \frac{4}{5} = \frac{32}{40} = 32 : 40$$

$$\text{Again } \frac{32}{40} = \frac{4}{5} = 4 : 5$$

(i) To find the ratio of 3 sq yds to 6 sq ft, we first reduce both the quantities to *same unit of measurement* i.e. sq ft

$$3 \text{ sq yds} = 27 \text{ sq ft}$$

Required ratio is 27 sq ft : 6 sq ft or 27 : 6 = 9 : 2

(Remember that ratio is a unitless quantity)

(ii) Find the ratio compounded of ratio $2m : 3n$ and duplicate ratio of $9n^2 : mn$

The duplicate ratio of $9n^2 : mn$ will be $= 81n^4 : m^2n^2$

Required ratio compounded of $2m : 3n$ and $81n^4 : m^2n^2$ will be

$$\frac{2m}{3n} \times \frac{81n^4}{m^2n^2} = \frac{54n}{m} = 54n : m$$

(vii) Find the ratio compounded of the duplicate ratio of

$$\frac{2x}{y} : \frac{\sqrt{6x^2}}{y^2} \text{ and the ratio } 3ax : 2by$$

$$\text{Given ratio } \frac{2x}{y} : \frac{\sqrt{6x^2}}{y^2} = \frac{2x}{y} \times \frac{y^2}{\sqrt{6x^2}} = 2y : \sqrt{6} x$$

$$\text{Duplicate ratio of } 2y : \sqrt{6} x = 4y^2 : 6x^2$$

Required ratio compounded of $4y^2 : 6x^2$ and $3ax : 2by$ will be

$$\frac{4y^2}{6x^2} \times \frac{3ax}{2by} = \frac{ay}{bx} = ay : bx$$

(viii) If $m+7 : 2(m+14)$ is the duplicate ratio of $5 : 8$ find the value of m

Duplicate ratio of $5 : 8$ is $25 : 64$

$$\frac{m+7}{2(m+14)} = \frac{25}{64} \quad \text{On solving we obtain } m = \frac{252}{14} = 18$$

(ix) If $a : b = 3 : 4$ find the ratio of $7a-4b : 3a+b$

$$\text{Since } \frac{a}{b} = \frac{3}{4} = \frac{3k}{4k} \text{ (assume) it means } a=3k \text{ and } b=4k$$

$$\text{With these values we get } 7a-4b=5k$$

$$\text{and } 3a+b=13k$$

$$7a-4b : 3a+b = 5 : 13$$

Proportion A proportion expresses equality between two ratios

If two equal ratios are given to be $x : y$ and $m : n$, then $x : y = m : n$ is called a proportion which is written in symbolic form as $x : y :: m : n$ (read as x is to y as m is to n) and written in fractional form as $\frac{x}{y} = \frac{m}{n}$

In the proportion $x : y :: m : n$, terms x and n are called *extremes*, y and m the *means*

Quantities, a, b, c, d are said to be in continued proportion if $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$ so that if only three quantities a, b, c are in continued proportion

$$\text{then } a : b :: b : c, \text{ or } \frac{a}{b} = \frac{b}{c} \text{ or } ac = b^2$$

In this case b is said to be mean proportional between a and c , and c is said to be a third proportional to a and b

Laws of Proportion

(1) *Invertendo* If $a : b :: c : d$ then $b : a :: d : c$

$$\text{Since } \frac{a}{b} = \frac{c}{d} \quad 1 + \frac{a}{b} = 1 + \frac{c}{d} \quad \text{or } \frac{b}{a} = \frac{d}{c}$$

$$\text{or } b : a = d : c$$

(2) *Alternando* If $a : b = c : d$, then $a : c = b : d$.

Since $\frac{a}{b} = \frac{c}{d}$ $ad = bc$, dividing both sides by the same quantity cd

we get $\frac{ad}{cd} = \frac{bc}{cd}$ or, $\frac{a}{c} = \frac{b}{d}$ or $a : c = b : d$

(3) *Componendo* If $a : b = c : d$ then $(a+b) : b = (c+d) : d$

Since $\frac{a}{b} = \frac{c}{d}$ $\frac{a}{b} + 1 = \frac{c}{d} + 1$

$\frac{a+b}{b} = \frac{c+d}{d}$ or $(a+b) : b = (c+d) : d$

(4) *Dividendo* If $a : b = c : d$, then $(a-b) : b = (c-d) : d$

Since $\frac{a}{b} = \frac{c}{d}$ $\frac{a}{b} - 1 = \frac{c}{d} - 1$

$(a-b) : b = (c-d) : d$

(5) *Componendo and Dividendo* If $a : b + c : d$ then

$$((a+b) : (a-b)) = ((c+d) : (c-d))$$

From (3) we have $(a+b) : b = (c+d) : d$

From (4) we have $(a-b) : b = (c-d) : d$

By division we get $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

$$(a+b) : (a-b) = (c+d) : (c-d)$$

Ex 1 Determine the value of x in the proportion

$$(x+4) : 1 = (2-x) : 2$$

$$\frac{x+4}{1} = \frac{2-x}{2} \text{ or } 2(x+4) = (2-x)$$

$$x = -2$$

Ex 2 Find the fourth proportional to -2, 5, 6

Let x be the fourth proportional then

$$-2 : 5 = 6 : x \text{ or } \frac{-2}{5} = \frac{6}{x} \quad x = 15$$

i.e. required fourth proportional to -2, 5, 6 is 15

Ex 3 Find the third proportional to ab , $\sqrt{ab}$

Let x be the third proportional then

$$ab : \sqrt{ab} = \sqrt{ab} : x$$

$$\frac{ab}{\sqrt{ab}} = \frac{\sqrt{ab}}{x} \quad \text{or } x = 1 \text{ which is the required third proportional}$$

to ab , $\sqrt{ab}$

Ex. 4 Find the mean proportion of between $360a^4$ and $250a^2b^2$
 [Let x be the mean proportion; then

$$\text{i.e.,} \quad \frac{360a^4}{x} = \frac{x}{250a^2b^2}$$

$$\text{or } x^2 = (360 \times 250)a^4b^2 \quad \text{or } x = 300a^3b$$

$300a^3b$ is the required mean proportion of between $360a^4$ and $250a^2b^2$

Ex. 5 Income of Rs. 500 is to be divided between two brothers in the ratio of 2 : 3 Find the share of each

Suppose one of them receives Rs. x then the other will receive Rs. $(500 - x)$ and these quantities must be in the ratio of 2 : 3

$$\frac{x}{500 - x} = \frac{2}{3} \quad \text{or } x = 200$$

The required shares are Rs. 200 and Rs. 300

Ex. 6. Divide income of Rs. 18,000 among four brothers in the proportion of 2, 3, 6, 7

Let the required four shares be $2x$, $3x$, $6x$ and $7x$

$$\text{So that total income } 2x + 3x + 6x + 7x = 18000 \\ (= 18000)$$

Hence the required shares are 2000, 3000, 6000 and 7000

Ex. 7 Two numbers are in the ratio of 6 : 7 if one is added to each of the numbers the resulting ratio is 7 : 8 Find the numbers

Let the numbers be x and y , then

$$\frac{x}{y} = \frac{6}{7} \tag{1}$$

$$\text{and} \quad \frac{x+1}{y+1} = \frac{7}{8} \tag{2}$$

Solving two equations we obtain

$$x = -\frac{1}{2} \quad \text{and} \quad y = -\frac{3}{7}$$

i.e., the two numbers are $-\frac{1}{2}$ and $-\frac{3}{7}$ which are in the ratio of $\frac{7}{6}$ and when 1

is added to each, the resulting ratio is $\frac{-\frac{1}{2}+1}{-\frac{3}{7}+1} = \frac{7}{8}$

Variation. When two quantities depend upon each other and if change

in one brings about change in the other in the same ratio then the two quantities are said to *vary directly*.

Suppose there are two quantities X and Y and if Y is changed X also changes in the same ratio then X is said to vary directly as Y .

The symbol $\propto$ is used to denote variation. Thus $X \propto Y$ will be read as X varies directly as Y . Remember in economic theory we say that supply (S) varies directly as price (P) or we can write $S \propto P$.

Constant Variation. If X varies as Y (generally word directly is ignored) then X is equal to Y multiplied by some constant quantity that is $X = kY$ or $\frac{X}{Y} = k$ where k is the constant of variation.

This constant of variation can be determined if any pair of corresponding values of two quantities (X and Y) are known.

For example if $X=4$ when $Y=3$ we have

$$4 = k \cdot 3 \quad \text{or} \quad k = \frac{4}{3}$$

Ex. 8 If X varies directly as Y and $X=6$ when $Y=15$ find X when $Y=10$.
Given $X=6$ $Y=15$ Since $X \propto Y$

$$X = kY \quad \text{or} \quad 6 = k \cdot 15 \quad \text{or} \quad k = \frac{6}{15}$$

$$\text{Thus if } Y = 10 \text{ then } X = \frac{6}{15} \times 10 = \frac{16}{3}$$

Ex. 9 If X varies as Y^2 and if $X=4$ when $Y=3$ find X when $Y=6$.
Given $X \propto Y^2$ $X = kY^2$

$$k = \frac{4}{9}$$

$$\text{hence } X = \frac{4}{9} \times (6) = 16$$

Inverse Variation. Two quantities are said to vary inversely if one varies as the reciprocal of the other.

$$\text{Thus if } X \text{ varies inversely as } Y \text{ then } X = k \cdot \frac{1}{Y} \\ k = XY$$

For example economic theory we say demand (D) varies inversely as price (P) $D \propto \frac{1}{P}$

Ex. 10 If X varies inversely as Y and $X=7$ when $Y=3$ find X when $Y=7$

$$\text{Given : } X \propto \frac{1}{Y} \quad \therefore \quad X = k \frac{1}{Y}$$

$$\text{i.e. } 7 = k \times \frac{1}{3} \quad \text{or, } k = 21$$

$$\text{that is, } X = 21 \times \frac{3}{7} = 9$$

Ex. 11. If the square of m varies as the cube of n , and $m=3$ when $n=4$, find the value of n when $m=\frac{1}{\sqrt{3}}$.

$$\text{Given : } m^2 \propto n^3, \quad m^2 = k.n^3, \quad \text{or, } k = \frac{m^2}{n^3} = \frac{9}{64}$$

$$\text{For } m = \frac{1}{\sqrt{3}}, \quad \frac{1}{3} = \frac{9}{64} n^3 \quad \text{or, } n = \frac{4}{3}$$

A quantity is said to vary jointly as a number of other quantities, when it varies directly as their product.

Thus quantity X will be said to vary jointly as Y and Z , when

$$X = k.YZ$$

For example, in economic theory we say that amount of interest (I) varies jointly as the principal (P), the time (t), and the rate of interest (r)

$$I = P \times t \times \frac{1}{100} r$$

Ex. 12. P varies as Q and R jointly, if $P=2$ when $Q=\frac{3}{5}$ and $R=\frac{10}{27}$, find R when $P=54$ and $Q=3$.

Since $P \propto$ jointly as Q and R , hence $P = k.QR$

$$\text{We are given : } 2 = k \frac{3}{5} \cdot \frac{10}{27}, \quad \text{hence } k = 9$$

$$54 = 9 \times 3 \times R, \quad \text{i.e., } R = 2.$$

Ex. 13. A varies directly as B and inversely as C . It is known that $A=\frac{2}{3}$ when $B=\frac{3}{7}$ and $C=\frac{9}{14}$. Find B when $A=48$ and $C=75$.

$$\text{Given : } A \propto \frac{B}{C} \quad \text{or } A = k \cdot \frac{B}{C}$$

$$\frac{2}{3} = k \cdot \frac{\frac{3}{7}}{\frac{9}{14}} \quad \text{that is, } k = 2$$

$$48 = 2 \times \frac{B}{75} \quad \text{i.e., } B = (24 \times 75) = 1800.$$

Ex. 14. If P varies directly as the square root of Q and inversely as the cube root of R , and if $P=3$ when $Q=16$ and $R=8$, find Q when $P=27$ and $R=\frac{1}{8}$.

$$P \propto \frac{\sqrt{Q}}{\sqrt[3]{R}}, \text{ i.e., } P = k \cdot \frac{\sqrt{Q}}{\sqrt[3]{R}}$$

$$3 = k \times \frac{4}{2}, \therefore k = \frac{3}{2}$$

$$\therefore 27 = \frac{3}{2} \cdot \frac{\sqrt{Q}}{\frac{1}{2}}, \text{ i.e., } Q = 6$$

Ex. 15. If X_1 varies as X_2 and X_3 jointly while X_2 varies as square of X_4 , and X_3 varies inversely as X_1 , show that X_1 varies directly as X_4 .

$$\text{Given } X_1 \propto X_2 X_3, \quad X_1 = k_1 X_2 X_3 \quad (1)$$

$$X_2 \propto X_4^2, \quad X_2 = k_2 X_4^2 \quad (2)$$

$$X_3 \propto \frac{1}{X_1}, \quad X_3 = k_3 \cdot \frac{1}{X_1} \quad (3)$$

Substituting Eqn (2) and (3) in (1),

$$X_1 = k_1 \cdot k_2 \cdot X_4^2 \cdot k_3 \cdot \frac{1}{X_1}$$

$$X_1^2 = k_1 \cdot k_2 \cdot k_3 \cdot X_4^2$$

$$X_1 = \sqrt{k_1 k_2 k_3} X_4 \text{ which shows that}$$

$$X_1 \propto X_4$$

Ex. 16. (i) If $x - y = 2 \cdot 3$, find the value of $\frac{3x-y}{4x-3y}$.

(ii) If $\frac{a}{b} = \frac{c}{d}$; prove that $\frac{a-b}{b} = \frac{c-d}{d}$

(i) $\frac{x}{y} = \frac{2}{3}, \frac{x}{2} = \frac{y}{3} = k$ (say)

$$x = 2k \text{ and } y = 3k$$

$$\frac{3x-y}{4x-3y} = \frac{6k-3k}{8k-9k} = \frac{3k}{-k} = -3$$

(ii) Given $\frac{a}{b} = \frac{c}{d}$, subtract one from both sides

$$\therefore \frac{a}{b} - 1 = \frac{c}{d} - 1$$

$$\text{i.e., } \frac{a-b}{b} = \frac{c-d}{d}$$

Ex. 17. The ratio of two numbers is 3 : 2. If each is increased by 10, the ratio becomes 4 : 3. Find the numbers.

Let x and y be two required numbers, then

$$\frac{x}{y} = \frac{3}{2}; 2x - 3y = 0 \quad \dots\dots(1)$$

$$\text{Also, } \frac{x+10}{y+10} = \frac{4}{3}, 3x-4y=10 \quad (ii)$$

Solving Eqn (i) and (ii), we obtain

$$x = 30 \text{ and } y = 20$$

Ex. 18 The income tax on an individual's income varies as the square of the excess of the individual's yearly income over Rs 6,000. The tax on the yearly income of Rs 10,000 is Rs 640. Find the tax on the yearly income of Rs. 12,000.

Let T represent income tax and I the income, then

$$T \propto (I - 6000)^2$$

$$T = K(I - 6000)^2$$

Given $T = 640$ when $I = 10,000$

$$640 = K(4000)^2$$

$$\text{i.e., } K = 4 \times 10^{-5}$$

We have to find T for $I = 12000$, hence

$$K = 4 \times 10^{-5} (12000 - 6000)^2$$

$$T = 4 \times 10^{-5} (6000)^2$$

$$T = 4 \times 10^{-5} \times 36 \times 10^6$$

$$T = 4 \times 36 \times 10 = 1440$$

Ex. 19. A man's savings are proportional to his income above Rs. 400. When his income is Rs. 500, he saves Rs. 20. What will be his savings if his income becomes Rs. 800? What must be his income if he wants to save Rs. 100?

Let S represent savings and I the income

$$S \propto (I - 400)$$

$$S = K(I - 400)$$

Given that, $I = 500$ then $S = 20$

$$20 = K(500 - 400)$$

$$\text{i.e., } K = \frac{1}{5}$$

We have to find S when $I = 800$

$$S = \frac{1}{5} (800 - 400) = 80$$

When $S = 100$, we find I ,

$$100 = \frac{1}{5} (I - 400),$$

$$I = 900$$

3.4. Logarithms

In mathematical simplifications we come across operations like multi-

multiplication, division, taking roots of power of numbers. Many a time the solution is rather difficult. Suppose we have to find the value of

$$\left\{ \frac{200000000}{(100)^{3.54}} \right\}$$

To find this, we shall have to find the value of $(100)^{3.54}$. From simple arithmetic we know to compute $(100)^3$ and $(100)^4$ but not $(100)^{3.54}$. To simplify such calculations, we use the technique of Logarithms.

Definition

If $a^x = M$, then $\log_a M = x$, or,

If $\log_a M = m$, then $a^m = M$

If a^x is equal to M , then x is defined to be equal to logarithm (short-form log) of M to the base a or simply $\log M$ to the base a .

In this way, the logarithm of any quantity M to the given base a (i.e., $\log_a M$) is equal to a number x , which is the index of power to which a should be raised (a^x) to produce the number M .

Thus we are able to express the relation between three numbers a , x and M in two alternative forms

and $a^x = M$ (Exponential form)
 and $\log_a M = x$ (Logarithmic form)

For example, we know that $(4)^2 = 16$

We may also write $\log_4 16 = 2$ which is log-form. This is read as 'log of 16 to the base 4 is equal to 2'.

It is to be noted that the base in the exponential form (i.e., 4) is also the base in the log-form. The term entering as index of power in the exponential form (i.e., 2) is the value on the R.H.S. in the log-form.

This can be explained as follows

If $\log_{10} 1000 = 3$, then

$10^3 = 1000$ is the exponential form

Similarly if $\log_{20} 400 = 2$, then

$(20)^2 = 400$ is the exponential form

We give more examples in the following table

Exponential form $a^x = M$	Log form $\log_a M = x$	Base
$10^2 = 1000$	$\log_{10} 1000 = 3$	10
$7^3 = 343$	$\log_7 343 = 3$	7
$2^1 = M$	$\log_2 M = 1$	2
$9^{\frac{1}{2}} = \frac{1}{2}$	$\log_9 \frac{1}{3} = -\frac{1}{2}$	9

Common Logarithms (Logarithms to the Base 10) In the above examples we find that base of logarithms is different in each case. In arithmetic simplifications we usually take 10 as base. The logarithm of any number M to base 10 ($\log_{10} M$) is termed as its common logarithm.

We give below the examples of common logarithms.

$$\log_{10} 1000 = 3 \quad \text{because } 10^3 = 1000$$

$$\log_{10} 100 = 2 \quad \text{because } 10^2 = 100$$

$$\log_{10} 1 = 0 \quad \text{because } 10^0 = 1$$

$$\log_{10} 0.1 = -1 \quad \text{because } 10^{-1} = 0.1$$

$$\log_{10} 0.01 = -2 \quad \text{because } 10^{-2} = 0.01$$

It is convention that when we are concerned with common logarithms, we do not mention the base 10, in other words, $\log_{10} 0.001$ is simply written as $\log 0.001$. But when we use base of logarithms other than 10, we should mention it.

It should be noted that we cannot take logarithm of zero or negative quantities.

Napierian Logarithms. Another system of logarithms employed by the discoverer of logarithms, Napier, and called the Napierian or Hyperbolic system is also used in calculations. The base of this system is denoted by the symbol e and is the number which is sum of the series

$$2 + \frac{1}{2} + \frac{1}{2 \times 3} + \frac{1}{2 \times 3 \times 4} + \dots$$

This sum to four decimals is 2.7183. The common logarithms or simply logarithms can be calculated from the Napierian logarithms and *vice versa*.

Let l and L be the logarithms of the same number, with common and Napierian systems respectively, then

$$l = \frac{1}{\log_e 10} (L)$$

$$\log_e 10 = 2.30258509$$

$$\text{and, } \frac{1}{2.30258} = 0.43429$$

Hence, the common logarithm of number may be obtained by multiplying the Napierian logarithm of the same number by 0.4343.

Again, to convert common into Napierian logarithms we multiply by 2.3026. For example hyperbolic (Napierian) logarithm of 8.43 can be obtained as follows

$$\log_e 8.43 = 0.9258 \quad \log_e 8.43 = 0.9258 \times 2.3026 = 2.1317$$

Characteristic and Mantissa

We have learnt that $\log_{10} 1 = 0$

$$\log_{10} 10=1,$$

$$\log_{10} 100=2 \text{ and } \log_{10} 1000=3 \text{ and so on}$$

Thus the logarithm of numbers between 1 and 10 will lie between 0 and 1 *i.e.*, it will be decimal number like 0.72 or 0.25. Similarly the logarithm of number between 10 and 100 will lie between 1 and 2 like 1.72 and 1.25. In this way the logarithm of numbers like 0.10, 100 will be integer powers of 10 and whole numbers. The logarithm of all the other numbers consists of the two parts

- (i) An integer—called the *Characteristic*
- (ii) A decimal part—called the *Mantissa*

(i) Characteristic is always one less than the digits in the number excluding the decimal part of the number. Thus the characteristic of 562.79, 562.63 and 562.00 will be 2, because these numbers lie between 100 and 1,000 and also they contain 3 digits (excluding the decimal parts). Characteristic of pure decimal numbers (*i.e.*, numbers between 0 and 1) is found on the following principle

We know that $\log 0.001 = -3$, $\log 0.01 = -2$, $\log 0.1 = -1$ and $\log 1.0 = 0$. Therefore, the logarithm of numbers lying between 1 and 0.1 will be between 0 and -1, say -0.75

But -0.75 can also be written as $(-1+0.25)$ which we conventionally write as $\bar{1}.25$ and characteristic will then be $\bar{1}$ or -1

Similarly, the logarithm of numbers lying between 0.1 and 0.01 will lie between -1 and -2, say -1.75, -1.75 can also be written as $(-2+0.25)$ which we write as $\bar{2}.25$ and characteristic = -2.

Thus we conclude that the negative characteristic is numerically one more than the number of zeros immediately after the decimal point.

(ii) The decimal part of a logarithm is called the mantissa. We have learnt that

$$\log_{10} 100.0 = 2.00$$

$$\log_{10} 1000.0 = 3.00$$

$$\log_{10} 0.1 = -1.00 \text{ or } \bar{1}.00$$

In all these cases the mantissa is same (*i.e.*, 0.00) though the numbers 0.1, 1000.0 and 100.0 are different. But it should be noted that though they are different, the difference is only in the position of the decimal point. We say, therefore, that if the numbers differ only in the position of the decimal point *i.e.* if the number is divided or multiplied by powers of 10 then the mantissa remains unchanged. Hence when we find the mantissa of any number (from the logarithm table), we remove its decimal point. Nevertheless the position of the decimal point is all important in determining the characteristic of the number.

The mantissa of any number is found by using logarithmic tables.

To obtain the logarithm of a number consisting of four figures, the way to proceed is as follows

Ex 1 Find the log of 3768

First look in the logarithmic table (given at the end of the book) for the number 37 in the column on the extreme left. The next figure which is 6 in this case is found at the top of the table, so that the mantissa of

$$\log 376 = 5752$$

At the extreme right of the table will be seen a column of differences as they are called. Thus under the figure 8 and in the row of 37 is found the number 9. This must be added to the mantissa previously obtained

Hence we have mantissa of $\log 376 = 5752$

Add difference $= 9$

Gives mantissa for $\log 3768 = 5861$

To find characteristic, we count the digits of the given number. The number of digits in 3768 is four hence characteristic of the number is three

Therefore $\log 3768 = 3.5861$

Ex 2 Find the log of 0.3768

For finding mantissa we neglect the decimal point placed anywhere in the number. Hence mantissa of 3768 is once again $= 5861$

What will be the characteristic? There is no figure on the left side of the decimal, i.e. zero number of digits. Since the characteristic is one less than the number of digits on the left hand side of the number in this case

$$\text{Characteristic} = 0 - 1 = \bar{1}$$

$$\text{i.e. } \log 0.3768 = \bar{1}.5861$$

$$\text{Similarly } \log 0.03768 = \bar{2}.5861$$

$$\log 0.003768 = \bar{3}.5861 \text{ and so on}$$

Rules for the Use of Logarithms

(i) *Rule of Multiplication* $\log_a MN = \log_a M + \log_a N$

Ex 3 Find log of (102 multiplied by 1067)

$$\log (102 \times 1067) = \log 102 + \log 1067$$

From the logarithmic tables $\log 102 = 2.0086$

and $\log 1067 = 3.0281$

i.e. $\log (102 \times 1067) = 5.0367$

(ii) *Rule of Division* $\log_a \frac{M}{N} = \log_a M - \log_a N$

Ex 4 Find log of (102÷1067)

$$\log \frac{102}{1067} = \log 102 - \log 1067$$

$$= 2.0086 - 3.0281 = -1.0195$$

$$= -1 - .0195$$

But mantissa can never be negative so to do away with the minus sign from the mantissa we write (-0.195) as $(-1 + 9805)$

$$\begin{aligned}\log \frac{102}{1067} &= -1.0195 = -2 + 9805 \\ &= \bar{2} 9805\end{aligned}$$

(iii) *Rule of power* $\log_r M^r = r \log_r M$

Ex 5 Find $\log$ of 121 and 100

$$\log 121 = \log (11)^2 = 2 \log 11 = 2 (1.0414) = 2.0828$$

$$\log 100 = \log (10)^2 = 2 \log 10 = 2 (1.0000) = 2.0000$$

Antilogarithms So far we have learnt how to find the logarithm of any given number but our ultimate objective is know the use of logarithms to simplify our calculations.

We shall illustrate this point by taking a few examples.

Ex 6 Find the value of (4.63×85.76)

Suppose value of $4.63 \times 85.76 = x$

Taking the $\log$ on both sides

$$\log x = \log (4.63 \times 85.76)$$

$$\log x = \log 4.63 + \log 85.76$$

$$\log x = 0.6656 + 1.9333$$

$$\log x = 2.5989$$

But we have to obtain the value of x and not of $\log x$

In fact 2.5989 is the $\log$ of the value which is required. Since characteristic = 2 we can say that the required value should be a three (2+1 = 3) digit number of which mantissa is .5989.

To find the exact value we make use of antilogarithm tables. The left hand side column of these tables contains the first two digits of mantissa in ascending order while the top row contains the third digit of the mantissa (*Characteristic is not seen in the antilog tables*).

In the above example we have to find the anti $\log$ of 2.5989 in order to obtain the value of x .

2 is characteristic while .5989 is mantissa.

We consider the mantissa part .5989. We look for .59 in first column on left side of the anti $\log$ table and then the next figure 8 is found at the top of the table so that anti $\log$ of .5989 = 3963. Fourth digit 9 is seen in the column of differences. Thus under the figure 9 we find the number 8. Hence

$$\text{anti } \log \text{ of } .5989 = 3963 + 8 = 3971$$

Now since the characteristics is 2, the required value of x should be of 3 digits, i.e. $x=3971$

Ex. 7. Find the value of $\sqrt[4]{0.0007}$

$$\text{Let } x = \sqrt[4]{0.0007} = (0.0007)^{1/4}$$

$$\log x = \frac{1}{4} \log 0.0007 = \frac{1}{4} (4 + 8451)$$

$$= -1.3333 + 2817$$

$$= -1.3333 + 2817 + 2 - 2 = 2.2817 - 1.3333 - 2$$

$$= \bar{2} + 99484 = \bar{2} 9484$$

Anti log of 9484 is = 8880, since the characteristics is $\bar{2}$, the number of zeros after the decimal point in the required number should be one less, i.e., 1

Hence, $x = 0.888$

Ex. 8. Find the value of $\left\{ \frac{579 - \sqrt[4]{25}}{\sqrt{7}} \right\}$

$$\text{Let } x = \frac{579 - \sqrt[4]{25}}{\sqrt{7}}$$

$$x = \frac{579}{(7)^{1/2}} - \frac{(25)^{1/4}}{(7)^{1/2}}$$

$$\text{Let } y = \frac{579}{(7)^{1/2}} \text{ and } z = \frac{(25)^{1/4}}{(7)^{1/2}}$$

$$x = y - z$$

We find the value of each part separately as follows

$$y = \frac{579}{(7)^{1/2}}$$

$$\log y = \log 579 - \frac{1}{2} \log 7 = 2.7627 - \frac{1}{2} (.8451)$$

$$= 2.7627 - .4226 = 2.3401$$

Therefore, $y = \text{antilog } 2.3401 = 218.9$

$$\text{Again, } z = \frac{(25)^{1/4}}{(7)^{1/2}}$$

$$\begin{aligned}\log z &= \frac{1}{3} \log 25 - \frac{1}{2} \log 7 = \frac{1}{3} (1.3979) - \frac{1}{2} (.8451) \\ &= .4660 - .4226 = .0434\end{aligned}$$

so that, $z = \text{antilog } .0434 = 1.105$, $x = y \cdot z = 217.795$

Ex. 9 Find the values of (i) $\log_7 49$ (ii) $\log_{\sqrt{2}} (16)$

$$\begin{aligned}\text{(i)} \quad & \text{Let } \log_7 49 = x \\ & (7)^x = 49 \text{ or, } (7)^x = (7)^2 \text{ or, } x = 2 \\ & \log_7 49 = 2\end{aligned}$$

$$\begin{aligned}\text{(ii)} \quad & \text{Let } \log_{\sqrt{2}} [16] = x \\ & \left(\frac{1}{\sqrt{2}}\right)^x = 16, \quad \left(\frac{1}{\sqrt{2}}\right)^x = (2)^4, (2)^{-\frac{x}{2}} = 2^4 \\ & -\frac{1}{2}x = 4, \text{ or } x = -8\end{aligned}$$

That is, $\log_{\sqrt{2}} (16) = -8$

Ex. 11. Using logarithmic tables, calculate

$$\frac{17.2 \times 24.5}{43.15}$$

Let the given expression be equal to X , then

$$\begin{aligned}\log X &= (\log 17.2 + \log 24.5) - \log (43.15) \\ &= (1.2355 + 1.3892) - 1.6350 \\ &= 0.9897\end{aligned}$$

Anti log of 0.9897 = 9.766

Ex. 12. Calculate by using logarithmic tables

$$\sqrt{\frac{628}{523} + \frac{16}{27}}$$

Let the given expression be equal to X

$$\begin{aligned}\therefore \log X &= \frac{1}{2} [(\log 628 + \log 27) - (\log 523 + \log 16)] \\ &= \frac{1}{2} [(2.7980 + 1.4314) - (2.7185 + 1.2041)] \\ &= \frac{1}{2} [(4.2294) - (3.9226)]\end{aligned}$$

$$= \frac{1}{2} (0.3068) = 0.1534$$

Taking into log $X = 1.423$

Ex 13 Calculate by using logarithmic tables

$$\sqrt{\frac{156.6}{225.3}}$$

Let the given expression be equal to X

$$\log X = \frac{1}{2} [\log 156.6 - \log 225.3]$$

$$= \frac{1}{2} [2.1947 - 2.3528]$$

$$= \frac{1}{2} [-1.1581]$$

$$= \frac{1}{2} [2 + 1.8419]$$

$$= 1.9209$$

Taking into log the value of the given expression is = 0.835

Ex 14 Find the values of

(i) $(0.0001)^{\frac{1}{4}}$ (ii) $\log_{\sqrt{6}} [216]$

$$(i) \quad (0.0001)^{\frac{1}{4}} = \left(\frac{1}{0001} \right)^{\frac{1}{4}} = (10000)^{\frac{1}{4}} \\ = (10^4)^{\frac{1}{4}} = (10) = 10000$$

(ii) Let $\log_{\sqrt{6}} 216 = x$

$$(\sqrt{6})^x = 216$$

$$(6)^{\frac{x}{2}} = (6)^3$$

$$\frac{x}{2} = 3 \quad \text{i.e.} \quad x=6 \quad \text{that is } \log_{\sqrt{6}} [216] = 6$$

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1 Find the logarithms of the following

(i) 56.789 (ii) 0.007894 (iii) 0.92007 (iv) 888.88

2 Find the value of the following by using logarithms

(i) 36.25×003

(ii) $\frac{5.0235}{\sqrt{(585)^2 \times 5.03}}$

(iii) $\sqrt[3]{235.67}$

(iv) $\frac{0.6834}{\sqrt{1.234 \times 0.8468}}$

(v) $5\sqrt{8.00125}$

- 3 If demand of me it is given by $D = \frac{2}{p^3}$ find the demand at $p=2$ and $p=4$
- 4 If the production function of a firm is $Y=100I^{0.4}K^{0.6}$ where X is output I is labour input and K is capital input find Y when
- $L=105$ units $K=1200$ units and
 - $L=140$ units $K=120$ units
- 5 Pareto distribution of incomes (Y) in a community is given by $N=1236000000/Y^{4.1}$ Find the number of people who have income of
- Rs 125200 or higher
 - Rs 2365567 or higher and
 - Rs 1567 or higher
- 6 Income distribution of a community is represented by a function
- $$N = \frac{813000000}{x^{1.48}}$$
- N = No. of people who have income x or higher
Obtain the lowest income of the 10 richest people

Progressions

A set of numbers arranged in a definite order with a definite rule is called a sequence. When the numbers in a sequence are connected by plus or minus these form a series.

Each number of series is called a term of the series.

Thus different sets of numbers or series can be formed with different rules.

3.5 Arithmetic Progression. An arithmetic progression (AP) is a set of numbers in which terms are formed by adding (or subtracting) a constant number to the preceding term. The constant number is called *common difference*.

For example 2, 5, 8, 11, ... is an AP since each term is obtained by adding 3 to the preceding number. Hence 3 is the common difference.

First term is always denoted by letter a and the common difference by d so if it were written an AP in the general form is under

a	$a+d$	$a+2d$	$a+3d$
1 st term	2 nd term	3 rd term	4 th term

We notice that in any term the coefficient of common difference (d) is less by one than the number of the term of the series. Thus in second term ($a+d$) coefficient of d is 1. In this way

in third term	$(a+2d)$ coefficient of d is 2
in fourth term	$(a+3d)$ coefficient of d is 3

What will be the coefficient of d in n^{th} term?

It will be $(n-1)d$ if the n^{th} term = $\{a+(n-1)d\}$

Thus we can find any term (i.e., n^{th} term) of AP by knowing first term (a) and common difference (d)

$$n^{\text{th}} \text{ term} = a + (n-1)d$$

Ex 1 Find the 10th term of the following series

$$\frac{1}{4}, 1, \frac{7}{4}, \frac{11}{4}$$

The given series is an AP , since $\left\{1 - \frac{1}{4}\right\} = \frac{3}{4} = \left\{\frac{7}{4} - 1\right\}$

Thus we have $a = \frac{1}{4}$ and $d = \frac{3}{4}$

$$\text{Required 10th term of series} = a + (10-1)d = \frac{1}{4} + 9\left(\frac{3}{4}\right) = 7$$

Ex 2 Find the 12th term of the series

$$(x-3y), (2x-5y), (3x-7y)$$

The given series is an AP since

$$(2x-5y) - (x-3y) = (x-2y)$$

$$\text{and } (3x-7y) - (2x-5y) = (x-2y)$$

Thus we have $a = (x-3y)$ and $d = (x-2y)$

$$\text{required 12th term} = a + 11d = (x-3y) + 11(x-2y) = 12x - 25y$$

Ex 3 Which term of the series $\frac{3}{\sqrt{5}}, \frac{4}{\sqrt{5}}, \sqrt{5}$ is $\frac{12}{\sqrt{5}}$

$$n^{\text{th}} \text{ term} = a + (n-1)d$$

$$\text{i.e. } \frac{12}{\sqrt{5}} = \frac{3}{\sqrt{5}} + (n-1)\frac{1}{\sqrt{5}}$$

$$n=10, \text{ i.e., } \frac{12}{\sqrt{5}} \text{ is the 10th term of the given series}$$

Sum of the first n terms of an AP

Suppose we are required to find the sum of first n numbers (S_n) of the series $a, a+d, a+2d, \dots, [a+(n-1)d]$

$$S_n = a + (a+d) + (a+2d) + \dots + [a+(n-2)d] + [a+(n-1)d]$$

Writing the above sum in the reverse order

$$S_n = [a + (n-1)d] + [a + (n-2)d] + \dots + (a+d) + a$$

Adding together these two sums

$$2S_n = [2a + (n-1)d] + [2a + (n-1)d] + [2a + (n-1)d] + \dots$$

$$2S_n = n[2a + (n-1)d]$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

Ex 4 Find the sum of series $\frac{11}{2}, \frac{27}{4}, 8$ up to 15 terms

The given series is an AP in which $a = \frac{11}{2}$ and $d = \frac{5}{4}$

$$\text{Hence } S_{15} = \frac{15}{2} \left[2 \left(\frac{11}{2} \right) + (15-1) \frac{5}{4} \right] = \frac{855}{4}$$

Lx 5 Find S_n (sum of first n terms) of the series

$$\frac{1}{8}, 1, \frac{7}{8}, \frac{3}{4}$$

Here $a = \frac{9}{8}$ and $d = -\frac{1}{8}$

$$\text{Hence } S_n = \frac{n}{2} \left[2 \left(\frac{9}{8} \right) + (n-1) \left(-\frac{1}{8} \right) \right] = \frac{n}{16} (19-n)$$

Ex 6 How many terms of the series $12, 16, 20$ will add up to 208

Suppose $S_n = 208$ therefore

$$208 = \frac{n}{2} [2a + (n-1)d]$$

$$208 = \frac{n}{2} [2(12) + (n-1)4] \quad \text{or} \quad n^2 + 5n - 104 = 0$$

i.e. $n = -13$ or $+8$ since number of terms cannot be negative the sum of first 8 terms of the given series will be 208

Lx 7 In an AP $S_{10} = 550$ and $a = 100$ What is the common difference?

$$S_{10} = 550 = \frac{10}{2} [2(100) + (10-1)d]$$

$$550 = 5[200 + 9d]$$

$$550 = 1000 + 45d$$

$$d = -10$$

Lx 8 In an AP $S_{15} = 600$ and $d = 5$ What is a ?

$$S_{15} = 600 = \frac{15}{2} [2a + 14 \times 5]$$

$$1200 = 30a + 1050$$

$$a = 5$$

Lx 9 The third term and seventh term of an AP are 18 and 30 respectively. What is S_{20} ?

$$\text{Third term} \quad 18 = a + 2d$$

$$\text{Seventh term} \quad 30 = a + 6d$$

On solving $d = 3$ and $a = 12$

$$S_{20} = \frac{20}{2} [2(12) + 19 \times 3] = 810$$

Ex 10 In an AP, $S_3 = 33$ and $S_5 = 50$. What is a and d ?

$$S_3 = 33 = \frac{3}{2} [2a + 2d] = 3a + 3d$$

$$S_5 = 50 = \frac{5}{2} [2a + 4d] = 5a + 10d$$

i.e. $3a + 3d = 33$

and, $5a + 10d = 50$, solving these equations we obtain

$$a = 12 \text{ and } d = -1$$

Ex 11. If the sum of first n , $2n$ and $3n$ terms of an AP is S_n , S_{2n} , S_{3n} respectively, prove that $S_{3n} = 3(S_{2n} - S_n)$

We are given $S_n = \frac{n}{2} [2a + (n-1)d]$ (1)

$$S_{2n} = \frac{2n}{2} [2a + (2n-1)d]$$
 (2)

$$S_{3n} = \frac{3n}{2} [2a + (3n-1)d]$$
 (3)

$$\begin{aligned} S_n - S_{2n} &= \left\{ \frac{n}{2} [2a + (n-1)d] \right\} - \frac{2n}{2} [2a + (2n-1)d] \\ &= \frac{n}{2} [2a + (3n-1)d] \end{aligned}$$

$$3(S_{2n} - S_n) = \frac{3n}{2} [2a + (3n-1)d] \text{ which is equal to } S_{3n}$$

Ex. 12 Find S_{10} of an AP whose n^{th} term is $(4n+1)$

Since n^{th} term is $4n+1$

first term will be $4(1)+1=5$

2nd term will be $4(2)+1=9$, therefore

$$d=4$$

Thus $a=5$, $d=4$ and we need S_{10}

$$\begin{aligned} S_{10} &= \frac{10}{2} [2(5) + 9(4)] \\ &= 50 + 180 = 230 \end{aligned}$$

Ex 13. The sum of three numbers of an AP is 15 and sum of their cubes is 645, find them

Since three numbers are in AP, they will be of the form

$a-d$, a and $a+d$, their cubes will be, $(a-d)^3$, a^3 and $(a+d)^3$

Thus $(a-d)+a+(a+d)=15$ or $a=5$

Again $(a-d)^3 + a^3 + (a+d)^3 = 645$

$$(5-d)^3 + 125 + (5+d)^3 = 645$$

$(5-d)^3 + (5+d)^3 = 645 - 125 = 520$ on simplifying we get $d=3$
required three numbers are 2, 5 and 8

Ex 14 An enterprise produced 600 units in the third year of its existence and 700 units in its seventh year. Assuming that it grows in AP find

(i) its production in the first year

(ii) its production in the fifth year

and (iii) its total production during the first five years

Since the enterprise grows in AP

Production in the third year $= a + 2d = 600$

Production in the seventh year $= a + 6d = 700$

$$a = 550$$

Production in the fifth year $= a + 4d = 650$ units

Production during the first five years $= \frac{5}{2} [2(550) + 4 \times 25] = 3000$ units

Ex 15 Firm A starts producing 400 units and decreases production by 50 units yearly. Firm B starts by producing 250 units and increases production by 25 units yearly. Assuming that both firms grow (or decay) in an AP find

(i) In what year both will produce the same amount?

(ii) When will firm A produce zero output?

(iii) What will be the production of firm B in the year when firm A produces nothing?

Firm A $a=400$ units $d=-50$ units

Firm B $a=250$ units $d=25$ units

(i) Let both firms be producing the same amount of production in the n th year

Production of firm A in the n th year $= a + (n-1)d = 400 + (n-1)(-50)$

Production of firm B in the n th year $= a + (n-1)d = 250 + (n-1)(25)$

$$400 + (n-1)(-50) = 250 + (n-1)(25)$$

$$\text{Hence } n=3$$

i.e. both the firms will produce equal amount of production (300 units) in the third year of their production

(ii) Assume that firm A produces zero products in the n th year

Therefore $0 = a + (n-1)d = 400 + (n-1)(-50)$

$$-400 = 50 - 50n$$

$$n = 9$$

Firm A will produce nothing in the ninth year. Production of firm B in the ninth year (when production of firm A is zero) will be

$$250 + 8(25) = 450 \text{ units}$$

Arithmetic Mean When three quantities are in AP the middle one is said to be the arithmetic mean (AM) of the other two

For instance, (i) In an AP 2, 8, 14, 8 is AM of 2 and 14

(ii) In an AP 2, 4, 6, 8, 4 and 6 are two arithmetic means between 2 and 8

(iii) In an AP $a, a+d, a+2d, \dots, a+(n-1)d, a+d, a+2d, \dots, a+(n-2)d$ there are the $(n-2)$ arithmetic means between a and $a+(n-1)d$

Ex 16 What is AM between 15 and 27?

Including the two given terms 15 and 27 the number of terms will be 3

We are given $a=15$ and $a+2d=27$

$$15+2d=27 \quad d = \frac{12}{2} = 6$$

Second term (AM between 15 and 27) $= a+d=15+6=21$

Ex 17 Insert five AM between $\frac{1}{2}$ and $\frac{1}{2}$

Including the extremes the number of terms will be 7 so that we have to find a series of seven terms in AP of which $\frac{1}{2}$ is the first term and $-\frac{1}{2}$ is the seventh term

$$a = \frac{1}{2} \text{ and } a + 6d = -\frac{1}{2}$$

$$d = -\frac{1}{6}$$

$$\text{2nd term} = a + d = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

$$\text{3rd term} = a + 2d = \frac{1}{2} - \frac{2}{6} = \frac{1}{6}$$

$$\text{4th term} = a + 3d = \frac{1}{2} - \frac{3}{6} = 0$$

$$\text{5th term} = a + 4d = \frac{1}{2} - \frac{4}{6} = -\frac{1}{6}$$

$$\text{6th term} = a + 5d = \frac{1}{2} - \frac{5}{6} = -\frac{1}{3}, \text{ hence}$$

$$\frac{1}{3}, \frac{1}{6}, 0, -\frac{1}{6} \text{ and } -\frac{1}{3} \text{ are the required arithmetic means}$$

Ex 18. If a, b, c are the $p^{\text{th}}, q^{\text{th}}$ and r^{th} terms respectively of an A.P. then show that

$$\frac{a(q-r)}{p} + \frac{b(r-p)}{p} + \frac{c(p-q)}{p} = 0$$

Let A and d be the first term and the common difference then

$$a = A + (p-1)d \quad (1)$$

$$b = A + (q-1)d \quad (2)$$

$$c = A + (r-1)d \quad (3)$$

From (1) and (2) we have

$$(a-b) = (p-q)d$$

Similarly,

$$(b-c) = (q-r)d, \text{ and}$$

$$(c-a) = (r-p)d$$

The given relation therefore becomes

$$\begin{aligned} a(b-c) + b(c-a) + c(a-b) \\ = ab - ac + bc - ab + ac - bc = 0 \text{ Proved} \end{aligned}$$

EXERCISE 3.4

1. Find the last terms and the sum of the following series

(i) $-3, -1, 1$ to 40 terms

(ii) $1, 2, 3$ to n terms

(iii) $x, 4x, 7x$ to n terms

(iv) $2\frac{1}{4}, 5\frac{1}{2}, 8\frac{3}{4}$ to n terms

(v) $\frac{3a}{7}, \frac{9a}{14}, \frac{6a}{7}$ to n terms

2. Find the AM between

(i) $(a-b), (a+b)$ (ii) $\frac{1}{x}, \frac{1}{y}$ (iii) $\frac{a^2+b^2}{a+b}, \frac{2ab}{a+b}$

3. Insert 16 arithmetic means between 19.2 and 5.6

4. Construct an AP of which the sum of two terms is the same as the sum to seven terms and whose common difference is three

5. How many terms must be taken of the series $-5, -1, 3$ to make the sum 897?

6. Three numbers are in the ratio of 1:2:4. If 4 is subtracted from the second, the resulting numbers form an AP. Determine the numbers and their common difference.

7. A firm produced nothing in its fifth year. Its total production during the first four years was 2,400 units. Find

- (i) production of the firm in the first and fourth year
 (ii) yearly increase or decrease in production

8 The sum of three consecutive terms in $A.P.$ is 18 and their product is 192 Find the numbers

9 If a^2, b^2, c^2 are in $A.P.$ then show that

$$\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b} \text{ are also in } A.P.$$

[Hint. $\frac{1}{c+a} - \frac{1}{b+c} = \frac{1}{a+b} - \frac{1}{c+a}$ will lead to $b^2 - a^2 = c^2 - b^2$]

10 The sum of three numbers in $A.P.$ is 21 and sum of their squares is 155 Find the numbers

11 The sum of 10 terms of a series in $A.P.$ is zero and the first term is $\frac{9}{2}$ Find the series

12 The first and last terms of n terms in $A.P.$ are 1 and 9 and the sum of the terms is 45 Find n and the common difference

3.6 Geometric progression Geometric progression (GP) is a set of numbers which are formed by multiplying the preceding term by a constant. This constant is called the *common ratio*. For instance in the series 2, 8, 32, 128, the first term is 2, the second term is $8=2 \times 4$, the third term is $32=8 \times 4$, the fourth term is $128=32 \times 4$, thus preceding term gets multiplied every time by the constant number 4. 4 is common ratio of the GP .

Let us denote first term of a GP by ' a ' and the common ratio by ' r '

$$\text{1st term} = a$$

$$\text{2nd term} = a \times r = ar$$

$$\text{3rd term} = ar \times r = ar^2$$

$$\text{4th term} = ar^2 \times r = ar^3$$

Thus, n th term will be $= ar^{n-1}$, so that if in a GP $a=2$ and $r=3$, fifth term will be $2 \times (3)^{5-1} = 2 \times (3)^4 = 162$

Sum of a number of terms in GP Let S_n denote the sum of first n terms of a GP of which a is first term and r is common ratio

$$S_n = a + ar + ar^2 + \dots + ar^{n-1} \quad \dots (1)$$

$$rS_n = ar + ar^2 + ar^3 + \dots + ar^n \quad \dots (2)$$

[on subtracting (1) from (2)]

$$\text{so that, } rS_n - S_n = ar^n - a$$

$$(r-1)S_n = a(r^n - 1)$$

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad \frac{a(1 - r^n)}{1 - r}$$

Sum of an infinite number of terms of a decreasing *G.P.* having common ratio r less than unity, is given by (Since r^n will tend to zero)

$$\text{Sum to Infinity } S_\infty = \frac{a}{1 - r}$$

Ex 19 Sum the series $\frac{2}{3}, -1, \frac{3}{2}$ to 5 terms

Common ratio = $\left(-1 + \frac{2}{3}\right) = \left(\frac{3}{2} + -1\right) = -\frac{3}{2}$ and $a = \frac{2}{3}$

$$S_5 = \frac{\frac{2}{3} \left[1 - \left(-\frac{3}{2}\right)^5 \right]}{1 - \frac{3}{2}} = \frac{761}{120}$$

Ex 20 Sum the series $1, \frac{1}{x}, \frac{1}{x^2}, \dots$ to n terms

Here $a=1$ and common ratio = $\frac{1}{x}$

$$S_n = \frac{\left[1 - \left(\frac{1}{x}\right)^n \right]}{1 - \frac{1}{x}} \quad \left(\text{when } \frac{1}{x} < 1 \right)$$

$$= \frac{x^n - 1}{x - 1} \quad (x^{1-n})$$

Ex 21 Find three numbers in *G.P.* whose sum is 14 and whose product is 64

Let the three numbers be $\frac{a}{r}, a, ar$

Therefore, $\frac{a}{r} \times a \times ar = 64$, or $a = 4$

The sum of the numbers

$$\frac{4}{r} + 4 + 4r = 14, \text{ or, } 4r^2 - 10r + 4 = 0$$

$$r = 8 \text{ or } 2$$

Required numbers are 2, 4, 8

Ex 22 Find S_∞ of the *G.P.* $\frac{1}{3}, \frac{1}{3^2}, \frac{1}{3^3}, \dots$

Here $a = \frac{1}{3}$ and $r = \frac{1}{3}$ so that sum to infinity

$$S_{\infty} = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{2}$$

Geometrical Mean When three quantities are in *GP*, the middle one is called the Geometrical Mean (*GM*) between the other two

For instance in three terms of a *GP* 2, 8, 32, 8 is *GM* between 2 and 32

Similarly there can be any number of geometrical means between the two quantities of a *GP*

Ex. 23 Find *GM* between these two terms $\frac{1}{\sqrt{2}}$ and $\frac{8}{\sqrt{2}}$

Including the *GM* between these two terms, we shall have three terms a, ar, ar^2

$$a = \frac{1}{\sqrt{2}} \text{ and } ar^2 = \frac{8}{\sqrt{2}}$$

$$\frac{ar^2}{a} = \frac{\frac{8}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}, \quad r^2 = 8, \quad r = \sqrt{8}$$

$$\text{GM (or 2nd term)} = ar = \frac{1}{\sqrt{2}} \times \sqrt{8} = \pm 2$$

i.e., *GM* between $\frac{1}{\sqrt{2}}$ and $\frac{8}{\sqrt{2}}$ can be +2 or -2

Ex. 24. Insert 5 *GMs* between $\frac{32}{9}$ and $\frac{81}{2}$

Including 5 *GMs*, between two terms we shall have 7 terms

$$a = \frac{32}{9} \text{ and } ar^6 = \frac{81}{2}$$

$$\frac{ar^6}{a} = \frac{\frac{81}{2}}{\frac{32}{9}}, \text{ that is } r = \frac{3}{2}$$

In this procedure	2nd term (first <i>GM</i>)	$= ar = \frac{32}{9} \times \frac{3}{2}$
	3rd term (2nd <i>GM</i>)	$= ar^2 = \frac{32}{9} \times \frac{9}{4}$
	4th term (3rd <i>GM</i>)	$= ar^3 = \frac{32}{9} \times \frac{27}{8}$

in this way we can find all the required G M

Applications

(i) *Population Growth* Under the assumption that population of a particular area increases or decreases by same percentage we can use G P for the purpose of interpolation or extrapolation

Suppose population of a country in the first year = a and it increases by $r\%$ each year

The rise in population during first year = ar

Total population at the end of the first year = $a + ar = a(1+r)$

Similarly total population at the end of second year

$$= a(1+r) + ar(1+r) = a(1+r)^2,$$

and total population at the end of n^{th} year = $a(1+r)^n$ [r is % terms]

Ex 25 If population of a country rises by 2.1% and it was 36 crores in 1950, what was the population in 1967 ?

In year 1950 is assumed to be first year, then 1967 will be the 17th year

$a=36$ crores $r=2.1\%$ and $n=17$

$$\begin{aligned} \text{Population at the end of 1967} &= 36 \left(1 + \frac{2.1}{100}\right)^{17} = 36(1.021)^{17} \\ &= 51.40 \text{ crores} \end{aligned}$$

(We make use of log tables in such problems)

(ii) *Investment Multiplier* Keynesian investment multiplier can also be explained with the help of G P

Suppose in a certain period investment is increased by Rs ΔI (Δ - is read as delta)

Assuming that the propensity to consume is c , the total rise in the income in the first year = ΔI Total income in the second year will be $c \times \Delta I$ In third year $c(c \times \Delta I)$ will become income. Thus we can write

$$\begin{aligned} \text{Total change in income} &= \Delta Y = \Delta I + \Delta I \times c + \Delta I \times c^2 \quad \text{to } \infty \\ &= \Delta I [1 + c + c^2 + c^3 \quad \text{to } \infty] \\ &= \Delta I \left[\frac{1}{1-c} \right] \end{aligned}$$

$\left[\frac{1}{1-c} \right]$ is called investment multiplier c is marginal propensity to consume which is always less than one

Thus we find that change in GNP is $\left(\frac{1}{1-c} \right)$ times the change in investment

Ex 26 By how much the income will rise if the initial change in the investment is Rs 100 crores (Assuming that $MPC=0.9$)

$$\Delta Y = \Delta I \left(\frac{1}{1-c} \right) = \frac{100}{1-0.9} = (1000 \text{ crores})$$

In other words income will rise by Rs. 1000 crores if the change in investment is made by Rs. 100 crores.

(iii) *Compound Interest.* A sum of Rs. 100 is invested and accumulated at compound rate of 4% a year.

If interest is added yearly then the amount at the end of the year will be Rs. $(100+4)=100(1+0.04)=104$. Next year interest at the rate of 4% will be paid on Rs. 104.

$$\text{Amount at the end of second year} = 104 + 104 \times \frac{4}{100}$$

$$= 104(1+0.04) = 100(1+0.04)(1+0.04) = 100(1+0.04)^2$$

$$\text{Similarly amount at the end of third year} = 100(1+0.04)^3$$

Generalising the problem the total amount at the end of t^{th} year with rate of interest $r\%$ will be $= 100(1+r)^t$

If P_t denotes the total amount and P_0 the principal then

$$P_t = P_0(1+r)^t \text{ when interest is added yearly}$$

$$P_t = P_0 \left(1 + \frac{r}{2} \right)^{2t} \text{ when interest is added half yearly}$$

$$\text{In general terms } P_t = P_0 \left(1 + \frac{r}{n} \right)^{nt} \text{ when interest is added } n \text{ times in a year}$$

(iv) *Present or Discounted Value.* From the above formula we know that if P_0 is invested at $r\%$ per year P_0 will amount to $P_t = P_0(1+r)^t$ after t years.

In other words we may say that if P_t amount at the end of the period t and rate of interest is known we can find the value of P_0 in the following way

$$\text{Since, } P_t = P_0(1+r)^t$$

$$P_0 = \frac{P_t}{(1+r)^t}$$

P_0 is nothing but the present or discounted value of P_t .

EXERCISE 3.5

1. Consider the GP $7, \sqrt{42}, 6, \dots$, find

(i) 9th term of the series

(ii) Which term will be $\frac{36}{7}$, and

(iii) S_4 and S_∞

2. Insert three GM between $\frac{9}{4}$ and $\frac{4}{9}$

3. A GP has a common ratio $= \frac{1}{2}$. If $S_3 = 10$, find the sixth term of

the series and also S_6

4 The product of three numbers in GP is 27 and sum of the first and the last is $\frac{51}{4}$ find the numbers

5 Assuming that MPC is 0.95 find by how much the income will rise if Government makes a net investment of Rs. 125 crores

6 If A and G denote the arithmetic and geometric means between a and b prove that $A > G$

7 Find three numbers in GP whose sum is 19 and product is 216

8 Find three consecutive numbers in GP whose sum is 26 and product 216

9 Show that the p th, q th and r th terms of a GP are also three consecutive terms of a GP if p, q, r are in AP

10 The population of India in 1971 was 54 crores. Predict the population in 1981 assuming that it increases at 2 per cent per year

11 Sum to n terms

$$(i) 4+44+444+\dots$$

$$(ii) 3+33+333+\dots$$

$$(iii) 2+22+222+\dots$$

12 If a, b, c be in AP and x, y, z in GP prove that $x^{b-c} y^{c-d} z^{a-b} = 1$

13 If the sum of n terms of a series in GP be S their product P and the sum of reciprocals R show that

$$P^2 = \left(\frac{S}{R}\right)^n$$

14 If $y = 1 + x + x^2 + \dots + \infty$ then show that $x = 1 - \frac{1}{y}$

15 If $x = 1 + a + a^2 + \dots + \infty$ and $y = 1 + b + b^2 + \dots + \infty$ then

prove that $1 + ab + a^2b^2 + \dots + \infty = \frac{xy}{x+y-1}$

3.7 Harmonic Progression.

A sequence reciprocal of whose terms form an AP is called harmonic progression

For instance $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$ is an HP because $3, 5, 7, \dots$ is an AP but

$2, 4, 6$ is not an HP since $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$ is not an AP

Ex. 27 Find the tenth term of the HP $\frac{1}{5}, \frac{1}{5}, \frac{1}{7}, \dots$

The corresponding AP will be 3, 5, 7, ..

$$10^{\text{th}} \text{ term of AP will be } = a + 9d = 21$$

$$10^{\text{th}} \text{ term of HP will be } = \frac{1}{21}$$

Ex. 28. Third and seventh terms of HP are $\frac{1}{12}$ and $\frac{1}{32}$ respectively Find 15^{th} terms

Suppose the first term of AP is a and its common ratio is d

$$3^{\text{rd}} \text{ term} = a + 2d = 12$$

$$7^{\text{th}} \text{ term} = a + 6d = 32$$

$$d = 5 \text{ and } a = 2.$$

$$15^{\text{th}} \text{ term of AP} = a + 14d = 72$$

$$15^{\text{th}} \text{ term of HP will be } \frac{1}{72}, \text{ hence the HP will be}$$

$$\frac{1}{2}, \frac{1}{7}, \frac{1}{12},$$

Harmonic Mean : If H denotes the harmonic mean between two quantities a and b , then a, H, b will be in HP and $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ will be in AP.

Ex. 29. Insert seven harmonic means between $\frac{1}{2}$ and 2.

Let $H_1, H_2, \dots, H_7$, denote the seven HM_s , then

$$2, H_1, H_2, \dots, H_7, \frac{1}{2} \text{ will form an AP.}$$

Total number of terms = 9

$$\text{Ninth term} = 2 = \frac{1}{2} + 8d,$$

$$d = \frac{3}{16} \text{ and } a = \frac{1}{2}.$$

$$\text{Second term} = a + d = \frac{1}{2} + \frac{3}{16}$$

$$\text{Third term} = a + 2d = \frac{1}{2} + \frac{6}{16}$$

$$\text{Fourth term} = a + 3d = \frac{1}{2} + \frac{9}{16}$$

... ..

$$\text{Eighth term} = a + 7d = \frac{1}{2} + \frac{21}{16}.$$

Therefore, A.P. is $= \frac{1}{2}, \frac{11}{16}, \frac{14}{16}, \frac{17}{16}, \dots$

Hence, the H.P. is $2, \frac{16}{11}, \frac{16}{14}, \frac{16}{17}, \frac{16}{20}, \frac{16}{23} \dots$

The seven H.M.s, therefore, are

$$\frac{16}{11}, \frac{16}{14}, \frac{16}{17}, \frac{16}{20}, \frac{16}{23}, \frac{16}{26} \text{ and } \frac{16}{29}.$$

Ex. 30. If A, G, H be the arithmetic, geometric and harmonic means between a and b , express them in terms of a and b and prove that

(i) $AH = G^2$ and (ii) $A > G > H$

We know that,

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab} \quad \text{and} \quad H = \frac{2ab}{a+b}.$$

$$AH = 2 + \frac{b}{2} \times \frac{2ab}{a+b} = ab = G^2;$$

$$\begin{aligned} \text{Since,} \quad A - G &= \frac{a+b}{2} - \sqrt{ab} = \frac{a+b-2\sqrt{ab}}{2} \\ &= \frac{(\sqrt{a}-\sqrt{b})^2}{2} \quad \text{is positive number,} \end{aligned}$$

therefore $A - G > 0$ which proves that
 $A > G$

Now, squaring both sides.

$$A^2 > G^2$$

$$A^2 > AH \quad (\because G^2 = AH)$$

$$\therefore A > H$$

Once again, $G^2 = AH$, $\therefore \frac{G}{A} = \frac{H}{G}$

Hence A, G, H are in G.P.

Again, $A > G$, $\frac{G}{A}$ will be < 1

$$\frac{H}{G} \text{ will also be } < 1 \left(\frac{G}{A} = \frac{H}{G} \right)$$

$$\therefore G > H$$

which proves that $A > G > H$

EXERCISE 3 6

1. Find the fifth and seventh terms of the following series

(i) $\frac{1}{3}, \frac{1}{5}, \frac{1}{7},$

(ii) $\frac{1}{3}, \frac{2}{7}, \frac{1}{4},$

(iii) $2, \frac{5}{2}, \frac{10}{3},$

2. Find the H M between
- $\sqrt{3}$
- and
- $\sqrt{2}$

3. Insert four H M 's between
- $\frac{2}{3}$
- and
- $\frac{2}{13}$

4. If a is A.M between b and c , b is G M between a and c , then prove that c is H M between a and b 5. Find the third term of (i) the A P, (ii) the G P and (iii) the H P of which a and b are the first two terms6. If the p th term of an H P is q and q th term is P , show that r th term is $\frac{Pq}{r}$ 7. If a, b, c are in H P, prove that $a(b+c), b(c+a), c(a+b)$ are in A P8. If a, b, c are in A P as well as in H P, show that they are also in G P9. If $a^p = b^q = c^r$ and a, b, c are in G P prove that p, q, r shall be in H P

3 8 Binomial Expansion

We know that

$$(x \pm a)^2 = x^2 \pm 2ax + a^2$$

$$(x \pm a)^3 = x^3 \pm 3x^2a + 3xa^2 \pm a^3, \text{ and}$$

$$(x \pm a)^4 = x^4 \pm 4x^3a + 6x^2a^2 \pm 4xa^3 + a^4$$

We also know the *combination* formula

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Replacing the coefficients of the above three expansions by *combination formula*

$$(x \pm a)^2 = x^2 \pm {}^2C_1ax + a^2 \quad ({}^2C_1 = 2)$$

$$(x \pm a)^3 = x^3 \pm {}^3C_1x^2a + {}^3C_2xa^2 \pm a^3 \quad ({}^3C_1 = {}^3C_2 = 3)$$

$$(x \pm a)^4 = x^4 \pm {}^4C_1x^3a + {}^4C_2x^2a^2 \pm {}^4C_3xa^3 + a^4 \quad ({}^4C_1 = {}^4C_3 = 4)$$

Thus we can now write the expansion of the term having n index

$$(x \pm a)^n = x^n \pm {}^nC_1x^{n-1}a + {}^nC_2x^{n-2}a^2 \pm {}^nC_3x^{n-3}a^3 \quad (1)$$

Definition. Any expression containing two terms is called a *binomial*. The general form of a binomial is $(x + a)$ where x is the first term and a is the second. The expansion given in (1) is called *Binomial expansion*.

Ex.31 Expand $(x + 3)^7$

Here $n = 7$ and $a = 3$

There will be $(7 + 1) = 8$ terms in the expansion and all the terms will be positive

$$(x + 3)^7 = x^7 + {}^7C_1 x^6 3 + {}^7C_2 x^5 (3)^2 + {}^7C_3 x^4 (3)^3 + {}^7C_4 x^3 (3)^4 \\ + {}^7C_5 x^2 (3)^5 + {}^7C_6 x (3)^6 + {}^7C_7 x^0 (3)^7$$

We can now solve for all combination terms to get the required expansion of $(x + 3)^7$

Ex 32 Expand $(2x + 3)^5$

Here $n = 5$ $a = 3$ and $x = 2x$

There will be six terms in the expansion and all will be positive

$$(2x + 3)^5 = (2x)^5 + {}^5C_1 (2x)^4 (3)^1 + {}^5C_2 (2x)^3 (3)^2 + {}^5C_3 (2x)^2 (3)^3 \\ + {}^5C_4 (2x) (3)^4 + {}^5C_5 (3)^5 \\ = 32x^5 + 240x^4 + 720x^3 + 1080x^2 + 810x + 243$$

Ex33 Expand $\left(\frac{x}{2} - 2y\right)^6$

Here $n = 6$ $a = 2y$ and $x = \frac{x}{2}$

There will be seven terms in the expansion with *minus and plus signs* occurring alternately

$$\left(\frac{x}{2} - 2y\right)^6 = \left(\frac{x}{2}\right)^6 - {}^6C_1 \left(\frac{x}{2}\right)^5 (2y) + {}^6C_2 \left(\frac{x}{2}\right)^4 (2y)^2 - {}^6C_3 \left(\frac{x}{2}\right)^3 (2y)^3 \\ + {}^6C_4 \left(\frac{x}{2}\right)^2 (2y)^4 - {}^6C_5 \left(\frac{x}{2}\right) (2y)^5 + {}^6C_6 (2y)^6$$

Simplify and get the expansion

General term How to find the general term in the expansion of

$$(x + a)^n ?$$

We have observed in the expansion of $(x + a)^n$ that the suffix of combination C in any term is one less than the number of the terms. The power of x is n minus the suffix of C while the power of a is the same as the suffix of C .

If we need r th term in the expansion

it will be ${}^nC_{r-1} x^{n-(r-1)} a^{r-1}$ while the $(r+1)^{th}$ term will be

$$= {}^nC_r x^{n-r} a^r$$

Denoting $(r+1)^{th}$ term of the expansion by T_{r+1} we have

$$T_{r+1} = {}^nC_r x^{n-r} a^r$$

T_{r+1} is called the general term because by giving suitable values to r , we can find any term of the expansion

Ex.34 Find the general term in the expansion of $\left(\frac{x}{2} - 2y\right)^6$

$$\text{General Term} = T_{r+1} = {}^nC_r x^{n-r} a^r$$

In our example $x = \frac{x}{2}$ $a = -2y$ and $n = 6$

$$\begin{aligned} T_{r+1} &= {}^6C_r \left(\frac{x}{2}\right)^{6-r} (-2y)^r \\ &= {}^6C_r (x)^{6-r} \left(\frac{1}{2}\right)^{6-r} (-1)^r (2)^r (y)^r \\ &= (-1)^r {}^6C_r (x)^{6-r} (y)^r \frac{2^r}{2^{6-r}} \\ &= (-1)^r {}^6C_r (x)^{6-r} (y)^r (2)^{2r-6} \end{aligned}$$

We can now find any term from the above expression

$$\begin{aligned} \text{Third term} = T_{2+1} &= (-1)^2 {}^6C_2 x^{6-2} (y)^2 (2)^{4-6} \\ &= \frac{15}{4} x^4 y^2 \end{aligned}$$

Thus we get the the third term $= \frac{15}{4} x^4 y^2$ directly from the general term without expanding the binomial

Middle term How to find the middle term in the expansion of $(x+a)^n$

We know that the total number of terms in the expansion exceed the index or power by unity Hence there can be two cases

(i) If index n is even, the total number of terms in the expansion will be odd and there will be one middle term which will be

$$= \frac{n+1+1}{2} = \left(\frac{n+2}{2}\right)^{th} \text{ term}$$

(ii) If index n is *odd* the total number of terms in the expansion will be even and there will be two middle terms

$$\left(\frac{n+1}{2}\right)^{\text{th}} \text{ and } \left(\frac{n+3}{2}\right)^{\text{th}} \text{ term}$$

Ex 35 Find the middle term of $(3x-2)^{10}$

Total terms in the expansion of $(3x-2)^{10}$ will be = 11

Hence there will be one middle term = $\left(\frac{n+2}{2}\right) = \left(\frac{10+2}{2}\right) = 6^{\text{th}}$ term

To get the sixth term of the expansion we make use of the general term

$$T_{r+1} = {}^nC_r x^r a^r$$

We want sixth term which will be = T_{5+1}

Thus $n = 10$ $x = 3x$ $a = -2$ and $r = 5$

$$\begin{aligned} T_{5+1} &= {}^{10}C_5 (3x)^5 (-2)^5 \\ &= (-1)^5 {}^{10}C_5 (3x)^5 (2)^5 \\ &= -1055x^5 \end{aligned}$$

Binomial Expansion for Any Index

So far the index in the Binomial was assumed to be a positive integer and hence it was possible to get the value of nC_r for different values of n and r . It will not be possible to expand nC_r if n is a fraction or a negative quantity. For example we cannot find the value of ${}^{1/5}C_r$ or ${}^{-5}C_r$. Hence it is difficult to write this expansion of $(x+a)^{-3}$.

We make use of expansion $(1+x)^n$ in such cases

$$(1+x) = 1 + nx + \frac{n(n-1)}{1 \cdot 2} x^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} x^3 + \dots$$

[Note that we have just replaced a by unity in the expansion of $(a+x)^n$]

We illustrate the use of this general expansion by the following examples

Ex.36. Expand $(4+3y)^{-4}$

We write the given Binomial in the form of $(1+x)^n$

$$(4+3y)^{-4} = \left\{ 4 \left(1 + \frac{3}{4} y \right) \right\}^{-4} = (4)^{-4} \left(1 + \frac{3}{4} y \right)^{-4}$$

We now expand $\left(1 + \frac{3}{4}y\right)^{-4}$ with the help of the expansion of $(1+x)^n$

In this case $x = \frac{3}{4}y$ and $n = -4$

$$\begin{aligned}\left(1 + \frac{3}{4}y\right)^{-4} &= 1 + (-4)\left(\frac{3}{4}y\right) + \left(\frac{-4 \times -5}{1 \times 2}\right)\left(\frac{3}{4}y\right)^2 \\ &\quad + \left(\frac{-4 \times -5 \times -6}{1 \times 2 \times 3}\right)\left(\frac{3}{4}y\right)^3 + \dots \\ &= 1 - 3y + \frac{45}{8}y^2 - \frac{135}{16}y^3 + \dots\end{aligned}$$

But

$$\begin{aligned}(4+3y)^{-4} &= (4)^{-4}\left(1 + \frac{3}{4}y\right)^{-4} \\ &= (4)^{-4}\left\{1 - 3y + \frac{45}{8}y^2 - \frac{135}{16}y^3 + \dots\right\} \\ &= 4^{-4} - (4)^{-4}(3y) + 4^{-4}\left(\frac{45}{8}y^2\right) - 4^{-4}\left(\frac{135}{16}y^3\right) + \dots \\ &= \frac{1}{256} - \frac{3y}{256} + \frac{45}{2048}y^2 - \frac{135}{4096}y^3 + \dots\end{aligned}$$

Ex. 37 Expand $(5 + 2x)^2$

$$\begin{aligned}(5 + 2x)^2 &= \left\{5\left(1 + \frac{2x}{5}\right)\right\}^2 \\ &= (5)^2\left(1 + \frac{2x}{5}\right)^2 \\ &= \frac{1}{25}\left\{1 - 2\left(\frac{2x}{5}\right) + \frac{(-2)(-3)}{1 \times 2}\left(\frac{2x}{5}\right)^2\right. \\ &\quad \left.+ \frac{(-2)(-3)(-4)}{1 \times 2 \times 3}\left(\frac{2x}{5}\right)^3 + \dots\right\} \\ &= \frac{1}{25}\left[1 - \frac{4x}{5} + \frac{12x^2}{25} - \frac{32}{625}x^3 + \dots\right]\end{aligned}$$

$$= \frac{1}{25} - \frac{4x}{125} + \frac{12x^2}{625} - \frac{32}{3125} x^3 +$$

3.9 A Few Important Expansions

Let us first find the expansion of $(1+x)^r$ and $(1-x)^r$

$$(1+x)^{-n} = 1 - nx + \frac{(-n)(-n-1)}{1 \times 2} (x)^2 + \frac{(-n)(-n-1)(-n-2)}{1 \times 2 \times 3} (x)^3$$

$$(1-x)^{-n} = 1 + nx + \frac{-n(-n-1)}{1 \times 2} x^2 + \frac{-n(-n-1)(-n-2)}{1 \times 2 \times 3} (x)^3$$

$$\text{i.e., } (1-x)^{-n} = 1 - nx + \frac{n(n+1)}{1 \times 2} x^2 - \frac{n(n+1)(n+2)}{1 \times 2 \times 3} (x)^3 +$$

$$\text{and } (1-x)^{-n} = 1 + nx + \frac{n(n+1)}{1 \times 2} x^2 + \frac{n(n+1)(n+2)}{1 \times 2 \times 3} (x)^3 +$$

(Note that the only difference between the two expansions is that of signs)

On the basis of above expansions we can have following important expansions

$$(i) \quad \frac{1}{1+x} = (1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^r x^r +$$

$$(ii) \quad \frac{1}{(1+x)^2} = (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots + (-1)^r (r+1)x^r +$$

$$(iii) \quad \frac{1}{1-x} = (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + x^r +$$

$$(iv) \quad \frac{1}{(1-x)^2} = (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots + (r+1)(x^r) +$$

$$(v) \quad \frac{1}{(1+x)^3} = (1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots + (-1)^r \frac{(r+1)(r+2)}{1 \times 2} x^r +$$

$$(vi) \quad \frac{1}{(1-x)^3} = (1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots + \frac{(r+1)(r+2)}{1 \times 2} x^r +$$

Exponential Function and Expansions

The sum of the series : $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ for real values of x is defined as e^x (read as exponential x) This is a convergent series

By substituting $x = 1$, we obtain the value of $e = 2.71828\dots$,

$$e^1 = e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots = 2.71828 \dots$$

and thus e is an irrational number.

Again, if we substitute $x = -1$, then ,

$$e^{-1} = \frac{1}{e} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \dots$$

Logarithmic Function and Expansion

Since $\log x$ is not defined at $x = 0$ (i.e., $\log 0$ does not exist), we consider expansion of $\log(1+x)$

We know that

$$\frac{1}{1+x} = 1 + x + x^2 - x^3 + \dots \quad (|x| < 1)$$

Hence,

$$\int_0^x \frac{1}{1+x} dx = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad \left[\begin{array}{l} -1 < x < 1 \\ \text{or } |x| < 1 \end{array} \right]$$

(See Chapter on integration for the evaluation of $\int_0^x \frac{1}{1+x} dx$)

$$\text{But, } \int_0^x \frac{1}{1+x} dx = \log(1+x)$$

$$\text{Therefore, } \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\text{Similarly, } \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots \text{ for all } |x| < 1.$$

Thus $(1 \pm x)$ should always be greater than zero.

EXERCISE 37

1 Find the expansion of the following binomials

$$(i) (x+y)^5 \quad (ii) (2x+3)^5 \quad (iii) \left(\frac{x}{3} - \frac{1}{2}\right)^{10}$$

2 Find the middle term of

$$(i) \left(\frac{x}{3} - 1\right)^{10} \quad (ii) \left(3 - \frac{x}{2}\right)^{11} \quad (iii) \left(x - \frac{1}{2}\right)^{13}$$

3 Find the 5th and 8th term of $\left(2x + \frac{1}{2}y\right)^{12}$ 4 Expand to 4 terms only $(1-2x)^5$, $(1-xy)^{1/2}$, $(2x-x)^{1/2}$, $(a^2ax)^{1/2}$ 5 Find the approximate value of $(412)^{1/3}$ and $\sqrt[3]{128}$ (Using Binomial Expansion)

6 Show that

$$(i) 1 + \frac{3}{4} + \frac{3}{4} \cdot \frac{5}{8} + \frac{3}{4} \cdot \frac{5}{8} \cdot \frac{7}{12} + \dots = 2\sqrt{2}$$

$$(ii) 1 - \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} - \frac{1}{23} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{23} + \dots = \sqrt{\frac{2}{3}}$$

$$(iii) 1 + \frac{x}{3} + \frac{1 \cdot 4}{3 \cdot 6} x^2 + \frac{1 \cdot 4 \cdot 7}{3 \cdot 6 \cdot 9} x^3 + \dots = \left(\frac{1}{1-x}\right)^{1/2}$$

Chapter 4

Elements of Trigonometry and Analytical Geometry

The word 'Trigonometry' comes from two Greek words which mean *triangle measurement*

Trigonometry is also defined as the science of indirect measurement. With the help of trigonometry we are able to compute the distance which could not be measured directly. This indirect computation of distances is made by means of six ratios called *Trigonometric Functions*.

4.1 Trigonometric Functions

Suppose two right angled triangles ACB and PRQ have been so constructed that $\angle A = \angle P$ (Fig. 1)

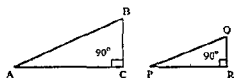


Fig. 1

In these Δ s, we have

$$\angle A = \angle P$$

$$\angle C = \angle R = 90^\circ$$

$$\angle B = \angle Q$$

Since the two Δ s are similar, their corresponding sides are in proportion

$$(i) \quad \frac{BC}{AB} = \frac{QR}{PQ}$$

$$(ii) \quad \frac{AC}{AB} = \frac{PR}{PQ}$$

$$(iii) \quad \frac{BC}{AC} = \frac{QR}{PR}$$

The ratios which occur in each of these proportions are called *Trigonometric Functions*.

Sine A. The ratio of proportion (i) is formed by dividing the side opposite $\angle A$ (i.e., perpendicular) by the hypotenuse of each triangle. This ratio is called the *sine of $\angle A$* (or $\angle P$) and abbreviated as *sin A*.

Cosine A. The ratio of proportion (ii) is formed by dividing the base of the triangle by the hypotenuse of each triangle. This ratio is called the *cosine of $\angle A$* (or $\angle P$) and abbreviated as *cos A*.

Tangent A. The ratio of proportion (iii) is formed by dividing the perpendicular by the base of the triangles this ratio is called the *tangent* of $\angle A$ (or $\angle P$) and abbreviated as $\tan A$

Consecant A, Secant A and Contangent A. If the ratios in all the above proportions are inverted, the resulting ratios are called cosecant A, secant A and contangent A respectively. These are abbreviated as *cosec A*, *sec A* and *cot A*

We summarise these six trigonometric functions (in terms of sides of Δs) in the following tabular form

$\sin A = \frac{BC}{AB} = \frac{\text{Perp}}{\text{Hyp}}$	$\text{cosec } A = \frac{AB}{BC} = \frac{\text{Hyp}}{\text{Perp}}$
$\cos A = \frac{AC}{AB} = \frac{\text{Base}}{\text{Hyp}}$	$\sec A = \frac{AB}{AC} = \frac{\text{Hyp}}{\text{Base}}$
$\tan A = \frac{BC}{AC} = \frac{\text{Perp}}{\text{Base}}$	$\cot A = \frac{AC}{BC} = \frac{\text{Base}}{\text{Perp}}$

We can observe from the above table that the value of a trigonometric function of an angle depends solely on the size of the angle and not on the size of the right-angled triangle ACB or PRQ

In Fig 1, $\angle A = \angle P$, the Δs ABC and PQR are similar and the ratio represented by $\sin A \left(= \frac{BC}{AC} \right)$ is equal to the ratio represented by $\sin P \left(= \frac{QR}{PQ} \right)$, whatsoever may be the size of Δs . The same is true of all the remaining five functions

Since the value of trigonometric functions depends on the size of the angle, these functions must always be mentioned along with the angle. These functions are meaningless without a symbol for the angle

4.2. Reciprocal and Complementary Functions

We observe in the above table that in case of $\sin A$ and $\text{cosec } A$, perpendicular (Perp.) and hypotenuse (Hyp) are used in both but their positions are reversed. We, therefore, say that the ratio $\frac{\text{Perp}}{\text{Hyp}} (= \sin A)$ is the *reciprocal* of the ratio $\frac{\text{Hyp}}{\text{Perp}} (= \text{cosec } A)$

(Two numbers are said to be reciprocal of each other if their product is equal to unity)

Thus following pairs of functions are reciprocal functions

$\sin A$ and $\text{cosec } A$,	$\cos A$ and $\sec A$	$\tan A$ and $\cot A$
$\therefore \frac{1}{\sin A} = \text{cosec } A$	$\therefore \frac{1}{\cos A} = \sec A$	$\therefore \frac{1}{\tan A} = \cot A$
and, $\frac{1}{\text{cosec } A} = \sin A$	and, $\frac{1}{\sec A} = \cos A$	and, $\frac{1}{\cot A} = \tan A$

hence $(\sin A \times \operatorname{cosec} A = 1)$, $(\cos A \times \sec A = 1)$, $(\tan A \times \cot A = 1)$

Complementary Functions The prefix 'Co-', which appears in terms *Cosine A*, *Cosecant A* and *Cotangent A* comes from the first two letters of the word 'Complementary'

In the right angled ΔPRQ (Fig. 2)

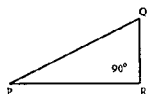


Fig. 2

and

$$\sin P = \frac{QR}{PQ}$$

$$\cos Q = \frac{QR}{PQ}$$

$$\sin P = \cos Q$$

$\angle P$ and $\angle Q$ are complementary angles
Similarly we can show that

$$\cos P = \sin Q$$

$$\tan P = \cot Q$$

$$\cot P = \tan Q$$

$$\sec P = \operatorname{cosec} Q,$$

$$\operatorname{cosec} P = \sec Q$$

4.3. Numerical Values of the Trigonometric Functions of 30° , 45° and 60°

We shall now learn the computation of numerical values of trigonometric functions of 30° , 45° , 60° and of integral multiples of these angles

Values of Trigonometric functions of 45° Construct $\angle P = 45^\circ$ in right-angled ΔPRQ with base $PR = 1$ unit (Fig. 3)

$$\angle R = 90^\circ, \quad \angle Q = 45^\circ$$

Hence, PQR is an isosceles rt. $\angle$ ed Δ

Hence $PR = QR = 1$ unit

From the Pythagorean theorem, we know that

$$PQ^2 = PR^2 + QR^2 = 2 \text{ units,}$$

which gives $PQ = \sqrt{2}$

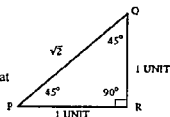


Fig. 3

It is now possible to determine the values of all the trigonometric functions of $\angle P = 45^\circ$

$$\sin P = \sin 45^\circ = \frac{1}{\sqrt{2}}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}}, \quad \tan 45^\circ = 1$$

$$\operatorname{cosec} 45^\circ = \sqrt{2}, \quad \sec 45^\circ = \sqrt{2}, \quad \cot 45^\circ = 1$$

Values of trigonometric functions of 30° and 60° In plane geometry we learn that in a rt $\angle$ ed triangle with two remaining angles being equal to 30° and 60° the side opposite the 30° angle is equal to one half of the hypotenuse

In rt $\angle$ ed $\triangle PRQ$ (Fig 4) $QR = 1$ unit, the hypotenuse PQ must be equal to 2 units. By applying hythagorean rule, we have

$$PQ^2 = PR^2 + QR^2$$

$$PR^2 = 4 - 1 = 3$$

$$PR = \sqrt{3}$$

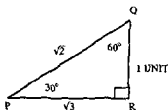


Fig 4

The values of the Trigonometric functions of 30° and 60° can now be obtained

$$\text{Functions of } 30^\circ \quad \sin P = \sin 30^\circ = \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\operatorname{cosec} 30^\circ = 2, \sec 30^\circ = \frac{2}{\sqrt{3}}, \cot 30^\circ = \sqrt{3}$$

$$\text{Functions of } 60^\circ \quad \sin Q = \sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}, \tan 60^\circ = \sqrt{3}$$

$$\operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}}, \sec 60^\circ = 2, \cot 60^\circ = \frac{1}{\sqrt{3}}$$

The values of the functions for some important angles are tabulated as under

Angle- θ	0°	30°	45°	60°	90°	180°	270°	360°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	0	∞	0
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	$-\infty$	0	$-\infty$
$\operatorname{cosec} \theta$	2	$\sqrt{2}$	2	$\frac{2}{\sqrt{3}}$	1	∞	-1	$-\infty$
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞	-1	1	1

Note All figures of the table can be derived from the figures of the box within the table

Values of other trigonometric functions when any one is given If we know the values of one of the trigonometric functions, we can, by means of definitions of these functions, calculate the value of each of the remaining functions.

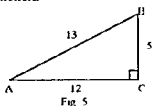


Fig 5

For example, suppose $\tan A$ is given to be $\frac{5}{12}$ and we are required to find the numerical values of remaining functions of $\angle A$.

We construct a right-angled $\triangle BCA$ in which $BC=5$ units and $AC=12$ units, so that $\tan A = \frac{5}{12}$ (Fig 5).

By Pythagorean rule we have,

$$\begin{aligned} AB^2 &= AC^2 + BC^2 \\ AB^2 &= 25 + 144 = 169 \\ AB &= 13 \end{aligned}$$

Now we are able to compute the values of all the functions from the given three sides of $\triangle BCA$

$$\begin{aligned} \sin A &= \frac{5}{13}, \quad \cos A = \frac{12}{13}, \quad \tan A = \frac{5}{12} \\ \operatorname{cosec} A &= \frac{13}{5}, \quad \sec A = \frac{13}{12} \quad \text{and} \quad \cot A = \frac{12}{5} \end{aligned}$$

4.4 Algebraic Signs of the Trigonometric Functions

A trigonometric function of an angle may be a negative number. We now examine the sign of functions by dividing a plane into four parts called quadrants by drawing a horizontal axis, XX' and vertical axis, YY' . The four quadrants thus formed are numbered in a counter-clockwise (anti-clockwise) direction as shown in Fig. 6.

If a line is rotated in a counter-clockwise direction from its original position OM along the X -axis to a new position OP , $\angle MOP$ is considered by definition to be a positive angle. Angles formed by rotation in a clockwise direction are said to be negative angles. Line OM is called the initial side of $\angle MOP$ and OP is the terminal side. Naturally, therefore, if an angle is acute its terminal side lies in quadrant I. As the angle increases from 90° to 180° , the terminal side lies in quadrant II; from 180° to 270° in quadrant III and 270°

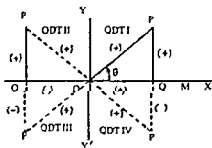


Fig 6

to 360° in quadrant IV. To examine the sign of functions of an angle in each of these quadrants, we drop a perpendicular from P to the X -axis forming thereby a rt. $\triangle$ ΔPOQ . Considering the signs of x and y co-ordinates in the usual way (i.e., x is -ive, y is +ive in quadrant II, x and y both -ive in quadrant III and x is +ive but y is -ive in quadrant IV) we tabulate the sign of $\angle \theta$ for three trigonometric functions in the four quadrants

Trigonometric functions	Sign in quadrant I	Sign in quadrant II	Sign in quadrant III	Sign in quadrant IV
$\sin \theta$	$\frac{(+PQ)}{(+OP)} = +ive$	$\frac{(+PQ)}{(+OP)} = +ive$	$\frac{(-PQ)}{(+OP)} = -ive$	$\frac{(-PQ)}{(+OP)} = -ive$
$\cos \theta$	$\frac{(+OQ)}{(+OP)} = +ive$	$\frac{(-OQ)}{(+OP)} = -ive$	$\frac{(-OQ)}{(+OP)} = -ive$	$\frac{(+OQ)}{(+OP)} = +ive$
$\tan \theta$	$\frac{(+PQ)}{(+OQ)} = +ive$	$\frac{(+PQ)}{(-OQ)} = -ive$	$\frac{(-PQ)}{(-OQ)} = +ive$	$\frac{(-PQ)}{(+OQ)} = -ive$

Note. Signs of the other three functions can be obtained with the help of these functions.

4.5. Numerical Values of Trigonometric Functions Greater than 90°

Thus far, we have considered the trigonometric functions of angles which are less than 90° and hence all such angles were in the first quadrant. Let us now consider the numerical values of functions of an angle greater than 90° , the terminal side of which will fall in the II, III or IV quadrant.

Functions of an angle in the II quadrant. Assume that we want to find the trigonometric functions of an obtuse angle $P'OQ$. We first relate this angle to an acute angle in the first quadrant as shown in Fig. 7. Suppose $\angle POQ = \theta$ (given). Extend OQ to Q' so that $OQ = OQ'$ and $\angle POQ = \angle P'OQ'$. Drop perpendicular from P on OY forming rt. $\triangle$ ΔPQO . Similarly, drop perpendicular from P' on OX , forming rt. $\triangle$ $\Delta P'Q'O$.

Since we have two angles and one side of one triangle equal to the corresponding angles and side of the other triangle, two triangles are congruent

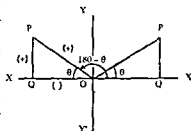


Fig. 7

$$\Delta PQO \cong \Delta P'Q'O.$$

$$PQ = P'Q', OP = OP'$$

$$OQ = OQ'$$

By definition, $\sin QOP' = \sin (180^\circ - \theta)$

$$\text{and, } \sin QOP' = \frac{(+P'Q')}{(+P'O)} \quad \text{Also } \sin \theta = \frac{PQ}{PO}$$

$$\sin QOP = \sin (180^\circ - \theta) = \sin \theta$$

$$(\quad PQ = P'Q \text{ and } PO = P'O)$$

Again $\cos QOP = \cos (180^\circ - \theta) = \frac{(-)OQ}{(+)OP}$ and $\cos \theta = \frac{OQ}{OP}$

$$\cos QOP = \cos (180^\circ - \theta) = \frac{(-)OQ}{(+)OP} = -\frac{OQ}{OP} = -\cos \theta$$

Similarly $\tan QOP = \tan (180^\circ - \theta) = \frac{(+)P'Q}{(-)OQ}$ and $\tan \theta = \frac{PQ}{OQ}$

$$\tan QOP = \tan (180^\circ - \theta) = -\tan \theta$$

From these functions similar values of remaining functions can be obtained easily

Ex 1 Find the numerical values of all the trigonometric functions of angle 120°

$$\text{We know, } 120^\circ = (180^\circ - 60^\circ)$$

$$\text{therefore } \sin 120^\circ = \sin (180^\circ - 60^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 120^\circ = \cos (180^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$$

$$\tan 120^\circ = \tan (180^\circ - 60^\circ) = -\tan 60^\circ = -\sqrt{3}$$

Functions of an angle in the III and IV quadrants To obtain the value of the trigonometric functions of an angle in the third and fourth quadrants we use a procedure similar to that of quadrant II

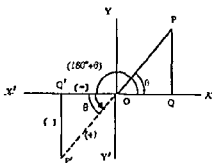


Fig 8

In Fig 8, $\angle QOP'$ lies in the III quadrant

$\Delta P'QO$ is constructed congruent to ΔPQO

$$\angle QOP' = \theta$$

$$\angle QOP' = \sin (180^\circ + \theta)$$

By definition,

$$\sin QOP' = \sin (180^\circ + \theta)$$

$$= \frac{(-)P'Q}{(+)OP'}$$

$$= -\sin \theta$$

Also,

$$\sin \theta = \frac{PQ}{OP} = \frac{P'Q}{OP'}$$

$$\sin (180^\circ + \theta) = -\sin \theta$$

Similarly,

$$\cos (180^\circ + \theta) = -\cos \theta$$

[Since, OQ is negative]

$$\text{and, } \tan (180^\circ + \theta) = \tan \theta$$

[$P'Q'$ and OQ' both are negative]

In Fig 9, $\angle QOP$ lies in the *IV* quadrant

$\triangle PQO$ is constructed congruent to $\triangle P'Q'O$.

By definition,

$$\sin QOP' = \sin(360^\circ - \theta) = \frac{(-)P'Q}{(+)OP'} \text{ and } \sin \theta = \frac{PQ}{OP}$$

$$\text{Thus gives, } \sin(360^\circ - \theta) = -\sin \theta$$

$$\cos(360^\circ - \theta) = \cos \theta, \text{ and}$$

$$\tan(360^\circ - \theta) = -\tan \theta \quad [\text{Since } P'Q \text{ is negative}]$$

We can now put all these results in a tabular form.

Qdt. II	Qdt. III	Qdt. IV
$\sin(180^\circ - \theta) = \sin \theta$	$\sin(180^\circ + \theta) = -\sin \theta$	$\sin(360^\circ - \theta) = -\sin \theta$
$\cos(180^\circ - \theta) = -\cos \theta$	$\cos(180^\circ + \theta) = -\cos \theta$	$\cos(360^\circ - \theta) = \cos \theta$
$\tan(180^\circ - \theta) = -\tan \theta$	$\tan(180^\circ + \theta) = \tan \theta$	$\tan(360^\circ - \theta) = -\tan \theta$

The remaining ratios can be obtained from these ratios

The above relationships are used as formulae whenever we wish to express the value of a function of an angle in the *II*, *III* or *IV* quadrants in terms of a function of an acute angle and thus the numerical value of the angle can be found

Ex. 2. Find the numerical values of all the trigonometric functions of following angles (i) 225° and (ii) 330° .

$$(i) \quad 225^\circ = (180^\circ + 45^\circ), \text{ hence } \theta = 45^\circ$$

$$\therefore \sin 225^\circ = -\sin 45^\circ = -\frac{1}{\sqrt{2}}$$

$$\cos 225^\circ = -\cos 45^\circ = -\frac{1}{\sqrt{2}}$$

$$\tan 225^\circ = \tan 45^\circ = 1$$

$$(ii) \quad 330^\circ = (360^\circ - 30^\circ) \text{ Hence } \theta = 30^\circ.$$

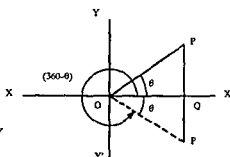


Fig 9

$$\sin 330^\circ = -\sin 30^\circ = -\frac{1}{2}$$

$$\cos 330^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 330^\circ = -\tan 30^\circ = -\frac{1}{\sqrt{3}}$$

4.6. Negative Angles

In considering trigonometric functions we have so far considered positive angles, i.e., angles generated by rotating a line in an anti-clockwise direction. If we rotate the initial line OP in a clockwise direction a negative angle is formed, which we designate as $(-\theta)$.

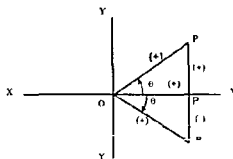


Fig. 10

We compare the values of the functions of $(-\theta)$ with those of a positive angle of same size $= \theta$. Since $(-\theta)$ lies in the fourth quadrant θ is also acute lying in first quadrant.

By dropping perpendicular from P_2 and P_1 on OX we have two congruent triangles (Fig. 10)

$$OP_2 = OP_1 \text{ and } PP_1 = PP_2$$

In $\triangle POP_1$	In $\triangle POP_2$	
$\sin(-\theta) = \frac{(-)PP_1}{(+)OP_1}$	$\sin \theta = \frac{(+)PP_2}{(+)OP_2}$	$\sin(-\theta) = -\sin \theta$
$\cos(-\theta) = \frac{(+)OP}{(+)OP_1}$	$\cos \theta = \frac{(+)OP}{(+)OP_2}$	$\cos(-\theta) = \cos \theta$
$\tan(-\theta) = \frac{(-)PP_1}{(+)OP}$	$\tan \theta = \frac{(+)PP_2}{(+)OP}$	$\tan(-\theta) = -\tan \theta$

Values of the reciprocal functions can be obtained similarly.

Note—The above results are based on negative angles in the fourth quadrant only. Nevertheless if we examine the functions of $(-\theta)$ when $(-\theta)$ is in the *I*, *II* and *III* quadrants we would obtain conclusions identical to those stated above.

Ex. 3 Find numerical values of $\sin(-45^\circ)$, $\tan(-225^\circ)$, $\sec(-300^\circ)$, $\sin(-240^\circ)$ and $\cos(-210^\circ)$

$$(i) \quad \sin(-45^\circ) = -(\sin 45^\circ) = -\left(\frac{1}{\sqrt{2}}\right) = -\frac{1}{\sqrt{2}}$$

$$(ii) \tan(-225^\circ) = -\tan(225^\circ) = -\tan(180^\circ + 45^\circ) = -(\tan 45^\circ) = -1$$

$$(iii) \sec(-300^\circ) = (\sec 300^\circ) = \sec(360^\circ - 300^\circ) = \sec 60^\circ = 2$$

$$(iv) \sin(-240^\circ) = -(\sin 240^\circ) = -\sin(180^\circ + 60^\circ) = -(-\sin 60^\circ) \\ = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$(v) \cos(-210^\circ) = (\cos 210^\circ) = \cos(180^\circ + 30^\circ) = -(\cos 30^\circ) = -\frac{\sqrt{3}}{2}$$

EXERCISE 4.1

1. By using the value of the trigonometric functions of 30° , 45° and 60° , show that the following statements are true

$$(i) \cos 45^\circ + \tan 45^\circ = \frac{1+\sqrt{2}}{\sqrt{2}}$$

$$(ii) \cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ = 0$$

$$(iii) \cos^2 30^\circ - \sin^2 30^\circ = \frac{1}{2}$$

$$(iv) \sec^2 45^\circ - \tan^2 45^\circ = 1$$

$$(v) 1 + \tan^2 30^\circ = \sec^2 30^\circ$$

$$(vi) \sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = 1$$

$$(vii) 1 - \sin^2 60^\circ = \cos^2 60^\circ$$

$$(viii) \operatorname{cosec}^2 45^\circ - \cot^2 45^\circ = 1$$

$$(ix) \tan 60^\circ \cot 240^\circ = 1$$

$$(x) \sin 30^\circ = \sqrt{\frac{1 - \cos 60^\circ}{2}}$$

$$(xi) 2 \sin 45^\circ \cos 45^\circ = 1$$

$$(xii) 2 \cos^2 45^\circ - 1 = 0$$

$$(xiii) \cos 30^\circ \sec 330^\circ = 1$$

$$(xiv) \sin 45^\circ \operatorname{cosec} 225^\circ = -1$$

$$(xv) \sin 120^\circ = 2 \sin 60^\circ \cos 60^\circ$$

$$(xvi) \cos 120^\circ = \cos^2 60^\circ - \sin^2 60^\circ$$

$$(xvii) \cos^2 120^\circ - \sin^2 120^\circ = \cos 240^\circ$$

$$(xviii) \sin 120^\circ = \sqrt{\frac{1 - \cos 240^\circ}{2}}$$

$$(xix) \cos 60^\circ = \frac{\cos 120^\circ}{2}$$

$$(xx) \quad \cos 150^\circ = -\sqrt{\frac{1+\cos 300^\circ}{2}}$$

(xxi) $\cot 30^\circ$, $\cot 45^\circ$ and $\cot 60^\circ$ are in G.P.

Note. The expression $\sin^2 \theta$ means square of $\sin \theta$ i.e. $\sin^2 \theta = (\sin \theta)^2$. This will be true for all the trigonometric functions.

2. In each of the following, θ is an acute angle. Find the values of other five functions.

$$(i) \quad \sin \theta = \frac{2}{3} \quad (ii) \quad \cos \theta = \frac{4}{5} \quad (iii) \quad \tan \theta = \frac{1}{2}$$

$$(iv) \quad \cot \theta = \frac{15}{8} \quad (v) \quad \sec \theta = \frac{4}{3} \quad (vi) \quad \operatorname{cosec} \theta = 1.2$$

3. Prove whether the following statements are true or false.

- (i) $\sin 30^\circ + \sin 60^\circ = \sin 90^\circ$
- (ii) $\sin 45^\circ \cos 45^\circ = \frac{1}{3} \sin 90^\circ$
- (iii) $\cos 30^\circ + 2 \cos 60^\circ = \operatorname{cosec} 150^\circ$
- (iv) $\tan 60^\circ = 2 \tan 30^\circ$
- (v) $\cos 60^\circ - \cos 30^\circ = \cos 30^\circ$

4. If θ is greater than 0° but less than or equal to 45° then prove.

- (i) $\sin \theta$ will be less than $\tan \theta$ and
- (ii) $\cos \theta$ will be greater than $\tan \theta$

5. If θ is greater than 0° but less than 90° then prove.

- (i) $\sec \theta$ will be greater than $\tan \theta$ and
- (ii) $\operatorname{cosec} \theta$ will be greater than $\cos \theta$

6. Find θ which satisfies $7 \sin^2 \theta + 3 \cos^2 \theta = 4$

4.7 Trigonometric Identities

A statement of equality between two quantities is an equation. An equation is called **identity** if the given statement is true for all possible values of the variables involved in the equation. Against this equation is satisfied by only certain values of the variables.

In trigonometry, as in algebra, identities as well as equations occur. We shall be dealing with trigonometric identities here.

There are three kinds of identities in trigonometry.

1 Reciprocal identities

$$(i) \quad \sin \theta = \frac{1}{\operatorname{cosec} \theta}, \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad \sin \theta \operatorname{cosec} \theta = 1$$

$$(ii) \cos \theta = \frac{1}{\sec \theta} \quad \sec \theta = \frac{1}{\cos \theta} \quad \sec \theta \cos \theta = 1$$

$$(iii) \tan \theta = \frac{1}{\cot \theta} \quad \cot \theta = \frac{1}{\tan \theta} \quad \tan \theta \cot \theta = 1$$

II Quotient identities

$$(i) \frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$(ii) \frac{\cos \theta}{\sin \theta} = \cot \theta$$

III Pythagorean identities

$$(i) \sin^2 \theta + \cos^2 \theta = 1$$

$$(ii) 1 + \tan^2 \theta = \sec^2 \theta$$

$$(iii) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Pythagorean identities Proofs

In rt $\angle$ ed ΔPQR

$$(PQ)^2 + (QR)^2 = (PR)^2 \quad (i)$$

If we divide both sides of (i) by $(PR)^2$ we have

$$\frac{(PQ)^2}{(PR)^2} + \frac{(QR)^2}{(PR)^2} = \frac{(PR)^2}{(PR)^2} \quad (ii)$$

In Fig 11 $\frac{PQ}{PR} = \cos \theta$ $\frac{QR}{PR} = \sin \theta$

from (ii) we have $\sin^2 \theta + \cos^2 \theta = 1$

If we divide both sides of (i) by $(PQ)^2$, we have

$$\frac{(PQ)^2}{(PQ)^2} + \frac{(QR)^2}{(PQ)^2} = \frac{(PR)^2}{(PQ)^2} \quad (iii)$$

In Fig 11, $\frac{PQ}{PR} = \cos \theta$, $\frac{QR}{PR} = \sin \theta$

from (iii), we have $1 + \tan^2 \theta = \sec^2 \theta$

If we divide both sides of (i) by $(QR)^2$, we have

$$\frac{(PQ)^2}{(QR)^2} + \frac{(QR)^2}{(QR)^2} = \frac{(PR)^2}{(QR)^2}$$

In Fig 11, $\frac{PQ}{QR} = \cot \theta$, $\frac{PR}{QR} = \operatorname{cosec} \theta$

from (iv) we have $\cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$

Trigonometric identities

The basic identities given above are useful in proving trigonometric identities in general. We illustrate this by giving a few examples below

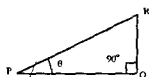


Fig 11

Ex 4 Prove that $\frac{\tan \theta}{1 + \sec \theta} - \frac{\tan \theta}{1 - \sec \theta} = 2 \csc \theta$

$$\begin{aligned} \text{L.H.S.} &= \frac{\tan \theta}{1 + \sec \theta} - \frac{\tan \theta}{1 - \sec \theta} \\ &= \frac{\tan \theta(1 - \sec \theta) - \tan \theta(1 + \sec \theta)}{(1 + \sec \theta)(1 - \sec \theta)} \\ &= \frac{-2 \tan \theta \sec \theta}{1 - \sec^2 \theta} = \frac{-2 \tan \theta \sec \theta}{-\tan^2 \theta} \\ &= \frac{-2 \sec \theta}{\tan \theta} = \frac{2}{\sin \theta} = \text{R.H.S.} \end{aligned}$$

Ex 5 Prove that $\frac{1 + \csc \theta}{\sec \theta} = \cos \theta + \cot \theta$

$$\begin{aligned} \text{L.H.S.} &= \frac{1 + \frac{1}{\sin \theta}}{\frac{1}{\cos \theta}} = \frac{\frac{(\sin \theta + 1)}{\sin \theta}}{\frac{1}{\cos \theta}} = \frac{(\sin \theta + 1) \cos \theta}{\sin \theta} \\ &= \frac{(\sin \theta \cos \theta + \cos \theta) \cos \theta}{\sin \theta} = \cos \theta + \cot \theta = \text{R.H.S.} \end{aligned}$$

Ex 6 Prove that $\frac{\sec \theta}{1 - \cos \theta} = \frac{\sec \theta + 1}{\sin^2 \theta}$

$$\begin{aligned} \frac{\sec \theta}{1 - \cos \theta} &= \frac{\sec \theta(1 + \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)} = \frac{\sec \theta + \sec \theta \cos \theta}{1 - \cos^2 \theta} \\ &= \frac{\sec \theta + \sec \theta \cos \theta}{\sin^2 \theta} = \frac{\sec \theta + 1}{\sin^2 \theta} = \text{R.H.S.} \end{aligned}$$

Ex 7 Prove that $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{\sin \theta}{1 + \cos \theta}$

$$\begin{aligned} \text{L.H.S.} &= \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \cdot \sqrt{\frac{1 + \cos \theta}{1 + \cos \theta}} = \sqrt{\frac{1 - \cos^2 \theta}{(1 + \cos \theta)^2}} \\ &= \sqrt{\frac{\sin^2 \theta}{(1 + \cos \theta)^2}} = \frac{\sin \theta}{1 + \cos \theta} = \text{R.H.S.} \end{aligned}$$

Ex 8. Prove that $\frac{\cos^4 \theta - \sin^4 \theta}{(1 - \tan^4 \theta)} = \cos^4 \theta$

$$\text{L.H.S.} = \frac{(\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta)}{(1 - \tan^2 \theta)(1 + \tan^2 \theta)} = \frac{\cos^2 \theta - \sin^2 \theta}{(1 - \tan^2 \theta)(\sec^2 \theta)}$$

$$\text{But } (1 - \tan^2 \theta) \sec^2 \theta = \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \cdot \frac{1}{\cos^2 \theta} = \frac{\cos^2 \theta - \sin^2 \theta}{\cos^4 \theta}$$

$$\therefore \frac{\cos^2 \theta - \sin^2 \theta}{(1 - \tan^2 \theta) (\sec^2 \theta)} = \frac{\cos^2 \theta - \sin^2 \theta}{\frac{(\cos^2 \theta - \sin^2 \theta)}{\cos^4 \theta}} = \cos^4 \theta = R.H.S$$

EXERCISE 4 2

1. Verify the following identities for the angles specified

- (i) $\sin^2 \theta + \cos^2 \theta = 1$ when $\theta = 45^\circ$
 (ii) $\tan^2 \theta + 1 = \sec^2 \theta$ when $\theta = 60^\circ$
 (iii) $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$ when $\theta = 300^\circ$
 (iv) $\frac{\sin \theta}{\cos \theta} = \tan \theta$ when $\theta = 240^\circ$
 (v) $\cos^2 \theta = 1 - \sin^2 \theta$ when $\theta = 315^\circ$

2. Prove the following identities

- (i) $\cos^2 \theta + \sin^2 \theta = \sec \theta \cos \theta$
 (ii) $(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1) = \cot^2 \theta$
 (iii) $\frac{2 \cot \theta}{1 + \cot^2 \theta} = 2 \sin \theta + \cos \theta$
 (iv) $(1 - \sin^2 \theta) \operatorname{cosec}^2 \theta = \cot^2 \theta$
 (v) $\frac{\sin \theta}{1 - \cos \theta} + \frac{1 - \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$
 (vi) $\frac{\tan \theta}{1 + \sec \theta} = \sec \theta - 1$
 (vii) $\frac{1 + 2 \sin \theta \cos \theta}{\sin \theta + \cos \theta} = \sin \theta + \cos \theta$
 (viii) $\frac{\sin^2 \theta + \cos^2 \theta}{2 \sin^2 \theta - 1} = \frac{\sec \theta - \cos \theta}{\tan \theta - 1}$

3. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, prove that $\tan \theta = \pm \frac{1}{\sqrt{3}}$

4. Show that :

$$(i) \quad (1 + \cot A - \operatorname{cosec} A) (1 + \tan A + \sec A) = 2$$

$$(ii) \quad \sqrt{\frac{1 - \sin A}{1 + \sin A}} = \sec A - \tan A$$

$$(iii) \quad \frac{1}{\sec A + \tan A} - \frac{1}{\cos A} = \frac{1}{\cos A} - \frac{1}{\sec A - \tan A}$$

- (vi) $\frac{(\tan A + \sec A) - 1}{(\tan A + \sec A) + 1} = \frac{1 + \sin A}{\cos A}$
 (v) $\sin \theta - \cos \theta = \pm \sqrt{1 - 2 \sin \theta \cos \theta}$
 (vi) $(1 + \tan^2 A) \cot A = \frac{1}{\sin A \cos A}$
 (vii) $\sec^4 A - \sec^2 A = \tan^2 A + \tan^4 A$

5 Show that

- (i) $\operatorname{cosec} A + \cot A = \pm \sqrt{\frac{1 + \cos A}{1 - \cos A}}$
 (ii) $\frac{\sin A}{1 - \cos A} = \frac{1 + \cos A}{\sin A}$
 (iii) $\frac{\cos^4 \theta - \sin^4 \theta}{1 - \tan^2 \theta} = \cos^2 \theta$
 (iv) $\frac{\cot \alpha + \tan \beta}{\cot \beta + \tan \alpha} = \cot \alpha \tan \beta$

6 Express all trigonometric functions in terms of $\sec \theta$

7 If (i) $2[\cos^2 \theta - \sin^2 \theta] = 1$ and (ii) $\cot^2 \theta + \sin^2 \theta = \frac{5}{4}$

$$\text{Show that } \theta = \frac{\pi}{6}$$

8 If $\tan A + \sin A = m$ and $\tan A - \sin A = n$
 show that $(m^2 - n^2) = 16mn$

4.8 Functions of the Sum and the difference

Until now we dealt with the trigonometric functions of one angle. Many a time it is desirable to express a trigonometric function of sum or difference of two angles in terms of the functions of the separate angles.

In Fig 12, $\angle QOS = \angle x$

and $\angle SOT = \angle y$

$\angle QOT = \angle (x+y)$

Take any point T on terminal side of $\angle y$ and draw perpendiculars to the OX and to the terminal side of $\angle x$ calling the feet of these perpendiculars P and S draw perpendiculars to OY and PT , calling the feet of these Q and R respectively

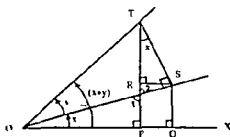


Fig 12.

In $\Delta s PRO$ and RTS (R is at the intersection of OS and RP)

$\angle TOS = \angle R'PO = 90^\circ$, $\angle 1 = \angle 2$, third angle $\angle PTS$ should be equal to $\angle x$.

$$\begin{aligned}\text{By definition, } \sin(x+y) &= \frac{TP}{OT} = \frac{PR+RT}{OT} = \frac{PR}{OT} + \frac{RT}{OT} \\ &= \frac{QS}{OT} + \frac{RT}{OT} \quad [PR = QS]\end{aligned}$$

We find that these two ratios $\left(\frac{QS}{OT} \text{ and } \frac{RT}{OT}\right)$ are not the function of either of angles (*i.e.* $\angle x$ and $\angle y$) involved in the figure. We multiply $\frac{PR}{OT}$ by $\frac{OS}{OS}$ and $\frac{RT}{TS}$ by $\frac{TS}{TS}$ to obtain the result

$$\begin{aligned}\sin(x+y) &= \frac{QS}{OT} \frac{OS}{OS} + \frac{RT}{OT} \frac{TS}{TS} \\ &= \frac{QS}{OS} \frac{OS}{OT} + \frac{RT}{TS} \frac{TS}{OT} \\ &= \sin x \cos y + \cos x \sin y\end{aligned}$$

$$\begin{aligned}\text{Again } \cos(x+y) &= \frac{OP}{OT} = \frac{OQ-PQ}{OT} = \frac{OQ}{OT} - \frac{PQ}{OT} \\ &= \frac{OQ}{OT} - \frac{RS}{OT} \quad [PQ = RS]\end{aligned}$$

We multiply $\frac{OQ}{OT}$ by $\frac{OS}{OS}$ and $\frac{RS}{OT}$ by $\frac{TS}{TS}$

$$\begin{aligned}\cos(x+y) &= \frac{OQ}{OT} \frac{OS}{OS} - \frac{RS}{OT} \frac{TS}{TS} \\ &= \frac{OQ}{OS} \frac{OS}{OT} - \frac{RS}{TS} \frac{TS}{OT} \\ &= \cos x \cos y - \sin x \sin y\end{aligned}$$

We can also find the sine and cosine of difference of two angles with the help of the above derivations

$$\sin(x-y) = \sin(x+[-y]) = \sin x \cos(-y) + \cos x \sin(-y)$$

$$\cos(x-y) = \cos(x+[-y]) = \cos x \cos(-y) - \sin x \sin(-y)$$

We know from Section 4.6 that $\sin(-y) = -\sin y$ and $\cos(-y) = \cos y$

$$\sin(x-y) = \sin x \cos y - \cos x \sin y$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y$$

Collecting all the results

$$\sin(x+y) = \sin x \cos y + \cos x \sin y \quad \dots(i)$$

$$\sin(x-y) = \sin x \cos y - \cos x \sin y \quad \dots(ii)$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y \quad \dots(iii)$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y \quad \dots(\text{iv})$$

Adding (i) and (iv), we obtain

$$\sin(x+y) + \sin(x-y) = 2 \sin x \cos y \quad \dots(\text{v})$$

Subtracting (ii) from (i), we obtain

$$\sin(x+y) - \sin(x-y) = 2 \cos x \sin y \quad \dots(\text{vi})$$

Similarly, from (iii) and (iv) we obtain

$$\cos(x+y) + \cos(x-y) = 2 \cos x \cos y \quad \dots(\text{vii})$$

$$\cos(x+y) - \cos(x-y) = -2 \sin x \sin y \quad \dots(\text{viii})$$

Tangent of sum and difference of two angles

$$\tan(x+y) = \frac{\sin(x+y)}{\cos(x+y)} = \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y}$$

Divide numerator and denominator by $(\cos x \cos y)$

$$\tan(x+y) = \frac{\frac{\sin x \cos y}{\cos x \cos y} + \frac{\cos x \sin y}{\cos x \cos y}}{\frac{\cos x \cos y}{\cos x \cos y} - \frac{\sin x \sin y}{\cos x \cos y}} = \frac{\tan x + \tan y}{1 - \tan x \tan y} \quad \dots(\text{ix})$$

By substituting $(-y)$ for y in the above derivation, we can prove

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y} \quad \dots(\text{x})$$

4.9. Functions of Twice and Half of an Angle

We have obtained,

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \text{ so that}$$

$$\sin(x+x) = \sin x \cos x + \cos x \sin x$$

$$\sin 2x = 2 \sin x \cos x$$

$$\text{Similarly, } \cos(x+x) = \cos x \cos x - \sin x \sin x$$

$$= \cos^2 x - \sin^2 x$$

$$= \cos^2 x - (1 - \cos^2 x) = 2\cos^2 x - 1$$

$$= (1 - \sin^2 x) - \sin^2 x = 2\cos^2 x - 1$$

$$\text{Thus, we have, } \sin 2x = 2 \sin x \cos x = 2 \tan x / (1 + \tan^2 x)$$

$$\text{and, } \cos 2x = 2\cos^2 x - 1 = 1 - 2\sin^2 x = (1 - \tan^2 x) / (1 + \tan^2 x)$$

To find the sine and cosine of half an angle, we let $2x = \theta$ in the above derivations:

$$\cos 2x = 1 - 2 \sin^2 x$$

$$\text{which gives } \cos \theta = 1 - 2 \sin^2 \theta/2$$

$$\text{or, } \sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2}$$

$$\text{Therefore, } \sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\text{Again } \cos 2x = 2 \cos^2 x - 1 \quad \text{Let } 2x = \theta$$

$$\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1$$

$$2 \cos^2 \frac{\theta}{2} = \cos \theta + 1$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

With the help of $\sin \frac{\theta}{2}$ and $\cos \frac{\theta}{2}$ we find the value of $\tan \frac{\theta}{2}$

$$\tan \frac{\theta}{2} = \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} = \frac{\pm \sqrt{\frac{1 - \cos \theta}{2}}}{\pm \sqrt{\frac{1 + \cos \theta}{2}}} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

Multiplying it by $\pm \sqrt{\frac{1 + \cos \theta}{1 + \cos \theta}}$, we can prove that $\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$

Thus we have following derivations

$$\cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \quad (i)$$

$$\sin 2\theta = 2 \sin \theta \cos \theta \quad (ii)$$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}} \quad (iii)$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}} \quad (iv)$$

$$\text{and } \tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta} \quad (v)$$

Ex 9 Using functions of 30° , 45° and 60° , find the value of
(i) $\sin 90^\circ$ and $\cos 90^\circ$, (ii) $\sin 15^\circ$ and $\cos 15^\circ$, (iii) $\sin 165^\circ$ and $\cos 165^\circ$

(i) We know that $\sin (x + y) = \sin x \cos y + \cos x \sin y$

$$\sin (90^\circ) = \sin (30^\circ + 60^\circ) = \sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$$

$$\sin 90^\circ = \frac{1}{2} \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = 1$$

$$\begin{aligned}\cos (90^\circ) &= \cos (30^\circ + 60^\circ) = \cos 30^\circ \cos 60^\circ - \sin 30^\circ \sin 60^\circ \\ &= \frac{\sqrt{3}}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = 0\end{aligned}$$

$$(ii) \sin (x - y) = \sin x \cos y - \cos x \sin y$$

$$\begin{aligned}\sin 15^\circ &= \sin (45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\end{aligned}$$

$$\frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{3} - 1}{2\sqrt{2}} = 0.259$$

$$\begin{aligned}\cos 15^\circ &= \cos (45^\circ - 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{\sqrt{3} + 1}{2\sqrt{2}}\end{aligned}$$

$$(iii) \sin 165^\circ = \sin (90^\circ + 75^\circ) = \sin 90^\circ \cos 75^\circ + \cos 90^\circ \sin 75^\circ$$

We must now obtain the value of $\sin 75^\circ$ and $\cos 75^\circ$ first

$$\begin{aligned}\sin 75^\circ &= \sin (45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = 0.966\end{aligned}$$

$$\begin{aligned}\cos 75^\circ &= \cos (45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = 0.259\end{aligned}$$

$$\begin{aligned}\sin 165^\circ &= \sin (90^\circ + 75^\circ) = \cos 90^\circ \cos 75^\circ + \sin 90^\circ \sin 75^\circ \\ &= 0 \left(\frac{\sqrt{3}}{2\sqrt{2}} \right) + \left(\frac{\sqrt{3} + 1}{2\sqrt{2}} \right) \\ &= \left(\frac{\sqrt{3} + 1}{2\sqrt{2}} \right) = 0.966\end{aligned}$$

Ex 10 If $\cos A = \frac{5}{13}$, $\sin B = \frac{3}{5}$, find $\cos (A + B)$ and $\cos (A - B)$ $\sin (A + B)$ and $\sin (A - B)$

From the given values $\sin B = \frac{3}{5}$ and $\cos A = \frac{5}{13}$, we construct two triangles DAC and FBE (Fig. 13)

Applying Pythagorean theorem, we obtain the third side in each triangle

$$\cos (A + B) = \cos A \cos B - \sin A \sin B$$

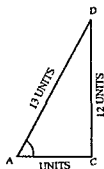


Fig 13



$$= \frac{5}{13} \cdot \frac{4}{5} - \frac{12}{13} \cdot \frac{3}{5}$$

$$= -\frac{16}{65} = -0.246$$

$$\text{and, } \cos(A - B)$$

$$= \cos A \cos B + \sin A \sin B$$

$$= \frac{5}{13} \cdot \frac{4}{5} + \frac{12}{13} \cdot \frac{3}{5}$$

$$= \frac{56}{65} = 0.862$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$= \frac{12}{13} \cdot \frac{4}{5} + \frac{5}{13} \cdot \frac{3}{5} = 0.969$$

$$\text{and, } \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$= \frac{12}{13} \cdot \frac{4}{5} - \frac{5}{13} \cdot \frac{3}{5} = 0.508$$

Ex. 11. If $\tan x = \frac{3}{4}$ and $\tan y = \frac{8}{15}$, find the value of $\tan(x + y)$ and $\tan(x - y)$

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y} = \frac{\frac{3}{4} + \frac{8}{15}}{1 - \frac{3}{4} \cdot \frac{8}{15}} = \frac{77}{36} = 2.139$$

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y} = \frac{\frac{3}{4} - \frac{8}{15}}{1 + \frac{3}{4} \cdot \frac{8}{15}} = \frac{13}{84} = 0.155$$

Ex. 12. If $\sin \theta = \frac{4\sqrt{5}}{9}$ and θ is an acute angle, find the value of $\sin 2\theta$, $\sin \frac{\theta}{2}$, $\cos 2\theta$ and $\cos \frac{\theta}{2}$

In Fig 14, we construct $\triangle ABC$ with $AC = 4\sqrt{5}$ units and $AB = 9$ units and find the third side of the triangle ($\angle ACB = 90^\circ$)

$$\sin \theta = \frac{4\sqrt{5}}{9} \text{ and } \cos \theta = \frac{1}{9}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$



Fig 2 14

$$\sin 2\theta = 2 \frac{4\sqrt{5}}{9} \cdot \frac{1}{9} = \frac{8\sqrt{5}}{81} = 0.221$$

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}} = \pm \sqrt{\frac{1 - \frac{1}{9}}{2}} = \frac{2}{3} = 0.667 \quad [\because \theta \text{ is acute}]$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = \frac{1}{81} - \frac{80}{81} = \left(\frac{-79}{80}\right) = -0.987$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}} = \pm \sqrt{\frac{1 + \frac{1}{9}}{2}} = \frac{\sqrt{5}}{3} = 0.745 \quad [\because \theta \text{ is acute}]$$

Ex. 13. Given $\cos \theta = \frac{4}{5}$ (θ is an acute angle), find the value of $\tan 2\theta$ and $\tan \frac{\theta}{2}$

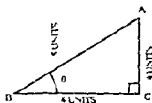


Fig. 15

In ΔABC , $\angle B = \theta$,

$$\cos \theta = \frac{4}{5}, \sin \theta = \frac{3}{5} \text{ and } \tan \theta = \frac{3}{4}$$

$$\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta} = \frac{\frac{3}{5}}{1 + \frac{4}{5}} = \frac{\frac{3}{5}}{\frac{9}{5}} = \frac{1}{3} = 0.334$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2 \cdot \frac{3}{4}}{1 - \frac{9}{16}} = \frac{24}{7} = 3.429$$

EXERCISE 4.3

1. Using functions of 30° , 45° and their multiples, find the values of each of the following functions:-

(i) $\sin 105^\circ$, $\cos 105^\circ$, and $\tan 105^\circ$.

(ii) $\sin 165^\circ$, $\cos 165^\circ$ and $\tan 165^\circ$

(iii) $\sin 15^\circ$, $\cos 15^\circ$ and $\tan 15^\circ$

2 If $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, A and B are both acute angles, find the values of

$$\sin(A+B), \cos(A-B), \sin(A-B) \text{ and } \cos(A+B)$$

3 If $\tan x = \frac{3}{4}$ and $\tan y = \frac{8}{15}$, find the value of $\tan(x+y)$ and $\tan(x-y)$

4. If $\tan x = \frac{8}{15}$ and $y = -\frac{3}{4}$, find $\tan(x+y)$ and $\tan(x-y)$

5. If $\sin A = \frac{3}{5}$, A is in quadrant I $\cos B = -\frac{5}{13}$, B is in quadrant II find

$$\sin(A+B), \sin(A-B), \cos(A+B) \text{ and } \cos(A-B)$$

Also prove that

$$\sin(A+B) \sin(A-B) = 2 \sin A \cos B$$

$$\text{and } \cos(A+B) - \cos(A-B) = -2 \sin A \sin B$$

6 If $A = 30^\circ \left(\frac{\pi}{6} \right)$ verify that $\sin$

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

7. Prove that $\cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$

8 Show that $\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$

9. If $\tan A + \sin A = m$, and $\tan A - \sin A = n$ show that $(m^2 - n^2)^2 = 16mn$.

4 10. Angular Measure (Radian)

If OA is a line movable about a centre O and in anti-clockwise direction then any angle such as BOA (Fig 16) may be measured in either degrees or radians. If the circumference of a circle be divided into 360 equal parts, then straight line joining any two consecutive points (of the consecutive parts) to the centre of the circle will enclose an angle of the degree written as 1° . A degree is sub-divided into 60 minutes and a minute into 60 seconds

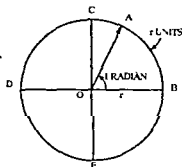


Fig 16

The relation between the abscissa, ordinate and radius vector

In Fig 17 OM, AM and OA are abscissa, x , the ordinate y and the radius vector, r of the point A. Since OMA is a right angled triangle, by Pythagorean rule

$$(OA)^2 = (OM)^2 + (AM)^2$$

From the given coordinates of point A we know that

$$OM = 3 \text{ units and } AM = 2 \text{ units}$$

$$OA^2 = 9 + 4 = 13$$

$$OA = \sqrt{13} = 3.606 \text{ units}$$

OA is radius vector or the distance of point A from the origin

To generalize the distance of the point P (x, y) from the origin will be

$$r = \sqrt{x^2 + y^2}$$

Ex 16 Find the distance of the point $(-5, -4)$ from the origin

Distance from the origin

4.12 Length of Line Joining the Points P (x_1, y_1) and Q (x_2, y_2)

Plot the points P and Q and draw PN and QM perpendicular to XX' as shown in Fig 18. Draw PS parallel to Y axis. Then by Pythagorean rule in $\triangle QSP$

$$\text{But } PS = NM = OM - ON$$

$$= x_2 - x_1$$

$$\text{and } QS = MQ - MS$$

$$= MQ - NP = y_2 - y_1$$

$$(PQ)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

PQ is the required distance between the two points P and Q

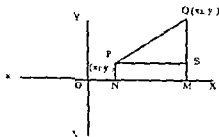


Fig 18

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Ex 17 Find the length of the line joining the points (1, 1) and (3, 2)

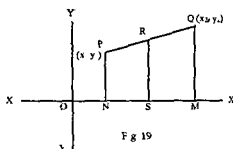
$$\begin{aligned} \text{Length} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{5} = 2.236 \text{ units} \end{aligned}$$

4.13 Coordinates of the Mid point of the Line joining the points P (x_1, y_1) and Q (x_2, y_2)

In Fig 19, let R with coordinate (x, y) be the mid point of PQ. Draw PN, QM and RS perpendiculars to X axis. Since R is the mid point of PQ therefore, S is the mid point of NM also

$$\text{Now } OS = ON + NS$$

$$\text{and } OS = OM - NS \text{ also}$$



But $NS = SM$

$$2OS = ON + OM$$

$$OS = x = \frac{ON + OM}{2}$$

$$= \frac{x_1 + x_2}{2}$$

Similarly if perpendicular be drawn on Y axis we can obtain

$$RS = y = \frac{y_1 + y_2}{2}$$

Hence

$$R(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Ex. 18. Find the coordinates of mid-point of the line joining $(-3, 4)$ and $(7, -7)$

$$\text{Mid point } R(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$x = \frac{x_1 + x_2}{2} = \frac{-3 + 7}{2} = \frac{4}{2} = 2$$

$$y = \frac{y_1 + y_2}{2} = \frac{4 - 7}{2} = -\frac{3}{2}$$

$$R(x, y) = \left(2, -\frac{3}{2} \right)$$

EXERCISE 45

1. Point out abscissa and ordinate of the following points

$(3, -2)$, $(5, 6)$, $(-8, 0)$, $(0, -5)$, $(6, -7)$

2. Plot the following points with coordinates

$(2, 3)$, $(3, -1)$, $(-4, 1)$, $(-3, -2)$, $(5, 18)$, $(-4, 3)$, $(-2, -5)$, $(0, 14)$, $(4, 0)$ and $(-3.5, 0)$

3. Plot a point $(3, 4)$ and find the distance from origin

4. Plot a point $A(2, -4)$. Drop perpendiculars AB and AC to the X axis and Y axis respectively. Calculate the lengths of the diagonals of $OBAC$

5. A straight line PQ cuts the axes at the point $M(4, 0)$ and $N(0, 1)$. Find the length of MN

6. Calculate the distances between the following pairs of points

$(1, 2)$ and $(3, 4)$, $(-1, 3)$ and $(2, -1)$, $(-3, 6, 4)$ and $(-5.2, -2.5)$, $(4, 5, 0)$ and $(1, 6, -3, 8)$, $(-4, -2)$ and $(6, -3)$

7. Find the coordinates of the mid points of the lines joining the following pairs of points

(1, 2) and (3, 4), (-3, 2) and (-1, -4), (0, 0) and (-2, 3), (2, 6, 0) and (1, 3, 8), (-3, 6, -2, 4) and (3, 6, 2, 4)

8. Plot a point $A(4, 2)$. Drop perpendiculars AB and AC on the axes. What is the mid point of BC ? Calculate its distance from the origin.

9. Let $ABCD$ be a parallelogram. If $A=(2, 3)$, $B=(-1, 4)$, $C=(5, -2)$ find the coordinates of the fourth point D .

4.14. The Straight Line

We have learnt that a point can be located in the plane by its coordinates. Let us now see if it is possible to represent any curve—say, a straight line by coordinates. For example, we take an equation $4y + 6x = 8$. This equation is not sufficient to determine both the unknown quantities x and y ; but by assuming any and every value of x (or y) there is corresponding value of y (or x). Thus this equation gives an infinite number of values of x and y which make up the infinite number of coordinates of different points. These points will satisfy the given equation $4y + 6x = 8$.

x	y	Coordinates of points
-2	5	$A(-2, 5)$
-1	$7/2$	$B(-1, 7/2)$
0	2	$C(0, 2)$
1	$1/2$	$D(1, 1/2)$
2	-1	$E(2, -1)$
3	$-5/2$	$F(3, -5/2)$

The values of x and y in the above table depend upon one another. A slight change in the value of any one of these will bring about a corresponding change in the value of other; and all these coordinates of points satisfy the equation: $4x + 6y = 8$. All these points when joined will give a straight line.

We say, therefore, that an equation can be plotted through its coordinates. On joining these coordinates of points we obtain a continuous curve. The simplest form of curve is a straight line, which is represented by a linear equation. The given equation in the above illustration is a linear one, thus it represents a straight line.

We shall now study more about the straight line.

To locate the position of straight line in a plane, we need any one of the following information

I. Any two points through which it passes

II The intercept on the Y axis and the angle it makes with X -axis

III Intercepts on both the axis

I Given two points Suppose a straight line AB goes through two points $M(x_1, y_1)$ and $N(x_2, y_2)$ and we have to locate the position of such a straight line. Take a point $R(x, y)$ (Fig 20)

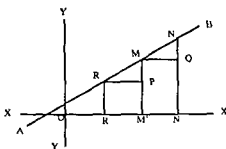


Fig 20

Now we have two similar right Δ s RPM and QNP

$$\frac{MP}{RP} = \frac{NQ}{PQ}$$

$$\text{or, } \frac{M'M - M'P}{OM' - OR'} = \frac{N'N - N'Q}{ON' - OM'}$$

$$\text{or } \frac{y_1 - y}{x_1 - x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{or } \frac{-(y - y_1)}{-(x - x_1)} = \left(\frac{y_2 - y_1}{x_2 - x_1} \right)$$

Hence the equation of the straight line is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

On substitution of the values of (x_1, y_1) and (x_2, y_2) in the above equation, we obtain the position of the straight line AB in the form of a linear equation

Ex. 19. Find the equation of straight line which passes through two points $(2, 2)$ and $(4, 8)$

Here we have : $x_1 = 2, y_1 = 2, x_2 = 4$ and $y_2 = 8$

$\therefore$ Required equation will be : $y - 2 = \left(8 - \frac{2}{4} - 2 \right) (x - 2)$ or $y = 3x - 4$ is the required equation.

Slope of a line : The expression $\left(\frac{y_2 - y_1}{x_2 - x_1} \right)$

in the derived general equation is known as the slope of the line and is symbolically represented by letter m

In Ex. 19, it is $= \frac{6}{2} = \frac{3}{1} = 3$

Slope (or gradient) of the line represents the ratio of the vertical rise (or fall) of the line to the horizontal line. From any point on the line we proceed horizontally by one unit to the right. Then from the new point we proceed

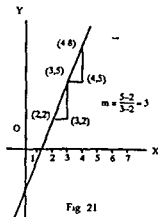


Fig 21

vertically by m units so that the slope of line $= m/1 = m$ (Fig. 21). Slope of the line at points (2, 2), (3, 5) and (4, 8) is equal to $\frac{5-2}{3-2} = \frac{8-5}{4-3} = \frac{11-8}{5-4}$, each equal to 3.

II. Intercept and slope given. Let the given intercept on Y axis be C units and the angle be θ with X axis of the st. line AB so that in Fig. 22 $\angle XMB = \theta$ and $ON = C$ units i.e. point N is $(0, C)$.

We take any point P on AB with coordinates (x, y) . From P draw PQ perpendicular to X axis and from N draw NQ perpendicular to PR .

In rt. $\triangle$ ed $\triangle PQN$, $\angle PNQ = \angle XMB = \theta$

$$PQ = PR - QR = PR - NO = y - C$$

$$NQ = OR = x$$

In rt. $\triangle$ ed $\triangle PQN$

$$\tan \theta = \frac{PQ}{NQ}$$

$$x \tan \theta = \frac{y - C}{x}$$

$$y = x \tan \theta + C$$

Since we can obtain the numerical value of $\tan \theta$ let it be denoted by m . $y = mx + C$ which is the slope intercept form of equation of line AB . m is also called the gradient (the numerical value of the slope) of the line AB .

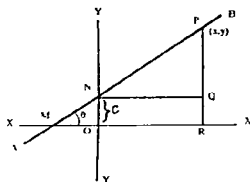


Fig. 22

Important points

1. If $C=0$ in other words if there is no intercept on Y axis, the equation will reduce to $y=mx$. It means that such a line will pass through the origin.

2. If $m=0$ the equation reduces to $y=C$. In other words, such a line will always remain at a constant distance of C units from X axis hence it will be parallel to X axis.

(Therefore, remember that gradient of the line parallel to X axis is zero)

Similarly if the equation of a straight line is given by $x=a$, it means that the straight line is parallel to Y axis at a distance of a units.

(Therefore, remember that the gradient of a line parallel to Y axis is infinity)

3. **Sign of m** m is positive slope in Fig. 22 and negative slope in Fig. 23 and 24.

4. **Sign of C** C will be positive or negative according as the point of

intersection of the line and the Y axis lies above or below the axis of X

Ex 20 Find the gradient and the intercept made by the following lines on the Y axis

(i) $\sqrt{3}x - y + 3 = 0$ (ii) $y - x + \sqrt{3} = 0$ (iii) $x - y - 6 = 0$

We must write each equation in the general form $y = mx + c$

(i) $\sqrt{3}x - y + 3 = 0$ $y = \sqrt{3}x + 3$

gradient $= \sqrt{3} = \tan 60^\circ$ and intercept on Y axis $= 3$ units. In other words the given equation represents a straight line which passes through $(0, 3)$ and makes an angle $= 60^\circ$ with the X axis

The given equation represents a straight line which passes through $(0, -3)$ and makes an angle of 45° with the X axis

(ii) $y - x + \sqrt{3} = 0$ $y = x - \sqrt{3}$ gradient $= 1 = \tan 45^\circ$

and intercept on Y axis $= -\sqrt{3}$ units

(iii) $x - y - 6 = 0$, $y = x - 6$

gradient $= 1 = \tan 45^\circ$, and intercept on Y axis is $= -6$ units

Ex 21 Write the equation of a straight line of gradient $2/3$ and which makes negative intercept of 4 units on the Y axis

Here, we have $m = 2/3$ and $C = -4$ units

equation of the line $y = mx + c$ or $y = (2/3)x - 4$

or required equation is $2x - 3y - 12 = 0$

III Intercepts are given. Let the straight line AB cut X axis at N and Y axis at M and let $ON = a$ units and $OM = b$ units. ON and OM are the intercepts of line on X axis and Y axis respectively, in other words line AB crosses the axes at the points $(a, 0)$ and $(0, b)$ and we want the equation of such a line

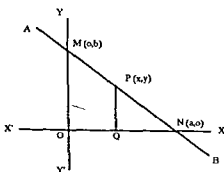


Fig 23

Take $P(x, y)$ any point on the line (Fig 23) and draw PQ perpendicular to XX' . Then by similar triangles ONM and QNP

$$\frac{PQ}{MO} = \frac{NQ}{NO} = \frac{NO - OQ}{NO}$$

$$NO = a \quad OQ = x \quad PQ = y \quad \text{and} \quad MO = b$$

$$\frac{y}{b} = \frac{a - x}{a}$$

or $\frac{y}{b} = 1 - \frac{x}{a}$ or $\frac{x}{a} + \frac{y}{b} = 1$ is the required general equation when intercepts are given

Note The denominators of x and y are the intercepts on the axes of x and y respectively

Parallel and perpendicular lines

(i) If the lines PQ and RS in Fig. 24 are parallel, then evidently $\theta = \theta'$ or $\tan \theta' = \tan \theta$ i.e., gradient of line PQ = gradient of line RS

Therefore if the equations to two parallel lines PQ and RS be represented by

$$y = mx + c$$

and $y = m'x + c'$

then $m = m'$

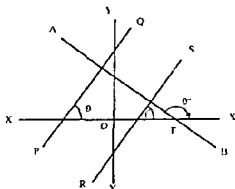


Fig. 24

(ii) Suppose AB is perpendicular to PQ then

$$\tan \theta' = -\tan \angle OTA = -\cot \theta$$

$$= -\frac{1}{\tan \theta}$$

i.e., $\tan \theta' \tan \theta = -1$, in other words, the product of the gradients of two perpendicular lines $= -1$

Therefore, if the equations to two lines perpendicular to each other are represented by $y = mx + c$, and

$$y = m''x + c'', \text{ then } nm'' = -1 \quad \text{or} \quad m'' = -\frac{1}{m}$$

Ex. 22. Find the intercepts on axes of X and Y in the case of following equations.

(i) $2x - 4y - 3 = 0$

(ii) $x - y + 1 = 0$

(i) First, we put the given equation in the intercept form :

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$2x - 4y - 3 = 0 \text{ can be put as } \frac{2}{3}x - \frac{4}{3}y = 1$$

or $\frac{x}{3/2} - \frac{y}{3/4} = 1$

Thus, intercept on X -axis $= \frac{3}{2}$ units

and " " Y axis $= -\frac{3}{4}$ units

In other words, the straight line passes through the points

$$\left(\frac{3}{2}, 0\right) \text{ and } \left(0, -\frac{3}{4}\right)$$

$$(ii) \quad x - y + 1 = 0, \quad x - y = -1$$

$$\text{or} \quad -x + y = +1$$

$$\text{or,} \quad \frac{x}{-1} + \frac{y}{1} = 1,$$

hence intercepts on X and Y axes are -1 and 1 respectively

Ex 23 Find the equation of a straight line whose intercept on the X axis is 3 times its intercept on the Y axis and which passes through the point $(-1, 3)$

Let its intercept on Y axis be a units, then intercept on X axis $= 3a$ units

$$\text{equation of line with these intercepts will be } \frac{x}{3a} + \frac{y}{a} = 1$$

This equation must also satisfy $(-1, 3)$

$$\frac{-1}{3a} + \frac{3}{a} = 1 \quad \text{or} \quad a = \frac{8}{3}$$

$$\text{the required equation is } \frac{x}{8} + \frac{y}{\frac{8}{3}} = 1 \quad \text{or} \quad x + 3y = 8$$

Ex 24 Find the equation of a straight line which makes a negative intercept of 4 units on the X axis and passes through the point $(2, 4.5)$

$$a = -4$$

$$\frac{x}{-4} + \frac{y}{b} = 1 \quad \text{But point } (2, 4.5) \text{ also lies on this line}$$

$$\frac{-2}{4} + \frac{4.5}{b} = 1 \quad \text{or} \quad b = 3$$

$$\text{The required equation must be } \frac{-x}{4} + \frac{y}{3} = 1$$

$$\text{or} \quad 3x - 4y + 12 = 0$$

Ex 25 Find the equation of the line which passes through the point $(5, 6)$ and has intercepts on the axes equal in magnitude but opposite in sign

Since the line has equal intercept with opposite signs on the axes

$$\frac{x}{a} + \frac{y}{-a} = 1 \quad \text{or} \quad x - y = a$$

This equation also satisfies the point $(5, 6)$

$$5 - 6 = a \quad \text{or} \quad a = -1$$

$$\text{the required equation is } x - y = -1 \quad \text{or} \quad x - y + 1 = 0$$

Ex 26 Show that the lines $4x - 5y + 3 = 0$ and $8x - 10y = 0$ are parallel

Gradient of line $4x - 5y + 3 = 0$ is $\frac{4}{5}$

Gradient of line $8x - 10y = 0$ is $\frac{8}{10} = \frac{4}{5}$

The given lines are parallel since they possess equal gradients

Ex. 27. Prove that the lines $2x - 3y + 4 = 0$ and $3x + 2y + 1 = 0$ are perpendicular to each other

Gradient of line $2x - 3y + 4 = 0$ is $\frac{2}{3}$

Gradient of line $3x + 2y + 1 = 0$ is $\left(-\frac{3}{2}\right)$

$$mm' = \frac{2}{3} \times -\frac{3}{2} = -1$$

Since the product of their gradients = -1 , lines are perpendicular to each other

Ex. 28. Write down the equation to a straight line cutting off an intercept 5 units from the negative axis of y and parallel to the bisector of angle XOY

Since the line is parallel to the bisector OM ,

m makes 45° with OX (Fig 25) Thus we have $m = \tan 45^\circ$ and $c = -5$

The required equation is, therefore, by substitution in $y = mx + c$,

$$y = x - 5 \text{ or } y - x + 5 = 0$$

Ex. 29. Find the equation to the straight line through the point $(3, -4)$ which is parallel to the line $2x - 4y = 3$

The gradient of $2x - 4y = 3$ is $\frac{1}{2}$

The gradient of the required line is, therefore, $\frac{1}{2}$ also

Hence, we have $x_1 = 3$, $y_1 = -4$ and $m = \frac{1}{2}$

$$\therefore y + 4 = \frac{1}{2}(x - 3) \text{ or } 2y - x + 11 = 0$$

Ex. 30. Find the equation to the straight line passing through the point $(2, -3)$ and perpendicular to $x - 2y = 8$

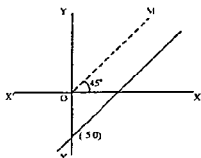


Fig. 25

The gradient of $x - 2y = 8$ is $1/2$, therefore, that of required line should be -2

The required equation is, therefore, $y + 3 = -2(x - 2)$

or, $2x + y - 1 = 0$

Ex. 31 The equation of a line is $3x - 2y = 6$

(i) Find whether the point $(0, 3)$ lies on the line

(ii) Find the slope and y -intercept of this line

(iii) Find the equation of the line passing through $(0, 3)$ and perpendicular to the above line

(i) No, since point $(0, 3)$ does not satisfy the given equation

(ii) $3x - 2y = 6$

or, $y = \frac{3}{2}x - 3$

i.e., slope $= \frac{3}{2}$ and intercept $= -3$

(iii) in other words we have to find the equation of a line with slope $= -\frac{2}{3}$ and passing through $(0, 3)$ This is given by

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -\frac{2}{3}(x)$$

∴ the required equation is $3y + 2x = 9$

Ex. 32. Give the mathematical equation of the supply function of a commodity such that the quantity supplied is zero when the price is Rs. 5 or below and it increases continuously at the constant rate of 10 units for each one rupee rise in price above Rs. 5

In Fig. 26, Point A is $(0, 5)$

Point B is $(10, 6)$

Point C is $(20, 7)$

Equation of st. line joining two points $(10, 6)$ and $(20, 7)$ is given by

$$y - y_1 = \left(\frac{x - x_1}{x_2 - x_1} \right) (y_2 - y_1)$$

$$y - 6 = \left(\frac{x - 10}{10} \right) (7 - 6)$$

$$\text{or } 10y - 60 = x - 10$$

$$\text{or } x = 10y - 50$$

$$\text{or } S = 10p - 50$$

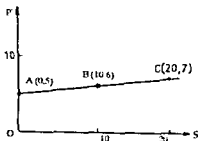


Fig. 26

is the required equation of the supply function

Ex. 33. It is known that when the price of a certain paper back book is Rs. 10 there will be no purchasers but it is thought for every Re. 1 drop in price, 12 new purchasers will appear. Find the demand function.

In other words, we are given

$$p = 10, \quad D = 0$$

$$p = 9, \quad D = 12$$

$$p = 8 \quad D = 24$$

Equation of straight line joining two points (Fig. 27)

(12, 9) and (24, 8)

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - 9}{-1} = \frac{x - 12}{12}$$

$$12y - 108 = 12 - x$$

$$12y + x = 120$$

or, $D = 120 - 12p$ is the required demand equation

Ex. 34. Points (1, 10), (2, 1) and (-7, 0) are the vertices of an isosceles right angled triangle. Find the area of the triangle.

Suppose PQR is an isosceles right angled triangle (Fig. 28)

Let the three sides of triangle be represented by d_1 , d_2 and d_3

Three vertices being

$$P(2, 1), Q(1, 10), R(-7, 0)$$

Distance between (1, 10) and

$$(2, 1) = d_1 = \sqrt{1+81} = \sqrt{82}$$

Distance between (2, 1) and

$$(-7, 0) = d_2 = \sqrt{81+1} = \sqrt{82}$$

Distance between (1, 10) and

$$(-7, 0) = d_3 = \sqrt{64+100} = \sqrt{164}$$

$\therefore$ Area of $\Delta PQR = 1/2 \text{ Base} \times \text{Altitude}$

$$= \frac{1}{2} \times \sqrt{82} \times \sqrt{82} = 41$$

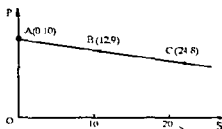


Fig. 27

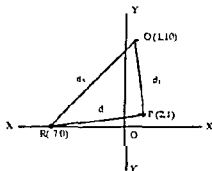


Fig. 28

EXERCISE 4.6

1. Find the equations to the straight lines joining the following pairs of points

$(0, 0)$ and $(8, 2)$, $(-3, 7)$ and $(-1, 5)$, $(2, 4)$ and $(-5, -3)$, $(1, 2)$ and $(3, -2)$
 $(-4, 0)$ and $(0, 2)$, $(-6, 0)$ and $(0, -9)$, $(1, -1)$ and $(4, 1)$, $(-1, 2)$ and $(7, -1)$
 $(-1, 2)$ and $(2, -1)$ and (a, b) and $(a+b, a-b)$

2. Write down the gradients of the lines joining the following pairs of points

$(0, 0)$ and $(2, -3.8)$, $(-2, 5)$ and $(3, 2.6)$, $(0, 0)$ and $(5, 4)$, $(0, 0)$ and $(-5, -4)$ and $(a, 0)$ and $(0, b)$

3. Write the equations to lines having gradients and y-intercepts as follows

	Gradient	Intercept
(i)	$2/3$	4
(ii)	$-8/5$	3
(iii)	-0.8	-2
(iv)	1.7	0
(v)	-1.5	-2.5

4. Find the slope and intercept of the following equations and then graph them appropriately.

(i) $2x - 3y + 4 = 0$ (ii) $8x + 5y - 15 = 0$ (iii) $5y + 4x + 10 = 0$
 (iv) $(\sqrt{3} + 1)y + 2x + 7 = 0$

5. Prove that the following pairs of lines are parallel

(i) $3x - 2y = 4$ (ii) $4x + 7y = 1$
 $6x - 4y = 3$ $20x + 35y = 6$

6. A straight line is drawn through the point $(3, 4)$ parallel to the line $x + 2y = 5$. Find its equation.

7. A straight line is drawn through the point $(1, 1)$ perpendicular to $5x + 7y = 2$. Find its equation.

8. Write the equations of the straight lines whose intercepts on the axes are ;

	<i>Intercept on X axis</i>	<i>Intercept on Y axis</i>
(i)	5 units	-3 units
(ii)	-4 "	3 "
(iii)	3 "	8 "
(iv)	-6 "	-9 "
(v)	x_1 "	$-y_1$ "

9 Put the following equations into intercept form and thus find the intercepts on axes

(i) $4x+2y=9$ (ii) $5x-6y+3=0$ (iii) $2x-2y=5$ (iv) $y=2x-4$

10 Find the equation of straight line which

(i) passes through the point (15 -3) and cuts off equal intercepts from the axes

(ii) passes through the point (-4 1) and has equal intercepts (but opposite in sign) on the axes

(iii) passes through the point (5 3) and whose intercept on the x axis is 4 times its intercept on y axis

11. Show that the points (-1 1) (5 -3) and (8 -5) lie on the same straight line. What is the equation of this line?

12 The demand for milk is given by

<i>Price</i>	1	2	3
<i>Demand</i>	100	50	0

Find the linear demand function and its slope

13 Find the equation of the straight line passing through the intersection of $3x+5y+2=0$ and $y=x+2$ and with -3 as its slope

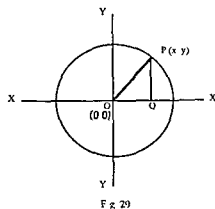
4.15 The Circle

If a point is constrained to move according to a particular law the path it follows is called the locus of the point. For example, fix one end of a piece of string by a tack to a flat piece of paper. To the other end fasten a pencil and move it so as to keep the string always tight during the movement of the pencil. The point of the pencil will leave a circular trace—called circle—the path along which the point has moved. Thus we say that circle is the locus of a point which moves in a plane so that its distance from the fixed point (called centre) always remains constant (called the radius).

Equation of a circle To fix the position of a circle in a plane either of the following should be known

- 1 Its radius and the coordinates of its centre or
- 2 A point through which it passes and two intercepts it cuts off on the axes or
- 3 Two points through which it passes and its radius or
- 4 Three points through which it passes

We shall find the equation of circle when its radius and centre are assumed to be given



(A) When centre is at the origin

Given (i) Coordinates of centre $O(0, 0)$

(ii) Radius a units Take a point $P(x, y)$

Draw perpendicular to OX

$OQ = x$ $PQ = y$

and $PO = a$ units (Fig 29)

In rt $\triangle OQP$

$$(PO)^2 = (OQ)^2 + (PQ)^2$$

$$\text{or } a^2 = x^2 + y^2$$

therefore $x^2 + y^2 = a^2$ is the equation to the locus of $P(x, y)$ and hence the required equation to the circle whose centre is at origin and radius = a units

Important remarks

1 If the length of radius be varied we shall obtain circles with different radii while centre is always at the origin (Fig 30) called concentric circles. If we continuously decrease the radius then circles will become smaller and ultimately when $a = 0$ we arrive at a point namely the origin. A degenerate circle of this kind is called a point circle. We therefore, say that $x^2 + y^2 = 0$ is a point at origin.

2 We have obtained the equation of circle $x^2 + y^2 = a^2$ which can be written as $x = \pm \sqrt{a^2 - y^2}$ or $y = \pm \sqrt{a^2 - x^2}$. This form of equation shows that

(i) x and y can never be numerically

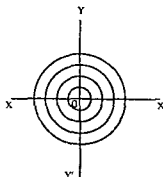


Fig 30

greater than a (which is radius of circle). If x or y is greater than a then $\sqrt{a^2 - x^2}$ and $\sqrt{a^2 - y^2}$ will become imaginary quantities.

(ii) For any value of x (or y) there are two corresponding values of y (or x) which are numerically equal but have opposite signs. In other words, these points would lie on either side of the x axis (or y axis) and at equal distances from it, one directly above the other.

For example, let us consider an equation $x^2 + y^2 = 4$.

Comparing it with the equation $x^2 + y^2 = a^2$ we find that the given equation is that of a circle whose centre is at origin with radius $a = \sqrt{4} = 2$ units.

Also $x = \pm \sqrt{4 - y^2}$ and $y = \pm \sqrt{4 - x^2}$. Therefore x and y cannot be greater than 2. If $x > 2$, say 3, then $y = \pm \sqrt{-5}$ or $y > 2$, say 2.5 then $x = \pm \sqrt{-2.25}$, both the quantities are imaginary quantities and cannot be graphed.

(B) When the centre is not at the origin

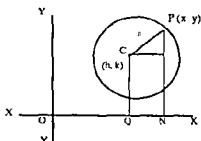


Fig. 31

Given (i) Coordinates of centre $C(h, k)$

(ii) radius $= a$

In rt. $\triangle$ ed $\triangle CMP$ (Fig. 31)

$CP = a$

$CM = QN = OQ - OQ = (x - h)$

$PM = PN - NM = PN - CQ$
 $= (y - k)$

By Pythagorean theorem,

$$(CP)^2 = (CM)^2 + (PM)^2$$

$$(x - h)^2 + (y - k)^2 = a^2 \quad (i)$$

$$x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - a^2 = 0 \quad (i)$$

If we put $h = -g, k = -f$ and $h^2 + k^2 - a^2 = c$ in equation (i) we get

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (iii)$$

which represents another form of general equation of circle. This equation is at times more handy than (i) or (ii).

In general equations (ii) and (iii) we observe that

(i) The equation of a circle is of second degree in x and y

(ii) The coefficients of x^2 and y^2 are equal

(iii) The term in xy is absent, i.e., the coefficient of xy is zero

These are the necessary conditions for an equation to represent a circle.

Let us now see whether these conditions are sufficient for an equation to represent a circle.

We take the equation of form $x^2 + y^2 + 2gx + 2fy + c = 0$ which can be written as

$$\begin{aligned}
 (x+g)^2 + (y+f)^2 - g^2 - f^2 + c &= 0 \\
 (x+g)^2 + (y+f)^2 &= g^2 + f^2 - c \\
 (x+g)^2 + (y+f)^2 &= \sqrt{(g^2 + f^2 - c)^2} \quad (iv)
 \end{aligned}$$

Three cases can arise

$$(i) \quad (g^2 + f^2 - c) > 0 \text{ say } g^2 + f^2 - c = k^2$$

then equation (iv) becomes $((x+g)^2 + (y+f)^2 = (\sqrt{k^2})^2$ which in fact is of the form (i) hence it does represent a circle with centre $(-g, -f)$ and radius $= k$

$$(ii) \quad (g^2 + f^2 - c) = 0$$

then equation (iv) becomes $(x+g)^2 + (y+f)^2 = 0$ which represents in reality only a point $(-g, -f)$

$$(iii) \quad (g^2 + f^2 - c) < 0$$

then expression the $(x+g)^2 + (y+f)^2$ of equation (iv) becomes a negative quantity as we know that the sum of two squares of real quantities can never be negative therefore, such a case is impossibility and will represent an imaginary circle

From the above discussion, we conclude that the three conditions given above are not sufficient. For a real circle that is circle with centre and radius that can be marked $(g^2 + f^2 - c)$ must also be positive quantity (other than zero) along with the three necessary conditions

Ex 35 What is represented by $3x^2 + 3y^2 - 12x - 30y + 100 = 0$?

In the given equation, we find

- (i) It is of second degree in x and y
- (ii) Coefficients of x^2 and y^2 are equal ($=3$)
- (iii) It contains no term in xy

Hence all the necessary conditions for representing a circle get satisfied by the equation

Let us examine whether $g^2 + f^2 - c > 0$

$$3x^2 + 3y^2 - 12x - 30y + 100 = 0$$

$$\text{or} \quad x^2 + y^2 - 4x - 10y + \frac{100}{3} = 0$$

$$\text{or} \quad (x^2 - 4x + 4) + (y^2 - 10y + 25) - 25 - 4 + \frac{100}{3} = 0$$

$$\text{or} \quad (x-2)^2 + (y-5)^2 = \left(25 + 4 - \frac{100}{3}\right)$$

i.e. $(g^2 + f^2 - c) = \left(29 - \frac{100}{3}\right) < 0$, hence the given equation is an

imaginary circle.

Ex 36 Find the equation of circle with centre at $(-2, 3)$ and radius equal to 2

Here $h = -2$, $k = 3$ and $a = 2$

Equation of circle $\left(x + \frac{2}{1}\right)^2 + \left(y - \frac{3}{1}\right)^2 = 4$

or $x^2 + y^2 + \frac{4}{3}x + \frac{3}{2}y - \frac{431}{144} = 0$

or $144x^2 + 144y^2 + 192x + 216y - 431 = 0$ is the required equation

Ex 37 Find the centre and radius of circle

$$x^2 + y^2 + 6x - 4y - 3 = 0$$

We must put the given equation in the general form

$$(x - h)^2 + (y - k)^2 = a^2 \quad x^2 + y^2 + 6x - 4y - 3 = 0$$

$$(x^2 + 6x) + (y^2 - 4y) = 3$$

We make each bracket a complete square by adding required quantity to each bracket and then adding the same quantity on R.H.S. of the equation

$$(x^2 + 6x) + (y^2 - 4y) = 3$$

or $(x^2 + 6x + 9) + (y^2 - 4y + 4) = 9 + 4 + 3$

$$(x + 3)^2 + (y - 2)^2 = 16$$

the centre of circle is at point $(-3, 2)$ and its radius is 4 units

Ex 38 The centre of a circle is at the point $(5, 6)$ and it passes through the origin find its equation

Let a be the radius of the circle since $(5, 6)$ is its centre

$$\text{equation should be } (x - 5)^2 + (y - 6)^2 = a^2$$

Again since it passes through the origin

$$(0 - 5)^2 + (0 - 6)^2 = a^2 \quad \text{or} \quad a^2 = 61$$

$$\text{Required equation of circle is } (x - 5)^2 + (y - 6)^2 = 61$$

or $x^2 + y^2 - 10x - 12y = 0$

Ex 39 Find the equation of a circle which passes through the three points $(0, 1)$, $(5, 1)$ and $(2, -3)$

First method. The general equation of circle with centre at (h, k) and radius = a units is represented by $(x - h)^2 + (y - k)^2 = a^2$

Since (h, k) is the centre of the circle naturally its distance from each of the three points must be equal to its radius

$$\text{Distance from } (0, 1) \quad (h - 0)^2 + (k - 1)^2 = a^2 \quad (i)$$

$$\text{Distance from } (5, 1) \quad (h - 5)^2 + (k - 1)^2 = a^2 \quad (ii)$$

$$\text{Distance from } (2, -3) \quad (h - 2)^2 + (k + 3)^2 = a^2 \quad (iii)$$

Solving these equations we obtain

$$h = \frac{5}{2}, k = -\frac{1}{4} \text{ and } a^2 = \frac{125}{16}$$

$$\text{Required equation of circle is } \left(x - \frac{5}{2}\right)^2 + \left(y + \frac{1}{4}\right)^2 = \frac{125}{16}$$

$$\text{i.e., } 2x^2 + 2y^2 - 10x + y - 3 = 0$$

Second method. Since point (0, 1), (5, 1) and (2, 3) are points on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, we have

$$1 + 2f + c = 0 \quad (i)$$

$$25 + 1 + 10g + 2f + c = 0 \quad (ii)$$

$$4 + 9 + 4g - 6f + c = 0 \quad (iii)$$

Solving these equations we obtain

$$g = -\frac{5}{2}, f = \frac{1}{4} \text{ and } c = -\frac{3}{2}$$

$$\text{Required equation is } x^2 + y^2 + 2\left(-\frac{5}{2}\right)x + 2\left(\frac{1}{4}\right)y - \frac{3}{2} = 0$$

$$\text{or, } 2x^2 + 2y^2 - 10x + y - 3 = 0$$

Ex 40 Find the equation of the circle of radius 2 units touching both the axes and having its centre in the first quadrant

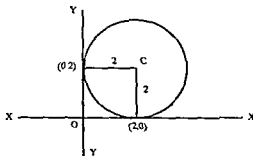


Fig 32

Equation of required circle will be $(x - 2)^2 + (y - 2)^2 = 4$ (Fig 32)

Ex 41. Find the co-ordinates of the centre and radius of the circle whose equation is $3x^2 + 3y^2 - 6x + 9y - 4 = 0$

The given equation can be re written as

$$x^2 + y^2 - 2x + 3y - \frac{4}{3} = 0$$

$$(x^2 - 2x + 1) + \left(y^2 + 3y + \frac{9}{4}\right) = \frac{55}{12}$$

$$(x-1)^2 + \left(y + \frac{3}{2}\right)^2 = \frac{55}{12}$$

which is the equation of a circle with centre at $\left(1, -\frac{3}{2}\right)$ and radius

$$= \sqrt{\frac{55}{12}} = 2.141 \text{ units}$$

EXERCISE 4.7

1. What is represented by

(i) $x^2 + y^2 + 8x - 18y - 97 = 0$

(ii) $4x^2 + y^2 - 8x + 16y - 5 = 0$

(iii) $x^2 + y^2 + 8x + 12y + 52 = 0$

(iv) $2x^2 + 2y^2 + 10x + 6y + 17 = 0$

(v) $9x^2 + 9y^2 + 30x + 12y + 38 = 0$

2. Find the equations of circles having centre at the origin and radii 1, 3, 7 and 10 units

3. What is the length of radius of the circle $x^2 + y^2 = \sqrt{16}$

4. Find the locus of a point which moves at a distance of 4 units from the point (5, 3)

5. Find the equation of the following circles

(i) centre (-4, -3) radius 5 units

(ii) centre (4, 2) radius 3 units

(iii) centre (-6, 5) radius 2 units

(iv) centre (0, 3) radius 5 units

(v) centre (-1, 2) radius 4 units

6. Obtain the centres and radii of the following circles

(i) $x^2 + y^2 = 25$ (ii) $x^2 + y^2 - 2x - 6y + 6 = 0$ (iii) $x^2 + y^2 + 6x + 8y + 25 = 0$

(iv) $x^2 + y^2 + 4x - 8y + 11 = 0$ (v) $x^2 + y^2 - 10x - 24y = 0$

(vi) $x^2 + y^2 - 6x + 8y = 0$

(vii) $4x^2 + 4y^2 - 8x - 21 = 0$ (viii) $6x^2 + 6y^2 - 16x + 20y = 15$

(ix) $7x^2 + 7y^2 - 4x - y = 3$ (x) $x^2 + y^2 - 2ax - 2by = 0$

7. The centre of a circle is at the point (-3, 4) and it passes through the origin. Find its equation

8. Show that the sum of the equations of any number of circles is the equation of a circle

9. Find the equations of circles passing through the following sets of points

- (i) $(1, 0)$ $(0, -6)$ and $(3, 4)$
 (ii) $(5, 7)$ $(8, 1)$ and $(1, 3)$
 (iii) $(2, 1)$, $(-2, 5)$ and $(-3, 2)$

4.16 Conic Sections

From the straight line and circle, we now pass on to the other curves which are obtained by cutting a cone in various ways. Hence they are called conic sections.

We have defined 'locus' under Section 4.15. We make use of it to define conic sections.

The conic section is the locus of a point which moves in such a way that its distance from another fixed point always remains in a constant ratio to its perpendicular distance from a fixed straight line.

- (i) The fixed point is called the *focus*.
 (ii) The fixed straight line is called the *directrix*.
 (iii) The straight line which passes through the *focus* and is perpendicular to the *directrix* is called the *axis*.
 (iv) The constant ratio is called the *eccentricity* and is symbolised by letter *e*.

The conic sections, represented by different types of curves, according as the value of eccentricity is unity, or not, are

- (a) If $e = 1$, the conic section is called a **Parabola**.
 (b) If $e > 1$, the conic section is called a **Hyperbola**.
 (c) If $e < 1$, the conic section is called an **Ellipse**.

1. 2.17. Parabola From the above definition of conic section, we can say that a parabola is a curve traced out by a point which moves in such a manner that its distance from a fixed point (called focus) is always equal to its distance from a fixed straight line (called directrix). Thus, $e = 1$ in the case of parabola.

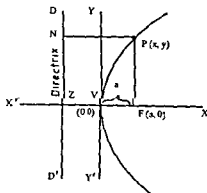


Fig. 33.

The above definition could be understood with the help of Fig. 33. Let *P* be the moving point at a particular position (x, y) . It moves in a manner that its distance from a fixed point $F(a, 0)$ is equal to its perpendicular distance from the directrix *DD'*. In other words, $PF = PN$. The path traced by point *P* will be Parabola depicted in Fig. 33.

Note that when point *P* reaches *V* $(0, 0)$ then *VF* should be

equal to ZV . Therefore, if $VF=a$ then ZV is also equal to a

Equation of a parabola (i)

Let P be the moving point at a particular position (x, y) (Fig. 34)

(ii) F be the focus with coordinates $(a, 0)$

(iii) DD' be the directrix. Draw FZ perpendicular from F on the directrix, Fig. 34

By the definition of parabola, since $VF = a$, $ZV = a$

Hence the equation of the directrix is $x = -a$ or $x + a = 0$

(iv) $V(0, 0)$ is called the vertex of parabola. Vertex is the intersection of the parabola and its axis

$$\text{In } \triangle PMF \quad (PF)^2 = (FM)^2 + (PM)^2$$

Now, $FM = VM - VF = x - a$, $PM = y$

$$(PF)^2 = (x - a)^2 + y^2 \quad (i)$$

Again, $NP = VM + ZV = (x + a)$ (ii)

By definition of parabola, $PF = NP$ or $(PF)^2 = (NP)^2$, substituting their values from (i) and (ii), we get

$$\begin{aligned} (x - a)^2 + y^2 &= (x + a)^2 \\ x^2 - 2ax + a^2 + y^2 &= x^2 + 2ax + a^2 \\ \text{or, } y^2 &= 4ax \end{aligned}$$

which is the general equation of a parabola with origin being the vertex, the focus at $(a, 0)$ and the directrix $x + a = 0$

Note. In order to obtain the above equation of parabola, the axis of parabola and its tangent at the vertex have been taken as the axes of x and y respectively

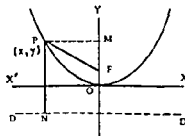


Fig. 34

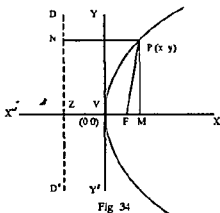


Fig. 34

The above equation has been derived with its concavity towards the positive side of the X -axis. Equation on Parabola with concavity towards negative side of X -axis will be $y^2 = -4ax$

Let us now obtain the equation of parabola with its concavity towards the positive side of Y -axis

In this case :

(i) F is $(0, a)$

(ii) Equation of directrix is

$$y = -a \text{ or } y + a = 0$$

By definition :

$$PF = PN$$

$$(PF)^2 = (PN)^2$$

$$(y - a)^2 + x^2 = (y + a)^2$$

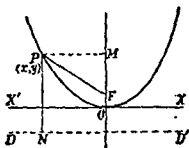
$$y^2 - 2ay + a^2 + x^2 = y^2 + 2ay + a^2$$

$$x^2 = 4ay$$

is the required equation

Similarly, the equations of parabola with its concavity towards the negative side of Y -axis will be $x^2 = -4ay$.

We can sum up the above derivations in the following way



(Fig. 35)

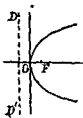


Fig. 36

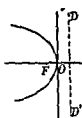


Fig. 37



Fig. 38

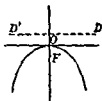


Fig. 39

Vertex	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
Focus	$(a, 0)$	$(-a, 0)$	$(0, a)$	$(0, -a)$
Directrix	$x + a = 0$	$x - a = 0$	$y + a = 0$	$y - a = 0$
Equation	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$

Latus rectum. LL' is called the latus rectum of the parabola (Fig. 40). It is the straight line which passes through the focus and is at right angles to the axis of the parabola.

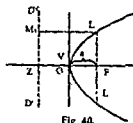


Fig. 40.

Assume that $FL = 1$ units ; then by definition of parabola :

$$(FL)^2 = (LM)^2$$

$$LM = VF + VZ = a + a = 2a$$

$$(LM)^2 = 4a^2$$

$$(FL)^2 = 4a^2$$

$$\therefore FL = 2a$$

Hence, the length of the latus rectum is $4a$ units. Just as all the circles possess the same shape, only their sizes differ according to their radii, similarly, all the parabolas have the same shape, they differ in their sizes according to their length of the latus rectum.

Shifting of vertex from origin

So far we have considered the parabola with its vertex at the origin. Assume that the vertex shifts to (h, k) and axis remains parallel to axis of X . By shifting the origin to the point

(h, k) as shown in Fig. 41, the equation of such parabola would be

$$(y-k)^2 = 4a(x-h)$$

while the equation to the corresponding directrix will be

$$x = h - a$$

Suppose vertex of the parabola $y^2 = 4ax$ ($a = 4$) gets shifted to $(2, 3)$. Therefore, all the points will also shift by $(2, 3)$. The focus will lie on the axis of the parabola at a distance $= 4$ units. Therefore, the coordinates of the focus will be $(2 + 4, 3)$ or $(6, 3)$. Directrix will be parallel to axis of Y also at a distance $a = 4$ units. Hence its equation will be

$$x = -4 + 2$$

$$x + 2 = 0$$

We tabulate below all such four possible cases of shifting of vertex against the general equations of parabola with vertex at the origin.

Parabola with $V(0, 0)$			Parabola with $V(h, k)$		
Equation	Focus	Directrix	Equation	Focus	Directrix
1. $y^2 = 4ax$	$(a, 0)$	$x = -a$	$(y-k)^2 = 4a(x-h)$	$(a+h, k)$	$x = h-a$
2. $y^2 = -4ax$	$(-a, 0)$	$x = a$	$(y-k)^2 = -4a(x-h)$	$(h-a, k)$	$x = h+a$
3. $x^2 = 4ay$	$(0, a)$	$y = -a$	$(x-h)^2 = 4a(y-k)$	$(h, k+a)$	$y = k-a$
4. $x^2 = -4ay$	$(0, -a)$	$y = a$	$(x-h)^2 = -4a(y-k)$	$(h, k-a)$	$y = k+a$

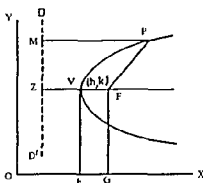


Fig. 41

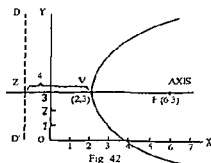


Fig. 42

Ex 42 Find the focus and directrix of the parabola

(i) $y^2 = 16x$

(ii) $x^2 = -8y$

(i) Comparing the given equation with $y^2 = 4ax$ we find

$a = 4$ ($y^2 = 4 \times 4 \times x$)

Hence directrix is at a distance equal to 4 units on other side of the vertex of parabola and parallel to the axis of Y . In other words, its equation is $x = -4$ or $x + 4 = 0$ (or $x + a = 0$ of the general form). Coordinates of focus = $(4, 0)$ (Fig 43)

(ii) Comparing it with $x^2 = 4ay$ we find that

$a = -2$

in the case [$x^2 = 4(-2)y$]

Hence the directrix is parallel to the axis of X situated at a distance of 2 units on the other side of the curve, i.e., directrix $y = 2$

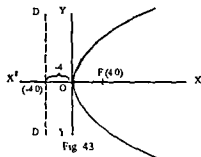


Fig 43

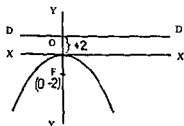


Fig 44

or $y - 2 = 0$ ($y - a = 0$ is the general form)

Coordinates of the focus and the directrix of the parabola

(i) $y^2 = 4x + 4y$

(ii) $x^2 + 4x + 2y = 0$

(i) $y^2 = 4x + 4y$

Changing this equation to the form

$y^2 = 4ax$ or $y^2 = -4ax$,

$y^2 - 4y = 4x$

$(y^2 - 4y + 4) = 4x + 4$

$(y-2)^2 = 4(x+1)$

which is of the form $y^2 = 4ax$. Thus vertex is at $(-1, 2)$ and focus will be at $(0, 2)$

$a = 1$

Directrix will be parallel to the Y axis

$x = -2$ or $x + 2 = 0$ (Fig 45)

(ii) $x^2 + 4x + 2y = 0$ To put it in the form

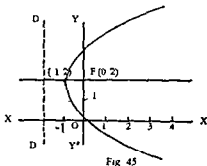


Fig 45

$$x^2 = 4ay \text{ or } x^2 = -4ay$$

$$x^2 + 4x = -2y$$

$$x^2 + 4x + 4 = -2y + 4$$

$$(x + 2)^2 = 2(y - 2)$$

which is of the form $x^2 = -4ay$

Vertex is at $(-2, +2)$ and focus will be at $(0, 2)$ Directrix will be parallel to y -axis $y + 2 = 0$

EXERCISE 4.8

Trace the following curves and then find their vertex, focus and directrix

1. $x^2 = -16y$

2. $x^2 + 6x + y = -8y + 16$

3. $y^2 - 4y + 4x = 0$

4. $x^2 - 8x + 2y + 7 = 0$

5. $x^2 + 4x + 2y = 0$

6. $9x^2 - 6x + 36y + 19 = 0$

7. $x^2 - 8x + 2y + 7 = 0$

8. $3x^2 - 4y + 6x - 5 = 0$

9. $x^2 + 4x + 3y = 0$

10. $80y^2 + 384x + 120y + 301 = 0$

11. Sketch the illustrative graphs of the following economic relationships mentioning their characteristic properties

(i) Fixed cost as a function of output

(ii) Average fixed cost as a function of output,

(iii) Total utility obtained from a commodity as a function of its quantity consumed under the assumption that the marginal utility diminishes and actually becomes negative.

4.18 Ellipse. The equation of the ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

$$b^2x^2 + a^2y^2 = a^2b^2$$

If traced the curve will be egg-shaped. Its centre will be at $(0, 0)$ and it will be symmetric with respect to both axes (Fig. 46).

It is to be noted that the equation of the ellipse looks similar to the circle except that the coefficients of x^2 and y^2 are not equal

If the centre lies at (h, k) the equation of the ellipse will be

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 0$$

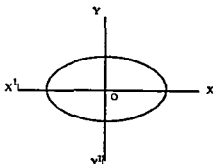


Fig. 46.

4.19. Hyperbola. What happens when the sign between $\frac{x^2}{a^2}$ and $\frac{y^2}{b^2}$ is negative instead of positive as in the case of ellipse?

That is if $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $\frac{x^2}{a^2} - 1 = \frac{y^2}{b^2}$ or $\frac{y^2}{b^2} = \frac{x^2 - a^2}{a^2}$

$$y^2 = \frac{b^2}{a^2}(x^2 - a^2) \text{ or } y = \pm \frac{b}{a}\sqrt{x^2 - a^2}$$

Let us trace the curve,

(i) when $x = \pm a$, $y = 0$

(ii) $x > a$, y values will be equal and opposite in sign. Same values will hold whether x values are negative or positive

Assume that $a = b = 2$, then $y = \pm \sqrt{x^2 - 4}$ so that if $x < 2$, value of y does not exist

Values of x		Values of y	
2	-2	0	0
3	-3	+5	-5
4	-4	+12	-12
5	-5	+21	-21

and the curve will be as in Fig 47 and is termed as hyperbola

Origin $(0, 0)$ is the centre of the hyperbola

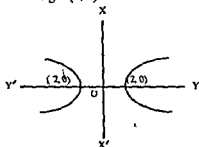


Fig. 47

If the centre is (h, k) the equation of the hyperbola will be

$$\frac{(x+h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Rectangular or equilateral hyperbola. A rectangular hyperbola is a curve such that for every point on it, the product of the distance of the point from two fixed perpendicular lines is a positive constant x^2 . The fixed lines are called the *asymptotes*

and their point of intersection the centre of the rectangular hyperbola

Take the simplest case when the origin is the centre of the curve, i.e., when the axes are the asymptotes, as in Fig 48

Let P be any point on the curve, then the distances of P from the asymptotes are $NP(x)$ and $MP(y)$, their product xy is the area $NPMO$, which is constant $= a^2$. Hence, the equation of the rectangular hyperbola is

$$xy = \alpha^2$$

$$\text{or } x = \frac{\alpha^2}{y} \quad \text{or } y = \frac{\alpha^2}{x}$$

Hence x goes on decreasing as y goes on increasing and *vice versa*

If the centre of the curve is at (h, k) the equation will be $(x - h)(y - k) = \alpha^2$

$$y = \frac{\alpha^2}{x - h} + k = \frac{\alpha^2 + kx - kh}{x - h}$$

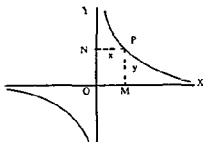


Fig 48

4.20 Variables and Functions*

Mathematical problems involve two kinds of quantities—Constants and Variables. The quantity which retains the same value all through the mathematical operations is called a constant. It has become customary to denote constant by earlier letters of the alphabet a, b, c .

A quantity which can assume different values at different times or in different cases is called a variable and thus may be defined as an unspecified number. Variables are symbolized by letters such as x, y or z . Suppose that the variable under consideration is called y and that y can take on any positive value from zero to one hundred. Thus $1, 5, 18\frac{1}{2}, 75\frac{3}{4}$ or any such arbitrary numbers are the values of y . So y is an unspecified number, a number which may assume any value between zero to one hundred.

Particular numbers for which the variable can stand are called the values of the variable and the whole set of possible values makes up the range of the variable. In the above example, range of the variable y is 1 to 100. At times, the range of variable may be unlimited.

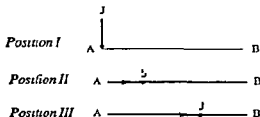


Fig 49

Continuous and Discrete Variables Let AB be a straight line and let a point start from one position A and move steadily towards B along the line AB (Fig 49). As the point J moves from A to B , the step AJ steadily increases (AJ is greater in position III than in II). AJ is a continuously varying

* Students should also refer to the appendix on Functions and Curves in Economics given at the end of this chapter.

magnitude during the motion of J . If we represent the magnitude of the the step AJ by x which is steadily increasing x is then a continuously varying number or more briefly stated x is a continuous variable. Let us take another example. Assume x to be continuous and again with a range from 1 to 100 x being assumed to be a continuous variable, it can assume the following values $1 \frac{1}{50}$ $1 \frac{9}{20}$ $99 \frac{1}{99}$ or any such real numbered values without gaps from 1 to 100. Thus a variable is said to be a continuous variable if its range is either the whole set of real numbers or any particular subset of that set. On the other hand a discontinuous or discrete variable has a range that is neither the whole set of real numbers nor any interval of that set.

For example the set of positive integers (1 2 3 ... 100) provides a range that is discontinuous. Now if it is assumed that variable can take all such values but no such value between 1 and 2 or 2 and 3 and so on then there is no continuity but there are gaps hence such variable is called a discrete variable.

In Economics all the empirical magnitudes are considered as variables—continuous or discrete and symbolised by different letters. For example magnitudes as price income profit production cost labour hours etc each of which can assume many values are called variables in economics.

Functions We are in fact not interested in variables for their own sake. What we need is the relationship between the variables. In economic theory we relate one attribute or variable to the other.

Let us assume two variables x (denoting quantity demanded) and y (denoting its price). It will usually happen that when one of the variables (say y i.e. price) is given a series of hypothetical values the other variable (x i.e. demand) will take a series of definite values one for each that the first is supposed to have been given. The second variable is then said to be a function of the first or to be a variable dependent on the first. The first variable is called the independent variable. In other words the variable y whose numerical value is arbitrarily assigned is called the independent variable. The variable x whose numerical value is determined after a value has been assigned to y is the dependent variable.

When there are more than two variables it may happen that when definite values are assigned to all but one of them the value of that one variable becomes determinate. This one variable is then said to be a function of or dependent on the other variables which are called the independent variables of the problem.

Generally therefore a variable y is said to be a function of another variable x when to every value of x corresponds a definite value of y . Also variable y is said to be a function of two or more variables $x, z, \dots$ when to each set of values of the variables $x, z, \dots$ corresponds a definite value of y .

It is clear from the above discussion that the notion of a function follows upon the idea of a relationship amongst the dependent and independent variables

Notation for Function. Every function is defined by a specific equation and can be represented by a graph. When we want to indicate that y is a function of x without fixing the form (equation) of the function, we denote it by enclosing the independent variable in a bracket and prefixing a letter, thus $y = f(x)$ or $y = F(x)$ or $y = (x)$. The letters f , F , are functional symbols, not multipliers. The symbol $f(x)$ or $F(x)$ or (x) must be taken as a whole, which means that y depends upon x in some unspecified way. Many a time, we write this functional form as $y = y(x)$ which is just a matter of convenience but it means the same as explained above.

Suppose we want to express the demand for a commodity in a functional form. Let p denote the price of the commodity and q its quantity demanded. Since the quantity demanded in the market depends in a same way (which we do not know) upon the price of the commodity, the demand function can be written as $q = f(p)$. Likewise we can write the consumption function in the notation form $C = f(y)$, where C denotes the total consumption of the household and y its disposable income. The notation $C = f(y)$ is taken to mean that consumption depends (or is function of) on the level of income.

Many other examples can be taken from economic theory, e.g., supply function of a commodity, production function of a firm or industry and cost function of the output of a firm—all can be expressed in the functional notation once we know the dependent and independent variables of the problem.

Types of Functions. The functional notation $y = f(x)$ only states that there exists a certain functional relationship between x and y , but what is needed is to state the exact way in which y depends on x . This is done by means of an equation. Assume that our function, that is, $y = f(x)$ is in the form of equation $y = f(x) = 4x + 2$. This equation states the exact functional relationship (which is linear) between the two variables. We can find the series of values of dependent variables y by assigning different values to independent variable x from the given linear function $y = f(x) = 4x + 2$. In other words, we are able to solve this equation for the dependent variable in terms of any value assigned to independent variable. Here we say that y is an explicit function of x . Thus

$$\begin{array}{lll} y = 5x^2 + 6 & s = \log t & p = \frac{4}{v} \\ y = f(x) & s = f(t) & p = f(v) \end{array}$$

are the equations which give y , s and p explicitly as a function of x , t and v respectively and hence are called explicit functions. On the other hand, when the equation could not be solved (for the dependent variable in terms of any

value assigned to the independent variable), the dependent variable is called an implicit function of the independent variable, or is said to be given implicitly.

The Implicit Relationship It is symbolised in a form $f(x, y) = 0$, of which particular form may be : $4xy + 2x + 3y + 8 = 0$. This equation when solved for y in terms of x gives $y = -\left(\frac{2x+8}{4x+3}\right)$ and hence now y becomes an explicit function of x . But one could also find x from the equation $4xy + 2x + 3y + 8 = 0$ in terms of y ; $x = -\left(\frac{3y+8}{4y+2}\right)$ and thus x becomes an explicit function of y .

As such, from the given implicit function either variable can be written as an explicit function of the other. In such cases each of the two functions is called the *Inverse* (function) of the other.

More generally, the equation $y = f(x)$, which defines y explicitly as a function of x , also defines x implicitly as a function of y , the two functions thus defined by the one equation are said to be *inverse* to each other.

For example, the equation $y = x^3$ which can also be put as $x = \sqrt[3]{y}$ defines two functions $y = f(x) = x^3$ and $x = f(y) = \sqrt[3]{y}$ which are inverse to each other. One is the cube and the other cube root.

Usually we employ f^{-1} as a symbol of the function inverse to that denoted by symbol f . Thus if $y = f(x)$, then we may write $x = f^{-1}(y)$ as the inverse of $y = f(x)$.

Also when a function is given explicitly (or implicitly) by an equation wherein one and only one value of the dependent variable corresponds to each given value of the independent variable, the function is called single valued function. Against this, functions in which each value of independent variable gives rise to more than one value for the dependent variable, are called multivalued functions.

For example : $y = 5x + 8$ is a single valued explicit function,
and $y = y \pm \sqrt{x}$, is a multivalued explicit function.

Value of Function at a Point : The functional symbol : $y = f(x)$ also affords a convenient means of indicating the substitution for x and thus finding the value of the function at that substituted value. The result of substituting " a " for x is denoted $y = f(a)$ or the result of substituting 4 for x by $y = f(4)$.

For example if : $y = f(x) = 5x + 8$; then
 $y = f(a) = 5a + 8$;
 $y = f(8) = 5(8) + 8 = 48$;
 $y = f(-1) = 5(-1) + 8 = 3$.

4.21. Limits

In using the limit concept we make statements such as the following :

- (a) $f(x)$ approaches 20 as x approaches 2
 (b) $f(x)$ approaches -10 as x approaches 0
 (c) $f(x)$ does not approach a limit as x approaches 3

Let us assume a function

$f(x) = \frac{3x}{x+1}$, it appears quite reasonable to say that $f(x)$ approaches 2 as

x approaches 2. This statement can be justified intuitively by the observation that when x is very nearly 2, $3x$ is very nearly 6, $x+1$ is very nearly 3 and the quotient is very nearly 2. But when we pass from such a simple case to general one, we must have a clear and explicit formal definition of what is meant by a statement that $f(x)$ approaches A (a constant) as x approaches x_0 (a particular constant).

In mathematics, for the sake of convenience we use brief notations for expressing the statement " $f(x)$ approaches A as x approaches x_0 " in the form of following notations

$$(i) f(x) \rightarrow A \quad \text{as} \quad x \rightarrow x_0$$

$$(ii) \lim_{x \rightarrow x_0} f(x) = A, \text{ which is read as 'limit of } f(x), \text{ as } x \text{ approaches } x_0, \text{ is } A'$$

Definition of limit When it is possible to make the independent variable of a given function so nearly to a definite number a that the function will differ from another definite number A by as little as we please, that difference remaining as small as we please when the independent variable is taken still nearer to a , then A is called the limit of the function.

This definition can be put in other form as follows

The function $f(x)$ is said to have the limit A , A being a definite number, as the independent variable x approaches a , if to every arbitrarily chosen positive number ϵ , however small, there corresponds a positive number such that $|A - f(x)| < \epsilon$ for every value of x and for which $0 < |x - a| \leq \eta$.

These definitions can be put in the form of notation.

$$\lim_{x \rightarrow a} f(x) = A$$

Observe that in the notation we put arrow notation below the word limit ($\lim$) by which we mean that "as x approaches a ". We say nothing about what happens when x is equal to a . Therefore, it should be clear that the limit of $f(x)$ as $x \rightarrow a$ is not necessarily the value of the function when $x = a$. It is for this reason that we mention clearly in the definition $0 < |x - a| \leq \eta$ to indicate the fact that $|A - f(x)| < \epsilon$ for every value of x in the interval of $(a - \eta)$ and $(a + \eta)$, except $x = a$.

Thus we notice that a limiting value of $f(x)$ is that quantity from which we may make $f(x)$ differ as little as we please by making x approach nearer and

nearer in magnitude to some particular value without actually becoming equal to it

Let us take a few examples to make the definition more clear

(i) Suppose $f(x) = x^2$, and $\lim_{x \rightarrow 3} x^2 = 9$, i.e., $f(x) \rightarrow 9$ as $x \rightarrow 3$

According to the definition, this means that for any positive number ϵ , howsoever small, a value of η can always be found, such that

$$|x^2 - 9| < \epsilon \text{ (since limit of function is 9) when}$$

$$0 < |x - 3| \leq \eta \text{ (.. limit is 9 only when } x \text{ approaches 3)}$$

(ii) Suppose $f(x) = \frac{x^2 - 9}{x - 3}$ and $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$

According to the definition this means that for any positive number ϵ , howsoever small, a value of η can always be found, such that

$$\left| \frac{x^2 - 9}{x - 3} - 6 \right| < \epsilon \text{ (limit of the function is 6), when}$$

$$0 < |x - 3| \leq \eta \text{ (.. limit is 6 only when } x \text{ approaches 3)}$$

(iii) $\lim_{x \rightarrow 2} (x^3 + 10) = 18$ means that for any positive number, howsoever small, a value for η can always be found such that

$$|x^3 + 10 - 18| < \epsilon \text{ when } 0 < |x - 2| < \eta$$

Let us now obtain ϵ and η relationship in the above example

$$|x^3 + 10 - 18| = |x^3 - 8| = |(x - 2)(x^2 + 2x + 4)|$$

We are interested in having x nearer to 2, so let us confine our attention to values of x between 1 and 3. For such values of x , we see that

$$|x^2 + 2x + 4| \leq 9 + 6 + 4 = 19$$

$$\text{Hence } |(x^3 + 10) - 18| \leq 19(x - 2) \text{ only and only if } 1 < x < 3$$

$$\text{On the other hand, we want to make } |(x^3 + 10) - 18| < \epsilon$$

$$\text{This can be done if we make } 19|x - 2| < \epsilon$$

$$\text{or, } |x - 2| < \frac{\epsilon}{19}$$

provided we also insist that $1 \leq x \leq 3$. We, therefore, take η to be the smaller of the two numbers $\frac{\epsilon}{19}$ and 1 and require

$$0 < |x - 2| < \eta$$

Then the conditions (i) $19|x - 2| < \epsilon$, and

(ii) $1 \leq x \leq 3$ are satisfied, and thus we have proved that $f(x) \rightarrow 18$ as $x \rightarrow 2$

4.22 Basic Theorems used in Finding Limits. In actual practice we do not always deal with limits by working through the details of

inequalities as given in the above illustrations. There are certain theorems about limits which are of great convenience, they relate to sum, product and quotients of functions.

Theorem I The limit of the sum of any number of functions is equal to the sum of the limits of functions. In notation

if $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} \psi(x) = B$ then this theorem

states that $\lim_{x \rightarrow a} [f(x) + \psi(x)] = A + B$

Theorem II The limit of the product of functions is equal to the product of the limits of the functions

If $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} \psi(x) = B$ then

$$\lim_{x \rightarrow a} [f(x) \psi(x)] = AB$$

Theorem III The limit of the quotient of two functions is equal to the quotient of the limits of the functions, provided the limit of the denominator is not zero

If $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} \psi(x) = B$ if $B \neq 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{\psi(x)} = \frac{A}{B}$$

To understand the proofs of the above theorems, some ability in reasoning with inequalities is needed. Let us, therefore, not bother ourselves with the proofs at this stage.

Ex. 44 Find $\lim_{x \rightarrow 2} \frac{3x^3 - 2x + 7}{x^2 + 5x}$

We make use of the above theorems in finding the limit of the given function

(i) $3x^2 \rightarrow 3 \cdot 2 \cdot 2 = 12$ as $x \rightarrow 2$

This is by an application of Theorem II

likewise $-2x \rightarrow -2 \cdot 2 = -4$

$x^3 \rightarrow 2 \cdot 2 \cdot 2 = 8$

$5x \rightarrow 5 \cdot 2 = 10$ as $x \rightarrow 2$

(ii) The rule of sums shows that,

$3x^2 - 2x + 7 \rightarrow 12 - 4 + 7 = 15$

$x^3 + 5x \rightarrow 8 + 10 = 18$ as $x \rightarrow 2$

(iii) The theorem for quotient gives,

$$\lim_{x \rightarrow 2} \frac{3x^2 - 2x + 7}{x^3 + 5x} = \frac{15}{18} = \frac{5}{6}$$

Ex. 45 Find $\lim_{x \rightarrow 3} \frac{x^3 - x^2 - 9x + 9}{x^2 - x - 6}$

Here, we find that the numerator as well as the denominator of the function happens to be zero when $x=3$. This also indicates that $(x-3)$ is one of the factors of both numerator and denominator. Before attempting to find the limit of the function as x approaches 3, this factor is cancelled out

$$\begin{aligned} \frac{x^3 - x^2 - 9x + 9}{x^2 - x - 6} &= \frac{(x-3)(x+3)(x-1)}{(x-3)(x+2)} \\ &= \frac{(x+3)(x-1)}{x+2} \quad \therefore (x-3) = 0 \end{aligned}$$

Since we require $x \neq 3$ in considering the limit as $x \rightarrow 3$

$$\therefore \lim_{x \rightarrow 3} \frac{x^3 - x^2 - 9x + 9}{x^2 - x - 6} = \lim_{x \rightarrow 3} \frac{(x+3)(x-1)}{x+2} = \frac{12}{5}$$

Ex. 46. Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

$$\frac{x^2 - 9}{x - 3} = \frac{(x-3)(x+3)}{x-3} = x+3$$

We require $x \neq 3$ in considering the limit as $x \rightarrow 3$

$$\therefore \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x+3) = 6$$

Ex. 47. Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$ and $\lim_{x \rightarrow \infty} \frac{\sqrt{x+1} - 1}{x}$

Here, we find that in both cases the function turns out to be indeterminate as x takes its limit value. Hence to seek the limits of the given functions we must divide out x from the denominator of both the functions

$$\begin{aligned} (i) \quad \frac{\sqrt{x+1} - 1}{x} &= \frac{(\sqrt{x+1} - 1)(\sqrt{x+1} + 1)}{x(\sqrt{x+1} + 1)} \\ &= \frac{(x+1) - 1}{x(\sqrt{x+1} + 1)} = \frac{1}{\sqrt{x+1} + 1} \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+1} + 1} = \frac{1}{2}$$

$$(ii) \quad \lim_{x \rightarrow \infty} \frac{\sqrt{x+1} - 1}{x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x+1} + 1} = 0$$

Ex. 48 Find $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} = \frac{(\sqrt{1+x} - \sqrt{1-x})(\sqrt{1+x} + \sqrt{1-x})}{x(\sqrt{1+x} + \sqrt{1-x})}$$

$$\begin{aligned}
 &= \frac{(1+x) - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} \\
 &= \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \\
 \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} &= \lim_{x \rightarrow 0} \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \\
 &= \frac{2}{2} = 1
 \end{aligned}$$

Ex 49 $\lim_{x \rightarrow -1} \frac{x^2 + 4x + 3}{x^2 - 7x - 8}$

$$\frac{x^2 + 4x + 3}{x^2 - 7x - 8} = \frac{(x+1)(x+3)}{(x+1)(x-8)} = \frac{x+3}{x-8}$$

$$\lim_{x \rightarrow -1} \frac{x^2 + 4x + 3}{x^2 - 7x - 8} = \lim_{x \rightarrow -1} \frac{x+3}{x-8} = -\frac{2}{9}$$

Ex 50 Find the limits of the following functions

$$\lim_{x \rightarrow 0} \frac{\sqrt{2-x} - \sqrt{2+x}}{x}$$

Relationalising the numerator, we have

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{(\sqrt{2-x} - \sqrt{2+x})(\sqrt{2-x} + \sqrt{2+x})}{x(\sqrt{2-x} + \sqrt{2+x})} \\
 &= \lim_{x \rightarrow 0} \frac{(2-x) - (2+x)}{x(\sqrt{2-x} + \sqrt{2+x})} \\
 &= \lim_{x \rightarrow 0} \frac{-2x}{x(\sqrt{2-x} + \sqrt{2+x})} \\
 &= \lim_{x \rightarrow 0} \frac{-2}{\sqrt{2-x} + \sqrt{2+x}} \quad (x \neq 0) \\
 &= \frac{-2}{\sqrt{2-0} + \sqrt{2+0}} \\
 &= -\frac{2}{2\sqrt{2}} = -\frac{1}{\sqrt{2}}
 \end{aligned}$$

Ex 51 $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 2x - 3}$

$$\begin{aligned}
 &= \lim_{x \rightarrow 3} \frac{(x-3)(x+2)}{(x-3)(x+1)} \quad (x \neq 3) \\
 &= \lim_{x \rightarrow 3} \frac{x+2}{x+1}
 \end{aligned}$$

$$= \frac{3+2}{3+1} = \frac{5}{4}$$

Two standard limits :

$$(i) \quad \lim_{x \rightarrow 0} \left(1 + \frac{1}{x}\right)^x = e = 2.718 \text{ (approximately)}$$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$(ii) \quad \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

EXERCISE 4.10

Verify the limits of the following functions

$$1. \quad \lim_{x \rightarrow -1} (x^2 - 3x^2 - 2x + 4) = 2$$

$$2. \quad \lim_{x \rightarrow 1} (4x^2 - 5x^2 + x - 2) = -2$$

$$3. \quad \lim_{x \rightarrow 2} \frac{(2x-1)^2}{x+1} = 3$$

$$4. \quad \lim_{x \rightarrow 2} \frac{3x^2 - 4x + 7}{3x - 5} = 11$$

$$5. \quad \lim_{x \rightarrow 0} \frac{2x^2 - 5x + 6}{x^2 + x - 3} = 2$$

$$6. \quad \lim_{x \rightarrow 0} \frac{2x^2 + 3}{3x^2 + x + 9} = 0$$

$$7. \quad \lim_{x \rightarrow \infty} \frac{3x^2 + x - 9}{2x^2 + 3} = \infty$$

$$8. \quad \lim_{x \rightarrow \infty} (2x^2 + 1)^{1/2} = 0$$

$$9. \quad \lim_{x \rightarrow \infty} \frac{5x^2 + 4x^4}{5x^2 - 4x^4} = 1$$

$$10. \quad \lim_{x \rightarrow \infty} \frac{2x^2 - 5}{3x^2 - 17x + 4} = \frac{2}{3}$$

$$11. \quad \lim_{x \rightarrow -2} \frac{x^4 - 2x^3 + 4x - 20}{x^3 + 5x^2 + 3x - 4} = 2$$

$$12. \quad \lim_{x \rightarrow 1} \frac{(x^2 + 2x - 3)^2}{(x-1)(x+5)^2} = 0$$

$$13. \quad \lim_{x \rightarrow 2} \frac{x^4 - 3x^2 - 4}{x^2 + x - 6} = 4$$

$$14. \quad \lim_{x \rightarrow a} \frac{x-a}{x^2 - a^2} = \frac{1}{2a}$$

$$15. \quad \lim_{x \rightarrow \infty} \frac{x^2 + a^2}{x^4 - a^4} = -\frac{3}{4a}$$

$$16. \quad \lim_{x \rightarrow \infty} \frac{2x^2 - 5x + 6}{x^2 + x - 3} = 2$$

$$17. \quad \lim_{x \rightarrow \infty} \frac{x^2 - 6x + 8}{3x^2 + 2x^2 - 5x + 4} = \frac{1}{2}$$

$$18. \quad \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{x^3} = 1$$

$$19. \quad \lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}} = -8$$

$$20. \quad \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x^2} \sqrt{1-x^2}} = \infty$$

$$21. \quad \lim_{x \rightarrow 0} \frac{\sqrt{x+a} - \sqrt{x}}{\sqrt{a}} = \frac{2}{2\sqrt{x}}$$

$$22. \quad \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} = 1$$

$$23. \quad \lim_{x \rightarrow \infty} \frac{2x-1}{x+1} = \lim_{x \rightarrow \infty} \frac{2 - \frac{1}{x}}{1 + \frac{1}{x}} = 2$$

4.23 Continuity of a Function

The function $y = f(x)$ is continuous at $x = a$ if

- (i) $f(a)$ exists and is finite.
- (ii) $\lim_{x \rightarrow a} f(x)$ exists and
- (iii) $\lim_{x \rightarrow a} f(x) = f(a)$

Continuity represents the agreement between limit and value where both exist i.e. $f(x)$ assumes a definite value $f(a)$ at the point and that $f(x)$ tends to the same value $f(a)$ as x approaches a from either side. Hence the curve has no gaps or jumps at $x=a$.

Ex 52 Examine whether the function $y = \frac{(1-x^2)}{1-x}$ is a continuous one

The given function is not defined at $x=1$ but

$$\lim_{x \rightarrow 1} \frac{1-x^2}{1-x} = 2$$

hence the function is not continuous at $x=1$

Ex 53 $y = x(x+1)$ for $x > 0$ examine the continuity of this function

Now at $x=2$ $y=6$ and $\lim_{x \rightarrow 2} x(x+1) = 6$ hence the value and limit of the function at $x=2$ are the same

Thus the function is continuous at point $x=2$

Therefore, we say that a function $y=f(x)$ is continuous over a range of value of x if it is continuous at every point of the range. Obviously function $y=x(x+1)$ is continuous at all points. Conversely, if a function is continuous over a range then it will be continuous at every point of the range i.e., there are no gaps in the curve within that range. It does not mean that the curve is necessarily smooth or it has no sharp points. The only condition is that one should be able to draw the curve without lifting the pencil.

EXERCISE 4.11

Examine the continuity of the following functions

- 1 $y = x - 1$ at $x = 1$
- 2 $y = 1 - (x-1)^2$, $0 < x < 2$
- 3 $y = \frac{1-x}{1+x}$
- 4 $y = (x-1)^2(x-2)$
- 5 Show that the function

$$f(x) = \frac{2-x}{2-\sqrt{2}x} \quad x \neq 2$$

$$= 3 \quad x = 2$$

is not continuous at the point $x = 2$

6 Show that the function

$$f(x) = \frac{\sqrt{x-2}}{x-4} \quad x \neq 4$$

and $\quad = \frac{1}{4} \quad \text{at } x = 4$

is continuous at the point $x=4$

4.24 Convergence and Divergence of a Series

Suppose there is a sequence of terms $U_1, U_2, U_3, \dots, U_n$ and if we want to add them i.e. we want S_n

$$S_n = U_1 + U_2 + U_3 + \dots + U_n$$

Now the question is What happens to the sum S_n when $n \rightarrow \infty$

If $S_n = \sum_{r=1}^n U_r \rightarrow$ to some limit (S) as $n \rightarrow \infty$ then the infinite series $\sum U_r$ is convergent and it converges to S otherwise they form an infinite series)

Ex 54 Show that an arithmetic series is divergent

An arithmetic series is

$$a + (a + d) + (a + 2d) + \dots + (a + (n-1)d) +$$

where d is common difference and a is the first term

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

Here $S_n \rightarrow \infty$ as $n \rightarrow \infty$, thus arithmetic series is not convergent and does not possess sum to infinity

Ex 55 Discuss the convergence of a geometric series

A general geometrical series with common ratio r is

$$a + ar + ar^2 + ar^3 + \dots + ar^{n-1} +$$

and
$$S_n = \frac{a(1-r^n)}{1-r} = \left[\frac{a}{1-r} - \frac{ar^n}{1-r} \right]$$

Now if $r < 1$ $S_n \rightarrow \frac{a}{1-r}$ and the series is convergent, if $r = 1$ or $r > 1$ $S_n \rightarrow \infty$ or is indeterminate, and thus the series is not convergent /

Ex 56 Show that the following series is convergent

$$\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{n(n+2)}$$

Sum to n terms of the above series

$$S_n = \sum_{r=1}^n \frac{1}{r(r+2)}$$

$$S_n = \sum_{r=1}^n \frac{1}{2} \left[\frac{1}{r} - \frac{1}{r+2} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1} - \frac{1}{3} \right] + \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} \right] + \frac{1}{2} \left[\frac{1}{3} - \frac{1}{5} \right]$$

$$+ \dots + \frac{1}{2} \left[\frac{1}{n} - \frac{1}{n+2} \right]$$

$$= \frac{1}{2} \left[1 - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \dots + \frac{1}{n} - \frac{1}{n+2} \right]$$

$$= \frac{1}{2} \left[1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right]$$

$$S_n \rightarrow \frac{1}{2} \left[1 + \frac{1}{2} \right] \text{ as } n \rightarrow \infty$$

i.e., $S_n \rightarrow \frac{3}{4}$ as $n \rightarrow \infty$ which proves that the given series is convergent

Ex 57 Discuss the convergence of the series

$$1 + \frac{1}{2^k} + \frac{1}{3^k} + \frac{1}{4^k} + \dots$$

Here it is not possible to obtain the sum of the series up to n terms. To avoid this we make the following manipulation

We rewrite the series as follows

$$S_n = 1 + \frac{1}{2^k} + \left(\frac{1}{3^k} + \frac{1}{4^k} \right) + \left(\frac{1}{5^k} + \frac{1}{6^k} + \frac{1}{7^k} + \frac{1}{8^k} \right) + \dots$$

Case I $k < 1$, In case we substitute for each term within the bracket by the smallest term occurring in the bracket itself, then the sum of the terms on R.H.S. will be less than S_n

$$S_n > 1 + \frac{1}{2^k} + \left(\frac{1}{4^k} + \frac{1}{4^k} \right) + \left(\frac{1}{8^k} + \frac{1}{8^k} + \frac{1}{8^k} + \frac{1}{8^k} \right) + \dots$$

$$S_n > 1 + \frac{1}{2^k} + \frac{2}{4^k} + \frac{4}{8^k} + \dots$$

$$k < 1$$

$(k-1)$ is negative

$$\left(\frac{1}{2} \right)^{(k-1)n} \rightarrow \infty \text{ as } n \rightarrow \infty \text{ and thus the given series is divergent.}$$

Case II $k > 1$ In this case we replace each term in the bracket that makes the R.H.S. larger than L.H.S., so that

$$S_n < 1 + \left(\frac{1}{2^k} + \frac{1}{2^k}\right) + \left(\frac{1}{4^k} + \frac{1}{4^k} + \frac{1}{4^k} + \frac{1}{4^k}\right) +$$

$$S_n < 1 + \frac{1}{2^{k-1}} + \frac{1}{4^{k-1}} +$$

$$S_n < \left[\frac{1 - \left(\frac{1}{2^{k-1}}\right)^n}{1 - \frac{1}{2^{k-1}}} \right]$$

$$k > 1,$$

$k - 1$ is positive

$$\left(\frac{1}{2}\right)^{(k-1)n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

$S_n \rightarrow 0$ as $n \rightarrow \infty$ and thus the given series is convergent

Necessary and Sufficient Conditions for Convergence

(i) *Necessary Condition*

$$U_n \rightarrow 0 \text{ as } n \rightarrow \infty$$

(ii) *Sufficient Condition*

$$\frac{U_{n+1}}{U_n} \leq k \text{ for } k < 1$$

The sufficiency condition is D'Alembert's test for convergence

For instance, the series $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots$ is such that

$$U_n = \frac{1}{n} \rightarrow 0 \text{ as } n \rightarrow \infty \text{ (i.e., necessary condition is fulfilled) but}$$

$$\frac{U_{n+1}}{U_n} = \frac{U_{n+1}}{U_n} = \frac{n}{n+1} = \frac{1}{1 + \frac{1}{n}}$$

$$\therefore \frac{U_{n+1}}{U_n} \rightarrow 1 \text{ as } n \rightarrow \infty \text{ whereas } \frac{U_{n+1}}{U_n} \rightarrow < 1 \text{ for convergence}$$

Hence the sufficient condition is not satisfied

Absolute and Conditional Convergence When an infinite series has positive as well as negative terms, testing for convergence becomes difficult. In such cases two devices are usually useful. First, we ignore signs and examine $\sum |U_n|$ for convergence, and $\sum U_n$ is conditionally convergent if it is convergent

but $\sum |U_n|$ is not convergent

Now, $\sum U_n$ is necessarily convergent if it is conditionally convergent, i.e., $\sum |U_n|$ is convergent

Second, if U_n is positive and decreases steadily to zero as $n \rightarrow \infty$ then the alternating series $U_1 - U_2 + U_3 - U_4 + U_5 - \dots$ is convergent

Ex. 58. Since in the series

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

$$U_n = \frac{1}{2n-1} \text{ and } U_n \rightarrow 0 \text{ as } n \rightarrow \infty,$$

it is convergent by the above result. But for $1 + \frac{1}{2} + \frac{1}{5} + \frac{1}{7} + \dots$

$$\frac{U_{n+1}}{U_n} = \frac{2n+1}{2n-1}$$

$$\frac{U_{n+1}}{U_n} \rightarrow 1 \text{ as } n \rightarrow \infty, \text{ thus the series is not convergent}$$

Hence we say that the given series is conditionally convergent

Ex. 59. Test the convergency of the series

$$1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots$$

Here, $U_n = \frac{1}{n!}, U_n \rightarrow 0 \text{ as } n \rightarrow \infty$

Also, $\sum |U_n| = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots$

and, $\frac{U_{n+1}}{U_n} = \frac{\frac{1}{(n+1)!}}{\frac{1}{n!}} = \frac{n!}{(n+1)!} = \frac{1}{n+1}$

$$\frac{U_{n+1}}{U_n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Hence $\sum |U_n|$ is convergent and series is absolutely and conditionally convergent

EXERCISE 4.12

Test the convergence or divergence of the following series

1. $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

2. $1 + \frac{3}{4} + \frac{9}{16} + \frac{27}{64} + \dots$

$$3. \quad 1 - 1 + 1 - 1 + 1$$

$$4. \quad 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} +$$

$$5. \quad 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} +$$

$$6. \quad 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5}$$

$$7. \quad \frac{1}{2^2} + \frac{1}{2} + \frac{1}{2^4} + \frac{1}{2^3} + \frac{1}{2^6} + \frac{1}{2^5} +$$

$$8. \quad x + 2x^2 + 3x^2 + 4x^4 + \dots + rx^r$$

$$9. \quad \frac{1}{1 \cdot 2} + \frac{1}{3 \cdot 4} + \frac{1}{5 \cdot 6} + \dots + \frac{1}{(2n+1)(2n+2)} +$$

$$10. \quad \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} +$$

$$11. \quad 1 + \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} +$$

$$12. \quad 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} +$$

$$13. \quad \Sigma \frac{1}{np}$$

$$14. \quad \sum n^k k^n \text{ (for position } k)$$

$$15. \quad 1 + \sqrt{2} \cdot \frac{3}{4} + \sqrt{3} \left(\frac{3}{4} \right)^2 +$$

Appendix to Chapter 4

FUNCTIONS AND CURVES IN ECONOMICS*

In economics, we deal with relationships between variables which can be measured. These relationships can be put in various functional forms. This is done on the basis of our understanding of the nature of the relationship. Below we discuss some important functional relationships discussed and analysed in economics.

Demand functions and curves

We know that demand curve for normal goods is negatively sloped, i.e., demand is a negative function of price. For illustrative purposes, we take demand curves of the form *A* or *B* in Fig. 1. *A* is a straight line. *B* is a curve which can be taken as a parabola or hyperbola.

The generally used demand functions are

- | | |
|---|---|
| (i) $P = a - bx$ | Downward sloping straight line |
| (ii) $p = (a - bx)^2$
$p = a - bx^2$ | Parabola (to be taken in the positive quadrant only) |
| (iii) $(p + c)(x + b) = a$ | Rectangular hyperbola with centre at $x = -b, p = -c$ |

We take some specific demand curves

- (1) $p = 20 - x$, the curve is traced in Fig. 2

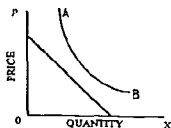


Fig. 1

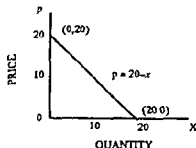


Fig. 2.

* This appendix should be read after the chapters on Differential Calculus.

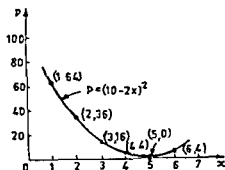


Fig 3

(2) $p = (10 - 2x)^2$	
x	p
0	100
1	64
2	36
3	16
4	4
5	0
6	4

See Fig 3 Obviously only the falling part of the curve (parabola) represents the demand curve. The demand curve is convex to the origin

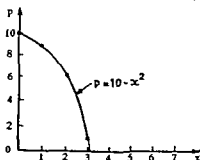


Fig 4

(3) $p = 10 - x^2$	
x	p
0	10
1	9
2	6
3	1
4	0

See Fig 4 Though the curve has negative slope, it is not the usual type of demand curve (convex to the origin)

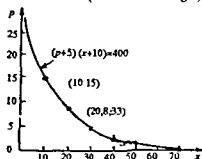


Fig 5

$$(4) (p + 5) = 400 / (x + 10) \\ \text{or } p = [400 / (x + 10)] - 5$$

x	p
0	35
10	15
20	8.33
30	5
40	3
50	1.7
60	0

See Fig 5 This is the usual demand curve. In fact this is a rectangular hyperbola with axes $x = -10$ and $p = -5$. The demand curve will cut both the axes ox and oy but will be asymptotic to its axes given above. The centre of the hyperbola is at $(-10, -5)$

(5) In economics, we usually deal with demand curves having unitary elasticity of demand at all the points on it. This type of demand curve is a rectangular hyperbola with the axes OX and OY as its axes and $(0,0)$ its centre. It is thus a special case of demand function explained and plotted below.

Assume $p \cdot x = 30$ or, $p = 30/x$, i.e. total expenditure remains constant.

x	p
0	—
1	30
2	15
3	10
4	7.5
∞	0

See Fig. 6

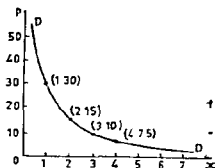


Fig. 6

Total Revenue Curve

We can easily obtain the total revenue curve for a firm from its demand curve or revenue curve.

$$TR = p \cdot x,$$

Hence, $TR = x(a - bx)$ is the total revenue function if the demand function is linear. It

is a parabola with maximum TR at $x = \frac{a}{2b}$.

It can be written as

$$TR - \frac{a^2}{4b} = -b \left(x - \frac{a}{2b} \right)^2$$

$TR = x(a - bx)^2$ is the cubic total revenue function if the demand function is quadratic $p = (a - bx)^2$.

$TR = x \cdot a/x = a$, if the demand curve is a rectangular hyperbola with OX and OY as axes, i.e., if $p = \frac{a}{x}$.

If we think that total revenue increases at first at an increasing rate, then at a constant rate and eventually at a decreasing rate, then the total revenue function must be a cubic function of quantity sold. That is

$$TR = a + bx + cx^2 + dx^3$$

with appropriate signs assigned to the constants as in the case of total revenue curve $TR = x(a - bx)^2$ above.

Ex 1. Plot the total revenue function $TR = x(10 - 2x)^2$

Verify that a straight line will cut the curve at not more than 3 points

Since $TR = x(10 - 2x)^2 = 100x - 40x^2 + 4x^3$

$$\frac{d}{dx}(TR) = 100 + 12x^2 - 80x = 0 \text{ (for an extreme point)}$$

or, $(10 - 6x)(10 - 2x) = 0$, therefore $x = 5/3, x = 5$,

and $\frac{d^2}{dx^2}(TR) = 24x - 80$

$$\begin{array}{lll} < 0 & \text{at} & x = 5/3, \quad \text{maxima} \\ > 0 & \text{at} & x = 5, \quad \text{minima} \end{array}$$

Total revenue rises very rapidly reaches maximum at $x = 5/3$ and then declines

Ex 2 Compare the form of demand curves and the total revenue curves derived from them in the following cases

$$p = \left(3 - \frac{x}{40}\right)^2 \text{ and } p = \left(4 - \frac{x^2}{100}\right)$$

We have already considered the first case. Let us discuss the second case,

$$TR = x \left(4 - \frac{x^2}{100}\right) = 4x - \frac{x^3}{100}$$

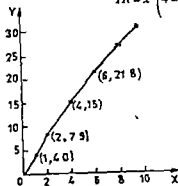


Fig 8.

x	TR	MR
0	0	
2	$8 - 0.08 = 7.92$	$\frac{7.92}{2} = 3.96$
4	$16 - 0.64 = 15.36$	$\frac{7.44}{2} = 3.72$
6	$24 - 2.16 = 21.84$	$\frac{5.48}{2} = 2.74$
8	$32 - 5.12 = 26.88$	$\frac{5.04}{2} = 2.52$
10	$40 - 10.0 = 30.00$	$\frac{3.12}{2} = 1.56$
12	$48 - 17.28 = 30.72$	$\frac{0.72}{2} = 0.36$

TR is increasing at decreasing rate, reaches maximum value at $x = 11.56$ and then decreases. Hence marginal revenue is monotonically decreasing (See Fig. 8). This can be derived with the help of calculus too.

$$\frac{d}{dx}(TR) = MR = 4 - \frac{3x^2}{100} = 0 \text{ (for maxima or minima.)}$$

Hence, $x = \sqrt{\frac{400}{3}} = 11.56$ Examine second order condition

i.e., $\frac{d^2}{dx^2}(TR) = \frac{d}{dx}(MR) = -\frac{6x}{100}$, hence total revenue is maximum at

$x = 11.56$ and MR is monotonically decreasing

Ex. 3 Trace the demand curves given below and also the corresponding TR and MR curves. Discuss the nature of the curves.

(i) $p = \sqrt{225 - 9x}$

(ii) $p = \frac{a}{x + b}$

(iii) $p = \frac{a}{x} - c$ (a, b and c are constants)

Cost Curves

We know that total cost increases at decreasing rate with increase in output at first then at a constant rate and eventually at an increasing rate. Such a curve cannot be shown as straight line or parabola or hyperbola. In fact, it (TC) should be a cubic function of output.

$$C = a + bx + cx^2 + dx^3$$

Take, for example,

$$C = \frac{1}{10}x^3 - 3x^2 + 50x + 300$$

Obviously, 300 is the total fixed cost and the rest is the total variable cost. The curve has been traced in Fig. 9.

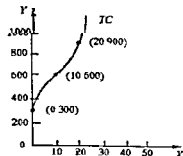


Fig. 9

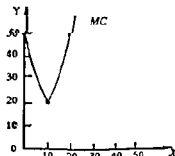


Fig. 10

Trace also the MC curve (as shown in Fig. 10)

$$\text{Now, } MC = \frac{dC}{dx} = \frac{3}{10}x^2 - 6x + 50$$

which is a quadratic function of output. It is parabola

$$\begin{aligned} MC &= \frac{3}{10} \left(x^2 - 20x + \frac{500}{3} \right) \\ &= \frac{3}{10} \left[(x-10)^2 + \frac{500}{3} - 100 \right] \\ &= \frac{3}{10} (x-10)^2 + 20 \end{aligned}$$

$$MC - 20 = \frac{3}{10} (x-10)^2$$

Hence, when the vertex of the parabola is $x=10$, $MC = 20$. This will be the lowest point of the parabola. The MC is a U -shaped curve (Fig. 10). From calculus, we know that MC is minimum at a point where

$\frac{d}{dx}(MC)$ is 0 and $\frac{d^2}{dx^2}(MC)$ is positive. Applying these conditions, we have

$\frac{6}{10}x = 6$, or $x = 10$, and $\frac{d^2}{dx^2}(MC) = \frac{6}{10}$ which is always positive, hence MC is minimum at $x = 10$. There is point of inflexion at $x=10$ on the TC curve.

$$AC = \frac{1}{x} \left(\frac{1}{10}x^3 - 3x^2 + 50x + 300 \right)$$

$$\begin{aligned} AVC &= \frac{1}{x} \left(\frac{1}{10}x^3 - 3x^2 + 50x \right) \\ &= \frac{1}{10}x^2 - 3x + 50 \end{aligned}$$

These are U -shaped curves. The AVC has a minimum when

$$\frac{d}{dx}(AVC) = 0, \quad \frac{2}{10}x = 3, \text{ i.e., when } x = 15$$

Ex. 4 Trace the TC curve $C = ax \left[\frac{x+b}{x+c} \right]$

a, b, c are positive and $b > c$ (take $a = 2, b = 200$ and $c = 100$). Show that average cost decreases with output. Discuss the nature of the cost curves. Trace also the MC curve.

Ex. 5. Trace the TC curve

$$C = 100 + 5x - 0.3x^2 + 0.02x^3$$

Trace also the AC and MC curves

Indifference Curves and Maps

An indifference curve or preference function gives us the combinations of two commodities for achieving a given level of satisfaction. It is thus a relation between quantities of two commodities (we consider only the two-commodity case here). We also know that indifference curves are convex to the origin.

Now a curve convex to the origin can be represented by several functional forms: relevant portions of a parabola, hyperbola, circle, ellipse or logarithmic function. Let us take a few illustrations.

Ex 6. Let $U = (x + 3)(y + 1)$

$$\text{i.e. } y = \frac{U}{x+3} - 1$$

It is clearly a rectangular hyperbola. The hyperbola has $x = -3$ and $y = -1$ as axes and vertex $(-3, -1)$. By giving different values to U we can obtain different indifference curves. We derive two indifference curves for $U = 6$ and $U = 9$ as shown in Fig. 11.

The hyperbola has $x = -3$ and $y = -1$ as axes and vertex $(-3, -1)$.

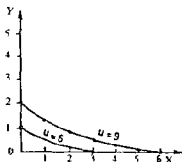


Fig. 11

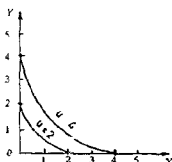


Fig. 12

Ex 7. $U = x + y + 2xy$

This is equivalent to $U - x - y = \sqrt{2xy}$

Squaring both sides and simplifying, we have

$$(x - U)^2 + (y - U)^2 = U^2$$

which is a system of circles with radius U and centre (U, U) .

The circles will touch the axes at $(U, 0)$ and $(0, U)$. We have drawn the circles for $U = 2$ and $U = 4$ as shown in Fig. 12.

Ex 8. $U = (x + 2)\sqrt{y + 1}$

$$\text{i.e. } y = \frac{U^2}{(x+2)^2} - 1$$

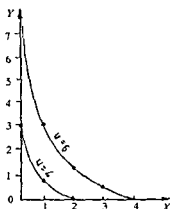


Fig 13

	for $U = 4$	for $U = 6$
x	y	y
0	3	8
1	0.8	3
2	0	1.25
3		0.44
4		0

Thus we obtain an indifference map as shown in Fig 13

Ex. 9 Stone-Geary function This function is of the form

$$U = (x_1 - a_1)^{b_1} (x_2 - a_2)^{b_2} \dots (x_n - a_n)^{b_n}, \text{ where } \sum b_i = 1$$

This leads to normal indifference curves. Take the two-commodity case

$$U = (x - 1)^{0.5} (y - 2)^{0.5} \quad \text{i.e.,} \quad U^2 = (x - 1)(y - 2)$$

or, $y = \frac{U^2}{x - 1} + 2$

We trace the curve for

$$U = 2 \text{ and } U = 3$$

x	y when $U=2$	y when $U=3$
2	6	11.0
3	4	7.5
4	3.3	5.0
5	3	4.25
6	2.8	3.8

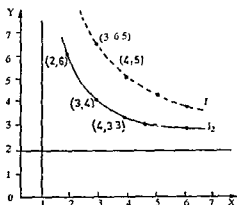


Fig 14

Value of y approaches infinity at $x = 1$ and to 2 as $x \rightarrow \infty$

Hence $x = 1$ and $y = 2$ are the asymptotes (see Fig 14)

Ex.10. Show that the MRS in the above function = $\frac{(y-2)}{(x-1)}$

Production Functions

Production functions and the related isoquants are widely used in theoretical

cal and empirical work in economics. The most popular production function is the Cobb Douglas function, which is also termed as the unitary elasticity production function or the double log function

$$Y = AL^a K^b, L \text{ and } K \text{ are labour and capital}$$

$$= 10 L^{1.5} K^{0.5} \text{ (say) Let } K \text{ be fixed at } 100,$$

$$\text{then, } Y = 10 L^{1.5} \sqrt{100} = 100 L \sqrt{L}$$

Hence, if	$L = 0,$	$Y = 0$	$L = 6$	$Y = 1470$
	$L = 1,$	$Y = 100$	$L = 9$	$Y = 2700$
	$L = 2,$	$Y = 282$		
	$L = 4,$	$Y = 800$		

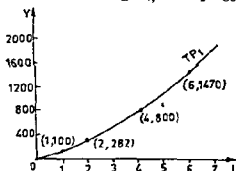


Fig 15 (a)

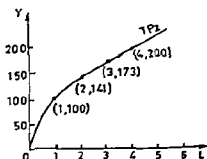


Fig 15 (b)

Output is increasing at an increasing rate [See Fig 15 (a)]

But if the production function is

$Y = 10 L^{0.5} K^{0.5}$ and $K = 100$, there will be diminishing returns to labour as depicted by the following table

L	Y
0	0
1	100
2	141
3	173
4	200
5	224
6	245

Plotting these points, we obtain curve [Fig 15 (b)]

Here output is increasing at diminishing rate as labour input is increased

[Note - There will be diminishing returns to capital for both the above production functions if L is taken to be fixed, say, $L = 100$]

Average product curves Let us consider the average and marginal product curves corresponding to these production functions

Production Functions

$$Y = 100 L^{1.5} \quad (1)$$

$$Y = 100 L^{0.5} \quad (2)$$

Average Product

$$\frac{Y}{L} = \frac{100 L^{1.5}}{L} = 100 L^{0.5} \quad (3)$$

$$\frac{Y}{L} = \frac{100}{L^{0.5}} \quad (4)$$

Hence curve 2 is the A P curve of the first function. Also A P for the first production function is rising. It shows increasing returns to labour throughout (see Fig. 16)

But A.P. for the second production function shows falling A P with labour input

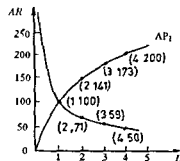


Fig. 16

L	A P of second function
0	
1	100
2	71
3	59
4	50

Marginal product curves M P curves can be obtained directly from the tables prepared for the total product curves

If you know differential calculus then M P function corresponding to the production function can be obtained by differentiation with respect to labour input or capital input

$$\left. \begin{aligned} Y &= 100 L^{1.5} \\ MP_1 &= \frac{dY}{dL} = 100 (1.5 L^{0.5}) \end{aligned} \right\}$$

$$MP_1 = 150 L^{0.5}$$

$$Y = 100 L^{0.5}$$

$$MP_2 = \frac{dY}{dL} = \frac{100 \times 0.5}{L^{0.5}}$$

$$MP_2 = 50 \times \frac{1}{L^{0.5}}$$

$$\text{that is } MP_1 > AP_1$$

$$\text{and } MP_2 < AP_2$$

Hence when AP is rising MP is above (greater than) AP (production function 1) and when AP is falling MP is below (is less than) AP .

Ex. Draw the MP curves for the above two production functions.

If U-shaped AP and MP curves are to be obtained then the function (1) and MP functions) should be quadratic, for this the production function should be such that the output is a cubic function of the input.

Ex. 11. Obtain AP and MP of capital, keeping labour input constant ($L = 100$) for the two production functions given above. Compare the MP curves so obtained with AP and MP curves of labour. Interpret.

Returns to Scale :

Suppose the initial labour and capital inputs are $L = 1$ and $K = 1$. Output will be

Prod f_1 1

$$Y = 10 \times 1 \times 1 = 10$$

$$\text{If } L = K = 2 \quad Y = 10 \times 2^{1.5} \times 2^{.5} \\ = 10 \times 2 = 40 > 2 \times 10$$

$$\text{If } L = K = 3 \quad Y = 10 \times 3^{1.5} \times 3^{.5} = 90 > 3 \times 10$$

$$\text{If } L = K = 4 \quad Y = 10 \times 4^{1.5} \times 4^{.5} = 160 > 4 \times 10$$

Prod f_2 2

$$Y = 10 \times 1 \times 1 = 10$$

$$= 10 \times 2 \times 2 = 40$$

$$= 10 \times 2 = 20$$

$$= 10 \times 3 \times 3 = 90$$

$$= 10 \times 4 \times 4 = 160$$

Thus, in the case of the first production function if both the inputs are multiplied by 2, 3, 4 times, the output is multiplied by more than 2, 3, 4 times. It shows increasing returns to scale. In contrast, in the case of the second production function output increases by 2, 3, 4 times. It shows constant returns to scale (See Ex. 11). Note that in case 2 the production function is a straight line.

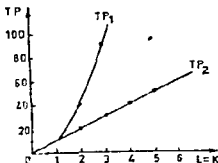


Fig. 17

Ex. 12. Obtain TP , AP , MP curves for the production function $Y = 10L^{1.25}K^{.50}$ taking $K = 100$.

Ex. 13. Obtain TP , MP and AP curves when the production function is $Y = 4LK^{.25}$ ($L, K > 0$).

Show the returns to scale on the graph paper

Ex 14 If the production function is given to be as

$$Y = 3KL + 4K^2L + 2L + 2K$$

Trace the *MP* and *AP* curves of labour when $K = 1$

and *MP* and *AP* curves of capital when $L = 2$

Show that the *MPs* of labour and capital when $L = 2$ and $K = 1$ are 18 and 30 respectively

Examine the returns to scale on the graph paper

Ex 15 Obtain *MP* and *AP* curves of labour for the production function

$$Y = 10L - 0.1L^2 + 15K - 0.2K^2 + 3KL$$

Find *MP* and *AP* when $L = K = 10$

Ex 16 Obtain the isoquants from the production functions given in the solved examples and other exercises (*Proceed exactly as in the case of deriving indifference curves from the utility function*)

Liquidity Preference Function and Schedule

Keynes in theory tells us that demand for money is a function of nominal income as well as the rate of interest. It is an increasing function of income and a decreasing function of the rate of interest. Let us take the following functional form

$$= \left[\frac{1}{2} Y \right] + \left[\frac{20}{r-2} \right]$$

Demand for money = Transaction demand + Speculative demand where Y is money (nominal) income, r is the rate of interest, income velocity of money is 2

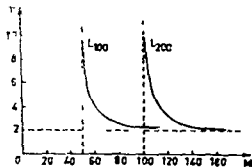


Fig. 13

Let us take different values of Y and trace the liquidity preference schedules for each of them

$$Y = 100$$

$$I_{100} = 50 + \frac{20}{r-2}$$

$$Y = 200$$

$$I_{200} = 100 + \frac{20}{r-2}$$

r	I	r	I
2	$\rightarrow \infty$	2	$\rightarrow \infty$
3	70	3	120
4	60	4	110
5	56.7	5	106.7
6	55.0	6	105
$\rightarrow \infty$	50	$\rightarrow \infty$	100

The first curve is asymptotic to $r=2$ and $M=50$ and the second curve is asymptotic to $r=2$ and $M=100$. There is liquidity trap at $r=2$.

EX. 17 Trace the liquidity preference schedules for the following I/P function

$$I = 60Y + \frac{10}{2r-5}$$

Chapter 5

Matrices and Determinants

5.1 Matrices

In real life, we do not have to deal with a single magnitude but with a set of magnitudes. In analysis, we treat such a set of magnitudes as a single entity or a single object of thought abstracted by numbers. Let us consider a problem where we treat a set as single entity and a number of such sets as single system. We will then examine how the four fundamental operations of arithmetics can be suitably defined to deal with such a system.

Suppose there are 3 brothers, Ajit (A), Bhanu (B) and Chander (C) in a family and that

A has a set of 3 Pants, 3 Shirts, 2 Bush shirts and 1 Tie.

B has a set of 5 Pants, 5 Shirts, No Bush shirt and 2 Ties.

C has a set of 6 Pants, 8 Shirts, 5 Bush shirts and No Tie.

We can arrange this data in the following convenient system

Brothers	Pants	Shirts	Bush shirts	Ties	
	(P)	(S)	(B)	(T)	
A	3	3	2	1	1st Row
B	5	5	0	2	2nd Row
C	6	8	5	0	3rd Row
	1st Col	2nd Col	3rd Col	4th Col	

This system comprises 3 rows and 4 columns. (Note that we write rows first and columns afterwards.) We have written belongings of Ajit in the first row, belongings of Bhanu in the second row and the belongings of Chander in the third row. Clearly, therefore, 1st column gives us the total number of Pants that A, B and C together have while second, third and fourth columns enumerate respectively the number of Shirts, Bush shirts and Ties that three brothers together have in their family.

Numbers written in such a particular form of rows and columns and enclosed by { } → square brackets or () → large parentheses or || || → double bars are called a matrix.

A matrix is defined as a rectangular array of elements arranged in rows and columns.

Its general form is

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{1n} \\ a_{21} & a_{22} & a_{2n} \\ a_{m1} & a_{m2} & a_{mn} \end{bmatrix}$$

This is a matrix of m rows and n columns. It is denoted by (a_{ij}) and is read m by n . The first subscript refers to the number of row and the second subscript refers to the number of column to which the element belongs. Thus a_{13} belongs to the 1st row and 3rd column and a_{34} to the 3rd row and 4th column, in general a refers to i th row and j th column.

Order of a Matrix. If the matrix has total number of elements = mn arranged in m rows and n columns, it is said to be of the order of m by n , which is often written as $m \times n$ matrix. *Also remember*

1 That a matrix is not just an aggregate of numbers, because in a given matrix each element has its assigned position in a particular row and column.

It is to be remembered that

$$\text{matrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 2 & 1 \end{bmatrix} \text{ is not same as } \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 1 & 1 \end{bmatrix}$$

as in a matrix each element has an assigned position.

2 Matrix A = Matrix B if they have the same order and each element of A is equal to the corresponding element of B .

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

then $A = B$ if and only if

$$a_{11} = b_{11}, a_{12} = b_{12}, a_{21} = b_{21} \text{ and } a_{22} = b_{22}, \text{ i.e., } a_{ij} = b_{ij}$$

3 A matrix of m rows and n column is an $m \times n$ matrix.

4 If $m = n$, the matrix is a square matrix of order n .

5 If the matrix consists of only one column as $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, it is a *column matrix*

or *column vector*.

6 If a matrix consists of only one row as $[a_1 \ b_1 \ c_1]$, it is a *row matrix* or *row vector*.

7 A matrix is said to be a zero or Null matrix if and only if each of its elements is zero.

5.2 Addition and Subtraction of Matrices

If the matrices A and B are of the same order ($m \times n$) then they can be added (or subtracted)

If $A = [a_{ij}]$ and $B = [b_{ij}]$ then

$$A + B = [a_{ij} + b_{ij}]$$

$$A - B = [a_{ij} - b_{ij}]$$

In other words corresponding elements are added (or subtracted). The new matrix will be of the same order $m \times n$.

In particular if

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix} \text{ and } A - B = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} \\ a_{21} - b_{21} & a_{22} - b_{22} \end{bmatrix}$$

also (i) $A + B = B + A$ (Commutative law of addition)

and (ii) $(A + B) + C = A + (B + C)$ where C is the third matrix (giving us the associative law of addition)

(iii) $A + 0 = 0 + A = A$ thus zero matrix plays the same role in matrix algebra as 0 in ordinary algebra

By extending this definition of addition we can define

Scalar multiplication as

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

then $A + A + A + \dots + A$ times

$$= kA = k \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = \begin{bmatrix} ka_{11} & ka_{12} & \dots & ka_{1n} \\ ka_{21} & ka_{22} & \dots & ka_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ka_{m1} & ka_{m2} & \dots & ka_{mn} \end{bmatrix}$$

i.e. each element is multiplied by k

The following results can be easily established now

(i) $A + (-A) = (-A) + A = 0$

(ii) $A + A = 2A$

$$(iii) \lambda(A+B) = \lambda A + \lambda B$$

$$(iv) (\lambda_1 + \lambda_2)A = \lambda_1 A + \lambda_2 A$$

$$\text{Ex 1. If } A = \begin{bmatrix} 2 & 0 \\ -5 & 6 \end{bmatrix} \text{ and } B = \begin{bmatrix} -3 & 6 \\ 4 & 1 \end{bmatrix} \text{ find } A+B$$

$$A+B = \begin{bmatrix} 2 & 0 \\ -5 & 6 \end{bmatrix} + \begin{bmatrix} -3 & 6 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} (2-3) & (0+6) \\ (-5+4) & (6+1) \end{bmatrix} = \begin{bmatrix} -1 & 6 \\ -1 & 7 \end{bmatrix}$$

$$\text{Ex 2 If } A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -1 & 2 \\ 3 & 0 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 3 & 4 \\ 6 & 2 & 0 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\begin{aligned} \text{then } A+B &= \begin{bmatrix} 1 & 2 & -3 \\ 0 & -1 & 2 \\ 3 & 0 & 4 \end{bmatrix} + \begin{bmatrix} -1 & 3 & 4 \\ 6 & 2 & 0 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} (1-1) & (2+3) & (-3+4) \\ (0+6) & (-1+2) & (2+0) \\ (3+2) & (0+1) & (4+3) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 5 & 1 \\ 6 & 1 & 2 \\ 5 & 1 & 7 \end{bmatrix} \end{aligned}$$

$$\text{Ex 3 If } A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & x & x^2 \\ 0 & -2 & -4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 3 & -2 \\ -4 & y & x \\ 0 & 2 & 4 \end{bmatrix}$$

$$\text{then } A+B = \begin{bmatrix} (1-1) & (-3+3) & (2-2) \\ (4-4) & (x+y) & (x^2+x) \\ (0+0) & (-2+2) & (-4+4) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & x+y & (x^2+x) \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Ex. 4 If } A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ and, } B = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{then } A+B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Ex. 5. If } A = \begin{bmatrix} 1 & -3 & 0 \\ 4 & 0 & -1 \\ 0 & -2 & -4 \end{bmatrix} \text{ and, } B = \begin{bmatrix} -1 & 3 & -2 \\ -4 & 0 & 1 \\ 0 & 2 & 4 \end{bmatrix}$$

$$\text{then } A-B = \begin{bmatrix} [1-(-1)] & (-3-3) & [0-(-2)] \\ [4-(-4)] & (0-0) & (-1-1)] \\ (0-0) & (-2-2) & (-4-4)] \end{bmatrix} = \begin{bmatrix} 2 & -6 & 2 \\ 8 & 0 & -2 \\ 0 & -4 & 8 \end{bmatrix}$$

Ex 6 If $A = \begin{bmatrix} x & -2 & y \\ 3 & z & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & x & z \\ x & y & 4 \end{bmatrix}$

then $A - B = \begin{bmatrix} (x-2) & (-2-x) & (y-z) \\ (3-x) & (z-y) & (3-4) \end{bmatrix} = \begin{bmatrix} (x-2) & (-2-x) & (y-z) \\ (5-x) & (z-y) & -1 \end{bmatrix}$

Ex 7 If $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 9 \\ 7 & 6 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$A - 2B = \begin{bmatrix} 1 & (3-2) & 2 \\ (2-2) & 6 & 9 \\ 7 & 6 & (2-2) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 6 & 9 \\ 7 & 6 & 0 \end{bmatrix}$$

Ex 8 Explain the meaning of

$$\begin{bmatrix} x+y & x+z \\ z-y & x+w \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$$

The equality between two matrices of the same order implies that each element of one matrix is equal to the corresponding element of the second matrix. Hence,

$$\begin{aligned} x + y &= 1 \\ z - y &= 0 \\ x + z &= 2 \\ x + w &= 3 \end{aligned}$$

Thus the matrix relationship represents a system of simultaneous equations

5.3 Matrix Multiplication

Multiplication of a matrix by a column matrix. Suppose the three brothers now intend to purchase some more clothes (or some items). They find that prices of the different items of their clothes at a particular shop S_1 are as follows:

Prices	1	Rs 50 per Pant	3	Rs 25 per Bush shirt
	2	Rs 30 per Shirt	4	Rs 20 per Tie

Let us write these prices in the form of a column so that column of prices becomes

$$\begin{bmatrix} 50 \\ 30 \\ 25 \\ 20 \end{bmatrix} \begin{array}{l} \longrightarrow \text{price per Pant} \\ \longrightarrow \text{" " Shirt} \\ \longrightarrow \text{" " Bush shirt} \\ \longrightarrow \text{" " Tie} \end{array}$$

$P S B T$

Supposing that A wishes to purchase $[2 \ 2 \ 1 \ 2]$ at these prices how much will he have to pay for this whole lot?

This can be obtained by multiplying the number of Pants, Shirts, Bush shirts and Ties by respective prices and then adding up the results of these multiplications, i.e.

A will have to pay

$$\begin{array}{c} \text{P S B T} \\ = [2 \ 2 \ 1 \ 2] \times \begin{bmatrix} 50 \\ 30 \\ 25 \\ 20 \end{bmatrix} \begin{array}{l} \longrightarrow \text{price per Pant} \\ \longrightarrow \text{" " Shirt} \\ \longrightarrow \text{" " Bush-shirt} \\ \longrightarrow \text{" " Tie} \end{array} \end{array}$$

which can be written as $(2 \times 50) + (2 \times 30) + (1 \times 25) + (2 \times 20) = [225]$

P S B T

Supposing B wishes to purchase $[1 \ 2 \ 2 \ 1]$ for which (at the same prices) he will pay

$$\begin{array}{c} \text{P S B T} \\ [1 \ 2 \ 2 \ 1] \times \begin{bmatrix} 50 \\ 30 \\ 25 \\ 20 \end{bmatrix} \begin{array}{l} \longrightarrow \text{price per Pant} \\ \longrightarrow \text{" " Shirt} \\ \longrightarrow \text{" " Bush-shirt} \\ \longrightarrow \text{" " Tie} \end{array} \end{array}$$

which can be written as $[(1 \times 50) + (2 \times 30) + (2 \times 25) + (1 \times 20)] = [180]$

P S B T

and supposing C wishes to purchase $[3 \ 3 \ 0 \ 5]$ at the same prices so that he will have to pay

$$\begin{array}{c} \text{P S B T} \\ [3 \ 3 \ 0 \ 5] \times \begin{bmatrix} 50 \\ 30 \\ 25 \\ 20 \end{bmatrix} \begin{array}{l} \longrightarrow \text{price per Pant} \\ \longrightarrow \text{" " Shirt} \\ \longrightarrow \text{" " Bush-shirt} \\ \longrightarrow \text{" " Tie} \end{array} \end{array}$$

which can be written as $[(3 \times 50) + (3 \times 30) + (0 \times 25) + (5 \times 20)] = [340]$

Or, we may also write

$$\begin{array}{l} \text{P S B T} \\ \text{A} \rightarrow [2 \ 2 \ 1 \ 2] \\ \text{B} \rightarrow [1 \ 2 \ 2 \ 1] \\ \text{C} \rightarrow [3 \ 3 \ 0 \ 5] \end{array} \times \begin{bmatrix} 50 \\ 30 \\ 25 \\ 20 \end{bmatrix} \begin{array}{l} \longrightarrow \text{price per Pant} \\ \longrightarrow \text{" " Shirt} \\ \longrightarrow \text{" " Bush-shirt} \\ \longrightarrow \text{" " Tie} \end{array}$$

50	30	25	20
P	S	B	T
2	2	1	2
1	2	2	1
3	3	0	5

$$\begin{bmatrix} (2 \times 50) + (2 \times 30) + (1 \times 25) + (2 \times 20) \\ (1 \times 50) + (2 \times 30) + (2 \times 25) + (2 \times 20) \\ (3 \times 50) + (3 \times 30) + (0 \times 25) + (5 \times 20) \end{bmatrix} = \begin{bmatrix} 225 \\ 180 \\ 340 \end{bmatrix}$$

This shows that

$$\begin{bmatrix} 2 & 2 & 1 & 2 \\ 1 & 2 & 2 & 1 \\ 3 & 3 & 0 & 5 \end{bmatrix} \begin{bmatrix} 50 \\ 30 \\ 25 \\ 20 \end{bmatrix} = \begin{bmatrix} 225 \\ 180 \\ 340 \end{bmatrix}$$

Using symbols in place of numbers we have

$$\begin{bmatrix} b_1 & b_2 & b_3 & b_4 \\ a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

$$\text{or } AB = \begin{bmatrix} a_{11}b_1 + a_{12}b_2 + a_{13}b_3 + a_{14}b_4 \\ a_{21}b_1 + a_{22}b_2 + a_{23}b_3 + a_{24}b_4 \\ a_{31}b_1 + a_{32}b_2 + a_{33}b_3 + a_{34}b_4 \end{bmatrix}$$

The product is obtained easily by putting the column of b 's in row form over the matrix A . Then the elements of row 1 of AB is obtained by multiplying elements of the first row of A by the corresponding elements of B and summing. For this it is necessary that the number of elements in each row of A is equal to that in the column B .

Note For each calculations and as an operational rule, it is convenient to write the prices in the form of a row above the matrix representing the different items of clothes which the brothers wish to purchase.

Multiplication of a matrix by another matrix Let us now consider a slightly more difficult case. Suppose that the brothers find different prices of the Pants, Shirts, Bush shirts and Ties at the different shops S_1, S_2, S_3 and S_4 as given under

Prices in shops		S_1	S_2	S_3	S_4
Per Pant	→	50	45	40	35
Per Shirt	→	30	35	30	40
Per Bush shirt	→	25	20	30	30
Per Tie	→	20	25	2	20

We assume that the three brothers want to spend minimum amount on the whole lot of clothes (consisting of P, S, B and T) which also remains fixed as given above. Our question now is, "From which shop A, B and C should purchase their respective lots assuming that the whole lot has got to be purchased from

the same shop. As such A will have 4 alternatives. He can purchase his lot consisting of $\begin{bmatrix} 2 & 2 & 1 & 2 \end{bmatrix}$ from either of shops S_1, S_2, S_3 or S_4 .

(1) If he purchases from shop S_1 , he would pay

$$A \rightarrow \begin{array}{c} P \ B \ S \ T \\ \hline \begin{bmatrix} 2 & 2 & 1 & 2 \end{bmatrix} \end{array} \times \begin{array}{c} S_1 \\ \downarrow \\ \begin{bmatrix} 50 \\ 30 \\ 25 \\ 20 \end{bmatrix} \end{array} \begin{array}{l} \longrightarrow P \\ \longrightarrow B \\ \longrightarrow S \\ \longrightarrow T \end{array}$$

$$= [(2 \times 50) + (2 \times 30) + (1 \times 25) + (2 \times 20)] = [225]$$

(2) If he purchases from shop S_2 , he would pay

$$A \rightarrow \begin{array}{c} P \ B \ S \ T \\ \hline \begin{bmatrix} 2 & 2 & 1 & 2 \end{bmatrix} \end{array} \times \begin{array}{c} S_2 \\ \downarrow \\ \begin{bmatrix} 45 \\ 35 \\ 20 \\ 25 \end{bmatrix} \end{array} \begin{array}{l} \longrightarrow P \\ \longrightarrow B \\ \longrightarrow S \\ \longrightarrow T \end{array}$$

$$= [(2 \times 45) + (2 \times 35) + (1 \times 20) + (2 \times 25)] = [230]$$

(3) If he purchases from shop S_3 , he would pay

$$A \rightarrow \begin{array}{c} P \ B \ S \ T \\ \hline \begin{bmatrix} 2 & 2 & 1 & 2 \end{bmatrix} \end{array} \times \begin{array}{c} S_3 \\ \downarrow \\ \begin{bmatrix} 40 \\ 30 \\ 30 \\ 20 \end{bmatrix} \end{array} \begin{array}{l} \longrightarrow P \\ \longrightarrow B \\ \longrightarrow S \\ \longrightarrow T \end{array}$$

$$= [(2 \times 40) + (2 \times 30) + (1 \times 30) + (2 \times 20)] = [210]$$

(4) If he purchases from shop S_4 , he would pay

$$A \rightarrow \begin{array}{c} P \ B \ S \ T \\ \hline \begin{bmatrix} 2 & 2 & 1 & 2 \end{bmatrix} \end{array} \times \begin{array}{c} S_4 \\ \downarrow \\ \begin{bmatrix} 35 \\ 40 \\ 30 \\ 30 \end{bmatrix} \end{array} \begin{array}{l} \longrightarrow P \\ \longrightarrow B \\ \longrightarrow S \\ \longrightarrow T \end{array}$$

$$= [(2 \times 35) + (2 \times 40) + (1 \times 30) + (2 \times 30)] = [240]$$

Therefore A should purchase his lot from shop S_3 because it is at this shop that he pays the minimum amount of Rs. 210.

Exactly in the same way we can find out the total amount spent by B and C for their respective lots at different shops—by the multiplication of two matrices.

$$\begin{array}{l} A \longrightarrow \\ B \longrightarrow \\ C \longrightarrow \end{array} \left[\begin{array}{cccc} P & S & B & T \\ 2 & 2 & 1 & 2 \\ 1 & 2 & 2 & 1 \\ 3 & 3 & 0 & 5 \end{array} \right]$$

and

$$\left[\begin{array}{cccc} S_1 & S_2 & S_3 & S_4 \\ 50 & 45 & 40 & 35 \\ 30 & 35 & 30 & 40 \\ 25 & 20 & 30 & 30 \\ 20 & 25 & 20 & 30 \end{array} \right] \begin{array}{l} \longrightarrow \text{price per Pant} \\ \longrightarrow \text{price per Shirt} \\ \longrightarrow \text{price per Bush shirt} \\ \longrightarrow \text{price per Tie} \end{array}$$

Proceeding exactly as shown above we can compute the total amounts to be spent by each brother at each shop as follows

$$\begin{array}{l} S_4 \longrightarrow \\ S_2 \longrightarrow \end{array} \left[\begin{array}{cccc} 35 & 40 & 30 & 30 \\ 45 & 35 & 20 & 25 \end{array} \right] \quad \begin{array}{l} S_3 \longrightarrow \\ S_1 \longrightarrow \end{array} \left[\begin{array}{cccc} 40 & 30 & 30 & 20 \\ 50 & 30 & 25 & 20 \end{array} \right]$$

$$\begin{array}{l} A \longrightarrow \\ B \longrightarrow \\ C \longrightarrow \end{array} \left[\begin{array}{cccc} P & S & B & T \\ 2 & 2 & 1 & 2 \\ 1 & 2 & 2 & 1 \\ 3 & 3 & 0 & 5 \end{array} \right]$$

$$\begin{array}{l} A \longrightarrow \\ B \longrightarrow \\ C \longrightarrow \end{array} \left[\begin{array}{cccc} S_1 & S_2 & S_3 & S_4 \\ 2 \times 50 & 2 \times 45 & 2 \times 40 & 2 \times 35 \\ +2 \times 30 & +2 \times 35 & +2 \times 30 & +2 \times 40 \\ +1 \times 25 & +1 \times 20 & +1 \times 30 & +1 \times 30 \\ +2 \times 20 & +2 \times 25 & +2 \times 20 & +2 \times 30 \\ 1 \times 50 & 1 \times 45 & 1 \times 40 & 1 \times 35 \\ +2 \times 30 & +2 \times 35 & +2 \times 30 & +2 \times 40 \\ +2 \times 25 & +2 \times 20 & +2 \times 30 & +2 \times 30 \\ +1 \times 20 & +1 \times 25 & +1 \times 20 & +1 \times 30 \\ 3 \times 50 & 3 \times 45 & 3 \times 40 & 3 \times 35 \\ +3 \times 30 & +3 \times 35 & +3 \times 30 & +3 \times 40 \\ +0 \times 25 & +0 \times 20 & +0 \times 30 & +0 \times 30 \\ +5 \times 20 & +5 \times 25 & +5 \times 20 & +5 \times 30 \end{array} \right]$$

$$\begin{array}{l} A \longrightarrow \\ B \longrightarrow \\ C \longrightarrow \end{array} \left[\begin{array}{cccc} 225 & 230 & 210 & 240 \\ 180 & 180 & 180 & 205 \\ 340 & 365 & 310 & 375 \end{array} \right] \begin{array}{l} \longrightarrow \text{Row 1} \\ \longrightarrow \text{Row 2} \\ \longrightarrow \text{Row 3} \end{array}$$

In other words, row 1 gives the different amounts which A will pay for the same lot of four different shops and so also the row 2 and row 3 represent

the amounts to be paid by B and C at the four shops

Naturally therefore A and C would like to purchase from S_3 while B can purchase from either of two shops S_2 and S_3 (since at these two shops he pays the same amount viz. Rs 180)

This helps us in defining matrix multiplication. It would, however, be seen that matrix product as explained is somewhat different from ordinary algebraic multiplication. First all the matrices cannot be multiplied by each other and secondly, matrix multiplication is generally not commutative, i.e. $AB \neq BA$.

Multiplication of two matrices is possible only when they are conformable.

Given a matrix A of order $m \times n$ and matrix B of order $p \times q$ then the ordered pair (A, B) of the matrices A and B is said to be conformable in $n=p$ i.e. the number of columns of A = the number of rows of B .

Note here the order of multiplication. In this case we say that we pre multiply matrix B by matrix A . The matrices A and B are conformable in the order A, B .

If $p \neq n$, then the A and B are not conformable in order A, B and we cannot pre multiply matrix B by matrix A .

It is also easy to see that AB may be defined (since $p = n$), in other words pre multiplication by matrix A may be defined but BA —post multiplication by A —may not be defined. It is because in order to post multiply B by A , the number of columns in B must be equal to number of rows in matrix A , or $q=m$. Hence AB may exist ($p = n$) and BA may not exist (q may not be equal to m).

Thus we say that applicability of commutative law of multiplication does not arise here.

$$\text{Suppose } A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 4 & 3 \\ 3 & 5 & 0 \end{bmatrix} \quad \text{and, } B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 2 & 6 \end{bmatrix}$$

then A is 3×3 matrix and B is 3×2 matrix

Ordered pair (A, B) is conformable since $n=p$ (here $n=3, p=3$). Therefore AB is defined. But is BA defined? No, because number of columns in B ($q=2$) is not equal to number of rows in A ($m=3$).

Therefore, we have the rule that if A is $m \times n$ matrix and B is $n \times q$ matrix then the product AB (not BA) is said to be defined.

If $A = [a_{ij}]$ is $m \times n$ matrix and

$B = [b_{ij}]$ is $n \times q$ matrix then the product

$AB = [C_{ij}]$ is $m \times q$ matrix, where C_{ij} is the product of the i th row of A and j th column of B .

$$\begin{aligned} \text{That is, } C_{ij} &= \sum_{k=1}^n a_{ik} b_{kj}, \quad i = 1, 2, 3, \dots, m, \quad j = 1, 2, 3, \dots, q \\ &= a_{i1} b_{1j} + a_{i2} b_{2j} + a_{i3} b_{3j} + \dots + a_{in} b_{nj} \end{aligned}$$

$$1 \times 9 \quad A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow 3 \times 3 \text{ matrix}$$

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \rightarrow 3 \times 2 \text{ matrix}$$



Here $A \times B = C$ will be a 3×2 matrix [$m = 3$ and $n = 2$]

$$AB = C = [C_{ij}]$$

$$= \sum_{k=1}^{n=3} a_{ik} b_{kj}$$

$$= a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j}$$

where i indicates the row of (A) and j the column of (B)

$$C = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \\ C_{31} & C_{32} \end{bmatrix}$$

where C_{11} is obtained by having product of 1st row of A with 1st col of B i.e. $a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}$

C_{12} is the product of 1st row of A and 2nd column of B
 $a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32}$

Similarly, C_{31} = product of 3rd row of A and 1st column of B
 $= a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31}$

More generally if

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{1j} & a_{1n} \\ a_{21} & a_{22} & a_{23} & a_{2j} & a_{2n} \\ a_{31} & a_{32} & a_{33} & a_{3j} & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{i1} & a_{i2} & a_{i3} & a_{ij} & a_{in} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ C_{11} & C_{12} & C_{13} & C_{1j} & C_{1n} \end{bmatrix}$$

or

$$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}$$

$a_{13} \quad a_{23}$

$a_{12} \quad b_{22}$

$a_{13} \quad a_{31}$

$$BA = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \times \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$BA = \begin{bmatrix} a_{11}b_{11} + a_{21}b_{12} & a_{12}b_{11} + a_{22}b_{12} & a_{13}b_{11} + a_{23}b_{12} \\ a_{11}b_{21} + a_{12}b_{22} & a_{12}b_{21} + a_{22}b_{22} & a_{13}b_{21} + a_{23}b_{22} \\ a_{11}b_{31} + a_{21}b_{32} & a_{12}b_{31} + a_{22}b_{32} & a_{13}b_{31} + a_{23}b_{32} \end{bmatrix}$$

Hence $AB \neq BA$.

Ex. 14. If $A = \begin{bmatrix} 3 \\ 0 \\ 5 \\ 1 \end{bmatrix}$ and $B = [4 \quad 2 \quad -1 \quad 0]$

find AB and BA

Since A is 4×1 matrix and B is 1×4 matrix

AB is defined and is a 4×4 matrix

0

-1

2

4

$$AB = \begin{bmatrix} 3 \\ 0 \\ 5 \\ 1 \end{bmatrix} [4 \quad 2 \quad -1 \quad 0]$$

$$= \begin{bmatrix} (3 \times 4) (3 \times 2) (3 \times -1) (3 \times 0) \\ (0 \times 4) (0 \times 2) (0 \times -1) (0 \times 0) \\ (5 \times 4) (5 \times 2) (5 \times -1) (5 \times 0) \\ (1 \times 4) (1 \times 2) (1 \times -1) (1 \times 0) \end{bmatrix} = \begin{bmatrix} 12 & 6 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 20 & 10 & -5 & 0 \\ 4 & 2 & -1 & 0 \end{bmatrix}$$

Can we find BA ? Yes, because the order of B is 1×4 and that of A is 4×1

$\therefore BA$ is defined and is of order 1×1 .

$$BA = \begin{bmatrix} 3 & 0 & 5 & 1 \\ 4 & 2 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} 3 \\ 0 \\ 5 \\ 1 \end{bmatrix}$$

$$= [(4 \times 3) + (2 \times 0) + (5 \times -1) + (1 \times 0)] = [12 + 0 - 5 + 0]$$

$$BA = [7]$$

Ex 15 A and B both are square matrices and of the same order as follows

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Show that $AB \neq BA$.

$$BA = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 2 \\ 3 & 2 & 7 \end{bmatrix}$$

and $AB = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 5 & 2 \end{bmatrix}$

Hence $AB \neq BA$

5.4 Associative Law Holds for Conformable Matrices

If A is $m \times n$ matrix

B is $n \times q$ matrix

C is $q \times r$ matrix, then the product (A, B) is conformable product (B and C) is also conformable.

And $ABC = A(BC) = (AB)C$ and as such associative law of multiplication holds.

(AB) is $m \times q$ matrix

C is $q \times r$ matrix

(AB)C is $m \times r$ matrix

Similarly, (BC) is $n \times r$ matrix

A is $m \times n$ matrix

(ABC) is $m \times r$ matrix

Ex 16. Find value of ACB if

$$A = [x_1 \ x_2 \ x_3] \quad B = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

There can be 2 ways of getting the product ACB

(1) First, find AC , A is 1×3 matrix, C is 3×3 matrix, AC is defined and is 1×3 matrix

and then find $(AC)B$, AC is 1×3 matrix, B is 3×1 matrix, ACB is defined and is 1×1 matrix

(2) First find CB , C is 3×3 matrix, B is 3×1 matrix

CB is defined and is 3×1 matrix

and then find $A(CB)$, A is 1×3 matrix, CB is 3×1 matrix

$A(CB)$ is defined and is 1×1 matrix

By method (1)

$$AC = \begin{bmatrix} 0 & 0 & a_{33} \\ 0 & a_{22} & 0 \\ a_{11} & 0 & 0 \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \times \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

$$AC = [x_1 a_{11} + 0 + 0 \quad 0 + x_2 a_{22} + 0 \quad 0 + 0 + x_3 a_{33}]$$

$$= [x_1 a_{11} \quad x_2 a_{22} \quad x_3 a_{33}]$$

$$\text{Now } ACB = [a_{11}x_1 \ a_{22}x_2 \ a_{33}x_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= [a_{11}x_1x_1 + a_{22}x_2x_2 + a_{33}x_3x_3]$$

By method (2)

$$CB = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11}x_1 \\ a_{22}x_2 \\ a_{33}x_3 \end{bmatrix}$$

$$\text{Now } ACB = [x_1 \ x_2 \ x_3] \begin{bmatrix} a_{11}x_1 \\ a_{22}x_2 \\ a_{33}x_3 \end{bmatrix} = [a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2]$$

By both methods, we get the same result

$$\text{Ex. 17. } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Compute (i) $2A-3B$, (ii) ABC

$$(i) \quad 2A = 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

$$3B = 3 \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -3 & 6 \\ 6 & -3 \end{bmatrix}$$

$$\therefore 2A - 3B = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} - \begin{bmatrix} -3 & 6 \\ 6 & -3 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 0 & 11 \end{bmatrix}$$

$$(ii) \quad AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 5 & 2 \end{bmatrix}$$

$$ABC = \begin{bmatrix} 3 & 0 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 17 \end{bmatrix}$$

Ex. 18 (i) If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, find $A^2 - 5A + 7I$

(ii) If $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and $(A+B)^2 = A^2 + B^2$

find a and b

$$(i) \quad A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

$$\therefore A^2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

$$5A = 5 \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix}$$

$$\therefore A^2 - 5A = \begin{bmatrix} 8 & 5 \\ 5 & 3 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} = \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix}$$

$$\therefore A^2 - 5A + 7I = \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix} + 7I$$

$$(ii) \quad A + B = \begin{bmatrix} (1+a) & 0 \\ (2+b) & -2 \end{bmatrix}$$

$$(A+B)^2 = \begin{bmatrix} (1+a) & 0 \\ (2+b) & -2 \end{bmatrix} \begin{bmatrix} (1+a) & 0 \\ (2+b) & -2 \end{bmatrix} = \begin{bmatrix} (1+a)^2 & 0 \\ (2a-b+ab-2) & 4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} = \begin{bmatrix} (a^2 + b) & (a-1) \\ (ab-b) & (a+1) \end{bmatrix}$$

$$\begin{aligned} A^2 + B^2 &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} (a^2 + b) & (a-1) \\ (ab-b) & (a+1) \end{bmatrix} \\ &= \begin{bmatrix} (a^2 + b - 1) & (a-1) \\ (ab-b) & b \end{bmatrix} \end{aligned}$$

$$\text{Since } (A+B)^2 = A^2 + B^2$$

$$\begin{bmatrix} (1+a)^2 & 0 \\ (2a-b+ab-2) & 4 \end{bmatrix} = \begin{bmatrix} (a^2 + b - 1) & (a-1) \\ (ab-b) & b \end{bmatrix}$$

$$\text{i.e., } (a-1) = 0 \quad \text{or, } a = 1 \quad \text{and } b = 4$$

5.5. Transpose of a Matrix

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

then the transpose of A is denoted by A' — which is a matrix where the rows and columns have been interchanged.

$$\text{that is, } A' = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \\ a_{14} & a_{24} & a_{34} \end{bmatrix}$$

This is sometimes also denoted by A^T . But in this book we will denote it by A' .

$$\text{Similarly, if } X = [x_1 \quad x_2 \quad x_3 \quad x_4 \dots x_n]$$

$$\text{then the transpose of } X \text{ is } X' = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

In other words,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = [x_1 \quad x_2 \quad x_3 \quad \dots \quad x_n]' \text{ and vice versa.}$$

5.6. Properties of Transpose of a Matrix

(1) Transpose of transpose of a matrix is the original matrix

$$(A')' = (A)$$

Ex. 19. Let $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$ then $A' = \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$ and

$$(A')' = \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} = A$$

(2) Transpose of the sum of matrices is the sum of the transposes of the individual matrices

Ex. 20. Let $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$

$$A + B = \begin{bmatrix} 3 & 4 \\ 6 & 4 \end{bmatrix} \text{ and } (A + B)' = \begin{bmatrix} 3 & 6 \\ 4 & 4 \end{bmatrix}$$

Now $A' = \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$ and $B' = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$

and, $A' + B' = \begin{bmatrix} 3 & 6 \\ 4 & 4 \end{bmatrix} = (A + B)'$

Hence in this case we verify that $(A + B)' = A' + B'$

(3) Transpose of a product of matrices is the product of the transposes of the matrices, taken in the reverse order

$$(AB)' = B'A'$$

Ex. 21. If $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}$ then

$$AB = \begin{bmatrix} 4 & 8 \\ 10 & 18 \\ 1 & 3 \end{bmatrix} \text{ and, } (AB)' = \begin{bmatrix} 4 & 10 & 1 \\ 8 & 18 & 3 \end{bmatrix}$$

Again, $B' = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ and, $A' = \begin{bmatrix} 2 & 4 & 1 \\ 1 & 3 & 0 \end{bmatrix}$

We find here that AB does not exist but $B'A'$ does exist

$$\text{We find } BA' = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 1 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 1 \\ 8 & 18 & 3 \end{bmatrix}$$

$$\text{Thus, } (AB) = BA'$$

5.7 Some Special Forms of Square Matrices

(a) *Diagonal matrix*. If all off-diagonal elements of a matrix are zero the matrix is called diagonal matrix

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

in a diagonal matrix

(b) *Identity matrix*. If all the diagonal elements are equal to 1, it is defined as a unit or identity matrix. Identity matrix must always be a square matrix

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

is an identity matrix. Here we also note that

$$I = I$$

An identity matrix (of proper order) is such that pre-multiplying or post multiplying any conformable matrix A by it yields the same matrix. For example, if A and I , I and A are conformable, then $AI = IA$. This will happen when A and I are square matrices

(c) *A triangular matrix* is a square matrix of the form

$$T = \begin{bmatrix} a_{11} & \times & \times & \times \\ 0 & a_{22} & \times & \times \\ 0 & 0 & a_{33} & \times \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

where symbol $\times$ indicates an element which can take any value.

$$\text{Ex. 22 } A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 2 \end{bmatrix} \text{ and } I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AI = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 2 \end{bmatrix}$$

$$IA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 2 \end{bmatrix}$$

$AI = IA$, in other words, pre multiplication and post multiplication with an identity matrix leave the square matrix A unchanged

(d) *Symmetric matrix.* It is square matrix which has same elements in the arrangement on both sides of the diagonal running from north west to south east

For example, in matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

a_{11} , a_{22} , a_{33} are the diagonal elements of the square matrix. The matrix would be termed as symmetric matrix if the terms over the diagonal are symmetrically situated with respect to the terms under the diagonal, i.e. if $a_{12} = a_{21}$, $a_{23} = a_{32}$ and $a_{13} = a_{31}$. Thus, in the matrix given below

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 2 \\ 0 & 2 & 3 \end{bmatrix}$$

1, 2, 3 form the diagonal and the elements above the diagonal are the exact reflection of the elements under the diagonal.

It can be easily seen that transpose of a symmetric matrix is the original matrix itself, i.e., if A is a symmetric matrix, $A' = A$. The converse is also true, i.e., if $A' = A$, then A is a symmetric matrix

5.8 Trace

The trace of a square matrix is the sum of its diagonal elements

Thus, in the matrix

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 3 & 3 & -1 \end{bmatrix} \quad \text{the trace (tr)} = 1 + 0 - 1 = 0$$

$$\text{In general, } tr(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn} = \sum_{i=1}^n a_{ii}$$

We can also verify that

$$tr(A + B) = tr(A) + tr(B)$$

$$\text{tr}(A') = \text{tr}(A)$$

$$\text{tr}(AB) = \text{tr}(BA) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ji}$$

In general the trace of an identity matrix of order $n \times n$ is $1 + 1 + 1 + \dots$ up to n terms $= n$

$$\text{If, } I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which is the identity matrix of order 3×3 , then

$$\text{tr}(I) = 1 + 1 + 1 = 3$$

5.9. Idempotent Matrix

A symmetric matrix that reproduces itself when multiplied by itself is termed as an idempotent matrix.

That is, A will be termed as idempotent if $AA = A$.

Ex. 23. Show that the identity matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is idempotent matrix

$$\text{Now } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Note that, since A is symmetric, $\therefore A' = A$.

Ex. 24. Show that the matrix,

$$A = \frac{1}{6} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

is an idempotent matrix.

Here,

$$\frac{1}{6} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} \cdot \frac{1}{6} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

$$= \frac{1}{36} \begin{bmatrix} 1+4+1 & -2-8-2 & 1+4+1 \\ -2-8-2 & 4+16+4 & -2-8-2 \\ 1+4+1 & -2-8-2 & 1+4+1 \end{bmatrix}$$

$$= \frac{1}{36} \begin{bmatrix} 6 & -12 & 6 \\ -12 & 24 & -12 \\ 6 & -12 & 6 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} = A.$$

EXERCISE 5.1

1. Compute the indicated operations

$$(a) \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 6 & 4 \\ 2 & -3 \end{bmatrix} \quad (b) \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(c) \begin{bmatrix} 10 & 13 & 1 & 7 \\ 2 & 3 & 1 & -1 \\ 2 & -4 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 0 \\ 3 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

$$(d) \begin{bmatrix} x & -2 & y \\ 3 & z & 3 \end{bmatrix} - \begin{bmatrix} 2 & -x & z \\ x & y & 4 \end{bmatrix} + \begin{bmatrix} 2-x & 2-x & x-y \\ x-3 & y-z & 1 \end{bmatrix}$$

2. Find the value of:

$$(a) \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \quad (b) \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

$$(c) \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ 2 & -3 \end{bmatrix} \quad (d) \begin{bmatrix} 6 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$(e) \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} 6 & 7 & 8 \\ 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix}$$

$$(f) \begin{bmatrix} 3 & 4 & 2 \\ 6 & 5 & 7 \end{bmatrix} \begin{bmatrix} 6 & 7 & 8 \\ 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix}$$

$$(g) [3 \ 4 \ 2] \begin{bmatrix} 6 & 7 & 8 \\ 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix} \quad (h) [2 \ 1 \ 3] \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

$$(i) [a \ b \ d] \begin{bmatrix} d \\ c \\ f \end{bmatrix} \quad (j) \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$(k) \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad (l) \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

3. For the following pairs of matrices compute AB and BA

$$(i) A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 0 & 3 \\ 6 & -3 & 4 & -2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 6 & 5 \\ 7 & 0 \end{bmatrix}$$

$$(ii) A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 6 & -3 & 0 \end{bmatrix}, B = \begin{bmatrix} 6 & 3 & 1 \\ 2 & 4 & -8 \\ 3 & -6 & 1 \end{bmatrix}$$

$$(iii) A = [2 \ 3 \ 4 \ 5], B = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$(iv) A = [0 \ 3 \ 1 \ 5], B = \begin{bmatrix} 4 \\ 2 \\ -1 \\ 0 \end{bmatrix}$$

$$(v) A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(vi) A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 3 & -2 & 1 \\ 0 & 1 & -2 \\ 3 & -4 & 5 \end{bmatrix}$$

$$(vii) A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 2 & 0 & 3 \\ 0 & 2 & 0 & 3 & 1 \\ 2 & 0 & 3 & 1 & 0 \\ 0 & 3 & 1 & 2 & 2 \end{bmatrix}$$

$$(viii) A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

$$(ix) \quad A = \begin{bmatrix} 2 & 6 & 24 \\ 6 & 12 & 36 \\ 24 & 36 & 48 \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{5}{2} & 2 & -\frac{1}{4} \\ 2 & -\frac{5}{3} & \frac{1}{4} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{24} \end{bmatrix}$$

4 Compute the following products

$$(i) \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$$

$$(ii) \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$$

$$(iii) \quad \begin{bmatrix} 1 & -1 & 0 \\ 2 & 0 & -3 \\ 3 & 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 & -1 \\ -2 & 0 & 2 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$(iv) \quad \begin{bmatrix} 1 & 2 & -3 \\ -4 & 3 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 2 & 3 & 1 \\ 2 & -1 & 4 & 0 & -3 \\ -3 & 0 & 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(v) \quad \begin{bmatrix} 1 & 2 & -3 \\ -4 & 3 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 2 & 3 & 1 \\ 2 & -1 & 4 & 0 & -3 \\ -3 & 0 & 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$(vi) \quad [x \ y \ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$(vii) \quad [x \ y \ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$(viii) \quad [a_{11} \ b_{11} \ c_{11}] \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix}$$

(ix) A manufacturer produces three products P , Q and R which he sells in two markets

Annual sales volumes are indicated as follows

Markets	Products		
	<i>P</i>	<i>Q</i>	<i>R</i>
I	10,000	2,000	18,000
II	6,000	20,000	8,000

If unit sales prices of *P*, *Q* and *R* are Rs. 2.50, 1.25 and 1.50 respectively, find the total revenue in each market with the help of Matrix Algebra

Hint Total revenue in each market is obtained from the matrix product

$$\begin{bmatrix} 2.50 & 1.25 & 1.50 \end{bmatrix} \begin{bmatrix} 10000 & 6000 \\ 2000 & 20000 \\ 18000 & 8000 \end{bmatrix} = \begin{bmatrix} 54500 & 52000 \end{bmatrix}$$

(x) If the unit costs of the above 3 commodities are Rs. 1.80, 1.20 and 0.80 respectively, find his gross profits

$$\begin{aligned} \text{Hint Total cost} &= \begin{bmatrix} 1.80 & 1.20 & 0.80 \end{bmatrix} \begin{bmatrix} 10000 & 6000 \\ 2000 & 20000 \\ 18000 & 8000 \end{bmatrix} \\ &= \begin{bmatrix} 34800 & 41200 \end{bmatrix} \end{aligned}$$

Now find his profits

5. If $A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 2 & 4 & -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & 0 & 3 \\ 1 & -1 & 2 & 3 \end{bmatrix}$

(a) Find a 2×4 matrix *X* such that

(i) $A - X = 3B$ (ii) $A - 2X = 3B$

(b) Find a 2×4 matrix *Y* such that

$A + 2Y = 4B$

(c) If $A = \begin{bmatrix} 9 & 1 \\ 4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix}$

Find the matrix *X* such that $3A + 5B + 2X = 0$

6. In example above, are *A* and *B* conformable? What about *A* and *B'*? Form *AB'*

7. Find whether *AB*, *BA*, *AB'* and *BA'* [products] are defined. If so find the products

(i) $A = \begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$

$$(ii) \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$(iii) \quad A = [2 \quad 4 \quad 3], \quad B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$(iv) \quad A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 0 & 1 \\ 2 & 0 & 4 & 2 \\ 1 & 0 & 5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 & -7 & -1 \\ 2 & 5 & 8 & 2 \\ 3 & 6 & -9 & 3 \end{bmatrix}$$

8. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 2 \\ 3 & 1 & 3 \end{bmatrix}$ then show that A and B , as well as B and C are

conformable. Also show that $ABC = A(BC) = (AB)C$

9. If $A = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Find (a) AB (b) AB' (c) $A'B$ (d) $(AB)'$ (e) $(A+B)'$

10. Find in the above problem No 9

(a) $A(BC)$ (b) $(ABC)'$ (c) $C'BA'$

and show that $(ABC)' = C'B'A'$

11. Prove that if A, B, C are matrices such that $AC = CA$ and $BC = CB$, then $(AB + BA)C = C(AB + BA)$ where are scalar

12. If $V = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ find V^2 and V^3 . Examine some products of the kind $V I, V I, V^2 I$

13. Prove that if A and B are such that AB and BA exist, then AB and BA have the same trace

14. Define (i) a scalar matrix, (ii) a zero matrix, (iii) the sum of two matrices when it exists

15. If $A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ 3 & 0 \\ 0 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$

Find $(AB)C$, hence or otherwise write down $A(BC)$

16 If $A = \begin{bmatrix} 3 & 4 \\ -2 & -3 \end{bmatrix}$

show that A^2 and A^4 are identity matrices

[This is an interesting example though A is not I , $A^2, A^4, \dots, A^{2n}$ are all identity matrices]

17 $A = \begin{bmatrix} 7 & 1 \\ 0 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$
 $C = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}$

work out the following

(i) $(A + B)(C + D)$ (ii) $(C + D)(A + B)$

(iii) $A^2 - B^2$ (iv) $C^2 + D^2$

(v) Under what circumstances can we write

$$(A + B)(A - B) = A^2 - B^2$$

where A and B are square matrices?

18 If $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ find $B^{-1}B$ and $(B^{-1}B)^{-1}$

Hence find $A = I_3 - B(B^{-1}B)^{-1}B$ where I_3 (3×3) is an identity matrix and show that A is an idempotent matrix

19 Given $A^2 + B^2 = (A + B)^2$ where $A = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$

and $B = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$ find a and b

20 Given $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}$

compute $A = I_4 - X(X^T X)^{-1}X^T$

Show that A is an idempotent matrix

5.10 Determinants

Determinant of a square matrix. To each square matrix there corresponds a determinant. This determinant is written by enclosing the elements of the matrix by vertical bars.

Thus the third order determinant of the matrix

MATRICES AND DETERMINANTS

$$\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \text{ is } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Value of a determinant The value of a determinant can be expressed as a single number

The second order determinant

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1$$

The third order determinant

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1 (b_2 c_3 - b_3 c_2) - b_1 (a_2 c_3 - a_3 c_2) + c_1 (a_2 b_3 - a_3 b_2)$$

Minors and Cofactors It is to be noted that a_1 is multiplied by a lower order determinant $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ obtained by deleting the column and row containing a_1 . Similarly, for the other terms, $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ is termed as minor of a_1 in the original matrix. There will be as many minors as are the elements of the matrix.

Thus, the minor of b_2 will be obtained by deleting the 2nd row and 2nd column

$$\text{In } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \quad \text{minor of } b_2 = \begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$$

The *cofactor* of an element is the minor of that element multiplied by $(-1)^{i+j}$ when i th row and j th column have been deleted since they contain the element.

Now, a_1 lies in the 1st row and 1st column, hence its cofactor is

$$(-1)^{1+1} \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} \quad \text{Similarly } b_3 \text{ lies in 3rd row and 2nd column, hence}$$

$$\text{its cofactor is } (-1)^{3+2} \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

$$\text{thus } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 A_{11} + b_1 A_{12} + c_1 A_{13}$$

$$\text{Where } A_{11} = \text{cofactor of } a_1 = (-1)^{1+1} \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$$

$$A_{12} = \text{cofactor of } b_1 = (-1)^{1+2} \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix}$$

$$A_{13} = \text{cofactor of } c_1 = (-1)^{1+3} \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

More generally the determinant can be written as

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Hence its value is $a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$

This is when we take elements of the 1st row for expansion. We can take any column or row and find the value of the determinant.

Thus when we take 2nd row and 3rd column, the value of the determinant will be $a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$ and $a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33}$ respectively where A_{21} is the cofactor of a_{21} and so on.

Value of a fourth order determinant will be

$$a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + a_{14}A_{14}$$

where A_{11} is the cofactor (of 3rd order) of a_{11} and so on.

Ex 25 Find the value of the determinant

$$\begin{vmatrix} 3 & 4 \\ 10 & -2 \end{vmatrix}$$

The value of the determinant is $(3) \times (-2) + (-4) \times (10) = -46$

Ex 26 Find the value of the determinant

$$\begin{vmatrix} 3 & 4 & 7 \\ 2 & 1 & 3 \\ 7 & 2 & 1 \end{vmatrix}$$

Expanding the determinant by 1st column we have the value of determinant

$$D = 3 \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 7 \\ 2 & 1 \end{vmatrix} + 7 \begin{vmatrix} 4 & 7 \\ 1 & 3 \end{vmatrix}$$

$$= 3(1-6) - 2(4-14) + 7(12-7) = -15 + 20 + 35 = 40$$

We can expand the determinant by any other row or column, value of the determinant will come out to be same. Let us expand it by the first row. Then

$$D = 3 \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 4 \begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix} + 7 \begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix}$$

$$= 3(1-6) - 4(2-21) + 7(4-7) = -15 + 76 - 21 = 40$$

EXERCISE 5.2

Find the value of the following determinants

1. $\begin{vmatrix} 4 & 2 \\ 3 & 4 \end{vmatrix}$

2. $\begin{vmatrix} -4 & 2 \\ -3 & -4 \end{vmatrix}$

3. $\begin{vmatrix} x & x \\ 1 & -x \end{vmatrix}$

4. $\begin{vmatrix} 3 & -2 \\ 1 & 0 \end{vmatrix}$

5. $\begin{vmatrix} \frac{2}{1-x} & 3 \\ \frac{1}{1-x^2} & \frac{1}{1+x} \end{vmatrix}$

6. $\begin{vmatrix} a-b & a+b \\ a+b & a-b \end{vmatrix}$

7. $\begin{vmatrix} 3 & 4 & 8 \\ 2 & 1 & 3 \\ 7 & -2 & 0 \end{vmatrix}$

8. $\begin{vmatrix} 1 & 2 & 3 \\ a & -a & b \\ -a & 0 & -b \end{vmatrix}$

9. $\begin{vmatrix} 1 & 3 & 1 \\ 2 & 5 & 4 \\ 6 & 1 & 0 \end{vmatrix}$

10. $\begin{vmatrix} x & 1 & 2 \\ 2 & x & 2 \\ 3 & 1 & x \end{vmatrix}$

11. $\begin{vmatrix} 3 & 4 & 7 \\ 2 & 1 & 3 \\ -5 & -1 & 2 \end{vmatrix}$

12. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

13. $\begin{vmatrix} 2 & 2 & 4 & 9 \\ 4 & 1 & 0 & 2 \\ 4 & 1 & 0 & 0 \\ 3 & 2 & 1 & 1 \end{vmatrix}$

14. $\begin{vmatrix} 2 & 5 & -3 & -2 \\ -2 & -3 & 2 & -5 \\ 1 & 3 & -2 & 2 \\ -1 & -6 & 4 & 3 \end{vmatrix}$

Some Properties of Determinants

Property 1 If all the rows of a determinant are changed into columns or vice versa, value of determinant remains unchanged.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{23} & a_{33} \end{vmatrix}$$

If the matrix of original determinant is A then that of the changed one is transpose of A i.e. A' hence $|A| = |A'|$

That is the value of the determinants of matrix and its transpose are same

For example

$$\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 1 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & 0 \\ 0 & 0 & 3 \end{vmatrix}$$

Property 2 If any two rows (or columns) are interchanged the sign of determinant changes. For example,

$$\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 1 & 0 & 3 \end{vmatrix} = (-) \begin{vmatrix} 3 & 1 & 0 \\ 4 & 2 & 0 \\ 0 & 1 & 3 \end{vmatrix} \quad \text{[by interchanging columns (1) and (2)]}$$

$$(-) \begin{vmatrix} 3 & 1 & 0 \\ 4 & 2 & 0 \\ 0 & 1 & 3 \end{vmatrix} = (-) \begin{vmatrix} 4 & 2 & 0 \\ 3 & 1 & 0 \\ 0 & 1 & 3 \end{vmatrix} \quad \text{[by interchanging rows (1) by (2)]}$$

Property 3 If any column (or row) is shifted over any adjacent columns (or row) then the absolute value of the determinant does not change but its sign changes if the jumped columns (or rows) are odd in number

$$\begin{vmatrix} 2 & 1 & -2 & 2 \\ 4 & 3 & 1 & 1 \\ 0 & 2 & 0 & 1 \\ 3 & 4 & 3 & 1 \end{vmatrix} = (-)^2 \begin{vmatrix} 1 & -2 & 2 & 2 \\ 3 & 1 & 4 & 1 \\ 2 & 0 & 0 & 1 \\ 4 & 3 & 3 & 1 \end{vmatrix}$$

(The sign does not change because jumped columns are 2 in number)

$$= (-)^3 \begin{vmatrix} -2 & 2 & 2 & 1 \\ 1 & 4 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 3 & 3 & 1 & 4 \end{vmatrix}$$

[Sign changes since jumped columns are 3 (odd)]

Property 4 If there is any common factor in the elements of any one column or row, it can be taken out. For example

$$\begin{vmatrix} 4 & 1 & 2 \\ 6 & 2 & 2 \\ 12 & 3 & 4 \end{vmatrix} = 3 \begin{vmatrix} 1 & 1 & 2 \\ 2 & 2 & 2 \\ 4 & 3 & 4 \end{vmatrix}$$

[3 is common factor in the 1st column]

$$= 2 \times 3 \begin{vmatrix} 1 & 1 & 2 \\ 1 & 1 & 1 \\ 4 & 3 & 4 \end{vmatrix} \quad [2 \text{ is common factor in the 2nd column}]$$

Property 5 If all the elements of any one row (or column) are multiplied (or divided) by K , then value of the determinant is also multiplied (or divided) by K . This property follows from Property 4

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 3 & 4 & 2 \end{vmatrix} \quad \text{multiply first column by 2 we have}$$

$$\begin{vmatrix} 2 & 2 & 3 \\ 4 & 0 & 1 \\ 6 & 4 & 3 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 3 & 4 & 2 \end{vmatrix}$$

Property 6 If any two columns (or any two rows) of a determinant are exactly identical, the value of the determinant is zero. For example

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{vmatrix} = 0 \quad [\text{since row 1 and 3 are identical}]$$

$$\text{and } \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 1 & 2 & 4 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 & 3 \\ 2 & 2 & 1 \\ 1 & 1 & 4 \end{vmatrix} \\ = 0 \quad [\text{columns 1 and 2 are identical}]$$

Property 7 If any column (or row) or a multiple of any column (or row) is added or subtracted from any other column (or row) the value of the determinant remains unchanged. Let

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 0 \\ 6 & 2 & 12 \end{vmatrix} \quad \begin{array}{l} \text{multiply column (1) by 2 and subtract} \\ \text{from column (2)} \end{array}$$

$$\begin{vmatrix} 1 & 2-2 & 3 \\ 2 & 4-4 & 0 \\ 6 & 2-12 & 12 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 3 \\ 2 & 0 & 1 \\ 6 & -10 & 12 \end{vmatrix} \quad \begin{array}{l} (\text{multiply col 1 by 3 and} \\ \text{subtract from col 3}) \end{array}$$

$$\begin{vmatrix} 1 & 0 & 3-3 \\ 2 & 0 & 0-6 \\ 6 & -10 & 12-18 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 0 & -6 \\ 6 & -10 & -6 \end{vmatrix}$$

Now, expanding by first column, we have

$$\begin{aligned} |A| &= 1 \begin{vmatrix} 0 & -6 \\ -10 & -6 \end{vmatrix} + 0 + 0 \\ &= 1(0 - 60) = -60 \end{aligned}$$

It is to be noted that the column (row) which is multiplied remains as it is.

Property 8. If a determinant is of the form

$$\begin{vmatrix} a_1+b_1 & c_1 & d_1 \\ a_2+b_2 & c_2 & d_2 \\ a_3+b_3 & c_3 & d_3 \end{vmatrix} \quad \text{it can be written as the sum of two determinants;}$$

$$\begin{vmatrix} a_1 & c_1 & d_1 \\ a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \end{vmatrix} + \begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix}$$

$$\text{For example; } \begin{vmatrix} 4 & 2 & 1 \\ 7 & 3 & 2 \\ 10 & 4 & 3 \end{vmatrix} = \begin{vmatrix} 3+1 & 2 & 1 \\ 5+2 & 3 & 2 \\ 7+3 & 4 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 2 & 1 \\ 5 & 3 & 2 \\ 7 & 4 & 3 \end{vmatrix} + \begin{vmatrix} *1 & 2 & 1* \\ 2 & 3 & 2 \\ 3 & 4 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 2+1 & 2 & 1 \\ 3+2 & 3 & 2 \\ 4+3 & 4 & 3 \end{vmatrix} + \text{zero}$$

$$= \begin{vmatrix} *2 & *2 & 1 \\ 3 & 3 & 2 \\ 4 & 4 & 3 \end{vmatrix} = \begin{vmatrix} *1 & 2 & 1* \\ 2 & 3 & 2 \\ 3 & 4 & 3 \end{vmatrix}$$

$$= 0 + 0 = 0.$$

[The *columns are identical]

These properties, specially property 7, provide very powerful methods for calculating the value of a determinant.

Ex. 27. Prove without expanding,

$$\begin{vmatrix} 19 & 17 & 15 \\ 9 & 8 & 7 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 19 & 17 & 15 \\ 9 & 8 & 7 \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow[\text{Col (2) - Col (3)}]{\text{Col (1) - Col (3)}} \begin{vmatrix} 4 & 2 & 15 \\ 2 & 1 & 7 \\ 0 & 0 & 1 \end{vmatrix}$$

(Note, column (3) is kept as it is)

$$\begin{array}{l} \text{Taking 2 com-} \\ \text{mon from} \\ \text{column (2),} \end{array} \quad 2 \begin{vmatrix} 2 & 2 & 15 \\ 1 & 1 & 7 \\ 0 & 0 & 1 \end{vmatrix} = 0$$

Since Col (1) and Col (2) are exactly same. The same result can be obtained by row operations

Ex. 28 Evaluate $\begin{vmatrix} 2 & 45 & 55 \\ 1 & 29 & 32 \\ 3 & 68 & 87 \end{vmatrix}$

Expanding the given determinant by first column we get

$$= 2 \begin{vmatrix} 29 & 32 \\ 68 & 87 \end{vmatrix} - 1 \begin{vmatrix} 54 & 55 \\ 68 & 87 \end{vmatrix} + 3 \begin{vmatrix} 45 & 55 \\ 29 & 32 \end{vmatrix}$$

$$= 2(347) - 1(175) + 3(-155)$$

$$= 694 - 175 - 465 = 54$$

Ex 29 Evaluate $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} \xrightarrow[\text{Col (2) - Col (1)}]{\text{Col (3) - Col (1)}} \begin{vmatrix} 1 & 1 & 1 \\ a & b & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix}$$

Expanding by row (1), we have

$$\begin{aligned} (1) \quad & \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix} = (b-a)(c-a) \begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix} \\ &= (b-a)(c-a)(c+a-b-a) \\ &= (b-a)(c-a)(c-b) \\ &= (a-b)(b-c)(c-a) \end{aligned}$$

Ex 30 Find the value of

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

Here Add row (2) and row (3) and subtract from row (1),

$$\begin{vmatrix} (b+c)-(b+c) & a-(2c+a) & a-(2b+a) \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = \begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$= -2 \begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} \xrightarrow[r(3)-r(1)]{r(2)-r(1)} -2 \begin{vmatrix} 0 & c & b \\ b & a & 0 \\ c & 0 & a \end{vmatrix}$$

$$\xrightarrow[\text{Co (2) \& Col (3)}]{\text{Interchange}} 2 \begin{vmatrix} 0 & b & c \\ b & 0 & a \\ c & a & 0 \end{vmatrix}$$

$$= 2 \left[0 - b \begin{vmatrix} b & a \\ c & 0 \end{vmatrix} + c \begin{vmatrix} b & 0 \\ c & a \end{vmatrix} \right]$$

$$= 2 [-b \times -c - a + ba] = 4abc.$$

Ex. 31. Evaluate $\begin{vmatrix} 0 & ab^2 & ac^2 \\ a^2b & 0 & bc^2 \\ a^2c & b^2c & 0 \end{vmatrix}$

$$\begin{vmatrix} 0 & ab^2 & ac^2 \\ a^2b & 0 & bc^2 \\ a^2c & ab^2 & 0 \end{vmatrix} = abc \begin{vmatrix} 0 & b^2 & c^2 \\ a^2 & 0 & c^2 \\ a^2 & b^2 & 0 \end{vmatrix}$$

$$= (abc) (a^2b^2c^2) \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$a^3b^3c^3 [-1(0-1)+1(1-0)] = 2a^3b^3c^3.$$

Ex. 32 Prove that $\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3$

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = \begin{vmatrix} 2a+2b+2c & a & b \\ 2a+2b+2c & b+c+2a & b \\ 2a+2b+2c & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c) \begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c) \begin{vmatrix} 1 & a & b \\ 0 & b+c+a & 0 \\ 0 & 0 & c+a+b \end{vmatrix}$$

$$= 2(a+b+c) [(b+c+a)^2] = 2(a+b+c)^3$$

EXERCISE 5.3

Prove that :

$$1. \begin{vmatrix} 1 & a & 1 \\ 0 & a+b & b \\ 0 & a & a+b \end{vmatrix} = \begin{vmatrix} 0 & 1 & 1 \\ 1 & a & 0 \\ 1 & 0 & b \end{vmatrix}$$

$$2. \begin{vmatrix} 1 & b & c \\ -b & 0 & a \\ -c & -a & 0 \end{vmatrix} = 0$$

$$3. \begin{vmatrix} 1 & 1 & 1 \\ x & 2y & 3z \\ x^2 & 4y^2 & 9z^2 \end{vmatrix} = (x-2y)(2y-3z)(3z-x)$$

$$4. \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc+ab+bc+ca$$

$$5. \begin{vmatrix} y & a & a \\ a & y & a \\ a & a & y \end{vmatrix} = (y-a)^2 \begin{vmatrix} 1 & 0 & a \\ 0 & 1 & a \\ -1 & -1 & y \end{vmatrix}$$

$$6. \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^2$$

[Hint. Add row (2) and row (3) to row (1) and take out (a+b+c) common]

$$7. \begin{vmatrix} 1 & ab & c^2 \\ 1 & bc & a^2 \\ 1 & ca & b^2 \end{vmatrix} = (b-c)(c-a)(b-a)(a+b+c)$$

$$8. \begin{vmatrix} x^3 & x^2 & 1 \\ y^3 & y^2 & 1 \\ z^3 & z^2 & 1 \end{vmatrix} = (y-z)(z-x)(x-y)(x^2+y^2+xy)$$

[Hint. $(x^3-y^3) = (x-y)(x^2+y^2+xy)$]

9. If A is the $n \times n$ matrix whose diagonal elements are all 'a' and whose off diagonal elements are all b, where a and b are any two real numbers, find the value of $|A|$

10. If (x_1, y) and (x_2, y_2) are points on the x, y plane, show that the equation

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

represents a straight line through these two points

11. Prove that

$$\begin{vmatrix} a_{11} & a_{12} & a_{1n} \\ 0 & a_{22} & a_{2n} \\ \vdots & \vdots & \vdots \\ 0 & a_{m2} & a_{mn} \end{vmatrix} = a_{11} a_{22} \dots a_{mn}$$

12. Prove that

$$\begin{vmatrix} a+3 & -1 & 1 \\ 5 & a-3 & 1 \\ 6 & -6 & a+4 \end{vmatrix} = (a+2)(a-2)(a+4)$$

13. Show that $x = 2$ is a root of the equation

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+4 \end{vmatrix} = 0$$

find the other roots

14. Show that $x = 1$ is a root of

$$\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$$

Prove that

$$15. \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$16. \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

$$17. \begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = (x+y+z)(x-z)^2$$

$$18. \begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix} = 0$$

$$19 \quad \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & c-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

$$20 \quad \begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$$

5.11 Rank of Matrix

Non singular and Singular Matrices The square matrix A is called a *singular* matrix if the corresponding determinant $|A| = 0$

The square matrix $[A]$ is called a *non singular* matrix if corresponding determinant $|A| \neq 0$ [is not equal to zero]

Rank* of a Matrix The rank of matrix A denoted by $\rho(A)$ is equal to the order of the highest order non singular square matrix contained in A

Ex. 33 The rank of matrix $\begin{pmatrix} 5 & 2 \\ -2 & -3 \end{pmatrix}$ is 2, because this second order matrix corresponds to the determinant $\begin{vmatrix} 5 & 2 \\ -2 & -3 \end{vmatrix}$ which is not zero hence this given matrix contains a non singular matrix of order 2

Ex. 34 The rank of matrix $\begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix}$ is 1, the highest order determinant $\begin{vmatrix} 3 & 2 \\ 6 & 4 \end{vmatrix}$ is zero, next to this would be of the 1st order which is non zero

Ex. 35 The rank of matrix $\begin{pmatrix} 3 & 3 & -2 \\ -2 & 0 & 5 \end{pmatrix}$ is 2, for it contains no 3×3 matrix but contains at least one non singular 2×2 matrix

Ex. 36. The rank of

$$(i) \quad \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \text{ is 4, because } \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \neq 0$$

$$(ii) \quad \begin{vmatrix} 6 & 0 & 2 & 2 \\ 3 & 1 & 0 & 2 \\ 4 & 1 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{vmatrix} \text{ is 3, because } \begin{vmatrix} 6 & 0 & 2 & 2 \\ 3 & 1 & 0 & 2 \\ 4 & 1 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{vmatrix} = 0$$

* Rank of a matrix assumes much importance in solving a system of simultaneous equations. See section 5.14

but at least one 3rd order determinant is not zero

$$(iii) \begin{vmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 3 & 6 & 12 \end{vmatrix} \text{ is } 1, \text{ for } \begin{vmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 3 & 6 & 12 \end{vmatrix} = 0$$

and, also, all the 2nd order determinant are zero

$$(iv) \text{ Rank of } \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ is also zero}$$

From the above examples, we can make the following generalizations

1 The rank of matrix is not related in any way to the number of zero elements in it

2 The rank of matrix cannot exceed the number of its rows or number of its column—whichever is less (since a determinant has equal number of rows and columns)

3 The rank of a column matrix of any number of rows (matrix 3×1) is at the most 1, while rank of a matrix of 3×50 is at the most 3

4 The rank of a matrix is at least one, unless the matrix has all zero elements, i.e., when it is a null matrix. The rank of null matrix is zero

5 Rank of the transpose of a matrix A is the same as the rank of A

$$\text{Ex. 37. Find the rank of } A = \begin{pmatrix} 1 & 3 & 4 & -2 \\ 2 & 6 & 8 & -4 \\ 3 & 0 & 3 & 3 \end{pmatrix}$$

The straightforward way to find the rank of any matrix is to look for non zero determinant of the highest order which the given matrix contains. Here, the highest order determinant can be of order 3, hence the rank of the given matrix can at the most be 3. To see whether it is actually 3 we form all the possible determinants of this matrix—which are 4 in number—as under

$$(1) \begin{vmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 3 & 0 & 3 \end{vmatrix} \quad (2) \begin{vmatrix} 3 & 4 & -2 \\ 6 & 8 & -4 \\ 0 & 3 & 3 \end{vmatrix}$$

$$(3) \begin{vmatrix} 1 & 3 & -2 \\ 3 & 6 & -3 \\ 3 & 0 & 3 \end{vmatrix} \text{ and, } (4) \begin{vmatrix} 1 & 4 & -2 \\ 2 & 8 & -4 \\ 3 & 3 & 3 \end{vmatrix}$$

It can be shown by expansion that each of these determinants is zero. Hence the rank of the given matrix can now be at the most 2. To see whether it is actually 2, consider all the possible determinants of 2nd order

All the 2nd order determinants formed from the first two rows of the given

matrix are zero, nevertheless 2nd order determinants formed from the 1st and

3rd rows are not zero e.g., $\begin{vmatrix} 1 & 3 \\ 3 & 0 \end{vmatrix} = -9 \neq 0$

Hence, the rank of the given matrix is 2, i.e. $\rho(A) = 2$

Ex. 38. consider the rank of the following matrices

$$(i) \begin{bmatrix} 8 & 7 & 0 \\ 0 & 7 & 3 \\ 2 & 5 & 2 \\ -8 & 2 & 3 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 6 & 3 & 5 \\ -10 & 2 & 8 \\ 5 & 2 & 3 \end{bmatrix}$$

$$\text{and (iii)} \begin{bmatrix} 1 & 2 & 3 \\ -2 & 4 & 6 \\ -3 & -6 & -9 \end{bmatrix}$$

(i) Atleast one non zero determinant of highest order of the square matrix,

is i.e., $\begin{vmatrix} 8 & 7 & 0 \\ 0 & 7 & 3 \\ 2 & 5 & 2 \end{vmatrix}$ is not zero

Rank of the given matrix is 3

(ii) Determinant of the matrix, i.e., $\begin{vmatrix} 6 & 3 & 5 \\ -10 & 2 & 8 \\ 5 & 2 & 3 \end{vmatrix} = 0$, therefore, rank

can at most be 2. To see this, we find that there is at least one non zero determinant of order $\begin{vmatrix} 2 & 8 \\ 2 & 3 \end{vmatrix} = -10 \neq 0$. Hence the rank of the given matrix is 2

(iii) Here $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ -3 & -6 & -9 \end{vmatrix} = 0$. Also each of the 2nd order determinant

is zero. Again, the matrix is not a null matrix, hence the rank of this matrix is 1

5.12. Linear Dependence (Independence) and Rank of a Matrix

The vectors or rows (columns) of a matrix are said to be linearly dependent if and only if some linear combination of them is a null vector or null row (column). The rows (columns) which are not linearly dependent are said to be linearly independent.

Consider the matrix,

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

A linear combination of the rows 1, 2, 3 is obtained by multiplying the 1st, 2nd and 3rd rows by any constants K_1 , K_2 and K_3 and adding them, where atleast one K is not zero. Thus the new row will be

$$= [\begin{matrix} K_1 a_{11} + K_2 a_{21} + K_3 a_{31} & K_1 a_{12} + K_2 a_{22} + K_3 a_{32} & K_1 a_{13} + K_2 a_{23} + K_3 a_{33} \\ C_1 & C_2 & C_3 \end{matrix}]$$

The rows are linearly dependent if all the three elements C_1 , C_2 and C_3 are equal to zero for some values of K_1 , K_2 , K_3 . Of course, K_1 , K_2 , K_3 should all not be equal to zero, at least one of these must be non-zero.

Ex. 39 Test for linear independence and find the rank of the matrix,

$$\begin{pmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 3 & 6 & 12 \end{pmatrix}$$

Multiplying row 1 (R_1) and 2 (R_2) by -1 and adding to row 3 (R_3), we have

$$-1R_1 - 1R_2 + R_3 = 0 \quad \dots (1)$$

$$\text{Similarly,} \quad -2R_1 + R_2 = 0 \quad \dots (2)$$

$$-3R_1 + R_3 = 0 \quad \dots (3)$$

Hence (1) shows that rows R_1 , R_2 , R_3 are linearly dependent, (2) shows that R_1 and R_2 are linearly dependent, (3) shows that R_1 and R_3 are linearly dependent. Considering columns, we have

$$2C_1 - C_2 + C_3 = 0$$

$$-2C_1 + C_2 = 0$$

$$-4C_1 + C_3 = 0$$

Hence, C_1 , C_2 , C_3 are all linearly dependent

Thus

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 3 & 6 & 12 \end{bmatrix} \text{ can be written as } \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, determinants up to the 2nd order will be zero, only the 1st order determinant will be non-zero. Thus, the rank is 1.

Definition of Rank.

The rank of a matrix $\rho(A)$, is the maximum number of linearly independent rows (columns) of A . As we have already seen the rank of a matrix cannot exceed the number of rows or columns whichever is smaller.

Ex. 40. Test for linear dependence and find rank of (A), if

$$A = \begin{pmatrix} 6 & 3 & 5 \\ -10 & 2 & 8 \\ 5 & 2 & 3 \end{pmatrix}$$

$\therefore C_1 R_1 + C_2 R_2 + C_3 R_3 = 0$ for linear dependence, i.e., if R_1, R_2, R_3 are linearly dependent then

$$6C_1 - 10C_2 + 5C_3 = 0$$

$$3C_1 + 2C_2 + 2C_3 = 0$$

$$5C_1 + 8C_2 + 3C_3 = 0$$

These equations will be satisfied for

$$C_1 = -10, C_2 = 1, C_3 = 14$$

hence R_1, R_2, R_3 are linearly dependent. However, no linear combination of any two rows will be zero. Hence $\rho(A) = 2$.

EXERCISE 5.4

1. Show that the following matrices are singular,

$$(a) \begin{bmatrix} 6 & 3 \\ 8 & 4 \end{bmatrix} \quad (b) \begin{bmatrix} 5 & 2 & 1 \\ -6 & 1 & 4 \\ 6 & 3 & 0 \end{bmatrix}$$

2. Determine the rank of the following matrices

$$(a) \begin{bmatrix} 1 & 3 & 3 \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix} \quad (b) \begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 0 \\ -3 & 1 & 2 \end{bmatrix} \quad (c) \begin{bmatrix} 5 & 7 & 3 \\ 4 & 8 & 9 \\ 8 & 4 & -15 \end{bmatrix}$$

$$(d) \begin{bmatrix} 4 & 0 & 1 \\ 19 & 1 & 3 \\ 5 & 4 & 7 \end{bmatrix} \quad (e) \begin{bmatrix} 7 & 0 & 3 \\ 8 & -45 & 2 \\ 4 & 9 & 2 \end{bmatrix} \quad (f) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ -2 & -3 & -1 \end{bmatrix}$$

$$(g) \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix} \quad (h) \begin{bmatrix} 2 & -1 & 3 & 1 & 5 \\ 3 & 2 & 0 & 2 & 1 \end{bmatrix}$$

$$(i) \begin{bmatrix} 1 & -2 & 3 & 7 & 6 \\ -1 & -1 & -3 & 7 & 4 \\ 0 & 3 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

3. If $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ find the ranks of A^4, A^2, A^3 and A^4

4. Find the rank of a matrix whose every element is unity
5. If A is a non zero $(n \times 1)$ column matrix, while B is a non zero row matrix, show that $\rho(BA)$ is unity and $\rho(AB)$ is n
6. Show that the rank of a matrix does not alter on affixing any number of additional rows or columns of zeros
7. Find the ranks of $A, B, A + B, AB$ and BA in the following cases

$$(i) \quad A = \begin{bmatrix} 1 & 5 & 4 \\ 2 & 3 & 2 \\ 3 & 13 & 10 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 3 \end{bmatrix}$$

$$(ii) \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 13 & 5 \end{bmatrix}$$

8. Show that $n + 1$ vectors each of dimensions n must necessarily be linearly dependent

5.13. Inverse of a Matrix.

The operation of dividing one matrix directly by another does not exist in matrix theory, but equivalent of division of a unit matrix by any square matrix can be accomplished (in most cases) by a process known as "Inversion of a Matrix".

In ordinary algebra, if $x \times y = 1$ then $x = \frac{1}{y}$ or we say that y is inverse of x or x is inverse of y . The product of quantity x and its inverse is one (if $\frac{1}{x}$ exists or $x \neq 0$)

Similarly, if A is a matrix and A^{-1} its inverse, then their product must be equal to the identity matrix

$$A \times A^{-1} = I \quad (I \text{ is identity matrix})$$

$$A^{-1} = \frac{1}{A}$$

The concept of inverse matrix is very useful in economics in solving simultaneous equations in input output analysis and regression analysis. Hence the process of finding the inverse of a matrix should be understood well.

We give two methods of finding the inverse of a given square matrix.

- (i) Gauss elimination method
- (ii) Using Adjoint matrices—Co-factor method

I Gauss Elimination Method of Matrix Inversion

If A is a square matrix of order $[A/I]$ and I is an identity matrix of order n then a particular form of new matrix $[A | I]$ (by placing I matrix by the side of matrix A) is of order $n \times 2n$. A has inverse only and only if $[A/I]$ can be transformed to $[I/A^{-1}]$.

The method, thus, consists in placing an identity matrix of the same order alongside the original matrix A which is required to be inverted. Then by performing the same row elementary transformations on both A and I portions, we can transform [reduce] A into an identity matrix, the transformed identity matrix will then become the inverse matrix.

Ex. 41. Obtain inverse of matrix, $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$

$$[A | I] = \begin{bmatrix} 3 & 4 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 3 & 4 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & -1 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2} \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\xrightarrow{R_1 - \frac{1}{2}R_2} \begin{bmatrix} 1 & 0 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

which is, in fact, $[I | A^{-1}]$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}$$

Check $A \times A^{-1}$ should be I (Identity matrix)

$$\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Hence, $\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}$ is inverse of $\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ and vice versa.

Ex 42 Find the inverse of $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 0 & -3 \\ 3 & 4 & 0 \end{bmatrix}$

$$[A | I] = \begin{bmatrix} 2 & -2 & 3 & 1 & 0 & 0 \\ 1 & 0 & -3 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} R \\ \\ 2 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & 3/2 & 1/2 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} R \\ R_1 \\ 3R_1 \end{matrix} \quad \begin{matrix} R \\ -3R \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -1 & 1/2 & 1/2 & 0 & 0 \\ 0 & +1 & 9/2 & -1/2 & 1 & 0 \\ 0 & 7 & 9/2 & -3/2 & 0 & 1 \end{bmatrix} \quad \begin{matrix} R + R \\ R_1 \\ 7R_1 \end{matrix} \quad \begin{matrix} R + R \\ \\ 7R \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & -3 & 0 & 1 & 0 \\ 0 & 1 & 9/2 & 1/2 & 1 & 0 \\ 0 & 0 & 27 & 2 & -7 & 1 \end{bmatrix} \quad \begin{matrix} R \\ \\ 7 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 9/2 & 1/2 & +1 & 0 \\ 0 & 0 & 1 & 2/27 & 7/27 & 1/27 \end{bmatrix} \quad \begin{matrix} R + 3 \\ R + 9 \\ R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 6/27 & 6/27 & 1/9 \\ 0 & 1 & 0 & 1/6 & 1/6 & 1/6 \\ 0 & 0 & 1 & 2/27 & -7/27 & 1/27 \end{bmatrix} = [I | A^{-1}]$$

$$A^{-1} = \begin{bmatrix} 6/27 & 6/27 & 1/9 \\ -1/6 & -1/6 & 1/6 \\ 2/27 & -7/27 & 1/27 \end{bmatrix} = \frac{1}{27} \begin{bmatrix} 6 & 6 & 3 \\ -9 & -9 & 9 \\ 2 & -7 & 1 \end{bmatrix}$$

(Check If we multiply $A A^{-1}$, i.e.

$$A A^{-1} = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 0 & -3 \\ 3 & 4 & 0 \end{bmatrix} \begin{bmatrix} 6/27 & 6/27 & 1/9 \\ -1/6 & -1/6 & 1/6 \\ 2/27 & -7/27 & 1/27 \end{bmatrix}, \text{ we get}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence $\frac{1}{27} \begin{bmatrix} 6 & 6 & 3 \\ -9 & -9 & 9 \\ 2 & -7 & 1 \end{bmatrix}$ is inverse of $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 0 & -3 \\ 3 & 4 & 0 \end{bmatrix}$

(Always remember that only row transformations are to be performed for finding the inverse matrix by Gauss elimination method)

Co-Factor Method of Matrix Inversion

Suppose we are given matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

and that the determinant of the matrix, i.e. $|A| \neq 0$ (i.e. matrix A is non singular) then,

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$\text{i.e., } A^{-1} = \frac{\text{adj } A}{|A|}$$

where adjoint matrix of A is the transpose of the matrix of co factors of the elements of matrix A

when, $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$$\text{Adj } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

Where A_{11} is the co-factor of element a_{11} in A , i.e., $\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

Where A_{21} is the co-factor of element a_{21} in A , i.e., $-\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$

Where A_{31} is the co-factor of element a_{31} in A , i.e., $\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$

Exactly in the same way,

A_{12} and A_{13} are the co-factors of elements a_{12} and a_{13} in A

A_{22} and A_{23} are the co-factors of elements a_{22} and a_{23} in A

and A_{32} and A_{33} are the co-factors of elements a_{32} and a_{33} in A

From the above definition of "Inverse of matrix", it is clear that inversion of matrix by this method is possible if and only if

- (a) Matrix (whose inversion is required) is a square matrix or else we shall not be able to form the determinant of the matrix.
 (b) The determinant of the matrix should not be zero.

Let us explain this method of inversion by taking an example.

Ex 43 Find A^{-1} if

$$A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix} = 2$$

$$A_{11} = 2 \quad A_{12} = -1$$

$$A_{21} = -4 \quad A_{22} = 3$$

Hence matrix of co-factors is

$$\begin{bmatrix} 2 & -1 \\ -4 & 3 \end{bmatrix}$$

$$\text{Therefore, } \text{Adj } A = \begin{bmatrix} 2 & -4 \\ -1 & 3 \end{bmatrix}$$

We know, $A^{-1} = \frac{\text{Adj } A}{|A|}$, therefore

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 2 & -4 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}$$

which is the same result as obtained by method I

$$\text{Ex 44} \quad \text{If } A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, C = \begin{bmatrix} 2 \\ -3 \end{bmatrix} \text{ and } I$$

is the unit matrix of order 2,

find (a) $B - 4A - 2I$ (b) X if $AX = C$

$$(a) \quad \text{Given } B = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$$

$$4A = 4 \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 8 & -4 \\ 16 & 12 \end{bmatrix}$$

$$2I = 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$B - 4A - 2I = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} - \begin{bmatrix} 8 & -4 \\ 16 & 12 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & 4 \\ -18 & 13 \end{bmatrix}$$

$$(b) \quad AX = C \quad X = A^{-1}C$$

$$\text{Given } A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix} \quad A^{-1} = \frac{1}{10} \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$$

Since $A_{11} = 3$ $A_{12} = -4$ $A_{21} = 1$ $A_{22} = 2$ and $|A| = 10$

$$\begin{aligned} \text{therefore } X &= \begin{bmatrix} \frac{3}{10} & \frac{1}{10} \\ \frac{-2}{5} & \frac{1}{5} \end{bmatrix} \begin{bmatrix} 2 \\ -3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{3}{5} - \frac{3}{10} \\ \frac{-4}{5} - \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{3}{10} \\ \frac{7}{5} \end{bmatrix} \end{aligned}$$

Ex 45 Obtain the inverse of matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$$

We have defined $A^{-1} = \frac{1}{|A|} [\text{Adj } A]$

$$\text{Here } |A| = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{vmatrix} = 3$$

$$A_{11} = \begin{vmatrix} 3 & 5 \\ 5 & 12 \end{vmatrix} = 11 \quad A_{12} = (-) \begin{vmatrix} 1 & 5 \\ 1 & 12 \end{vmatrix} = -7 \quad A_{13} = \begin{vmatrix} 1 & 3 \\ 1 & 5 \end{vmatrix} = 2$$

$$A_{21} = (-) \begin{vmatrix} 2 & 3 \\ 5 & 12 \end{vmatrix} = -9 \quad A_{22} = \begin{vmatrix} 1 & 3 \\ 1 & 12 \end{vmatrix} = 9 \quad A_{23} = (-) \begin{vmatrix} 1 & 2 \\ 1 & 5 \end{vmatrix} = 3$$

$$A_{31} = \begin{vmatrix} 2 & 3 \\ 3 & 5 \end{vmatrix} = 1 \quad A_{32} = (-) \begin{vmatrix} 1 & 3 \\ 1 & 5 \end{vmatrix} = -2$$

$$\text{and } A_{33} = \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1$$

The value of the determinant can be found by any row or column. The matrix of co-factors is given by

$$\begin{bmatrix} 11 & -7 & 2 \\ -9 & 9 & -3 \\ 1 & -2 & 1 \end{bmatrix}$$

Therefore, Adjoint $A = \begin{bmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{bmatrix}$

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 11/3 & -3 & 1/3 \\ -7/3 & 3 & -2/3 \\ 2/3 & -1 & 1/3 \end{bmatrix}$$

Ex 46 Calculate the inverse of matrix

$$A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 0 & -3 \\ 3 & 4 & 0 \end{bmatrix}$$

$$A_{11} = 12, A_{12} = 29, A_{13} = 4, A_{21} = 12, A_{22} = -9, A_{23} = -14, A_{31} = 6, A_{32} = 9, A_{33} = 1$$

Hence, $|A| = \begin{vmatrix} 2 & -2 & 3 \\ 1 & 0 & -3 \\ 3 & 4 & 0 \end{vmatrix} = 54$

and, $\text{Adj } A = \begin{bmatrix} 12 & 12 & 6 \\ 9 & -9 & 9 \\ 4 & -14 & 2 \end{bmatrix}$

$$A^{-1} = \frac{1}{54} \begin{bmatrix} 12 & 12 & 6 \\ 9 & -9 & 9 \\ 4 & -14 & 2 \end{bmatrix} = \frac{1}{27} \begin{bmatrix} 6 & 6 & 3 \\ -9/2 & -9/2 & 9/2 \\ 2 & -7 & 1 \end{bmatrix}$$

which is the same result as obtained by method 1

Ex 47. Find the inverse of $A = \begin{bmatrix} 1 & 4 & 3 \\ 4 & 2 & 1 \\ 3 & 2 & 2 \end{bmatrix}$

Here $|A| = \begin{vmatrix} 1 & 4 & 3 \\ 4 & 2 & 1 \\ 3 & 2 & 2 \end{vmatrix} = -12$

$$\text{Adj } A = \begin{bmatrix} \begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix} - \begin{vmatrix} 4 & 3 \\ 2 & 2 \end{vmatrix} & \begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 4 & 2 \\ 3 & 2 \end{vmatrix} \\ \begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} \\ \begin{vmatrix} 4 & 2 \\ 3 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 4 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 4 \\ 4 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 & -2 \\ -5 & -7 & 11 \\ 2 & 10 & -14 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj } A = -\frac{1}{12} \begin{bmatrix} 2 & 2 & -2 \\ -5 & -7 & 11 \\ 2 & 10 & -14 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -1/6 & 1/6 & 1/6 \\ 5/12 & 7/12 & -11/12 \\ -1/6 & -5/6 & 7/6 \end{bmatrix}$$

EXERCISE 5.5

1. Prove that if

$$(a) \quad A = \begin{pmatrix} 2 & 3 \\ 4 & 2 \end{pmatrix} \text{ then } A^{-1} = \begin{pmatrix} -1/4 & 3/8 \\ 1/2 & 1/4 \end{pmatrix}$$

$$(b) \quad A = \begin{bmatrix} -1 & -5 \\ -2 & 3 \end{bmatrix} \text{ then } A^{-1} = \frac{1}{13} \begin{bmatrix} 3 & 5 \\ 2 & 1 \end{bmatrix}$$

$$(c) \quad A = \begin{pmatrix} 1 & 7 \\ 2 & 9 \end{pmatrix} \text{ then } A^{-1} = \begin{pmatrix} -9/5 & -7/5 \\ 2/5 & 1/5 \end{pmatrix}$$

$$(d) \quad A = \begin{pmatrix} 2 & -3 & 0 \\ 3 & 1 & -2 \\ -1 & 0 & -4 \end{pmatrix} \text{ then } A^{-1} = -\frac{1}{50} \begin{pmatrix} -4 & -12 & 6 \\ 14 & -8 & 4 \\ 1 & 3 & 11 \end{pmatrix}$$

$$(e) \quad A = \begin{bmatrix} 1 & -2 & 3 \\ 3 & -1 & 4 \\ 2 & 1 & -2 \end{bmatrix} \text{ then } A^{-1} = \begin{bmatrix} 2/5 & 1/5 & 1/5 \\ -14/5 & 8/5 & -1/5 \\ -1/5 & 1/5 & -1/5 \end{bmatrix}$$

$$(f) \quad A = \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 3 & 0 & 1 \end{bmatrix} \text{ then } A^{-1} = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$$

2. Find the inverse of the following matrices

$$(a) \quad A = \begin{pmatrix} 5 & 1 & 1 \\ 0 & 2 & 2 \\ 3 & 1 & 4 \end{pmatrix}$$

$$(b) \quad A = \begin{pmatrix} 4 & -1 & 1 \\ -3 & 1 & 1 \\ 1 & 1 & -1 \end{pmatrix}$$

$$(c) \quad A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 0 & -3 \\ 3 & 4 & 0 \end{bmatrix}$$

$$(d) \quad A = \begin{bmatrix} 50 & 5 & -5 \\ 25 & -17 & 10 \\ -5 & 10 & -5 \end{bmatrix}$$

$$(e) \quad A = \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 2 & 1 & 2 & 1 \\ 3 & -2 & 1 & 6 \end{bmatrix}$$

$$(f) \quad A = \begin{bmatrix} \frac{1}{6} & \frac{2}{3} & \frac{1}{2} \\ 1 & -\frac{1}{2} & \frac{1}{3} \\ \frac{1}{3} & 1 & \frac{1}{2} \end{bmatrix}$$

$$(g) \quad A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

[Hint for (f) First remove the fraction by scalar multiplication]

3 Prove that $(AB)^{-1} = B^{-1}A^{-1}$ where A and B are any two square, non singular matrices

5.14 Application of Matrices to the Solution of Linear Equations

Suppose we are required to solve the following set of simultaneous equations for x_1 , x_2 and x_3

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = h_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = h_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = h_3$$

Let us write the given equations in matrix form

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 \end{bmatrix} = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

but since

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Therefore, the given set of equations can be put as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

or,

$$\begin{array}{ccc} \downarrow & & \downarrow \\ A & & X \\ & & \downarrow \\ & & B \end{array}$$

where A is a 3×3 matrix

X is a 3×1 matrix

and, H is a 3×1 matrix

This way, the three simultaneous given equations are written in the form of a single matrix equation as $AX=H$

We can find x_1, x_2 and x_3 if we write this single matrix equation in the form

$$X = A^{-1}H$$

In general, if there are n variables and n equations then

$$AX = H$$

is the matrix presentation of the n simultaneous equations, where

A = matrix of order $n \times n$

X = col. matrix of variables or order $n \times 1$

H = col. matrix of constants of order $n \times 1$

and, then $X = A^{-1}H$,

where A is the inverse of matrix A . Remember here that only the product $A^{-1}H$ exists and the product HA^{-1} does not exist. Hence, $X \neq HA^{-1}$.

Therefore, what we now need is the inversion of the matrix formed by coefficients of x_1, x_2 and x_3 in the given equations. We directly get the required values of x_1, x_2 and x_3 by post multiplying this inverse matrix by the column of constants on the right hand side of the equality sign of the equations.

We know that computation of A^{-1} requires division of the Adjoint by the determinants of A , $|A|$. Now if the A matrix is $n \times n$ matrix and the rank of the matrix is less than A , the value of $|A|$ will be zero, and the method fails to give any unique solution to the problem.

The given system of linear equations will be consistent if the rank of the coefficient matrix is equal to the rank of the augmented matrix. If however the rank of the coefficient matrix is not equal to the rank of the augmented matrix, then the given system of equations will be inconsistent. Again in a consistent system, let r be the rank and n the number of unknowns, then the system of equations will have a unique set of solution if $r=n$ and the given system of equations will have an infinite number of solutions if $r < n$. Actually we know that the value of the determinant is zero only if any two rows or columns can be shown to be identical after scalar manipulations. Thus, the equations (at least 2) are not mutually independent. Hence an equation system involving r linear equations can give a unique solution only if the rank of the matrix A is equal to n .

Ex. 48. Solve for x and y

$$2x + 3y = 7$$

$$4x + 2y = 10$$

Putting the given equations in matrix form, we have

$$\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$$

$$\mathbf{A} \quad \mathbf{X} \quad \mathbf{B}$$

$$\mathbf{X} = \mathbf{A}^{-1} \mathbf{B}.$$

We have $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$

and $\mathbf{A}^{-1} = \begin{bmatrix} -1/4 & 3/8 \\ 1/2 & -1/4 \end{bmatrix}$

$$\mathbf{X} = \begin{pmatrix} -1/4 & 3/8 \\ 1/2 & -1/4 \end{pmatrix} \begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

$$\mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

i.e., $x = 2, y = 1$

Ex 49 Solve for x, y and z from the following set of equations

$$x - 2y + 3z = 1$$

$$3x - y + 4z = 3$$

$$2y + y - 2z = -1$$

Putting the given equations in the matrix form

$$\begin{bmatrix} 1 & -2 & 3 \\ 3 & -1 & 4 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$$

$$\mathbf{AX} = \mathbf{B}, \quad \text{i.e., } \mathbf{X} = \mathbf{A}^{-1} \mathbf{B}$$

Here, $\mathbf{A}^{-1}$ comes to $\begin{bmatrix} 2/15 & 1/15 & 1/3 \\ -14/15 & 8/15 & -1/3 \\ -1/3 & 1/3 & -1/3 \end{bmatrix}$

Hence, $\mathbf{X} = \begin{bmatrix} 2/15 & 1/15 & 1/3 \\ -14/15 & 8/15 & -1/3 \\ -1/3 & 1/3 & -1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$$\mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$x = 0, y = 1, z = 1$$

5.15 Solution of Simultaneous Equations By Determinants Cramer's Rule

Let the system of equations be

$$a_1x + b_1y + c_1z = m_1$$

$$a_2x + b_2y + c_2z = m_2$$

$$a_3x + b_3y + c_3z = m_3$$

Then the solution is given by Cramer's Rule*:

$$\frac{1}{\text{Determinant formed by the coefficients of } x, y, z \text{ (variables)}} = \frac{x}{\text{Determinant d with the column of coefficients of } x \text{ replaced by column of constants}}$$

Let this determinant be termed as D

$$\frac{y}{\text{Determinant formed by the coefficients of } x, y, z \text{ (variables)}} = \frac{z}{\text{Determinant d with the column of coefficients of } z \text{ replaced by column of constants}}$$

Let this determinant be termed as D

Hence

$$\frac{1}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}} = \frac{x}{\begin{vmatrix} m_1 & b_1 & c_1 \\ m_2 & b_2 & c_2 \\ m_3 & b_3 & c_3 \end{vmatrix}} = \frac{y}{\begin{vmatrix} a_1 & m_1 & c_1 \\ a_2 & m_2 & c_2 \\ a_3 & m_3 & c_3 \end{vmatrix}} = \frac{z}{\begin{vmatrix} a_1 & b_1 & m_1 \\ a_2 & b_2 & m_2 \\ a_3 & b_3 & m_3 \end{vmatrix}}$$

Ex 50 solve Ex 49 by Cramer's Rule

$$\frac{1}{\begin{vmatrix} 1 & -2 & 3 \\ 3 & -1 & 4 \\ 2 & 1 & -2 \end{vmatrix}} = \frac{x}{\begin{vmatrix} 1 & -2 & 3 \\ 3 & -1 & 4 \\ -1 & 1 & -2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} 1 & 1 & 2 \\ 3 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix}}$$

$$= \frac{z}{\begin{vmatrix} 1 & -2 & 1 \\ 3 & -1 & 3 \\ 2 & +1 & -1 \end{vmatrix}}$$

$$\text{Now } \begin{vmatrix} 1 & -2 & 1 \\ 3 & -1 & 3 \\ 2 & 1 & -1 \end{vmatrix} = 1 \begin{vmatrix} -1 & 4 \\ 1 & -2 \end{vmatrix} + 2 \begin{vmatrix} 3 & 4 \\ -1 & -2 \end{vmatrix} + 3 \begin{vmatrix} 3 & -1 \\ -1 & 1 \end{vmatrix}$$

* For proof see M.V. Murthy Mathematical Formulations of Microeconomics
Asia Pacific Publishing House, pp. 47-49.

$$= -2 - 28 + 15 = -15$$

$$\begin{vmatrix} 1 & -2 & 3 \\ 3 & -1 & 4 \\ -1 & 1 & -2 \end{vmatrix} = 1 \begin{vmatrix} -1 & 4 \\ 1 & -2 \end{vmatrix} + 2 \begin{vmatrix} 3 & 4 \\ -1 & -2 \end{vmatrix} + 3 \begin{vmatrix} 3 & -1 \\ -1 & 1 \end{vmatrix}$$

$$= 2 - 4 + 6 = 0$$

$$\begin{vmatrix} 1 & 1 & 3 \\ 3 & 3 & 4 \\ 2 & -1 & -2 \end{vmatrix} = 1 \begin{vmatrix} 3 & 4 \\ -1 & -2 \end{vmatrix} - 1 \begin{vmatrix} 3 & 4 \\ 2 & -2 \end{vmatrix} + 3 \begin{vmatrix} 3 & 3 \\ 2 & -1 \end{vmatrix}$$

$$= -2 + 14 - 27 = -15$$

$$\begin{vmatrix} 1 & -2 & 1 \\ 3 & -1 & 3 \\ 2 & 1 & -1 \end{vmatrix} = 1 \begin{vmatrix} -1 & 3 \\ -1 & -1 \end{vmatrix} + 2 \begin{vmatrix} 3 & 3 \\ 2 & -1 \end{vmatrix} + 1 \begin{vmatrix} 3 & -1 \\ 2 & 1 \end{vmatrix}$$

$$= -2 - 18 + 5 = -15$$

Hence $\frac{1}{-15} = \frac{x}{0} = \frac{y}{-15} = \frac{z}{-15}$

that is, $x = \frac{0}{-15} = 0$ $\left[\begin{array}{l} \frac{x}{0} = \frac{1}{-15} \end{array} \right]$

$y = \frac{-15}{-15} = 1$ $\left[\begin{array}{l} \frac{y}{-15} = \frac{1}{-15} \end{array} \right]$

$z = \frac{-15}{-15} = 1$ $\left[\begin{array}{l} \frac{z}{-15} = \frac{1}{-15} \end{array} \right]$

EXERCISE 5.6

Solve the following set of equations by Cramer's rule and matrix method

1. $3x + 4y = 5$

$3x - 4y = 2$

3. $7x - 5y - 11$

$3x + 2y = 13$

$-5x + y + 10z = 14$

5. $2x - 3y + 5z = 11$

$5x + 2y - 7z = -12$

$-4x + 3y + z = 5$

2. $3x + 3y - 5 = 0$

$3x - 4y = 2$

4. $2x + 3y + 5z = -9$

$x + 10y + 7z = -13$

$-5x + y + 10z = 14$

6. $5x + 3y = 65$

$2y - z = 11$

$3x + 4z = 57$

7. $2x + 3y + 4z = 29$
 $3x + 2y + 5z = 32$
 $5x_1 + x_2 + 2x_3 = 25$
8. $2x_1 + x_2 - x_3 = 0$
 $x_1 - 5x_2 + 3x_3 = 31$
 $6x_1 + 2x_2 - 5x_3 = 3$
9. $3x_1 + x_2 + x_3 = 1$
 $2x_1 + 2x_3 = 0$
 $5x_1 + x_2 + 2x_3 = 2$
10. $3x + 3y - z = 11$
 $2x - y + 2z = 9$
 $4x + 3y + 2z = 25$

11. Given $A = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}$, $C = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

Find X if $AX = C$

12. Solve : $2x - 3y + 4z = 8$
 $3x + 4y - 5z = -4$
 $4x - 5y + 6z = -12$
13. Solve : $2x - 3y + 4z = 8$
 $3x + 4y - 5z = -4$
 $4x - 5y + 6z = 12$
14. Solve $5x - 9y + 4z = 14$
 $2z - 3y = 3$
 $y + 27z - 11x = -12$

15. If supply and demand functions of a firm are :

$$3p + 2x = 27 \quad (\text{Demand})$$

$$6p - 2x = 9 \quad (\text{Supply})$$

Find equilibrium price and supply. If a tax of Rs $3/2$ per unit of commodity is imposed, find the new equilibrium price.

16. If $A = \begin{bmatrix} 2 & 4 \\ 5 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $B = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$ and

$$AX = B$$

Find x_1 and x_2

17. For what value of k , $A = \begin{pmatrix} 3 & -4 & 5 \\ -6 & 7 & -8 \\ 9 & 0 & k \end{pmatrix}$ will be singular

18. For what value of k , do equations

$$2x - 3y + 7z = 0$$

$$5x + 4y - 2z = -3$$

$$x - 13y + kz = 9$$

not have a unique solution ?

19. Show that the equations

$$5x + 2y - 7z = 7$$

$$3x + y - 3z = 5$$

$$2x + y - 4z = 10$$

are incompatible

(Hint. Subtract equation (2) from (1), left hand side of the new equation is same but not the right-hand side of the equation)

20. Solve the equations

$$x + y + z = 1$$

$$x + 2y + 3z = 6$$

$$x + 3y + 4z = 6$$

21. A set of linear equations in which the right hand side consisting of constants has zeros is called a set of homogeneous equations

Show that, $ax + by + cz = 0$

$$bx + cy + az = 0$$

$$cx + ay + bz = 0$$

$$\text{with } D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

have the trivial solution ($x = y = z = 0$) if $D \neq 0$ and has a non-trivial solution if $a^3 + b^3 + c^3 - 3abc = 0$

Find the solution treating x to be constant

22. Prove that
- $X = A^{-1}Y$
- as the solution of
- $AX = Y$
- is equivalent to Cramer's rule

(Hint. Write down A^{-1} in terms of co-factors)

23. Examine the system.

$$2x + 4y + 4z = 2$$

$$2x + 4y + z = -1$$

$$x + 2y + 2z = 1$$

Show that the matrix has rank 2. Find a solution for z and show that x can be obtained in terms of z (and conversely). What happens when the constant of the second equation is changed to some other value?

24. Comment on the following system of equations:

$$(i) \quad 2x + 5y = 10$$

$$4x + 5y = 10$$

$$(ii) \quad x + 3y = 6$$

$$2x + 6y = 12$$

5.16 Partitioned Matrices

If there are many rows and columns of a matrix and computation aids are not adequate, it is found useful to partition the matrix in submatrices.

Suppose we have a 3×4 matrix

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 3 & 2 & 1 & 2 \\ 1 & 1 & 3 & 3 \end{bmatrix} \text{ it can be partitioned by drawing a line between}$$

any columns or rows or both (as shown above). The line should be drawn across the entire matrix. Let us partition the above matrix vertically between columns 2 and 3. Let the matrix on the left H.S. be represented by A_1 and the one on R.H.S. by A_2 , then the given matrix will be

$$A = [A_1 \ A_2] \text{ where } A_1 = 3 \times 2 \text{ matrix and } A_2 = 3 \times 2 \text{ matrix}$$

Transpose of Partitioned Matrix

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 3 & 2 & 1 & 2 \\ 1 & 1 & 3 & 3 \end{bmatrix} = [A_1 \ A_2]$$

$$\text{Thus } A^T = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix}$$

The upper sub matrix in A^T is clearly the transpose of A_1 while lower sub matrix in A^T is transpose of A_2 .

Therefore the rule $A^T = [A_1^T \ A_2^T]$

$$= \begin{bmatrix} A_1^T \\ A_2^T \end{bmatrix}$$

Partitioned Matrix Multiplication

Let $A = [A_1 \ A_2]$ and that A is $m \times n$ matrix
 A_1 is $m \times n_1$ matrix
 and A_2 is $m \times n_2$ matrix
 So that $n = n_1 + n_2$

Let $B = \begin{bmatrix} B_1 \\ \vdots \\ B_2 \end{bmatrix}$ and that B is $n \times p$ matrix
 B_1 is $n_1 \times p$ matrix
 and B_2 is $n_2 \times p$ matrix
 So that $n = n_1 + n_2$ again

Since A is $m \times n$ matrix and B is $n \times p$ matrix the product $[AB]$ is defined — and will be $m \times p$ matrix

Here we note two important things

- (i) Matrix A has been divided vertically and Matrix B horizontally, and
- (ii) The columns of matrix A are divided into n_1 and n_2 while rows of B are divided into n_1 and n_2 such that the number of columns in $A_1 =$ number of rows in B_1 and the number of columns in $A_2 =$ number of rows in B_2

As such now the same rule of multiplications will apply for partitioned matrices as for the original matrices A and B

$$AB = [A_1 \begin{smallmatrix} \vdots \\ A_2 \end{smallmatrix}] \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

$$[A_1 B_1 + A_2 B_2]$$

Ex 51 Multiply A and B by partition device

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & 3 & 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Since A is 3×4 matrix and B is 4×2 matrix hence AB is defined. Let us partition A vertically between column 2 and 3, and B horizontally between rows 2 and 3, so that

$$A_1 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$B_1 = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}, B_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$A_1 B_1 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A_2 B_2 = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Since $AB = A_1B_1 + A_2B_2$

$$AB = \begin{pmatrix} 1 & 0 \\ 4 & 2 \\ 7 & 6 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 4 & 3 \\ 8 & 6 \end{pmatrix}$$

We will get the same result if we multiply matrices A, B in their original form

This method of partitioning matrices can be extended further as under

Let A be $m \times n$ matrix and B be $n \times p$ matrix so that AB exists

Partition both A and B vertically as well as horizontally

$$A = \begin{matrix} m_1 \\ \vdots \\ m_2 \end{matrix} \begin{bmatrix} A_{11} & \vdots & A_{12} \\ \vdots & \ddots & \vdots \\ A_{21} & \vdots & A_{22} \end{bmatrix}$$

So that;

A_{11} is $m_1 \times n_1$ matrix A_{12} is $m_1 \times n_2$ matrix

A_{21} is $m_2 \times n_1$ matrix A_{22} is $m_2 \times n_2$ matrix

Again partition matrix B is the same way,

$$\text{so that ; } B = \begin{matrix} n_1 \\ \vdots \\ n_2 \end{matrix} \begin{bmatrix} B_{11} & \vdots & B_{12} \\ \vdots & \ddots & \vdots \\ B_{21} & \vdots & B_{22} \end{bmatrix}$$

Then $(AB) = C$ (say)

$$= \begin{bmatrix} C_{11} & \vdots & C_{12} \\ \vdots & \ddots & \vdots \\ C_{21} & \vdots & C_{22} \end{bmatrix}$$

$$\begin{bmatrix} A_{11} & \vdots & A_{12} \\ \vdots & \ddots & \vdots \\ A_{21} & \vdots & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & \vdots & B_{12} \\ \vdots & \ddots & \vdots \\ B_{21} & \vdots & B_{22} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & \vdots & A_{11}B_{12} + A_{12}B_{22} \\ \vdots & \ddots & \vdots \\ A_{21}B_{11} + A_{22}B_{21} & \vdots & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

Ex 52. Let us take the same example 51 and apply this method of partitioning

$$A = \begin{array}{cc|cc} A_{11} & & A_{12} & & \\ \hline 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 1 \\ \hline & & \dots & & \\ 1 & 3 & 1 & 1 \\ \hline A_{21} & & A_{22} & & \end{array} \quad \text{and} \quad B = \begin{array}{cc|cc} B_{11} & & B_{12} & & \\ \hline 1 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 \\ \hline & & \dots & & \\ 1 & 1 & 1 & 1 \\ \hline B_{21} & & B_{22} & & \end{array}$$

Find $C = AB$

$$\text{Then } C_{11} = A_{11} B_{11} + A_{12} B_{21}$$

$$= \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$C_{12} = A_{11} B_{12} + A_{12} B_{22} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$C_{21} = A_{21} B_{11} + A_{22} B_{21} = [1 \quad 3] \begin{pmatrix} 1 \\ 2 \end{pmatrix} + [1 \quad 1] \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= [7] + [1] = [8]$$

$$C_{22} = A_{21} B_{12} + A_{22} B_{22} = [1 \quad 3] \begin{pmatrix} 0 \\ 2 \end{pmatrix} + [1 \quad 1] \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= [6] + [0] = [6]$$

$$\therefore AB = C = \begin{pmatrix} C_{11} & C_{12} \\ \hline C_{21} & C_{22} \end{pmatrix}$$

$$= \begin{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} & \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\ \hline [8] & [6] \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 3 \\ 8 & 6 \end{bmatrix}$$

Chapter 6

Differential Calculus : One Variable Case

Most of the economic decisions are based on the solution to the following basic question

"Whether a particular line of action would comparatively add more to our benefits than the efforts spent for pursuing it."

For a decision making authority, whosoever it may be (a consumer or a producer) such a question is of vital importance for making final decision or taking any project or solving any economic problem. In economics, this question is the core of "Marginal Analysis"—which analyses the change in the over all performance due to marginal change consciously made in the value of the variable in question

As we shall see, marginal analysis is closely related to a mathematical technique known as Differential Calculus. Because of this, marginal analysis has become *operative, precise and exact in economic decisions.*

6.1 Geometry of Marginal Analysis

Let us assume a hypothetical total utility curve as represented in Fig. 6.1. This curve indicates how the total utility will vary with consumer's intake of different amounts of units of certain commodity. The average and marginal utilities can also be read off from this total utility curve (Fig. 6.1)

For example, point P indicates that with the consumption of OX_1 units, the total utility amounts to OY_1 units or the average utility per unit of consump-

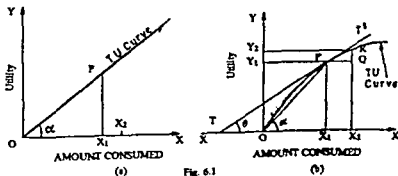


Fig. 6.1

ion is $\frac{OY_1}{OX_1}$. But since $OX_1 = PX_1$, average utility = $\frac{PX_1}{OX_1}$. This ratio is called the slope of the line OP (i.e., $\frac{PX_1}{OX_1} = \tan POX_1$). Exactly in the same way at point R , the average utility = $\frac{OY_2}{OX_2} = \frac{RX_2}{OX_2}$ which is the slope of the line OR .

But how to read the marginal utility from the total utility curve? Let us take the simple case when total utility is steadily increasing as represented by a straight line. Let this straight line pass through origin [Fig. 6.1 (a)]. This means that initially when no units were consumed, utility was zero. As the slope of any straight line at any point is the same, the slope of this line at any point will be constant. The slope of any line which joins the origin with any point on this line will also be the same but this slope measures the average utility at that point. Therefore, the average utility at any point will be the same at all points. Here the same additions are made per unit of consumption, and thus the average does not change. Since the additions made per unit are the same, the marginal utility will be uniform and constant. It is also given by the slope of this very straight line. However, total utility is not always represented by a straight line. We have, therefore, to see what happens to marginal utility when the total utility is represented by a curve and not by a straight line. Let P be any point on the curve [Fig. 6.1 (b)]. We know the average utility at this point is given by the slope of the line OP —join of origin with this point, i.e., $\tan \alpha$. The marginal utility at this point is given by the straight line which is tangent to the curve at P , i.e., PT' . Thus the marginal utility at any point is given by the slope of the tangent to the curve at that point, i.e., $\tan \theta$.

This is not only true of marginal utility, but is also true of marginal product, marginal cost, marginal revenue, etc., when these are to be found out from their total curves. At any point on a curve representing total y , the marginal y can be found by evaluating the slope of the tangent to the curve at that point.

6.2. Differential Calculus—Its Relation to Marginal analysis

Let us now translate the marginal concept defined above in the form of slope and in terms of derivatives. Suppose we denote small increase in the consumption X_1, X_2 by (Δx) . This changes total utility by small amount $Y_1, Y_2 = (\Delta y)$, therefore, the change in utility per unit will be

$$\frac{Y_1, Y_2}{X_1, X_2} = \frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Since $y = f(x)$ and $y + \Delta y = f(x + \Delta x)$

This is slope of line joining P and R ($= \tan RPQ$) and depends upon the value of Δx . But line PR is not tangent to the curve at R . However, if P approaches R on the curve PR will shift and coincide with OR , PR will become tangent to the curve

at R and will measure rate of change of Y with respect of x

Now as P approaches R $PQ (= \Delta x)$ approaches zero and the slope $\frac{f(x + \Delta x) - f(x)}{\Delta x}$ approaches the slope of the tangent at R

If $\frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$ or $\frac{\Delta y}{\Delta x}$ tends to a finite limit as $\Delta x \rightarrow 0$ this limit is called the derivative of $f(x)$ at $x = x_1$ and the function $f(x)$ is said to be differentiable at $x = x_1$. If $f(x)$ is differentiable at every value of x , then $f(x)$ is said to be a differential function. This fact is written as $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$

In terminology of calculus, $\frac{dy}{dx}$ stands for differentiation of y with respect to x and the value of $\frac{dy}{dx}$ is called the derivative of y with respect to x . The process through which we find the value of $\frac{dy}{dx}$ is known as *process of differentiation*.

Finally, let us say a few words about this notation

We should always remember that $\frac{dy}{dx}$ does not mean "dy divided by dx" as it appears to be. In fact, dy and dx have no separate entity. It is just a notation which saves our time in writing the whole expression "the limit to which the ratio $\frac{\Delta y}{\Delta x}$ tends when Δx approaches zero". Hence it must always be remembered that $\frac{dy}{dx}$ appears as a notation or symbol only to mean that it is the limit of a ratio but not a ratio.

6.3 Process of Differentiation

We shall now consider how to find the value of $\frac{dy}{dx}$. Let us suppose that the total y is related to x by $y = 16x^2$. Our objective is to find $\frac{dy}{dx}$ in this particular case.

As emphasised above, $\frac{dy}{dx}$ just means the determination of the limit of the ratio $\frac{\Delta y}{\Delta x}$ when Δx is made to approach zero. Therefore, we shall first find out the ratio $\left(\frac{\Delta y}{\Delta x}\right)$ and then we shall find what happens to this in limit if Δx approaches zero.

We are given $y = 16x^2$

Let us suppose that we increase x by a small value Δx so that x changes to $(x + \Delta x)$ and with this change in x , total y change to $(y + \Delta y)$. Therefore, given relationship takes the form

$$\begin{aligned}(y + \Delta y) &= 16(x + \Delta x)^2 = 16 \{x^2 + 2(x)(\Delta x) + (\Delta x)^2\} \\ &= 16x^2 + 32x(\Delta x) + 16(\Delta x)^2\end{aligned}$$

But since our purpose is to find $\frac{\Delta y}{\Delta x}$, we transfer y to the right hand side

$$\Delta y = 16x^2 + 32x(\Delta x) + 16(\Delta x)^2 - y$$

Substituting $y = 16x^2$ on the right hand side,

$$\Delta y = 32x(\Delta x) + 16(\Delta x)^2$$

Dividing throughout by Δx , to get the required fraction,

$$\frac{\Delta y}{\Delta x} = \frac{32x(\Delta x) + 16(\Delta x)^2}{\Delta x} = 32 + 16(\Delta x)$$

Since $\frac{dy}{dx}$ stands for the value of the ratio $\frac{\Delta y}{\Delta x}$ when Δx tends to zero, we let $\Delta x \rightarrow 0$ in the above ratio,

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} (32x + 16\Delta x) \qquad \frac{dy}{dx} = 32x + 0 = 32x$$

Hence, $\frac{dy}{dx}$ in case of $y = 16x^2$ is $32x$.

In a particular case, if $x = 10$, $y = 16(10)^2 = 1600$, $\frac{dy}{dx} = 32(10) = 320$ units, while average x is equal to $\frac{1600}{10} = 160$

6.4 Rules of Differentiation

We took a particular relationship between x and y , viz., $y = 16x^2$, but the relationship between x and y can take any form. Rules of differentiation are given below to deal with different forms of functions :

Rule 1. Functions in power terms (ax^n)

Let us suppose that the functional relationship between x and y is of the form : $y = ax^n$, where a is some constant term

$$\text{In this case } \frac{dy}{dx} = anx^{n-1}$$

Thus, our first rule of Differentiation about power terms is as under :

$$\text{If } y = ax^n, \quad \frac{dy}{dx} = anx^{n-1}$$

Rule 2. Differentiation of a constant

Suppose the functional relationship is of the form

$$y = C \quad (C \text{ being constant})$$

We can write this as $y = Cx^0$ (Since $x^0 = 1$)

Applying Rule 1, $\frac{dy}{dx} = C(0)(x)^{0-1} \therefore \frac{dy}{dx} = 0$

$$\text{If } y = C, \text{ then } \frac{dy}{dx} = 0$$

Rule 3. Differentiation of sum and difference.

Let $y = 7x^3 + 5x^5 - 3x^6 + 8$

We obtain $\frac{dy}{dx}$ by differentiating each term separately by applying Rules 1 and 2

$$\frac{dy}{dx} = \frac{d}{dx}(7x^3) + \frac{d}{dx}(5x^5) - \frac{d}{dx}(3x^6) + \frac{d}{dx}(8) = 21x^2 + 25x^4 - 18x^5$$

$$\text{If } Y = u + v + w, \text{ then } \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx}$$

Rule 4 Differentiation of a product.

Supposing that y is given by the product of two expressions

$$y = (4x^2 + 2x)(8x^3 + 3x^2)$$

We can differentiate this relationship by finding the product of the two terms and then differentiating term by term on the basis of above rules. But there is an easier method which we describe below

Let us suppose $4x^2 + 2x = U$ and $8x^3 + 3x^2 = V$

So that, $y = UV$ (U is first term, V is second term)

Then $\frac{dy}{dx}$ can be obtained by the following rule.

$$\text{If } y = UV; \text{ then } \frac{dy}{dx} = V \frac{dU}{dx} + U \frac{dV}{dx}$$

Since $V = 8x^3 + 3x^2 \therefore \frac{dV}{dx} = 24x^2 + 6x$

$$\text{and } U = 4x^2 + 2x \quad \frac{dU}{dx} = 8x + 2$$

$$\frac{dy}{dx} = V \frac{dU}{dx} + U \frac{dV}{dx} = (8x^3 + 3x^2)(8x + 2) + (4x^2 + 2x)(24x^2 + 6x)$$

If y is expressed as a product of three functions U , V and W as $y = UVW$, then $\frac{dy}{dx} = UV \frac{dW}{dx} + UW \frac{dV}{dx} + VW \frac{dU}{dx}$

More generally, if y is expressed as a product of four functions of x , i.e., if $y = SUVW$,

$$\text{then, } \frac{dy}{dx} = SUV \frac{dW}{dx} + SUW \frac{dV}{dx} + SWV \frac{dU}{dx} + UVW \frac{dS}{dx}$$

Rule 5 Differentiation of a quotient

Supposing that y is expressed as a quotient of two functions of x ,

$$\text{that is, } y = \frac{8x^8 + 6x^2 - 2x}{3x^2 + 5x}$$

then, $\frac{dy}{dx}$ can be found from the following rule

$$\boxed{\text{If } y = \frac{U}{V}, \text{ then } \frac{dy}{dx} = \frac{V \left(\frac{dU}{dx} \right) - U \left(\frac{dV}{dx} \right)}{V^2}}$$

The same result will be obtained by applying product rule to

$$y = UV^{-1} = (8x^8 + 6x^2 - 2x)(3x^2 + 5x)^{-1}$$

$$\text{Here } U = 8x^8 + 6x^2 - 2x \quad \therefore \frac{dU}{dx} = 64x^7 + 12x - 2$$

$$\text{and, } V = 3x^2 + 5x \quad \therefore \frac{dV}{dx} = 6x + 5$$

$$\frac{dy}{dx} = \frac{(3x^2 + 5x)(64x^7 + 12x - 2) - (8x^8 + 6x^2 - 2x)(6x + 5)}{(3x^2 + 5x)^2}$$

Rule 6. Differentiation of a function of a function (chain rule)

Let us suppose that consumption is not related to total utility directly but indirectly via income (I)

And our problem is what will be the change in total utility if consumption changes by one unit. Here we have to apply the rule for differentiation of a function of a function

$$\text{Assume that } y = \frac{I}{2}, \text{ and } I = \frac{x}{5}.$$

Take the expression $I = \frac{x}{5}$, here differentiating I w r t x (which represents change in I due to a small change in x), we get

$$\frac{dI}{dx} = \frac{1}{5} \quad (i)$$

Also when $y = \frac{I}{2}$, then $\frac{dy}{dI}$ (which represents change in y due to a small change in I) = $\frac{1}{2}$ (ii)

Thus now we have two derivatives,

$$\frac{dI}{dx} = \frac{1}{5} \text{ and, } \frac{dy}{dI} = \frac{1}{2}$$

But we need the change in y with small change in x , i.e., we need $\frac{dy}{dx}$. The result can be obtained by the following Rule

$$\frac{dy}{dx} = \frac{dy}{dI} \cdot \frac{dI}{dx} = \frac{1}{2} \cdot \frac{1}{5} = \frac{1}{10}$$

This particular way of getting the desired derivative in case of a function of a function is known as Chain Rule

Suppose, $y = I^4 + 3I^3$, and $I = x^2$, we obtain $\frac{dy}{dx}$,

by Chain Rule : $\frac{dy}{dx} = \frac{dy}{dI} \cdot \frac{dI}{dx}$

$$\frac{dy}{dx} = \left[\frac{d}{dI} (I^4 + 3I^3) \right] \left[\frac{d}{dx} (x^2) \right] = (4I^3 + 9I^2) (2x) = 8x I^3 + 18x I^2 \cdot x$$

$$\begin{aligned} \text{or, } \frac{dy}{dx} &= 8x(x^2)^3 + 18x(x^2)^2 \\ &= 8x \cdot x^6 + 18x \cdot x^4 = 8x^7 + 18x^5. \end{aligned}$$

(a) *Differentiation of power on expressions in brackets.*

Suppose, $y = (3 + 2x^2)^3$

Assuming $(3 + 2x^2) = U$, we write $y = U^3$.

$$\therefore \frac{dU}{dx} = 4x, \text{ and } \frac{dy}{dU} = 3U^2. \text{ But } \frac{dy}{dx} = \frac{dy}{dU} \cdot \frac{dU}{dx} = (3U^2) (4x)$$

$$\text{But since } U = (3 + 2x^2), \quad \therefore \frac{dy}{dx} = 3(3 + 2x^2)^2 4x = 12x(3 + 2x^2)^2$$

(b) *Differentiation of roots of expressions*

The above chain rule is also useful when the relationship involves roots

such as $y = \frac{1}{\sqrt{8x^3 + 5x}}$

$$\text{Let, } 8x^3 + 5x = u \quad y = \frac{1}{\sqrt{u}} = u^{-\frac{1}{2}}$$

$$\frac{dy}{du} = -\frac{1}{2} u^{-\frac{3}{2}} \text{ and } \frac{du}{dx} = 24x^2 + 5$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = -\frac{1}{2} u^{-\frac{3}{2}} (24x^2 + 5)$$

$$= -\frac{1}{2} (8x^3 + 5x)^{-\frac{3}{2}} (24x^2 + 5)$$

$$= -\frac{24x^2 + 5}{2(8x^3 + 5x)^{\frac{3}{2}}} = -\frac{24x^2 + 5}{2\sqrt{(8x^3 + 5x)^3}}$$

Rule 7 *Differentiation of implicit functions*

If y and x are related in a form such as $y = \frac{2x}{x^2 + 1}$ then y is said to be an explicit function of x , because the R.H.S. contains expression in x alone (no- y) and L.H.S. contains only y . In such cases y is expressed explicitly in terms of x .

An equation $x^3y + y - 2x = 0$, does not express y in terms of x directly, we say that the given relation is an implicit one.

In such case we differentiate term by term with respect to (w.r.t) x in the following way

$$\text{Given implicit relation } x^3y + y - 2x = 0$$

$$\frac{d}{dx} (x^3y) + \frac{d}{dx} (y) - \frac{d}{dx} (2x) = 0$$

$$x^3 \frac{d}{dx} (y) + y \frac{d}{dx} (x^3) + \frac{d}{dx} (y) - \frac{d}{dx} (2x) = 0$$

Application of Rule 4

$$x^3 \frac{dy}{dx} + y (3x^2) + \frac{dy}{dx} - 2 = 0 \text{ or, } x^3 \frac{dy}{dx} + 3x^2y + \frac{dy}{dx} - 2 = 0$$

Now, this equation can be solved for $\frac{dy}{dx}$

$$\frac{dy}{dx}(x^3 + 1) = 2 - 3x^2y \quad \frac{dy}{dx} = \frac{2-3x^2y}{x^3+1}$$

Consider another problem, $x^2 - y^2 + 3x = 5y$ Differentiating term by term w.r.t x , we get

$$\frac{d}{dx}(x^2) - \frac{d}{dx}(y^2) + \frac{d}{dx}(3x) = \frac{d}{dx}(5y)$$

$$2x - 2y \frac{dy}{dx} + 3 = 5 \frac{dy}{dx}$$

Collecting $\left(\frac{dy}{dx}\right)$ on one side of the equation

$$2y \frac{dy}{dx} + 5 \frac{dy}{dx} = 2x + 3 \quad \frac{dy}{dx} = \frac{2x+3}{2y+5}$$

Let us now summarize all the rules of differentiation for the different functional relationships between y and x considered above. On the basis of these rules, we shall be able to find the derivative of functional relationships expressed in any form

Form of function	Function	Derivative $\left(-\frac{dy}{dx}\right)$
Power terms	$y = ax^n$	anx^{n-1}
Constant	$y = C$	
Sum and difference	$y = U + V$	$\frac{dU}{dx} \pm \frac{dV}{dx}$
Product	$y = UV$	$U \frac{d}{dx}(V) + V \frac{d}{dx}(U)$
Quotient	$y = \frac{U}{V}$	$\frac{V \frac{d}{dx}(U) - U \frac{d}{dx}(V)}{V^2}$

Along with these basic rules some special cases are given below

Function of a function $y = f_2(t), t = f_1(x) \quad \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

Powers of expressions

Roots of expressions

Implicit functions

These cases involve substitutions.

These require differentiation of each term

Some standard results :

Many a time, we have functional relationship in trigonometric, exponential or logarithmic forms. The results given under will be useful for finding the derivatives of such functions. Some of the standard results are derived by the first principal below, others are given without proof

$$\begin{aligned}
 (i) \quad y &= \frac{1}{x} = x^{-1} \\
 y + \Delta y &= (x + \Delta x)^{-1} \\
 &= x^{-1} \left(1 + \frac{\Delta x}{x} \right)^{-1} \\
 &= x^{-1} \left(1 - \frac{\Delta x}{x} + \frac{(\Delta x)^2}{x^2} + \dots \right) \\
 &\quad \text{(using binomial theorem)}
 \end{aligned}$$

using $y = x^{-1}$ we obtain

$$\begin{aligned}
 \Delta y &= x^{-1} \left(1 - \frac{\Delta x}{x} + \frac{(\Delta x)^2}{x^2} + \dots \right) - x^{-1} \\
 &= -\frac{\Delta x}{x^2} + \frac{(\Delta x)^2}{x^3} + \dots \\
 \frac{\Delta y}{\Delta x} &= -\frac{1}{x^2} + \frac{\Delta x}{x^3} + \dots
 \end{aligned}$$

$$\text{Lt } \Delta x \rightarrow 0 \quad \frac{\Delta y}{\Delta x} = -\frac{1}{x^2}$$

Hence

$$\boxed{\frac{d\left(\frac{1}{x}\right)}{dx} = -\frac{1}{x^2}}$$

(ii)

$$\begin{aligned}
 y &= \log x \\
 y + \Delta y &= \log(x + \Delta x) \\
 \text{and } y &= \log x
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} y + \Delta y &= \log(x + \Delta x) \\ y &= \log x \end{aligned}} \right\} \text{ Subtracting, we obtain}$$

$$\begin{aligned}
 \Delta y &= \log(x + \Delta x) - \log x \\
 &= \log \frac{x + \Delta x}{x} \\
 &= \log \left(1 + \frac{\Delta x}{x} \right) \\
 &= \frac{\Delta x}{x} - \frac{\left(\frac{\Delta x}{x}\right)^2}{2} + \dots
 \end{aligned}$$

$$\text{Since, } \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

$$\frac{\Delta y}{\Delta x} = \frac{1}{x} - \frac{\Delta x}{\Delta x^2} +$$

$$\text{Lt } \Delta x \rightarrow 0, \frac{\Delta y}{\Delta x} = \frac{1}{x}$$

Therefore, $\boxed{\frac{d(\log x)}{dx} = \frac{1}{x}}$

(iii) $y = a^x$ where a is any positive constant

Taking logarithms on both sides we have

$$\log_e y = x \log_e a$$

Differentiating w.r.t. x , we have

$$\frac{1}{y} \frac{dy}{dx} = 1 \cdot \log_e a$$

$$\frac{dy}{dx} = y \log_e a = a^x \log_e a$$

Hence $\boxed{\frac{d(a^x)}{dx} = a^x \log_e a}$

(iv) $y = e^x$

$$y + \Delta y = e^{x+\Delta x}$$

$$y = e^x$$

$$\Delta y = e^{x+\Delta x} - e^x$$

$$= e^x (e^{\Delta x} - 1)$$

$$= e^x \left(1 + \Delta x + \frac{\Delta x^2}{2} + \dots - 1 \right)$$

$$\frac{dy}{dx} = \text{Lt } \Delta x \rightarrow 0, \frac{\Delta y}{\Delta x} = e^x$$

Hence $\boxed{\frac{d}{dx} (e^x) = e^x}$

The above functions along with other important functions are described in the following tabular form

Form of Functions	Function	Derivative $\left(= \frac{dy}{dx} \right)$
Trigonometric	$y = \sin x$	$\cos x$
	$y = \cos x$	$-\sin x$
	$y = \tan x$	$\sec^2 x = \frac{1}{\cos^2 x}$
	$y = \cot x$	$-\operatorname{cosec}^2 x = -\frac{1}{\sin^2 x}$
	$y = \sec x$	$\sec x \tan x$
	$y = \operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
Exponential	$y = e^x$	e^x
	$y = e^{-x}$	$-e^{-x}$
	$y = e^{ax}$	ae^{ax} (a is constant)
	$y = a^x$	$a^x \log_e a$
Logarithmic	$y = \log x$	$1/x$

Ex. 1. Differentiate the following functions :

$$(i) (\sqrt[3]{x})^4 \quad (ii) \frac{2x^3 - x^2 + x - 2}{x^2} \quad (iii) x^2(x-3)$$

$$(i) y = (\sqrt[3]{x})^4 = x^{4/3}$$

$$\therefore \frac{dy}{dx} = \frac{4}{3} x^{4/3-1} = \frac{4}{3} x^{1/3}$$

$$(ii) y = \frac{2x^3 - x^2 + x - 2}{x^2}$$

$$= 2x - 1 + \frac{1}{x} - \frac{2}{x^2}$$

$$= 2x - 1 + x^{-1} - 2x^{-2}$$

$$\therefore \frac{dy}{dx} = 2 - x^{-1-1} + 4x^{-2-1}$$

$$= 2 - x^{-2} + 4x^{-3} = 2 - \frac{1}{x^2} + \frac{4}{x^3}$$

$$(iii) y = x^2(x-3) = x^3 - 3x^2$$

$$\therefore \frac{dy}{dx} = 3x^2 - 6x$$

Ex. 2. Find $\frac{dy}{dx}$ if $y = (1 - \sqrt{x})(1 + \sqrt{x})$

Derivative can be obtained by two methods given below :

Method I.

$$y = (1 - \sqrt{x})(1 + \sqrt{x})$$

i.e., $y = uv$, where $u = (1 - \sqrt{x})$ and $v = (1 + \sqrt{x})$

$$\begin{aligned}\frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} \\ &= (1 - \sqrt{x}) \frac{d(1 + \sqrt{x})}{dx} + (1 + \sqrt{x}) \frac{d(1 - \sqrt{x})}{dx} \\ &= (1 - \sqrt{x}) \left(\frac{1}{2} x^{-1/2} \right) + (1 + \sqrt{x}) \left(-\frac{1}{2} x^{-1/2} \right) \\ &= \frac{1}{2} (1 - \sqrt{x}) x^{-1/2} - \frac{1}{2} (1 + \sqrt{x}) x^{-1/2} \\ &= \frac{1}{2} x^{-1/2} [1 - \sqrt{x} - 1 - \sqrt{x}] = -x^{-1/2} x^{-1/2} = -1\end{aligned}$$

Method II

$$\begin{aligned}y &= (1 - \sqrt{x})(1 + \sqrt{x}) \\ &= (1 - x) \quad [(a - b)(a + b) = a^2 - b^2] \\ \frac{dy}{dx} &= -1\end{aligned}$$

Ex 3 If $y = \sqrt{\frac{1-x}{1+x}}$, find $\frac{dy}{dx}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{\sqrt{1+x} \left(\frac{d}{dx} \sqrt{1-x} \right) - \sqrt{1-x} \left(\frac{d}{dx} \sqrt{1+x} \right)}{(\sqrt{1+x})^2} \\ &= \frac{(\sqrt{1+x}) \left[\frac{1}{2} (1-x)^{-1/2} \times -1 \right] - (\sqrt{1-x}) \left[\frac{1}{2} (1+x)^{-1/2} \times 1 \right]}{(1+x)} \\ &= \frac{-\frac{1}{2} \frac{\sqrt{1+x}}{(1-x)^{1/2}} - \frac{1}{2} \frac{\sqrt{1-x}}{(1+x)^{1/2}}}{1+x} \\ &= -\frac{1}{2(1+x)} \left[\frac{(1+x) + (1-x)}{(1-x)^{1/2} (1+x)^{1/2}} \right] = -\frac{1}{(1+x)^{1/2} (1-x)^{1/2}} \\ &= -\frac{1}{2(1+x)} \left[\frac{(1+x) + (1-x)}{(1-x)^{1/2} (1+x)^{1/2}} \right] = -\frac{1}{(1+x)^{1/2} (1-x)^{1/2}}\end{aligned}$$

Ex 4 If $y = (x^2 + 3)(2x^2 + 7)$, find $\frac{dy}{dx}$

$$\begin{aligned}\frac{dy}{dx} &= (x^2 + 3) \frac{d}{dx} (2x^2 + 7) + (2x^2 + 7) \frac{d}{dx} (x^2 + 3) \\ &= (x^2 + 3) [3(2x^2 + 7) \times 4x] + (2x^2 + 7) \times 2x\end{aligned}$$

$$\begin{aligned}
 &= 12x(x^2 + 3)(2x^2 + 7)^2 + 2x(2x^2 + 7)^3 \\
 &= 2x(2x^2 + 7)^2 [6(x^2 + 3) + (2x^2 + 7)] \\
 &= 2x(2x^2 + 7)^2 (8x^2 + 25)
 \end{aligned}$$

Ex. 5. If $x^3 + 5x^2y + yx = 5$, find $\frac{dy}{dx}$

Differentiating implicitly, we have

$$3x^2 + \left[5x^2 \frac{dy}{dx} + 5 \cdot 2xy \right] + x \frac{dy}{dx} + y \cdot 1 = 0$$

$$5x^2 \frac{dy}{dx} + x \frac{dy}{dx} = -3x^2 - 10xy - y$$

$$\frac{dy}{dx} [5x^2 + x] = -3x^2 - 10xy - y$$

so that,
$$\frac{dy}{dx} = -\frac{(3x^2 + 10xy + y)}{x(5x + 1)}$$

Ex. 6. If $y = (2x^2 + 3)e^{-3x^2}$, find $\frac{dy}{dx}$

$$\frac{dy}{dx} = (2x^2 + 3) \frac{d}{dx} (e^{-3x^2}) + e^{-3x^2} \frac{d(2x^2 + 3)}{dx}$$

$$\frac{dy}{dx} = (2x^2 + 3) (e^{-3x^2} \times -3 \times 3x) + e^{-3x^2} (4x)$$

$$= e^{-3x^2} [-6x(2x^2 + 3) + 4x]$$

$$= e^{-3x^2} [-12x^3 - 14x]$$

$$= -2x(6x^2 + 7) e^{-3x^2}$$

Ex. 7 Differentiate : $y = \frac{(x-2)(2x+3)}{(x+7)(1-x)}$

Taking logarithms on both sides, we have

$$\log y = \log(x-2) + \log(2x+3) - \log(x+7) - \log(1-x)$$

Differentiating w.r.t x , we have

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x-2} + \frac{1}{2x+3} \times 2 - \frac{1}{x+7} - \frac{1}{1-x} \times -1$$

$$\frac{dy}{dx} = y \left[\frac{1}{x-2} + \frac{2}{2x+3} - \frac{1}{x+7} + \frac{1}{1-x} \right]$$

where $y = \frac{(x-2)(2x+3)}{(x+7)(1-x)}$

Ex. 8 Find the derivative of the following function $z = 1 - y^2$ with respect to x , given that

$$2z = 1 - y^2 \quad \text{and} \quad y = x^2 + 3$$

We know that $\frac{dz}{dx} = \frac{dz}{dy} \times \frac{dy}{dx}$

Hence, differentiating first function z w.r.t y we get $\frac{dz}{dy}$ and second function y with respect to x , we have

$$\frac{dz}{dy} = \frac{1}{2}(-2y), \text{ and } \frac{dy}{dx} = 2x$$

$$\frac{dz}{dx} = -y \times 2x = -2xy = -2x(x^2 + 3)$$

Ex. 9. Differentiate $y = \sqrt{\frac{(x-1)(x+2)}{(2x-1)(x-3)}}$

Taking logarithms on both sides, we have

$$\log y = \frac{1}{2} \cdot \log \frac{(x-1)(x+2)}{(2x-1)(x-3)}$$

$$= \frac{1}{2} [\log(x-1) + \log(x+2) - \log(2x-1) - \log(x-3)]$$

Differentiating both sides w.r.t x , we have

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \left[\frac{1}{x-1} + \frac{1}{x+2} - \frac{1}{2x-1} \times 2 - \frac{1}{x-3} \right]$$

$$\frac{dy}{dx} = \frac{y}{2} \left[\frac{1}{x-1} + \frac{1}{x+2} - \frac{1}{2x-1} - \frac{1}{x-3} \right]$$

Ex. 10. If $y = a^{x^2}$, find $\frac{dy}{dx}$

Taking logarithms on both sides,

$$\log y = x^2 \log a$$

Differentiating both sides w.r.t x , we have

$$\frac{1}{y} \frac{dy}{dx} = 2x \log_e a$$

$$\frac{dy}{dx} = y \cdot 2x \log_e a = 2x \cdot a \cdot \log_e a$$

Ex 11 If $y = \sqrt{\frac{1-x}{1+x}}$ show that $(1-x^2) \frac{dy}{dx} + y = 0$

Taking logarithms on both sides

$$\log_e y = \frac{1}{2} \{ \log(1-x) - \log(1+x) \}$$

Differentiating we get

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \left(\frac{1}{1-x} - \frac{1}{1+x} \right) = \frac{1}{2} \left(\frac{1}{1-x} - \frac{1}{1+x} \right)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \left(\frac{1+x+1-x}{(1-x)(1+x)} \right)$$

$$\text{or } \frac{1}{y} \frac{dy}{dx} = \frac{1}{1-x^2}$$

$$\text{or } \frac{dy}{dx} = \frac{y}{1-x^2}$$

$$\text{or } (1-x^2) \frac{dy}{dx} + y = 0 \text{ hence proved}$$

Ex 12 If $(x-y) \frac{x}{x-y} = a$ then prove that

$$y \frac{dy}{dx} + x = 2y$$

Taking logarithms we have

$$\log(x-y) + \frac{x}{x-y} \log e = \log a$$

Differentiating w.r.t. x we have

$$\frac{d}{dx} \left(\frac{x}{x-y} \right) + \frac{d}{dx} \log(x-y) = 0 \quad [\log e = 1]$$

$$\text{or, } \frac{(x-y) - x \left(1 - \frac{dy}{dx} \right)}{(x-y)^2} + \left(\frac{1}{x-y} - \frac{\frac{dy}{dx}}{(x-y)} \right)$$

$$\text{or, } \frac{-y}{(x-y)^2} + \frac{1}{(x-y)} = \frac{dy}{dx} \left(\frac{1}{x-y} - \frac{x}{(x-y)^2} \right)$$

$$\text{or, } \frac{-y+x-y}{(x-y)^2} = \frac{dy}{dx} \left[\frac{x-y-x}{(x-y)^2} \right]$$

$$\text{i.e., } x-2y = -y \frac{dy}{dx}$$

$$\text{or, } y \frac{dy}{dx} + x = 2y, \text{ hence proved}$$

Ex. 13. Find $\frac{dy}{dx}$, if $x = at^3$ and $y = 3at$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\text{Given : } x = at^3, \quad y = 3at$$

$$\frac{dx}{dt} = 3at^2, \quad \frac{dy}{dt} = 3a$$

$$\frac{dy}{dx} = \frac{3a}{3at^2} = \frac{1}{t^2}$$

Ex. 14. $y = \frac{1-x^2}{x^3+2x}$ find $f'(-1)$, $f'(1)$ and the point at which $f'(x) = 0$

$$\frac{dy}{dx} = \frac{(x^3+2x) \cdot \frac{d(1-x^2)}{dx} - (1-x^2) \frac{d}{dx}(x^3+2x)}{(x^3+2x)^2}$$

$$\frac{dy}{dx} = \frac{(x^3+2x)(-2x) - (1-x^2)(3x^2+2)}{(x^3+2x)^2} = \frac{x^4-5x^2-2}{(x^3+2x)^2}$$

$$\left. \begin{array}{l} f'(-1), \text{ i.e., obtain} \\ \frac{dy}{dx} \text{ when } x = -1 \end{array} \right\} = \frac{(-1)^4 - 5 \times (-1)^2 - 2}{\{(-1)^3 + 2 \times -1\}^2} = \frac{-6}{(-3)^2} = -\frac{2}{3}$$

$$\text{Also } f'(1) = \frac{(1)^4 - 5 \times (1)^2 - 2}{(1^3 + 2 \times 1)^2} = -\frac{2}{3}$$

$$\text{hence, } f'(-1) = f'(1)$$

$$\text{If } \frac{dy}{dx} = 0 \text{ i.e., } \frac{x^4 - 5x^2 - 2}{(x^3 + 2x)^2} = 0$$

$$\text{or, } x^4 - 5x^2 - 2 = 0$$

$$\text{or, } x^2 = \frac{+5 \pm \sqrt{25+8}}{2}$$

$$\text{or, } x^2 = \frac{+5 \pm 5.74}{2} = 5.37 \text{ or } 0.37$$

Two values of $x (\approx \pm \sqrt{-0.37})$ are imaginary. Other two values are $x (\approx \pm \sqrt{5.37})$. Putting the values of x in the original equation, we get the corresponding values of y and, thus, the required points.

Ex. 15 Let $Y = 3x^2 + 9x + 8$ and $Z = x^3 + 10$. Evaluate the derivatives with respect to x at $x=1$ of the following

$$(i) \quad Y + Z \quad (ii) \quad YZ \quad (iii) \quad Z^2 \quad (iv) \quad Z/Y$$

$$(i) \quad Y + Z = (3x^2 + 9x + 8) + (x^3 + 10)$$

$$\frac{d}{dx}(Y + Z) = 6x + 9 + 3x^2,$$

its value at $x=1$ will be

$$= 6(1) + 9 + 3(1)^2 = 18$$

$$(ii) \quad YZ = (3x^2 + 9x + 8)(x^3 + 10)$$

$$\begin{aligned} \frac{d}{dx}(YZ) &= (x^3 + 10) \frac{d}{dx}(3x^2 + 9x + 8) + (3x^2 + 9x + 8) \frac{d}{dx}(x^3 + 10) \\ &= (x^3 + 10)(6x + 9) + (3x^2 + 9x + 8)(3x^2) \end{aligned}$$

its value at $x=1$ will be

$$\begin{aligned} &= (1 + 10)(6 + 9) + (3 + 9 + 8)(3) \\ &= (11)(15) + (20)(3) = 225. \end{aligned}$$

$$(iii) \quad Z^2 = (x^3 + 10)^2$$

$$\frac{d}{dx}(Z^2) = 2(x^3 + 10)(3x^2);$$

its value at $x=1$ will be

$$\begin{aligned} &= 2(1 + 10)(3) \\ &= 2 \times 33 = 66 \end{aligned}$$

$$(iv) \quad \frac{Z}{Y} = \frac{x^3 + 10}{3x^2 + 9x + 8}$$

$$\begin{aligned} \frac{d}{dx} \left(\frac{Z}{Y} \right) &= \frac{(3x^2 + 9x + 8) \frac{d}{dx}(x^3 + 10) - (x^3 + 10) \frac{d}{dx}(3x^2 + 9x + 8)}{(3x^2 + 9x + 8)^2} \\ &= \frac{(3x^2 + 9x + 8)(3x^2) - (x^3 + 10)(6x + 9)}{(3x^2 + 9x + 8)^2} \end{aligned}$$

Its value at $x = 1$ will be

$$\begin{aligned}
 &= \frac{(3+9+8)(3) - (1+10)(6+9)}{(3+9+8)^2} \\
 &= \frac{60 - 165}{(20)^2} = \frac{105}{400} = -\frac{21}{80}
 \end{aligned}$$

Ex 16 Find $\frac{dy}{dx}$ in the following

(i) $y = 10^x + \log(2x+1) + x^2 - 6$

(ii) $y = (3x-1)^2 \sqrt{x+3}$

(i) $\frac{dy}{dx} = 10^x \log 10 + \left(\frac{2}{2x+1}\right) + 2x$

(ii) $\frac{dy}{dx} = (3x-1)^2 \cdot \frac{1}{2}(x+3)^{-1/2} + (x+3)^{1/2} \cdot 2(3x-1)(3)$
 $= \frac{(3x-1)^2}{2\sqrt{x+3}} + 6\sqrt{x+3}(3x-1)$
 $= \frac{(3x-1)^2 + 12(x+3)(3x-1)}{2\sqrt{x+3}}$

Ex. 17. Find $\frac{dy}{dx}$ in the following

(i) $y = \frac{x^2+4}{x+2}$ (ii) $y = e^{3x} + \sqrt{x^2-4} + 5$

(i) $\frac{dy}{dx} = \frac{(x+2) \frac{d}{dx}(x^2+4) - (x^2+4) \frac{d}{dx}(x+2)}{(x+2)^2}$
 $= \frac{(x+2)(2x) - (x^2+4)(1)}{(x+2)^2}$
 $= \frac{x^2+4x-4}{(x+2)^2}$

(ii) $\frac{dy}{dx} = 3e^{3x} + \frac{1}{2}(x^2-4)^{-1/2}(2x) + 0$
 $= 3e^{3x} + \frac{x}{\sqrt{x^2-4}}$

Ex. 18 If $y = x^2 + 2xz - z^2$, calculate the approximate change in y if x increases from 2 to 2.1 and z increases from 3 to 3.2

Approximate change in $y = \Delta y$ (say)

$$\begin{aligned}
 \Delta y &= [(2.1)^2 + 2(2.1)(3.2) - (3.2)^2] - [2^2 + 2(2)(3) - 3^2] \\
 &= (7.61 - 7.0) = 0.61
 \end{aligned}$$

EXERCISE 6.1

A Find $\frac{dy}{dx}$ for the following values of y

- | | | | | | | | |
|----|----------------------------------|----|-------------------------------------|----|-----------------------------|----|-------------------|
| 1 | x | 2 | $x^{5/2}$ | 3 | $x^{5/3}$ | 4 | x^{-5} |
| 5 | $3\sqrt{x}$ | 6 | $(4\sqrt{x})^5$ | 7 | $3\sqrt[3]{x}$ | 8 | $\frac{1}{x}$ |
| 9 | $\frac{1}{x}$ | 10 | $\frac{1}{x}$ | 11 | $16x^5 + 14x$ | 12 | $\frac{5+x^2}{x}$ |
| 13 | $3x + x^2$ | 14 | $x(1+5-3x)$ | 15 | $3x + 7x^2 + x^3$ | 16 | $x^2 + 3x + 2$ |
| 17 | $\frac{7x^2 + 4x^3 + 9x + 3}{x}$ | 18 | $\frac{7x^2 + 3}{x} - \frac{5}{3x}$ | 19 | $\frac{1}{3\sqrt{3x} - 2x}$ | 20 | $\sqrt{4x - 3}$ |

B Find $\frac{d}{dx}$ for the following values of y

- | | | | |
|---|--|----|----------------------------------|
| 1 | $(x + 1)(x + 2)$ | 2 | $(x^2 - x - 1)(x^2 + x + 1)$ |
| 3 | $(x + 2)(x^2 - 1)$ | 4 | $(ax^2 + bx + cx)(px + q)$ |
| 5 | $\frac{1}{x}(2x + 3)$ | 6 | $(x + 1)^3(x - 1)$ |
| 7 | $\sqrt{x - 2x - 1}(x^2 + x + 1)$ | 8 | $(x^3 - 1)(x - 1)(x - 2x^2 - 1)$ |
| 9 | $\sqrt{x}(\sqrt{x} + 1)(2 + \sqrt{x})$ | 10 | $(1 + 2x)(3 + 4x)^2(2 + 3x)^3$ |

C Find $\frac{dy}{dx}$ for the following values of y

- | | | | | | |
|----|---|----|---|----|----------------------------------|
| 1 | $\frac{2x}{x-5}$ | 2 | $\frac{x^2+5}{x^2-x}$ | 3 | $\frac{4x^2-7x^2-8}{4x+3}$ |
| 4 | $\frac{x+c}{x-c}$ | 5 | $\frac{\sqrt{x}}{x+1}$ | 6 | $\frac{1+\sqrt{x}}{1-\sqrt{x}}$ |
| 7 | $\sqrt{\frac{1+x}{1-x}}$ | 8 | $\frac{\sqrt{a+x}-\sqrt{a-x}}{\sqrt{a+x}+\sqrt{a-x}}$ | 9 | $\frac{3\sqrt{x}}{\sqrt{x^2-2}}$ |
| 10 | $\frac{2x-1}{\sqrt{x}}$ | 11 | $\frac{x-1}{x+1}$ | 12 | $\frac{x+1}{x^2+2x+2}$ |
| 13 | $\left(\frac{x+1}{2x+3}\right)(2x-5)$ | 14 | $\frac{x^2+1}{x^2-1}$ | | |
| 15 | $\left(\frac{x^2-x+6}{x^2+1}\right)(x^2+x+1)$ | | | | |

D Find $\frac{dy}{dx}$ for the following functions

1 $(2x^2 + 2x + 2)^2$

2 $(3x + 8)^{1/3}$

3 $\sqrt{3x + 5}$

4 $\sqrt{a + x^2}$

5 $\sqrt{4x + x^2}$

6 $(3x^2 + 2x^3 - x)^{-1}$

7 $\frac{\sqrt{1+x}}{1-x}$

8 $\frac{x}{\sqrt{1-x}}$

9 $(x^2 + 2x)^4 (x^3 + 3x + 7)^3$

10 $y = \sqrt{\log x}$

11 It is given that $y = t + t^2$ where $t = 1 - x$ find $\frac{dy}{dx}$ is a function of x (that is $\frac{dy}{dx}$). Write down the derivatives of y with respect to x and t and $\frac{dy}{dx}$ with respect to z . Also verify that $\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

E Find $\frac{dy}{dx}$

1 $x^2y = 10$

2 $x^4 + y^5 = 3x^2y^2$

3 $x^2 + y^2 + 2x + 2y - 2 = 0$

4 $4x^2 + 2xy + y^2 = 12$

5 $2x^2 - 3xy + y^3 + x + 2y - 8 = 0$

6 $xy = 4x$

7 $x^2 - 2xy = 3$

8 $xy - 4 = 0$

9 $x^m + y^n = a^k$

10 $2x^2 - y^2 = 1$

11 $x^p y^q = (x + y)^{p+q}$

12 $e^x = e^y$

13 $x^y = y^x$

F Find $\frac{dy}{dx}$ (Trigonometrical functions)

1 $y = 4x + \sin x$

2 $y = \sin x + \cos 2x$

3 $y = \cos \frac{x}{2} + \sin \frac{x}{2}$

4 $y = \sin^3(2x)$

5 $y = \cos^2(2x)$

6 $y = x^2 + 3 \sin \frac{x}{2}$

7 $y = \tan 3x - \tan 2x$

8 $y = x (\sin x - \cos x)$

9 $y = (x \cos x) (\cos x + \tan x)$

10 $y = \frac{\sin^2 x}{1 + \sin x}$

11 $y = \frac{\sin x}{\cos x}$

12 $y = \sin \sqrt{x}$

13 $y = \cot^{-1} \frac{x}{y}$

G Find $\frac{dy}{dx}$ when

1 $y = e^{4x}$

2 $y = e^{-x}$

3. $x^2 e^x$
4. $y = e^x$
5. $y = e^{kx}$
6. e^{ax+b}
7. $e^{\sin x}$
8. $y = e^{ax+b}$
9. $2^x - 3^{4x}$
10. By writing $e^{(ax+b)}$ as $e^b(e^x)^a$, show that $\frac{d}{dx} e^{(ax+b)} = a e^{(ax+b)}$
11. $y = x^3 e^{3x}$
12. $y = x^2 - e^{-x}$
13. $y = \frac{1}{(3+x^2)}$
14. $y = e^{\frac{x}{a}} \cos \frac{x}{a}$
15. $y = (ax+b) e^{-x}$
16. Show that $\frac{d}{dx} \frac{1}{2} \{e^{2x} + e^{-x}\}^2$
17. $\sqrt{x} e^{x/2}$
18. $y = \frac{1+e^x}{1-e^x}$

II. Find $\frac{dy}{dx}$

1. $y = \log(ax^2 + bx + c)$
2. $y = \log(x^2 + 2x)$
3. $y = \log \sqrt{1-x^2}$
4. $y = \log(x+1)(x+2)$
5. $y = \log \left(\frac{1+x^2}{1-x^2} \right)$
6. $y = 2 \log \sin x$
7. $y = \log(x \sqrt{x^2 + a^2})$
8. $(x^2 + 1) \log(x^2 + 1)$
9. $\log y = x$
10. $y = \log(x + \sqrt{a^2 + x^2})$
11. Find the derivatives of the following terms by taking logarithms before derivation :
 - (a) e^{x^2+2x+2}
 - (b) $e^x (6x^2 + 6x + 7)$
 - (c) $\frac{(x+2)(x+3)}{(x-2)(x-3)}$
 - (d) $x^2(2x-1)^2$
 - (e) x^x
 - (f) $x^{\sin x}$

- I. (1) Find the derivative of the function z with respect to x ; given :
 $y = x^2 + 3x$ and $z = y^2 + 1$
- (2) Find the derivative of the following function of t with respect to x ; given :
 $y = x^2 + 5x$ and $t = y^2 + 5$

J. Differentiate the following functions :

1. $(x^2 + 1) \log_e(x^2 + 1)$
2. $\sqrt{x^2 - 2x^3}$
3. $\frac{1 + 4e^x}{2 + 3e^x}$
4. $\frac{3x+2}{4x^2+3}$

$$5. \quad \sqrt{\frac{a^2+x^2}{a^2-x^2}}$$

$$6. \quad \frac{x+2}{x^2+x-6}$$

K. Find $\frac{dy}{dx}$ by the first principle

$$1. \quad y = x^2$$

$$2. \quad y = \log x$$

$$3. \quad y = e$$

$$4. \quad y = \frac{1}{x}$$

$$5. \quad y = \frac{1}{\sqrt{x}}$$

L 1. If $f(x) = \frac{\cos^2 x}{1 + \sin^2 x}$

$$\text{Show that } f\left(\frac{\pi}{4}\right) - 3f'\left(\frac{\pi}{4}\right) = 3$$

2. Show that $y = \frac{x - e^{-x^2}}{2x^2}$ satisfies the differential equation

$$y \frac{dy}{dx} + 2y = e^{-x^2}$$

3 Find $\frac{dy}{dx}$ if (i) $y = x^x$

[Hint. Take log twice, then $\log \log y = x \log x + \log (\log x)$]

(ii) $y = (\log x)^{\log x}$

4. If $y \sqrt{x^2 + 1} = \log (x + \sqrt{x^2 + 1})$

show that : $(x^2 + 1) \frac{dy}{dx} + xy = 1$

6.5. Derivatives of Higher Order

Let us consider a functional relationship between two variables x and y , viz. $y = 8x^4$, if we want to measure the rate of change in y due to a small change in x , we find $\frac{dy}{dx}$ of this expression which is given by

$$\frac{dy}{dx} = 32x^3$$

This expression $(32x^3)$ can once again be differentiated as

$$\frac{d}{dx} (32x^3) = 96x^2$$

which is in fact the derivative of $\frac{dy}{dx}$ ($= 32x^3$). In the terminology of calculus we call this expression ($96x^2$) is the second differential coefficient. It is to represent this symbolically as follows:

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2} = 96x^2$$

Here again it is to be noted that $\frac{d^2y}{dx^2}$ does not mean d^2y divided by dx^2 . Also that d^2 here in no way means multiplied by square of d and that dx in no way means d multiplied by square of x . The ratio $\frac{d^2y}{dx^2}$ just signifies that (the original function which is $8x^4$) has been differentiated *two times* and each time with x .

More specifically as $\frac{dy}{dx}$ measures the rate at which y is changing with small changes in x similarly $\frac{d^2y}{dx^2}$ measures the rate at which $\frac{dy}{dx}$ is changing with small changes in x . In other words $\frac{dy}{dx}$ measures the rate of change in $y = 8x^4$ as x changes while $\frac{d^2y}{dx^2}$ measures the rate of change of the rate of change (that is $32x^3$) as x changes.

We can go further to find the third, fourth and other derivatives of $y = 8x^4$ as follows. The successive differential coefficients of $y = 8x^4$ will be as follows:

Given function $y = 8x^4$

$$\text{First differential coefficient} = \frac{dy}{dx} = 32x^3$$

$$\text{Second} \quad \quad \quad = \frac{d}{dx} (32x^3) = \frac{d^2y}{dx^2} = 96x^2$$

$$\text{Third} \quad \quad \quad = \frac{d}{dx} (96x^2) = \frac{d^3y}{dx^3} = 192x$$

$$\text{Fourth} \quad \quad \quad = \frac{d}{dx} (192x) = \frac{d^4y}{dx^4} = 192$$

$$\text{Fifth} \quad \quad \quad = \frac{d}{dx} (192) = \frac{d^5y}{dx^5} = 0$$

Important Alternative Notations The successive differential coefficients or derivatives of the original functions are also conveniently denoted in a particular short form as follows:

If the given function is denoted by $y = f(x)$, then

First differential coefficient or first derivative is denoted by $f'(x)$ or $\frac{dy}{dx}$

Second " " second " " " " " $f''(x)$

Third " " third " " " " " $f'''(x)$

Fourth " " fourth " " " " " $f^{(4)}(x)$

and so on

Ex. 18 If $f(x) = x^2 - 5x^2 + 7$, find $f'(x)$ and then find $f'(1)$ and $f'(2)$ & what value of x does $f'(x)$ vanish?

Since $f(x) = x^2 - 5x^2 + 7$

$$f'(x) = 3x^2 - 10x \quad \text{Substituting } x=1, f'(1) = -7$$

$$f''(x) = 6x - 10 \quad \text{Substituting } x=2, f''(2) = 2$$

If $f'(x)$ is to be zero, then $3x^2 - 10x = 0$

$\therefore f'(x)$ would be zero (or it would vanish) if and only if $x=0$ or $\frac{10}{3}$

EXERCISE 6.2

1. Write all the derivatives of the following functions

$$(a) \ y = x^3 - 6x^2 + 9x \quad (b) \ y = 10x^5 - 4x^3 + 5x - 7$$

2. Write the third derivatives of

$$(a) \ \frac{1}{\sqrt{x}} \quad (b) \ \sqrt{1-x^2}$$

3. If $y = \sqrt{4-x^2}$ find f'' and value of x when $f'' = 0$

4. If $y = (4-x^2)^3$, find all the values of x when $f' = 0$

5. If $y = x^3 - 9x^2 + 9x$, evaluate all derivatives for $x = 1$ and $x = 3$

6. Find first and second derivatives of $y = \left(\frac{1}{ax+b}\right)$ and $y = \left(\frac{ax+\beta}{ax+b}\right)$

Prove that the successive derivatives of these functions differ only by a constant factor.

7. If $xy = a$, show that $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = 0$

Should the same result hold for $a + bx = xy$?

8. If $y = \left(x^2 + \frac{1}{x^2}\right)$, show that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} = 4$

Should the same result hold for the relation $y = \left(ax^2 + \frac{b}{x^2}\right)$?

9. Find the second derivatives of the following

$$(a) \ y = \log e^{ax^2+bx+c} \quad (b) \ y = \log (ax^2 + bx + c)$$

$$(c) \quad y = \log \left(\frac{(ax+b)^2}{(ax-b)} \right) \quad (d) \quad y = x^2 \log x$$

10. If $y = \log(x + \sqrt{x^2 + 1})$
prove that

$$(x^2 + 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = 0$$

11. If $xy = a + bx$ then show that

$$x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = 0$$

12. If $y = \log_e x$, show that

$$x^2 \frac{d^2y}{dx^2} + 1 = 0$$

13. Find the derivative of each of the following

$$(i) \quad e^{x^a} \quad (ii) \quad \log x \quad (iii) \quad \sin x \quad (iv) \quad \sqrt{x}$$

14. If $f(x) = (1+x)^n$, find

$$f(0), f'(0), f''(0), f'''(0).$$

15. Show that the function $y = xe^x$ satisfies the equation

$$x^2y - xy' + y = 0$$

16. If $f(x) = x^2$ show that

$$f(1) + \frac{f'(1)}{1} + \frac{f''(1)}{2} + \frac{f'''(1)}{3}$$

17. Deduce

$$(uv)'' = u''v + 2u'v' + uv''$$

$$\text{and} \quad (uv)''' = u'''v + 3u''v' + 3u'v'' + uv'''$$

(This is termed as Leibniz rule)

18. Using Leibniz rule find the second derivative of

$$(i) \quad y = e^2 \cos x \quad (ii) \quad y = x^2 \log x$$

6.6 Sign of the Differential Coefficient

Let us once again go back to the actual meaning of differential coefficient. As explained above $\frac{dy}{dx}$ is the

limit of a ratio and it means $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$
or in general terms if we are given a functional relationship between x and

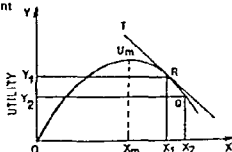


Fig. 6.2

$y, \frac{dy}{dx}$ is the rate of change of y when x changes. If we take y as total utility and x as consumption, $\frac{dy}{dx}$ will be the rate of change in total utility y if the consumption x changes by small amounts. Here with small increments in consumption the total utility also rises by small amounts.

Nevertheless as we know after a point of saturation has been reached the increment in consumption may cause fall in the total utility. At this juncture we have to be particular about the sign of Δx and Δy .

For this let us now consider a point R on the right side of U_m on the total utility curve in Fig. 6.2

At point R the total utility is OY_1 with the consumption amount equal OX_1 . If the consumption increases to OX_2 the total utility falls OY_2 .

In other words with the increment of consumption viz. $OX_2 - OX_1 = X_1X_2$ the total utility changes by an amount $OY_2 - OY_1 = -Y_1Y_2$ and the marginal utility

$$= \lim_{x \rightarrow x_1} \frac{Y_1Y_2}{X_1X_2} \text{ (which is a negative amount)}$$

If we denote the change in the consumption X_1, Y_2 by Δx (which has a positive sign) the change in total utility will be denoted by $(-\Delta y)$.

Now to find the rate of change in total utility y at point R , we must put the limiting value of Δx

$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$ will be as before $\frac{dy}{dx}$ but the value will be negative. In other words the rate of change in y is negative (note that the slope of the tangent line TRQ is also negative).

At this stage we may conclude that if the change in the dependent variable

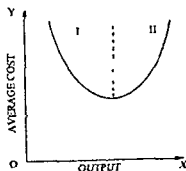


Fig. 6.3

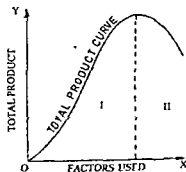


Fig. 6.4

(y in our example) is not associated with the change in the independent variable (x in our example) in the same direction the sign of the differential coefficient becomes negative. It gives us the notion of the nature of the curve represented by the given function.

Sign of Derivative and Nature of Function Let us now relate the sign of derivative $\frac{dy}{dx}$ with the nature of the curve represented by the given function.

Suppose we are given two functions. Average cost and total production which when plotted give us the graphs as given in Fig. 6.3 and 6.4. We divide each curve into two parts at the minimum average cost point and maximum product point and consider each part of the each curve separately.

Fig. 6.5 gives Part I of Fig. 6.3 and Fig. 6.6 gives Part II of Fig. 6.4. In both the above cases (Fig. 6.5 and 6.6) the increment in variable factor x is associated with the fall in the average cost as well as the total product. Hence the sign of differential coefficient will be negative (i.e. the rate of change in

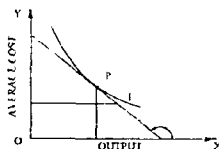


Fig. 6.5

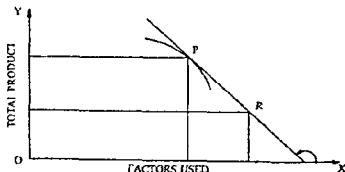


Fig. 6.6

average cost or total product is negative with respect to changes in variable x on these parts of curves)

In case of curves in Figs 6.7 and 6.8 total cost and total product are rising. Hence the sign of the differential coefficient will be positive.

Also note that tangent (or slope of the curve) on the points of these part of the curve is rising (positive) from left to right; thus the slope of the curve is positive.

Relation of function with respect to its derivative gives us the following important properties of the derivatives.

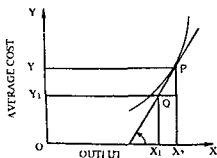


Fig. 6.7

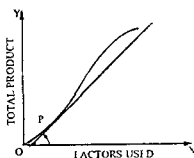


Fig. 6.8

- (1) If $\frac{dy}{dx} > 0$ i.e. when first derivative of the given function is positive the curve rises from left to right.
- (2) If $\frac{dy}{dx} < 0$ i.e. when first derivative of the given function is negative the curve falls from left to right.

There still remains one question to be answered.

What will be the nature of the curve if $\frac{dy}{dx}$ is neither > 0 nor < 0 but exactly equal to zero?

In fact when $\frac{dy}{dx} = 0$ at any point it means that the rate of change in y with respect to a small increment in x at that particular point is zero.

To examine the nature of curve when $\frac{dy}{dx} = 0$ let us consider the following three functions.

$$1. \quad y = x^2 - 4x + 10$$

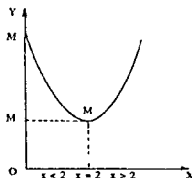
$\frac{dy}{dx} = 2x - 4$ If $\frac{dy}{dx}$ is to be zero, then $2x - 4 = 0$ i.e. $x = 2$ i.e. $\frac{dy}{dx}$ of the

given function will be equal to zero if x takes the value two. But what happens when x is less than or greater than 2?

(2) If $x < 2$, $\frac{dy}{dx} (= 2x - 4)$ turns out to be negative, which in turn means that curve should be falling from left to right.

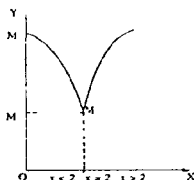
(3) If $x > 2$, $\frac{dy}{dx} (= 2x - 4)$ turns out to be positive, which in turn means that curve should be rising from left to right.

But when $x = 2$, the curve neither rises nor falls, it is stationary. This point is called stationary point since when $x = 2$, the value of y is not changing for a moment, but the very next moment its (y 's) value rises as soon as $x > 2$.



TURNING OR STATIONARY
POINT

Fig 6.9



TURNING OR STATIONARY
POINT

Fig 6.10

Thus, so far $x < 2$, the function gives a falling curve and so far $x > 2$ the function gives a rising curve.

Obviously, therefore, the curve changes its course from falling to rising at $x = 2$; hence this point is termed as "turning point".

On the basis of the above discussion the two possible shapes of the given curve $y = x^2 - 4x + 10$ can be as given in Fig 6.9 or Fig 6.10. (The exact shape can be determined on the basis of a discussion of convexity of the curve.)

Both Fig. 6.9 and Fig 6.10 show that till x attains the value 2, the curve is falling from left to right (since $\frac{dy}{dx} < 0$). After point M has been reached (that is when $x = 2$) curve rises from left to right for all values of $x > 2$ (since $\frac{dy}{dx} > 0$).

Clearly, therefore, point M (when $x = 2$) becomes the point of the curve

where the value of function is minimum

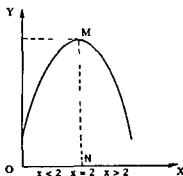
II. Consider another function

$$y = 10 + 4x - x^2, \quad \frac{dy}{dx} = 4 - 2x$$

If $\frac{dy}{dx}$ is to be zero, $4 - 2x = 0$, that is $\frac{dy}{dx}$ of the given function will be equal to zero if x takes the value 2

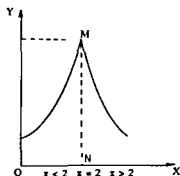
Here, again, we have two cases, but just opposite to those in the above example

(i) If $x < 2$, $\frac{dy}{dx} = 4 - 2x$ turns out to be positive, which means that the curve should be a rising one from left to right



TURNING OR STATIONARY
POINT

Fig 6 11



TURNING OR STATIONARY
POINT

Fig 6 12.

(ii) If $x > 2$, $\frac{dy}{dx} = 4 - 2x$ turns out to be negative which means that the curve should be a falling one from left to right.

But when $x = 2$ the curve neither rises nor falls but is stationary

Exactly in the above manner, the two possible shapes of the curve $y = 10 + 4x - x^2$ could be as given in Fig 6 11 or Fig 6 12 (the exact shape remains to be determined)

In Figs. 6 11 and 6 12, till x attains the value 2 the curve rises up to point M and falls when x takes any value greater than 2.

Here at M , the function attains the maximum value.

III We now consider yet another function -

$$y = 6x - 3x^2 + \frac{1}{2}x^3$$

$$\frac{dy}{dx} = 6 - 6x + \frac{3}{2}x^2 = \frac{3}{2}(x-2)^2$$

In this case the value of $\frac{dy}{dx}$ becomes zero as soon as $x = 2$

That is at $x = 2$ $\frac{dy}{dx}$ of the given function will be zero hence $x = 2$ is a stationary point

If $x < 2$ $\frac{dy}{dx}$ of the given function has positive value and hence curve will again be a rising curve

If $x > 2$ $\frac{dy}{dx}$ of the given function has still positive value and hence curve will again be a rising one

Therefore in the possible shapes of the given curve

$$y = 6x - 3x + \frac{1}{2}x^3$$

there will be a bend at $x = 2$

The curve always rises even before or after $x = 2$, but at $x = 2$ the curve has stationary point at M . Here stationary point happens to be neither maximum nor minimum. In this particular case, the point M is called a 'point of inflexion'.

From the above three illustrations we learn one more property of $\frac{dy}{dx}$ which is

when $\frac{dy}{dx} = 0$,

(1) the rate of change is zero, and

(2) the rising or falling course of the curve changes after the point at which

$$\frac{dy}{dx} = 0$$

In our first two illustrations, the curve changed the nature of rising (or falling) to falling (or rising) one. Therefore, the point at which $\frac{dy}{dx} = 0$, the function is either maximum or minimum. In the case of third illustration the curve keeps rising even after a point when $\frac{dy}{dx} = 0$. Nevertheless the nature of the curve definitely changes.

Summarising, we have now three important properties of $\frac{dy}{dx}$ which help us to say something about the nature of the curve

- (1) If $\frac{dy}{dx} > 0$ The curve rises from left to right
- (2) If $\frac{dy}{dx} = 0$ The curve remains stationary for a moment and then changes its course. Therefore, this point is minimum, maximum or point of inflexion
- (3) If $\frac{dy}{dx} < 0$ The curve falls from left to right

We thus know where the curve is rising, falling or stationary. Even though we are sure about the rising of the curve when $\frac{dy}{dx} > 0$, or about its falling when $\frac{dy}{dx} < 0$, the particular nature (or shape) of curve still remains undecided.

For example, if $\frac{dy}{dx} > 0$, the rising curve can take either of two shapes given in Fig. 6.13 and 6.14, and if $\frac{dy}{dx} < 0$ falling curve can take either of two shapes given in Fig. 6.15 and 6.16.



$$\frac{dy}{dx} > 0$$

Fig. 6.13



$$\frac{dy}{dx} > 0$$

Fig. 6.14



$$\frac{dy}{dx} < 0$$

Fig. 6.15



$$\frac{dy}{dx} < 0$$

Fig. 6.16

Also, how are we to decide that a point where $\frac{dy}{dx} = 0$ is a maximum or minimum point of the curve? For example, in each of the first two illustrations $\frac{dy}{dx} = 0$ at point M but in the first illustration point M becomes a minimum point (see Fig. 6.9 and 6.10) while second illustration shows M as a maximum point of the curve (see Fig. 6.11 and 6.12).

Thus, with the three properties of the first derivative summarized above, we are still faced with the following problems:

1. If $\frac{dy}{dx} > 0$, How to decide about the rising nature of the curve (i.e., whether the rising shape of curve is concave or convex to x -axis)?
2. If $\frac{dy}{dx} < 0$, How to decide about the falling nature of curve (i.e., whether the falling shape of curve is concave or convex to x -axis)?

- 3 If $\frac{dy}{dx} = 0$, How to decide whether it is a maximum point or a minimum point?

We have to make use of the second derivative to solve these problems

6.7. Second Derivative and Nature of Curve

We have learnt that as a first derivative $f'(x)$ measures the rate of change in y with respect to x , in the same way the second derivative $f''(x)$ measures the rate of change of first derivative $f'(x)$. In other words, the second derivative measures the rate of change of the rate of change original function $y = f(x)$

To put this in a comparative way

First derivative :

If $f'(x) \left(= \frac{dy}{dx} \right) > 0$, It means that the value of function tends to increase (or that the rate of change of y w.r.t x is positive)

If $f'(x) \left(= \frac{dy}{dx} \right) < 0$, It means that the value of function tends to decrease (or that the rate of change of y w.r.t x is negative)

Second derivative :

If $f''(x) \left(= \frac{d^2y}{dx^2} \right) > 0$, It means that the slope of curve tends to increase (or that the rate of change of dy/dx is positive)

If $f''(x) \left(= \frac{d^2y}{dx^2} \right) < 0$; It means that the slope of curve tends to decrease (or that the rate of change of dy/dx is negative)

Thus, the first derivative refers to the absolute value of function and the second derivative refers to the slope of the curve. We shall, therefore, make use of the second derivative for deciding the nature of the curve.

In the plane let us consider a single valued differential function $y = f(x)$. We graph this function

A plane curve is called convex up or convex upwards on the interval (b, c) if all points of the curve lie above any tangent to it on the interval. The reverse is the case when the curve is convex down or downward. A convex down curve is also called a concave down or simply concave curve. Similarly, a convex up curve is simply called convex or concave down curve if at all points of an interval (a, b) , the second derivative of the function $f(x)$ is negative, i.e., $f''(x) < 0$, the curve on this interval is convex upwards (concave downwards). Similarly, when $f''(x) > 0$, the curve $y = f(x)$ is convex downwards (or concave upwards or simply concave). When $f''(x) = 0$, it is a necessary condition of a point of inflexion and the sufficient condition is that $f''(x)$ changes sign. Thus, a point that separates the convex part of a curve its concave part.

We consider the following four possible cases :

Case 1

$$f'(x) > 0 \text{ and } f''(x) > 0$$

This would mean that the value of function is increasing at an increasing rate or that the slope of the curve at this point is positive and increasing as the value of x increases (Fig. 6 17)

Case 2

$$f'(x) > 0 \text{ and } f''(x) < 0$$

This would mean that the value of function is increasing at a decreasing rate or that the slope of the curve is positive but decreasing as the value of x increases (Fig. 6 18)

Case 3

$$f'(x) < 0 \text{ and } f''(x) > 0$$

This means that the value of function is decreasing at a decreasing rate or that the slope of the curve must be negative and decreasing as x increases (Note that it means that the negat.ve slope of the curve tends to become steeper and steeper as the value of x increases) (Fig. 6 19)

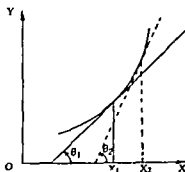


Fig. 6 17

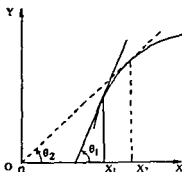


Fig. 6 18

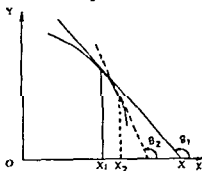


Fig. 6 19

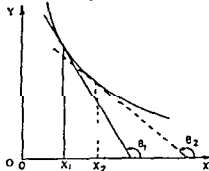


Fig. 6 20

Case 4

$$f'(x) < 0 \text{ and } f''(x) > 0$$

This means that the value of function is decreasing at an increasing rate or that the slope of the curve is negative but increasing as the value of x increases. Remember that this means that the negative slope tends to become less steep as x increases (Fig. 6.20)

Let us describe the above cases in different ways

$$\text{Case 1. } f'(x) > 0 \text{ and } f''(x) > 0$$

The curve will have the shape as given in Fig. 6.17. It is concave upwards (simply concave and thus convex downwards)

Since $f'(x) > 0$, the slope of curve is positive and since $f''(x) > 0$, the slope of curve tends to become steeper and steeper as x increases

(i.e., θ_2 becomes greater than θ_1 as X_1 rises to X_2)

$$\text{Case 2. } f'(x) > 0 \text{ and } f''(x) < 0$$

The curve will have shape as given in Fig. 6.18. It is concave downwards or convex upwards or simply convex

Since $f'(x) > 0$, the slope of curve is positive and since $f''(x) < 0$, the slope of the curve goes on decreasing as x increases

(i.e., θ_2 becomes less than θ_1 as x increases)

$$\text{Case 3. } f'(x) < 0 \text{ and } f''(x) < 0$$

The curve will have the shape as given in Fig. 6.19. It is also concave downwards (and thus convex upwards)

Since $f'(x) < 0$, the slope of curve is negative and since $f''(x) < 0$, the negative slope of the curve goes on decreasing as x increases

(i.e., θ_2 becomes less than θ_1 as x increases)

$$\text{Case 4. } f'(x) < 0 \text{ and } f''(x) > 0$$

The curve will have the shape as given in Fig. 6.20. It is concave upwards (and thus convex downwards)

Since $f'(x) < 0$, the slope of curve is negative and since $f''(x) > 0$, the negative slope of the curve goes on increasing as x increases.

(i.e., θ_2 becomes greater than θ_1 as x increases)

Thus with the help of second derivative we are able to decide about the rising and falling nature of the curve.

The only problem which now remains is about maximum or minimum point of the curve when $\frac{d^2y}{dx^2} = 0$. Before we deal with this problem, let us first obtain the minimum and the maximum points of the function.

Ex. 19. Examine the curve $y = x^3$ for convexity

Given $y = x^3$

$$y' = 3x^2$$

$$y'' = 6x$$

$$y' < 0 \quad \text{for } x < 0$$

$$\text{and } y' > 0 \quad \text{for } x > 0$$

$$\text{also } y'' = 0 \quad \text{for } x = 0$$

Hence, for $x < 0$ the curve is convex upwards (concave downwards) and for $x > 0$ it is convex downwards $x = 0$ is the point of inflexion

Ex. 20. Examine the curve $y = x^3 - 3x^2 - 9x + 6$ for convexity

Here, $y = 3x^2 - 6x - 9$

$$y' = 6x - 6$$

$$\text{Hence } y' < 0 \quad \text{for } x < 1$$

$$> 0 \quad \text{for } x > 1$$

$$= 0 \quad \text{for } x = 1$$

Thus, $x = 1$ is the point of inflexion. The curve is convex upwards for $x < 1$ and convex downwards for $x > 1$

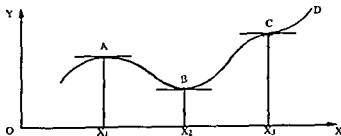


Fig 6.21

6.8. Maximum and Minimum Values of a Function

Suppose a given function $y = f(x)$ when plotted takes a form as given in Fig 6.21. We consider three points A, B and C where $\frac{dy}{dx} = 0$ in each case

Consider Point A We call this point a maximum point because y has a maximum value when $x = OY_1$ (Fig 6.22) but all the same we know that the value of Y at this point is not higher than

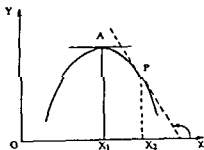


Fig 6.22.

the value of Y at point D of the same curve. Therefore, when we say that at A , Y is maximum it only means that the value of Y at A is higher than any value on either side of A .

This also means that value of Y increases with the increase in X up to a point A , but it must fall after point A has been reached. With this definition of maximum point we shall be able to solve the problem posed above. Consider a part of curve involving maximum point A separately as shown in Fig. 6.22. As value of x increases from X_1 to X_2 the slope of the curve changes from zero to negative quantity, or that slope of curve is decreasing as x increases. Thus if A is to be maximum, then

$$f'(x) = 0 \text{ and also } f''(x) < 0 \text{ at the point } A$$

Consider Point B We call this point a point where Y is minimum because value of Y at B is smaller than any value on either side of B . In fact, it

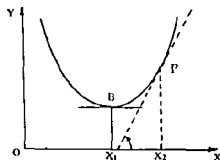


Fig. 6.23

is a point where Y is minimum. This also means that value Y decreases as the value of X increases up to a point B , but it must increase after point B (minimum) has been attained. In Fig. 6.23, as X increases from X_1 to X_2 the slope of the curve increases from zero (since $\frac{dy}{dx} = 0$ at B) to positive values or that the slope of the curve is increasing. Therefore, if B is to be minimum point then

$$f'(x) = 0 \text{ and also } f''(x) > 0 \text{ at point } B$$

In both the above cases, $f'(x) = 0$, but we are able to decide about the maximum or minimum point with the help of second derivative.

Consider Point C On both sides of point C the curve slopes upwards. Therefore, at all the points to the right or left of C , the value of first derivative must be positive (except at point C when $f'(x) = 0$). This point is called point of inflexion because of the bend in the curve. We need a more rigorous treatment for deciding the shape of the curve under this case. However, we give below simple criteria for points of inflexion:

(1) $f'(x) = 0$ but $f''(x) \neq 0$ there is a point of inflexion

(2) $f'(x) = 0$, $f''(x) = 0$ but $f'''(x) \neq 0$ then, if $f'''(x) < 0$ it is maximum and if $f'''(x) > 0$ it is minimum

The conditions imply that $f'(x) = \frac{dy}{dx}$ is maximum or minimum at the point of inflexion since $f''(x) = 0$ and $f'''(x) \neq 0$. It is not necessary that $f''(x) = 0$ at the point of inflexion. It may be (Fig. 6.21) or it may not be (Figs 6.24 and 6.25)

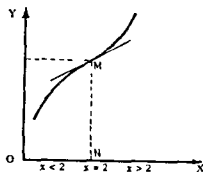


Fig 6.24

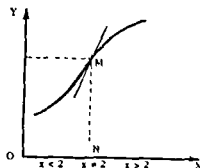


Fig 6.25

6.9. Order Conditions for Maximum-Minimum (Extreme) Values

We have learnt that for a function $y = f(x)$ if its first derivative at a certain point (say at $x = a$) is equal to zero then the value of the given function at this point ($x = a$) will either be

- (i) minimum if $f''(a) > 0$ or,
- (ii) maximum if $f''(a) < 0$

Thus, we observe that the given function should satisfy two conditions in order to decide about maximum or minimum value at a particular point. These conditions are known as *Order Conditions*

(a) Conditions for minimum value

- (1) First order condition (Necessary condition) $\left[f'(x) = 0, \text{ i.e., } \frac{dy}{dx} = 0 \right]$
- (2) Second order condition (Sufficient condition) $\left[f''(x) > 0, \text{ i.e., } \frac{d^2y}{dx^2} > 0 \right]$

(b) Condition for maximum value

- (1) First order condition (Necessary condition) $\left[f'(x) = 0, \text{ i.e., } \frac{dy}{dx} = 0 \right]$
- (2) Second order condition (Sufficient condition) $\left[f''(x) < 0, \text{ i.e., } \frac{d^2y}{dx^2} < 0 \right]$

Note that first order condition is the same for both maximum and minimum values

(c) In certain cases when $f''(x) = 0$ the above conditions fail to give the maximum or minimum values of the given function

If $f'(x) = 0$ it means there may be a point of inflexion

Ex 21 Find the maximum and minimum values of the following two functions

(i) $y = x^3 - 3x + 1$

(ii) $y = 3x^4 - 10x^3 + 6x^2 + 5$

(i) Given $y = x^3 - 3x + 1$

$$\frac{dy}{dx} = f'(x) = 3x^2 - 3 \text{ and } \frac{d^2y}{dx^2} = f''(x) = 6x$$

First order condition for maximum and minimum value is $f'(x) = 0$

Therefore if the given function has maximum or minimum value then $3x^2 - 3 = 0$ i.e. $x = \pm 1$

To distinguish between maximum and minimum values we apply second order condition

(i) At $x = +1$ $f'(1) = 6 \times 1 = 6 > 0$ i.e. Minimum value

(ii) At $x = -1$ $f'(-1) = 6 \times (-1) = -6 < 0$ i.e. Maximum value

Hence the function y has a minimum value 1 at $x = 1$ and maximum value 3 at $x = -1$

(ii) Given $y = 3x^4 - 10x^3 + 6x^2 + 5$

$$f'(x) = 12x^3 - 30x^2 + 12x$$

and $f''(x) = 36x^2 - 60x + 12$

First order condition $12x^3 - 30x^2 + 12x = 0$

$$\text{or } 3x(4x^2 - 10x + 4) = 0$$

i.e. either $x = 0$ or $4x^2 - 10x + 4 = 0$

i.e. either $x = 0$ or $(4x - 2)(x - 2) = 0$

i.e., either $x = 0$ or $x = 2$ or $x = \frac{1}{2}$

Second order condition

(i) At $x = 0$ $f''(0) = 36(0) - 60(0) + 12 = 12 > 0$

Hence function has a minimum value 5 at $x = 0$

(ii) At $x = 2$, $f''(2) = 36(4) - 60(2) + 12 = 36 > 0$

Hence the function has minimum value -3 at $x = 2$

(iii) At $x = \frac{1}{2}$ $f''(\frac{1}{2}) = 36(\frac{1}{4}) + 12 = -9 < 0$

Hence the function has maximum value $\frac{87}{16}$ at $x = \frac{1}{2}$

Ex. 22. Show that the curve $y = 2x - 3 + \frac{1}{x}$ is a convex from below for all positive values of x

$$y = 2x - 3 + \frac{1}{x}$$

$$f'(x) = 2 - x^{-2} = 2 - \frac{1}{x^2}$$

$$\text{and, } f''(x) = 2x^{-3} = \frac{2}{x^3}$$

If this curve is to be convex from below, then it must possess minimum value for all positive values of x

$$\text{Hence } f''(x) > 0$$

First order condition

$$f'(x) = 0, \quad \left(2 - \frac{1}{x^2}\right) = 0, \quad \therefore x = \pm \sqrt{\frac{1}{2}}$$

Second order condition

$$f''(x) > 0$$

Substituting, $x = \sqrt{\frac{1}{2}}$ in $f''(x) = 2x^3$, we obtain

$$f''(x) = 2 \left(\frac{1}{\sqrt{\frac{1}{2}}} \right)^3 > 0$$

Hence condition for minimum value is satisfied

Substituting, $x = -\sqrt{\frac{1}{2}}$ in $f''(x)$, we obtain,

$$f''(x) = 2 \left(\frac{1}{\sqrt{\frac{1}{2}}} \right)^3 > 0$$

Hence, the condition for minimum value is not satisfied

Thus the given curve $y = 2x - 3 + \frac{1}{x}$ is convex from below for all positive values of x [$\because$ any positive value of x will make $f''(x) > 0$]

Ex. 23. Show that $y = x + \frac{1}{x}$ has one maximum and one minimum value and the latter is larger than the former

$$\text{Given } y = x + \frac{1}{x}$$

$$f'(x) = 1 - x^{-2} = 1 - \frac{1}{x^2} \text{ and } f''(x) = 2x^{-3} = \frac{2}{x^3}$$

First order condition

When $f'(x) = 0$, then $1 - \frac{1}{x^2} = 0$ or $x = \pm 1$

Second order condition

$$(i) \quad \text{At } x = 1, f''(x) = \frac{2}{x^3} = 2 > 0$$

Hence curve has minimum value $y = 2$ at $x = 1$

$$(ii) \quad \text{at } x = -1, f''(x) = -2 < 0$$

Hence curve has maximum value $y = -2$ at $x = -1$

As such minimum value is greater than the maximum value

Note. Though it seems peculiar that maxima is less than minima nevertheless this may happen because the maxima and minima are defined as only local turning points

Ex. 24. Show that $y = \frac{1}{2x+1}$ and $y = \frac{1-x}{1+x}$ are monotonic functions of x

Monotonic property could be examined from the first derivative of the function. If the value of first derivative is positive for all values of x , we say that the given function is monotonically increasing. If it is negative we say that the given function is monotonically decreasing.

$$(i) \quad y = \frac{1}{2x+1} \quad f'(x) = \frac{-2}{(2x+1)^2} \quad (\text{By quotient rule})$$

For all values of x , $f'(x) < 0$. Hence given function is monotonically decreasing.

$$(ii) \quad y = \frac{1-x}{1+x} \quad f'(x) = \frac{-2}{(1+x)^2} \quad (\text{By quotient rule})$$

For all values of x , $f'(x) < 0$

Hence the function is monotonically decreasing.

EXERCISE 6.3

Find the maximum, minimum and inflexion values (wherever the case may be) of the following functions and state the corresponding values of x :

1. $y = x^3 - 6x^2 + 9x$

2. $y = x^3 - 3x^2 + 20$

3. $y = 3x^2 - x + 1$

4. $y = \frac{1}{3}x^3 - 2x^2 + 4x + 1$

5. $y = -x^3 + \frac{9}{2}x^2 - 6x + 6$

$$6. \quad y = x^4 - 4x^3 + 6x^2 - 4x - 3 \qquad 7. \quad y = \frac{1-x^2}{1+x^2}$$

$$8. \quad y^2 = x(x^2 - 1)$$

$$9. \quad y = (x+1)(x-2)^2$$

$$10. \quad y = 4x + \frac{1}{x}$$

$$11. \quad y = x^3 + x - x^2 + 1$$

12. Show that $y = x\sqrt{1+x}$ has a minimum value and $y = -x\sqrt{1+x}$ has maximum value. Why is it so?

13. Show that the curves $y = \frac{a}{x-b}$ and $y = (a-bx)^2$ are downward sloping and convex from below

14. Show that the curves $y = \sqrt{a-bx}$ and $y = \sqrt{a-bx^2}$ are downward sloping and concave from above

15. Show that the curve $y = \sqrt{x-b} + c$ is upward sloping and concave from below

16. Show that the curve $y = 4a^4 - 3ax^3 + x^3$ ($x < 2$) is downward sloping with a point of inflexion

17. Show that the first derivative of the curves

$$\sqrt{ax+b} + c \text{ and } y = ax \left(\frac{x+b}{x+c} \right) + d \quad (b > c)$$

fall continuously as the value of x increases

18. Show that the function $y = 3x^3 + 3x^2 + x - 1$ is a monotonic increasing function

If $y = ax^3 + bx^2 + cx + d$, express its first derivative in the form of a square plus a constant term

Deduce that the function is monotonically increasing if $b^2 < 3ac$

(Note: In all the examples from 13 to 18, a, b, c and d are positive constants)

19. Which of the following quadratic functions are concave upwards

$$(i) \quad y = 6x^2 - 4x + 11 \qquad (ii) \quad W = -3x^2 + 7$$

$$(iii) \quad U = 3 - x^2 \qquad (iv) \quad V = 2 + 3x + x^2$$

20. Does the function $y = 3 + 5x + 4x^2$ possess an inflexion point?

21. Establish the nature of the function $y = e^x$

22. Determine whether the following utility functions would give rise to indifference curves of usual nature

$$(i) \quad u = q_1^2 + q_2^2 \qquad (ii) \quad u = (q_1 + q_2)^2$$

$$(iii) \quad u = q_1 q_2$$

Chapter 7

Applications of Simple Derivatives

7.1 Differential Coefficient and Elasticity of Demand

Price elasticity of demand is defined as the value of the ratio of the relative (or proportionate) change in the demand to the relative (or proportionate) change in the price

Precisely if we suppose that the demand changes from (x) to $(x + \Delta x)$ when the price changes from (p) to $(p + \Delta p)$ then elasticity of demand as per definition is given by

$$\frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in price}} = \frac{\frac{\Delta x}{x}}{\frac{\Delta p}{p}} = \frac{p}{x} \times \frac{\Delta x}{\Delta p}$$

In fact the ratio $\frac{p}{x} \times \frac{\Delta x}{\Delta p}$ gives us the average elasticity of demand over the price range of (p) to $(p + \Delta p)$. But since we are interested in the point elasticity i.e., elasticity of demand at a particular price level, say (p) then Δp should be made to approach zero. In terms of calculus we should take limit of this ratio as follows

Elasticity of demand at price p is

$$e_d = \lim_{\Delta p \rightarrow 0} \left\{ \frac{p}{x} \times \frac{\Delta x}{\Delta p} \right\} = \frac{p}{x} \lim_{\Delta p \rightarrow 0} \frac{\Delta x}{\Delta p}$$

According to the law of demand if price increases (Δp is positive) demand falls (Δx is negative). Thus,

$$e_d = (-) \frac{p}{x} \lim_{\Delta p \rightarrow 0} \frac{\Delta x}{\Delta p}$$

The absolute positive value of e_d is denoted by $|e_d|$

$$\text{Therefore price elasticity} = |e_d| = \left| \frac{p}{x} \lim_{\Delta p \rightarrow 0} \frac{\Delta x}{\Delta p} \right|$$

Since we know that $\lim_{\Delta p \rightarrow 0} \frac{\Delta x}{\Delta p}$ can be written mathematically in the form

or $\frac{dx}{dp}$, hence

$$e_d = \frac{-p}{x} \left(\frac{dx}{dp} \right) \text{ but } |e_d| = \frac{p}{x} \frac{dx}{dp}$$

Observe that in this expression $\frac{dx}{dp}$ is the derivative of the given demand function $x = f(p)$ with respect to price. For each price, this expression will give the elasticity of demand.

Again e_d can also be written as $\frac{-dy/dp}{x/p}$

Since $\frac{\Delta x}{\Delta p}$ represents the marginal quantity demanded and $\frac{x}{p}$ represents the average quantity demanded of the given demand function $x = f(p)$, we can, therefore, put the price elasticity of demand as

$$e_d = \frac{(-)dy/dp}{x/p} = (-) \frac{\text{Marginal quantity demanded}}{\text{Average quantity demanded}}$$

We write e_d in the form $|e_d|$ which means that we only consider the absolute value of e_d irrespective of its sign. In the following pages, we shall use e_d for $|e_d|$; i.e., we shall always take absolute (positive) value of the elasticity.

Ex. 1 Find e_d if the demand function is $x = 25 - 4p + p^2$, where x is the demand for commodity at price p .

Since $x = 25 - 4p + p^2$

$$\text{Marginal function} = \frac{dx}{dp} = -4 + 2p$$

$$\text{Average function} = \frac{x}{p} = \frac{25 - 4p + p^2}{p}$$

$$|e_d| = \frac{\text{Marginal function}}{\text{Average function}} = \frac{(-4 + 2p)p}{(25 - 4p + p^2)}$$

the expression gives the elasticity as a function of p . We can, therefore, easily find the point elasticity at any particular price level.

(1) Suppose p is given to be 8, then elasticity at price = 8 is,

$$|e_d| = \frac{(-4 + 16)8}{25 - 32 + 64} = 1.7 \text{ (approx.)}$$

which is greater than 1; so that demand is said to be elastic, at $p = 8$.

(2) Suppose $p = 4$, then $|e_d| = \frac{(-4 + 8)4}{25 - 16 + 16} = 0.64$ which is < 1

$\therefore$ Demand at $p = 4$ is inelastic.

(3) Suppose $p = 5$, then $|e_d| = \frac{(-4 + 10)5}{25 - 20 + 25} = 1.$

at $p = 5$, the elasticity of demand is *unitary*

Ex 2 The demand law is given by $x = 10 - p$ near the point $x = 4$ and $p = 6$. If the price increases by 5% determine the per cent decrease in demand and hence an approximation to the elasticity of demand

Compare the result with the correct value of elasticity of demand corresponding to $x = 4$

At $p = 6$, price increases by 5%, hence increased price is

$$= \left(6 + \left[6 \times \frac{5}{100} \right] \right) = 6.30$$

Rise in price = $6.30 - 6.0 = 0.30$

Corresponding to this increased price, the new demand is

$$x_1 = 10 - 6.30 = 3.70$$

so that the fall in demand is $(4 - 3.70) = 0.30$

Thus we have now

(i) Rise in price = 0.30 so that per cent rise in price
 $= \left\{ + (0.30/6) \times 100 \right\} = 5\%$

(ii) Fall in demand = 0.30, so that per cent fall in demand
 $= \left\{ + (0.30/4) \times 100 \right\} = 7.5\%$

$$\begin{aligned} |e_d| &= \frac{\text{Percentage change in demand}}{\text{Percentage change in price}} \\ &= \left| \frac{-7.5\%}{5\%} \right| = |1.5| = 1.5 \end{aligned}$$

The same result can be arrived at with the help of calculus

$$|e_d| = \left| \frac{p}{x} \frac{dx}{dp} \right| = \left| \frac{6}{4} \frac{d}{dp} [(10 - p)] \right|$$

Since we have to find elasticity at price = 6 when demand = 4

$$|e_d| = \left| \frac{6}{4} (-1) \right| = \left| -\frac{6}{4} \right| = |(-) 1.5| = 1.5$$

Ex. 3 If the demand law is given by $p = 10e^{-x/2}$, find the elasticity of demand. Also show that if the demand law is in the form $p = ae^{-kx}$, then

$$e_d = \frac{(-)1}{kx} \quad (p \text{ being price and } x \text{ being demand})$$

(a) We know $|e_d| = \frac{p}{x} \frac{dx}{dp}$

In the example since demand function is in terms of x [in fact we need x

in terms of p to get (dx/dp) , we shall treat the given demand function as an implicit function of x to get (dx/dp)

$$p = 10e^{-x/2}$$

$$1 = 10e^{-x/2} \left(-\frac{1}{2}\right) \frac{dx}{dp}, \text{ that is, } \frac{dx}{dp} = \left\{-\frac{1}{5e^{-x/2}}\right\}$$

substituting this in e_d we get,

$$e_d = \frac{p}{x} \left\{(-)\frac{1}{5e^{-x/2}}\right\} = \frac{10e^{-x/2}}{x} \left\{-\frac{1}{5e^{-x/2}}\right\} = -\frac{2}{x}, \text{ since } p = 10e^{-x/2}$$

(b) If $p = ae^{-kx}$, then differentiating with respect to p , we have

$$1 = ae^{-kx}(-k) \frac{dx}{dp} \text{ or } \frac{dx}{dp} = \frac{1}{-ak e^{-kx}}$$

$$\text{Since, } e_d = \frac{p}{x} \frac{dx}{dp} \quad e_d = \frac{ae^{-kx}}{x} \left(-\frac{1}{ak e^{-kx}}\right) = -\frac{1}{kx}$$

Hence proved

Ex. 4 If the demand law is $x = \frac{20}{p+1}$, find e_d with respect to price at the point where $p = 3$

$$\text{We are given } x = \frac{20}{p+1} = 20(p+1)^{-1}$$

$$\text{Therefore, } \frac{dx}{dp} = \frac{-20}{(p+1)^2}$$

$$\text{At } p = 3, \quad \frac{dx}{dp} = \frac{-20}{(4)^2} = \frac{-5}{4},$$

$$\text{So that, } e_d = \frac{p}{x} \cdot \frac{dx}{dp} = \frac{3}{5} \times \frac{-5}{4} = \frac{(-)3}{4}$$

$$|e_d| = \frac{3}{4}$$

EXERCISE 7.1

Find the elasticity of demand in terms of x for each of the following demand laws :

$$1. \quad p = \frac{1}{1+x^2}$$

$$2. \quad p = \frac{10}{(x+1)^2}$$

$$3. \quad p = \sqrt{a-bx}$$

$$4. \quad p = a-bx^2$$

$$5. \quad x = \frac{8}{p^2}$$

$$6. \quad x = \frac{27}{p^2}$$

7 $y = x^n e^{-b(x+c)}$

8 $y = xe^x$

9 $y = xe^x$

10 The demand function at point $x = 0$ and $p = 3$ is given by $x = 18 - 2p^2$. If the price decreases by 5% determine the relative increase in demand and hence an approximation to the elasticity of demand.

Do you find the result comparable with elasticity of demand at $p = 3$?

11 The demand function is $x = 48 - 3p^2$ at point $p=3$ and $x=21$. If the price decreases by 4%, determine the relative increase in demand and hence an approximation to the elasticity of demand. Compare this with the elasticity of demand at price $p=3$.

12 If the demand function is $p = (4 - 5x)^2$ or what value of x the elasticity of demand at point $p = 2$ in each case using first explicit differentiation and then implicit differentiation.

13 If the demand law is $x = \sqrt{10 - p^2}$ or $x^2 + p^2 = 10$ find the elasticity of demand at point $p = 2$ in each case using first explicit differentiation and then implicit differentiation.

14 Prove that $e_d = \frac{AR}{AR - MR}$ (AR is average revenue and MR is the marginal revenue). Verify this relation for demand law $p = a - bx$.

15 The supply of certain goods is given by $x_s = a\sqrt{p-b}$ where p is price, a and b are positive constants ($p > b$). Find an expression for e_s , i.e. elasticity of supply as function of price. Show that e_s decreases as price and supply increases and becomes unity at price $= 2b$.

[Hint: Proceed exactly in the same way as in case of estimation of e_d .

Here e_s will be $\frac{p}{x_s} \frac{dx_s}{dp}$ where x_s is supply (not demand) and $\frac{dx_s}{dp}$ will be marginal function of supply with price. In the above equation $x_s = a\sqrt{p-b}$

$$\frac{dx_s}{dp} = \frac{a}{2\sqrt{p-b}} \text{ and } e_s = \frac{p}{2(p-b)}]$$

16 If the demand function is $pf^n = b$, calculate price elasticity of demand. Discuss the case when $n=1$.

5.2 Total, Average and Marginal Cost Curves

We know that the term marginal cost represents the change in the total cost for each additional unit of production.

Let us assume that the total cost is represented as a function of total output Q , $C = f(Q)$ and that the change in the total cost is ΔC when the change in quantity produced is ΔQ .

In order to find the change in total cost with the smallest change in the

output, assuming the infinite divisibility and continuity in the output is possible, amount ΔQ is made to approach to zero then in the limit

$$\lim_{\Delta Q \rightarrow 0} \frac{\Delta C}{\Delta Q} = \frac{dC}{dQ} \text{ is defined as marginal cost} = MC$$

Thus, MC is the first derivative of the total cost function $C = f(Q)$ exactly in the same way as we defined and calculated marginal utility $= \frac{dU}{dx}$ from total utility function $U = f(x)$ where x is the amount of consumption

Again, since average cost (AC) represents the cost per unit of production, it can be easily calculated from the given total cost function

Total cost function is $C = f(Q)$ where Q is total output

$$\text{Therefore } AC = \frac{C}{Q} = \frac{f(Q)}{Q}$$

Ex 5 Suppose we are given a short run total cost function as

$$C = f(Q) = Q^3 - 3Q^2 + 15Q + 27$$

Obtain AC and MC functions

$$\text{Given } TC = C = Q^3 - 3Q^2 + 15Q + 27$$

$$(i) \quad AC = \frac{C}{Q} = Q^2 - 3Q + 15 + \frac{27}{Q}$$

$$(ii) \quad MC = \frac{d}{dQ}(TC) = \frac{d}{dQ}(Q^3 - 3Q^2 + 15Q + 27) = 3Q^2 - 6Q + 15$$

It can be observed that the *fixed cost* in the above total cost is represented by the constant 27, for even if the quantity produced is zero, the cost equal to 27 will have to be incurred by the firm

Again since this constant 27 drops out during the process of deriving marginal cost, obviously therefore, the magnitude of fixed cost does not affect the marginal cost

Let us now generalize the total cost function

$$TC = C = f(Q) + b \quad (b \text{ is fixed cost})$$

From this total cost function other cost functions can be derived as follows

$$\text{Average total cost} = ATC = \frac{f(Q) + b}{Q}$$

$$\text{Average variable cost} = AVC = \frac{f(Q)}{Q}$$

$$\text{Average fixed cost} = AFC = \frac{b}{Q}$$

$$\text{Marginal cost} = MC = \frac{dC}{dQ}$$

$$\left. \begin{array}{l} ATC = \frac{f(Q) + b}{Q} \\ AVC = \frac{f(Q)}{Q} \\ AFC = \frac{b}{Q} \end{array} \right\} \quad ATC = AVC + AFC$$

7.3. Relation Between Average and Marginal Cost Curves

Although cost functions may assume many different shapes under different circumstances, yet, usually under natural economic limitations, we assume average and marginal costs to have U-shapes. The relation between them is assumed to be as follows

- (i) When AC falls, MC is below AC .
- (ii) When AC is minimum, $AC = MC$
- (iii) When AC rises, MC is more than AC

We will make use of calculus to prove these relationships. Assume that we are given total cost C for the output Q

$$\text{Therefore, average cost} = \frac{C}{Q}$$

We have learnt that rising or falling nature of any function can be ascertained from its derivatives

- (1) Consider that AC is declining

If AC is declining, then $\left\{ \frac{d}{dQ} \left(\frac{C}{Q} \right) \right\}$ should possess negative sign, that is,

$$\frac{d}{dQ} \left(\frac{C}{Q} \right) < 0$$

$$\frac{d}{dQ} \left(\frac{C}{Q} \right) = \frac{Q \cdot \frac{dC}{dQ} - C \cdot \frac{d}{dQ} (Q)}{Q^2}, \text{ which should be } < 0$$

$$\text{i.e., } \frac{Q \cdot \frac{dC}{dQ} - C}{Q^2} < 0 \left[\text{Since } \frac{dQ}{dQ} = 1 \right]$$

$$\left(\frac{1}{Q} \cdot \frac{dC}{dQ} - \frac{C}{Q^2} \right) < 0 \text{ or, } \frac{1}{Q} \left(\frac{dC}{dQ} - \frac{C}{Q} \right) < 0$$

$$\text{or } \frac{dC}{dQ} - \frac{C}{Q} < 0 \text{ that is, } \frac{dC}{dQ} < \frac{C}{Q} \text{ [Since } Q > 0 \text{]}$$

But, since $\frac{C}{Q}$ is the AC and $\frac{dC}{dQ}$ is MC , it shows $MC < AC$

In other words, if AC falls, then MC should be below AC curve.

- (2) Consider that AC is rising

If AC rises, then $\left\{ \frac{d}{dQ} \left(\frac{C}{Q} \right) \right\}$ should possess positive sign,

that is, $\frac{d}{dQ} \left(\frac{C}{Q} \right) > 0$ or $\frac{1}{Q} \left(\frac{dC}{dQ} - \frac{C}{Q} \right) > 0$

or $\frac{dC}{dQ} - \frac{C}{Q} > 0$ [Since $Q > 0$]

or, $MC > AC$

that is if AC rises MC is greater than AC

(3) Consider that AC is minimum

AC will be minimum where $\left\{ \frac{d}{dQ} \left(\frac{C}{Q} \right) \right\} = 0$ and

$$\frac{d}{dQ} \left\{ \frac{d}{dQ} \left(\frac{C}{Q} \right) \right\} > 0$$

First consider $\frac{d}{dQ} \left(\frac{C}{Q} \right) = 0$ or, $\frac{1}{Q} \left(\frac{dC}{dQ} - \frac{C}{Q} \right) = 0$

which means $\frac{dC}{dQ} = \frac{C}{Q}$, that is to say that $AC = MC$

Therefore, AC will be minimum when $AC = MC$

We consider an example to verify these general marginal average relationships for a particular cost function

Ex. 6. Suppose a total variable cost function is

$$C = Q^3 - 3Q^2 + 15Q,$$

examine the general marginal and average cost relations

From the given cost function

$$AC = \frac{C}{Q} = Q^2 - 3Q + 15$$

$$\text{and } MC = \frac{dC}{dQ} = 3Q^2 - 6Q + 15$$

(a) Consider when AVC declines

$$\text{then, } \frac{d}{dQ} (Q^2 - 3Q + 15) < 0 \quad \text{or, } 2Q - 3 < 0 \quad \dots (1)$$

But when AC declines, $MC < AC$

$$3Q^2 - 6Q + 15 < Q^2 - 3Q + 15$$

$$2Q^2 - 3Q < 0$$

$$Q(2Q - 3) < 0$$

But $Q > 0$,

therefore $2Q - 3 < 0$ or $Q < \frac{3}{2}$ (2)

that is, inequality must hold good if MC is to be below AC , the same inequality as in (1) suggests that AC is declining

(b) Consider when AVC is rising

$$\frac{d}{dQ}(Q^2 - 3Q + 15) > 0$$

$$2Q - 3 > 0$$
 (3)

But when AC rises MC should be above AC ,

Which means $3Q^2 - 6Q + 15 > Q^2 - 3Q + 15$

$$2Q^2 - 3Q > 0$$

since $Q > 0$

we get, $2Q - 3 > 0$ or, $Q > \frac{3}{2}$ (4)

which again is the same inequality as (3)

(c) Consider when AVC is minimum

$$\frac{d}{dQ}(Q^2 - 3Q + 15) = 0 \quad \text{or, } 2Q - 3 = 0$$
 (5)

but when AC is minimum, $AC = MC$

$$3Q^2 - 6Q + 15 = Q^2 - 3Q + 15$$

$$2Q - 3 = 0 \quad \text{or, } Q = \frac{3}{2}$$
 (6)

which is the same result as in (5)

7.4. Minimum Average Cost

We can also determine the minimum average cost from the derivatives (by applying 1st and 2nd order conditions) of the average cost function

Ex. 7. If the total cost function is $C = \frac{1}{3}Q^3 - 3Q^2 + 9Q$,

find at what level of output AC be minimum and what level will it be ?

$$TC = C = \frac{1}{3}Q^3 - 3Q^2 + 9Q$$

$$AC = \frac{C}{Q} = \frac{1}{3}Q^2 - 3Q + 9$$

$$MC = \frac{dC}{dQ} = Q^2 - 6Q + 9$$

First Method. To find minimum of the average cost function we must apply first and second order conditions

$$(i) \quad \frac{d}{dQ} (AC) = 0, \text{ and } (ii) \quad \frac{d^2}{dQ^2} (AC) > 0$$

$$\text{when } \frac{d}{dQ} \left(\frac{1}{2} Q^2 - 3Q + 9 \right) = 0$$

$$\frac{2}{3} Q - 3 = 0 \quad \text{or,} \quad Q = 9/2$$

At this level of output AC will be minimum if second order condition is also satisfied

$$(ii) \quad \frac{d}{dQ} \left(\frac{2}{3} Q - 3 \right) = \frac{2}{3} \text{ which is greater than zero,}$$

it proves that AC is minimum when $Q = 9/2$

Therefore, AC will be minimum at an output level of $9/2$ and its value will be $9/4$

Second Method. We know AC is minimum where $AC = MC$

$$\frac{1}{3} Q^2 - 3Q + 9 = Q^2 - 6Q + 9 \quad \text{or,} \quad Q = 9/2,$$

which is the same level of output as obtained by first method

7.5. Cost Function in Cubic Form

Consider general example of a cost function in cubic form :

$$TC = AQ^3 + BQ^2 + CQ + D$$

We want to examine with what restrictions on constants A, B, C and D , the given function would appear 'Normal' under certain natural economic limitations which are .

- (1) If no goods are produced, the total cost must still be positive (since there is always some fixed cost).
- (2) The total cost (TC) must increase as the output (Q) increases so that $\frac{dC}{dQ}$ (marginal cost) is always positive.

Our assumed total cost function is : $TC = AQ^3 + BQ^2 + CQ + D$

(i) Even if output $Q = 0$; $TC = D$, which is fixed cost.

Hence, first restriction is that D should always be positive,

That is ; $D > 0$

(ii) If marginal cost is to be always positive, it means that absolute

minimum of MC should remain positive. In other words, we can be sure of the positive value of MC at all levels of output if the absolute minimum of the MC is positive.

$$TC = C = AQ^3 + BQ^2 + CQ + D$$

$$MC = \frac{dC}{dQ} = 3AQ^2 + 2BQ + C$$

On the basis of these conditions, the minimum of the MC will occur where

$$\text{First order condition } \frac{d}{dQ}(MC) = 0, \text{ i.e., } 6AQ + 2B = 0$$

$$\text{Second order condition } \frac{d^2}{dQ^2}(MC) > 0, \text{ i.e., } 6A > 0$$

The second order condition is fulfilled if $A > 0$ and MC is minimum when $6AQ = -2B$ or, $Q = \frac{-2B}{6A} = \frac{-B}{3A}$

Since Q can never be a negative quantity, therefore to make $\frac{-B}{3A}$ always a positive quantity B should always be a negative quantity or $B < 0$ ($A > 0$ from second order condition)

What will be the absolute value of this minimum MC ?

$$MC = 3A \left(\frac{-B}{3A} \right)^2 + 2B \left(\frac{-B}{3A} \right) + C$$

$$MC = \frac{B^2}{3A} - \frac{2B^2}{3A} + C$$

$$MC = \frac{3AC - B^2}{3A} \text{ which should always be positive}$$

$$\text{or, } 3AC - B^2 > 0 \quad (\text{Since } A > 0)$$

$$3AC > B^2$$

$$\text{i.e., } B^2 < 3AC \quad [C > 0]$$

Therefore, under the normal economic conditions, the coefficients A , B , C and D in the given cubic cost function should be restricted as under

$$(i), A > 0 \text{ and } D > 0 \quad (ii) \quad B < 0 \quad \text{and,} \quad (iii) \quad B^2 < 3AC$$

7.6 Total, Average, Marginal Revenue Curves

For any demand function $p = f(Q)$, the total revenue (TR) is the product

of quantity demanded (Q) and the price per unit of output

$$TR = Q \times p \text{ but since } p = f(Q)$$

$$TR = R = Q \times f(Q)$$

Again since the marginal revenue represents the change in TR for each additional unit of sale, MR will be defined as

$$\frac{d}{dQ}(TR) \text{ which describes derivative of } TR \text{ w.r.t. quantity demanded}$$

$$\text{Thus we have } TR = R = Q \times f(Q)$$

$$AR = \frac{Qp}{Q} \quad \text{and} \quad MR = \frac{dR}{dQ}$$

Relationship between revenue curves

$$TR = R = Qp$$

$$MR = \frac{dR}{dQ} = p \left\{ \frac{d}{dQ}(Q) \right\} + Q \left\{ \frac{d}{dQ}(p) \right\}$$

$$MR = p + Q \left\{ \frac{dp}{dQ} \right\}$$

Assuming that at any level of Q , MR lies below AR then

$$p + Q \frac{dp}{dQ} < p \quad (\text{since } AR = p)$$

$$Q \frac{dp}{dQ} < 0 \quad \text{or}$$

$$\frac{dp}{dQ} < 0 \quad (\text{since } Q > 0)$$

$$\text{which means } \frac{d}{dQ}(AR) < 0$$

In other words, if MR is to lie below AR , then $\left\{ \frac{d}{dQ}(AR) \right\}$ should be less than zero or AR curve should be declining. This gives us the general marginal average revenue relationship: MR lies below AR when AR is downward sloping.

There are other two important relationships between AR and MR curves which we consider below.

(1) Rate of fall of MR curve is twice the rate of fall of AR curve

Let us take a point D on AR curve in Fig. 7.1. At this point we have to prove that $AC = 2AB$.

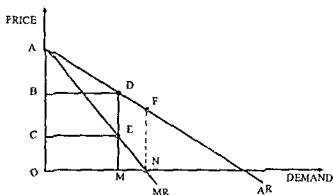


Fig. 7.1

We know, $TR = R = p \times Q$

$$MR = \frac{dR}{dQ} = p + Q \frac{dp}{dQ}$$

At point D , we have

$$DM = p \text{ and } OM = Q \text{ (demand)}$$

$$EM = MR = \frac{dR}{dQ}$$

Substituting these values in $MR = p + Q \frac{dp}{dQ}$

$$EM = DM + OM \left(\frac{dp}{dQ} \right)$$

But $\frac{dp}{dQ}$ is the slope of AR at point D ,

$$\frac{dp}{dQ} = -\frac{AB}{BD} \quad (\text{since } AR \text{ has negative slope})$$

$$EM = DM + OM \left(-\frac{AB}{BD} \right)$$

But $OM = BD$

$\therefore EM = DM - AB$

Since $EM = DM - DE$

$\therefore DM - AB = DM - DE$

$$AB = DE \text{ which is also } = BC$$

$\therefore AC = 2AB$

(2) *AR and MR curves are also related in the form :*

$$MR = AR \left(1 - \frac{1}{|e_d|} \right)$$

$$TR = R = pQ$$

$$MR = \frac{dR}{dQ} = p + Q \frac{dp}{dQ} = p \left(1 + \frac{Q}{p} \frac{dp}{dQ} \right)$$

Since $e_d = -\frac{p}{Q} \frac{dQ}{dp}$

$$MR = p \left(1 - \frac{1}{|e_d|} \right)$$

that is, $MR = AR \left(1 - \frac{1}{|e_d|} \right) = AR \left(1 - \frac{1}{e_d} \right)$

7.7. Maximum Total Revenue

We can also determine the level of output Q , where TR will be maximum, since we know that TR is maximum when $MR=0$. Thus, by equating MR function to zero we find the output level Q and then substituting this value of Q in the given demand function we estimate price p . Finally, the price of product and level of output gives us the maximum level of TR .

We take an example to understand this

Ex. 8 If the demand function is $p = \sqrt{9-Q}$, find at what level of output Q the TR will be maximum and what will it be?

Suppose the level of output that maximises the TR is Q

With price = p in the market, $TR = Qp$

$$TR = Q \sqrt{9-Q} = Q(9-Q)^{1/2}$$

TR is maximum when $MR = 0$, thus making $MR = 0$,

$$MR = \frac{dR}{dQ} = \frac{18-3Q}{2\sqrt{9-Q}} = 0$$

so that, $6-Q=0$ i.e., Q must be = 6

In other words, TR will be maximum when $Q = 6$ (Examine second order condition)

Substituting value of Q in $p = \sqrt{9-Q}$ we have $p = \sqrt{3} = 1.732$

Hence the value of $TR = pQ = 6\sqrt{3} = 10.392$

7.8. Conditions for Profit Maximisation

We know how to estimate the total revenue maximising output level from the demand function (which is AR -function) irrespective of total cost function

of the firm. But in fact we are interested in firm's output level which would maximise its total profit and not total revenue. We will now see that in order to maximise profit a firm must choose the output level such that its $MC = MR$.

Assuming that we are given the total cost function along with the total revenue function—both in terms of output i.e. given functions are

$$\text{Total cost function } C = f(Q)$$

$$\text{Total revenue function } R = \Phi(Q)$$

$$\text{Total profit } \pi = R - C = \Phi(Q) - f(Q)$$

For π to be maximum the conditions are

First order condition

$$\frac{d\pi}{dQ} = \frac{dR}{dQ} - \frac{dC}{dQ} = 0$$

$$\text{or } \frac{dR}{dQ} = \frac{dC}{dQ}$$

which means $MR = MC$

Thus the first order condition for profit maximisation gives us the level of output where MR is equal to MC .

Again we also check the second order condition, which is

$$\frac{d^2\pi}{dQ^2} = \frac{d^2R}{dQ^2} - \frac{d^2C}{dQ^2} < 0$$

that is,

$$\frac{d^2R}{dQ^2} < \frac{d^2C}{dQ^2}$$

This would mean that rate of change of MR (i.e., slope of MR) should be less than the rate of change of MC (i.e., slope of MC) at the profit maximising output level given by $MR = MC$.

These two conditions can be illustrated graphically in Fig 7.2

In both the figures we find that the first order condition, i.e. $MR = MC$ gets satisfied at two output levels, i.e. at Q_1 and Q_2 . Which of the two output levels will earn max. net revenue to the firm? It is Q_2 . Why? Since at point G where output = Q_1 the slope of MC is > the slope of MR . But second order condition states just the opposite. It is at

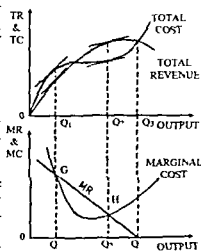


Fig 7.2.

point H the rate of change in MR (which is negative) < the rate of change in MC (which is now positive)

We take some examples to make these points more clear

Ex 9 A firm has the following total cost and demand functions

$$C = \frac{1}{3}Q^3 - 7Q^2 + 111Q + 50$$

$$\text{and } Q = 100 - p$$

Find profit maximising level of output also find profit at this level of output

$$\text{We are given } Q = 100 - p$$

$$p = 100 - Q$$

Assuming that the profit maximising level of output is Q

$$TR = R = Q \times p = Q(100 - Q) = 100Q - Q^2$$

$$\text{Profit} = TR - TC$$

$$= (100Q - Q^2) - \left(\frac{1}{3}Q^3 - 7Q^2 + 111Q + 50\right)$$

$$= -\frac{1}{3}Q^3 + 6Q^2 - 11Q - 50$$

First order condition for maximising total profit

$$\frac{d\pi}{dQ} = -Q^2 + 12Q - 11 = 0$$

$$(Q - 1)(11 - Q) = 0$$

This means that profit will be maximum when either $Q = 1$ or $Q = 11$

To decide between the two quantities we apply second order condition

$$\frac{d^2\pi}{dQ^2} = -2Q + 12 \text{ which should be } < 0$$

(a) Substitute $Q = 1$, the inequality does not hold good

(b) Substitute $Q = 11$, the inequality holds good

Therefore output level of 11 units is the only profit maximising level
What will be the total profit at this level of output?

[Note The maximizing level of output can also be determined by equating MR with MC]

In example 9, if the firm is assumed to act as monopolist with the object of maximising net revenue, it fixes the output Q and leaves the price to be determined by the demand conditions, alternatively, the firm can also fix its price and let demand condition determine the approximate output

Let us examine here how the result of the above example is affected if the firm is assumed to fix the price

When the firm fixes the price

$$\text{Given Total cost} = C = \frac{1}{3}Q^3 - 7Q^2 + 111Q + 50$$

$$\text{Demand } Q = 100 - p$$

Total revenue = $p \times Q$ but now since Q is not known we must substitute Q in terms of p (which in present case is assumed to be known)

$$TR = p(100 - p) = 100p - p^2$$

Similarly TC which is given in terms of Q must now be changed in terms of p

$$TC = \frac{1}{3}(100 - p)^3 - 7(100 - p)^2 + 111(100 - p) + 50$$

$$\begin{aligned} \pi = TR - TC &= (100p - p^2) - \left\{ \frac{1}{3}(100 - p)^3 - 7(100 - p)^2 \right. \\ &\quad \left. + 111(100 - p) + 50 \right\} \end{aligned}$$

First order condition for maximising total profit

$$\frac{d\pi}{dp} = (100 - 2p) + (100 - p)^2 - 14(100 - p) + 111 = 0$$

$$8811 - 168p + p^2 = 0$$

$$(p - 89)(p - 99) = 0$$

that is either p be fixed at 89 or at 99 units to maximise the net revenue

If $p = 89$ and since $Q = 100 - p$ then $Q = 11$. This is the same level of output as we got by the first method

If $p = 99$ then $Q = 1$. This is not possible as has been explained in first method

Ex 10 A monopolist firm has the following total cost and demand functions $C = aQ^2 + bQ + c$, $p = \beta - \alpha Q$

What is the profit maximising output level when the firm is assumed to fix the output? Verify that this is the same result as when firm fixes the price

(1) When firm fixes the output level

$$TR = Qp = Q(\beta - \alpha Q)$$

$$MR = \frac{dR}{dQ} = \beta - 2\alpha Q$$

$$TC = C = aQ^2 + bQ + c$$

$$MC = \frac{dC}{dQ} = 2aQ + b$$

The condition for profit maximising output level is $MR = MC$

$$\text{i.e., } \beta - 2\alpha Q = 2aQ + b$$

$$\text{or, } \beta - b = 2aQ + 2\alpha Q$$

$$Q = \frac{\beta - b}{2(\alpha + a)}$$

Hence the profit maximising level of output is $\left\{ \frac{\beta - b}{2(\alpha + a)} \right\}$

(2) When firm fixes the price

Now TR and TC should be put in terms of price p

Since $p = \beta - \alpha Q$

$$Q = \frac{\beta - p}{\alpha}$$

$$\text{Hence } TR = Qp = \left(\frac{\beta - p}{\alpha} \right) p = \frac{\beta p - p^2}{\alpha}$$

$$MR = \frac{dR}{dp} = \frac{1}{\alpha} (\beta - 2p)$$

$$\text{and } TC = a \left(\frac{\beta - p}{\alpha} \right)^2 + b \left(\frac{\beta - p}{\alpha} \right) + c$$

$$MC = \frac{dC}{dp} = \left[\frac{-2a\beta + 2ap}{\alpha^2} - \frac{b}{\alpha} \right]$$

For profit maximisation

$$\frac{1}{\alpha} (\beta - 2p) = \left(\frac{-2a\beta + 2ap}{\alpha^2} - \frac{b}{\alpha} \right)$$

$$\text{This gives, } p = \frac{2a\beta + \alpha\beta + \alpha b}{2(\alpha + a)}$$

Since the demand function is given by $p = \beta - \alpha Q$

$$\frac{2a\beta + \alpha\beta + \alpha b}{2(\alpha + a)} = \beta - \alpha Q,$$

$$\text{or, } \alpha Q = \beta - \frac{2a\beta + \alpha\beta + \alpha b}{2(\alpha + a)}$$

$$\text{i.e., } Q = \frac{\beta - b}{2(\alpha + a)}$$

Which is the same level as when the firm fixes the output

Ex. 11. A radio manufacturer produces 'x' sets per week at a total cost of Rs $\left(\frac{x^2}{25} + 3x + 100 \right)$. He is a monopolist and the demand for his market is

$x = 75 - 3p$, where p is the price in rupees per set. Show that the maximum net revenue is obtained when about 30 sets are produced per week. What is the monopoly price?

Let x be the number of sets which maximises the net revenue of the monopolist

$$TC \text{ for } x \text{ sets is given } C = \frac{x^2}{25} + 3x + 100$$

$$MC = \frac{dC}{dx} = \frac{2x}{25} + 3$$

$$\text{Demand function is } x = 75 - 3p \quad p = \left(\frac{75 - x}{3} \right) \text{ per set}$$

$$TR \text{ for } x \text{ sets} = \left(\frac{75 - x}{3} \right) \times x$$

$$MR = \frac{dR}{dx} = 25 - \frac{2x}{3}$$

Net revenue will be maximum at the level of output where $MR = MC$

$$\text{i.e., } 25 - \frac{2x}{3} = \frac{2x}{25} + 3 \quad \text{or, } \frac{56}{75}x = 22 \text{ or } x = 30 \text{ (approx.)}$$

$$\text{Since, } p = \frac{75 - x}{3} \quad \text{Hence when } x = 30 \text{ } p \text{ will be } \frac{75 - 30}{3} = \text{Rs } 15 \text{ per set}$$

Alternatively

$$\text{we have net revenue} = \pi = TR - TC = p \cdot x - TC$$

$$= x \left(\frac{75 - x}{3} \right) - \left(\frac{x^2}{25} + 3x + 100 \right)$$

$$\text{For maxima or minima } \frac{d\pi}{dx} = 0 \quad (\text{First order condition})$$

$$\text{which gives } 25 - \frac{2x}{3} - \left(\frac{2x}{25} + 3 \right) = 0$$

$$\text{or } x = 30 \text{ (approx.)}$$

[Also examine whether second order condition is satisfied at this output level]

Ex. 12. The total cost C of output x is given by

$$C = \frac{2}{3}x + \frac{35}{2}$$

- Find (i) cost when output is 4 units ,
 (ii) average cost of output of 10 units ,
 (iii) marginal cost when output is 3 units

$$(i) \quad C = \frac{2}{3}x + \frac{35}{2}$$

$$C \text{ for 4 units} = \frac{2}{3}(4) + \frac{35}{2} = 20.67 \text{ units}$$

$$(ii) \quad C \text{ for 10 units} = \frac{2}{3}(10) + \frac{35}{2} = \frac{145}{6} = 24.16 \text{ units}$$

$$AC = \frac{145}{6} \times \frac{1}{10} = \frac{29}{12} = 2.42 \text{ units}$$

$$(iii) \quad MC = \frac{d}{dx} \left(\frac{2}{3}x + \frac{35}{2} \right) = \frac{2}{3} = 0.67 \text{ units}$$

(MC is constant here)

Ex. 13. The demand function faced by a firm is $p=500-0.2x$ and its cost function is $C=25x+10000$ (p = price, x = output and C = cost). Find the output at which the profits of the firm are maximum. Also find the price it will charge.

Condition for maximum profits $MR = MC$

$$TC = 25x + 10000$$

Therefore, $MC = 25$

$$TR = p \cdot x = (500 - 0.2x)x$$

$$TR = 500x - 0.2x^2$$

$$MR = 500x - 0.4x$$

$$500 - 0.4x = 25 \quad (\text{By equaling } MC \text{ with } MR)$$

$$x = \frac{4750}{4} = 1187.50$$

i.e., profit maximising output level is = 1187.50 units

$$\text{Given } p = 500 - 0.2x$$

$$p \text{ at } x = 1187.50 \text{ will be } = 500 - 0.2(1187.50)$$

$$p = 262.50$$

Ex. 14 If the MR is Rs. 25 and the elasticity of demand with respect to price is 2, find AR

$$\text{We know, } MR = AR \left(1 - \frac{1}{e} \right)$$

Therefore $25 = AR \left(1 + \frac{1}{2}\right)$ which gives

$$AR = 50$$

Ex 15 The production function of a commodity is given by

$$Q = 40F + 3F^2 - \frac{F^3}{3}$$

where Q is the total output and F depicts the units of inputs.

(i) Find the number of units of input required to give maximum output

(ii) Find the maximum value of marginal product

(iii) Verify that when the average product is maximum it is equal to marginal product

$$\begin{aligned} \text{(i) } MP \text{ of } F = MP_F &= \frac{dQ}{dF} = 40 + 6F - \frac{3F^2}{3} \\ &= 40 + 6F - F^2 \end{aligned}$$

$$\begin{aligned} \text{For maximum or minimum } 40 + 6F - F^2 &= 0 \\ (F + 4)(F - 10) &= 0 \\ F &= -4 \text{ or } 10 \end{aligned}$$

$$\text{and } \frac{d^2Q}{dF^2} = 6 - 2F$$

$$\text{At } (-4) \quad \frac{d^2Q}{dF^2} = 14 > 0$$

$$\text{At } (10) \quad \frac{d^2Q}{dF^2} = -14 < 0$$

$\therefore$ output is maximum when 10 units of input are used

$$\text{(ii) } MP_F = \frac{dQ}{dF} = 40 + 6F - F^2$$

$$\text{For } MP_F \text{ to be maximum } \frac{d}{dF} (40 + 6F - F^2) \text{ should be } = 0$$

$$\text{that is } 6 - 2F = 0, \text{ or } F = 3$$

$$\text{Again } \frac{d^2}{dF^2} (40 + 6F - F^2) = -2 < 0$$

MP_F is maximum when $F = 3$

$$\text{(iii) } AP = 40 + 3F - \frac{F^2}{3}$$

For AP_F to be maximum $\frac{d}{dF}(AP) = 3 - \frac{2F}{3}$ should be $= 0$

that is $F = \frac{9}{2}$

and $\frac{d^2}{dF^2}(AP) = -\frac{2}{3} < 0$

AP is max when $F = \frac{9}{2}$

$$AP \left(\text{when } F = \frac{9}{2} \right) = 40 + 3 \left(\frac{9}{2} \right) - \frac{81}{12} = \frac{187}{4} = 46.75$$

$$MP \left(\text{when } AP \text{ is max i.e. } F = \frac{9}{2} \right) = 40 + 27 - \frac{81}{4} = \frac{187}{4} = 46.75$$

Hence proved

Ex 16 Given the demand function $Q = 150 - 15P$ where P is the price, find the elasticity of demand at $P = 4$

Will the elasticity of demand change if the price changes?

When $P = 4$ we obtain $Q = 150 - 60 = 90$

Substituting in the formula

$$e_d = \frac{P}{Q} \cdot \frac{dQ}{dP} = \frac{4}{90} (-15) \quad \left(\frac{dQ}{dP} = -15 \right)$$

so that $e_d = -\frac{2}{3}$

Yes elasticity will change with the changes in the price, though $\frac{dQ}{dP}$ will remain unchanged

Ex 17 The total revenue (R) and total cost (C) functions of a firm are given by

$$R = 30Q - Q^2 \quad C = 20 + 4Q$$

where Q is the output. Find the equilibrium output of the firm

$$\text{Given } R = 30Q - Q^2 \quad C = 20 + 4Q$$

$$MR = 30 - 2Q \text{ and } MC = 4$$

For equilibrium output $MR = MC$

$$30 - 2Q = 4$$

that is

$$2Q = 26 \quad Q = 13$$

EXERCISE 7.2

1. The total cost function is $C = 2Q - 2Q^2 + Q^3$
(C —total cost and Q —total quantity produced)

Find (i) Average cost function

(ii) Marginal cost function

Also verify that at the minimum of average cost, average cost is equal to marginal cost

2. The total cost function is $C = 4Q - Q^2 + 2Q^3$

(i) Find the average cost function

(ii) At what level of output average cost will be minimum?

(iii) Find the marginal cost function

(iv) Show that at the minimum of average cost, average cost is equal to marginal cost

3. Given the following demand curves demonstrate the relationship between marginal revenue and elasticity of demand

(i) $p = 14.7 - 0.1x$

(ii) $p = 86 - 25x$

(iii) $p = \frac{120}{x^{1.5}}$

(iv) $p = \frac{280}{x^2}$

(x represents demand and p represents price)

4. Given the following total cost functions investigate the nature of average and marginal cost curves

(i) $C = 0.0003Q^2 + 6.75Q - 10495$

(ii) $C = 0.8Q + 3000$

(iii) $Q = 0.002Q^2 + 1.052Q + 16.8$

Considering the following demand and total cost functions estimate the level of output which maximises the profit. Compute the values of maximum profit and price (x = demand, p = price, C = cost, AC = average cost)

Demand functions

Total cost functions

5. $p = 12 - 4x$

$C = 8x - x^2$

6. $x = 75 - 3p$

$C = 100 + 3x$

7. $p = 10e^{-2x}$

$AC = \frac{1}{x}$

8. $p = \sqrt{16 - 3x}$

$C = x$

9. $p = (80 - x)^2$

$AC = 55 - 8x$

10. If a monopolist has total cost function $C = ax^2 + bx + c$ and if the demand law is $p = \beta - \alpha x^2$, show that the output for maximum net revenue is,

$$\sqrt{a^2 + \frac{3\alpha^2(p-b)}{3\alpha}} = a$$

11. A soft set manufacturer can manufacture x soft sets per week at a total cost of Rs. $(\frac{1}{2}x^3 + 3x + 90)$. How many sets per week should he manufacture for maximum monopoly revenue when the demand law of his product is $x = 10\sqrt{25-p}$ sets per week — assuming that he is a monopolist?

12. A plant produces Q tonnes of steel per week at a total cost of Rs. $(\frac{1}{3}Q^3 - Q^2 + 452Q + 50)$. If the market price is fixed at Rs. 500 per tonne, show that the plant should produce 8 tonnes per week.

13. State the conditions of a maximum and minimum. Find the profit maximising output level given

$$Q = 200 - 10p \quad \text{and} \quad AC = 10 + \frac{Q}{25}$$

where Q represents units of output and p represents price and AC represents average cost.

14. Derive the first and second order conditions of a firm. Interpret in the case of perfect competition and monopoly.

15. Show that MC and AC curves in each of the following functions fall continuously with increasing output.

$$(i) \quad C = \sqrt{aQ} + b + c \quad (ii) \quad C = \left(\frac{aQ^2 + abQ}{Q + c} \right) + d$$

16. Show, if necessary, by imposing suitable conditions on the constants that the demand curves,

$$p = (a - bQ)^2 \quad \text{and} \quad p = a - bQ^2$$

are downward sloping and convex from below.

Do the same properties hold for the marginal revenue curves?

17. Determine the maximum profit and the corresponding price and quantity for a monopolist whose demand and cost functions are

$$p = 20 - 0.5q \quad \text{and} \quad C = 0.4q^3 - 1.94q^2 + 32.96q$$

18. A revenue maximising monopolist requires profit of at least 1500. Its demand and cost functions are $p = 304 - 2q$ and $C = 500 + 4q + 8q^2$. Determine his output level and price. Compare these values with those that would be achieved under profit maximisation.

19. The demand and cost functions of a monopolist are $p = 100 - 3q + 4\sqrt{A}$ and $C = 4q^2 + 10q + A$ (A is the level of advertising expenditure). Find the values of A , q and p that maximise profit.

20 A monopolist uses only labour X to produce his output Q which he sells in a competitive market at the fixed price $p = 2$. His production and labour supply functions are $q = 6x + 3x^2 - 0.2x^3$ and $r = 60 + 3x$ respectively. Determine the values of x , q and r at which he maximises his profit.

21 A company has examined its cost and revenue structure and has determined that C the total cost, R the total revenue and X the number of units produced are related as

$$C = 100 + 0.015x^2 \quad \text{and} \quad R = 3X$$

Find the product level X at which it will maximise profit for this company. How much will it be?

Find also the profit when $X = 120$ units.

22 A radio manufacturer produces x sets per week. The total cost of Rs $(x^2 + 78x + 2500)$. He is a monopolist and the demand function for his product is $x = \frac{(100 - p)}{8}$ when the price is Rs p per set. Show that maximum net revenue (i.e. profit) is obtained when 29 sets are produced per week. What is the monopoly price?

23 Let the cost function of a firm be given by the following equation where C stands for cost and x for output

$$C = 300x - 10x^2 + \frac{1}{3}x^3$$

Calculate (i) output at which marginal cost is minimum (ii) output at which average cost is minimum and (iii) output at which $1C = MC$.

24 The total cost functions are

$$(i) \quad C = 2Q - 2Q^2 + Q^3$$

$$(ii) \quad C = 4Q - Q^2 + 2Q^3$$

Find (i) AC function

(ii) MC function

(iii) At what level of output AC is minimum?

(iv) Verify that at minimum average cost $1C = MC$.

25 A company finds that it can sell out all the products that it produces at the rate of Rs. 2 per unit. It estimates the cost function for the product to be

$$\text{Rs. } \left[1000 + \frac{1}{2} \left(\frac{Q}{50} \right)^2 \right] \text{ for } Q \text{ units produced}$$

(i) Find the expression for the total profit if Q units are produced and sold.

(ii) Find the number of units produced that will maximise the company's profit and

(iii) What is the amount of the maximum profit?

7.9. Effects of Taxation and Subsidy on Monopoly

Let us now examine the effects of lump-sum tax and taxes based on per unit price on the monopolist's optimum price-quantity combination

Ex. 18 A monopolist firm has following types of total cost and demand functions

$$C = aQ^2 + bQ + c \quad (a, b, c > 0)$$

$$p = \beta - \alpha Q \quad (\alpha, \beta > 0)$$

Our Problem is If the government plans to impose a tax t per unit quantity upon the commodity produced by the monopolist, what tax rate (t) should the government choose in order to maximise total tax revenue from this source?

The imposition of a sales or profits tax whether based upon quantity sold or value of sales will reduce the profit and output level and increase the price level of the monopolist.

Supposing Q is the amount produced, then the imposition of tax (t) per unit of quantity will raise his total cost by $t \times Q$

Hence new total cost function (C_N) shall be

$$C_N = C + tQ$$

$$C_N = aQ^2 + bQ + c + tQ, \text{ and}$$

$$\text{New marginal cost } MC_N = \frac{dC_N}{dQ} = 2aQ + b + t$$

$$TR \text{ for } Q \text{ output} = (\beta - \alpha Q)Q = \beta Q - \alpha Q^2$$

$$MR = \frac{dR}{dQ} = \beta - 2\alpha Q$$

The level of output that maximises his net revenue will be at $MR = MC$, that is,

$$Q = \frac{\beta - b - t}{2(a + \alpha)}$$

This level of output is the level after the tax (t) has been imposed. Let it be denoted by Q_N ,

$$Q_N = \frac{\beta - b - t}{2(a + \alpha)}$$

Since the government imposes tax (t) per unit of output, therefore, the total tax revenue for this (Q_N) level of output will be $t \times Q_N$

$$= \left\{ \frac{\beta - b - t}{2(a + \alpha)} \right\} \times t$$

Let this be denoted by T $T = \frac{\beta t - bt - t^2}{2(a + \alpha)}$

Our Problem is To find the value of rate of tax (t) which would maximise the total tax revenue (T)

For (T) to be maximum, apply order conditions

First order condition .

$$\frac{dT}{dt} = 0, \text{ that is, } \frac{\beta - b - 2t}{2(a + \alpha)} = 0$$

$$\text{i.e., } \beta - b - 2t = 0$$

$$\text{or, } t = \frac{\beta - b}{2}$$

i.e., the rate of tax (t) should be $\left(\frac{\beta - b}{2} \right)$ This would maximise the total tax revenue of the government

Let us now check the *second order condition*

$$\frac{d^2T}{dt^2} < 0, \text{ that is, } \frac{-2}{2(a + \alpha)} < 0$$

which is true; therefore, second order condition is also satisfied

Case of subsidy. *Subsidy* (say, s) is a negative tax and thus the general analysis will remain the same.

New total cost function will now be $C_N = C - sQ$

(s being the subsidy per unit of output)

$$\text{New total cost . } C_N = aQ^2 + bQ + c - sQ$$

$$\text{New marginal cost . } MC_N = 2aQ + b - s$$

$$TR = T = (\beta - \alpha Q)Q = \beta Q - \alpha Q^2$$

$$\text{Hence, } MR = \beta - 2\alpha Q$$

For the necessary condition .

$$MR = MC_N$$

$$\therefore 2\alpha Q + b - s = \beta - 2\alpha Q$$

$$\text{that is, } Q_V = \frac{\beta - b + s}{2(a + \alpha)}$$

Since the government grants subsidy s per unit of output, the total subsidy granted for this new level of output (Q_N) will be

$$\left\{ \frac{b}{2(a+c)} \right\} \times s = \frac{bs - bs + s^2}{2(a+c)}$$

[Note: There is no theoretical maximum amount which the government might spend upon subsidy.]

Ex. 19 Suppose that the demand and total cost functions of a monopolist are $p = 20 - 4x$ and $C = 4x$ respectively.

(a) If a tax of $\frac{1}{2}$ per unit quantity produced is imposed, find the quantity and the price that would correspond to the maximum profit of monopolist. What will be maximum profit?

(b) If the tax of (i) per unit is imposed, find the quantity and the price that correspond to the maximum profit and find the maximum profit.

(c) Determine the tax that maximises the tax revenue and determine the maximum tax revenue.

(d) Determine the total tax revenue if the tax is 20% of the sales.

(e) Assuming that x units are produced with imposition of tax, the total cost will rise by $\frac{1}{2}x$.

$$\text{New total cost} \quad TC_N = 4x + \frac{1}{2}x, \text{ and } TR = 20x - 4x^2$$

$$\text{New marginal cost} \quad MC_N = \frac{9}{2}, \text{ and } MR = 20 - 8x$$

$$MC_N = MR \quad (\text{for profit maximisation output})$$

$$\text{that is} \quad 20 - 8x = \frac{9}{2}, x = \frac{31}{16}$$

i.e., quantity produced will be $\frac{31}{16}$ units

$$\text{Price will be } p = 20 - 4x = \left\{ 20 - \left(\frac{31}{16} \times 4 \right) \right\} = 12.25$$

$$\text{Maximum profit} = TR - TC = (20x - 4x^2) - \left(\frac{9}{2} \right)x$$

$$= \frac{31}{2}x - 4x^2 \approx \left(\frac{31}{2} \times \frac{31}{16}\right) - \left(\frac{31}{16}\right)^2 \quad \text{is (approx.)}$$

(b) If the tax (t) is imposed, then with x units produced

$$TC_N = 4x + tx, \quad MC_N = 4 + t$$

$$TR = 20x - 4x^2, \quad MR = 20 - 8x$$

To maximise profits, $MR = MC_N$

$$4 + t = 20 - 8x, \quad x = \frac{16 - t}{8}$$

i.e., quantity produced will be $\left(2 - \frac{t}{8}\right)$

$$\text{Price will be } p = 20 - 4x = 20 - 4\left(2 - \frac{t}{8}\right) = \left(12 + \frac{t}{2}\right)$$

$$\text{Profit} = TR - TC = (20x - 4x^2) - (4x + tx)$$

$$\pi = x(20 - 4x - 4 - t) = x(16 - 4x - t)$$

Substituting $x = \left(\frac{16 - t}{8}\right)$, we obtain

$$\text{maximum profit} \quad \pi = \frac{16 - t}{8} \left[16 - t - \frac{16 - t}{2}\right] = \frac{(16 - t)^2}{16}$$

(c) Suppose tax (t) per unit output maximises the total tax revenue T with the output level x

Therefore ; $T = tx$

But we have found out above that when (t) is levied

$$x = \frac{16 - t}{8},$$

$$\text{Therefore,} \quad T = \left(\frac{16 - t}{8}\right) \times t = \frac{16t - t^2}{8}$$

T will be maximum where $\frac{dT}{dt} = 0$ and $\frac{d^2T}{dt^2} < 0$

$$\text{which gives,} \quad \frac{16 - 2t}{8} = 0, \quad \text{i.e., when } t = 8$$

That is, if the rate of tax is 8 per unit of quantity produced, the government will be able to maximise its tax revenue which will be :

$$T_{(max)} = tx = 8 \left(\frac{16-8}{8} \right) = 8$$

(d) the sales tax is levied on the sales proceeds, i.e. on Total Revenue. With the imposition of sales tax of 20% the tax will amount to $= 0.20(20-4x)x$ or $\frac{1}{5}(20-4x)x$ and $TC_N = 4x + \frac{1}{5}(20-4x)x$

$$MC_N = 4 + \frac{1}{5}(20-8x) = 8 - \frac{8}{5}x$$

$$MR = 20 - 8x \quad (\text{Since } TR \text{ remains unaltered})$$

For profit maximization $MR = MC_N$

$$8 - \frac{8}{5}x = 20 - 8x \quad \text{or } x = \frac{15}{8}$$

$$\text{and } t = \frac{1}{5}(20-4x)x = \frac{75}{16}$$

[Note Prof. Musgrave disfavours addition of *ad valorem* tax to cost. Since tax $= 0.20p$ post tax average revenue $= p - 0.20p = 0.8(20-4x)$. Thus, $TR = 0.8(20-4x)x$ and $MR = 0.8(20-8x)$. Hence $0.8(20-8x) = 4$, giving $x = \frac{15}{8}$.]

EXERCISE 7.3

If the demand and cost functions of the monopolist are as under and an additive tax of t per unit output is imposed, determine the change in price, the maximum profit and revenue received by the government as function of t . Also determine the maximum revenue that can be obtained by the government.

1 $x_d = p/4 - 3$

$TC = x^2 + 2x$

2 $p = 12 - 4x_d$

$TC = 8x + x^2$

3 $x_d = 75 - 3p$

$TC = 100 + 3x$

4. If the demand law is $p = \beta + \alpha x$ of a monopolist with total cost function $C = ax^2 + bx + c$, determine the quantity and price that maximise the profit of monopolist if tax of 2 per unit of output is levied by the government.

5. If the demand law is $p = 12 - 5Q$ and total cost is $C = -Q^3 + 3Q^2$, determine the change in the price due to imposition of sales tax of 20%. Find the corresponding profit. Determine the price if government grants subsidy (instead of imposing tax) of 1% per unit of output.

What will be the profit if a sales tax of 10% is imposed?

6 Given the demand function $p = (10 - x)^2$ and the cost function $C = (55 - 8x^2)$, find the maximum profit. What would be the effect of an imposition of a tax of Rs 9 per unit quantity on price?

7 Given the demand function $p = 100 - 0.5x^2$ and the cost function $C = \frac{1}{3}x^3 - 6x^2 + 108x$ determine the equilibrium output. Determine the change in price, output and profit after the imposition of a tax of Re 1 per unit of output.

8 Given the demand function $p = 12 - x^2$ and the cost function $C = -\frac{4}{3}x^3 + 4x^2 + 10$ where p denotes price and x the number of units of the product determine the profit maximising output of the firm. What will be the impact of a tax of Rs 12 per unit of product on price and profit? Interpret the results.

Determine the tax per unit that maximises the tax revenue if $C = x^3 + 4x^2 + 10$.

9 Given the demand function $p = 90 - x^2$ and the cost function $C = 10 + 2x^2 + 3x^3$, determine the profit maximising output of a monopolist firm. What would be the impact of a tax of Rs 10 per unit of output on price and profit?

10 Prove that a tax proportional to the output sold by a monopolist will result in an increase of the monopoly price. What will be the effect on monopoly price if the tax is a given fraction of the monopoly profit?

[Hint. If monopolist pays tax per unit of product sold – say ' t ' per unit of output Q , then his profit becomes

$$\pi = R(Q) - C(Q) - tQ$$

For maximisation of π

$$\frac{d\pi}{dQ} = R'(Q) - C'(Q) - t = 0$$

$$\frac{d\pi}{dQ} \propto MR - MC - t = 0$$

that is, $MR = MC + t$

The imposition of such tax results in a smaller quantity sold and a higher price, thus tax is shifted in this case.

But monopolist cannot avoid a tax paid by him in the form of fraction of his total profits. This type of tax will reduce his profits but will not affect his optimum price quantity combination.

Under such circumstances, if he is asked to pay t per cent out of his total

profits = profits after tax payment becomes

$$\pi = q - t[R(Q) - C(Q)]$$

For maximisation of π

$$\frac{d\pi}{dQ} = R(Q) - C(Q) - t = 0$$

$$\text{or} \quad \frac{d\pi}{dQ} \text{ or } MR - MC = 0$$

$$\text{i.e.} \quad MR = MC$$

Since t is a constant, it vanishes upon differentiation and the monopolist's output level and price are determined by the equality of MR and MC as would be the case if no tax were imposed.)

Chapter 8

Partial and Total Derivatives

8.1 Introduction

So far we concerned ourselves with functions of one independent variable for example while introducing the technique of differentiation we thought of total utility u as a function of consumption of one commodity x so that $u = f(x)$. Also all our rules of differentiation assumed y as function of only one variable x .

But the total utility of any quantity may in fact be function of two or more independent variables. We can cite many such examples in economics. If tonnes of wheat is produced on y acres of land with a number of labourers x is the function of x and y both that is $z = f(x, y)$. If consumer purchases meat x and bread y for his lunch then his total utility u will depend on amounts of both x and y consumed by him so that $u = f(x, y)$. Similarly we can think of total utility U being function of amounts of different commodities $x_1, x_2, x_3, \dots, x_n$ consumed by the consumer in a form of function

$$U = f(x_1, x_2, x_3, \dots, x_n)$$

where $x_1, x_2, x_3, \dots, x_n$ are all independent of one another and that each can vary by itself without affecting others.

We consider a simple case of U being dependent on two variables

$$U = f(x, y)$$

If the variable x undergoes change (assuming that consumption of only one commodity undergoes change) while y remains constant (as it was assumed previously) then there will be a corresponding change in U say it is ΔU then

$$U + \Delta U = f(x + \Delta x, y)$$

$$\text{or } \Delta U = f(x + \Delta x, y) - U$$

$$\text{or } \Delta U = f(x + \Delta x, y) - f(x, y)$$

$$\text{or } \frac{\Delta U}{\Delta x} = \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

Now we take limit of $\frac{\Delta U}{\Delta x}$ as Δx tends to zero to find the derivatives

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta U}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

we call this specific derivative as '*Partial Derivative*' of U with respect to x . We call it "Partial" derivative to indicate that y (or all other variables in case there are more than two variables) in the function has been held constant.

Such partial derivative is assigned different symbol to indicate that this is the partial derivative with respect to x and that other variable y has been regarded as fixed. In place of letter 'd', we use symbol ∂ and write partial derivative of U with respect to x as

$$\frac{\partial U}{\partial x} \quad \left(\text{not as } \frac{dU}{dx} \right)$$

$$\text{Thus,} \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta U}{\Delta x} = \frac{\partial U}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

Similarly, we can have the partial derivative of U with respect to y as

$$\lim_{\Delta y \rightarrow 0} \frac{\Delta U}{\Delta y} = \frac{\partial U}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Partial derivatives are also depicted by symbols as f_x, f_y, f_z where subscript indicates which independent variable (alone) is being allowed to vary.

For example, $\frac{\partial U}{\partial x}$ can also be represented by f_x and $\frac{\partial U}{\partial y}$ by f_y . [Remember that in case of $\frac{dy}{dx}$ we used the symbol $f'(x)$]

8.2. Technique of Partial Differentiation

The process of taking partial derivative is called partial differentiation and it differs from previously discussed differentiation primarily in that we hold and treat all the independent variables constant except the one which is assumed to vary.

The practical technique of partial differentiation is illustrated by the following examples.

Ex. 1. Given function is $U = 5x - 6y + 8$ and we are required to find partial derivatives. There can be only two partial derivatives

- (1) U with respect to x when y is held constant

$$\text{that is, } \frac{\partial U}{\partial x} \quad \text{or, } f_x$$

- (2) U with respect to y when x is held constant

$$\text{that is, } \frac{\partial U}{\partial y} \quad \text{or, } f_y$$

In case of (1), since y is held constant it is treated as a constant term during

differentiation, while in case of (2), since x is held constant it is treated as a constant term during differentiation, thus we have

$$(1) \quad \frac{\partial U}{\partial x} = f_x = 5 - 0 + 0 = 5 \quad (6y \text{ and } +8 \text{ are treated as constant terms})$$

$$(2) \quad \frac{\partial U}{\partial y} = f_y = 0 - 6 + 0 = -6 \quad (5x \text{ and } +8 \text{ are treated as constant terms})$$

Similarly if $Z = ax + by + c$ then

$$\frac{\partial Z}{\partial x} = a, \quad \frac{\partial Z}{\partial y} = b$$

Ex. 2. Find the partial derivatives of $Z = 4x^2 + 4xy + y^2$

We have, $\frac{\partial Z}{\partial x} = 8x + 4y$ and,

$$\frac{\partial Z}{\partial y} = 4x + 2y$$

Ex. 3 Given $Z = (x+4)(2x+5y)$, find partial derivatives

Partial derivatives can be obtained by the use of product rule

Thus we have, $f_x = (x+4)(2) + (1)(2x+5y)$

$$f_x = 2x + 8 + 2x + 5y = 4x + 5y + 8$$

Similarly, by holding x as constant

$$f_y = (x+4)(5) + (0)(2x+5y) = 5x + 20$$

Ex. 4. $Z = x^3 e^{2y}$; find partial derivatives

$$f_x = 3x^2 e^{2y}$$

$$f_y = x^3 2e^{2y} = 2x^3 e^{2y}$$

Ex. 5. $Z = \frac{x+y}{2x+5y}$; find partial derivatives.

$$\frac{\partial Z}{\partial x} = f_x = \frac{(1)(2x+5y) - (2)(x+y)}{(2x+5y)^2} = \frac{3y}{(2x+5y)^2}$$

$$\frac{\partial Z}{\partial y} = f_y = \frac{(1)(2x+5y) - (5)(x+y)}{(2x+5y)^2} = \frac{-3x}{(2x+5y)^2}$$

Ex. 6 $Z = \frac{5x^2}{5x-y+4}$; find partial derivatives.

$$f = \frac{10x(5x - y + 4) - (1)5(5x - y)}{(5x - y + 4)^2} = \frac{25x^2 - 10xy + 40x}{(5x - y + 4)^2}$$

$$f_y = \frac{(0)(5x - y + 4) - 5(5x^2)}{(5x - y + 4)^2} = \frac{-5x^2}{(5x - y + 4)^2}$$

8.3 Partial Derivatives of Second Order

We can also find the partial derivatives of second or higher orders.

The process of partial differentiation can be repeated till the partial derivative happens to be the function of any of the independent variables in the original function. For example

(1) If $\frac{\partial f}{\partial x}$ (the partial derivative of f with respect to x) happens to be the function of x and y , it could be differentiated partially with respect to x and written as

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \text{ and denoted by } \frac{\partial^2 f}{\partial x^2} \text{ or } f_{xx}$$

In other words $f_{xx} \left(= \frac{\partial^2 f}{\partial x^2} \right)$ is the second partial derivative obtained by partial derivation first with respect to x and then again with respect to x .

(2) If $\frac{\partial f}{\partial x}$ is a function of x and y , it can be differentiated partially with respect to y . This is denoted by $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ or $\frac{\partial^2 f}{\partial y \partial x}$ or f_{yx} , in other words f_{yx} is the second order partial derivative obtained by partial derivation first with respect to x and then with respect to y .

(3) Similarly, if $\frac{\partial f}{\partial y}$ (the partial derivative of f with respect to y) happens to be the function of both x and y , it could be partially differentiated with respect to x and y , so that when it is partially differentiated with respect to x we have $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$ denoted by $\frac{\partial^2 f}{\partial x \partial y}$ or f_{xy} .

$$(4) \text{ We can also have } \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \text{ denoted by } \frac{\partial^2 f}{\partial y^2} \text{ or } f_{yy}$$

Thus, it may be noted that we obtain two partial derivatives from each of the partial derivatives of f , or we obtain 4 second order partial derivatives of the original function.

These are f_{xx}, f_{xy}, f_{yx} and f_{yy} .

8.4 Cross Partial Derivatives

Second order partial derivatives f_{xy} and f_{yx} are known as cross partial derivatives. It may be noticed that the cross partial derivatives

$$f_{xy} \left(= \frac{\partial^2 Z}{\partial x \partial y} \right) \text{ and } f_{yx} \left(= \frac{\partial^2 Z}{\partial y \partial x} \right)$$

are different only in the order in which Z is being differentiated partially.

- (1) $\frac{\partial^2 Z}{\partial y \partial x}$ or f_{xy} indicates that the given function Z is first partially differentiated with respect to x and then with respect to y .
- (2) $\frac{\partial^2 Z}{\partial x \partial y}$ or f_{yx} indicates that given function Z is first partially differentiated with respect to y and then with respect to x .

However, it can be shown that under certain conditions (relating to the continuity of the function) the cross partial derivatives are identical in value that is

$$\frac{\partial^2 Z}{\partial y \partial x} = \frac{\partial^2 Z}{\partial x \partial y} \quad \text{or} \quad f_{yx} = f_{xy}$$

In other words, the order of the partial derivation does not make any difference in the process of partial differentiation.

We take a few examples to show this property of partial differentiation.

Ex. 7 $Z = \frac{x+4}{2x+5y}$

$$\frac{\partial Z}{\partial x} = f_x = \frac{5y-8}{(2x+5y)^2}$$

Since $\frac{5y-8}{(2x+5y)^2}$ is the function of x and y both, we can find second order derivatives as follows

$$\frac{\partial^2 Z}{\partial x \partial x} = f_{xx} = \frac{0(2x+5y)^2 - 2(2x+5y)(2)(5y-8)}{(2x+5y)^4} = \frac{-4(5y-8)}{(2x+5y)^3}$$

$$\frac{\partial^2 Z}{\partial y \partial x} = f_{xy} = \frac{5(2x+5y)^2 - 2(2x+5y)(5)(5y-8)}{(2x+5y)^4} = \frac{10x-25y+80}{(2x+5y)^3}$$

Also $\frac{\partial Z}{\partial y} = f_y = \frac{-x-20}{(2x+5y)^2}$

$$\frac{\partial^2 Z}{\partial x \partial y} = f_{xy} = \frac{-5(2x+5y)^2 - 2(2x+5y)(2)(-5x-20)}{(2x+5y)^4} = \frac{10x-25y+80}{(2x+5y)^3}$$

Hence $f_{xy} = f_{yx}$ and

$$\frac{\partial^2 z}{\partial y \partial y} = f_{yy} = \frac{0(2x+5y)^2 - 2(2x+5y)(5)(-5x-20)}{(2x+5y)^4} = \frac{50(x+4)}{(2x+5y)^3}$$

Ex. 8. Given $Z = x^3 e^{2y}$ Find all the partial derivatives of second order.

$$f_x = 3x^2 e^{2y}$$

$$f_{xx} = 6x(e^{2y}) = 6xe^{2y}$$

$$f_{yx} = 6x^2 e^{2y} + (0)(e^{2y}) = 6x^2 e^{2y}$$

$$f_y = 2x^3 e^{2y}$$

$$f_{xy} = 2x^3(0) + 6x^2 e^{2y} = 6x^2 e^{2y}$$

$$f_{yy} = 2x^3 e^{2y}(2) + (0)(e^{2y}) = 4x^3 e^{2y}$$

EXERCISE 8.1

Find first and second order partial derivatives of each of the following functions, also verify that:

$$\frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial^2 Z}{\partial y \partial x} \quad (\text{in the examples up to 8})$$

1. $Z = 2x^2 + 5x^2y + xy^2 + y^2$

2. $Z = 12 - x^2 - y^2 + xy$

3. $Z = (2x+5y)e^x$

4. $Z = 3\sqrt{xy}$

5. $Z = (x^2 + 2xy + y^2)e^2$

6. $Z = x + ye^{-x}$

7. $Z = \frac{2x^2}{2x-3y+1}$

8. $Z = \frac{x^3+y^3}{x^2+y^2}$

9. If $Z = \log(1+x^2) + y^2$, find f_{xx} , f_{yy} and f_{xy} .

10. If $Z = \frac{x^2}{y^3} + \frac{y^2}{x^3}$ and $Z = 3x^2 + \frac{x}{y}$.

Find the values of f_{xx} , f_{yy} , f_{yx} and f_{xy} in each of the above cases.

8.5 Partial Derivatives of Functions of More Than Two Variables

When defining the partial derivatives we considered a simple case of functions involving only two variables e.g., $Z = f(x, y)$. But we can have a function involving more than two variables, $[U = f(x, y, z)]$. Partial derivatives in this case may be defined as

- (1) $f_x \left(= \frac{\partial U}{\partial x} \right)$ represents the change in U with respect to x when y and z are being held as constants
- (2) $f_y \left(= \frac{\partial U}{\partial y} \right)$ represents the rate of change in U with respect to y when x and z are being held as constants
- (3) $f_z \left(= \frac{\partial U}{\partial z} \right)$ represents the rate of change in U with respect to z when x and y are being held as constants

The second order partial derivatives can be obtained as in the case of functions involving two variables

Since each of the three partial derivatives of first order will give rise to three second order partial derivatives, we shall have a group of nine second order partial derivatives as follows

Original function $U = (x, y, z)$

First order partial derivatives

f_x
 f_y
 f_z

Second order partial derivatives

f_{xx} f_{xy} f_{xz}
 f_{yy} f_{yz} f_{zy}
 f_{zx} f_{xz} f_{zy}

Total 9 partial derivatives

Of which f_{xy} , f_{yx} , f_{yz} , f_{zy} , f_{zx} and f_{xz} are cross partial derivatives. Again the following will be identical

$$f_{xy} = f_{yx}, f_{yz} = f_{zy} \text{ and } f_{zx} = f_{xz}$$

Therefore, we will have only six distinct values of the nine second order partial derivatives, three of which will be direct partial derivatives, viz. f_{xx} , f_{yy} and f_{zz} and three cross partial derivatives each equal, respectively to f_{xy} , f_{yz} and f_{zx} .

8.6 Application of Partial Derivatives in Economics

In the case of a function of one variable $y = f(x)$ we observed that the derivative $\frac{dy}{dx}$ depicts the rate of change in y due to marginal change in x

Similarly, partial derivative $\frac{\partial U}{\partial x}$ in a function $U = f(x, y, z)$ depicts the rate of

change in U with the marginal change in x when y and z do not change $\frac{\partial U}{\partial x}$ and $\frac{\partial U}{\partial z}$ give the rate of change in U with respect to marginal change in y and z respectively—when the remaining two variables are assumed to be held constant.

We consider the application of the above technique in the following analysis:

(A) *Demand analysis*, (B) *Utility Analysis*, and (C) *Production analysis*.

(A) **Demand Analysis** Let us assume that there are n number of consumer goods $X_1, X_2, \dots, X_n$ which are sold at prices $p_1, p_2, \dots, p_n$ with given tastes and incomes. Then definitely amount x of the good X demanded in the market is dependent on the prices of all the goods sold in the market. So that $x = f(p_1, p_2, \dots, p_n)$ which is termed as the demand function of the consumer good X . If the problem is to be restricted to two goods X_1 and X_2 with prices p_1 and p_2 , then the demand x_1 and x_2 of X_1 and X_2 will be written as

$$x_1 = f_1(p_1, p_2) \quad \text{and} \quad x_2 = f_2(p_1, p_2)$$

The partial derivatives of these two functions will indicate the variations in the demand as one of the prices vary while other price remains constant. For

example, partial derivative $\frac{\partial x_1}{\partial p_1}$ shows the rate at which the demand of x_1

changes as the price (p_1) is changed or $\frac{\partial x_1}{\partial p_1}$ represents the marginal (partial)

demand for x_1 with respect to p_1 . Similarly, partial derivative $\frac{\partial x_1}{\partial p_2}$ shows the

rate of change in the demand for x_1 when the price of x_2 (i.e. p_2) is changed. In other words, $\frac{\partial x_1}{\partial p_2}$ represents the marginal demand of x_1 with respect to p_2 .

In this way, we can obtain four partial derivatives of these two functions $x_1 = f_1(p_1, p_2)$ and $x_2 = f_2(p_1, p_2)$ as follows:

1. $\frac{\partial x_1}{\partial p_1}$ is the (partial) marginal demand of x_1 with respect to p_1 .

2. $\frac{\partial x_1}{\partial p_2}$ is the (partial) marginal demand of x_1 with respect to p_2 .

3. $\frac{\partial x_2}{\partial p_1}$ is the (partial) marginal demand of x_2 with respect to p_1 .

4. $\frac{\partial x_2}{\partial p_2}$ is the (partial) marginal demand of x_2 with respect to p_2 .

4 $\frac{dx_2}{dp_1}$ is the (partial) marginal demand of x_2 with respect to p_1

Nature of Commodities and Partial Derivatives The commodities are said to be *complementary* when an increase in the demand for one corresponds to an increase in the demand for other, and they are said to be *competitive* (substitutes) if an increase in demand for one corresponds to decrease in the demand for another.

We can easily decide about the nature of commodities with the help of partial derivatives in the following way

(a) If rise in p_2 not only tends to decrease the amount of x_2 but x_1 as well then the commodities x_1 and x_2 are said to be complementary i.e. $\frac{\partial x_2}{\partial p_2} < 0$ as well as $\frac{\partial x_1}{\partial p_2} < 0$ (since x_1 and p_2 change in opposite direction)

(b) If rise in p_1 not only tends to decrease the amount of x_1 but x_2 as well the commodities are again said to be complementary i.e. $\frac{\partial x_1}{\partial p_1} < 0$ as well as $\frac{\partial x_2}{\partial p_1} < 0$

(c) If rise in p_1 tends to decrease the amount x_1 but increase the amount of x_2 the commodities are said to be competitive i.e. $\frac{\partial x_2}{\partial p_1} > 0$ (since p_1 and x_2 change in the same direction) and $\frac{\partial x_1}{\partial p_1} < 0$

(d) Also if rise in p_2 tends to decrease the amount of x_2 but increase the amount of x_1 the commodities x_1 and x_2 are said to be competitive i.e. $\frac{\partial x_1}{\partial p_2} > 0$ and $\frac{\partial x_2}{\partial p_2} < 0$

Thus the partial derivatives help us to decide about the nature of the commodities

If all the partial derivatives of $x_2, x_3, \dots, x_n$ w.r.t. p_1 are positive (in value), the commodities $x_2, x_3, \dots, x_n$ will be competitive to commodity x_1 if the partial derivatives $x_2, x_3, \dots, x_n$ w.r.t. p_1 are negative the commodities $x_2, x_3, \dots, x_n$ are complementary to x_1

In both the cases the own partial derivatives $\frac{\partial x_1}{\partial p_1}$ and $\frac{\partial x_2}{\partial p_2}$ should always

be negative under the normal conditions because x_1 must increase if its corresponding price p_1 decreases and x_2 must decrease if its corresponding price p_2 rises.

Partial Elasticities Partial derivatives are also used to estimate the partial elasticities of demand.

On the basis of formula derived for point elasticity of demand, we can define

- 1 $\left| \frac{p_1}{x_1} \left(\frac{\partial x_1}{\partial p_1} \right) \right|$ as the *partial elasticity of demand* for x_1 with respect to its own price p_1
- 2 $\left| \frac{p_2}{x_1} \left(\frac{\partial x_1}{\partial p_2} \right) \right|$ as the *cross elasticity of demand* for x_1 with respect to price p_2
- 3 $\left| \frac{p_1}{x_2} \left(\frac{\partial x_2}{\partial p_1} \right) \right|$ as the *cross partial elasticity of demand* for x_2 with respect to price p_1
- 4 $\left| \frac{p_2}{x_2} \left(\frac{\partial x_2}{\partial p_2} \right) \right|$ as the *partial elasticity of demand* for x_2 with respect to own price p_2

Ex. 9. The demand function for commodity A is given by :

$$x_A = f(p_A, p_B, p_C) = 63.3 - 1.9p_A + 0.2p_B + 0.5p_C$$

(a) Find the partial elasticities of demand for A with respect to prices of other commodities B and C.

(b) Estimate the partial elasticities for $p_A = 10$, $p_B = 8$ and $p_C = 7$

(a) (i) Partial elasticity of A w.r.t. p_B is given by :

$$\frac{p_B}{x_A} \cdot \frac{\partial x_A}{\partial p_B} = \frac{0.2p_B}{63.3 - 1.9p_A + 0.2p_B + 0.5p_C}$$

(ii) Partial elasticity of A w.r.t. p_C is given by :

$$\frac{p_C}{x_A} \cdot \frac{\partial x_A}{\partial p_C} = \frac{0.5p_C}{63.3 - 1.9p_A + 0.2p_B + 0.5p_C}$$

(b) Evaluate the above cross partial elasticities for $p_A = 10$, $p_B = 8$ and $p_C = 7$.
We get ;

$$\frac{p_B}{x_A} \cdot \frac{\partial x_A}{\partial p_B} = \frac{(0.2)(8)}{63.3 - 1.9(10) + 0.2(8) + 0.5(7)} = 0.024$$

$$\frac{p_C}{x_A} \cdot \frac{\partial x_A}{\partial p_C} = \frac{0.5(7)}{63.3 - 1.9(10) + 0.2(8) + 0.5(7)} = 0.053$$

Ex. 10. Following are the demand functions for two commodities x_1 and x_2 :

$$x_1 = p_1^{-1.7} p_2^{0.8} \text{ and } x_2 = p_1^{0.5} p_2^{-0.2}$$

Determine whether the commodities are complementary or competitive, also find four partial elasticities of demand

In order to find the nature of commodity we must find the cross partial derivatives $\cdot \frac{\partial x_1}{\partial p_2}$ and $\frac{\partial x_2}{\partial p_1}$.

For demand function $x_1 = p_1^{-1.7} p_2^{0.8}$,

$$\frac{\partial x_1}{\partial p_2} = 0.5 p_1^{-1.7} p_2^{-0.2} \text{ which is also a positive quantity, hence}$$

$$\frac{\partial x_1}{\partial p_2} > 0$$

For demand function $x_2 = p_1^{0.5} p_2^{-0.2}$

$$\frac{\partial x_2}{\partial p_1} = 0.5 p_1^{-0.5} p_2^{-0.2} \text{ which is also a positive quantity, hence}$$

$$\frac{\partial x_2}{\partial p_1} > 0$$

Therefore, the commodities X_1 and X_2 are competitive

Partial elasticities of demand for demand function : $x_1 = p_1^{-1.7} p_2^{0.8}$ are given by

$$(i) \quad \frac{p_1}{x_1} \cdot \frac{\partial x_1}{\partial p_1} = \frac{p_1}{p_1^{-1.7} p_2^{0.8}} \left[(-1.7) p_1^{-2.7} p_2^{0.8} \right] = (-1.7)$$

$$(ii) \quad \frac{p_2}{x_1} \cdot \frac{\partial x_1}{\partial p_2} = \frac{p_2}{p_1^{-1.7} p_2^{0.8}} \left[(0.8) p_1^{-1.7} p_2^{-0.2} \right] = (+0.8)$$

Elasticities for demand function $x_1 = p_1^{0.8} p_2^{0.2}$ are given by

$$(iii) \quad \frac{p_2}{x_2} \cdot \frac{dx_2}{dp_2} = \frac{p_2}{p_1^{0.8} p_2^{0.2}} \cdot \left\{ (-0.2) p_1^{0.8} p_2^{-1.2} \right\} = (-0.2)$$

$$(iv) \quad \frac{p_1}{x_1} \cdot \frac{dx_1}{dp_1} = \frac{p_1}{p_1^{0.8} p_2^{0.2}} \cdot \left\{ (0.8) p_1^{-0.2} p_2^{0.2} \right\} = (+0.8)$$

EXERCISE 8.2

Determine the partial elasticities and nature of commodities for the following demand functions

1 $x_1 = 1 - 2p_1 + p_2$ and $x_2 = 5 - 2p_1 - 3p_2$

2 $x_1 = \frac{p_1}{p_2}$ and $x_2 = \frac{p_2}{p_1}$

3 $x_1 = \frac{1}{p_1 p_2}$ and $x_2 = \frac{1}{p_1 p_2}$

4 $x_1 = p_1^{-1.5} p_2^{0.4}$ and $x_2 = p_1^{0.3} p_2^{0.5}$

5 $x_1 = p_1^{-1} p_2^{1.4}$ and $x_2 = p_1^{0.8} p_2^{0.4}$

6 When are the two goods x_1 and x_2 said to be (a) competitive and (b) complementary in demand?

Examine the variation between x_1 and x_2 in the case of following demand function

(a) $x_1 = a_1 - a_{11}p_1 + a_{12}p_2$, $x_2 = a_{21}p_1 - a_{22}p_2$

(b) $x_1 = \frac{a_1}{p_1 + a_{11}} + a_{12}p_2$, $x_2 = \frac{a_2}{p_2 + a_{22}} + a_{21}p_1$

(c) $x_1 = p_1^{a_1} e^{(a_2 p_1 + a_3)}$, $x_2 = p_2^{a_2} e^{(a_1 p_1 + a_4)}$

7. If x_1 and p_1 are demand and price of tea and x_2 and p_2 are demand and price of coffee, it was found approximately that

$$x_1 = p_1^{-1.5} p_2^{0.5}$$

$$\text{and } x_2 = p_1^{0.3} p_2^{-0.5}$$

Show that the two commodities are competitive

8 If the demand laws for two goods are given by

$$x_1 = p_1^{a_{11}} e^{a_{12}p_2 + a_{13}}$$

$$\text{and } x_2 = p_2^{a_{21}} e^{a_{22}p_1 + a_{23}}$$

Show that the 'direct' price elasticities of demand are independent of the prices while 'cross' price elasticities are determined in sign by the constants a_1 and a_2 respectively

(B) Utility Analysis If U represents the total utility derived from consumption of goods $X_1, X_2, \dots, X_n$, then the total utility function will be

$$U = f(x_1, x_2, \dots, x_n)$$

where $x_1, x_2, \dots, x_n$ are the amounts of goods $X_1, X_2, \dots, X_n$ consumed at any time

The partial derivative $\frac{\partial U}{\partial x_1}$ will then depict the rate of change in total utility U with respect to change in the amount of consumption X_1 of X_2 when the amount of consumption of other goods remains unchanged. Thus $\frac{\partial U}{\partial x_1}$ is the marginal utility of X_1 . Similarly the partial derivatives $\frac{\partial U}{\partial x_2}, \frac{\partial U}{\partial x_3}, \dots, \frac{\partial U}{\partial x_n}$ represent the marginal utilities of various goods to the individual for a set of purchases $(x_1, x_2, \dots, x_n)$.

Even when the utility is regarded as immeasurable (hence there is no meaning in finding out the marginal utility) the ratio of marginal utilities has definite meaning in the ordinal approach of utility analysis.

Ex. 11. Suppose one form of the given utility function is

$$U = (ax_1 + bx_2 + c\sqrt{x_1x_2})$$

where only two goods X_1 and X_2 are considered to be consumed by a person and U is the utility derived therefrom, obtain the ratio of marginal utilities

$$f_{x_1} = \frac{\partial U}{\partial x_1} = a + \frac{1}{2}c(x_1x_2)^{-\frac{1}{2}} \quad x_2 = \frac{2a\sqrt{x_1x_2} + cx_2}{2\sqrt{x_1x_2}}$$

$$f_{x_2} = \frac{\partial U}{\partial x_2} = b + \frac{1}{2}c(x_1x_2)^{-\frac{1}{2}} \quad x_1 = \frac{2b\sqrt{x_1x_2} + cx_1}{2\sqrt{x_1x_2}}$$

So that the ratio of marginal utilities of two goods X_1 and X_2 is

$$\frac{f_{x_1}}{f_{x_2}} = \frac{\partial U / \partial x_1}{\partial U / \partial x_2} = \frac{2a\sqrt{x_1x_2} + cx_2}{2b\sqrt{x_1x_2} + cx_1}$$

Ex. 12. If the utility function is

$U = \log(ax_1 + bx_2 + c\sqrt{x_1x_2})$ obtain the ratio of marginal utilities

Here,
$$\frac{\partial U}{\partial x_1} = \frac{2a\sqrt{x_1x_2} + cx_2}{2\sqrt{x_1x_2}(ax_1 + bx_2 + c\sqrt{x_1x_2})}$$

$$\frac{\partial U}{\partial x_2} = \frac{2b\sqrt{x_1x_2} + cx_2}{2\sqrt{x_1x_2}(ax_1 + bx_2 + c\sqrt{x_1x_2})}$$

So, the ratio of marginal utilities is again the same as in the previous example

(C) **Production Analysis.** The production of certain commodities requires the use of two or more factors of production. These factors might be land, labour, capital, material or machines. The amount of production of any commodity, therefore, uniquely depends on the amounts of different factors (of production) used under certain technical conditions.

Supposing that quantity x of commodity X is produced by using the amount of $a_1, a_2, \dots, a_n$ of different factors $A_1, A_2, \dots, A_n$, then the amount of production is $x = f(a_1, a_2, \dots, a_n)$. Now, if in this particular combination of factors $\{a_1, a_2, \dots, a_n\}$ we assume that only factor A_1 is varied (while others are held constant), then the ratio $\left(\frac{x}{a_1}\right)$ will depict the average product of factor A_1 ,

and partial derivative $\frac{\partial x}{\partial a_1}$ will depict the rate of change in the total amount of production x with respect to marginal change in a_1 . In other words, $\frac{\partial x}{\partial a_1}$ will represent the marginal product of factor A_1 .

Summarising, therefore, if $x = f(a_1, a_2, \dots, a_n)$

and it is assumed that only factor A_1 is varied while others are kept constant, then

$$\text{Average product of } A_1 = \frac{x}{a_1}$$

$$\text{Marginal product of } A_1 = \frac{\partial x}{\partial a_1}$$

We can now depict average and marginal products diagrammatically and can also specify the famous three stages of production for any production function $z = f(x, y)$

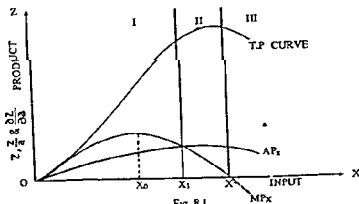


Fig 8.1

Graphical Representation of the Three Stages

The total product curve OM in Fig 8.1 depicts the amount of total product Z with different levels of input X while the other factor (input) Y is fixed at level y_1 .

We are already familiar with the three stages of production which can be summarized as follows

Stage No	Nature of total, average and marginal product curves
I (First)	Marginal product begins to decline but average product reaches its maximum and marginal product curve cuts average product curve at its maximum
II (Second)	Average product and marginal product both fall and when marginal product = 0 total product is maximum
III (Third)	Marginal product is negative total product and average product both fall

Let us give the mathematical interpretation of the points where the first two stages end

End point of stage *First* is where AP_x is maximum and at this maximum $AP_x = MP_x$. End of stage *Second* is where $MP_x = 0$ and total product is maximum

Ex 13 Supposing under a given technical condition the quantity z of commodity Z is produced by using the amount x and y of two factors X and Y and the production function is of the form

$$z = 2Hxy - Ax^2 - By^2$$

(II, A and B are constants and $H^2 > AB$)

Assuming factor Y is fixed at y_1 and factor X is varied at different levels, then the production function at fixed level of $Y=y_1$ will be equal to

$$z = 2Hxy_1 - Ax^2 - By_1^2$$

Average product of factor X is

$$AP_x = \frac{z}{x} = 2Hy_1 - Ax - \frac{By_1^2}{x}$$

Marginal product of factor X will be equal to

$$MP_x = \frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (2Hxy_1 - Ax^2 - By_1^2) = 2(Hy_1 - Ax)$$

so that for a fixed amount of factor Y (at a level of y_1) we have been able to estimate the average and the marginal products of factor X from the given production function

AP_x will be maximum where

$$\frac{\partial}{\partial x} (AP_x) = 0 \text{ and } \frac{\partial^2}{\partial x^2} (AP_x) < 0$$

$$\frac{\partial}{\partial x} \left(2Hy_1 - Ax - B \frac{y_1^2}{x} \right) = -A + B \frac{y_1}{x^2} = 0$$

$$A = B \frac{y_1^2}{x^2} \text{ i.e. } x^2 = \frac{B}{A} y_1^2$$

$$x = \sqrt{\frac{B}{A}} y_1$$

$$\text{At } x = \sqrt{\frac{B}{A}} y_1$$

$$\frac{\partial^2}{\partial x^2} (AP_x) = \left(-\frac{2By_1}{x^3} \right) = \frac{-2By_1}{\left(\sqrt{\frac{B}{A}} y_1 \right)^3} < 0$$

Hence AP_x is maximum at $x = \sqrt{\frac{B}{A}} y_1$ when factor Y is fixed at y_1 . Also

AP_x increases up to $x = \sqrt{\frac{B}{A}} y_1$ and then decreases monotonically. What will be the absolute value of AP when it is at the maximum?

Substituting $x = \sqrt{\frac{B}{A}} y_1$ in AP_x

$$AP_x = 2Hy_1 - Ax - B \frac{y_1^2}{x}$$

$$\text{i.e. } AP_{(max)} = 2Hy_1 - A \sqrt{\frac{B}{A}} y_1 - By_1^2 \sqrt{\frac{A}{B}} \frac{1}{y_1}$$

$$AP_{(max)} = 2Hy_1 - 2\sqrt{AB} y_1 = 2y_1 (H - \sqrt{AB})$$

Since AP must be positive, hence $H - \sqrt{AB} > 0$ $H^2 > AB$

But at $x = \sqrt{\frac{B}{A}} y_1$ not only AP is maximum but $AP_x = MP_x$

Wh it is MP when $x = \sqrt{\frac{B}{A}} y_1$,

$$MP_x = 2Hy_1 - 2\lambda x$$

Substituting $x = \sqrt{\frac{B}{A}} y_1$

$$MP_x = 2Hy_1 - 2\lambda = \sqrt{\frac{B}{A}} y_1$$

$$MP_x = 2Hy_1 - 2\sqrt{AB} y_1 = 2y_1 (H - \sqrt{AB})$$

which is equal to AP_x

This proves that at $x = \sqrt{\frac{B}{A}} y_1$ AP is maximum and $AP_x = MP_x$

End of stage II is when $MP_x = 0$ ($AP_x \neq 0$) and total product is maximum

$$MP_x = 2Hy_1 - 2\lambda x$$

which must be equated to zero to find whether total product is maximum or not

It may be noted that MP_x is a monotonic decreasing function of x . Hence output is increasing at a decreasing rate.

$$\text{So} \quad 2Hy_1 - 2\lambda x = 0 \text{ or } x = \frac{H}{\lambda} y_1$$

What will be the value of total product at this level of $x = \frac{H}{\lambda} y_1$?

$$\text{Total product } z = 2Hxy_1 - \lambda x^2 - By_1^2$$

$$\text{Substituting, } x = \frac{H}{\lambda} y_1$$

$$\begin{aligned} z &= 2H \left(\frac{H}{\lambda} \right) y_1^2 - \lambda \left(\frac{H}{\lambda} \right)^2 y_1^2 - By_1^2 \\ &= y_1^2 \left(\frac{2H^2}{\lambda} - \frac{H^2}{\lambda} - B \right) = y_1^2 \frac{H^2 - \lambda B}{\lambda} \end{aligned}$$

In order to prove that this absolute value is the maximum value of total

In other words, the end of second stage is at $a = 12 \times 100$ man-hours and at this point total product is maximum = 640 tonnes of wheat

We can now depict three stages diagrammatically on the fixed area of land = 10. The three stages of production with varied levels of labour in terms of '100a' man hours will be as follows

It should be noted that output x is negative for $a < 4$ and zero at $a = 4$. Hence the production function is defined only for values of $a < 4$

Stage I—from $a = 4$ to $a = 80$ (x is negative, for $a < 4$)

Stage II—from $a = \sqrt{80}$ to $a = 12$

Stage III—from $a = 12$ onwards

These stages have been marked in Fig. 8.2

Ex. 15. Find the average and marginal product curves for labour employed on (a) 10 hectares and (b) 100 hectares when

$$x = \frac{40}{a+b} (12ab - 5a^2 - 4b^2)$$

is the wheat product for '100a' man hours on b hectares of land

$$\text{Production function is } x = \frac{40}{a+b} (12ab - 5a^2 - 4b^2)$$

$$\text{Therefore, } AP_a = \frac{x}{a} = \frac{40(12ab - 5a^2 - 4b^2)}{a(a+b)}$$

$$MP_a = \frac{\partial x}{\partial a} = \frac{40}{(a+b)^2} [-5a^2 - 10ab + 16b^2]$$

$$AP_a \text{ (when } b = 10 \text{ hectares)} = \frac{40}{(a+10)} [1,200a - 5a^2 - 40,000]$$

$$MP_a \text{ (when } b = 10 \text{ hectares)} = \frac{40}{(a+10)^2} [-5a^2 - 100a - 1,600]$$

$$AP_a \text{ (when } b = 100 \text{ hectares)} = \frac{40}{(a+100)} [1,200a - 5a^2 - 4,000]$$

$$MP_a \text{ (when } b = 100 \text{ hectares)} = \frac{40}{(a+100)^2} [-5a^2 - 1000a - 160,000]$$

Ex. 16. For the production function: $x = Aa^{\alpha}b^{\beta}$;

show that marginal product of a is $\left(\frac{\alpha}{a}\right)$ times the product while marginal product of b is $\left(\frac{\beta}{b}\right)$ times the product

Given production function

$$\text{Total product } x = Aa^\alpha b^\beta$$

If b is assumed to be fixed

$$MP_a = \frac{\partial x}{\partial a} = A\alpha a^{\alpha-1} b^\beta = \frac{\alpha}{a} A a^\alpha b^\beta = \left(\frac{\alpha}{a}\right) x$$

If a is assumed to be fixed,

$$MP_b = \frac{\partial x}{\partial b} = Aa^\alpha b^{\beta-1} = A\beta a^\alpha b^{\beta-1} = \left(\frac{\beta}{b}\right) Aa^\alpha b^\beta = \left(\frac{\beta}{b}\right) x$$

Ex 17. Given the production function

$$x = \sqrt{2Hab - Aa^2 - Bb^2}$$

Show that the maximum value of average product of a is a constant

$$\sqrt{\frac{H^2 - AB}{B}} \text{ which is independent of fixed amount of } b \text{ used}$$

$$\text{Given } x = \sqrt{2Hab - Aa^2 - Bb^2}$$

$$AP_a = \frac{x}{a} = \frac{1}{a} \sqrt{2Hab - Aa^2 - Bb^2}$$

AP_a will be maximum, if

$$\frac{\partial}{\partial a} (AP_a) = 0 \text{ and } \frac{\partial^2}{\partial a^2} (AP_a) < 0$$

$$\text{that is, } \frac{\partial}{\partial a} \left(\frac{1}{a} \sqrt{2Hab - Aa^2 - Bb^2} \right) = 0$$

$$\frac{\partial}{\partial a} (AP_a) = \frac{1}{a^2} \left\{ \frac{a(2Hb - 2Aa)}{2\sqrt{2Hab - Aa^2 - Bb^2}} - \sqrt{2Hab - Aa^2 - Bb^2} \right\}$$

$$\text{Hence, } \frac{1}{a^2} \left\{ \frac{-2Hab + 2Bb^2}{2\sqrt{2Hab - Aa^2 - Bb^2}} \right\} = 0$$

$$-2Hab + 2Bb^2 = 0, \text{ or } a = \frac{B}{H} b$$

$$\left[\text{Check that } \frac{\partial^2}{\partial a^2} (AP_a) < 0 \text{ at } a = \frac{B}{H} b. \right]$$

That is, AP_a will be maximum when $a = \frac{Bb}{H}$.

What will be AP_a at this level of a ?

$$\begin{aligned}
 AP_{m, \lambda} &= \frac{H}{Bb} \sqrt{2H \frac{Bb}{H} B - A \frac{Bb}{H} Bb} \\
 &= \frac{1}{\sqrt{B}} \sqrt{H^2 - AB} = \sqrt{\frac{H^2}{B} - \frac{AH}{B}}
 \end{aligned}$$

which was required to be proved

EXERCISE 8.3

1 The production function of a commodity is given by

$$\lambda = f(Q, L, K) = 100Q + 200L + 50(K - Q^2 - 2L - 3K - 5QL + 3QK - 1K)$$

(Q , L and K represent *land*, *labour* and *capital* respectively)

Find the marginal productivities of land, labour and capital. Determine these marginal productivities if $Q=1$, $L=2$ and $K=3$ units.

2 The production function of a particular commodity was estimated as

$$\lambda = L^{0.64} K^{0.36}$$

where λ is the production of that commodity, L is labour and K is capital.

Find the marginal productivities of L and K . Determine the marginal productivities for

$$L = 1.5 \text{ and } K = 1.1 \text{ units}$$

3 The production function of agricultural farm in a particular district was estimated to be

$$X = L^{2.13} K^{0.34}$$

where X is the total product, L labour and K the fixed capital.

Find the marginal productivities of labour and capital.

Estimate the marginal productivities for $L = 1.7$ and $K = 2$ units.

4 Given the production function

$$Q = f(L, K) = LK + 0.2L^2 - 0.8K^2$$

Construct the average and marginal productivities for labour (L).

Assume that $K=10$. At what respective values of L will the AP and MP of labour equal zero?

5 Given the production function

$$\lambda = 10a^{0.28} b^{0.22} c^{0.02} d^{0.16} f^{0.18}$$

where a , b , c , d and f represent factors of production.

(i) Construct the average and marginal productivities of each factor.

(ii) Estimate the marginal productivities for $a = b = c = d = f = 1$.

6 It is known that $x = Aa^u b^{1-u}$ is the product (in tonnes) of wheat when

100 a man hours of labour are employed on b hectares of land and the constants A and a given that 1500 tonnes can be obtained from 100 hectares when 10 000 man hours of labour are employed and that 2120 tonnes can be obtained from the same area when 20 000 man hours of labour are employed. Draw a graph of the variation of product as varying amounts of labour are applied to 100 hectares. What is the product when 20,000 man hours of labour are employed on 150 hectares?

7. It is given that $x = 40 \left\{ \frac{12ab - 5a^2 - 4b^2}{a + b} \right\}$ tonnes of wheat are obtained when 100 a man hours of labour are employed on b hectares of land. If 10 hectares are cultivated, draw a graph to show the variation of product as a varies. Show that the greatest product is obtained from this area when approximately 1050 man hours are employed.

8. For the linear homogeneous production function

$$x = \frac{2Hab - Aa^2 - Bb^2}{C(a + b)}$$

show that average and marginal products of the factors depend only on the ratio of the factors and verify that the product is always a times the marginal product of A and b times the marginal product of B .

9. If a men are employed in planting b hectares with timber, the amount of timber cut after t years is $x = f(a, b, t)$. What meaning can be attached to

$$\frac{\partial x}{\partial a}, \quad \frac{\partial x}{\partial b} \quad \text{and} \quad \frac{\partial x}{\partial t}?$$

If $x = Aa^{\alpha} b^{1-\alpha} t^{\beta}$ where α and β are fixed positive functions, show that there are constant returns to land and labour after a fixed time t and that for given employment of land and labour the timber product increases but in a decreasing proportion as time goes on.

8.7 Total Differential

The technique of partial derivation gave us the rate of change in $U\{U=f(x,y)\}$ when small change was made in x (or in y) while y (or x) was assumed to remain unchanged. But in case it is assumed that x and y both changed simultaneously in the given function $U = f(x, y)$ how are we to find the corresponding change in U ?

This is the problem which we deal under the *technique of finding total differential*.

Let us consider a problem from Economics. z tonnes of wheat are produced on y hectares of land with x number of labourers so that we have function $z = f(x, y)$. If small changes are made in x (labourers) and y (land), how much of a change will there be in z (wheat production)?

Let us try to find the solution through partial derivation technique. We have learnt that partial derivative $\frac{\partial z}{\partial x}$ will give us the change in z when there is a small change in x holding y as constant.

If we suppose that x changes by an amount Δx , the change in z will be $\left[\frac{\partial z}{\partial x} \Delta x \right]$. Similarly, since $\frac{\partial z}{\partial y}$ represents the change in z for small change in y , hence change in z will be $\left[\frac{\partial z}{\partial y} \Delta y \right]$ if amount of change in y is Δy .

Therefore, the total change in z (say, it is equal to Δz) due to small change in x and y (equal to Δx and Δy) will be

$$\Delta z = \left(\frac{\partial z}{\partial x} \Delta x \right) + \left(\frac{\partial z}{\partial y} \Delta y \right)$$

and alternatively when these small changes tend to approach zero,

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \quad (i)$$

Since the expression for dz gives the sum of changes in z due to changes in both the independent variables (labour x and land y), it is called the total differential of the production function $z=f(x, y)$.

This expression can also be written in abbreviated form

$$dz = f_x dx + f_y dy \quad \dots (ii)$$

We should remember that the expressions dz and dx can stand alone and are known as differentials, but notations ∂x and ∂y are meaningless if they stand alone.

We consider one more example of the utility function :

$$U = (x_1, x_2, x_3, \dots, x_n)$$

Total differential of the function can be written as :

$$dU = \frac{\partial U}{\partial x_1} \cdot dx_1 + \frac{\partial U}{\partial x_2} \cdot dx_2 + \dots + \frac{\partial U}{\partial x_n} dx_n$$

in which right hand side indicates the amount of change in utility resulting from a small change in consumption of each commodity separately. The first term $\left\{ \frac{\partial U}{\partial x_1} \cdot dx_1 \right\}$ means the marginal utility of first commodity times the increment in consumption of the commodity. We give similar meaning to the other terms and as such right hand side depicts the total change in utility originating from the changes in consumption of all the commodities.

We shall now study the rules of 'Total Differential' with the help of following examples :

Rule 1 Total differential of the sum of terms

Ex 18 Find total differential of $z = 3x^2 + xy - 2y^3$

$$dz = d(3x^2) + d(xy) - d(2y^3) = 6x dx + (y dx + x dy) - 6y^2 dy$$

$$dz = 6x dx + y dx + x dy - 6y^2 dy = (6x + y) dx + (x - 6y^2) dy$$

The same result can be obtained via partial derivative.

$$z = 3x^2 + xy - 2y^3$$

$$f_x = \frac{\partial z}{\partial x} = 6x + y \quad \text{and} \quad f_y = \frac{\partial z}{\partial y} = x - 6y^2$$

But $dz = f_x dx + f_y dy$

$$dz = (6x + y) dx + (x - 6y^2) dy$$

Rule 2 Total differential of the product

Ex 19 Find total differential of $z = (x^2 + y)(2x - y^2)$

$$\begin{aligned} dz &= (2x - y^2) d(x^2 + y) + (x^2 + y) d(2x - y^2) \\ &= (2x - y^2)(2x dx + dy) + (x^2 + y)(2 dx - 2y dy) \\ &= (6x^2 - 2xy^2 + 2y) dx + (2x - 3y^2 - 2x^2y) dy \end{aligned}$$

Rule 3 Total differential of the quotient

Ex 20 Find the total differential of $z = \frac{x}{x+y}$

$$dz = \frac{1}{(x+y)^2} [-y d(x+y) + (x+y) d(x)]$$

$$dz = \frac{1}{(x+y)^2} [-y(dx+dy) + (x+y)dx]$$

$$dz = \frac{-y dy + y dx}{(x+y)^2}$$

The same result is obtained via partial derivatives

$$f_x = \frac{\partial z}{\partial x} = \left(\frac{1}{x+y} \right)^2 [-y + (x+y)] = \frac{+y}{(x+y)^2}$$

$$f_y = \frac{\partial z}{\partial y} = \left(\frac{1}{x+y} \right)^2 (-x) = \frac{-x}{(x+y)^2}$$

Since $dz = f_x dx + f_y dy$

$$\begin{aligned} dz &= \frac{+y}{(x+y)^2} dx + \frac{-x}{(x+y)^2} dy \\ &= \frac{-y dy + y dx}{(x+y)^2} \end{aligned}$$

Ex. 21 Find the total differential of $Z = \frac{x^2}{x^2 + y^2} \cdot \frac{y^2}{x^2 + y^2}$

$$dZ = \frac{(x^2 + y^2) d(x^2 - y^2) - (x^2 - y^2) d(x^2 + y^2)}{(x^2 + y^2)^2}$$

$$dZ = \frac{(x^2 + y^2)(2xdx - 2ydy) - (x^2 - y^2)(2xdx + 2ydy)}{(x^2 + y^2)^2}$$

$$dZ = \frac{-4x^2 dy + 4y^2 dx}{(x^2 + y^2)^2}$$

Rule 4 Chain rule in total differential

Ex. 22. Find the total differential of $Z = \sqrt{x + y}$

Let $u = x + y$

$$Z = \sqrt{u} = u^{1/2} \text{ and } dZ = \frac{1}{2} u^{-1/2} du \quad (i)$$

$$du = d(x) + d(y) = dx + dy \quad (ii)$$

Substitute (i) in (ii), we get

$$dZ = \frac{1}{2} u^{-1/2} (dx + dy)$$

$$dZ = \frac{1}{2\sqrt{x+y}} (dx + dy)$$

Ex. 23 Find the total differential of $Z = \sqrt{3x^2 + y^2}$

Let $u = 3x^2 + y^2$

$$Z = \sqrt{u} = u^{1/2} \text{ and } dZ = \frac{1}{2} u^{-1/2} du \quad (i)$$

Since $u = 3x^2 + y^2$

$$du = d(3x^2) + d(y^2) = 6xdx + 2ydy \quad (ii)$$

Substitute (ii) in (i),

$$dZ = \frac{1}{2} u^{-1/2} (6xdx + 2ydy)$$

$$dZ = \frac{1}{2\sqrt{3x^2 + y^2}} (6xdx + 2ydy)$$

Ex. 24. Find the total differential of $Z = \log(x^2 - 2y)$

Let $u = x^2 - 2y$

Therefore, $Z = \log u$, $dZ = d(\log u) = \frac{1}{u} du \quad (i)$

$$du = d(x^2) - d(2y) = 2xdx - 2dy \quad (ii)$$

Substitute (ii) in (i)

$$dZ = \frac{1}{u} (2xdx - 2dy)$$

$$dZ = \frac{1}{x^2 - 2y} (2xdx - 2dy)$$

Ex. 25. Given the production function $P = (\beta K^{-\rho} + \alpha L^{-\rho})^{-1/\rho}$

Find dP

$$\text{Let } \beta K^{-\rho} + \alpha L^{-\rho} = u$$

$$\text{so that } P = (u)^{-1/\rho}$$

$$dP = (-) \frac{1}{\rho} (u)^{(-\frac{1}{\rho} - 1)} du \quad (i)$$

$$\text{Since } u = \beta K^{-\rho} + \alpha L^{-\rho}$$

$$du = d(\beta K^{-\rho}) + d(\alpha L^{-\rho})$$

$$du = \beta(-\rho)K^{-\rho-1} dK + \alpha(-\rho)L^{-\rho-1} dL$$

$$du = -\beta\rho K^{-\rho-1} dK - \alpha\rho L^{-\rho-1} dL \quad (ii)$$

Substitute (ii) in (i),

$$dP = (-) \frac{1}{\rho} (u)^{(-\frac{1}{\rho} - 1)} [-\beta\rho K^{-\rho-1} dK - \alpha\rho L^{-\rho-1} dL]$$

$$dP = (-) \frac{1}{\rho} [\beta K^{-\rho} + \alpha L^{-\rho}]^{(-\frac{1}{\rho} - 1)} [-\beta\rho K^{-\rho-1} dK - \alpha\rho L^{-\rho-1} dL]$$

EXERCISE 84

Find the total differential of the following functions

1 $Z = x^2 + 3x^2y + 6xy^2 + 2y^3$

2 $Z = x^5y^4 - x^4y^5$

3 $Z = (2x^2 + y)(x + 2y^2)$

4 $Z = (x + y)(x - 2y)$

5 $Z = \frac{x^2 + y^2}{x + y}$

6 $Z = \frac{xy}{x^2 + y^2}$

7 $Z = 7x \log(1 + y)$

8 $Z = \log \sqrt{x^2 + y}$

9 $Z = \sqrt{x^3 + 2y + 1}$

10 $Z = e^x e^y$

11. Given a utility function $U = f(q_1, q_2)$ where U is utility, q_1 and q_2 are consumption amounts of two commodities, Find du .

12. Given production function $P = f(x_1, x_2)$ where P is the output, x_1 and x_2 are the amounts of inputs utilized. Find dP .

13. Given the production function

$$P = \alpha L^\alpha K^\beta$$

where P is product, L is labour, K is capital and α, β are constants. Find dP .

14. Given production function

$$P = \gamma [\alpha K^\beta + (1 - \alpha) L^\beta]^{1/\beta}$$

where P is the output, K is capital and L is labour. γ, α and β are constants. Find dP .

8.8 Second Order Total Differential

We have observed above that for a given function $z = f(x, y)$ total differential can be written in terms of partial derivatives as

$$dz = f_x dx + f_y dy$$

The expression on right hand side of total differential (dz) contains partial derivatives of first order.

Since we have learnt to find the second order partial derivatives we can now find the second order total differential in terms of second order partial derivatives as under

$$d(dz) = d(f_x dx) + d(f_y dy)$$

$$\text{We know } f_x = \frac{\partial}{\partial x}(x, y) \text{ and } f_y = \frac{\partial}{\partial y}(x, y)$$

$$d(f_x dx) = d(f_x) dx$$

$$= \left(\frac{\partial}{\partial x} f_x dx + \frac{\partial}{\partial y} f_x dy \right) dx = (f_{xx} dx + f_{xy} dy) dx$$

$$\text{Similarly, } d(f_y dy) = d(f_y) dy$$

$$= \left(\frac{\partial}{\partial x} f_y dx + \frac{\partial}{\partial y} f_y dy \right) dy = (f_{xy} dx + f_{yy} dy) dy$$

$$\text{Again, } d(dz) = d^2 z = (f_{xx} dx + f_{xy} dy) dx + (f_{xy} dx + f_{yy} dy) dy$$

$$d^2 z = f_{xx} (dx)^2 + f_{xy} dy dx + f_{xy} dx dy + f_{yy} (dy)^2$$

$$d^2 z = f_{xx} (dx)^2 + 2f_{xy} dx dy + f_{yy} (dy)^2$$

$$(\text{Since } f_{xy} = f_{yx})$$

It can be noticed that the derived form of above total differential of second order, $d^2 z$ is similar to $(x + y)^2 = x^2 + 2xy + y^2$.

$$\text{Similarly, we have } d^2 z = (f_x dx + f_y dy)^{(2)}$$

$$\text{or, } d^2z = f_{xx}(dx)^{(2)} + 2f_{xy}dxdy + f_{yy}(dy)^{(2)}$$

In case we need third order total differential, it can be written on the same line as :

$$\begin{aligned} d^3z &= (f_x dx + f_y dy)^{(3)} \\ &= f_{xxx}(dx)^{(3)} + 3f_{xxy}(dx)^{(2)}dy + 3f_{xyy}dx(dy)^{(2)} + f_{yyy}(dy)^{(3)} \end{aligned}$$

In this way n th order of the total differential will be

$$d^n z = (f_x dx + f_y dy)^{(n)}$$

The index n is not the power of the expression but has a behaviour similar to that and has therefore been distinguished by keeping it in brackets

If the function involves more than two variables the same rule will apply
 $z = f(x, y, w)$

$$\begin{aligned} \text{so that, } d^2z &= (f_x dx + f_y dy + f_w dw)^{(2)} \\ &= f_{xx}(dx)^{(2)} + f_{xy}dxdy + f_{xw}dx dw + f_{yx}dy dx + f_{yy}(dy)^{(2)} + f_{yw}dy dw \\ &\quad + f_{wx}dw dx + f_{wy}dw dy + f_{ww}(dw)^{(2)} \end{aligned}$$

and n th order of the total differential shall be depicted as .

$$d^n z = (f_x dx + f_y dy + f_w dw)^{(n)}$$

Ex. 26. Given : $z = x^2 - 2xy + y^2$

find first and second order total differentials

$$\begin{aligned} \text{Since } z &= x^2 - 2xy + y^2 \\ f_x &= 2x - 2y ; f_{xx} = 2 \\ f_y &= -2x + 2y ; f_{yy} = 2, \text{ and } f_{xy} = -2 = f_{yx} \\ dz &= f_x dx + f_y dy = (2x - 2y)dx + (2y - 2x)dy \\ d^2z &= (f_x dx + f_y dy)^{(2)} = f_{xx}(dx)^{(2)} + 2f_{xy}dxdy + f_{yy}(dy)^{(2)} \\ &= 2(dx)^{(2)} + 2(-2)dxdy + 2(dy)^{(2)} \\ &= 2(dx)^{(2)} - 4dxdy + 2(dy)^{(2)} \end{aligned}$$

The same result can be obtained without finding partial derivatives .

$$\begin{aligned} z &= x^2 - 2xy + y^2 \\ \therefore dz &= d(x^2) - d(2xy) + d(y^2) \\ &= 2xdx - 2ydx - 2xdy + 2ydy \end{aligned}$$

On further differentiating dz (remembering that dx and dy are just constants),

$$d^2z = d(2x)dx - d(2y)dy - d(2x)dy + d(2y)dy$$

$$d^2z = (2dx)dx - (2dy)dy - (2dx)dy + (2dy)dy$$

$$= 2(dx)^2 - 4dxdy + 2(dy)^2$$

Ex 27 Find the second order total differential of the function

$$z = 7x \log(1+y)$$

In this case $f_x = 7 \log(1+y)$ $f_{xx} = 0$ $f_y = 7x(1+y)$ $f_{yy} = \frac{(7x)}{(1+y)^2}$

and $f_{xy} = \frac{7(1+y)}{(1+y)^2} = \frac{7}{(1+y)}$

Since $d^2z = f_{xx}dx^2 + 2f_{xy}dxdy + f_{yy}dy^2$

$$d^2z = 0 + 2\left(\frac{7}{1+y}\right)dxdy + \frac{7x}{(1+y)^2}dy^2$$

Therefore, $d^2z = \frac{-7x}{(1+y)^2}dy^2 + \left(\frac{14}{1+y}\right)dxdy$

EXERCISE 8.5

Find the first and second order total differentials of the following functions

1 $f = x^4 + y^4 + 4xy$

2 $f = x^3y^2 + y^3x^2$

3 $f = x^2 + y^2$

4 $f = xy$

5 $u = \sqrt{x^2 + y^2}$

6 $u = 7y \log(1+x)$

7 $u = 5x e^{5x}$

8 $u = f(x, y, z) = xyz$

8.9 Derivatives of Implicit Functions

We have already explained the technique of finding the derivative $\left(\frac{dy}{dx}\right)$ of the implicit functions

Here we shall learn to find out the derivative of the implicit functions (involving more than one variable) with the help of total differential method

$y = f(x) = \frac{4x}{2+x}$ is an explicit function, since y is explicitly expressed as a function of x

$2y + xy - 4x = 0$ is an implicit function, since when x changes y must also change in order to make the expression $(2x + xy - 4x)$ equal to zero.

Therefore explicit function (of one variable) is expressed in the form

$$y = f(x)$$

and implicit function is expressed in the form $f(x, y) = 0$

Similarly explicit function (of two variables) is expressed as $z = f(x, y)$ and implicit function is expressed in the form $F(x, y, z) = 0$

Here our problem is to find the derivatives of implicit functions

$$F(x, y) = 0 \text{ or } F(x, y, z) = 0$$

We now explain the method of finding derivatives of such functions

Lx 28 Find $\frac{dy}{dx}$ of the implicit function $2y + xy - 4 = 0$

We apply total differential rule of differentiating each function of the left and right hand side,

i.e. $d(2y) + d(xy) - d(4) = d(0)$

$$2dy + xdy + ydx - 4dx = 0$$

$$(2 + x)dy + (y - 4)dx = 0$$

or $\frac{dy}{dx} = \frac{4 - y}{2 + x}$

Lx 29 Find $\frac{dy}{dx}$ of implicit function $x^2 - xy - 2x + 3 = 0$

Given $x^2 - xy - 2x + 0 = 0$

$$d(x^2) - d(xy) - d(2x) + d(3) = d(0)$$

$$2xdx - ydx - xdx - 2dx + 0 = 0$$

$$(2x - y - 2)dx - xdy = 0$$

$$\frac{dy}{dx} = \frac{2x - y - 2}{x}$$

General Rule for finding (1st Order) Derivatives of Implicit Function

On the basis of above two illustrations we can develop a general rule for finding derivative $\left(\frac{dy}{dx}\right)$ for an implicit function of one variable which is given in the form $F(x, y) = 0$

Take the total differential of both sides of $F(x, y) = 0$

i.e. $F_x dx + F_y dy = d(0) = 0$

which gives $F_y dy = -F_x dx$

So that $\frac{dy}{dx} = -\frac{F_x}{F_y}$

In other words the first order derivative $\frac{dy}{dx}$ of the implicit function $F(x, y) = 0$ is the negative of the ratio of the two partial derivatives F_x and F_y ,

Let us again take example 29 and solve it by this general rule

$$x^2 - xy - 2x + 3 = 0 = F(x, y)$$

$$F_x = \frac{\partial F}{\partial x} = 2x - y - 2 \quad \text{and} \quad F_y = \frac{\partial F}{\partial y} = -x$$

So that
$$\frac{dy}{dx} = \frac{-F_x}{F_y} = \frac{2x - y - 2}{x}$$

which is the same result as obtained earlier

Ex 30 Find $\frac{dy}{dx}$ of the function $ax^3 + bx^2y + cxy^2 + d = 0$

$$F(x, y) = ax^3 + bx^2y + cxy^2 + d = 0$$

$$F_x = 3ax^2 + 2bxy + cy^2 \quad \text{and} \quad F_y = bx^2 + 2cxy$$

Therefore
$$\frac{dy}{dx} = \frac{-F_x}{F_y} = \frac{-(3ax^2 + 2bxy + cy^2)}{bx^2 + 2cxy}$$

$$\frac{dy}{dx} = \frac{3ax^2 + 2bxy + cy^2}{-bx^2 - 2cxy}$$

Implicit Functions of more than One Variable The above general implicit function rule can be easily extended to a case where there are (say) three variables in the given function $F(x, y, z) = 0$. By differentiation of both sides of this equation we get

$$F_x dx + F_y dy + F_z dz = 0$$

If we have n number of variables $(x_1, x_2, \dots, x_n)$ in the function F , then by differentiation of both sides, we get $F_1 dx_1 + F_2 dx_2 + \dots + F_n dx_n$

Now, if only two variables x_1 and x_2 are allowed to vary, all the differentials except dx_1 and dx_2 will become zero and the equation will reduce to

$$F_1 dx_1 + F_2 dx_2 + 0 + 0 + \dots + 0 = 0$$

that is, $F_1 dx_1 = -F_2 dx_2$

$$\left(\frac{dx_1}{dx_2} \right)_{\text{other variables constant}} = -\frac{F_2}{F_1}$$

In the same way

$$\left(\frac{dx_i}{dx_j} \right)_{\text{other variables constant}} = -\frac{F_j}{F_i}$$

EXERCISE 8.6

Find $\frac{dy}{dx}$ by making use of implicit differentiation

1. $xy^2 - y + 2 = 0$

2. $x^2 - 2x^2y + 3xy^2 = 0$

3. $x^3 + 4xy + y^3 = 0$

4. $x^2 + y^2 = a^2$ (a is constant)

5. $2x^3 - xy^2 + y^2 + 12 = 0$

6. $x^2 + y^2 + z^2 = 16$

7. $xe + ze^y - ze = 0$

8. $x^2 \log z + ze^x - y - 1 = 0$

9. A demand function is defined implicitly as $X^2 p^3 = 1$ (X is the quantity demanded and p is its per unit price). Find the elasticity of demand by differentiating the function implicitly and then by making the function explicit and differentiating it.

10. Demand function of commodity A is given in the implicit form as

$$X_a p_a^2 p_b^3 = 0$$

Find the partial elasticities of demand by differentiating the implicit function and then by making the function explicit.

8.10 Maxima and Minima of a Function of Two Variables

While dealing with functions of one variable $y = f(x)$ we derived the order conditions for deciding the maximum or minimum values of y .

We once again state these conditions in the following tabular form

Conditions	For given function $y = f(x)$ to be	
	Maximum	Minimum
First order	$f'(x) = \frac{dy}{dx} = 0$	$f'(x) = \frac{dy}{dx} = 0$
Second order	$f''(x) = \frac{d^2y}{dx^2} < 0$	$f''(x) = \frac{d^2y}{dx^2} > 0$

Our problem now is to find the extreme values (i.e., maximum or minimum) of the function if the given function involves two variables. With two independent variables, the graph of the function $Z = f(x, y)$ becomes a surface in a three-dimensional space.

On the lines of the above table for one variable we can write the order conditions for extreme values of a function involving two variables

		Alternative representation
First order condition for both maximum and minimum values of function $Z = f(x, y)$	if (i) $f_x \left(= \frac{\partial Z}{\partial x} \right) = 0$ (ii) $f_y \left(= \frac{\partial Z}{\partial y} \right) = 0$	$f_1 = 0$ $f_2 = 0$
Second order condition for maximum value of function $Z = f(x, y)$	if (i) $f_{xx} \left(= \frac{\partial^2 Z}{\partial x^2} \right) < 0$ (ii) $f_{yy} \left(= \frac{\partial^2 Z}{\partial y^2} \right) < 0$	$f_{11} < 0$ $f_{22} < 0$
Second order condition for minimum value of function $Z = f(x, y)$	(i) $f_{xx} > 0$ or $f_{11} > 0$ (ii) $f_{yy} > 0$ or $f_{22} > 0$ and (iii) $f_{xx} f_{yy} > (f_{xy})^2$ or $f_{11} f_{22} > f_{12}^2$	

This can be put in tabular form for easy understanding

Conditions	For function $Z = f(x, y)$ to be	
	Maximum	Minimum
First order	$f_x = f_y = 0$	$f_x = f_y = 0$
Second order	$f_{xx} < 0, f_{yy} < 0$ and $f_{xx} f_{yy} > f_{xy}^2$	$f_{xx} > 0, f_{yy} > 0$ and $f_{xx} f_{yy} > f_{xy}^2$

In case $f_x = f_y = 0$ and $f_{xx} f_{yy} = (f_{xy})^2$ we are not in a position to give any information about the nature of the given function

Ex 31 Find whether the following function has minimum or maximum values and at what point?

$$Z = y^3 + y^2 - xy + x^2 + 4$$

$$f_x = -y + 2x \text{ and } f_{xx} = 2$$

$$f_y = 3y^2 + 2y - x \text{ and } f_{yy} = 6y + 2$$

$$f_{xy} = -1 \text{ and, hence } (f_{xy})^2 = 1$$

$$(i) \text{ First order condition } f_x = f_y = 0$$

$$\text{i.e., } 2x - y = 0 \text{ and } 3y^2 + 2y - x = 0$$

Solving the equations

$2x - y = 0$ and $3y^2 + 2y - x = 0$ we get

$$x = -\frac{1}{4} \text{ and } y = -\frac{1}{2}$$

(ii) Second order condition

$$\text{At } x = -\frac{1}{4} \quad f_{xx} = 2 > 0 \text{ and } f_{yy} = -1 < 0$$

Thus f_{xx} and f_{yy} have opposite signs and so the second order condition fails to give any solution. In fact, this suggests (when f_{xx} and f_{yy} have opposite signs) that the surface in question shapes upward in one direction but downward in another thereby giving rise to a *saddle point* (corresponding to this we had point of inflexion in case of one variable function)

Ex 32 Examine the following function for maximum and minimum values $Z = \frac{4}{3}x^3 + y^2 - 4x + 8y$

$$\text{Here } f_x = 4x^2 - 4 \text{ and } f_{xx} = 8x$$

$$f_y = 2y + 8 \text{ and } f_{yy} = 2$$

$$f_{xy} = 0$$

(i) First order condition $f_x = f_y = 0$

$$\text{i.e. } 4x^2 - 4 = 0 \text{ and } 2y + 8 = 0$$

Solving the equations we get $x = +1$ or -1 and $y = -4$

Thus here we obtain two points $(1, -4)$ and $(-1, -4)$ where the given function can take maximum or minimum value

Let us examine it by second order condition

(ii) Second order condition

$$\text{At point } (1, -4) \quad f_{xx} = 8 \quad f_{yy} = 2 \text{ and } f_{xy} = 0$$

$$\text{So that } f_{xx} > 0 \quad f_{yy} > 0 \text{ and } f_{xx} \cdot f_{yy} = 16 > 0$$

Thus, $(1, -4)$ satisfies the condition for a *minimum* value. Hence function $Z = \frac{4}{3}x^3 + y^2 - 4x + 8y$ has a *minimum* value at point $(1, -4)$ which will be

$$Z = \frac{4}{3}(1)^3 + (-4)^2 - 4(1) + 8(-4) = -\frac{56}{3}$$

$$\text{At point } (-1, -4) \quad f_{xx} = -8 \quad f_{yy} = 2 \text{ and } f_{xy} = 0$$

$$\text{So that } f_{xx} < 0 \text{ but } f_{yy} > 0$$

So, as explained above, there is a *saddle point* at $(-1, -4)$.

Ex 33 Show that $Z = x^2 + xy + y^2$ has minimum value and $Y = x^2 + xy - y^2$ has a saddle point at $x = y = 0$

(1) Given $Z = x^2 + xy + y^2$
 $f_x = 2x + y$, $f_{xx} = 2$, $f_{xy} = 1$
 $f_y = x + 2y$, $f_{yy} = 2$

(i) First order condition $f_x = f_y = 0$

i.e., $2x + y = 0$ and, $x + 2y = 0$

Solving the equation we get $x = 0$, $y = 0$

(ii) Second order condition

At point $(0, 0)$ $f_{xx} = 2 > 0$, $f_{yy} = 2 > 0$ and $(f_{xy})^2 = 1$

Hence, at the point $(0, 0)$, the given function $Z = x^2 + xy + y^2$ has a minimum value which is zero

(2) Given $Z = x^2 + xy - y^2$
 $f_x = 2x + y$, $f_{xx} = 2$
 $f_y = x - 2y$, $f_{yy} = -2$, $f_{xy} = 1$

(i) First order condition $f_x = f_y = 0$

$2x + y = 0$ and, $x - 2y = 0$

Solving the equations, we get $x = 0$ and $y = 0$

(ii) Second order condition

At point $(0, 0)$, $f_{xx} = 2$ and $f_{yy} = -2$

So that f_{xx} and f_{yy} have opposite signs, hence there is a saddle point at $(0, 0)$

Ex. 34. Examine the following function for maximum and minimum values

$$Z = 16 - (x + 2)^2 - (y - 2)^2$$

$$f_x = -2(x + 2), \text{ and } f_{xx} = -2$$

$$f_y = -2(y - 2); \text{ and } f_{yy} = -2; f_{xy} = 0$$

(i) First order condition: $f_x = f_y = 0$

$$-2(x + 2) = 0 \text{ or } x = -2$$

and, $-2(y - 2) = 0 \text{ or } y = 2$

(ii) Second order condition:

At point $(-2, 2)$ $f_{xx} = -2 < 0$; and $f_{yy} = -2 < 0$

$$f_{xx}f_{yy} = 4 > 0 = (f_{xy})^2$$

Hence, the given function has maximum value at $(-2, 2)$ which is $= 16$.

EXERCISE 8.7

Examine the following functions for maximum, minimum or saddle points:

1. $Z = 8x^3 + 2xy - 3x^2 + y^2 + 1$ 2. $Z = 4x^2 - xy + y^2 - x^2$

3 $Z = e^{2x} - 2x + 2y^2 + 3$

4 $Z = 4x^3 - xy + y^3 - x^2$

5 $Z = 4UV - U^2 - 3V^2$

6 $Z = 3x^2 + y^2 - 3xy$

7 $Z = 2x^2 + 2y^2 + 9 - 6x - 6y - 2xy$

8 $Z = x e^x e^{2y} + 2e$

9 Show that $Z = ax^2 + by^2 + c$ has minimum, maximum or saddle point according as a and b are positive, negative or of opposite signs.

10 Show that $Z = 48 - 4x^2 - 4xy - 2y^2 + 16x + 12y$ has a maximum value and find this value.

8.11 Alternative Method for Obtaining Extreme Values of Functions Involving Two or More Variables (Hessian Determinant)

We discussed above the order conditions for occurrence of maxima or minima values for a given function with two variables using partial derivatives. In case the function contains more than two variables this method becomes complicated, especially when we check the second order condition which involves partial derivatives of higher order along with the required cross partial derivatives.

There is an alternative method for finding the extreme values of functions involving any number of variables and is much easier than the one discussed above. This method makes use of **Total differential** and **Matrix algebra**.

We shall first confine ourselves to a simple two-variable case to understand this method and then take up the case of three and more variables.

Consider a function of two variables $Z = f(x, y)$.

By method of total differential $dZ = f_x dx + f_y dy$.

The first order condition for Z to have an extreme value should be that $dZ=0$. This implies that $f_x = 0$ and $f_y = 0$.

This condition, therefore, is consistent with first order condition of the previous method that is $f_x = 0$ and $f_y = 0$.

Exactly as in the case of function of one variable $d^2Z > 0$ would imply that Z is increasing at an increasing rate and $d^2Z < 0$ would imply that Z is decreasing at a decreasing rate.

Thus we have the following order conditions in terms of total differential for a function involving any number of variables.

Condition	Maximum	Minimum
First order condition	$dZ = 0$	$dZ = 0$
Second order condition	$d^2Z < 0$	$d^2Z > 0$

The first order condition only involves partial derivatives which are to be

equated to zero

For the second order condition what we need to know is not the absolute value of d^2Z but only its sign

If d^2Z is to be greater than zero it must be a positive quantity And if d^2Z is to be less than zero it must be a negative quantity

It is on this account that we make use of matrices and determinants to decide about the sign of d^2Z without finding its absolute value

Consider $Z = f(x, y)$

$$dZ = f_x dx + f_y dy$$

$$d^2Z = f_{xx} (dx)^2 + 2f_{xy} dx dy + f_{yy} (dy)^2 \quad (i)$$

which in turn can be represented in a quadratic form say

$$q = aX^2 + 2hXY + bY^2 \quad (ii)$$

On comparing (i) and (ii) we find $dx = X$, $dy = Y$, $f_{xx} = a$ and $f_{yy} = b$. Our problem is what restrictions must be placed on f_{xx} , f_{xy} and f_{yy} so that for any values of dx and dy the value of d^2Z will turn out to be negative or positive?

$$q = aX^2 + 2hXY + bY^2$$

can be rewritten in the form

$$q = a \left(X + \frac{h}{a} Y \right)^2 + \left(\frac{ab - h^2}{a} \right) Y^2$$

Since the variable X and Y appear only in squares we can predict the sign of q entirely on the basis of coefficients a , b and h as follows

(1) q will have positive sign if $a > 0$ and $ab - h^2 > 0$

(2) q will have negative sign if $a < 0$ and $ab - h^2 > 0$

These two conditions can also be put in determinant form by making use of matrix algebra

Since, $q = aX^2 + 2hXY + bY^2$

$$q = a(X)^2 + h(XY) + h(XY) + h(Y^2)$$

which may be put in following matrix form

$$q = [X \ Y] \begin{bmatrix} a & h \\ h & b \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

Since we are not concerned with the value of X and Y , matrix of order 2×2 viz., $\begin{bmatrix} a & h \\ h & b \end{bmatrix}$ supplies us the required conditions for q to be positive or negative

(1) q will have positive sign

$$\text{if } |a| = a > 0 \text{ and } \begin{vmatrix} a & h \\ h & b \end{vmatrix} > 0$$

and, (2) q will have negative sign

$$\text{If } |a| = a < 0 \text{ and } \begin{vmatrix} a & h \\ h & b \end{vmatrix} > 0$$

But, what is the position of $|a|$ and $\begin{vmatrix} a & h \\ h & b \end{vmatrix}$ in the matrix $H = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$?

The determinant $|a| = a$ is sub-determinant of H that consists of the first element of the principal diagonal and is called first principal minor of H .

The determinant $\begin{vmatrix} a & h \\ h & b \end{vmatrix}$ can also be considered as sub-determinant of H and since it involves the first and second elements on the principal diagonal of H it is called second principal minor of H . In the case of function of two variables, only two principal minors are available but there will be three if the function involves three variables and so on.

Let us now put the coefficients of these two determinants in terms of second order total differential d^2Z .

For the quadratic form $q = aX^2 + 2hXY + bY^2$,

two determinants $|a|$ and $\begin{vmatrix} a & h \\ h & b \end{vmatrix}$ decide the sign of q .

Similarly, the quadratic form $d^2Z = f_{xx}dx^2 + 2f_{xy}dxdy + f_{yy}dy^2$ the two determinants $|f_{xx}|$ and $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}$ will decide the sign of d^2Z which will enable us to say about the maximum and minimum values of the given function.

If both determinants > 0 , d^2Z will have positive sign the given function will have minimum value, and if first determinant < 0 while other determinant > 0 , d^2Z will have negative sign or given function will have maximum value.

As a determinant $|f_{xx}|$ and $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}$ is called Hessian Determinant and (as it can be seen), Hessian is formed with the second order partial derivatives of the function.

Let us take one example say at a certain point on function $Z = f(x, y)$.

$$f_{xx} = -2.5; f_{xy} = 1.5 \text{ and } f_{yy} = -2$$

we have to examine whether d^2Z has negative sign or positive sign regardless of values of x and y .

The determinant of the quadratic form d^2Z is $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ which by substituting the respective values = $\begin{vmatrix} -2.5 & 1.5 \\ 1.5 & 2 \end{vmatrix}$

The principal minors of this determinant are

$$(1) \quad -2.5 < 0$$

$$(2) \quad \begin{vmatrix} -2.5 & 1.5 \\ 1.5 & 2 \end{vmatrix} > 0$$

Hence, d^2Z must possess negative sign and therefore the given function has a *maximum value*

Function of Three Variables

What will be the order conditions if the function involves three variables?

Assume $Z = f(x, y, w)$ then

$$dZ = f_x dx + f_y dy + f_w dw \quad \text{and}$$

$$d^2Z = f_{xx}dx^{(2)} + f_{xy}dx dy + f_{yw}dx dw + f_{yx}dy dx + f_{yy}dy^{(2)} + f_{yw}dy dw \\ + f_{wx}dw dx + f_{wy}dw dy + f_{ww}dw^{(2)}$$

But d^2Z can also be expressed in the form of matrices

$$d^2Z = [d_x \ d_y \ d_w] \begin{bmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{bmatrix} \begin{bmatrix} d_x \\ d_y \\ d_w \end{bmatrix}$$

The determinant of the matrix

$$\begin{bmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{bmatrix}$$

shall have three principal minors from this determinant (instead of two when the function involved two variables), namely

$$D_1 = |f_{xx}|, D_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}, \text{ and } D_3 = \begin{vmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{vmatrix}$$

Now, our problem is, what restrictions should be placed upon D_1 , D_2 and D_3 so that, for any value of dx and dy , the value of d^2Z will turn out to be positive or negative

For this, we once again put the quadratic form of three variables in a way that the three variables always appear as components of some square

Let $q = ax^2 + by^2 + cw^2 + 2hxy + 2fyx + 2gwx$ represent

$$d^2f = f_{xx}(dx^2) + f_{yy}(dy^2) + f_{ww}(dw^2) + 2f_{xy}dxdy + 2f_{yw}dydw + 2f_{wx}dxdw$$

By algebraic manipulation we can rewrite q in the form

$$q = a \left(x + \frac{h}{a}y + \frac{g}{a}w \right)^2 + \frac{ab - h^2}{a} \left(y + \frac{af - gh}{ab - h^2} \right)^2 + \left(\frac{abc - af^2 - bg^2 - ch^2 + 2fgh}{ab - h^2} \right) w^2$$

So that

(1) For q to be positive, the following three conditions must hold good

$$a > 0, ab - h^2 > 0, \text{ and } abc - af^2 - bg^2 - ch^2 + 2fgh > 0$$

But since $q = ax^2 + by^2 + cw^2 + 2hxy + 2fyx + 2gwx$ can be put in the form of

$$q = [x \ y \ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

the above three conditions (for q to be positive) can be put as the three principal minors of the matrix

$$\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$$

$$D_1 = a > 0, D_2 = \begin{vmatrix} a & h \\ h & b \end{vmatrix} > 0, \text{ and } D_3 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} > 0$$

Translating these elements in terms of partial derivatives, we have the following three conditions for d^2Z to be positive for the given function to have minimum value ;

$$D_1 = |f_{xx}| > 0; D_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0, \text{ and } D_3 = \begin{vmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{vmatrix} > 0$$

(2) For q to be negative, following three conditions must hold good,

$$D_1 = a < 0; D_2 = \begin{vmatrix} a & h \\ h & b \end{vmatrix} > 0, \text{ and } D_3 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} < 0$$

Or, in terms of partial derivatives, for d^2Z to be negative (or the given function to have maximum value), following conditions must hold good :

$$D_1 = |f_{xx}| < 0, D_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0, \text{ and } D_3 = \begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{xz} & f_{yz} & f_{zz} \end{vmatrix} < 0$$

i.e., the principal minors of the determinant of d^2Z should be successively negative, positive and negative

Function of 'n' Variables

When there are n independent variables the objective function may be expressed as $Z = f(x_1, x_2, x_3, \dots, x_n)$

The total differential will then be

$$dZ = f_1 dx_1 + f_2 dx_2 + f_3 dx_3 + \dots + f_n dx_n$$

so that the first order condition for minimum or maximum shall be that all the n first order partial derivatives must vanish (since $dZ = 0$) and, therefore, $f_1 = f_2 = f_3 = \dots = f_n = 0$

For second order condition

(1) d^2Z will have positive sign (i.e. function will have minimum value) if all the n principal minors of the Hessian determinant :

$$|H| = \begin{vmatrix} f_{11} & f_{12} & \dots & f_{1n} \\ f_{21} & f_{22} & \dots & f_{2n} \\ \dots & \dots & \dots & \dots \\ f_{n1} & f_{n2} & \dots & f_{nn} \end{vmatrix} \text{ are positive.}$$

(2) d^2Z will have negative sign (i.e., function will have maximum value) if n principal minors of $|H|$ possess alternate signs — the first being negative.

Let us now summarize extremum conditions in tabulated form :

Conditions	For given function : $Z = f(x_1, x_2, \dots, x_n)$ to be	
	Maximum	Minimum
First order	$dZ = 0$, which means $f_1 = f_2 = \dots = f_n = 0$	$dZ = 0$, which means $f_1 = f_2 = \dots = f_n = 0$
Second order	$d^2Z < 0$, which means $ H_1 < 0$, $ H_2 > 0$ $ H_3 < 0$	$d^2Z > 0$, which means $ H_1 > 0$, $ H_2 < 0$ $ H_3 > 0$

Ex 35 Indicate whether the following function

$$Z = f(x_1, x_2, x_3) = x_1^2 - 3x_1x_2 + 3x_2^2 + 4x_2x_3 + 6x_3^2$$

has maximum or minimum value ?

For first order condition

$$f_1 = 2x_1 - 3x_2 = 0$$

$$f_2 = -3x_1 + 6x_2 + 4x_3 = 0 \text{ and}$$

$$f_3 = 4x_2 + 12x_3 = 0$$

Solving these three equations we get

$$x_1 = 0 \quad x_2 = 0 \quad \text{and} \quad x_3 = 0$$

At point (0 0 0), what is the sign of $|H|$?

$$|H| = \begin{vmatrix} f_{x_1x_1} & f_{x_1x_2} & f_{x_1x_3} \\ f_{x_2x_1} & f_{x_2x_2} & f_{x_2x_3} \\ f_{x_3x_1} & f_{x_3x_2} & f_{x_3x_3} \end{vmatrix} = \begin{vmatrix} 2 & -3 & 0 \\ 3 & 6 & 4 \\ 0 & 4 & 12 \end{vmatrix}$$

The principal minors which are

$$|H_1| = 2 > 0 \quad |H_2| = \begin{vmatrix} 2 & -3 \\ 3 & 6 \end{vmatrix} = 3 > 0 \text{ and}$$

$$|H_3| = \begin{vmatrix} 2 & -3 & 0 \\ -3 & 6 & 4 \\ 0 & 4 & 12 \end{vmatrix} > 0$$

i.e., all principal minors are positive hence the given function has minimum value equal to zero at point (0 0 0)

Ex. 36. Determine whether the function -

$$U = f(x, y, z) = -x^3 + 3xyz + 2y - y^2 - 3z^2$$

possess a maximum or minimum value.

(i) First order condition

$$\begin{aligned} f_x &= -3x^2 + 3z = 0 & \text{or} & \begin{cases} -3x^2 & +3z = 0 \\ + & -2y & = -2 \\ 3x & & -6z = 0 \end{cases} \\ f_y &= 2 - 2y = 0 & \text{or} & \\ f_z &= 3x - 6z = 0 & \text{or} & \end{aligned}$$

Solving these equations, we get

$$x = 0 \text{ or } \frac{1}{2}, y = 1 \text{ and } z = 0 \text{ or } \frac{1}{4}$$

Thus, we have two points (0, 1, 0) and $(\frac{1}{2}, 1, \frac{1}{4})$ and we have to examine whether the given function has maximum or minimum value at these points

(ii) Second order condition

Hessian determinant of this function is

$$|H| = \begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{xy} & f_{yy} & f_{yz} \\ f_{xz} & f_{yz} & f_{zz} \end{vmatrix} = \begin{vmatrix} -6x & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{vmatrix}$$

At point $(0, 1, 1)$

$$|H| = \begin{vmatrix} 0 & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{vmatrix}$$

of which first element $f_{xx} = 0$ and therefore this does not fulfil the second order condition (since second order condition is valid only if first element $|H|$ is either greater or less than zero)

At point $\left(\frac{1}{2}, 1, \frac{1}{4}\right)$

$$|H| = \begin{vmatrix} 3 & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{vmatrix}$$

of which $|H_1| = 3 > 0$, $|H_2| = 6 > 0$ and $|H_3| = -18 < 0$ which alternate in sign

Hence at point $\left(\frac{1}{2}, 1, \frac{1}{4}\right)$ the given function

$$U = f(x, y, z) = -x^3 + 3xz + 2y - y^2 - 3z^2$$

possesses maximum value which will be $\frac{17}{16}$

EXERCISE 8.8

Examine the following functions for *maxima* and *minima*

1 $Z = f(x, y) = 3x^2 + y^3 - 3xy$

2 $Z = f(x, y) = 2x^2 + 3y^2 + 9 - 6x - 6y - 2xy$

3 $Z = f(x, y) = 8x^3 + 2xy - 8x^2 + y^2 + 1$

4 $U = f(x, y, z) = 2x^2 + xy + 4y^2 + xz + z^2 + 2$

5 $U = f(x, y, z) = 5 - (x^2 + y^2 + z^2)$

6 $U = f(x, y, z) = e^{2x} - e^y + e^{2z} - 2(x + e^y) + y$

8.12 Maxima and Minima Under Given Conditions (Constrained Extreme Values)

So far we concerned ourselves with the extreme values of functions

assuming that variables of the given function can take any values. For example for a hypothetical utility function of two variables $U = f(x, y)$ to get U maximized, we took it for granted implicitly that the consumer can purchase even an infinite amount of both the goods. But such an assumption has no relevance in reality because the consumption of two goods also depends on his purchasing power (income of the consumer). In other words, the problem of utility optimisation should also take into consideration the purchasing power of the consumer. As such what we need to find is 'How much of x and y should the consumer buy with the given purchasing power to maximise his utility?' We also know that with the given purchasing power if the consumer buys more of x he will have to buy less of y or vice versa and, therefore, the amounts of x and y are not independent of each other.

Most of economic problems concerning maxima and minima are of this nature. There is always a constraint on the variables and as such the variables x and y are not independent.

Let us take a concrete example. Consider a consumer with a simple utility function $U = f(x, y) = 4xy - y^2$. If this consumer can at most spend only Rs. 6 on two goods X and Y , and if the current prices are Rs. 2 per unit of x and Re. 1 per unit of y , then his budget constraint can be expressed by a linear equation $2x + y = 6$. Now, the problem is to maximise the function $U = 4xy - y^2$ subject to the constraint $2x + y - 6 = 0$.

Earlier our problem was merely to maximise the given function

$U = 4xy - y^2$ without any constraint placed on it. As could be seen from Fig. 8.3, such a constraint narrows the domain of the objective function to area OAB only.

Our interest now is in the location of the point representing maximisation of utility function in this area only.

Utility is maximized at the point P where the budget constraint line touches the indifference curve 2.

The solutions to such problems can easily be found without any new technique.

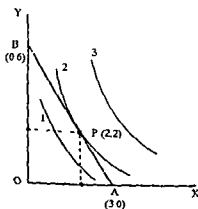


Fig. 8.3

Budget constraint is $2x + y = 6$

$$y = 6 - 2x$$

Substituting value of y in the utility function

$U = 4xy - y^2$, we obtain

$$U = 4x(6 - 2x) - (6 - 2x)^2$$

$$U = 24x - 8x^2 - (36 - 24x + 4x)$$

$$U = 36 + 48x - 12x^2$$

In this way we are able to put the utility function in terms of one variable only

Now to maximise U we apply order conditions

$$(i) \quad \text{First order condition } \frac{\partial U}{\partial x} = 0$$

$$\text{i.e. } 48 - 24x = 0 \text{ or } x = 2$$

$$(ii) \quad \text{Second order condition at } x = 2$$

$$\frac{\partial^2 U}{\partial x^2} = -24 < 0$$

Hence (U) is maximum when $x=2$. But when $x = 2$ then $y = 2$. In other words, if consumer purchases two units of x and two units of y he will be able to derive maximum utility of 12 units from the money he spends on these two commodities.

In the method described above we were able to find the solution by eliminating one variable from the given function. But when such elimination, as in the case of more than two variables, is difficult the above method fails. In such a case a new technique is used. This is known as the *Lagrange method*.

Use of Lagrange Multiplier We shall explain this method in the form of two conditions

(1) **First order condition.** We combine the given function and the constraint through a new variable in a way such that first order condition can still be applied.

For example Given utility function $U = 4xy - y^2$

and constraint $2x + y - 6 = 0$

Combining both through new variable λ , known as Lagrange's multiplier we get

$$Z = f(x, y) + \lambda (2x + y - 6)$$

$$\text{or, } Z = 4xy - y^2 + \lambda (2x + y - 6)$$

Treating λ as an additional variable, we have Z as a quadratic in variables x , y and λ

Apply first order condition which states $f_x = f_y = f_\lambda = 0$, we get

$$\left. \begin{aligned} f_x \left(= \frac{\partial Z}{\partial x} \right) &= 4y + 2\lambda = 0 \\ f_y \left(= \frac{\partial Z}{\partial y} \right) &= 4x - 2y + \lambda = 0 \\ f_\lambda \left(= \frac{\partial Z}{\partial \lambda} \right) &= 2x + y - 6 = 0 \end{aligned} \right\} \quad \begin{array}{l} \text{Solving three equations,} \\ x = 2, y = 2, \lambda = -4 \end{array}$$

The value of x and y agree with the solution already obtained by the first method and the value of U comes to 12 which is again identical to the value obtained by previous method of elimination of one variable

Let us consider one more example to understand first order condition more clearly

Ex 37 Find the point where the utility function

$$U = 48 - (x-5)^2 - 3(y-4)^2$$

will have maximum or minimum value, if it is subject to the condition $x+3y=9$

Combining the two functions through Lagrange's multiplier

$$Z = 48 - (x-5)^2 - 3(y-4)^2 + (x+3y-9)\lambda$$

First order condition gives the following results

$$\left. \begin{aligned} f'_x = \frac{\partial Z}{\partial x} &= 2(x-5) + \lambda = 0 \\ f'_y = \frac{\partial Z}{\partial y} &= -6(y-4) + 3\lambda = 0 \\ f'_\lambda = \frac{\partial Z}{\partial \lambda} &= x+3y-9 = 0 \end{aligned} \right\} \quad \begin{array}{l} \text{Solving three equations} \\ x = 3, y = 2, \lambda = -4 \end{array}$$

In other words, the given function $U = 48 - (x-5)^2 - 3(y-4)^2$ will have either maximum or minimum value (since we have considered only first order condition) at point (3, 2) and the value of the given function will be $48 - (3-5)^2 - 3(2-4)^2 = 32$.

Whether this value 32 at (3, 2) is the maximum or minimum (or neither) value) value will be tested by the second order condition

We have considered only functions of two variables, let us now generalise the first order condition to 'n' variable case.

Suppose the objective function is in the form

$$U = f(x_1, x_2, \dots, x_n) \text{ which is subject to constraint } \phi(x_1, x_2, \dots, x_n) = 0$$

Combining both the functions through Lagrangian multiplier

$$Z = f(x_1, x_2, \dots, x_n) + \lambda \phi(x_1, x_2, \dots, x_n)$$

for which first order condition will consist of following $(n + 1)$ simultaneous equations

$$f_x \left(= \frac{\partial Z}{\partial x_1} \right) = f_1 + \lambda \phi_1 = 0$$

$$f_x \left(= \frac{\partial Z}{\partial x_2} \right) = f_2 + \lambda \phi_2 = 0$$

$$f_{x_n} \left(= \frac{\partial Z}{\partial x_n} \right) = f_n + \lambda \phi_n = 0$$

$$f_\lambda \left(= \frac{\partial Z}{\partial \lambda} \right) = \phi(x_1, x_2, x_3, \dots, x_n) = 0$$

Solving these $(n+1)$ equations we get the required point where the given function can either have a maximum or minimum value

Lagrange's method can be applied even when there is more than one constraint. We introduce as many Lagrangian multipliers as there are number of constraints. However we shall only deal with problems which involve one constraint.

(2) Second order condition Had the function been free from any constraint we could have expressed second order condition in terms of Hessian determinant. Let us now see as to what happens to this Hessian when the function is constrained by some side condition.

We take the same quadratic form of function involving two variables x and y $q = ax^2 + 2hxy + by^2$

which now (suppose) is subject to constraint $\alpha x + \beta y = 0$

$$\text{This gives } y = -\left(\frac{\alpha}{\beta}\right)x$$

Substituting y in q we get

$$q = ax^2 + 2hx \left(-\frac{\alpha}{\beta}\right)x + b \left(-\frac{\alpha}{\beta}x\right)^2$$

$$\text{i.e. } q = (a\beta^2 - 2h\alpha\beta + b\alpha^2) \frac{x^2}{\beta^2}$$

The value of q will be positive or negative if $(a\beta^2 - 2h\alpha\beta + b\alpha^2)$ is positive or negative

$$\text{But since, } (-) (a\beta^2 - 2h\alpha\beta + b\alpha^2) = \begin{vmatrix} 0 & \alpha & \beta \\ \alpha & a & h \\ \beta & h & b \end{vmatrix}$$

$$\text{The value of } q \text{ will be } > 0 \text{ if and only if } \begin{vmatrix} 0 & \alpha & \beta \\ \alpha & a & h \\ \beta & h & b \end{vmatrix} < 0$$

$$\text{and value of } q \text{ will be } < 0 \text{ if and only if } \begin{vmatrix} 0 & \alpha & \beta \\ \alpha & a & h \\ \beta & h & b \end{vmatrix} > 0$$

It can be noticed that this determinant is different from the determinant $\begin{vmatrix} a & h \\ h & b \end{vmatrix}$ when the quadratic function was of the form $q = ax^2 + 2hxy + by^2$ and when it was considered without any side condition

Comparing the two determinants of two different cases, we find that determinant of constrained function contains one extra row and one extra column. Furthermore, the extra row and column is merely composed of the two coefficients of α and β from the constraint plus a zero element in the principal diagonal

Since this extra row and column is on the border of the determinant

$$\begin{vmatrix} a & h \\ h & b \end{vmatrix} \text{ we call the Hessian of the constraint function}$$

$$= \begin{vmatrix} 0 & \alpha & \beta \\ \alpha & a & h \\ \beta & h & b \end{vmatrix} \text{ as Bordered Hessian which is symbolised as } |\bar{H}|$$

Thus for $d^2Z = f_{xx}dx^2 + 2f_{xy}dxdy + f_{yy}dy^2$ subject to constraint $\varphi(x, y) = 0$ we shall have the second order condition in terms of the bordered Hessian in the following form

$$\left. \begin{array}{l} (1) \ d^2Z \text{ will be positive (or the} \\ \text{given function will have maximum} \\ \text{value)} \end{array} \right\} \text{ If } \begin{vmatrix} 0 & \varphi_x & \varphi_y \\ \varphi_x & f_{xx} & f_{xy} \\ \varphi_y & f_{xy} & f_{yy} \end{vmatrix} < 0$$

$$\left. \begin{array}{l} (2) \ d^2Z \text{ will be negative (or the} \\ \text{given function will have maximum} \\ \text{value)} \end{array} \right\} \text{ If the above determinant is } > 0$$

Bordered Hessian in n -Variable Case

If we have objective function involving n variables

$$U = f(x_1, x_2, x_3, \dots, x_n)$$

$$\text{subject to } \varphi(x_1, x_2, x_3, \dots, x_n) = 0$$

the second order condition can be put in the form of Bordered Hessian as follows

$$\text{Bordered Hessian } |H| = \begin{vmatrix} 0 & \varphi_1 & \varphi_2 & \varphi_n \\ \varphi_1 & f_{11} & f_{12} & f_{1n} \\ \varphi_2 & f_{21} & f_{22} & f_{2n} \\ \varphi_n & f_{n1} & f_{n2} & f_{nn} \end{vmatrix}$$

To decide about the sign of $|H|$ we must state the second order, condition in terms of the bordered principal minors of the Hessian

With the border of co-efficients of the constraints $(0, \varphi_1, \varphi_2, \dots, \varphi_n)$ over the Hessian determinant the principal minors become

$$|\bar{H}_1| = \begin{vmatrix} 0 & \varphi_1 \\ \varphi_1 & f_{11} \end{vmatrix}, \quad |\bar{H}_2| = \begin{vmatrix} 0 & \varphi_1 & \varphi_2 \\ \varphi_1 & f_{11} & f_{12} \\ \varphi_2 & f_{21} & f_{22} \end{vmatrix}, \text{ and}$$

$$|\bar{H}_3| = \begin{vmatrix} 0 & \varphi_1 & \varphi_2 & \varphi_3 \\ \varphi_1 & f_{11} & f_{12} & f_{13} \\ \varphi_2 & f_{21} & f_{22} & f_{23} \\ \varphi_3 & f_{31} & f_{32} & f_{33} \end{vmatrix}, \quad ,$$

$$\text{Since the first minor} = |\bar{H}_1| = \begin{vmatrix} 0 & \varphi_1 \\ \varphi_1 & f_{11} \end{vmatrix}, = -\varphi_1^2$$

which will always be negative, the second order conditions for d^2Z to be positive and negative will be

(1) d^2Z is positive [or the function $Z = f(x_1, x_2, \dots, x_n)$ will have minimum value] if $|\bar{H}_2| < 0$ $|\bar{H}_3| < 0$ $|\bar{H}_4| < 0, \dots, |\bar{H}_n| < 0$

(2) d^2Z is negative (or the given function will have maximum value) $|\bar{H}_2| > 0$ $|\bar{H}_3| < 0, |\bar{H}_4| > 0, \dots$

We can now summarize the order conditions for the constrained functions involving any number of variables in the form of the following table

Conditions	Maxima	Minima
First order	$dZ = 0$, which means $f_1 = f_2 = f_3 = \dots = f_n = 0$	$dZ = 0$, which means $f_1 = f_2 = f_3 = \dots = f_n = 0$
Second order	$d^2Z < 0$ which means $ \bar{H}_2 > 0, \bar{H}_3 < 0, \bar{H}_4 > 0, \dots$	$d^2Z < 0$ which means $ \bar{H}_2 > 0, \bar{H}_3 < 0, \bar{H}_4 > 0, \dots$

Ex. 38. Given the function $U = x^2 + y^2 + w^2$,
subject to a linear constraint $x + y + w = 1$,
find at what point U has a maximum or minimum value and also the value
of U .

Combining the function and constraint through Lagrange's multiplier we form $Z = x^2 + y^2 + w^2 + \lambda(x + y + w - 1) = 0$

(I) First order condition

$$\begin{aligned} f_x \left(\frac{\partial Z}{\partial x} \right) &= 2x + \lambda = 0 & f_y \left(\frac{\partial Z}{\partial y} \right) &= 2y + \lambda = 0 \\ f_w \left(\frac{\partial Z}{\partial w} \right) &= 2w + \lambda, & f_\lambda \left(\frac{\partial Z}{\partial \lambda} \right) &= x + y + w - 1 = 0 \end{aligned}$$

Solving these equations, we obtain

$$x = y = w = \frac{1}{3} \text{ and } \lambda = -\frac{2}{3}$$

In other words, function U can have either minimum value or maximum value at point $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. We apply second order condition to decide about this

(ii) Second order condition

$$\text{Borderd Hessian } |H| = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{vmatrix}$$

The principal minors are :

$$|H_2| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} = -4 < 0$$

$$\text{also, } |H_3| = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{vmatrix} = -12 < 0$$

That is, all the principal minors are < 0 , therefore d^2Z will have positive value

Since d^2Z possesses positive value, the given function will have minimum value at point $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. What will it be ?

$$U = x^2 + y^2 + w^2 = \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \frac{1}{3}$$

Ex 39 Determine the point which maximises or minimises the function $U = x^2 + xy + y^2 + 3z^2$ subject to $x + 2y + 4z = 60$

Incorporating Lagrange's multiplier λ we have

$$Z = x^2 + xy + y^2 + 3z^2 + \lambda(x + 2y + 4z - 60) = 0$$

(i) First order condition

$$\begin{aligned} f_y = 2x + y + \lambda &= 0 \\ f_x = x + 2y + 2\lambda &= 0 \\ f_z = 6z + 4\lambda &= 0 \\ f_0 = x + 2y + 4z - 60 &= 0 \end{aligned} \quad \left[\begin{array}{ccc} 2x + y & & + \lambda = 0 \\ x + 2y & & + 2\lambda = 0 \\ & 6z & + 4\lambda = 0 \\ x + 2y + 4z & & + 0 = 60 \end{array} \right.$$

Solving the equations we get $x = 0$, $y = \frac{90}{7}$, $z = \frac{60}{7}$ and $\lambda = -\frac{90}{7}$

Thus, the given function will have minimum or maximum value at this point

(ii) Second order condition

$$|B| = \begin{vmatrix} 0 & 1 & 2 & 4 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 0 & 0 & 6 \end{vmatrix}$$

The principal minors are

$$|B_2| = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{vmatrix} < 0$$

$$\text{also, } |B_3| = \begin{vmatrix} 0 & 1 & 2 & 4 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 0 & 0 & 6 \end{vmatrix} < 0$$

As all the principal minors are < 0 , d^2Z will have positive value. In other words, the given function will have minimum value at the point $(0, 90/7, 60/7)$. What it will be?

Since $U = x^2 + xy + y^2 + 3z^2$, substituting values, we obtain

$$(0)^2 + 0\left(\frac{90}{7}\right) + \left(\frac{90}{7}\right)^2 + 3\left(\frac{60}{7}\right)^2 = 385.71$$

We can make use of all the techniques learnt so far in the following topics of micro-economic theory

- 1 *The Theory of consumer behaviour*
- 2 *The Theory of firm.*
- 3 *The Theory of market equilibrium*

Use of these techniques in the above topics has been discussed in chapters 9, 10 and 11

EXERCISE 8.9

- 1 Find the minimum of the function $L = f(x, y) = 2x^3 + 3y^2$ under the condition that $5x + 6y = 0$
- 2 Find the maximum of the function $L = f(x, y) = 10x + 20y - x^3 - y^2$ under the condition that $2x + 5y = 10$
- 3 Find the minimum of the function $M = f(x, y, z) = x^3 + 3y^2 + 5z^2$ under the condition that $2x + 3y + 5z = 100$
- 4 For what values of x , y and z the given function $C = xyz$ will possess maximum value under the condition $2x - 3y + z = 10$
Compare it with the unconstrained maximum
- 5 For what values of q_1 and q_2 the given function $U = q_1^{1.5} q_2$ will have maximum value under the condition $3q_1 + 4q_2 = 100$

Chapter 9

Applications of Differential Calculus—I

(Theory of Consumer Behaviour)

9.1 Indifference Curve

Indifference curve may be defined as the locus of a point which represents the combinations of commodities such that the consumer derives the same level of satisfaction (utility) from any of these combinations. The indifference map is a collection of such indifference curves each of which represents a different level of satisfaction.

The indifference curve analysis is based on three basic assumptions

- (i) Non-satiation
- (ii) Transitivity and
- (iii) Diminishing Rate of Commodity Substitution (RCS)

All these characterise the particular nature of indifference curve.

Let us now examine the nature of indifference curves.

9.2 Rate of Commodity Substitution (RCS)

Consider a simple case in which consumer's purchases are limited to two commodities Q_1 and Q_2 , and that he derives a particular level of satisfaction U_1 (since he is on a particular indifference curve) with the consumption of q_1 and q_2 amounts of these two commodities respectively. As such his utility function becomes $U_1 = f(q_1, q_2)$.

What will be the change in total utility U if both q_1 and q_2 change simultaneously?

This can be obtained with the help of total differentiation of consumer's utility function.

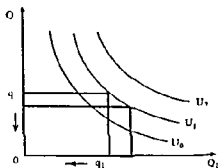


Fig. 9.1

Now since $U_1 = f(q_1, q_2)$ on a kink, tot it differentials we get

$$dU_1 = \frac{\partial U}{\partial q_1} dq_1 + \frac{\partial U}{\partial q_2} dq_2$$

$$dU_1 = f_1 dq_1 + f_2 dq_2$$

$$\left(f_1 = \frac{\partial U}{\partial q_1} \text{ and } f_2 = \frac{\partial U}{\partial q_2} \right)$$

Suppose that consumer remains on the same indifference curve even after making small changes in the amounts of q_1 and q_2 (Fig 9.1)

This would mean that change in total satisfaction $dU_1 = 0$

$$f_1 dq_1 + f_2 dq_2 = 0$$

$$\text{or } (-) \frac{f_1}{f_2} = \frac{dq_2}{dq_1} \quad (1)$$

Eq (1) gives us two important results

Since $\frac{dq_2}{dq_1}$ depicts the change in the amount of commodity Q_2 due to small change in the amount of Q_1 it represents the rate at which the consumer is willing to substitute Q_2 for Q_1 (or Q_1 for Q_2) in order to remain on the same indifference curve. In other words $\frac{dq_2}{dq_1}$ is the marginal rate of substitution (MRS) between the two commodities. This MRS is equal to the negative value of the ratio of f_1 to f_2 .

But what do f_1 and f_2 represent here?

$f_1 = \frac{\partial U}{\partial q_1}$ which represents the marginal utility of q_1 and

$f_2 = \frac{\partial U}{\partial q_2}$ which is marginal utility of q_2

Hence R.H.S. of Eq (1) represents the rate of commodity substitution (RCS or MRS as is generally known) and the L.H.S. of Eq (1) depicts the ratio between the marginal utilities of q_1 and q_2

$$\text{We have } \text{RCS} \left(= \frac{dq_2}{dq_1} \right) = (-) \frac{MU_{q_1}}{MU_{q_2}} \left(= - \frac{f_1}{f_2} \right)$$

Again $\frac{dq_2}{dq_1}$ represents the value of the slope of indifference curve. But

since $\frac{dq_2}{dq_1} = (-) \frac{f_1}{f_2}$, the indifference curve is negatively sloped

We shall have to examine second order condition in order to find the nature of the indifference curve

Second order condition is given by

$$\frac{d^2 q_2}{dq_1^2} = (-) \frac{1}{f_2^2} \left\{ f_2 \frac{d}{dq_1} (f_1) - f_1 \frac{d}{dq_1} (f_2) \right\} \quad [\text{By quotient rule}]$$

$$\text{Now } \frac{d}{dq_1} (f_1) = f_{11} + f_{21} \left(\frac{dq_2}{dq_1} \right) \quad \text{and} \quad \frac{d}{dq_1} (f_2) = f_{12} + f_{22} \left(\frac{dq_2}{dq_1} \right)$$

On substituting these values in $\frac{d^2 q_2}{dq_1^2}$ we obtain

$$\begin{aligned} \frac{d^2 q_2}{dq_1^2} &= (-) \frac{1}{f_2^2} \left\{ f_2 \left[f_{11} + f_{21} \left(-\frac{f_1}{f_2} \right) \right] - f_1 \left[f_{12} + f_{22} \left(-\frac{f_1}{f_2} \right) \right] \right\} \\ &= (-) \frac{1}{f_2^2} [f_2 f_{11} - f_{21} f_1] - \left[f_1 f_{12} - \frac{f_{22} f_1^2}{f_2} \right] \\ &= -\frac{1}{f_2^2} [f_2^2 f_{11} - f_2 f_{21} f_1 - f_2 f_{12} f_1 + f_{22} f_1^2] \\ \frac{d^2 q_2}{dq_1^2} &= -\frac{1}{f_2^2} [f_{11} f_2^2 - 2f_{21} f_1 f_2 + f_{22} f_1^2] \quad (II) \end{aligned}$$

The sign of $\frac{d^2 q_2}{dq_1^2}$, therefore, will depend on whether what is contained in bracket of (II) is positive or negative.

(i) If it is negative, the sign of $\frac{d^2 q_2}{dq_1^2}$ will turn out to be positive (there is negative sign outside the bracket) or, $\frac{d^2 q_2}{dq_1^2} > 0$. Also, therefore, the indifference curve will be concave upwards or convex downwards

(ii) If the expression in the bracket is positive, the sign of $\frac{d^2 q_2}{dq_1^2}$ will be

negative or $\frac{d^2 q_2}{dq_1^2} < 0$ and the indifference curve will be concave downwards or convex upwards

(The sign of the expression in the bracket of Eq. II has been considered in the following pages in the problems of utility maximisation under budget constraint.)

9.3. Maximisation of Utility

Supposing that a consumer is able to purchase a limited amount of two commodities Q_1 and Q_2 depending on the income (or amount) which he desires to spend on these two commodities. He also desires to choose such combinations of Q_1 and Q_2 so as to derive the highest level of satisfaction from the amount spent. If the amount which he wants to spend is given to be y rupees and the prices of Q_1 and Q_2 are p_1 and p_2 , our problem will be that of maximisation of consumer's utility function :

$$U = f(q_1, q_2)$$

subject to income constraint : $y = p_1 q_1 + p_2 q_2$

Apply Lagrange's method to obtain the amounts of q_1 and q_2 which would give maximum satisfaction to consumer under income constraint.

$$\therefore Z = U(q_1, q_2) + \lambda (y - p_1 q_1 - p_2 q_2)$$

$$[\text{or, } Z = U(q_1, q_2) - \lambda (p_1 q_1 + p_2 q_2 - y)]$$

First order condition :

$$\frac{\partial Z}{\partial q_1} = \frac{\partial U}{\partial q_1} - \lambda p_1 = 0$$

$$\frac{\partial Z}{\partial q_2} = \frac{\partial U}{\partial q_2} - \lambda p_2 = 0$$

$$\frac{\partial Z}{\partial \lambda} = y - p_1 q_1 - p_2 q_2 = 0$$

Letting $\frac{\partial U}{\partial q_1} = f_1$ and, $\frac{\partial U}{\partial q_2} = f_2$, we write

$$\frac{\partial Z}{\partial q_1} = f_1 - \lambda p_1 = 0 \quad \dots(1)$$

$$\frac{\partial Z}{\partial q_2} = f_2 - \lambda p_2 = 0 \quad \dots(2)$$

$$\frac{\partial Z}{\partial \lambda} = y - p_1 q_1 - p_2 q_2 = 0 \quad \dots(3)$$

Solution of the above three equations would give us a point at which the utility would either be maximum or minimum. Second order condition would enable us to say whether the function $U = f(q_1, q_2)$ would be maximum at that point. Before examining the second order condition we would like to say something more about the first order condition.

Interpretation of Order Conditions

(A) *First Order Condition* Solving (1) and (2) for λ , we find that

$$\frac{f_1}{p_1} = \frac{f_2}{p_2} = \lambda \quad (4)$$

But since $f_1 = MU_{q_1}$ and $f_2 = MU_{q_2}$ we write (4) as

$$\frac{MU_{q_1}}{\text{price of } Q_1} = \frac{MU_{q_2}}{\text{price of } Q_2} = \lambda \quad (III)$$

Equation (III) gives the equality between two ratios and states a familiar proposition of consumer's theory that in order to maximise utility the consumer must allocate his budget so as to equalise the ratio of marginal utility to the price of every commodity. These ratios must also be equal to the marginal utility of money which is represented by λ in equation (III).

We can also write

$$\frac{f_1}{p_1} = \frac{f_2}{p_2} = \lambda \text{ in the form } -\frac{f_1}{f_2} = -\frac{p_1}{p_2} \quad (IV)$$

What does $-\frac{f_1}{f_2}$ indicate?

It represents the slope of indifference curve (see Eq. I). What does

$(-\frac{p_1}{p_2})$ indicate? It represents the slope of budget line (or the constraint)

$$\left[y = p_1 q_1 + p_2 q_2 = \frac{y}{p_2} - \left(\frac{p_1}{p_2} \right) q_1 \therefore q_2 = \left(-\frac{p_1}{p_2} \right) q_1 + \frac{y}{p_2} \right]$$

Therefore, equation (III) in terms of (IV) also states another proposition of consumer's theory, consumer is in equilibrium at a point where the slope of price line is equal to the slope of indifference curve.

Since the budget line is a straight line with negative slope and the utility function is a curve convex from below (we will prove this later), the tangency point between the two can be the only point where the slopes of both would be equal. This point is M as shown in Fig. 9.2

Let us now examine the second order condition and see whether point M is the only point where the given utility function $U = f(q_1, q_2)$ takes the maximum value, consistent with the given constraint

(B) Second Order Condition

Point M can give maximum value of the function $f(q_1, q_2)$ if and only if bordered Hessian $\cdot |H| > 0$

Apply the second order condition to equation (1), (2) and (3), we make use of Bordered Hessian technique learnt in the previous chapter

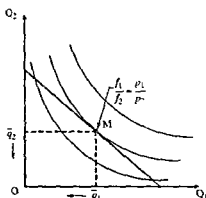


Fig. 9.2

$$|H| = \begin{vmatrix} 0 & -p_1 & -p_2 \\ -p_1 & f_{11} & f_{12} \\ -p_2 & f_{21} & f_{22} \end{vmatrix} > 0.$$

$$\text{i.e., } |H| = (2p_1p_2f_{12} - p_2^2 \cdot f_{11} - p_1^2 \cdot f_{22}) > 0$$

$$= (-) (f_{11}p_2^2 - 2f_{12}p_1p_2 + f_{22}p_1^2) > 0 \quad (V)$$

$$= (-) (f_{11} \cdot p_2^2 - 2f_{12} \cdot p_1 \cdot p_2 + f_{22} \cdot p_1^2) > 0 \quad (VI)$$

Inequality (VI) gets satisfied (i.e., second order condition will get satisfied) if and only if the indifference curve is convex downwards

that is, if $\frac{d^2q_2}{dq_1^2} > 0$.

$$\text{But from (II), } \frac{d^2q_2}{dq_1^2} = -\frac{1}{f_2^3} (f_{11}f_2^2 - 2f_{21}f_1f_2 + f_{22}f_1^2)$$

This should be positive quantity

$$\therefore (-) \frac{1}{f_2^3} (f_{11} \cdot f_2^2 - 2f_{21} \cdot f_1 \cdot f_2 + f_{22}f_1^2) > 0$$

On substituting $f_1 = \frac{p_1}{p_2} \cdot f_2$ from (IV), we get

$$(-) \frac{1}{f_2^3} \left\{ f_{11} \cdot f_2^2 - 2f_{21} \left(\frac{p_1}{p_2} \cdot f_2 \right) f_2 + f_{22} \left(\frac{p_1}{p_2} \cdot f_2 \right)^2 \right\} > 0$$

$$\text{or, } (-) \frac{f_2^2}{p_2^2 f_2^2} \left| f_{11} p_2^2 - 2f_{12} p_1 p_2 + f_{22} p_1^2 \right| > 0$$

$$\text{i.e. } (-) \frac{1}{f_2 p_2^2} \left| f_{11} p_2^2 - 2f_{12} p_1 p_2 + f_{22} p_1^2 \right| > 0$$

$$f_{11} p_2^2 - 2f_{12} p_1 p_2 + f_{22} p_1^2 < 0$$

$$(-) (f_{11} p_2^2 - 2f_{12} p_1 p_2 + f_{22} p_1^2) > 0$$

$$|H| > 0 \text{ (see equation V)}$$

In other words, second order condition gets satisfied only if the indifference curve is convex from below at point M ($\bar{q}_1, \bar{q}_2$). It is this point where the given function takes the maximum value.

Let us now take a few examples

Ex. 1 Assume that the utility function of a consumer is given by

$$U = (q_1, q_2) = q_1 q_2$$

Let us suppose $p_{Q1} = 1$ and $p_{Q2} = 2$. The consumer in question wants to spend amount of K units only. What will be his demand for Q_1 and Q_2 ?

The budget constraint is

$$K = q_1 + 2q_2$$

Lagrangian function $Z = q_1 q_2 + \lambda (q_1 + 2q_2 - K)$,

where λ is an undetermined constant (Lagrange multiplier). Let the partial derivatives equal to zero

$$\frac{\partial Z}{\partial q_1} = q_2 + \lambda = 0 \quad \text{---(i)}$$

$$\frac{\partial Z}{\partial q_2} = q_1 + 2\lambda = 0 \quad \text{---(ii)}$$

$$\frac{\partial Z}{\partial \lambda} = q_1 + 2q_2 - K = 0 \quad \text{---(iii)}$$

The solution of the above system of equations is

$$q_1 = \frac{K}{2} \text{ and } q_2 = \frac{K}{4}$$

Ex. 2. Suppose that a consumer has Rs. 90 to be divided between two commodities X and Y , and suppose the unit price of Y is fixed at Re. 0.20. What will be his demand equation for X if his utility function is .

$U = \log Q_x + 2 \log Q_y$ (where Q_x and Q_y are the amounts of X and Y consumed by him) ?

We have derived the first order condition for constrained utility maximisation which is

$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

We can derive MU_x and MU_y from utility function while p_y is given to be Re 0.20

$$U = \log Q_x + 2 \log Q_y$$

$$MU_x = \frac{\partial U}{\partial Q_x} = \frac{1}{Q_x} \quad \text{and} \quad MU_y = \frac{\partial U}{\partial Q_y} = \frac{2}{Q_y}$$

Substituting all the values in the above condition we get

$$\frac{1/Q_x}{2/Q_y} = \frac{P_x}{0.20} \quad \text{or} \quad \frac{Q_y}{2Q_x} = \frac{P_x}{0.20}$$

$$2P_x Q_x = 0.20 Q_y \quad (i)$$

Substitute the value of Q_y from the given budget constraint

$$P_x Q_x + P_y Q_y = 90$$

$$0.20 Q_y = 90 - Q_x P_x$$

Eq (i) therefore, becomes

$$2P_x Q_x = 90 - Q_x P_x$$

$$Q_x = \frac{30}{P_x} \quad \text{is the required demand function}$$

9.4 Slutsky Equation Income and Substitution Effects

We learn in economic theory that fall in the price of a commodity induces the consumer to increase its demand via two effects—substitution and income effects

Assuming that only price p_1 of commodity Q_1 changes these effects are represented in Fig 9.3

Movement from P_1 to S_1 along curve 1 is due to substitution effect while from S_1 to R_1 is due to income effect

Let the demand function of Q_1 be $q_1 = f(p_1, p_2, y)$ so that on the basis of the definition of these two effects

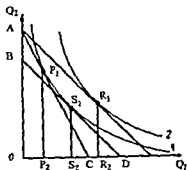


Fig 9.3

$\left(\frac{\partial q_1}{\partial p_1}\right)_{\text{constant}}$ will represent the rate of change of consumer's purchase of Q_1 with respect to changes of p_1 such that consumer remains neither better off nor worse-off. In other words $\left(\frac{\partial q}{\partial p}\right)_{\text{constant}}$ would correspond to the *substitution effect*.

We should remember here that $\left(\frac{\partial q}{\partial p}\right)_{\text{constant}} = P.S. \text{ fig. } 9.3 \frac{\partial q}{\partial p}$ is the rate of change and not the absolute change. $\left(\frac{\partial q}{\partial p}\right)_{\text{constant}}$ will represent the rate of change of consumer's purchases of Q_1 with respect to income y while prices p_1 and p_2 do not change. Hence this would correspond to *income effect*.

$$\text{Here also } \left(\frac{\partial q_1}{\partial p_1}\right)_{\text{constant}} = S.R.$$

While maximising the given utility function $U = f(q_1, q_2)$ under the budget constraint the first order condition give us the following three equations:-

$$f_1 - \lambda p_1 = 0 \quad \text{(I)}$$

$$f_2 - \lambda p_2 = 0 \quad \text{(II)}$$

$$y - p_1 q_1 - p_2 q_2 = 0 \quad \text{(III)}$$

These equations must always be satisfied whatever utility function and prices are.

Our present object is to obtain the rule of change in the consumer's purchases with respect to change in p_1 , p_2 and his income y . This can be found by finding total differentials of the above equations as follows:

$$(f_{11}dq_1 + f_{12}dq_2) - (p_1d\lambda + \lambda dp_1) = 0$$

$$(f_{21}dq_1 + f_{22}dq_2) - (p_2d\lambda + \lambda dp_2) = 0$$

$$(dy) - (p_1dq_1 + p_2dq_2) - (p_1dq_1 + p_2dq_2) = 0$$

These can be rearranged as

$$f_{11}dq_1 + f_{12}dq_2 - p_1d\lambda = \lambda dp_1 \quad \text{(IV)}$$

$$f_{21}dq_1 + f_{22}dq_2 - p_2d\lambda = \lambda dp_2 \quad \text{(V)}$$

$$-p_1dq_1 - p_2dq_2 = -dy + p_1dq_1 + p_2dq_2 \quad \text{(VI)}$$

In order to solve equations (IV), (V) and (VI) for dq_1 , dq_2 and $d\lambda$, we make use of matrix algebra

We put these equations in matrix notation

$$\begin{bmatrix} f_{11} & f_{12} & -p_1 \\ f_{21} & f_{22} & -p_2 \\ -p_1 & -p_2 & 0 \end{bmatrix} \begin{bmatrix} dq_1 \\ dq_2 \\ d\lambda \end{bmatrix} = \begin{bmatrix} dp_1 \\ dp_2 \\ -dy + q_1 dp_1 + q_2 dp_2 \end{bmatrix}$$

$$\begin{bmatrix} dq_1 \\ dq_2 \\ d\lambda \end{bmatrix} = \begin{bmatrix} f_{11} & f_{12} & -p_1 \\ f_{21} & f_{22} & -p_2 \\ -p_1 & -p_2 & 0 \end{bmatrix}^{-1} \begin{bmatrix} dp_1 \\ dp_2 \\ -dy + q_1 dp_1 + q_2 dp_2 \end{bmatrix}$$

Representing $\begin{bmatrix} f_{11} & f_{12} & -p_1 \\ f_{21} & f_{22} & -p_2 \\ -p_1 & -p_2 & 0 \end{bmatrix}$ by $[D]$ we can write

$$[D]^{-1} = \frac{1}{|D|} \begin{bmatrix} D_{11} & D_{21} & D_{31} \\ D_{12} & D_{22} & D_{32} \\ D_{13} & D_{23} & D_{33} \end{bmatrix}$$

where, $D_{11} = \begin{vmatrix} f_{22} & -p_2 \\ -p_2 & 0 \end{vmatrix}$, $D_{12} = -\begin{vmatrix} f_{21} & -p_2 \\ -p_1 & 0 \end{vmatrix}$ and so on

$$dq_1 = \frac{D_{11}\lambda dp_1 + D_{21}\lambda dp_2 + D_{31}(-dy + q_1 dp_1 + q_2 dp_2)}{D} \quad (\text{VII})$$

$$dq_2 = \frac{D_{12}\lambda dp_1 + D_{22}\lambda dp_2 + D_{32}(-dy + q_1 dp_1 + q_2 dp_2)}{D} \quad (\text{VIII})$$

$$d\lambda = \frac{D_{13}\lambda dp_1 + D_{23}\lambda dp_2 + D_{33}(-dy + q_1 dp_1 + q_2 dp_2)}{D} \quad (\text{IX})$$

Dividing (VII) by dp_1 to get $\left(\frac{dq_1}{dp_1}\right)$,

$$\frac{dq_1}{dp_1} = \frac{D_{11}\lambda + D_{21}\lambda \frac{dp_2}{dp_1} + \frac{D_{31}}{dp_1}(-dy + q_1 dp_1 + q_2 dp_2)}{D}$$

But, since we are to study only the rate of change in Q_1 with respect to p_1 keeping y and p_2 constant (i.e. $dy = 0$ and $dp_2 = 0$) we shall get

$$\frac{\partial q_1}{\partial p_1} = \frac{D_{11}}{D} + \frac{D_{31}}{D} q_1 \quad (\lambda)$$

Income Effect

Income effect corresponds to $\left(\frac{\partial q_1}{\partial y}\right)$ which we can now obtain by dividing (VII) by (dy)

$$\frac{dq_1}{dy} = \frac{1}{dy} \left[\frac{D_{11}\lambda dp_1 + D_{21}\lambda dp_2 + D_{31}(-dy + q_1 dp_1 + q_2 dp_2)}{D} \right]$$

Prices p_1 and p_2 are to remain unchanged in income effect

$$dp_1 = dp_2 = 0$$

$$\left(\frac{dq_1}{dy}\right) \text{ changes to } \left(\frac{\partial q_1}{\partial y}\right)_{\text{prices constant}}$$

$$\left(\frac{\partial q_1}{\partial y}\right)_{p=\text{constant}} = (-) \frac{D_{31}}{D} \quad \dots \text{(XI)}$$

Substitution Effect

This effect corresponds to changes in commodity purchases such that consumer remains neither better off nor worse off. In other words, the consumer stays on the same indifference curve

Therefore, from

$$U = f(q_1, q_2), \text{ we get}$$

$$dU = p_1 dq_1 + p_2 dq_2 = 0 \quad [\because f_1 = \lambda p_1 \text{ and } f_2 = \lambda p_2]$$

$$\text{Equation (VI), } -p_1 dq_1 - p_2 dq_2 = -dy + q_1 dp_1 + q_2 dp_2$$

$$-(p_1 dq_1 + p_2 dq_2) = -dy + q_1 dp_1 + q_2 dp_2$$

$$0 = -dy + q_1 dp_1 + q_2 dp_2$$

Substituting this in equation VII, we get

$$dq_1 = \frac{D_{11}\lambda dp_1 + D_{21}\lambda dp_2 + D_{31}(0)}{D} = \frac{D_{11}\lambda dp_1 + D_{21}\lambda dp_2}{D}$$

Dividing by dp_1

$$\frac{dq_1}{dp_1} = \frac{1}{dp_1} \left[\frac{D_{11}\lambda dp_1 + D_{21}\lambda dp_2}{D} \right] = \frac{D_{11}}{D} \lambda + \frac{D_{21}}{D} \cdot \frac{dp_2}{dp_1}$$

But since $dp_2 = 0$

$$\therefore \left(\frac{\partial q_1}{\partial p_1}\right)_{U=\text{constant}} = \frac{D_{11}}{D} \lambda \quad \dots \text{(XII)}$$

Substituting (XI) and (XII) in (X), we get

$$\left(\frac{\partial q_1}{\partial p_1}\right) = \left(\frac{\partial q_1}{\partial p_1}\right)_{U=\text{constant}} - q_1 \left(\frac{\partial q_1}{\partial y}\right)_{p=\text{constant}} \quad (\text{XIII})$$

Equation (XIII) is the Slutsky Equation. This equation gives us the rate of change in the purchases of commodity Q_1 due to changes in its price (p_1) in terms of substitution and income effects.

9.5 Important Results from Slutsky Equation

We derive following three important results from the above Slutsky equation :

(i) *The sign of substitution effect is always negative*

$$\begin{aligned} \text{Substitution effect} &= \left(\frac{\partial q_1}{\partial p_1}\right)_{U=\text{constant}} = \frac{\lambda D_{11}}{D} \\ &= \frac{-\lambda p_2^2}{2f_{12} \cdot p_1 \cdot p_2 - p_2^2 \cdot f_{11} - p_1^2 \cdot f_{22}} \end{aligned}$$

We have proved that the expression in the denominator is a positive quantity and (λ) is also a positive quantity, therefore, $\left(\frac{\partial q_1}{\partial p_1}\right)_{U=\text{constant}} < 0$

In other words, if the price of commodity Q_1 rises (falls) and if the consumer is to remain neither better-off nor worse-off, his purchases of Q_1 will decrease (rise).

(ii) *The sign of income effect may or may not be negative*

The sign of income effect depends on $\frac{\partial q_1}{\partial y}$. If $\frac{\partial q_1}{\partial y}$ is negative (i.e., with

the rise in income, purchases of Q_1 decrease), income effect $\left(-q_1 \cdot \frac{\partial q_1}{\partial y}\right)$ will have positive sign. If income effect is positive and is also large enough to make $\frac{\partial q_1}{\partial p_1}$ positive, Q_1 is said to be an *inferior (Giffen) good*. This would mean that as the price of commodity Q_1 falls (rises) its purchase also falls (rises).

(iii) *The extent of income effect depends on how important the particular commodity (Q_1) is in the consumption.*

The positive or negative effect of income will be significantly depending upon the quantity of Q_1 purchased by the consumer (observe Slutsky equation).

The generalized form of the Slutsky equation may be given as :

$$\frac{\partial q_j}{\partial p_i} = \left(\frac{\partial q_j}{\partial p_i} \right)_{U-\text{constant}} - q_j \left(\frac{\partial q_i}{\partial y} \right)_{p-\text{constant}}$$

Commodity Q_j is defined as

(a) Normal good if $\frac{\partial q_j}{\partial p_i} < 0$

(b) Giffen good if $\frac{\partial q_j}{\partial p_i} > 0$

(c) Inferior good if $\frac{\partial q_j}{\partial y} < 0$

To describe the changes in the demand for one commodity resulting from changes in the price of the other commodity, the generalized form of the Slutsky equation may be given as

$$\frac{\partial q_j}{\partial p_i} = \left(\frac{\partial q_j}{\partial p_i} \right)_{I-\text{constant}} - q_i \left(\frac{\partial q_j}{\partial y} \right)_{p_1 \text{ and } p_2 \text{ constant}}$$

Two commodities Q_i and Q_j are ,

(a) Substitutes if $\frac{\partial q_j}{\partial p_i} < 0$

(b) Complementary if $\frac{\partial q_j}{\partial p_i} < 0$

9.6. Elasticity form of the Slutsky Equation

$$\frac{\partial q_1}{\partial p_1} = \left(\frac{\partial q_1}{\partial p_1} \right)_{U-\text{constant}} - q_1 \left(\frac{\partial q_1}{\partial y} \right)_{p_1 \text{ and } p_2 \text{ constant}}$$

Multiply throughout by $\left(\frac{p_1}{q_1} \right)$

$$\frac{p_1}{q_1} \cdot \frac{\partial q_1}{\partial p_1} = \frac{p_1}{q_1} \left(\frac{\partial q_1}{\partial p_1} \right)_U - \frac{p_1}{q_1} \left(\frac{\partial q_1}{\partial p_1} \right)_{p_1 \text{ and } p_2}$$

But, $\left(\frac{\partial_1}{q_1} \cdot \frac{\partial q_1}{\partial p_1} \right)$ is the price elasticity of demand, so let it be represented by

$$E_{11} = E_{11}^* - \frac{p_1}{q_1} q_1 \left(\frac{\partial q_1}{\partial p_1} \right)_{p_1 \text{ and } p_2} \quad \text{which may be rewritten as}$$

$$E_{11} = E_{11}^* - \frac{p_1 q_1}{y} \frac{y}{q_1} \left(\frac{\partial q_1}{\partial y} \right)_{p_1 \text{ and } p_2}$$

But $\frac{y}{q_1} \left(\frac{\partial q_1}{\partial y} \right)_{p_1 \text{ and } p_2}$ is income elasticity, let it be denoted by E_y

$$\therefore \text{Therefore, } E_{11} = E_{11}^* - \left(\frac{p_1 q_1}{y} \right) E_y$$

$\left(\frac{p_1 q_1}{y} \right)$ is the proportion of income (y) spent on commodity Q_1 . Let it be denoted by a_1 , then, $E_{11} = E_{11}^* - a_1 E_y$ (XIV)

This is the elasticity form of Slutsky Equation

This equation states that the price elasticity of the ordinary demand curve (E_{11}) equals the price elasticity of the compensated demand curve (E_{11}^*) less the corresponding income elasticity (E_y) multiplied by the proportion of total expenditure spent on Q_1 . Naturally, therefore, the ordinary demand curve will have a greater demand elasticity than the compensated demand curve, i.e., E_{11} will be more negative than E_{11}^* so long as income elasticity is positive

EXERCISE 9.1

1. Give a mathematical elucidation of income and substitution effects of a single price change on demand
2. Derive Slutsky's condition from the equations of equilibrium attained by a consumer with a given budget equation and a preference field. Also prove that Slutsky's expression for compensated change:

$$\frac{\partial x_i}{\partial p_i} + x_i \frac{\partial x_i}{\partial I};$$

where x , p and I denote quantity, price and income, is invariant under any monotonic transformation of the utility index

3. A consumer's ordinal utility function is given by $U = u(q_1, q_2, \dots, q_n)$ where $q_1, q_2, \dots, q_n$ are quantities of n commodities purchased at prices $p_1, p_2, \dots, p_n$ respectively. He seeks to maximise it subject to his money income M

Derive the necessary and sufficient conditions for consumer equilibrium. State Slutsky conditions clearly and discuss the implications for Giffen good

4 Derive the Slutsky demand equation of a consumer with a two commodity fixed income budget. Write it in the elasticity form and comment on its significance.

5 Write the Slutsky equation for a 2-commodity budget (without proof). Convert it into elasticity form. Prove that in a n commodity case the negative of the income elasticity of demand equals the sum of the own and cross price elasticities of demand.

6 Provide mathematical demonstration of the following economic propositions:

- (i) when price falls, consumer expenditure will rise if demand is elastic.
- (ii) Giffen's paradox occurs when negative income effect outweighs the substitution effect.

7 Find the optimum commodity purchases for a consumer whose utility function is $U = f(q_1, q_2) = e^{q_1} q_2$, prices of q_1 and q_2 are 1 and 5 respectively and income = 10 units.

8 Find the demand for the commodities q_1 and q_2 for the consumer given the following data:

$$U = f(q_1, q_2) = q_1^{\frac{1}{2}} q_2^{\frac{1}{3}}$$

$$p_{q_1} = 3, p_{q_2} = 4 \text{ and } I = 100$$

9 Find the optimum commodity purchases for a consumer whose utility function and budget constraints are:

$$U = q_1^2 q_2^3 \text{ and } q_1 + 4q_2 = 10 \text{ respectively}$$

10 Suppose that a consumer has Rs. 50 to be divided between commodities X and Y and suppose that the unit price of Y is fixed at Re. 0.30. What will be his demand equation for X if his utility function is $(U = \log q_x + 3 \log q_y)$ where q_x is the amount consumed of X and q_y is the amount consumed of Y .)

11 Give two alternative proofs of the proposition that the substitution effect of a change in the price of a consumer good is always negative. Bring out the nature of the assumption on which your proofs are based.

12 "Tangency between the price line and an indifference curve is the expression in terms of indifference curves of the proportionality between marginal utilities and prices."

Is this a correct representation of the logic and utility of indifference curve analysis? Give full reasons.

13 Explain the condition of maximisation of the (ordinal) utility of a consumer subject to a budget constraint.

14 Develop mathematically the argument in the theory of consumer's behaviour which leads to the conclusion that the sum of elasticities of a good

with respect to each and every price is equal in absolute value but opposite sign to the income elasticity of demand for that good

15 Show that in a 2-commodity budget complementarity is impossible and also both the goods cannot be inferior goods

16 If $U = x^2y$ is a utility function find the equilibrium bundle if $x + 2y = 4$

Chapter 10

Applications of Differential Calculus-II

(Theory of the Firm)

10.1 Production Function

With given technological conditions of production, the amount q of the commodity 'Q' produced depends on the amount of the factors of production, say $X_1, X_2, X_3, \dots, X_n$. If q be the amount produced and for this $x_1, x_2, x_3, \dots, x_n$ be the amounts of factors employed, the production function is written as

$q = f(x_1, x_2, \dots, x_n)$ with the assumption that the factors and production are continuous variables

For the sake of simplicity, we shall assume the production to be dependent on the amount of two variable factors. The production function will then be described as $q = f(x_1, x_2)$

10.2. Constant Product Curves : Isoquants

Production function $q = f(x_1, x_2)$ states that q is the amount of output of Q which the firm can produce if it uses exactly x_1 units of X_1 and x_2 units of X_2 with particular technological knowledge or process. However, there can be

many other alternative ways in which the combination of X_1 and X_2 can be used to produce a given output q^0 with different technological processes available for its manufacture

Thus, the output q^0 will be possible with different input combinations, so that production $q^0 = f(x_1, x_2)$ can be represented by a curve called the 'constant product' curve or "Isoquant" as in Fig. 10.1

With increased input combinations the output level will also increase. Each

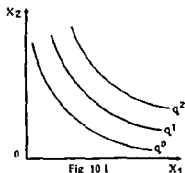


Fig 10.1

such increased output level will be given by a curve farther from the origin. Thus farther the isoquant lies from the origin, the greater the output level. Accordingly in Fig. 10.1

$$q^0 < q^1 < q^2$$

10.3 Shape of Isoquant

The equation of an isoquant is $q^0 = f(x_1, x_2)$ (1)

To find the rate of change in total production q^0 with respect to change in inputs used, we find total differential

$$dq^0 = \frac{\partial q^0}{\partial x_1} dx_1 + \frac{\partial q^0}{\partial x_2} dx_2$$

On being the same isoquant, output remains unaltered, therefore

$$\frac{\partial q^0}{\partial x_1} dx_1 + \frac{\partial q^0}{\partial x_2} dx_2 = 0$$

$$0 = f_1 dx_1 + f_2 dx_2 \quad \left(\text{on letting } \frac{\partial q^0}{\partial x_1} = f_1, \text{ and } \frac{\partial q^0}{\partial x_2} = f_2 \right)$$

$$(-) \frac{f_1}{f_2} = \frac{dx_2}{dx_1} \quad (2)$$

We derive two important conclusions from (2)

(i) $\frac{dx_2}{dx_1}$ is the rate of change in input X_2 due to a small change in input

X_1 or in other words, $\frac{dx_2}{dx_1}$ represents the rate at which the entrepreneur can substitute input X_2 for X_1 (or X_1 for X_2) in order to maintain specific output level q^0 . Therefore, the ratio $\frac{dx_2}{dx_1}$ is the *Rate of Technological Substitution (RTS)* which is equal to negative ratio between marginal productivities of two inputs

$$RTS = \frac{dx_2}{dx_1} = (-) \frac{f_1}{f_2} = (-) \frac{\partial q^0 / \partial x_1}{\partial q^0 / \partial x_2} = (-) \frac{MP_{x_1}}{MP_{x_2}} \quad (3)$$

(ii) $\frac{dx_2}{dx_1}$ also represents the slope of the isoquant. This has a negative

sign. Thus isoquant slopes downwards from left to right. It will also be shown to be convex downwards.

10.4. Isoquants and Ridge Lines

Isoquant map represents a family of curves and the negative slope of any

isoquant $\frac{dx_2}{dx_1}$ is restricted within a certain range. In other words, it is only within a certain range that the total product can be maintained when less of one factor is used by increased use of other factor.

Why is it so?

Let the factors used for production of commodity Q be A and B , so that in the present case q^0, q^1, q^2 are the different levels of output of Q with different input combinations of factors A and B as seen in Fig. 10.2

For isoquant q^1 it is only in the range from L to P within which the total output q^1 can be maintained when either of inputs can be decreased (increased) with the increased (decreased) use of other (since slope of q^1 is negative within this range). Beyond L and P points, such type of substitution between the inputs is not possible. On the contrary, beyond these points larger amounts of both inputs are needed to maintain the level of output q^0 .

This characteristic of isoquant may be translated in terms of marginal productivities of the inputs A and B .

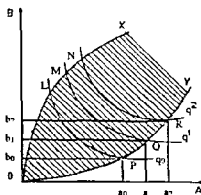


Fig. 10.2

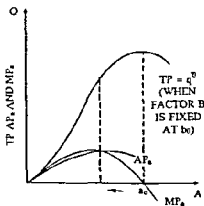


Fig. 10.3

At point P , the marginal product of input A is zero and the output level q^0 is maximum at this point corresponding to particular input combination (a_0, b_0) . This has been shown separately in Fig. 10.3.

Beyond point P , the marginal product of factor A becomes negative and hence more amounts of both inputs are needed to maintain the output level q^0 .

Similarly, beyond points Q and R , the marginal products of factor A become negative corresponding to output level q^1 and q^2 (Fig. 10.2).

On the other hand, beyond points L, M and N the marginal products of factor B become negative (after the application of this factor beyond b_0, b_1 and

b_2 respectively) corresponding to output q_0 , q_1 and q_2

Naturally, therefore, the area of rational operation is XOY, enclosed by lines OX and OY which are called 'Ridge lines'

The ridge points will represent the peak points of the total product curves corresponding to the different fixed levels of input B with varied amounts of input A (in the case of points L, M and N). In this way at points P, Q and R the marginal product of factor A is zero (but average product of factor B is maximum—refer to Euler's theorem) with the combination of b_0 , b_1 and b_2 amounts of factor B. Fig. 10.3 depicts only the case of point P.

Exactly in the same way, it can be shown with horizontal axis representing the factor B, that at points L, M and N the marginal product of factor B is zero (but average product of factor A is maximum—refer to Euler's theorem) with the combination of a_0 , a_1 and a_2 amounts of factor A.

10.5. Least Cost Combination (Constrained Cost Minimisation)

We now apply the technique of *constrained maxima-minima* to the problem of an entrepreneur who desires to minimise the total cost for a prescribed output level, say $q^0 = f(x_1, x_2)$ given that the prices of input x_1 and x_2 in the market are r_1 and r_2 .

The problem can now be put as follows :

Minimise total cost : $C = x_1 r_1 + x_2 r_2 + b$ (where b is fixed cost)

Subject to (output) constraint : $q^0 = f(x_1, x_2)$

Combining the two functions and using Lagrange's multiplier

$$Z = x_1 r_1 + x_2 r_2 + \lambda [q^0 - f(x_1, x_2)]$$

For Z to be minimum, we apply order conditions :

First order condition gives :

$$\frac{\partial Z}{\partial x_1} = r_1 - \lambda f_1 = 0 \quad \dots (1)$$

$$\frac{\partial Z}{\partial x_2} = r_2 - \lambda f_2 = 0 \quad \dots (2)$$

$$\frac{\partial Z}{\partial \lambda} = q^0 - f(x_1, x_2) = 0 \quad \dots (3)$$

where $f_1 = \frac{\partial q^0}{\partial x_1} = MP_{x_1}$ and $f_2 = \frac{\partial q^0}{\partial x_2} = MP_{x_2}$

From the first two equations, we have :

$$\frac{r_1}{f_1} = \frac{r_2}{f_2} = \lambda \text{ or, } \frac{f_1}{f_2} = \frac{r_1}{r_2} \quad \dots (4)$$

Since f_1 , f_2 , r_1 and r_2 represent the marginal productivities and prices of x_1 and x_2 respectively the above equality (4) means that in order to minimise (or maximise) the total cost the ratio of the marginal productivities of two inputs must be equated with the ratio of their respective prices

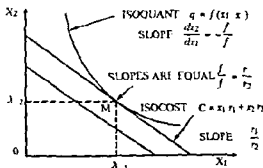


Fig 10.4

Furthermore $\frac{f_1}{f_2}$ also represents the negative slope of isoquant while $\frac{r_1}{r_2}$ represents the negative slope of isocost curve since $C = x_1r_1 + x_2r_2$ or $x_2 = \frac{C}{r_2} - \left(\frac{r_1}{r_2}\right)x_1$ represents straight line with slope $-\frac{r_1}{r_2}$ and vertical intercept $= \frac{C}{r_2}$.

Hence Eq (4) amounts to the equality between the slope of isoquant and that of isocost. Therefore the isocost line touches the isoquant at the point of cost minimisation.

In Fig 10.4 it is at the point $M(\bar{x}_1, \bar{x}_2)$ that the slopes of two curves are equal.

Second order condition To assume that total cost will be minimum (and not maximum) the value of the relevant bordered Hessian ought to be positive. Let us examine this

$$\text{In our example } |B| = \begin{vmatrix} 0 & -f_1 & -f_2 \\ -f_1 & -\lambda f_{11} & -\lambda f_{12} \\ -f_2 & -\lambda f_{21} & -\lambda f_{22} \end{vmatrix}$$

From Eq (4) we can substitute $f_1 = \frac{r_1}{\lambda}$ and $f_2 = \frac{r_2}{\lambda}$ in the above determinant, therefore

$$|B| = \begin{vmatrix} 0 & -r_1/\lambda & -r_2/\lambda \\ -r_1/\lambda & -\lambda f_{11} & -\lambda f_{12} \\ -r_2/\lambda & -\lambda f_{21} & -\lambda f_{22} \end{vmatrix}$$

Taking $(-\lambda)$ common from row (2) and row (3)

$$|H| = \lambda^2 \begin{vmatrix} 0 & -r_1/\lambda & -r_2/\lambda \\ r_1/\lambda^2 & f_{11} & f_{12} \\ r_2/\lambda^2 & f_{21} & f_{22} \end{vmatrix}$$

Taking out $\left(-\frac{1}{\lambda^2}\right)$ from column (1) and $\left(+\frac{1}{\lambda}\right)$ from row (1) we have

$$|H| = -\frac{1}{\lambda} \begin{vmatrix} 0 & -r_1 & -r_2 \\ -r_1 & f_{11} & f_{12} \\ -r_2 & f_{21} & f_{22} \end{vmatrix}$$

which is equal to $\left\{-\frac{1}{\lambda} (2f_{12}r_1r_2 - f_{11}r_2^2 - f_{22}r_1^2)\right\}$ and this expression should be less than zero in order to ensure that M is the point of minimum cost

Since $\lambda > 0$ (by Eq 4), therefore, the condition becomes,

$$\{-(2f_{12}r_1r_2 - f_{11}r_2^2 - f_{22}r_1^2)\} < 0 \text{ or } (2f_{12}r_1r_2 - f_{11}r_2^2 - f_{22}r_1^2) > 0$$

Assuming p_1 for r_1 and p_2 for r_2 , the above expression turns into

$$(2f_{12} p_1 p_2 - f_{11} p_2^2 - f_{22} p_1^2)$$

which we have already proved to be greater than zero (see Eq VI of the chapter 9, Section 9.3)

However, the solution for unconstrained profit maximisation will be different

$$\pi = p f(x_1, x_2) - (x_1r_1 + x_2r_2 + b)$$

$$\text{Hence, } \frac{\partial \pi}{\partial x_1} = p f_1 - r_1 = 0, \text{ or } \frac{f_1}{r_1} = p$$

$$\frac{\partial \pi}{\partial x_2} = p f_2 - r_2 = 0, \frac{f_2}{r_2} = p$$

Second order conditions are,

$$\frac{\partial^2 \pi}{\partial x_1^2} = p f_{11} < 0, \frac{\partial^2 \pi}{\partial x_2^2} = p f_{22} < 0$$

$$\text{and } H > 0 \text{ that is, } \begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix} > 0$$

These conditions are satisfied by a well behaved production function

Hence, the second order condition is also satisfied at point M where the isocost is tangent to the isoquant, i.e., where the slopes of both the curves are negative and equal

10.6. Constrained Profit Maximisation

The ultimate objective of the firm is the maximisation of profit. The point M in Fig. 10.4 involves minimum cost to the firm. This point would also mean a maximum profit to the firm—assuming that there exists perfect competition in output market.

Let us examine this point.

Assume that q^0 level of output maximises the profit of the firm which uses x_1 and x_2 amounts of inputs.

$$\text{Total profit } \pi = TC - TC = p q^0 - (x_1 r_1 + x_2 r_2 + b)$$

(p is the fixed unit output price)

The firm would maximise subject to the existence of the production function (given technology) $q^0 = f(x_1, x_2)$

Combining the functions with Lagrange multiplier

$$Z = p q^0 - r_1 x_1 - r_2 x_2 - b + \lambda [q^0 - f(x_1, x_2)]$$

First order conditions gives

$$\frac{\partial Z}{\partial q^0} = p + \lambda = 0 \quad (1)$$

$$\frac{\partial Z}{\partial x_1} = -r_1 - \lambda f_1 = 0 \quad (2)$$

$$\frac{\partial Z}{\partial x_2} = -r_2 - \lambda f_2 = 0 \quad (3)$$

$$\frac{\partial Z}{\partial \lambda} = q^0 - f(x_1, x_2) = 0 \quad (4)$$

Equations (2) and (3) give the same condition $\cdot \frac{r_1}{r_2} = \frac{f_1}{f_2}$ which is satisfied only at point M of Fig. 10.4

Ex. 1 A firm's production function is $Q = 5L^{0.7} K^{0.3}$. The price of labour is Re 1 per unit and the price of capital is Rs 2 per unit. Find the minimum cost combination of capital and labour for an output rate of 20.

The Cost equation : $C = L + 2K$

Production function : $Q = 5L^{0.7} K^{0.3}$

For cost minimisation under the constraint : $Q = 5L^{0.7} K^{0.3} = 20$

$$Z = L + 2K + \lambda \{20 - 5L^{0.7} K^{0.3}\}$$

First order condition gives :

$$\frac{\partial Z}{\partial L} = 1 - 3.5L^{-0.3}K^{0.3} = 1 - 3.5\lambda \left(\frac{K}{L}\right)^{0.3} = 0 \quad (i)$$

$$\frac{\partial Z}{\partial K} = 2 - 1.5L^{0.7}K^{-0.7} = 2 - 1.5\lambda \left(\frac{K}{L}\right)^{-0.7} = 0 \quad (ii)$$

$$\frac{\partial Z}{\partial \lambda} = 20 - 5L^{0.7}K^{0.3} = 0 \quad (iii)$$

$$\begin{aligned} \text{Equations (i) and (ii) give } 3.5 \left(\frac{K}{L}\right)^{0.3} &= \frac{1.5}{2} \left(\frac{K}{L}\right)^{-0.7} \\ \text{i.e. } \frac{L}{K} &= \frac{14}{3} \quad (iv) \end{aligned}$$

$$\text{Equation (iii) gives } L^{0.7}K^{0.3} = 4$$

$$\text{i.e. } L \left(\frac{3}{14}\right)^{0.3} = 4 \quad [\text{by substituting from (iv)}]$$

$$L = 4 \left(\frac{14}{3}\right)^{0.3} = 4(4.6)^{0.3}$$

$$K = \frac{6}{7} \left(\frac{14}{3}\right)^{0.3} = 0.86(4.6)^{0.3}$$

In other words the firm should use $\left\{4 \left(\frac{14}{3}\right)^{0.3}\right\}$ and $\left\{\frac{6}{7} \left(\frac{14}{3}\right)^{0.3}\right\}$ units of labour and capital respectively for an output rate of 20. This will incur the firm minimum cost (and bring maximum profit).

Ex 2 A firm's production function is $q = 12 - \frac{1}{LK}(L + K)$. The prices of labour, capital and output are Rs 1, 4 and 9 respectively. Find the maximum profit combination of capital, labour and output.

The cost equation $C = L + 4K$. Assuming level of output $= q$ that maximises the firm's profit $\pi = 9q - L - 4K$.

For constrained profit maximisation we construct a new function

$$Z = 9q - L - 4K + \lambda \left\{ q - 12 + \frac{1}{LK}(L + K) \right\}$$

$$\text{First order condition } \frac{\partial Z}{\partial q} = 9 + \lambda = 0$$

$$\frac{\partial Z}{\partial L} = -1 - \lambda L^{-2} = 0$$

$$\frac{\partial Z}{\partial K} = -4 - \lambda L^{-2} = 0$$

The above system of equations gives $L = 3$, $K = \frac{3}{2}$, $q = 11$ and $\pi = 90$

10.7 Homogeneous Functions

Special types of functions which are often used in Economics because of their peculiar characteristics are '*Homogeneous functions*'

We shall first give few examples of these functions before making an attempt to define them

Consider a function involving two independent variables x and y

$$Z = f(x, y)$$

Let us now examine how the original function changes when the independent variables are increased by equal proportion

$$1 \quad \text{Suppose, } Z = f(x, y) = \frac{x^2}{y^2} + \frac{y^2}{x^2}$$

$$\begin{aligned} \text{so that, } f(Kx, Ky) &= \frac{(Kx)^2}{(Ky)^2} + \frac{(Ky)^2}{(Kx)^2} \\ &= \frac{K^2 x^2}{K^2 y^2} + \frac{K^2 y^2}{K^2 x^2} = \frac{x^2}{y^2} + \frac{y^2}{x^2} \\ &= K^0 f(x, y) = f(x, y) = Z \end{aligned}$$

which shows that when the independent variables are increased by a fixed proportion (K) there is no change in the original function

$$\begin{aligned} 2 \quad \text{Suppose, } Z &= f(x, y) = \sqrt{ax^2 + bxy + cy^2} \\ f(Kx, Ky) &= \sqrt{a(Kx)^2 + b(Kx)(Ky) + c(Ky)^2} \\ &= \sqrt{K^2 ax^2 + bK^2 xy + K^2 cy^2} \\ &= K \sqrt{ax^2 + bxy + cy^2} = K f(x, y) = K(Z), \end{aligned}$$

which shows that when the independent variables are increased by fixed proportions (K) the original function is also increased by the same proportion

$$\begin{aligned} 3 \quad \text{Suppose, } Z &= f(x, y) = x^3 + 3x^2y + 3xy^2 + y^3 \\ f(Kx, Ky) &= (Kx)^3 + 3(Kx)^2(Ky) + 3(Kx)(Ky)^2 + (Ky)^3 \end{aligned}$$

$$\begin{aligned}
 &= K^3 x^3 + 3K^3 x^2 y + 3K^3 x y^2 + K^3 y^3 \\
 &= K^3 (x^3 + 3x^2 y + 3x y^2 + y^3) = K^3 f(x, y) \\
 &= K^3 (Z)
 \end{aligned}$$

which shows that when the independent variables are increased by fixed proportion (K), the original function is increased by (K^3) times

$$4 \quad \text{Assume that, } Z = f(x, y) = \frac{x}{y^2}$$

$$\text{therefore, } f(Kx, Ky) = \frac{Kx}{(Ky)^2} = \frac{Kx}{K^2 y^2} = \frac{x}{Ky^2}$$

$$\frac{1}{K} \left(\frac{x}{y^2} \right) = \frac{1}{K} f(x, y) = K^{-1} (Z)$$

which shows that when independent variables are increased by K times the function gets decreased by the same proportion, i.e., K .

In all these examples, one thing is common when the inputs are changed by the same proportion K , the function Z changes by some power of K . In particular, in the above examples, K times change in x and y has changed the function Z by K^0, K^1, K^3 and K^{-1} times respectively. This property renders these functions to be homogeneous functions.

The power of the multiplicand K in the transformed function defines the degree of homogeneity of the function. Thus the functions in the above examples are homogeneous of degree 0, 1, 3 and -1 respectively.

As can be seen from all these functions, the sum of the powers of the variables in each term on the right hand side is constant in a homogeneous function. If the sum is zero the degree of homogeneity of the function is zero.

All functions are not homogeneous. For example, the function $Z = x^2 + 2y$ is not a homogeneous function, since if x and y change to Kx and Ky , then

$$Z = K^2 x^2 + 2Ky = K(Kx^2 + 2y) = KZ$$

It is only when the degree of homogeneity is one that the function is referred to as linearly homogeneous function. Again, we must remember that the word "linear" should not be confused with the *linear relation* between the variables of the function. A function can be linearly homogeneous yet may not be linear in relation, for example function

$$Z = f(x, y) = \sqrt{x^2 + 3xy + y^2}$$

is linearly homogeneous, but not linear in relation. Thus, the word "linear" here should always be interpreted only as a synonym "of the first degree".

To quote a few more examples, following functions though not linear in relation are homogeneous of degree one or linearly homogeneous

$$1 \quad Z = \sqrt[3]{x^2y}$$

Changing each variable by K times

$$Z = \sqrt[3]{(K^2x^2)(Ky)} = (K^3x^2y)^{1/3} = K \sqrt[3]{x^2y}$$

$$Z = K(Z)$$

$$2 \quad Z = \frac{ax^2 + 2hxy + by^2}{cx + dy}$$

Changing each variable by K times

$$Z = \frac{aK^2x^2 + 2h(Kx)(Ky) + b(Ky)^2}{c(Kx) + d(Ky)} = \frac{K^2(ax^2 + 2hxy + by^2)}{K(cx + dy)}$$

$$= \frac{K(ax^2 + 2hxy + by^2)}{cx + dy} \quad Z = K(Z)$$

$$3 \quad P = (\beta K^{-\epsilon} + \alpha L^{-\epsilon})^{-1/\epsilon}$$

Changing each variable (K and L) by t times

$$P = (\beta(tK)^{-\epsilon} + \alpha(tL)^{-\epsilon})^{-1/\epsilon} = (\beta t^{-\epsilon} K^{-\epsilon} + \alpha t^{-\epsilon} L^{-\epsilon})^{-1/\epsilon}$$

$$P = (t^{-\epsilon})^{-1/\epsilon} [\beta K^{-\epsilon} + \alpha L^{-\epsilon}]^{-1/\epsilon} = t(P)$$

$$P = t(P)$$

Linear homogeneous functions are frequently used in the theory of production

Suppose, we have a production function $P = f(x_1, x_2, x_3)$ where, P is the total output and x_1, x_2 and x_3 are inputs used for production. In economic theory, we have learnt that if output changes by the same proportion as that of inputs, the returns to scale are said to be constant. Returns to scale are said to be increasing or decreasing if the proportional increase in the total output is more or less than the proportional increase in the inputs.

As such, the mathematical assumption of the linearly homogeneous production function amounts to the economic assumption of constant returns to scale in production. Similarly, the mathematical assumption of homogeneous function of more or less than degree one amounts to the economic assumption of increasing or decreasing returns to scale respectively.

10.8 Properties of Linearly Homogeneous Functions

Since other than linearly homogeneous functions are seldom assumed for production functions (because of complexities in calculation and analysis), we shall first describe the properties of linearly homogeneous function and then derive the properties of function of degree ' n '.

If $Z = f(x, y)$ is assumed to be a linearly homogeneous function the following three properties hold good for all values of independent variables x and y

Property I The given linearly homogeneous function can be written in either of the following two forms

$$Z = x \times \Phi\left(\frac{y}{x}\right) \quad \text{or,} \quad Z = y \times \Psi\left(\frac{x}{y}\right)$$

where Φ and Ψ are some functions of a single variable

Proof The given function $Z = f(x, y)$ is linearly homogeneous

$$f(Kx, Ky) = K f(x, y)$$

since K can be any value, let $K = \frac{1}{x}$,

$$f\left(1, \frac{y}{x}\right) = \frac{1}{x} f(x, y)$$

$$f(x, y) = x f\left(1, \frac{y}{x}\right) = x \Phi\left(\frac{y}{x}\right)$$

Similarly, if $K = \frac{1}{y}$ then

$$f\left(\frac{x}{y}, 1\right) = \frac{1}{y} f(x, y)$$

$$f(x, y) = y \Psi\left(\frac{x}{y}\right)$$

Ex 3. $Z = f(x, y) = \sqrt[3]{x^2 y}$

$$f(x, y) = (x^2)^{\frac{1}{3}} y^{\frac{1}{3}} = x^{\frac{2}{3}} y^{\frac{1}{3}}$$

$$= x (x^{\frac{1}{3}} y^{\frac{1}{3}}) = x \left(\frac{y}{x}\right)^{\frac{1}{3}}$$

$$= x \Phi\left(\frac{y}{x}\right) \text{ where, } \Phi\left(\frac{y}{x}\right) = \left(\frac{y}{x}\right)^{\frac{1}{3}}$$

Property II The partial derivatives of the dependent variable of the given linearly homogeneous function (i.e., $\frac{\partial Z}{\partial x}$ and $\frac{\partial Z}{\partial y}$) are functions of the ratio of x to y .

Proof From Property I, we have $Z = x \Phi\left(\frac{y}{x}\right)$

$$\frac{\partial Z}{\partial x} = \Phi\left(\frac{y}{x}\right) + x \cdot \Phi'\left(\frac{y}{x}\right) \cdot \frac{\partial}{\partial x} \left(\frac{y}{x}\right) \quad (\text{by product rule})$$

$$\frac{\partial Z}{\partial x} = \Phi\left(\frac{y}{x}\right) - \frac{y}{x} \Phi'\left(\frac{y}{x}\right)$$

i.e., partial derivative of Z with respect to x is function of ratio of the independent variables. Hence, $\frac{\partial Z}{\partial x}$ is homogeneous of degree 0 in y and x .

Again, since $Z = x \cdot \Phi\left(\frac{y}{x}\right)$

$$\frac{\partial Z}{\partial y} = 0 + x \cdot \Phi'\left(\frac{y}{x}\right) \cdot \frac{\partial}{\partial y} \left(\frac{y}{x}\right) = \Phi'\left(\frac{y}{x}\right)$$

So that $\frac{\partial Z}{\partial y}$ is also a function of ratio of y to x .

Ex 4 Given $Z = x^{\frac{2}{3}} y^{\frac{1}{3}}$

Here, $\frac{\partial Z}{\partial x} = \frac{2}{3} x^{-\frac{1}{3}} y^{\frac{1}{3}} = \frac{2}{3} \left(\frac{y}{x}\right)^{\frac{1}{3}}$ and

$$\frac{\partial Z}{\partial y} = \frac{1}{3} x^{\frac{2}{3}} y^{-\frac{2}{3}} = \frac{1}{3} \left(\frac{x}{y}\right)^{\frac{2}{3}} = \frac{1}{3} \left(\frac{y}{x}\right)^{-\frac{2}{3}}$$

Property III. The linearly homogeneous functions satisfy "Euler's theorem" which states that values of homogeneous functions can always be written as a sum of terms, each of these being the product of the first partial derivative and the corresponding independent variable.

Thus, if we have a linear homogeneous function $Z = f(x, y)$, then

$$x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} = Z$$

Proof Substitute values of $\frac{\partial Z}{\partial x}$ and $\frac{\partial Z}{\partial y}$ obtained in property II in

$$\begin{aligned} & \left[x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} \right] \\ &= x \left\{ \Phi\left(\frac{y}{x}\right) - \left(\frac{y}{x}\right) \Phi'\left(\frac{y}{x}\right) \right\} + y \Phi'\left(\frac{y}{x}\right) \\ &= x \cdot \Phi\left(\frac{y}{x}\right) - y \cdot \Phi'\left(\frac{y}{x}\right) + y \cdot \Phi'\left(\frac{y}{x}\right) = x \cdot \Phi\left(\frac{y}{x}\right) \end{aligned}$$

$$= f(x, y) = Z \quad (\text{by Property I})$$

Ex 5 Apply Euler's Theorem to $Z = \sqrt[3]{x^2 y}$

$$\frac{\partial Z}{\partial x} = \frac{2}{3} x^{-\frac{1}{3}} x^{\frac{2}{3}} y^{\frac{1}{3}} \text{ and } y^{\frac{1}{3}} \frac{\partial Z}{\partial y} = \frac{1}{3} x^{\frac{2}{3}} y^{-\frac{2}{3}}$$

$$\begin{aligned} \text{Therefore, } x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} &= x \left(\frac{2}{3} x^{-\frac{1}{3}} y^{\frac{1}{3}} \right) + y \left(\frac{1}{3} x^{\frac{2}{3}} y^{-\frac{2}{3}} \right) \\ &= x^{\frac{2}{3}} y^{\frac{1}{3}} \\ &= f(x, y) = Z \end{aligned}$$

Ex. 6 For the Cobb Douglas function

$$Q = aK^a L^{1-a}$$

where K is capital, L is labour and a is constant, verify Euler's theorem

$$K \frac{\partial Q}{\partial K} + L \frac{\partial Q}{\partial L} = Q$$

Given production function $Q = aK^a L^{1-a}$

$$(i) \quad \frac{\partial Q}{\partial K} = a^2 K^{a-1} L^{1-a}$$

$$(ii) \quad \frac{\partial Q}{\partial L} = a(1-a) K^a L^{-a}$$

$$\begin{aligned} \text{Therefore, } K \frac{\partial Q}{\partial K} + L \frac{\partial Q}{\partial L} &= K(a^2 K^{a-1} L^{1-a}) + L[a(1-a) K^a L^{-a}] \\ &= a^2 K^a L^{1-a} + a(1-a) K^a L^{1-a} \\ &= aK^a L^{1-a} [a + (1-a)] \\ &= Q, \text{ hence proved} \end{aligned}$$

Note The above three properties can be extended to homogeneous functions of any degree

Consider a two-variable function $Z = f(x, y)$ which is homogeneous of degree ' n ' then the function will have the following properties

$$\text{Property I} \quad Z = x^n \Phi\left(\frac{y}{x}\right) = y^n \Psi\left(\frac{x}{y}\right)$$

$$\text{Property II} \quad \frac{\partial Z}{\partial x} \text{ and } \frac{\partial Z}{\partial y} \text{ are functions of } \left(\frac{y}{x}\right) \text{ then}$$

$$\frac{\partial Z}{\partial x} = nx^{n-1} \Phi\left(\frac{y}{x}\right) - x^{n-2} y \Phi'\left(\frac{y}{x}\right)$$

[In other words, if the function is homogeneous of degree n the first partial derivative of function is homogeneous function of degree $(n-1)$ the

second partial derivative is the homogeneous function of degree $(n-2)$ and so on so that the k th partial derivative is homogeneous function of $(n-k)$ th degree]

Property III It will satisfy Euler's theorem

$$x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} = n \cdot Z$$

[So that if the given function is of degree three then

$$x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} = 3 \cdot Z \text{ and so on}]$$

The above properties of linearly homogeneous production function can be interpreted in economics as follows

Property I It depicts constant returns to scale

Property II Average and marginal products are homogeneous of degree zero. In other words the proportional increase in all the factors leaves the average and marginal products of each factor unaltered

Consider a linearly homogeneous function

$$P = f(L, K) = L^\alpha K^{1-\alpha}$$

$$MP_L = \frac{\partial P}{\partial L} = \alpha L^{\alpha-1} K^{1-\alpha} \quad (V)$$

$$MP_K = \frac{\partial P}{\partial K} = (1-\alpha) L^\alpha K^{-\alpha} \quad (VI)$$

We can now see that when inputs are increased by (t) times the product also increases by (t) times but MP_L and MP_K do not change

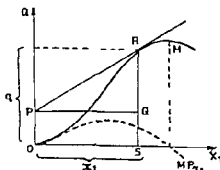
$$t \cdot P = f(Lt, Kt) = (tL)^\alpha (tK)^{1-\alpha}$$

$$MP_L = \alpha L^{\alpha-1} K^{1-\alpha} t^{1-\alpha} K^{1-\alpha}$$

$$MP_L = \alpha L^{\alpha-1} K^{1-\alpha} \quad [\text{which is same as (V)}]$$

Clearly MP_L is homogeneous of degree zero

Property III Euler's Theorem Linearly homogeneous function satisfies the Euler's theorem. In economics this property means that under the condition of constant returns to scale the total product will get exhausted if each factor is paid according to its marginal product (since partial derivative depicts the marginal product). This is a well



known property in theory of production in connection with the marginal productivity theory — usually under the name of '*Adding up Theorem*'

Euler's theorem can be illustrated in Economics' as follows (See Fig 10.5)

Let us assume that we get a total production curve of commodity Q with the application of variable amounts of input X_1 to a fixed amount of input $X_2 = x_2$.

Supposing at point R, the total product of commodity Q is q when the amount of $X_1 = x_1$

The slope of total product curve at point R = $\frac{RQ}{PQ}$. This slope also depicts the marginal product of input X_1 at point R

$$MP_{x_1} = \frac{\partial q}{\partial x_1} = \frac{RQ}{PQ}$$

$$\text{but } PQ = OS = x_1$$

$$\frac{\partial q}{\partial x_1} = \frac{RQ}{x_1} \quad \text{or, } x_1 \frac{\partial q}{\partial x_1} = RQ$$

According to Euler's theorem, for the linearly homogeneous function $q = f(x_1, x_2)$

$$x_1 \frac{\partial q}{\partial x_1} + x_2 \frac{\partial q}{\partial x_2} = q$$

Substituting, $x_1 \frac{\partial q}{\partial x_1} = RQ$ in this equation, we have

$$RQ + x_2 \frac{\partial q}{\partial x_2} = q$$

$$\text{or, } x_2 \cdot \frac{\partial q}{\partial x_2} = q - RQ$$

$$\text{or, } x_2 \frac{\partial q}{\partial x_2} = RS - RQ = QS$$

$$\text{or, } x_2 \frac{\partial q}{\partial x_2} = QS$$

In other words, total product $RS (= q)$ at point R is divided between

$$RQ \left(= x_1 \frac{\partial q}{\partial x_1} \right) \text{ and } QS \left(= x_2 \frac{\partial q}{\partial x_2} \right)$$

Furthermore, at point M where the total product of Q is maximum, say

$$MP_x = \frac{\partial q}{\partial x_1} = 0$$

or in the form of Euler's theorem $q_m = 0 + \lambda_2 \frac{\partial q}{\partial x_2}$

$$\text{or } \frac{q_m}{x_2} = \frac{\partial q}{\partial x_2} \text{ or } AP_x = MP_x.$$

which means that under constant returns to scale at a point where total product is maximum, MP_x is zero while marginal product of other factor $X_2 (= MP_x)$ is equal to average product of $X_2 (= AP_x)$. But when can the marginal and average products be equal?

It is only when average product is maximum. Hence we derive one of the most important conclusions from this property of linearly homogeneous function that *when the total product is maximum the marginal product of factor which is varying is zero while the average product of the factor which is fixed is maximum*. Thus, in Fig. 10.5 when total product is at the maximum at point M , $MP_x = 0$ and AP_x is at maximum. Similarly, when average product of x_1 is maximum marginal product of $x_2 = 0$.

In the case of isoquants we referred to the points P , Q and R in Fig. 10.2 as the points where though, marginal product of $A=0$ (total product is at maximum), average product of factor B will be maximum. In the same way at points L, M and N , marginal product of B is zero (total product is at maximum) but average product of factor A will be the maximum.

Ex. 7. Assume the demand functions for the three commodities X_A, X_B and X_C in the following forms

$$X_A = \frac{1}{p_C} (2p_C - p_A)$$

$$X_B = \frac{1}{p_C} (5p_C - p_A - 2p_B)$$

$$X_C = \frac{1}{p_C} (10p_A - p_B)$$

(p_A, p_B and p_C are the per unit prices of three commodities) Establish that the three demand functions are homogeneous functions of zero degree in the three prices.

Let each price be increased by t^{th} times so that the demand functions become

$$X_A = \frac{1}{p_C t} (2p_C t - p_A t) = \frac{1}{p_C} (2p_C - p_A) = t^0 \cdot X_A = X_A$$

$$X_B = \frac{1}{p_C t} (5p_C t - p_A t - 2p_B t)$$

$$= \frac{1}{p_C} (5p_C - p_A - 2p_B) = t^0 \quad X_B = X_B$$

$$X_C = \frac{1}{p_C t} (10p_A t - p_B t) = \frac{1}{p_C} (10p_A - p_B) = t^0 \quad X_C$$

This proves that all the demand functions are homogeneous of degree zero

Ex 8. Given the function $U = Ax^b y^c$, A , b and c are constants

(a) Find the conditions under which this is a linear homogeneous function,

(b) Apply Euler's theorem if these conditions hold true

(a) Let each variable be increased by constant t times the new function becomes

$$\begin{aligned} &= A (xt)^b (yt)^c \\ &= A t^{b+c} x^b y^c \\ &= t^{b+c} A x^b y^c = t^{b+c} (U) \end{aligned}$$

in other words the given function is homogeneous function of $(b+c)$ degree
The given function will be linearly homogeneous only when $b+c = 1$

(b) Euler's theorem states

if $f(x, y)$ is the homogeneous of the K th degree, then

$$K f(x, y) = x f_x(x, y) + y f_y(x, y)$$

In our case given function is $U = Ax^b y^c$ which is homogeneous of degree $(b+c)$

$$\begin{aligned} \text{Hence, } (b+c) U &= x \frac{\partial U}{\partial x} + y \frac{\partial U}{\partial y} \\ &= x (A b x^{b-1} y^c) + y (A c x^b y^{c-1}) \\ &= x \left(b \frac{U}{x} \right) + y \left(c \frac{U}{y} \right) \\ &= U (b+c), \text{ which proves the theorem} \end{aligned}$$

10.9. Cobb-Douglas Production Function

C.W Cobb and P.H. Douglas proposed a general production function

$$Q = AK^\alpha L^\beta$$

as a law of production for manufacturing industries. Here, Q represents total output of the industry or firm, K represents capital input, L represents labour input, while A , α and β are positive constants

Some major features of this function are

- 1 It is homogeneous of degree $(\alpha + \beta)$. In the special case of $\alpha + \beta = 1$, the function turns into linearly homogeneous function

(Note here that the function itself is not linear.)

- 2 Its isoquants are negatively sloped and convex downwards
- 3 Its exponents α and β represent

(a) elasticity of output with respect to total capital and labour

(b) share of each input in total output (Q)

These three features of the function may be proved in the following way

- (1) Homogeneity of the function can be examined by making proportionate changes in each input

$$\begin{aligned}\text{Then } Q &= A (tK)^\alpha (tL)^\beta = A t^\alpha K^\alpha t^\beta L^\beta \\ &= A t^{\alpha+\beta} K^\alpha L^\beta = t^{\alpha+\beta} (AK^\alpha L^\beta) = t^{\alpha+\beta} (Q)\end{aligned}$$

that is, the function is homogeneous of degree $(\alpha + \beta)$

In case $\alpha + \beta = 1$, the function will be linearly homogeneous and then it (function) will depict constant returns to scale.

When $\alpha + \beta = 1$

$$Q = AK^{\alpha-1} L^\beta$$

$$\text{or, } Q = AK^\alpha L^{1-\alpha}$$

It will then satisfy all the three properties of a linearly homogeneous function

- (2) That its isoquants have negative slopes can be verified from signs of its first and second derivatives :

$$\left\{ \frac{dQ}{dK} \text{ and } \frac{d^2Q}{dK^2} \right\} \text{ or } \left\{ \frac{dQ}{dL} \text{ and } \frac{d^2Q}{dL^2} \right\}$$

$$\text{Since, } Q = f(K, L) = AK^\alpha L^\beta$$

$$\text{Total differential : } dQ = \frac{\partial Q}{\partial K} \cdot dK + \frac{\partial Q}{\partial L} \cdot dL$$

$$\text{We have, } \frac{\partial Q}{\partial K} = A\alpha K^{\alpha-1} L^\beta \quad \dots(\text{VII})$$

$$\text{and, } \frac{\partial Q}{\partial L} = A\beta K^\alpha L^{\beta-1} \quad \dots(\text{VIII})$$

Since output remains unchanged on a particular isoquant, the total change in output $dQ = 0$.

$$\text{Therefore, } dQ = (A\alpha K^{\alpha-1} L^\beta) dK + (A\beta K^\alpha L^{\beta-1}) dL = 0$$

So that,
$$\frac{dK}{dL} = - \left(\frac{\beta}{\alpha} \frac{K}{L} \right) \quad (\text{IX})$$

Hence, the isoquant should possess negative slope

Again,
$$\begin{aligned} \frac{d^2K}{dL^2} &= \frac{d}{dL} \left(- \frac{\beta}{\alpha} \frac{K}{L} \right) = - \frac{\beta}{\alpha} \frac{d}{dL} \left(\frac{K}{L} \right) \\ &= - \frac{\beta}{\alpha} \cdot \frac{1}{L^2} \left(L \frac{dK}{dL} - K \right) \quad (\text{By quotient rule}) \\ &= - \frac{\beta}{\alpha} \cdot \frac{1}{L} \frac{dK}{dL} + \frac{\beta}{\alpha} \frac{K}{L^2} \end{aligned}$$

By (IX), $\frac{dK}{dL}$ has negative sign, therefore, whole of expansion of $\frac{d^2K}{dL^2}$ will be positive (as α , β , L and K can never take negative values)

Therefore, $\frac{dK}{dL} < 0$ and $\frac{d^2K}{dL^2} > 0$,

this proves that the isoquants of the function will have negative slope and will be convex downwards

(3) (a) Labour (input) elasticity of output (Q) is given by $\frac{L}{Q} \frac{\partial Q}{\partial L}$

Substituting value of $\frac{\partial Q}{\partial L}$ (from VIII) and value of L in this, we get

$$\frac{L}{Q} \frac{\partial Q}{\partial L} = \frac{L}{AK^\alpha L^\beta} \cdot \alpha \beta AK^\alpha L^{\beta-1} = \beta$$

Similarly capital (input) elasticity of output Q is given by $\frac{K}{Q} \frac{\partial Q}{\partial K}$

Substituting values of $\frac{\partial Q}{\partial K}$ (from VII) and of $Q = AK^\alpha L^\beta$

$$\frac{K}{Q} \frac{\partial Q}{\partial K} = \frac{K}{AK^\alpha L^\beta} \cdot \alpha AK^{\alpha-1} L^\beta = \alpha$$

(b) $\frac{\partial Q}{\partial K} = \alpha AK^{\alpha-1} L^\beta$ (from VII)

$$\frac{\partial Q}{\partial K} = \frac{\alpha}{K} (AK^\alpha L^\beta) = \alpha \frac{Q}{K} = \alpha (AP_K)$$

But $\frac{\partial Q}{\partial K} = MP_K$

$$\text{Therefore } MP_K = \alpha (AP_K) \quad (X)$$

$$\text{Similarly } MP_L = \beta (AP_L) \quad (XI)$$

In economics of competitive markets we know that for profit maximising behaviour marginal product of labour should equate with real wage rate i.e. $MP_L = \frac{W}{p}$ where W is the current wage rate and p is the per unit price of output

$$MP_L = \beta \frac{Q}{L} = \frac{W}{p}$$

$$\text{So that } \beta = \frac{WL}{Qp}$$

Similarly $\alpha = \frac{rK}{Qp}$ where r is the price per unit of capital input

$\frac{WL}{Qp}$ represents percentage share of labour in total output and $\frac{rK}{Qp}$ represents percentage share of capital in total output

10.10 Expansion Path for Cobb-Douglas Function

We once again go back to the theory of isoquant curves and examine the successive increases of total product of Q (as we move on higher and higher isoquants) assuming that input prices remain unchanged

Each shift of the isoquant from q^0 to q^1 to q^2 and so on will result in a new point of tangency with a higher isocost (so that slopes of two curves are each time equal and total cost is minimum). The locus of such contact points of tangents known as "Expansion path" of firms describes the varying levels of total output with minimum cost. OE is the expansion path in Fig 10.6

At M_0 the slope of isoquant q_0 = slope of isocost C_1

At M_1 the slope of isoquant q_1 = slope of isocost C_2 and so on

Each point M_0 , M_1 and M_2 must satisfy the first order condition which states that

$$\frac{f_1}{f_2} = \frac{r_1}{r_2}$$

Since input prices r_1 and r_2 are assumed to be fixed their ratio

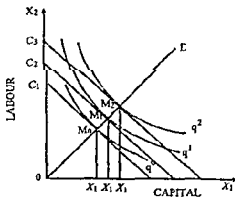


Fig 10.6

$\left(\frac{r_1}{r_2}\right)$ will always be the same constant for all the points, $M_0 M_1 M_2$ expansion path will, therefore, be a straight line emanating from the origin. This has been illustrated in Fig. 10.6, showing that the input ratios at these contact points of tangency are all equal.

$$\frac{OM_0}{OX'_1} = \frac{OM_1}{OX''_1} = \frac{OM_2}{OX'''_1}$$

We can show that the expansion path generated by Cobb-Douglas function of degree one is also linear (a straight line).

Linearly homogeneous Cobb-Douglas function is

$$Q = AK^\alpha L^{1-\alpha}$$

First order condition is $\frac{r_1}{r_2} = \frac{f_L}{f_K}$

In this case, $f_L = f_K = A\alpha K^{\alpha-1} L^{1-\alpha}$

and, $f_L = f_K = A(1-\alpha) K^\alpha L^{-\alpha}$

$$\frac{f_L}{f_K} = \frac{f_L}{f_K}$$

$$\frac{f_L}{f_K} = \frac{A\alpha K^{\alpha-1} L^{1-\alpha}}{A(1-\alpha) K^\alpha L^{-\alpha}} = \frac{\alpha L}{(1-\alpha) K}$$

$$\text{that is, } \frac{r_1}{r_2} = \frac{\alpha L}{(1-\alpha) K}$$

$$\text{or, } \frac{L}{K} = \frac{1-\alpha}{\alpha} \frac{r_1}{r_2},$$

which is a constant quantity (assuming that relative prices of inputs do not change). This proves that expansion path of linearly homogeneous Cobb-Douglas function is a linear one.

Similarly, prove that expansion path will still be linear even if the Cobb-Douglas function is not linearly homogeneous.

10.11. Elasticity of Substitution

Isoquant or equal product curve represents the fixed product for different levels of factors used in the production, its slope represents the RTS between the factors. The slope of the isoquant after a particular point—when MP of the factor on the relevant axis becomes zero—becomes positive which in turn means that the substitution between the factors is no more possible. In other

words, we may say that the slope or the curvature of the isoquant reflects the complementarity and substitutability between the two factors of production.

Every firm is interested in knowing the extent of substitutability between the factors of production in order to maximise profit or minimise cost looking to the prices which the factors command in the market. The measure of substitution between the factors is known as *elasticity of technical substitution*. This is a pure number and measures the extent to which the substitution between the factors can take place. Since this substitution depends mainly on the slope or RTS, elasticity of substitution is defined as the proportionate change in the ratio between the factors due to proportionate change in RTS.

Thus, the elasticity of substitution between the two factors—say labour (L) and capital (K) will be

$$\text{Elasticity of substitution } (\sigma) = \frac{\text{Proportionate change in the ratio between the factors}}{\text{Proportionate change in RTS}}$$

Thus, the elasticity of substitution between labour (L) and capital (K) will be

$$\begin{aligned}\sigma &= \frac{\Delta \left(\frac{K}{L} \right) / \frac{K}{L}}{\Delta \text{RTS} / \text{RTS}_{LK}} \text{ or } \frac{\Delta \left(\frac{L}{K} \right) / \frac{L}{K}}{\Delta \text{RTS} / \text{RTS}_{KL}} \\ \sigma &= \frac{d \left(\frac{K}{L} \right)}{\frac{K}{L}} \cdot \frac{\text{RTS}}{d(\text{RTS})} \text{ (Here } d \text{ stands for derivative)} \\ &= \frac{d \left[\log \left(\frac{K}{L} \right) \right]}{d[\log(\text{RTS})]}\end{aligned}$$

Since the RTS of capital for labour is the rate of marginal product of labour to that of capital, we can write the above formula as

$$\frac{d \left[\log \left(\frac{K}{L} \right) \right]}{d \left[\log \left(\frac{r_1}{r_2} \right) \right]}$$

(where r_1 and r_2 are the per unit prices of labour and capital respectively)

Let us find σ in the production function $Q = AK^\alpha L^\beta$

$$\text{Since } \frac{r_1}{r_2} = \frac{f_L}{f_K} = \frac{A\beta K^\alpha L^{\beta-1}}{A\alpha K^{\alpha-1} L^\beta} = \frac{\beta}{\alpha} \cdot \frac{K}{L}$$

③

$$\sigma = \frac{\left(\frac{L}{K}\right) d\left(\frac{K}{L}\right)}{\left(\frac{r_2}{r_1}\right) d\left(\frac{r_1}{r_2}\right)} = \frac{\left(\frac{L}{K}\right) d\left(\frac{K}{L}\right)}{\left(\frac{\alpha}{\beta} \frac{L}{K}\right) d\left(\frac{\beta}{\alpha} \frac{K}{L}\right)}$$

$\sigma = 1$ Hence for C-D function $\sigma = 1$

Important points in connection with σ

1 Since the above derivation of σ in no way relies on the exponents of the Cobb Douglas production function this type of function is characterised by unitary elasticity

2 The magnitude of σ is an indication of the ease with which product can be maintained by substituting one factor for another

3 Higher the value of σ the higher the degree of substitutability between the factors

4 Value of σ may vary between 0 and ∞

As $\sigma \rightarrow \infty$, the shape of isoquant tends to be flatter

As $\sigma \rightarrow 0$, the shape of isoquant tends to be an angle of 90°

5 If $\sigma = \infty$, the two factors are perfect substitutes

if $\sigma = 0$, the two factors are incapable of substitution

10.12 Elasticity of Substitution In a Linearly Homogeneous Function

In case the function is linearly homogeneous $Q = f(x_1, x_2)$ the elasticity of substitution (between x_1 and x_2) can be obtained by the following simple formula

$$\sigma = \frac{f_1 f_2}{Q f_{12}} \left(- \frac{\frac{\partial Q}{\partial x_1} \frac{\partial Q}{\partial x_2}}{Q \frac{\partial^2 Q}{\partial x_1 \partial x_2}} \right)$$

For example if the Cobb Douglas function is assumed to be linearly homogeneous, for $Q = AK^\alpha L^{1-\alpha}$

$$f_K = \frac{\partial Q}{\partial K} = \alpha A K^{\alpha-1} L^{1-\alpha}$$

$$f_L = \frac{\partial Q}{\partial L} = A(1-\alpha) K^\alpha L^{-\alpha}$$

$$f_{KL} = \frac{\partial^2 Q}{\partial K \partial L} = -\alpha A(1-\alpha) K^{\alpha-1} L^{1-\alpha}$$

Substituting all the values in the above formula of σ we find that $\sigma = 1$

Ex 9 Show that production function $Q^{-\beta} = aK^{-\beta} + bL^{-\beta}$ which is linearly homogeneous does not possess unitary elasticity of substitution between the factors K and L

Since the given function is linearly homogeneous (check it) we can obtain σ by the formula $\frac{f_L f_K}{Q_{LK}}$

$$\text{Given } Q^{-\beta} = aK^{-\beta} + bL^{-\beta}$$

$$\text{Therefore } Q = (aK^{-\beta} + bL^{-\beta})^{\frac{1}{\beta}}$$

$$f_L = \frac{\partial Q}{\partial L} = -\frac{1}{\beta} (aK^{-\beta} + bL^{-\beta})^{(\frac{1}{\beta}-1)} (b) (-\beta) L^{-\beta-1}$$

$$f_K = \frac{\partial Q}{\partial K} = -\frac{1}{\beta} (aK^{-\beta} + bL^{-\beta})^{(\frac{1}{\beta}-1)} (a) (-\beta) K^{-\beta-1}$$

$$\begin{aligned} f_{LK} &= \frac{\partial Q}{\partial K \partial L} = -\frac{1}{\beta} (b) (-\beta) L^{(-\beta-1)} \left(-\frac{1}{\beta} - 1 \right) \\ &\quad \times (aK^{-\beta} + bL^{-\beta})^{(\frac{1}{\beta}-2)} (-\beta) (a) K^{(-\beta-1)} \\ &= ab (-\beta) \left(-\frac{1}{\beta} - 1 \right) (L)^{-\beta-1} (K)^{-\beta-1} (aK^{-\beta} + bL^{-\beta})^{(\frac{1}{\beta}-2)} \end{aligned}$$

Substituting the values of f_L , f_K and f_{LK} in $\sigma = \frac{f_L f_K}{Q_{LK}}$ we obtain

$$\sigma = \frac{1}{1 + \beta} \text{ which is a constant quantity but not unity}$$

10.13 CES Function

There is function which though possesses the constant elasticity of substitution (CES) but may take a value other than unity. Ex 9 above is an example wherein we obtained $\sigma = \frac{1}{1 + \beta}$ (not unity) though the given function was linearly homogeneous. Such functions are usually known as CES functions. They are linearly homogeneous where $\sigma = 1$.

Consider the production function of the form

$$Q = A [\alpha K^{-\beta} + (1 - \alpha) L^{-\beta}]^{\frac{1}{\beta}}$$

in which K and L are the two factors of production. A , α and β are three

parameters. It can be proved that the above function is linearly homogeneous, consequently like all linearly homogeneous functions, it displays constant returns to scale, qualifies for the application of Euler's theorem and possesses average and marginal product that are homogeneous of degree zero.

We observe that though this function is linearly homogeneous the value of σ is not unity.

Value of σ may be obtained by two methods as explained in the following example 10.

Ex. 10. Obtain σ for $Q = A [\alpha K^{-\beta} + (1 - \alpha)L^{-\beta}]^{-1/\beta}$
 $(A > 0, 0 < \alpha < 1, \beta > -1)$

Method I

$$\begin{aligned}\frac{\partial Q}{\partial L} &= A \left(-\frac{1}{\beta} \right) [\alpha K^{-\beta} + (1 - \alpha)L^{-\beta}]^{\frac{1}{\beta}-1} \times (1 - \alpha)(-\beta)L^{-\beta-1} \\ \frac{\partial Q}{\partial L} &= f_L = \frac{1 - \alpha}{A^{\beta}} \left(\frac{Q}{L} \right)^{1+\beta} \\ \text{Again,} \quad \frac{\partial Q}{\partial K} &= f_K = \frac{\alpha}{A^{\beta}} \left(\frac{Q}{K} \right)^{1+\beta}\end{aligned}$$

In order to fulfil the least cost combination.

$$\frac{f_L}{f_K} = \frac{P_L}{P_K} \left(= \frac{r_1}{r_2} \right)$$

By substituting values of f_L and f_K we get

$$\begin{aligned}\frac{1 - \alpha}{\alpha} \left(\frac{K}{L} \right)^{1+\beta} &= \frac{P_L}{P_K} \\ \frac{K}{L} &= \left(\frac{\alpha}{1 - \alpha} \right)^{\frac{1}{1+\beta}} \left(\frac{P_L}{P_K} \right)^{\frac{1}{1+\beta}} = m \left(\frac{P_L}{P_K} \right)^{\frac{1}{1+\beta}} \\ \left[\text{where, } m &= \left(\frac{\alpha}{1 - \alpha} \right)^{\frac{1}{1+\beta}} \right]\end{aligned}$$

$$\text{Since, } \sigma = \frac{d \left(\frac{K}{L} \right) / d \left(\frac{P_K}{P_L} \right)}{\frac{K}{L} / \frac{P_L}{P_K}}$$

Therefore,
$$\sigma = \frac{\left(\frac{m}{1+\beta}\right) \left(\frac{p_L}{p_K}\right)^{\frac{1}{1+\beta}-1}}{m \left(\frac{p_L}{p_K}\right)^{\frac{1}{1+\beta}-1}} = \left(\frac{1}{1+\beta}\right)$$

Method II Since the given function is linearly homogeneous, we can apply formula $\sigma = \frac{f_L f_K}{Q f_{LK}}$ to obtain value of σ

Substituting the values f_L, f_K, f_{LK} and Q it can be shown that $\sigma = \frac{1}{1+\beta}$

Note (i) if $-1 < \beta < 0$ then $\sigma > 1$

(ii) if $\beta = 0$ then $\sigma = 1$

(iii) $0 < \beta < 1$ then $\sigma < 1$

in other words, when β takes value $= 0$ in the given function (which is CES in the present form), it will then correspond to Cobb-Douglas function. That is the CES function includes the Cobb-Douglas function as a special case.

EXERCISE 10.1

1. Establish the degree of homogeneity of the following functions

(i) $Z = f(x_1, x_2) = x_1^3 + 4x_1^2x_2 + x_2^3$

(ii) $Z = f(x_1, x_2, x_3, x_4) = \sqrt[3]{2x_1 - x_2 + 5x_3 - 3x_4}$

(iii) $Z = f(x_1, x_2, x_3) = \frac{2x_1 - 3x_2}{5x_3}$

(iv) $Q = f(L, K) = L^{0.64} K^{0.36}$

(v) $Q = f(n, K) = 0.68n^{0.53} K^{0.10}$

2. What is the degree of homogeneity of the following functions.

$$Q = (x_1, x_2) = \frac{ax_1x_2 - bx_1x_2^2 - cx_2^2}{ax_1 + \beta x_2}$$

$$Q = f(x_1, x_2) = x_1^a x_2^{1-a} + bx_1 + cx_2$$

(a, b and c are constants)

How would you interpret the zero degree of homogeneity of the marginal productivity of the inputs X_1 and X_2 in each production function

3. Establish the Euler theorem in case of $Q = 1$ for all the questions

4. The production function of a firm is given to be

$$Q = f(L, K) = 20 - \left\{ \frac{K^2 + L}{LK^2} \right\}$$

and prices per unit of K and L are $p_L = 20$ and $p_K = 40$. If the per unit output price is Rs 5, determine the amounts of the factors demanded, the amount of the product produced and the amount of profit made by the firm.

5 Find the necessary and sufficient conditions of maximum profit in case of Cobb-Douglas production function

$$Q = AL^a K^b$$

assuming that the prices of product, labour and capital are constant

6 Given the function $Z = f(a, b, c, d) = A a^{0.77} b^{0.22} c^{0.13} d^{0.54}$

Show that the conditions of maximisation of profits cannot be fulfilled

7 Derive the long-run supply function of a firm whose production function is $Y = \sqrt{LK}$. What is its elasticity of supply?

Also derive the short-run supply function

8 Given the production function

$$Q = L^{1/4} K^{1/2}$$

and price per unit of K and L Rs 2 and Rs 4 respectively and total cost of Rs 80, determine the maximum output subject to the cost constraint

9 The production function of a firm is

$$x = f(a, b) = 20 - a^2 - b^2$$

The prices are $P_a = 5$, $P_b = 20$, $P_x = 40$. Find the amount of the factors demanded and amount of the product produced corresponding to maximum profits

10 Suppose that a single-product firm has the production function

$$q = f(x_1, x_2, x_3)$$

where q denotes the quantity of the output and x_i ($i = 1, 2, 3$) denote quantities of the three factors which are required for producing q . Suppose the price of the product is denoted by p and the price of the three factors is denoted by p_i ($i = 1, 2, 3$). These prices do not vary with a variation in quantity produced. The firm has decided to spend a fixed amount of money = Rs. R for producing the product. Discuss the conditions which are necessary for the firm to obtain maximum profit under these circumstances.

11 Show mathematically that under condition of perfect competition the price of each factor of production must be equal to the value of its marginal product and factor payments will exhaust the product even if the production function be not homogeneous of first order.

12 Define Homogeneous Functions. State and prove Euler's theorem for homogeneous functions. Indicate the application of the theorem to production functions.

The production function is $x = A a^\alpha b^\beta$ where $\alpha + \beta < 1$

Show that there are decreasing returns to scale and deduce that the total product is $> a$ times the marginal product is a plus b times the marginal product of b

13. If the production function is given as

$$x = A.a^{1/2}.b^{1/4}.c^{1/4}$$

find expressions for the marginal products of a , b , and c .

Hence show that if these marginal products are all equal, then $x = \frac{Aa}{\sqrt{2}}$

14. What do you understand by 'an expansion path'?

Illustrate your answer by discussing the expansion path of the least cost combination of two factors, labour and capital, in a Cobb-Douglas model

Does the behaviour of the expansion path change if assumptions regarding returns to scale are varied?

15. Show that a production function homogeneous of degree one generates a linear production expansion path and also a linear total cost function

16. Explain the concept of elasticity of substitution

Investigate its value for the following production functions

(a) $x = AL^\alpha K^\beta$

(b) $x^{-\alpha} = AL^{-\alpha} + BK^{-\alpha}$

17. Prove that in the Cobb-Douglas production function

$$P = A.L^\alpha K^\beta$$

α and β are the elasticities of production with respect to labour and capital respectively

18. If $U = \log(X+a)^\alpha (X+b)^\beta$ is one form of a utility function, find the marginal substitution between the goods X and Y , and deduce the elasticity of substitution is:

$$n = 1 + \frac{b\alpha X + a\beta Y}{(\alpha + \beta)XY}$$

19. Explain the concept of a production function. State in what respects the Cobb-Douglas formulation of 2 production function differs from the more recent formulation of a production function given by Arrow, Chenery, Minhas and Solow.

20. Define the concept of factor substitution

Derive an expression for such an elasticity assuming a linear homogeneous production function

21. Prove that the elasticity of substitution of a Cobb-Douglas production function with constant returns to scale, is unity. Write out another production function in which this elasticity is constant but not necessary equal to unity.

22. The production function of a monopolist is $Q = AX_1^a X_2^b$ and the demand curve function is $Q = KP^d$ where Q is output and X_1 and X_2 are inputs, p is the price of output and p_1 and p_2 are fixed output prices. Derive his equilibrium output, price and profit.

23. What are the properties of a well behaved production function? Examine if $q = Ax_1^a x_2^{1-a}$ is a well behaved production function.

Chapter 11

Applications of Differential Calculus—III

(Theory of Market Equilibrium)

11.1 Multiple Production by Monopolist So far we have considered problems of profit maximisation, taxation and subsidy when producer produced only one product. We shall now consider a market situation in which the producer is a monopolist but does not produce a single product and does not sell his product at a uniform price in all the markets.

Consider a monopolist who produces two related commodities X and Y with known demand and joint cost functions

$$\text{Demand function of } X \quad x = f_1(p_1, p_2)$$

$$\text{Demand function of } Y \quad y = f_2(p_1, p_2)$$

$$\text{Total cost function} \quad C = f_3(x, y)$$

Eqs (1) and (2) represent the quantities of two commodities X and Y which monopolist produces as functions of their respective prices.

The monopolist with these three known functions seeks to adjust the production of each commodity so as to maximise his profits. How to obtain the quantities x and y of commodities X and Y which maximise his profits?

Assuming that he produces x amount of X commodity and y amount of Y commodity and sells at p_1 and p_2 prices respectively, his total revenue, TR , will be,

$$\text{Total revenue } TR = p_1 x + p_2 y$$

$$\text{Total cost (Joint) } TC = C = f_3(x, y)$$

$$\text{His profit will be } \pi = TR - TC$$

$$= \{p_1 x + p_2 y - f_3(x, y)\}$$

We can now easily find the values of x , y , p_1 and p_2 where his profit will be maximum with the help of Order Conditions.

As explained earlier in such types of questions we can either replace p_1 and p_2 by x and y (which would correspond to the method when monopolist

fixes output) or we can replace x and y and p_1 and p_2 (which would correspond to method when monopolist fixes price) in the objective functions π

Let us consider the problems in both these forms

Ex.1. The demand functions of two competitive commodities are given to be $x = 11 - 2p_1 - 2p_2$, and $y = 16 - 2p_1 - 3p_2$

The average cost of production of the commodities are constants, 3 and 1 respectively. Determine prices and quantities that maximise the profit of the monopolist

Suppose the monopolist in question produced x and y units of two commodities which are sold at p_1 and p_2 prices

$$\text{Total revenue} \quad TR = p_1 x + p_2 y$$

$$\text{Total cost} \quad TC = 3x + y$$

$$\begin{aligned} \text{Total profit} \quad \pi &= TR - TC \\ &= (p_1 x + p_2 y) - (3x + y) \end{aligned}$$

Substituting x and y in terms of p_1 and p_2 from given demand functions

$$\begin{aligned} \pi &= p_1(11 - 2p_1 - 2p_2) + p_2(16 - 2p_1 - 3p_2) \\ &\quad - \{3(11 - 2p_1 - 2p_2) + 16 - 2p_1 - 3p_2\} \\ \pi &= 19p_1 + 25p_2 - 4p_1 p_2 - 2p_1^2 - 3p_2^2 - 49 \end{aligned}$$

(i) *First order condition.*

$$\frac{\partial \pi}{\partial p_1} = 19 - 4p_2 - 4p_1 = 0 \quad (1)$$

$$\frac{\partial \pi}{\partial p_2} = 25 - 4p_1 - 6p_2 = 0 \quad \dots(2)$$

Solving Eqns (1) and (2), we get $p_1 = \frac{7}{4}$ and, $p_2 = 3$

To ensure that π is maximum at $\left(\frac{7}{4}, 3\right)$, we apply second order condition

(ii) *Second order condition :*

$$\frac{\partial^2 \pi}{\partial p_1^2} = -4 < 0$$

$$\frac{\partial^2 \pi}{\partial p_2^2} = -6 < 0$$

Hence, this condition is satisfied for maximum value.

Therefore, π is maximum at $\left\{\frac{7}{4}, 3\right\}$

Since $x = 11 - 2p_1 - 2p_2$

$$x = \frac{3}{4} \text{ (on substituting values of } p_1 \text{ and } p_2 \text{)}$$

Again, $y = 16 - 2p_1 - 3p_2 = \frac{7}{2}$

$$\begin{aligned}\pi_{\max} &= (p_1 x + p_2 y) - (3x + y) \\ &= \left\{\frac{21}{8} + \frac{21}{2}\right\} - \left[\frac{9}{2} + \frac{7}{2}\right] = \frac{41}{8} = 5.13\end{aligned}$$

That is, the monopolist should fix the prices at $\frac{7}{4}$ and 3 for two commodities X and Y to maximise his gross profit, which would be equal to 5.13 units

Ex. 2 The demand functions of two commodities x and y are and joint total cost function is $C = 10 + 4x + 2y$

Determine the quantities that maximise the profit of monopolist and also find the maximum profit

Profit $\pi = TR - TC$

$$\begin{aligned}&= (p_1 x + p_2 y) - (10 + 4x + 2y) \\ &= (8 - 2x)x + (14 - y^2)y - (10 + 4x + 2y) \\ &= -y^3 + 12y + 4x - 2x^2 - 10\end{aligned}$$

First order condition

$$\frac{\partial \pi}{\partial x} = 4 - 4x = 0$$

$$\frac{\partial \pi}{\partial y} = -3y^2 + 12 = 0$$

So we get $x = 1$ and $y = 2$

As x cannot be negative we consider the solution $x = 1, y = 2$.

To ensure the π is maximum at (1, 2), we apply second order condition

$$\frac{\partial^2 \pi}{\partial x^2} = -4 < 0$$

$$\frac{\partial^2 \pi}{\partial y^2} = -6y = -6 \times 2 = -12 < 0$$

$$\text{At } x = 1 \text{ and } y = 2, \quad p_1 = 6 \text{ and } p_2 = 12$$

$$\text{and } \pi_{\max} = p_1 x + p_2 y - (10 + 4x + 2y) = 26 - 18 = 8$$

The producer would fix the output levels of commodities X and Y at 1 and 2 respectively to maximise his gross profit, which would be 8 with given price 6 and 10 of the commodities in the market

Ex. 3. A monopolist firm produces two types of chocolate, X_1 and X_2 at constant average costs of Rs 2.50 and Rs 3.00 per kilo respectively. If p_1 and p_2 are prices charged (per kilo) and the market demands are given by

$$x_1 = 5(p_2 - p_1)$$

$$x_2 = 32 + 5p_1 - 10p_2$$

obtain the levels at which prices will be fixed for the two types of chocolates for maximum joint monopoly revenue.

Find also the prices of two types of chocolates fixed by two independent monopolists and show that they are less than the joint monopoly prices.

Suppose the monopolist produces x_1 quantities of X_1 chocolate and x_2 quantity of X_2 chocolate

$$\therefore \text{Total cost of } x_1 = C_{x_1} = AC_{x_1} \times x_1 = 2.50x_1$$

$$\text{Total Revenue of } x_1 = R_{x_1} = p_1 \times x_1 = p_1 \times 5(p_2 - p_1)$$

$$\therefore \text{Profit from } x_1 = R_{x_1} - C_{x_1} = 5p_1(p_2 - p_1) - 2.50x_1$$

$$\pi_{x_1} = 5p_1(p_2 - p_1) - 2.50[5(p_2 - p_1)]$$

$$\pi_{x_1} = 5p_1 p_2 - 5p_1^2 - 12.50p_2 + 12.50p_1$$

$$\text{Again, total cost of } x_2 \text{ is } C_{x_2} = AC_{x_2} \times x_2 = 3x_2$$

$$\text{Total Revenue of } x_2 \text{ is } R_{x_2} = p_2 \times x_2 = p_2(32 + 5p_1 - 10p_2)$$

$$\therefore \text{Profit from } x_2 \text{ is } \pi_{x_2} = R_{x_2} - C_{x_2}$$

$$\tau_{x_1} = p_2(32 + 5p_1 - 10p_2) - 3(32 + 5p_1 - 10p_2)$$

$$\pi_{x_2} = 32p_2 + 5p_1p_2 - 10p_2^2 - 96 - 15p_1 + 30p_2$$

Gross (joint) profit is $\pi = \tau_{x_1} + \tau_{x_2}$

$$\text{That is } \pi = 10p_1p_2 - 2.50p_1 + 49.50p_2 - 5p_1^2 - 10p_2^2 - 96$$

To maximise gross profit

(Assuming that he fixes prices to maximise profit)

$$\frac{\partial \pi}{\partial p_1} = 0 \quad \text{and} \quad \frac{\partial^2 \pi}{\partial p_1^2} < 0$$

$$\frac{\partial \pi}{\partial p_2} = 0 \quad \text{and} \quad \frac{\partial^2 \pi}{\partial p_2^2} < 0$$

$$\frac{\partial \pi}{\partial p_1} = 10p_2 - 2.50 - 10p_1 = 0 \quad (I)$$

$$\frac{\partial \pi}{\partial p_2} = 10p_1 + 49.50 - 20p_2 = 0 \quad (II)$$

Solving Eqns (I) and (II) for p_1 and p_2 , we get,

$$p_1 = 4.45 \text{ and } p_2 = 4.70$$

If there had been 2 monopolists each producing the chocolate types independently, π_{x_1} will have to be maximised assuming that the other monopolist will not change his price, i.e., p_2 will be taken as constant

$$\pi_{x_1} = 5p_1p_2 - 5p_1^2 - 12.5p_2 + 12.50p_1$$

$$\text{so that, } \frac{\partial \pi_{x_1}}{\partial p_1} = 5p_2 - 10p_1 + 12.50 = 0 \quad (III)$$

$$\text{Again, } \pi_{x_2} = 32p_2 + 5p_1p_2 - 10p_2^2 - 96 - 15p_1 + 30p_2$$

Assuming that p_1 remains constant, for maximising π_{x_2} , we get

$$\frac{\partial \pi_{x_2}}{\partial p_2} = 62 + 5p_1 - 20p_2 = 0 \quad \text{---(IV)}$$

Solving Equn (III) and (IV), we get

$$p_1 = 3.2 \text{ and } p_2 = 3.9 \text{ (approx.)}$$

Thus, it can be observed that prices in the present case are less than the joint monopoly prices

EXERCISE II 1

In the following problems the demand functions and joint cost function for 2 commodities X and Y are given. Determine quantities and prices that maximise the profit and find this profit.

(I) Assume that a monopolist fixes output level of commodities and allows prices to vary to maximise his profits

D_x	D_y	Joint cost function
1. $x = 7 - p_1 - p_2$	$y = 6 + p_1 - 2p_2$	$C = 3x + 2y$
2. $x = 1 - p_1 + 2p_2$	$y = 11 + p_1 - 3p_2$	$C = 4x + y$
3. $x = \sqrt{16 + p_1}$	$y = \sqrt{9 - p_2}$	$C = x^2 + 3y^3$
4. $p_1 = 36 - 3x$	$p_2 = 40 - 5y$	$C = 12x + 20y$
5. $p_1 = 100 - 2x_1$	$p_2 = 80 - x_2$	$C = 20(x_1 + x_2)$

(II) Assume that the monopolist fixes prices and allows output to vary

6. $p_1 = 16 - x^2$	$p_2 = 8 - 2y$	$C = 10 + 4x + 2y$
7. $p_1 = 2x - y$	$p_2 = 13 - x - y$	$C = 3x - 2y$
8. $x = p_1 - 12$	$y = p_2 - 15$	$C = 4x - 5y$

9. A monopolist manufactures razors and blades at a constant average cost of Rs 2 per razor and Rs 1 per dozen blades. The demand of the market per week is

$$x_1 = \frac{10}{p_1 p_2} \text{ thousand razors and } x_2 = \frac{20}{p_1 p_2} \text{ thousand dozen blades,}$$

p_1 and p_2 are the prices of per razor and per dozen blades respectively

Show that the monopoly prices, when fixed jointly, are Rs 4 razor and Rs 2 per dozen blades

10. Determine the joint monopoly prices in the above example if the demands are,

$$x_1 = 11 - 2p_1 - 2p_2 \text{ and } x_2 = \frac{67}{4} - 2p_1 - \frac{11}{4}p_2$$

11.2 Discriminating Monopoly

Under this kind of monopoly the firm charges different prices from different customers for the product which it produces to maximise its profit. The basic condition for such price discrimination is that the buyers are so split into different sectors (or markets) that they are not able to transfer the product from one sector to the other.

If a monopolist practises such price discrimination in two markets, his total profit from both the markets will be equal to the difference between his total revenue from the both markets and total cost of production.

Supposing he produces $x = x_1 + x_2$ amount of commodity X and sells x_1 in market I and remaining x_2 in market II so that his total cost of production is $C(x_1 + x_2)$.

$$\text{Total profits} = \pi = TR - TC = R_1(x_1) + R_2(x_2) - C(x_1 + x_2)$$

Where $R_1(x_1)$ is total revenue from market I, and $R_2(x_2)$ is the total revenue from market II.

Applying the order conditions,

First order condition

$$\frac{\partial \pi}{\partial x_1} = \frac{\partial R_1}{\partial x_1} + 0 - \frac{\partial C}{\partial x} = 0 \quad \left(x_1 + x_2 = x \right)$$

$$\frac{\partial R_1}{\partial x_1} - \frac{\partial C}{\partial x} = 0$$

$$\text{So that, } \frac{\partial R_1}{\partial x_1} = \frac{\partial C}{\partial x} \quad \text{or, } MR_1 = MC$$

$$\frac{\partial \pi}{\partial x_2} = \frac{\partial R_2}{\partial x_2} + 0 - \frac{\partial C}{\partial x} = 0$$

$$\text{So that, } \frac{\partial R_2}{\partial x_2} = \frac{\partial C}{\partial x} \quad \text{or, } MR_2 = MC$$

Thus conditions for π to be maximum are $MR_1 = MC = MR_2$.

In other words, MR of each market must be equal, and be also equal to the MC of the total output x .

What does the second order condition explain?

We know the relation between AR and MR

$$MR = AR \left(1 - \frac{1}{e} \right)$$

$$MR = p \left(1 - \frac{1}{e} \right)$$

If e_1 and e_2 , p_1 and p_2 are elasticities and prices in respective market I and II, then since, $MR_1 = MR_2$

$$p_1 \left(1 - \frac{1}{e_1} \right) = p_2 \left(1 - \frac{1}{e_2} \right)$$

$$\text{or, } \frac{p_1}{p_2} = \frac{\left(1 - \frac{1}{e_2} \right)}{\left(1 - \frac{1}{e_1} \right)}$$

If $e_1 > e_2$ i.e., if the demand is more elastic in market I, then $\frac{1}{e_2} > \frac{1}{e_1}$,

$$\text{Therefore, } \left(1 - \frac{1}{e_1} \right) > \left(1 - \frac{1}{e_2} \right), \quad \frac{1 - \frac{1}{e_2}}{1 - \frac{1}{e_1}} < 1$$

$$\text{Which means that } \frac{p_1}{p_2} < 1 \quad \text{or, } p_2 > p_1$$

This means that the monopolist will charge a lower price in market I (which is a comparatively elastic market). Prices will be equal if and only if the elasticities of demand are equal and hence there will be no possibility of price discrimination.

Let us take an example.

Ex 4 Under discriminating monopoly the demand curve of a monopolist is made up of two parts

$$p_1 = 140 - 7q_1 \quad \text{and} \quad p_2 = 90 - \frac{q_2}{2}$$

Total cost curve is given $C = 20 + 2q + 3q^2$, ($q = q_1 + q_2$)

Assuming that the firm maximises profit, determine the prices it will set in the two markets

Let TR_1 and TR_2 represent the total revenues in the markets I and II respectively

$$TR_1 = p_1 q_1 = 140q_1 - 7q_1^2$$

$$MR_1 = \frac{d}{dq} (TR_1) = 140 - 14q_1$$

$$TR_2 = p_2 q_2 = 90q_2 - \frac{q_2^2}{2}$$

$$MR_2 = \frac{d}{dq_2}(TR_2) = 90 - q_2$$

$$MC = \frac{d}{dq}(C) = 2 + 6q$$

Condition for profit maximisation for discriminating monopoly is $MR_1 = MR_2 = MC$, but since MC involves q in its function ($MC = 2 + 6q$) we shall not be able to calculate the value of q_1 and q_2 using this condition directly. We follow the method by which we first obtained the value of q and then of q_1 , q_2 , p_1 and p_2 .

$$MR_1 = 140 - 14q_1 \text{ and } MR_2 = 90 - q_2$$

$$q_1 = 10 - \frac{MR_1}{14}, \text{ and}$$

$$q_2 = 90 - MR_2$$

$$q = q_1 + q_2 = 100 - \frac{MR_1}{14} - MR_2$$

$$q = 100 - \frac{15}{14}(MR) \quad [MR_1 = MR_2 = MR]$$

For profit maximisation $MR = MC$,

$$q = 100 - \frac{15}{14}(MC)$$

$$q = 100 - \frac{15}{14}(2 + 6q) \quad [MC = 2 + 6q]$$

$$q = \frac{685}{52} = 13.17 \text{ units}$$

i.e., for profit maximisation the total output to be sold in the two markets should be 13.17 units

substituting in, $MC = 2 + 6q$

$$MC = 2 + 6\left(\frac{685}{52}\right) = \frac{2107}{26} = 81.04$$

For profit maximisation $MC = MR_1$

$$\text{i.e., } \frac{2107}{26} = 140 - 14q_1$$

$$\text{so that, } q_1 = \frac{1533}{364} = 4.21 \text{ units}$$

$$\text{substituting in } p_1 = 140 - 7p_1$$

$$\text{we obtain } p_1 = 140 - 7\left(\frac{1533}{364}\right) = \frac{5747}{52} = 110.52$$

Similarly, $MC = MR_2$

$$\frac{2107}{26} = 90 - q_2$$

$$\text{so that, } q_2 = \frac{233}{2} = 8.96 \text{ units}$$

$$\text{substituting in } p_2 = 90 - \frac{q_2}{2}$$

$$\text{we obtain, } p_2 = 90 - \frac{233}{52} = \frac{4447}{52} = 85.52$$

$$\text{Check } 1. \quad q_1 + q_2 = q$$

$$2. \quad e_2 > e_1$$

11.3. Duopoly

Duopoly is a two-firm industry hence it has peculiar features of interdependence in the decision making. The reason of the interdependence is that a major policy change on the other part duopolist. Naturally, therefore, price and profit of a duopolist depends upon the action of the other firm in his market.

There are different solutions for determining the price-profit combination of duopolists based upon different set of their behaviour assumptions and nature of product (i.e., Homogeneous or Heterogeneous). We shall deal in main with the following two

- When duopolists are assumed to ignore their interdependence.
- When duopolists make collusion in case of homogeneous products

(a) Ignoring Interdependence (Cournot's Solution in Duopoly)

Two firms produce a homogeneous product so that the demand function is

given by $p = f_1(x_1 + x_2)$, i.e., the price in market is a function of total quantity sold by the duopolist. Let the cost of producing output x_1 of duopolist I be $C_1(x_1)$, while that of duopolist II be $C_2(x_2)$.

Therefore, the total revenue of each duopolist can be obtained as under

$$\text{Duopolist I } R_1 = x_1 \times p = x_1 \times f_1(x_1 + x_2) = R_1(x_1, x_2)$$

$$\text{Duopolist II } R_2 = x_2 \times p = x_2 \times f_1(x_1 + x_2) = R_2(x_1, x_2)$$

The total profits of each duopolist will be equal to his revenue less his cost

$$\pi_1 = R_1(x_1, x_2) - C_1(x_1)$$

$$\pi_2 = R_2(x_1, x_2) - C_2(x_2)$$

Since we assume that each duopolist treats his rival's quantity produced as invariant with respect to his own quantity decision, π_1 can be maximised by treating x_2 as a constant quantity, while π_2 can be maximised by treating x_1 as constant

$$\frac{\partial \pi_1}{\partial x_1} = \frac{\partial R_1}{\partial x_1} - \frac{\partial C_1}{\partial x_1} = 0 \quad (\text{1st order condition})$$

$$\frac{\partial R_1}{\partial x_1} = \frac{\partial C_1}{\partial x_1}, \quad \text{i.e. } MR_1 = MC_1$$

$$\frac{\partial^2 \pi_1}{\partial x_1^2} = \frac{\partial^2 R_1}{\partial x_1^2} - \frac{\partial^2 C_1}{\partial x_1^2} < 0 \quad (\text{2nd order condition})$$

Similarly, we can have conditions for the other duopolist

This shows that each duopolist must equate his MC to MR to maximise his profit. Second order condition requires that at that point (where $MR = MC$), each duopolist's MR must be increasing less rapidly than his MC .

Ex. 5 Suppose there are only two firms in an industry producing a homogeneous product. Market demand function is $p = 140 - 0.6Q$ and the cost functions of the duopolists are, $C_1 = 7Q_1$ and $C_2 = 0.6Q_2^2$. Obtain the equilibrium output of each duopolist ignoring their interdependence.

Profits of duopolists can be obtained as under

$$\pi_1 = pQ_1 - C_1 = [140 - 0.6(Q_1 + Q_2)] Q_1 - 7Q_1$$

$$\pi_2 = pQ_2 - C_2 = [(140 - 0.6(Q_1 + Q_2))] Q_2 - 0.6Q_2^2$$

[Since total output sold Q will be equal to $(Q_1 + Q_2)$]

$$\pi_1 = 133Q_1 - 0.6Q_1^2 - 0.6Q_1Q_2 \quad \text{and,}$$

$$\pi_2 = 140Q_2 - 1.2Q_2^2 - 0.6Q_1Q_2$$

First order condition

$$\frac{\partial \pi_1}{\partial Q_1} = 133 - 1.2Q_1 - 0.6Q_2 = 0 \quad \text{and}$$

$$\frac{\partial \pi_2}{\partial Q_2} = 140 - 2.4Q_2 - 0.6Q_1 = 0$$

Thus we have two equations

$$1.2Q_1 + 0.6Q_2 = 133 \quad (1)$$

$$0.6Q_1 + 2.4Q_2 = 140 \quad (2)$$

The equilibrium output is given by the point of intersection, thus solving them for Q_1 and Q_2 , we obtain,

$$Q_1 = 93.3 \quad \text{and} \quad Q_2 = 35,$$

$$\pi_1 = 5227 \quad \text{and} \quad \pi_2 = 1417$$

The second order conditions are satisfied for the above solutions

$$\frac{\partial^2 \pi_1}{\partial Q_1^2} = -1.2 < 0 \quad \text{and} \quad \frac{\partial^2 \pi_2}{\partial Q_2^2} = -2.4 < 0$$

If mutual collusion is assumed, that is if duopolists recognize their mutual interdependence and agree to act unitedly in order to maximise their total profit jointly, it becomes a case of monopoly

Let us solve the above example with such an assumption

Since gross profits of both together are to be maximised therefore the objective function becomes

$$\pi = \pi_1 + \pi_2$$

$$\pi = (133Q_1 - 0.6Q_1^2 - 0.6Q_1Q_2) + (140Q_2 - 1.2Q_2^2 - 0.6Q_1Q_2)$$

$$\pi = 133Q_1 + 140Q_2 - 0.6Q_1^2 - 1.2Q_2^2 - 1.2Q_1Q_2$$

Now, we have to maximise π with respect to Q_1 and Q_2

First order conditions

$$\frac{\partial \pi}{\partial Q_1} = 133 - 1.2Q_1 - 1.2Q_2 = 0 \quad (3)$$

$$\frac{\partial \pi}{\partial Q_2} = 140 - 2.4 Q_2 - 1.2 Q_1 = 0 \quad (4)$$

Solving (3) and (4) we obtain $Q_1 = 105$ and $Q_2 = 5.8$

Output $Q_1 = 105$ and $Q_2 = 5.8$

Price $p = 140 - 0.6(Q_1 + Q_2) = 73.5$

Profit $\pi_1 = 133Q_1 - 0.6Q_1^2 - 0.6Q_1Q_2 = 6741$

$\pi_2 = 140Q_2 - 1.2Q_2^2 - 0.6Q_1Q_2 = 169.4$

On comparison with the situation when duopolists ignored their interdependence we observe that in the present situation the profits of the 1st duopolist have increased (from 5227 to 6741 = 1514) while that of 2nd duopolist have been reduced (from 1471 to 169.4 = -1301.6)

But still the collusion appears to be profitable to both since the reduction in the profit of 2nd duopolist (=1301.6) is less than the corresponding increase in the profit of 1st duopolist (=1514)

Thus the profit of both together has increased to 6910.4 from 6698 in the previous situation. Naturally therefore mutual distribution of profit will put each of them on higher indifference curve.

EXERCISE 11.2

1 Define the profit of a discriminating monopolist with home and foreign markets. Find out the first order and 2nd order conditions for maximising his profits. Also find out the relation between the prices and elasticities in the two markets.

Given the following demand functions for the separate markets and total cost function of a monopoly supplier what will be the profits, outputs, marginal revenues and elasticities in the two markets?

2 $p_1 = 17 - 2x_1$, $p_2 = 25 - 3x_2$ and $C = 2 + x_1 + x_2$

3 $p_1 = 2 - x_1$, $p_2 = 9 - 6x_2$ and $C = x_1 + x_2$

4 $p_1 = 80 - 5x_1$, $p_2 = 180 - 20x_2$ and $C = 50 + 20(x_1 + x_2)$

5 Examine the conditions in which price discrimination is possible and profitable. Suppose that monopolist is discriminating between two markets. In the first market the elasticity of demand for the first is equal -2, and in the second market it is equal to -10. In which market will the price be higher and why?

6 What conditions must be present for price discrimination to be possible?

How is optimum output determined in each market in case of price discrimination?

7. A profit maximising firm sells its product in two economically isolated market, the demand functions in which are given by

$$p_1 = a_1 q_1 + b_1 \quad \text{and} \quad p_2 = a_2 q_2 + b_2$$

(p_1, p_2 are prices q_1, q_2 quantities and a_1, a_2, b_1 and b_2 are constants)

Will the firm charge different prices in the two markets,

(i) if $b_1 = b_2$ and $a_1 \neq a_2$ (ii) if $a_1 = a_2$ and $b_1 \neq b_2$?

Given the following profits functions (i.e., π_1 and π_2) of each duopolists find the equilibrium output and maximum profit of each duopolist

(When duopolists ignore interdependence)

$$8. \pi_1 = 8Q_1 - Q_1^2 - 2Q_2^2 \quad \text{and} \quad \pi_2 = 10Q_2 - Q_2^2 - 4Q_1$$

$$9. \pi_1 = 12Q_1 - 2Q_1^2 - Q_2 \quad \text{and} \quad \pi_2 = 6Q_2 - Q_2^2 - Q_1$$

10 Find the equilibrium output and profits of Ex. 8 and Ex. 9 when duopolists recognize interdependence

11. Given the following demand and cost functions of two firms I and II determine equilibrium output, profits and prices of each firm

$$p = 90 - 0.5(q_1 + q_2), \quad C_1 = 0.5q_1 \quad \text{and} \quad C_2 = 0.5q_2$$

(a) When interdependence is ignored

(b) When duopolists recognize their mutual interdependence

12. For two producers with same total cost function, $T = ax^2 + bx + c$ and same demand function, $p = \beta - \alpha x$, discuss the equilibrium when producer assumes that the other producer's output is constant, (x = Total output, T = Total cost, p = price of goods and other are constants)

13. State mathematically the Cournot solution of the duopoly problem

14 Assume that the demand and cost functions of duopolists are

$$p_1 = 100 - 2q_1 - q_2 \quad c_1 = 2.5q_1^2$$

$$p_2 = 95 - q_1 - 3q_2 \quad c_2 = 2.5q_2^2$$

and currently established prices and quantities are

$$p_1 = 70, \quad q_1 = 10, \quad p_2 = 55 \quad \text{and} \quad q_2 = 10$$

Examine whether it would be in the interest of the first duopolist to change his price or leave at the current level and why?

Chapter 12

Simple Integration

12.1. We have seen that differentiation is the process of finding out rates of changes. If $Y = f(x)$ is a function of x , we can find the rate of change of y with respect to x by differentiating the function and thereby obtaining $\frac{dy}{dx}$ or $f'(x)$. Thus if the hypothetical total utility function is $U = q^2$, where q is the quantity consumed, then $\frac{dU}{dq} = 2q$ is the rate of change of total utility as given by the differential coefficient of U with respect to q , and is the marginal utility function.

But a more vital question can be asked: if the marginal utility function is $2q$, what is the total utility function? The problem is reversed here, because we are now given the marginal utility function and we are to find the total utility function, or in other words, we are given the change in quantities and we have to calculate the total effects of the change. This reverse process involves summation of the differences. Process of such calculations is called integration.

We know that if total utility function is $f(q) = q^2$, marginal utility function is $f'(q) = 2q$. Conversely, if the M U function is $f'(q) = 2q$, the total utility function is $f(q) = q^2$. The problem is thus to find out the function which on being differentiated produces the given function. If $f'(q)$ is given, our problem is to find out the function $f(q)$ which on differentiating will give $f'(q)$.

Definition. If the differential coefficient of $F(x)$ w.r.t. x is $f(x)$, then an integral of $f(x)$ with respect to x is $F(x)$.

In symbols: If $\frac{d}{dx} [F(x)] = f(x)$, then

$$\int f(x) dx = F(x)$$

Following points need to be remembered:

(i) $\int$ is used to denote the process of integration. In fact this symbol is an elongated 'S' denoting sum.

(ii) The differential symbol ' dx ' is written by the side of the function to be integrated to indicate the independent variable with respect to which the original differentiation was made and with respect to which we are now to integrate. In this way, $\int f(x) dx$ means integration of $f(x)$ w.r.t. x .

Suppose we have to integrate x^n

We know that $\frac{d}{dx}x^{n+1} = (n+1)x^n$

$$\frac{1}{n+1} \frac{d}{dx}x^{n+1} = x^n$$

Since x^n is differential coefficient of $\left(\frac{x^{n+1}}{n+1}\right)$

the integral of x^n is $\frac{x^{n+1}}{n+1}$ In symbols therefore

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

(It may be noted that differentiating $\frac{x^{n+1}}{n+1}$, we obtain x^n)

12.2 Constant of Integration (Indefinite Integration)

Let us have a function $F(x) = \frac{x^{n+1}}{n+1} + C$, then

$$\frac{d}{dx}F(x) = \frac{(n+1)x^n}{n+1} \quad [\text{Because differential coeff of constant is zero}]$$

Hence by the definition of an integral $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

where C is called the constant of integration. Since C can take any constant value, the integral is called *Indefinite Integral*.

Further elucidation of Integration of x^n

(1) $\int x^n dx = \frac{x^{n+1}}{n+1} + C$. On integration, we increase the exponent by 1 and divide by the new exponent thus formed

(2) We cannot find integral of x^{-1} by this formula.

Since $\int x^{-1} dx$ will be equal to $\frac{x^{-1+1}}{-1+1} + C$, which is indeterminate.

In fact, it is equal to $\log x$ (see Rule 3 of 12.3)

(3) If we have to integrate a power term which is in the denominator, we bring it to the numerator

$$\int \frac{1}{x^4} dx = \int x^{-4} dx = \frac{x^{-3}}{-3} + C$$

$$(4) \quad \int ax^n dx = \frac{ax^{n+1}}{n+1} + C$$

12.3 Basic Rules of Integration

Rule 1 Power Rule

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C$$

$$\text{Ex 1} \quad \int x^5 dx = \frac{x^6}{6} + C$$

$$\text{Ex 2} \quad \int x dx = \frac{1}{2} x^2 + C$$

$$\text{Ex 3} \quad \int 1 dx = \int x^0 dx = x + C$$

Rule 2 Exponential Rule

$$\int e^x dx = e^x + C, \text{ since } \frac{d}{dx} e^x = e^x, \text{ and}$$

$$\int a^x dx = \frac{a^x}{\log a}$$

Rule 3 Logarithmic Rule

$$\int \frac{1}{x} dx = \log x + C \text{ since } \frac{d}{dx} (\log x) = \frac{1}{x}$$

Rule 4 Integral of sum and difference

Integral of sum of a number of functions is equal to the sum of their separate integrals

$$\begin{aligned} \text{Ex 4} \quad \int (x^3 - x + 1) dx &= \int x^3 dx - \int x dx + \int 1 dx \\ &= \left(\frac{x^4}{4} + C_1 \right) - \left(\frac{x^2}{2} + C_2 \right) + (x + C_3) = \frac{x^4}{4} - \frac{x^2}{2} + x + C \end{aligned}$$

Note In the final answer we have added up all the constants into a single constant C

$$\begin{aligned}\text{Ex. 5.} \quad \int \left(e^x + \frac{1}{x^2} \right) dx &= \int e^x dx + \int x^{-2} dx \\ &= e^x + \left(\frac{x^{-1}}{-1} \right) + C = e^x - \frac{1}{x} + C\end{aligned}$$

Rule 5. Integral of a multiple

If a function is multiplied by a constant number, this number will remain a multiple of the integral of the function

$$\text{Ex. 6} \quad \int 4x^3 dx = 4 \int x^3 dx = \frac{4x^4}{4} + C = x^4 + C$$

$$\begin{aligned}\text{Ex. 7.} \quad \int \left(5e^x - x^2 + \frac{4}{x} \right) dx \\ = 5 \int e^x dx - \int x^2 dx + 4 \int \frac{1}{x} dx = 5e^x - \frac{x^3}{3} + 4 \log x + C\end{aligned}$$

The correctness of all the results can be easily verified by differentiation of results obtained

12.4. Standard Results : ($+C$ is implied in all the results)

Results are summarized below

$$1 \quad \frac{d}{dx} \left(\frac{x^n}{n+1} \right) = x^n \quad \therefore \quad \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$2 \quad \frac{d}{dx} (\log) = \frac{1}{x} \quad \therefore \quad \int \frac{1}{x} dx = \log_e x$$

$$3 \quad \frac{d}{dx} (e^x) = e^x \quad \therefore \quad \int e^x dx = e^x$$

$$4 \quad \frac{d}{dx} \frac{a^x}{\log a} = a^x \quad \therefore \quad \int a^x = \frac{a^x}{\log a}$$

$$5 \quad \frac{d}{dx} \sin x = \cos x \quad \therefore \quad \int \cos x dx = \sin x$$

$$6 \quad \frac{d}{dx} \cos x = -\sin x$$

$$\int \sin x \, dx = -\cos x$$

$$7 \quad \frac{d}{dx} \tan x = \sec^2 x$$

$$\int \sec^2 x \, dx = \tan x$$

$$8 \quad \frac{d}{dx} \sec x = \sec x \tan x$$

$$\int \sec x \tan x \, dx = \sec x$$

$$9 \quad \frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$\int \operatorname{cosec}^2 x \, dx = -\cot x$$

$$10 \quad \frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x$$

$$11 \quad \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x$$

$$12 \quad \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\int \frac{1}{1+x^2} \, dx = \tan^{-1} x$$

$$13 \quad \frac{d}{dx} \sec^{-1} x = \frac{1}{x \sqrt{x^2-1}}$$

$$\therefore \int \frac{1}{x \sqrt{x^2-1}} \, dx = \sec^{-1} x$$

$$14 \quad \frac{d}{dx} \sin^{-1} \frac{x}{a} = \frac{1}{\sqrt{a^2-x^2}}$$

$$\therefore \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a}$$

$$15 \quad \frac{d}{dx} \frac{1}{a} \tan^{-1} \frac{x}{a} = \frac{1}{a^2+x^2}$$

$$\therefore \int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$16 \quad \frac{d}{dx} \frac{1}{2a} \log \left(\frac{a+x}{a-x} \right) = \frac{1}{(a^2-x^2)}$$

$$\therefore \int \frac{dx}{a^2-x^2} = \frac{1}{2a} \log \left(\frac{a+x}{a-x} \right)$$

$$17 \quad \frac{d}{dx} (\log x + \sqrt{x^2 \pm a^2}) = \frac{1}{\sqrt{x^2 \pm a^2}}$$

$$\therefore \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \log[x + \sqrt{x^2 \pm a^2}]$$

Ex. 8.
$$\int \left(\frac{1}{x^2} + \frac{4}{x\sqrt{x}} + 2 \right) dx$$

$$= \int (x^{-2} + 4x^{-\frac{1}{2}} + 2) \, dx$$

$$\begin{aligned}
 &= \frac{x^{2+1}}{-2+1} + 4 \frac{x^{1/2} + 1}{-1/2+1} + 2 \frac{x^{0+1}}{0+1} + C \\
 &= \frac{x^3}{-1} + 4 \frac{x^{1/2}}{-1/2} + 2x + C \\
 &= -\frac{1}{x} - \frac{8}{\sqrt{x}} + 2x + C
 \end{aligned}$$

Ex 9.
$$\begin{aligned}
 \int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx \\
 &= \int \left(\frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x} \right) dx \\
 &= \int \tan x \sec x dx + \int \cot x \operatorname{cosec} x dx \\
 &= \sec x - \operatorname{cosec} x + C
 \end{aligned}$$

Ex. 10
$$\begin{aligned}
 \int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx &= \int \frac{\cos^2 x - \sin^2 x}{\cos^2 x \sin^2 x} dx \\
 &= \int \left(\frac{1}{\sin^2 x} - \frac{1}{\cos^2 x} \right) dx \\
 &= \int (\operatorname{cosec}^2 x - \sec^2 x) dx = -\cot x - \tan x \\
 &= -\left[\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \right] = -\left[\frac{\cos^2 x + \sin^2 x}{\sin x \cos x} \right] \\
 &= -\frac{1}{\sin x \cos x} = -\operatorname{cosec} x \sec x + C
 \end{aligned}$$

EXERCISE 12.1

1. $\int 4x dx$ 2. $\int 5 dx$

3. $\int \sqrt{t} \, dt$

4. $\int \frac{dx}{2\sqrt{2x^3}}$

5. $\int (x^2 - 4x^2 + x) \, dx$

6. $\int \left(ax^5 + \frac{a}{x} - 5 \right) dx$

7. $\int (3x)^7 \, dx$

8. $\int 6 \sin x \, dx$

9. $\int 5x^{-4} \, dx$

10. $\int \frac{x^4 - 8}{x} \, dx$

11. $\int x(1+x)(1-x) \, dx$

12. $\int \left(1 - \frac{1}{3}x^2 - \frac{1}{2\sqrt{x}} \right) dx$

13. $\int 13x^4 \, dx$

14. $\int \left(5e^x + \frac{3}{x^2} \right) dx$

15. $\int 2x(x^2 + 1) \, dx$

16. $\int \frac{4x^2 + 2 + \sqrt{x}}{x^2} \, dx$

17. $\int \frac{3x^2}{x^2} \, dx$

18. $\int \frac{x^2 - 3x - 2}{x - 1} \, dx$

19. $\int (e^{3x} + 3e^{2x} + x^4) \, dx$

20. $\int \left(\frac{1}{2} \right) \left(\frac{x^3}{3} + 5x^2 - 3x \right) dx$

21. $\int \sqrt{1 + \sin 2x} \, dx$

[Hint : Put $1 = \sin^2 x + \cos^2 x$ and $\sin 2x = 2 \sin x \cos x$]

22. $\int \left(x^2 + \frac{1}{3\sqrt{x}} \right)^2 dx$

23. $\int (2x^3 - 3\sin x + 5\sqrt{x}) \, dx$

24. $\int \frac{(1 - \sin^3 x)}{\sin^2 x} \, dx$

25. $\int \tan^2 x \, dx$

26. $\int \cot^2 x \, dx$

27. $\int \sin^2 \frac{x}{2} \, dx$

28. $\int \left(\frac{2}{1+x^2} - \frac{3}{\sqrt{1-x^2}} \right) dx$

29. $\int a^x \left(1 + \frac{a^{-x}}{x^5} \right) dx$

12.5 Methods of Integration

By Substitution Differentiation is a simple process but integration is often not easy. Many functions may not be integrated at all.

In order to integrate any given function we try to transform the function to our standard form by suitable substitution and then use the standard result.

The substitution has often to be guessed but after ample practice the process becomes easier.

Ex 11 Integrate $(5x + 7)^8$

We shall try to reduce it to the standard form $\int x^n dx$ for this the obvious way is to substitute $y = (5x + 7)$

$$\text{Given integral } I = \int y^8 dx$$

But dx too will have to be changed to dy otherwise integration will not be possible.

$$\text{Since } y = 5x + 7 \quad \frac{dy}{dx} = 5$$

$$\text{or } dy = 5x \, dx \quad dx = \frac{dy}{5}$$

$$\text{Therefore } I = \int y^8 dx = \int y^8 \frac{dy}{5}$$

$$= \frac{1}{5} \int y^8 dy = \frac{1}{5} \frac{y^{8+1}}{8+1} + C = \frac{y^9}{45} + C$$

$$\text{Putting back the value of } y = 5x + 7 \text{ we get } I = \frac{(5x + 7)^9}{45} + C$$

Ex 12 Evaluate $\int x \cos x^2 dx$

We know the integral of $\cos x$ but not of $\cos x^2$ hence substitute y for x^2

$$\text{Let } y = x^2 \quad dy = 2x \, dx$$

$$\text{or } dx = \frac{dy}{2x}$$

So then on substitution $I = \int x \cos x^2 dx$ becomes

$$= \int x \cos y \frac{dy}{2x}$$

$$\begin{aligned}
 &= \frac{1}{2} \int \cos y \, dy = \frac{1}{2} \sin y + C \\
 &= \frac{1}{2} \sin x^2 + C \quad (\text{Since } y = x^2)
 \end{aligned}$$

Ex 13. Evaluate $\int \frac{x \, dx}{2x^2 + 3}$

We clearly observe that in the numerator (x) is equal to $\frac{1}{4} \left[\frac{d}{dx} (2x^2 + 3) \right]$.
Under such cases, when the numerator (after adjustment of constant) is the differential coefficient of the denominator we substitute for the denominator

Here, we suppose $y = 2x^2 + 3$

$$dy = 4x \, dx \quad \text{or} \quad dx = \frac{dy}{4x}$$

Hence,

$$\begin{aligned}
 I &= \int \frac{x \, dx}{2x^2 + 3} \\
 &= \int \frac{x}{y} \cdot \frac{dy}{4x} = \frac{1}{4} \int \frac{dy}{y} \\
 &= \frac{1}{4} \log y + C = \frac{1}{4} \log (2x^2 + 3) + C
 \end{aligned}$$

Ex 14 Integrate $\tan x$

Suppose $I = \int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$

Here again, the numerator is differential coefficient of the denominator

Put $y = \cos x$, $dy = -\sin x \, dx$

$$dx = \frac{-dy}{\sin x}$$

$$\begin{aligned}
 I &= \int -\frac{dy}{\cos x} = - \int \frac{dy}{y} \\
 &= -\log y + C = -\log \cos x + C \\
 &= \log (\cos x)^{-1} + C = \log \left(\frac{1}{\cos x} \right) + C = \log \sec x + C
 \end{aligned}$$

Ex 15 Integrate $\sin(ax+b)$

$$\text{Let } I = \int \sin(ax+b) dx$$

$$\text{Let } y = ax+b \quad dy = a dx \text{ or } dx = \frac{dy}{a}$$

$$\text{Let } I = \int \frac{\sin y dy}{a} = -\frac{1}{a} \cos y + C = -\frac{1}{a} \cos(ax+b) + C$$

$$\text{Ex 16 Evaluate } \int \frac{dx}{a^2+x^2}$$

$$\text{Put } x = ay \quad dx = a dy$$

$$I = \int \frac{a dy}{a^2 + a^2 y^2} = \frac{1}{a} \int \frac{dy}{1+y^2}$$

$$= \frac{1}{a} \tan^{-1} y = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \quad [\text{From standard result 12}]$$

$$\text{Ex 17 Evaluate } \int \frac{dx}{\sqrt{16-9x^2}}$$

$$\text{Let } I = \int \frac{dx}{\sqrt{16-9x^2}} = \int \frac{dx}{\sqrt{4^2-(3x)^2}}$$

$$\text{Put } 3x = 4y \quad 3dx = 4dy \quad dx = \frac{4}{3} dy$$

$$I = \int \frac{\frac{4}{3} dy}{\sqrt{4^2-4y^2}} = \frac{4}{3} \int \frac{dy}{4\sqrt{1-y^2}} = \frac{1}{3} \sin^{-1} y$$

[From standard results 11 and 14]

$$= \frac{1}{3} \sin^{-1} \frac{3x}{4} + C \quad (\text{Since } 3x = 4y \quad y = \frac{3}{4}x)$$

$$\text{Ex 18 Integrate } \frac{x^2}{x^2-2} dx$$

$$\int \frac{x^2}{x^2-2} dx = \int \frac{x^2-2+2}{x^2-2} dx$$

$$\begin{aligned}
 &= \int \left(1 + \frac{2}{x^2 - 2} \right) dx \\
 &= \int dx + 2 \int \frac{1}{x^2 - 2} = dx \\
 &= x + 2 \cdot \frac{1}{2\sqrt{2}} \log \left(\frac{x + \sqrt{2}}{x - \sqrt{2}} \right)
 \end{aligned}$$

Using standard formula 16, we get

$$\int \frac{dx}{(a^2 - x^2)} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| \quad (\text{Here } a = \sqrt{2})$$

Ex. 19. Integrate $\frac{1}{\sqrt{x} + 2}$

Let, $I = \int \frac{dx}{\sqrt{x} + 2}$

Put $\sqrt{x} = u$, so that

$$x = u^2, \quad dx = 2u \, du$$

$$I = \int \frac{2u \, du}{u + 2}$$

$$= 2 \int \frac{u + 2 - 2}{u + 2} du = 2 \int \left(1 - \frac{2}{u + 2} \right) du$$

$$= 2 [u - 2 \log(u + 2)] + C = 2 [\sqrt{x} - 2 \log(\sqrt{x} + 2)] + C$$

Ex. 20. Integrate $x \sqrt[3]{x-2}$

Let, $\sqrt[3]{x-2} = y$

or, $x - 2 = y^3$, and $x = y^3 + 2$

$$\therefore dx = 3y^2 dy$$

Hence, $I = \int (y^3 + 2) \cdot y \cdot 3y^2 dy$

$$= 3 \int (y^6 + 2y^3) dy$$

$$= 3 \left(\frac{y^7}{7} + 2 \frac{y^4}{4} \right) + C = \frac{3}{7} (x-2)^{7/3} + \frac{3}{2} (x-2)^{4/3} + C$$

$$= 3 (x-2)^{7/3} \left[\frac{x-2}{7} + \frac{1}{2} \right] + C$$

$$= \frac{3}{14} (x-2)^{7/3} (2x+3) + C$$

Ex. 21. Integrate $\frac{x^2}{x+2}$

When the numerator is of equal or higher order than the denominator, numerator may be divided by denominator to yield simplified terms. Let us apply this in the present example

$$\begin{aligned} \text{Let, } I &= \int \left[(x-2) + \frac{4}{x+2} \right] dx && \left[\begin{array}{l} (x+2) \overline{) x^2} \quad (x-2) \\ \underline{x^2 + 2x} \\ -2x \\ \underline{-2x - 4} \\ +4 \end{array} \right. \\ &= \int (x-2) dx + \int \frac{4dx}{x+2} \\ &= \frac{x^2}{2} - 2x + 4 \log(x+2) + C \end{aligned}$$

Other Methods of Integration When the denominator is quadratic but not resolvable into linear factors when the quadratic form is $x^2 - x + 1$ and not $x^2 + 1$ i.e. it involves the terms of x also then the previous examples will not provide us any solution. Here we use a device of "completing the square"

Ex. 22 Integrate $\frac{1}{x^2 + x + 1}$

$$\begin{aligned} \text{Now, } x^2 + x + 1 &= x^2 + 2 \cdot \frac{x}{2} + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + 1 \\ x^2 + x + 1 &= \left(x + \frac{1}{2}\right)^2 + 1 - \frac{1}{4} = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \\ I &= \int \frac{1}{x^2 + x + 1} dx = \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} \\ &= \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \end{aligned}$$

so that now we have been able to put I in one of the standard forms, viz.,

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

In our example, a is $\frac{\sqrt{3}}{2}$ and x is $\left(x + \frac{1}{2}\right)$

$$I = \frac{2}{\sqrt{3}} \tan^{-1} \frac{(x + \frac{1}{2})}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}} \tan^{-1} \frac{2}{\sqrt{3}} \left(x + \frac{1}{2} \right) + C$$

Note (i) The coefficient of x^2 should always be made 1 before trying these operations

(ii) If the numerator contains terms of x or x^2 the method of partial fraction should be applied first

Ex 23 Integrate $\frac{1}{6x^2 + 4x - 2}$

Now $6x^2 + 4x - 2 = 6 \left(x^2 + \frac{4}{6}x - \frac{2}{6} \right) = 6 \left(x^2 + \frac{2}{3}x - \frac{1}{3} \right)$

Making it the perfect square

$$\begin{aligned} 6x^2 + 4x - 2 &= 6 \left[x^2 + \frac{2}{3}x + \left(\frac{1}{3} \right)^2 - \left(\frac{1}{3} \right)^2 - \frac{1}{3} \right] \\ &= 6 \left[\left(x + \frac{1}{3} \right)^2 - \frac{1}{9} - \frac{1}{3} \right] = 6 \left[\left(x + \frac{1}{3} \right)^2 - \frac{4}{9} \right] \end{aligned}$$

$$I = \frac{1}{6} \int \frac{dx}{\left(x + \frac{1}{3} \right)^2 - \left(\frac{2}{3} \right)^2}$$

which is the standard form i.e. $I = \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \frac{x-a}{x+a}$

In our example x is $\left(x + \frac{1}{3} \right)$ and a is $\frac{2}{3}$

$$I = \frac{1}{6} \cdot \frac{1}{2 \cdot \frac{2}{3}} \log \frac{\left(x + \frac{1}{3} \right) - \frac{2}{3}}{\left(x + \frac{1}{3} \right) + \frac{2}{3}} = \frac{1}{8} \log \frac{\left(x - \frac{1}{3} \right)}{\left(x + 1 \right)}$$

For solving the above type of questions we have to remember the following three standard results

1 $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \frac{x-a}{x+a}$

2 $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \frac{a+x}{a-x}$

$$3 \quad \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

Ex 24 Integrate $\frac{2x+5}{x^2+4x+5}$

Here the numerator should be first rearranged so that part of it is the differential coefficient of the denominator

We find $\frac{d}{dx}(x^2+4x+5) = 2x+4$ and $(2x+5)$ can be written as $(2x+4+1)$

$$I = \int \frac{2x+4}{x^2+4x+5} dx + \int \frac{dx}{x^2+4x+5}$$

$$= I_1 + I_2$$

$$I_1 = \int \frac{2x+4}{x^2+4x+5} dx, \quad \text{when } x^2+4x+5 = t \text{ and } (2x+4) dx = dt$$

$$= \int \frac{dt}{t} = \log t = \log(x^2+4x+5)$$

$$I_2 = \int \frac{dx}{x^2+4x+5} = \int \frac{dx}{x^2+4x+2^2-2^2+5}$$

$$= \int \frac{dx}{(x+2)^2+1} = \tan^{-1}(x+2)$$

Hence, $I = \log(x^2+4x+5) + \tan^{-1}(x+2) + C$

Ex. 25. Evaluate $\int \frac{dx}{\sqrt{x^2+4x+5}}$

In this question, the denominator is quadratic and irrational. We change it to the standard form.

$$I = \int \frac{dx}{\sqrt{x^2+4x+5}} = \int \frac{dx}{\sqrt{(x+2)^2+1}}$$

which is of the form $\int \frac{dx}{\sqrt{x^2+a^2}} = \log \left\{ x + \sqrt{x^2+a^2} \right\} + C$

In our example, $x = (x+2)$, $a = 1$

$$\begin{aligned} I &= \int \frac{dx}{\sqrt{x^2 + 4x + 5}} = \int \frac{dx}{\sqrt{(x+2)^2 + 1}} \\ &= \log \left[(x+2) + \sqrt{(x+2)^2 + 1} \right] = \log (x+2) + \sqrt{x^2 + 4x + 5} + C \end{aligned}$$

Ex. 26. Evaluate $\int \frac{x+1}{\sqrt{6x^2+4x-2}} dx$

Let, $I = \int \frac{x+1}{\sqrt{6x^2+4x-2}} dx$

$$I = \int \frac{x+1}{\sqrt{6(x^2 + \frac{4}{6}x - \frac{2}{6})}} dx = \frac{1}{\sqrt{6}} \int \frac{x+1}{\sqrt{(x^2 + \frac{2}{3}x - \frac{1}{3})}} dx$$

Differential coefficient of $(x^2 + \frac{2}{3}x - \frac{1}{3}) = 2x + \frac{2}{3}$

$$\begin{aligned} I &= \frac{1}{\sqrt{6}} \int \frac{\frac{1}{2}(2x+2)}{\sqrt{x^2 + \frac{2}{3}x - \frac{1}{3}}} \\ &= \frac{1}{2\sqrt{6}} \int \frac{2x + \frac{2}{3} - \frac{2}{3} + 2}{\sqrt{x^2 + \frac{2}{3}x - \frac{1}{3}}} \\ &= \frac{1}{2\sqrt{6}} \left[\int \frac{2x + \frac{2}{3}}{\sqrt{x^2 + \frac{2}{3}x - \frac{1}{3}}} dx + \int \frac{\frac{4}{3}}{\sqrt{x^2 + \frac{2}{3}x - \frac{1}{3}}} dx \right] \\ &= \frac{1}{2\sqrt{6}} \left[\underbrace{\int \frac{1}{\sqrt{t}} dt}_{\downarrow} + \underbrace{\int \frac{\frac{4}{3}}{\sqrt{x^2 + \frac{2}{3}x + (\frac{1}{3})^2 - (\frac{1}{3})^2 - \frac{1}{3}}} dx}_{\downarrow} \right] \end{aligned}$$

By putting $(x^2 + \frac{2}{3}x - \frac{1}{3}) = t$ *by completing square*

$$= \frac{1}{2\sqrt{6}} \left(\frac{t^{\frac{1}{2}}}{\frac{1}{2}} \right) + \int \frac{\frac{4}{3}}{\sqrt{(x + \frac{1}{3})^2 - (\frac{2}{3})^2}}$$

$$= \frac{1}{2\sqrt{6}} \left[2\sqrt{x^2 + \frac{2}{3}x - \frac{1}{3}} + \frac{4}{3} \log \left(x + \frac{1}{3} + \sqrt{x^2 + \frac{2}{3}x - \frac{1}{3}} \right) \right]$$

$$\left\{ \text{Since } \int \frac{dx}{\sqrt{x^2 - a^2}} = \log [x + \sqrt{x^2 - a^2}] \text{ and, here, } x = x + \frac{1}{3} \quad a = \frac{2}{3} \right\}$$

We have, therefore, to remember the following standard results

$$(i) \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \log |x + \sqrt{x^2 - a^2}|$$

$$(ii) \quad \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right)$$

$$(iii) \quad \int \frac{dx}{\sqrt{x^2 + a^2}} = \log \{x + \sqrt{x^2 + a^2}\}$$

EXERCISE 12.2

Integrate with respect to x

1. $\sqrt{5x+3}$

2. e^{7x}

3. $\frac{1}{(5x+3)^2}$

4. $\sec(2x+3) \tan(2x+3)$

5. $\frac{\log x}{x}$

6. $\frac{(\log x)^2}{x}$

7. $x \sin x^2$

8. $\frac{1}{x \log x}$

9. $\frac{\cos x}{a+b \sin x}$

10. $\cot x$

11. $\tan^3 x \sec^2 x$

12. $\sin^2 x \cos x$

[Hint. Put $\tan x = y$]

13. $\frac{e^{2x}}{1+e^{2x}}$

14. $\frac{(e^x+1)^2}{\sqrt{e^x}}$

15. $\frac{1+\cos x}{x+\sin x}$

16. $\frac{1}{9x^2+4}$

17. $\frac{1}{\sqrt{9-x^2}}$

18. $\frac{4x+5}{2x^2+5x+2}$

19. $\frac{1}{\sqrt{25-x^2}}$

20. $\frac{\sin \sqrt{x}}{\sqrt{x}}$

21. $(x^2+1)^3 \cdot x$

22. $(x^3+x)^2 (3x^2+1)$

23. $\frac{1}{x (\log x)^n}$

24. $\frac{\cos x}{\sin^3 x}$

25. $\frac{x}{(3x^2+5)^2}$

26. $e^{a \sin^{-1} x} \frac{1}{\sqrt{1-x^2}}$ (Put $\sin^{-1} x = t$)

27. $\frac{\cot x}{\log \sin x}$

28. $\frac{1}{x^2+a^2}$ (put $x = a \tan \theta$)

29. $\frac{1}{\sqrt{a^2-x^2}}$ (Put $x = a \sin \theta$
or $x = a \cos \theta$)

30. $4x \sqrt{x^2+8}$

31. $\frac{2x}{x^2+3}$

32. $\frac{x}{3x^2+5}$

33. $\frac{2x^3+1}{x^4+2x}$

34. $\frac{1+2e^{4x}}{1-3e^{2x}}$

35. $\sqrt{a^2-x^2}$ [Put $x = a \sin \theta \quad \therefore dx = a \cos \theta d\theta$

$$\text{and, } I = \int \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta d\theta$$

$$= a^2 \int \left(\frac{1 - \cos 2\theta}{2} \right) d\theta,$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2} \text{ and } \sin^2 \theta = \frac{1 - \cos 2\theta}{2} \text{ and proceed}]$$

36. $e^{\cos x} \sin x$

37. $\sqrt{x^2+1} \cdot x$

38. $\frac{x^2}{\sqrt{1+x^3}}$

39. $(e^x + e^{-x})^2$

40. $\frac{1 + \sin 2x}{\sin^2 x}$

41. $\frac{1-2 \sin x}{\cos^2 x} dx$

42. $\tan^2 x$

43. $\frac{1}{\sin x}$

44. $\frac{\sin^3 x}{\cos^2 x}$

45 $\sin 3x \sin 5x$

[Hint. Use the formulae for $\sin(A \pm B)$ and $\cos(A \pm B)$]

46 $\frac{(\sin x - \cos x)^2}{\sin 2x}$

47 $\frac{(1 + \sqrt{x})^n}{\sqrt{x}}$

48 xe^x

49 $x\sqrt{2x+1}$

50 $x^3\sqrt{x^2-2}$

51 $\frac{\sqrt{x}}{\sqrt{x}+1}$

52 $\frac{x^3}{\sqrt{x^2-1}}$

53 $\frac{1}{\sqrt{x}} \sin \sqrt{x}$

54 $\frac{\cos^2 x}{\sqrt{\sin x}}$

55 $\frac{1}{e^x + e^{-x}}$
[Put $e^x = t$]

56 $\frac{\sqrt{1 + \log x}}{x}$
[Put $1 + \log x = y$]

57 $\sqrt{\sin x \cos x}$

58 $\tan x \sec^2 x$

59 $\sin^2 x \cos x$

60 $\frac{1}{\cos^2 x \sqrt{\tan x - 1}}$

61 $\frac{1}{(1+x^2)} \tan^{-1} x \, dx$

62 $\frac{1}{x^2+9}$

63 $\frac{1}{x \sqrt{1 - \log^2 x}}$

64 $\frac{\sin 2x}{\sqrt{1 + \cos^2 x}}$

65 $\frac{1}{\sqrt{4-x^2}}$

66 $\frac{x}{\sqrt{3-x^4}}$

67 $\frac{1}{x^2+4x+5}$

68 $\frac{1}{\sqrt{2+3x-2x^2}}$

69 $\frac{1}{\sqrt{5-4x-x^2}}$

70 $\frac{1}{5+4\cos x}$

71 $\int \frac{dx}{(x^2-2x-2)}$

72 $\int \frac{dx}{(x^2+2x+2)}$

73 $\int \frac{dx}{(2x^2+4x+3)}$

74 $\int \frac{6x+7}{x^2-x+1} dx$

75 $\int \frac{3x+5}{1-2x-x^2} dx$

76 $\frac{x^2+1}{x^3+1} dx$ [Hint. Put $(x^3+1) = (x+1)(x^2-x+1)$]

77 $\int \frac{dx}{\sqrt{4+8x-5x^2}}$

78 $\int \frac{dx}{\sqrt{x-1-x^2}}$

12.6. Integration by Rational Partial Fractions

It will be shown below that every elementary function cannot be put into standard form. Similar is the case of many rational fractions, some of which can, however, be expressed in the elementary standard forms

A rational fraction is a ratio of the polynomials

$$= \frac{a_0 + a_1 x + a_2 x^2 + \dots}{b_0 + b_1 x + b_2 x^2 + \dots}$$

If the degree of the numerator is lower than that of the denominator then the fraction is called *proper fraction*, otherwise it is called *improper fraction*. If the fraction is an improper one, then by the division of the numerator by the denominator, the fraction can be converted into one of a sum of a non-fraction component and a proper fraction

It can be shown that every rational fraction may be represented as a sum of partial fractions

Let the proper fraction be

$$\frac{F(x)}{f(x)}, \text{ where } F(x) \text{ and } f(x) \text{ are polynomials in } x.$$

Let $f(x)$ be defined as

$f(x) = (x-a)^\alpha (x-b)^\beta (x^2+px+q)^\delta$ then the partial fraction can be represented as :

$$\begin{aligned} \frac{F(x)}{f(x)} &= \frac{A_1}{(x-a)^\alpha} + \frac{A_2}{(x-a)^{\alpha-1}} + \dots + \frac{A_\alpha}{(x-a)} \\ &+ \frac{B_1}{(x-b)^\beta} + \frac{B_2}{(x-b)^{\beta-1}} + \dots + \frac{B_\beta}{(x-b)} \\ &+ \frac{C_1 x + D_1}{(x^2+px+q)^\delta} + \frac{C_2 x + D_2}{(x^2+px+q)^{\delta-1}} + \dots \end{aligned}$$

Note that first all fractions must be converted into *proper fractions* by division.

Following four types of partial fractions of the fraction $\frac{F(x)}{f(x)}$ are possible

Type I : Roots of the denominator $f(x)$ are real and distinct (not repeated); that is $f(x)$ can be put as :

$$f(x) = (x-a)(x-b) \dots \dots \dots$$

then the partial fractions can be put as :

$$\frac{F(x)}{f(x)} = \frac{A}{(x-a)} + \frac{B}{(x-b)} + \dots \dots \dots$$

Type 2 Roots of the denominator $f(x)$ are real, and some of them are multiple (repeated), i.e.,

$$f(x) = (x-a)(x-b)^3 \quad \text{then,}$$

$$\frac{F(x)}{f(x)} = \frac{A}{(x-a)} + \frac{B}{(x-b)^3} + \frac{B_2}{(x-b)^2} + \frac{B_3}{(x-b)}$$

Type 3 The roots of the denominator include some complex distinct (non-repeated) roots, i.e.,

$$f(x) = (x-a)(x-b)^2(x^2+px+q)$$

Then

$$\frac{F(x)}{f(x)} = \frac{A}{(x-a)} + \frac{B_1}{(x-b)^2} + \frac{B_2}{(x-b)} + \frac{Cx+D}{x^2+px+q}$$

Type 4 The roots of the denominator include complex multiple (repeated) roots, i.e.,

$$f(x) = (x-a)(x^2+px+q)^2, \text{ then}$$

$$\frac{F(x)}{f(x)} = \frac{A}{(x-a)} + \frac{B_1x+C_1}{(x^2+px+q)^2} + \frac{B_2x+C_2}{x^2+px+q}$$

Ex. 27. Integrate $\frac{1}{4x^2-9}$

Here $(4x^2-9) = (2x-3)(2x+3)$ So we can write

$$\int \frac{1}{4x^2-9} dx = \int \frac{1}{(2x-3)(2x+3)} \cdot dx$$

It is a proper fraction of the form : $\frac{1}{(x-a)^a(x-b)^b}$ (type I)

We can, therefore, obtain partial fractions :

$$\begin{aligned} \frac{1}{4x^2-9} &= \frac{A}{2x-3} + \frac{B}{2x+3} \\ &= \frac{2Ax+2Bx+3A-3B}{(2x-3)(2x+3)} = \frac{2x(A+B)+3(A-B)}{(4x^2-9)} \end{aligned}$$

Comparing the numerators of L.H.S. and R.H.S., we find

$$1 = 2x(A+B) + 3(A-B)$$

Hence the term of x on R.H.S. = $2(A+B)$ must be equal to that on the L.H.S. = 0

$$\therefore 2(A+B) = 0 \quad \text{or,} \quad A = -B \quad \dots (i)$$

Similarly, $3(A - B) = 1$ or, $3A - 3B = 1$ (ii)

Putting $A = -B$ in (ii) we have

$$6A = 1 \quad \text{or} \quad A = \frac{1}{6} \quad \text{and} \quad B = -\frac{1}{6}$$

$$\begin{aligned} \int \frac{1}{4x^2 - 9} dx &= \int \frac{1}{6(2x - 3)} dx - \int \frac{1}{6(2x + 3)} dx \\ &= \frac{1}{6} \cdot \frac{1}{2} \log(2x - 3) - \frac{1}{6} \cdot \frac{1}{2} \log(2x + 3) \\ &\quad \text{(from standard formula 2 and 16)} \\ &= \frac{1}{12} [\log(2x - 3) - \log(2x + 3)] + C \end{aligned}$$

Ex 28 Integrate $\frac{x}{(x+1)^2(x-1)}$ or $\frac{x}{(x+1)(x^2-1)}$

Here $\frac{x}{(x+1)^2(x-1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x-1)}$ (Type II)

$$\begin{aligned} \frac{x}{D} &= \frac{A(x+1)(x-1) + B(x-1) + C(x+1)^2}{D} \\ &= \frac{A(x^2-1) + Bx - B + C(x^2+2x+1)}{D} \quad \text{(Type II)} \end{aligned}$$

Numerators $x = Ax^2 - A + Bx - B + Cx^2 + 2xC + C$
 $x = x^2(A+C) + x(B+2C) - A - B + C$

Comparing the terms of x^2 , x and without x on L.H.S. and R.H.S. we observe

$$\left. \begin{array}{ll} \text{(i)} & A + C = 0 \\ \text{(ii)} & B + 2C = 1 \\ \text{(iii)} & -A - B + C = 0 \end{array} \right\} \begin{array}{l} A = -C \\ B = 1 - 2C \\ A + B - C = 0 \end{array} \quad \left. \begin{array}{l} A = -\frac{1}{4} \\ B = \frac{1}{2} \\ C = \frac{1}{4} \end{array} \right\}$$

$$\begin{aligned} \int \frac{x}{(x+1)^2(x-1)} dx &= \int \frac{-\frac{1}{4}}{x+1} dx + \int \frac{\frac{1}{2}}{(x+1)^2} dx + \int \frac{\frac{1}{4}}{(x-1)} dx \\ &= -\frac{1}{4} \log(x+1) + \frac{1}{2} \frac{(x+1)^{-1}}{-1} + \frac{1}{4} \log(x-1) + C \\ &= \frac{1}{4} [\log(x-1) - \log(x+1)] - \frac{1}{2(x+1)} + C \end{aligned}$$

Ex. 29. Integrate $\frac{1}{(x^2 + 1)(x - 2)}$

Here suppose $\frac{1}{(x^2 + 1)(x - 2)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 1}$

Note Here $x^2 + 1$ is a quadratic factor which cannot be factorized. The numerator of the partial fraction with $x^2 + 1$ as denominator will be of the form $Bx + C$ (Type III)

$$\frac{1}{D} = \frac{A(x^2 + 1) + (Bx + C)(x - 2)}{D}$$

$$1 = Ax^2 + A + Bx^2 + Cx - 2Bx - 2C$$

$$1 = x^2(A + B) + x(C - 2B) + (A - 2C)$$

Comparing the terms, we have

$$\left. \begin{array}{ll} (i) & A + B = 0 \\ (ii) & C - 2B = 1 \\ (iii) & A - 2C = 1 \end{array} \right\} \begin{array}{l} \text{or, } A = -B \\ \text{or, } C = 2B \end{array} \quad \left. \begin{array}{l} A = -\frac{1}{5} \\ B = -\frac{1}{5} \\ C = -\frac{2}{5} \end{array} \right\} \text{ and}$$

$$\begin{aligned} \text{Therefore } I &= \int \frac{\frac{1}{5}}{x - 2} dx + \int \frac{\frac{1}{5}x - \frac{2}{5}}{(x^2 + 1)} dx \\ &= \frac{1}{5} \log(x - 2) - \frac{1}{5} \int \frac{x}{(x^2 + 1)} dx - \frac{2}{5} \int \frac{1}{x^2 + 1} dx \\ &= \frac{1}{5} \log(x - 2) - \frac{1}{10} \log(x^2 + 1) - \frac{2}{5} \tan^{-1} x \end{aligned}$$

Ex. 30 Integrate: $\frac{x^3 + x - 1}{(x^2 + 2)^2}$

$$\begin{aligned} \text{Here, } \frac{x^3 + x - 1}{(x^2 + 2)^2} &= \frac{B_1 x + C_1}{(x^2 + 2)^2} + \frac{B_2 x + C_2}{(x^2 + 2)} \\ &= \frac{B_1 x + C_1 + (B_2 x + C_2)(x^2 + 2)}{(x^2 + 2)^2} \end{aligned}$$

(It is of type IV)

$$(x^3 + x - 1) = B_1 x + C_1 + B_2 x^3 + C_2 x^2 + 2B_2 x + 2C_2$$

Hence, comparing terms, we have

$$1 = B_2 \qquad B_2 = 1$$

$$0 = C_2 \qquad C_2 = 0$$

$$1 = B_1 + 2B_2 \qquad B_1 = 1 - 2B_2 = -1$$

$$-1 = C_1 + 2C_2 \qquad C_1 = -1$$

$$\text{Therefore given } I = \int \frac{-x-1}{(x^2+2)^2} dx + \int \frac{x}{(x^2+2)} dx$$

$$\text{that is, } I = -\int \frac{x}{(x^2+2)^2} dx - \int \frac{1}{(x^2+2)^2} dx + \int \frac{x}{(x^2+2)} dx$$

$$I = I_1 + I_2 + I_3$$

$$\text{Here } I_1 = \int \frac{x}{(x^2+2)^2} dx \quad \left\{ \begin{array}{l} \text{put } (x^2+2) = t \\ 2x dx = dt \end{array} \right.$$

$$= \frac{1}{2} \int \frac{t^{-2+1}}{-2+1}$$

$$= -\frac{1}{2} \frac{1}{(x^2+2)}$$

$$\text{Similarly, } I_3 = \int \frac{x}{(x^2+2)} dx \quad \left\{ \begin{array}{l} \text{put } (x^2+2) = t \\ 2x dx = dt \end{array} \right.$$

$$= \frac{1}{2} \int \frac{1}{t} dt$$

$$= \frac{1}{2} \log t = \frac{1}{2} \log (x^2+2)$$

$$\text{and, } I_2 = \int \frac{1}{(x^2+2)^2} \cdot dx$$

(This seemingly easy integral is difficult to compute. Proceed with patience)

$$I_2 = \int \frac{1}{(x^2+2)^2} dx = \frac{1}{2} \int \frac{(x^2+2) - x^2}{(x^2+2)^2} dx$$

$$= \frac{1}{2} \int \frac{1}{(x^2+2)} dx - \frac{1}{2} \int \frac{x^2}{(x^2+2)^2} dx$$

$$= \frac{1}{2} \left[\frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} \right] - \frac{1}{2} \int \frac{x^2}{(x^2+2)^2} dx$$

Consider the last integral

$$\int \frac{x^2}{(x^2+2)^2} dx \quad \begin{array}{l} \text{In place of } dx \\ \text{put } \frac{1}{2} d(x^2+2) = x dx \end{array}$$

$$\begin{aligned} \text{Hence, } \int \frac{x^2}{(x^2+2)^2} dx &= \frac{1}{2} \int \frac{x}{(x^2+2)^2} d(x^2+2) \\ &= -\frac{1}{2} \int x d \left[\frac{1}{(x^2+2)} \right] \end{aligned}$$

$$\text{Since } d \left\{ \frac{1}{(x^2+2)} \right\} = -\frac{d(x^2+2)}{(x^2+2)^2}$$

$$\text{differentiating } \frac{1}{x^2+2} \text{ w.r.t. } (x^2+2)$$

Integrating by parts, we obtain (See Section 12.9)

$$\begin{aligned} \int x d \frac{1}{(x^2+2)} &= x \int \frac{dx}{(x^2+2)} - \int \left[\int d \frac{1}{(x^2+2)} dx \right] \\ &= x \frac{1}{(x^2+2)} - \int \frac{dx}{(x^2+2)} \\ &= \frac{x}{x^2+2} - \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} \end{aligned}$$

$$\text{Therefore, } I_2 = \frac{1}{2\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + \frac{1}{4} \cdot \frac{x}{x^2+2} - \frac{1}{4\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}}$$

$$\text{Now } I = I_1 + I_2 + I_3$$

$$= \frac{1}{2} \frac{1}{(x^2+2)} + \frac{1}{2} \log(x^2+2) - \frac{1}{4} \frac{x}{x^2+2} - \frac{1}{4\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C$$

$$= \frac{1}{4(x^2+2)} (2-x) + \log(x^2+2)^{1/2} - \frac{1}{4\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C$$

Ex. 31. Integrate $\frac{x^2 + 4}{(x^2 - 3x + 2)}$

Note The powers of the numerator and denominator are same, hence we must first divide the numerator by the denominator so that the numerator does not contain x^2

$$\frac{x^2 + 4}{x^2 - 3x + 2} = 1 + \frac{3x + 2}{x^2 - 3x + 2} = 1 + \frac{A}{x-1} + \frac{B}{x-2}$$

$$\frac{3x + 2}{x^2 - 3x + 2} = \frac{A(x-2) + B(x-1)}{x^2 - 3x + 2}$$

$$3x + 2 = Ax - 2A + Bx - B$$

So that, $3 = (A + B)$

$$2 = 2A - B$$

Again, $5 = -A \quad A = -5$

and $B = 3 - A = 3 + 5 = 8$

$$\begin{aligned} \text{Hence, } \int \frac{3x + 2}{x^2 - 3x + 2} &= \int \left[1 - \frac{5}{x-1} + \frac{8}{x-2} \right] dx \\ &= x - 5 \log(x-1) + 8 \log(x-2) + C \end{aligned}$$

Ex. 32. Evaluate $\int \frac{dx}{x(x^2 + 1)}$

Here, $I = \int \frac{dx}{x(x^2 + 1)}, \quad \left\{ \begin{array}{l} \text{put } (x^2 + 1) = t \\ 2x \, dx = dt \end{array} \right.$

$$= \int \frac{1}{2t^2} \cdot \frac{dt}{t}$$

$$= \int \frac{1}{2(t-1)t} \cdot dt$$

$$= \frac{1}{2} \int \left[\frac{1}{t-1} - \frac{1}{t} \right] dt$$

$$= \frac{1}{2} \log(t-1) - \frac{1}{2} \log t + C$$

$$= \frac{1}{2} \cdot \log \frac{t-1}{t} = \frac{1}{2} \log \frac{x^2}{x^2 + 1} + C$$

Ex. 33. Evaluate $\int \frac{dx}{x^3+1}$

Here, $x^3+1 = (x+1)(x^2-x+1)$

Hence $I = \int \frac{dx}{x^3+1} = \int \frac{1}{(x+1)(x^2-x+1)} dx$

$$\left\{ \frac{1}{(x+1)} - \frac{1}{(x^2-x+1)} \right\} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1}$$

$$I = Ax^2 - Ax + A + Bx^2 + Cx + Bx + C$$

$$= (A+B)x^2 + (-A+C+B)x + A+C$$

So that,

$$(A+B) = 0, \quad \text{or} \quad A = -B$$

$$-A+B+C = 0, \quad \text{or} \quad -2A+C = 0$$

$$A+C = 1, \quad \text{or} \quad A+C = 1$$

Hence $A = +\frac{1}{3}, B = -\frac{1}{3}$

$$C = 1 - A = \frac{2}{3}$$

$$\begin{aligned} \text{Thus,} \quad I &= \int \frac{1}{3} \cdot \frac{1}{(x+1)} dx - \frac{1}{3} \int \frac{x}{(x^2-x+1)} dx \\ &\quad + \frac{2}{3} \int \frac{1}{x^2-x+1} dx \end{aligned}$$

$$= I_1 + I_2 + I_3$$

Now it can be easily shown that

$$I_1 = \frac{1}{3} \int \frac{1}{x+1} \cdot dx = \frac{1}{3} \log(x+1)$$

$$I_2 = -\frac{1}{3} \int \frac{x}{x^2-x+1} \cdot dx = -\frac{1}{6} \cdot \log(x^2-x+1)$$

$$\text{and, } I_3 = \frac{2}{3} \cdot \int \frac{dx}{(x^2-x+1)} = \frac{2}{3} \int \frac{dx}{\{(x-\frac{1}{2})^2 + \frac{3}{4}\}}$$

$$= \frac{2}{3} \frac{1}{\sqrt{3/4}} \tan^{-1} \frac{(x + \frac{1}{2})}{\sqrt{3/4}}$$

$$= \frac{4}{3\sqrt{3}} \tan^{-1} \frac{2x + 1}{\sqrt{3}}$$

Hence $\int \frac{1}{(x^2 + 1)} dx = \frac{1}{6} \log \frac{(x+1)^2}{(x^2 - x + 1)} + \frac{4}{3\sqrt{3}} \tan^{-1} \frac{2x + 1}{\sqrt{3}} + C$

Ex 34 Evaluate $\int \frac{dx}{x [6 (\log x)^2 + 7 \log x + 2]}$

(Put $\log x = t$ or, $\frac{1}{x} dx = dt$)

$$I = \int \frac{dt}{6t^2 + 7t + 2} = \frac{dt}{(3t + 2)(2t + 1)}$$

$$= \int \left\{ \frac{A}{(3t + 2)} + \frac{B}{(2t + 1)} \right\} dt$$

$$= \int \frac{2At + A + 3Bt + 2B}{D} dt$$

Hence, $2A + 3B = 0$ or, $A = -\frac{3B}{2}$

and, $A + 2B = 1$, or, $\frac{-3B}{2} + 2B = 1$

or, $\frac{B}{2} = 1, \therefore B = 2$

Therefore, $A = -\frac{3 \times 2}{2} = -3$

Hence, $I = \int \frac{-3}{3t + 2} dt + 2 \int \frac{dt}{(2t + 1)}$

$$= \frac{3}{3} \log (3t + 2) + \frac{2}{2} \log (2t + 1)$$

$$= \log \left[\frac{2 \log x + 1}{3 \log (x + 2)} \right] + C$$

Ex. 35. Integrate $\frac{\sin 2x}{(\sin x + 1)(\sin x + 2)(\sin x + 3)}$

$$= \frac{2 \sin x \cos x}{(\sin x + 1)(\sin x + 2)(\sin x + 3)}$$

Putting $\sin x = t$, $\cos x \, dx = dt$

$$I = \int \frac{2t \, dt}{(t+1)(t+2)(t+3)}$$

Proceed by taking partial fraction

Result: $I = \log \frac{(\sin x + 2)^4}{(\sin x + 1)(\sin x + 3)^3}$

EXERCISE 12.3

Integrate

1. $\frac{x}{x+2}$
2. $\frac{1-x}{1+x}$ (Write it as $\frac{-(x+1)+2}{x+1}$, then divide)
3. $\frac{x^3}{9+x}$
4. $\frac{3x^2}{x+1}$
5. $\frac{x^2}{x+2}$
6. $\frac{x+35}{x^3-25}$
7. $\frac{1}{x^2-a^2}$
8. $\frac{2x-1}{(x-1)(x-2)}$
9. $\frac{x}{(x^2+1)(x-1)}$
10. $\frac{x^2+2}{(x+1)^3(x-2)}$
11. $\frac{x^5+x^4-8}{x^3-4x}$
12. $\frac{1}{\sin x + \sin 2x}$
13. $\frac{x-8}{x^3-4x^2+4x}$
14. $\frac{3x+2}{x(x+1)^3}$
15. $\frac{x^4+4x^3+11x^2+12x+8}{(x^2+2x+3)^2(x+1)}$
16. $\frac{x^2}{x^2+3x+2}$
17. $\frac{x^2-4}{x^2+3x+2}$
18. $\frac{x^3-2x^2-1}{(x^2-1)}$
19. $\frac{2x^3-2x^2-11x-8}{x^2-x-6}$
20. $\frac{2x+3}{x^2+2x-3}$
21. $\frac{1}{9x^2-16}$
22. $\frac{2x+1}{x^2-2x+1}$
23. $\frac{x+2}{3x^2+5x+2}$
24. $\frac{1}{x(x^2-1)}$
25. $\frac{1}{x^3(x+3)}$

- 26 $\frac{x^2 + 4x + 3}{x^2 + 6x^2 + 9x + 8}$ 27 $\frac{1}{x^2 + 4x + 3}$ 28 $\frac{3x}{(3x + 1)(x - 3)}$
- 29 $\frac{1}{x(x^2 - 1)}$ 30 $\frac{xe^x}{(1 + x^2)}$
- 31 $\frac{2x - 3}{(x^3 - 3x + 2)} + \frac{1}{(x - 1)(x - 2)}$ 32 $\frac{1}{x(x + 1)}$
- 33 $\frac{3x - 2a}{x^4 - ax^3}$ 34 $\frac{5x + 2}{x^2 + 2x + 10}$ 35 $\frac{11x + 6}{(x - 1)(x + 2)^2}$
- 36 $\frac{1}{x^4 - 1}$ 37 $\frac{x^2}{\sqrt{x^2 + 2}}$ 38 $x^2 \sqrt{4 - x^2}$
- 39 $\frac{x + 5}{(x + 1)(x + 2)^2}$ 40 $\frac{x^2 - 1}{(x^2 + 1)(x^2 + 2)(x^2 + 3)}$

12.9 Integration by Parts

Integration of a product of the functions U and V

$$\int U V \, dx = U \int V \, dx - \int \left[\frac{d}{dx} U \int V \, dx \right] dx$$

that is $\int U V \, dx =$ First function $\times$ Integral of the Second function minus
Integral of (Differential coefficient of the first function $\times$
Integral of the second function)

Remember that either of U and V can be taken as the first function. But it is to be noted that the second function is to be integrated and first is to be differentiated. Hence following points should be kept in mind

1. Second function should be such that it is easy to integrate
2. First function should be such that it is easy to differentiate and makes the resultant integration on the R.H.S. easier than the original one
3. Sometimes unity may be taken as second function it cannot be taken as first function since its differential coefficient is zero
4. The product formula can be applied more than once if necessary

The rule of integration by parts is widely used. Integration of the form

$$\begin{array}{lll} x^n \sin bx, & x^n \cos bx, & \\ e^{ax} \sin bx, & x^n e^{bx}, & x^n \log x \end{array}$$

and some expressions containing inverse trigonometric functions can be attempted by means of the rule of integration by parts

Ex 36 Integrate $x \sin x$

$$\text{Let } I = \int x \sin x \, dx$$

Suppose we take $\sin x$ as the first function and x as the second function

$$\begin{aligned} I &= \int \sin x \cdot x \, dx \\ &= \sin x \int x \, dx - \int \left[\frac{d}{dx} \sin x \int x \, dx \right] dx \\ &= \frac{x^2}{2} \sin x - \int \cos x \cdot \frac{x^2}{2} \, dx \end{aligned}$$

Now here $\left(\frac{x^2}{2} \cos x \right)$ itself is a product of $\frac{x^2}{2}$ and $\cos x$ which is more difficult to integrate than the original given function. Hence our selection of the first and second functions is not proper.

Let us now take (x) to be the first function and $(\sin x)$ as the second function.

$$\begin{aligned} I &= \int x \sin x \, dx \\ &= x \int \sin x \, dx - \int \left[\frac{d}{dx} x \int \sin x \, dx \right] dx \\ &= -x \cos x + \int 1 \cdot \cos x \, dx = -x \cos x + \sin x + C \end{aligned}$$

Ex 37 Evaluate $I = \int x^2 \log x \, dx$

We do not know integral of $\log x$. We therefore take it to be the first function

$$\begin{aligned} I &= \int \log x \cdot x^2 \, dx = \log x \int x^2 \, dx - \int \left[\frac{d}{dx} \log x \int x^2 \, dx \right] dx \\ &= \frac{x^3}{3} \log x - \int \frac{x^2}{3} \, dx = \frac{x^3}{3} \log x - \frac{x^3}{9} + C \end{aligned}$$

Ex 38 Integrate $\sin^{-1} x$

$$\text{Let } I = \int \sin^{-1} x \, dx = \int \sin^{-1} x \cdot 1 \, dx$$

$$= \sin^{-1} x \int dx - \int \left[\frac{d}{dx} \sin^{-1} x \int dx \right] dx$$

$$= x \sin^{-1} x - \int \frac{1}{\sqrt{1-x^2}} x \, dx$$

$$\text{Now for } \int \frac{x}{\sqrt{1-x^2}} \, dx \quad \text{put } 1-x^2 = t \quad x \, dx = -\frac{dt}{2}$$

$$\int \frac{x}{\sqrt{1-x^2}} \, dx = \int \frac{-\frac{dt}{2}}{\sqrt{t}} = -\frac{1}{2} \int t^{-\frac{1}{2}} \, dt = -\frac{1}{2} \cdot \frac{t^{\frac{1}{2}}}{\frac{1}{2}}$$

$$= -\sqrt{1-x^2}$$

$$I = x \sin^{-1} x + \sqrt{1-x^2} + C$$

Ex 39 Evaluate $I = \int e^{ax} \sin bx \, dx$

$$\int e^{ax} \sin bx \, dx = e^{ax} \int \sin bx \, dx - \int \left[\frac{d}{dx} e^{ax} \int \sin bx \, dx \right] dx$$

[Always take e^{ax} as the first term]

$$= \frac{-e^{ax} \cos bx}{b} - \int a e^{ax} \left(-\frac{\cos bx}{b} \right) dx$$

$$= \frac{-e^{ax} \cos bx}{b} + \frac{a}{b} \int e^{ax} \cos bx \, dx \quad (1)$$

As $\int e^{ax} \sin bx$ leads to $\int e^{ax} \cos bx$ and $\int e^{ax} \cos bx$ will lead to $\int e^{ax} \sin bx$, so that

$$\int e^{ax} \cos bx = e^{ax} \frac{\sin bx}{b} - \int a e^{ax} \frac{\sin bx}{b} \, dx$$

$$= e^{ax} \frac{\sin bx}{b} - \frac{a}{b} \int e^{ax} \sin bx \, dx \quad (2)$$

Substitute (2) in (1), so we get ,

$$\begin{aligned} I &= \int e^{ax} \sin bx \\ &= \frac{-e^{ax} \cos bx}{b} + \frac{a}{b} \left[e^{ax} \frac{\sin bx}{b} - \frac{a}{b} \int e^{ax} \sin bx \, dx \right] \\ \therefore \int e^{ax} \sin bx &= \frac{-e^{ax} \cos bx}{b} + \frac{a}{b^2} e^{ax} \sin bx - \frac{a^2}{b^2} I \end{aligned}$$

Now bringing I to L.H.S., we have

$$\begin{aligned} I + \frac{a^2}{b^2} I &= e^{ax} \left[\frac{-\cos bx}{b} + \frac{a}{b^2} \sin bx \right] \\ I \left[1 + \frac{a^2}{b^2} \right] &= \frac{e^{ax}}{b^2} [-b \cos bx + a \sin bx] \\ I \left[\frac{b^2 + a^2}{b^2} \right] &= \frac{e^{ax}}{b^2} [a \sin bx - b \cos bx] \end{aligned}$$

$$\text{That is, } I = \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx] + C$$

Ex. 40. Evaluate $I = \int x^3 \cos x \, dx$

$$\begin{aligned} \int x^3 \cos x \, dx &= x^3 \sin x - \int 3x^2 \sin x \, dx \quad (\text{Take } x^3 \text{ as first term}) \\ &= x^3 \sin x - 3 \left[-x^2 \cos x - \int -2x \cos x \, dx \right] \\ &= x^3 \sin x + 3x^2 \cos x - 6 \int x \cos x \, dx \\ &= x^3 \sin x + 3x^2 \cos x - 6 \left[x \sin x - \int 1 \cdot \sin x \, dx \right] \\ &= x^3 \sin x + 3x^2 \cos x - 6x \sin x + 6(-\cos x) \\ &= x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + C. \end{aligned}$$

Ex. 41. Evaluate $\int \sqrt{a^2 - x^2} \, dx$

Multiply and divide the given integrand by $\sqrt{a^2 - x^2}$

$$\begin{aligned} I &= \int \sqrt{a^2 - x^2} \, dx = \int \frac{a^2 - x^2}{\sqrt{a^2 - x^2}} \, dx \\ &= \int \frac{a^2}{\sqrt{a^2 - x^2}} \, dx - \int \frac{x^2}{\sqrt{a^2 - x^2}} \, dx \\ &= I_1 - I_2 \end{aligned}$$

$$I = a^2 \sin^{-1} \left(\frac{x}{a} \right) - \int x \frac{x}{\sqrt{a^2 - x^2}} \, dx$$

Now $\int x \frac{x}{\sqrt{a^2 - x^2}} \, dx$ (integrate by parts)

Take x as the first term,

$$\int x \frac{x}{\sqrt{a^2 - x^2}} = x \int \frac{x}{\sqrt{a^2 - x^2}} \, dx - \int \left[1 \int \frac{x}{\sqrt{a^2 - x^2}} \, dx \right]$$

[Put $t = (a^2 - x^2)$, $dt = -2x \, dx$]

$$\int \frac{x}{\sqrt{a^2 - x^2}} \, dx = -\frac{1}{2} \int \frac{dt}{\sqrt{t}} = -\frac{1}{2} \int t^{-\frac{1}{2}} \, dt$$

$$= -\frac{1}{2} (-2) t^{\frac{1}{2}} = t^{\frac{1}{2}}$$

$$I_2 = \int \frac{x^2}{\sqrt{a^2 - x^2}} \, dx = x \sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2} \, dx$$

that is $I = \int \sqrt{a^2 - x^2} \, dx = a^2 \sin^{-1} \frac{x}{a} + x \sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2} \, dx$

Transferring the integral from right to left hand side, we finally get

$$I = \frac{1}{2} a^2 \sin^{-1} \frac{x}{a} + \frac{x}{2} \sqrt{a^2 - x^2} + C$$

(Try directly by taking $\sqrt{a^2 - x^2}$ as the first function)

Ex 42 Evaluate $I = \int x^2 \cos 2x \, dx$

Let $U = x^2$ and $V = \cos 2x$

$$I = \int x^2 \cos 2x \, dx = x^2 \int \cos 2x \, dx - \int \left[2x \int \cos 2x \, dx \right]$$

$$\text{Now} \quad \int \cos 2x \, dx = \frac{1}{2} \sin 2x$$

$$I = \frac{1}{2} x^2 \sin 2x - \int x \sin 2x \, dx$$

$$= \frac{x^2}{2} \sin 2x - \frac{x}{2} (-\cos 2x) - \frac{1}{2} \int \cos 2x \, dx$$

$$= \frac{x^2}{2} \sin 2x + \frac{x}{2} (-\cos 2x) - \frac{1}{2} \int \cos 2x \, dx$$

$$= \frac{x^2}{2} \sin 2x + \frac{x}{2} \cos 2x - \frac{1}{2} \cdot \frac{1}{2} \sin 2x$$

$$= \frac{1}{4} [2x^2 \sin 2x - \sin 2x + 2x \cos 2x] + C$$

Ex 43 Find $I = \int x^n \log x \, dx$

Let the first function be $\log x$.

$$I = \int \log x \cdot x^n \, dx = \log x \cdot \frac{x^{n+1}}{n+1} - \int \frac{1}{x} \cdot \frac{x^{n+1}}{n+1} \, dx$$

$$= \frac{1}{n+1} x^{n+1} \log x - \int \frac{x^n}{n+1} \, dx$$

$$= \frac{1}{n+1} x^{n+1} \log x - \frac{x^{n+1}}{(n+1)^2} + C$$

$$= \frac{x^{n+1}}{n+1} \left[\log x - \frac{1}{n+1} \right] + C$$

Ex. 44. Obtain $I = \int e^x \frac{(1 + \sin x)}{(1 + \cos x)} \cdot dx$

$$\text{Given : } I = \int e^x \left\{ \frac{1 + 2 \sin \frac{x}{2} \cdot \cos \frac{x}{2}}{2 \cdot \cos^2 \frac{x}{2}} \right\} \cdot dx$$

$$= \int \left[\frac{1}{2} e^x \cdot \sec^2 \frac{x}{2} + e^x \cdot \frac{\sin \frac{x}{2} \cdot \cos \frac{x}{2}}{\cos^2 \frac{x}{2}} \right] dx$$

$$= \int \frac{1}{2} e^x \cdot \sec^2 \frac{x}{2} dx + \int e^x \tan \frac{x}{2} dx$$

$$\text{Now } I = I_1 + I_2$$

Integrating I_2 by parts

$$I_2 = \int e^x \tan \frac{x}{2} dx \quad (\text{take } \tan \frac{x}{2} \text{ to be the first function})$$

$$I_2 = \int \tan \frac{x}{2} e^x dx$$

$$= \tan \frac{x}{2} e^x - \int \frac{1}{2} \sec^2 \frac{x}{2} \cdot e^x dx$$

$$= \tan \frac{x}{2} \cdot e^x - I_1$$

Putting in I, we obtain :

$$I = I_1 + e^x \cdot \tan \frac{x}{2} - I_1$$

$$\text{Therefore, } I = e^x \tan \frac{x}{2} + C$$

Ex. 45 Integrate : $\int \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} \cdot dx$

$$[\text{Let } \tan^{-1} x = \theta \quad \tan \theta = x \text{ and } \sec^2 \theta d\theta = dx]$$

$$\begin{aligned}
 \therefore I &= \int \frac{\tan \theta \cdot \theta}{(1 + \tan^2 \theta)^{3/2}} \cdot \sec^2 \theta \cdot d\theta \\
 &= \int \frac{\theta \cdot \tan \theta}{(\sec^2 \theta)^{3/2}} \cdot \sec^2 \theta \cdot d\theta \\
 &= \int \theta \cdot \frac{\tan \theta}{(\sec \theta)} \cdot d\theta \\
 &= \int \theta \cdot \frac{\sin \theta}{\cos \theta \sec \theta} \cdot d\theta \\
 &= \int \theta \sin \theta \cdot d\theta
 \end{aligned}$$

Integrating by parts, taking θ to be the first function :

$$I = \theta \cdot (-\cos \theta) + \int \cos \theta \cdot d\theta$$

$$I = -\theta \cos \theta + \sin \theta$$

$$I = -\tan^{-1} x \cdot \frac{1}{\sqrt{1+x^2}} + \frac{x}{\sqrt{1+x^2}} + C$$

$$\left[\text{since } -\cos \theta = \frac{1}{\sqrt{1-\tan^2 \theta}} = \frac{1}{\sqrt{1-x^2}} \text{ and, } \sin \theta = \frac{\tan \theta}{\sqrt{1-\tan^2 \theta}} = \frac{x}{\sqrt{1-x^2}} \right]$$

EXERCISE 12.4

Integrate the following :

- | | | |
|-------------------------|----------------------|----------------------------|
| 1. $x^2 e^x$ | 2. $x^2 \sin x$ | 3. $x^3 \log x$ |
| 4. $\frac{\log x}{x^2}$ | 5. $e^x \cos x$ | 6. $e^{mx} \cos (bx + c)$ |
| 7. $e^x \sin x$ | 8. $x^2 \log x$ | 9. $x^3 e^x$ |
| 10. $\tan^{-1} x$ | 11. $x \tan^{-1} x$ | 12. $x \sin x \cos x$ |
| 13. $\sqrt{x} \log x$ | 14. $x^2 (\log x)^2$ | 15. $\frac{xe^x}{(x+1)^2}$ |

16. $e^x (\sin x + \cos x)$

17. $x^3 e^{x^2}$

18. $e^x \sin x$

19. $(\log x)^3$

20. $e^{-4x} \cos 2x$

21. $x e^{2x}$

22. $3x^2 e^{-4x}$

23. $e^{-4x} \cos x$

24. $\log(3x+1)$

25. $3x^2 (\log x)^2$

26. $x^3 e^{2x}$

27. $\frac{1}{x \log x}$

28. $\frac{\log x}{x}$

29. $(\log x)^3$

30. $\frac{x}{\cos^2 x}$

31. $x \cos^2 x$

32. $\log(x^2+1)$

33. $\frac{x^2}{(x \sin x + \cos x)^2}$

34. $\frac{\sqrt{a^2-x^2}}{x^2}$

35. $x \tan^{-1} \sqrt{x^2-1}$

36. $\log x + \sqrt{1+x^2}$

37. $(\log x)^2$

38. $\left(\frac{1+x \log x}{x}\right) e^x$

39. $\frac{1}{\sqrt{x}} \sin^{-1} \sqrt{x}$

40. $\sin^{-1} \sqrt{\frac{x}{1+x}}$

Miscellaneous Problems

41. $\frac{\sqrt{1+x}}{x}$

42. $\frac{1}{\sqrt{x}(1-x)}$

43. $\sqrt{\frac{1-x}{1+x}}$

44. $\frac{1}{(\sin x + \cos x)^2}$

45. $\frac{e^x - 2}{e^{2x} + 4}$

46. $\frac{\tan^{-1} x}{x^3}$

47. $\frac{1}{\sqrt{1-x} + \sqrt{2x}}$

48. $\frac{\tan e^{-\sqrt{x}}}{x^2}$

Chapter 13

Definite Integrals and Applications

13.1 Definition

Consider the function $y = f(x) = x$

Integrate and get the indefinite integral as explained in the previous chapter

$$\int f(x) dx = \int x dx = \frac{x^2}{2} + C$$

Suppose we are required to find the area of the portion lying between the given curve $y = x$ and the ordinates erected at $x = a$ and $x = b$ ($a < b$) in other words we want to find the area $ABCD$ of Fig 13.1

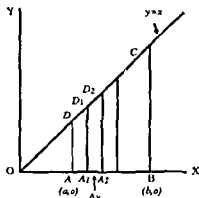


Fig 13.1

Let us divide the internal area from $x = a$ to $x = b$ into n equal parts, each equal to Δx so that $AA_1 = A_1A_2 = A_2A_3 = A_3A_4 = \Delta x$, and

$$\Delta x = \frac{b-a}{n} \text{ or } n \Delta x = (b-a) \quad (1)$$

If these parts are more in number Δx will be small and AA_1D_1D will be approximately rectangle area of which $AA_1D_1D = y \Delta x = a \Delta x$ (here $y = x = a$)

$$\text{Similarly } A_1A_2D_2D_1 = (y + \Delta y) \Delta x = (a + \Delta x) \Delta x$$

$$\text{Thus we have area } AA_1D_1D = a \Delta x$$

$$\text{area } A_1A_2D_2D_1 = (a + \Delta x) \Delta x$$

$$\text{area } A_2A_3D_3D_2 = (a + 2\Delta x) \Delta x \text{ and so on}$$

$$\text{Area } ABCD = a \Delta x + (a + \Delta x) \Delta x + (a + 2\Delta x) \Delta x + \dots + [a + (n-1) \Delta x] \Delta x$$

$$\begin{aligned}
 \text{i.e., Area } ABCD &= \Delta x \left[a + (a + \Delta x) + (a + 2\Delta x) + \dots + a + (n-1)\Delta x \right] \\
 &= \Delta x \left[na + \{1 + 2 + 3 + \dots + (n-1)\} \Delta x \right] \\
 &= \Delta x \left[na + \frac{n(n-1)}{2} \Delta x \right] \\
 &= a.n \Delta x + \frac{n^2 (\Delta x)^2}{2} - \frac{n}{2} (\Delta x)^2
 \end{aligned}$$

But $n \Delta x = (b-a)$ from (i)

$$\text{Area } ABCD = \left[a(b-a) + \frac{1}{2} (b-a)^2 - \frac{1}{2} (b-a) \Delta x \right]$$

Also from Fig 13.1 it is clear that as $n \rightarrow \infty$, $\Delta x \rightarrow 0$

$$\begin{aligned}
 \text{Area } ABCD &= \lim_{\Delta x \rightarrow 0} \left[a(b-a) + \frac{1}{2} (b-a)^2 + \frac{1}{2} (b-a) \Delta x \right] \\
 &= a(b-a) + \frac{1}{2} (b-a)^2 = ab - a^2 + \frac{1}{2} b^2 + \frac{1}{2} a^2 - ab \\
 &= \frac{1}{2} b^2 - \frac{1}{2} a^2 = \frac{1}{2} (b^2 - a^2)
 \end{aligned}$$

$$\text{Area of } ABCD = \frac{1}{2} (b^2 - a^2) \quad (ii)$$

$$\text{while we obtained} \quad \int f(x) dx = \frac{x^2}{2} + C \quad (iii)$$

Both these results have been derived from the same function $y = f(x) =$

Result (ii) is said to be the definite integral of (iii)

The definite integral is defined as the limit of sum of infinitesimal terms

$$\begin{aligned}
 &a, \Delta x, (a + \Delta x), \Delta x, \\
 \text{or,} &y, \Delta x, (y + \Delta y), \Delta x, \\
 \text{or,} &f(x), \Delta x, f(x + \Delta x), \Delta x, \\
 \text{i.e.,} &\lim_{n \rightarrow \infty} \sum_{r=1}^n f(x_r) (\Delta x_r)
 \end{aligned}$$

To specify that the summation extends within the limits $x = a$ to $x = b$, we write definite integral as

$$\int_a^b f(x) dx = \left[F(x) \right]_a^b = F(b) - F(a)$$

$$\text{Thus} \quad \int_a^b x dx = \left[\frac{x^2}{2} + C \right]_a^b = \left[\frac{b^2}{2} + C \right] - \left[\frac{a^2}{2} + C \right]$$

$$= \frac{b^2}{2} - \frac{a^2}{2} + C - C$$

$$= \frac{1}{2} (b^2 - a^2), \text{ which is the same result as}$$

obtained in (ii)

Summarizing If x is the differential coefficient of $F(x)$, we say that $\int f(x) dx = F(x) + C$ is an *indefinite integral*, because C is unknown here. But if integral of $f(x)$ exists for all values of x from a to b , we define

$$\int_a^b f(x) dx = F(b) - F(a), \text{ or } \left[F(x) \right]_a^b$$

as a *definite integral* of $f(x)$ between limits a and b , a is the lower limit and b is the upper limit

Important steps in evaluating a definite integral : $\int_a^b f(x) dx$

- (1) Find the indefinite integral $\int f(x) dx = F(x) + C$
- (2) Substitute for x in this integral the upper limit b , i.e., $= F(b) + C$
- (3) Substitute for x in the lower limit a , i.e., $= F(a) + C$,
- (4) Subtract (2) from (3) i.e., $F(a) + C$ from $F(b) + C$, so that

$$\int_a^b f(x) dx = F(b) - F(a)$$

Following examples will clarify the above steps

Ex 1. Find the value of

$$(i) \int_0^{\frac{\pi}{2}} \sin x dx, \quad (ii) \int_1^2 \frac{dx}{x(x+1)}, \quad (iii) \int_2^3 \frac{6x^2 + 1}{\sqrt{2x^3 + x - 2}} dx$$

$$(i) \int_0^{\frac{\pi}{2}} \sin x dx = \left[F(x) \right]_0^{\frac{\pi}{2}} = -\cos \frac{\pi}{2} + \cos 0 = 1$$

$$(ii) \int_1^2 \frac{dx}{x(x+1)} = \int_1^2 \left(\frac{1}{x} - \frac{1}{x+1} \right) dx$$

$$= \left[\log x \right]_1^2 - \left[\log (x+1) \right]_1^2$$

$$= \left[\log 2 - \log 1 - \log 3 + \log 2 \right] = \left[2 \log 2 - \log 3 \right]$$

$$\begin{aligned}
 \text{(iii)} \quad & \int_2^3 \frac{6x^3 + 1}{\sqrt{2x^3 + x - 2}} dx \qquad \begin{cases} \text{put } 2x^3 + x - 2 = t \\ (6x^2 + 1) dx = dt \end{cases} \\
 &= \int \frac{1}{\sqrt{t}} dt = \frac{t^{1/2}}{1/2} + C \\
 &= \left[2\sqrt{2x^3 + x - 2} + C \right]_2^3 = 2[\sqrt{54 + 3 - 2} - \sqrt{16 + 2 - 2}] \\
 &= 2[\sqrt{55} - 4] = 6.83
 \end{aligned}$$

Ex 2 Find (i) $\int_{-1}^{+1} (4-3x)^5 dx$

(ii) $\int_1^2 \frac{x^{5x+2x+1}}{x} dx$

(iii) $\int_2^5 2x dx$

(i) $\int (4-3x)^5 dx$

Let $4-3x = t$, $-3dx = dt$, $dx = -\frac{dt}{3}$

$$\int (4-3x)^5 dx = \int t^5 (-) \frac{dt}{3}$$

$$= -\frac{1}{3} \int t^5 dt$$

$$= -\frac{1}{3} \cdot \frac{t^6}{6} + C = -\frac{1}{3} \cdot \frac{(4-3x)^6}{6} + C$$

$$\int_{-1}^{+1} (4-3x)^5 dx = \left[\frac{(4-3x)^6}{18} + C \right]_{-1}^{+1} = 6549.61$$

(ii) $\int \frac{x^{5x+2x+1}}{x} dx$

$$= \int 5^x dx + \int 2dx + \int \frac{1}{x} dx$$

$$= [5^x \log 5 + 2x + \log x + C]_1^2$$

$$= [25 \log 5 + 4 + \log 2] - [5 \log 5 + 2 + 0]$$

$$= 20 \log 5 + 2 + \log 2$$

$$(iii) \int_2^6 2x \, dx = \left[2 \left(\frac{x^2}{2} \right) \right]_2^6 = 21.$$

Ex. 3. Evaluate (i) $\int_{-2}^3 \left(1 - \frac{2}{x^5} \right) dx$ (ii) $\int_1^5 \frac{12}{x} dx$

$$(i) \int \left(1 - \frac{2}{x^5} \right) dx = \int 1 \, dx - \int \frac{2}{x^5} dx$$

$$= x + \frac{1}{2x^4} + C$$

$$\int_{-2}^3 \left(1 - \frac{2}{x^5} \right) dx = 4.975$$

$$(ii) \int_1^5 \frac{12}{x} dx = \left[12 \log x \right]_1^5 = 12 \left[\log 5 - \log 1 \right]$$

$$= 12 \log 5$$

Ex. 4. Find the area above x axis bounded by

$$y = 4x - x^2 - 3, \quad x = 1 \text{ and } x = 3$$

Required area = $\int_1^3 (4x - x^2 - 3) dx$

$$= \left[2x^2 - \frac{x^3}{3} - 3x \right]_1^3$$

$$= (18 - 9 - 9) - \left(2 - \frac{1}{3} - 3 \right) = \frac{4}{3}$$

Ex. 5. Evaluate (i) $\int_a^b \frac{2x-3x^2}{x} dx$ (ii) $\int_{1/2}^{1/2} \frac{1}{(2x+1)^4} dx$

$$(i) \int \frac{2x-3x^2}{x} dx = \int (2-3x) dx$$

$$= \int 2 dx - \int 3x^2 dx = 2x - x^3 + C$$

$$\therefore \int_a^b (2-3x^2) dx = a^3 - b^3 - 2(a-b)$$

$$(ii) \int \frac{dx}{(2x+1)^4} = \int (2x+1)^{-4} dx$$

Let $2x + 1 = t$, $\therefore dx = \frac{dt}{2}$

$$\begin{aligned} \therefore \int (2x+1)^{-4} dx &= \int t^{-4} \frac{dt}{2} = -\frac{1}{6t^3} + C \\ &= -\frac{1}{6(2x+1)^3} + C \end{aligned}$$

$$\int_{-1/2}^{1/2} \frac{dx}{(2x+1)^4} = \infty$$

Such integrals are called improper integrals

Ex 6. Find the area between line $y = 4x$ and axis of x and the ordinate $x=4$

We have to find the area OAB (Fig 13.2)

Limits $x = 0$ to $x = 4$

Area required

$$\begin{aligned} &= \int_0^4 y \, dx \\ &= \int_0^4 4x \, dx \\ &= \left[2x^2 \right]_0^4 = 32 \end{aligned}$$

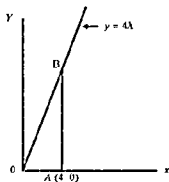


Fig 13.2

Ex 7 Find the entire area of circle

$x^2 + y^2 = r^2$ where r is radius and $(0, 0)$ is the centre of the circle

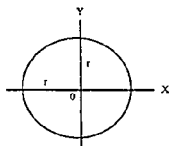


Fig 13.3

$$\text{Area of the circle} = \int y \, dx$$

$$\text{But, } y^2 = r^2 - x^2$$

$$y = \sqrt{r^2 - x^2}$$

$$\text{Area} = \int \sqrt{r^2 - x^2} \, dx$$

Area of circle as shown in Fig 13.3 is divided into four equal quadrants. We find the area of one quadrant and multiply it by 4 to get the area of the whole circle. The limits for one quadrant are

$$x = 0 \text{ to } x = r$$

$$\text{Area of one quadrant} = \int_0^r \sqrt{r^2 - x^2} \, dx \text{ and the}$$

$$\begin{aligned}
 \text{area of the circle} &= 4 \int_0^r \sqrt{r^2 - x^2} \, dx \\
 &= 4 \left[\frac{x}{2} \sqrt{r^2 - x^2} + \frac{r^2}{2} \sin^{-1} \frac{x}{r} \right]_0^r \\
 &= 4 \left[\frac{r}{2} \sqrt{r^2 - r^2} + \frac{r^2}{2} \sin^{-1} \frac{r}{r} \right] \\
 &\quad - 4 \left[\frac{0}{2} \sqrt{r^2 - 0} + \frac{r^2}{2} \sin^{-1} \frac{0}{r} \right] \\
 \text{Area} &= 4 \left[0 + \frac{r^2}{2} \sin^{-1} 1 - 0 - \frac{r^2}{2} \sin^{-1} 0 \right] \\
 \text{Area} &= 4 \frac{r^2}{2} \left(\frac{\pi}{2} \right) \quad \left\{ \begin{array}{l} \sin^{-1} 1 = \frac{\pi}{2} \\ \text{and } \sin^{-1} 0 = 0 \end{array} \right\} \\
 &= \pi r^2
 \end{aligned}$$

Therefore, area of circle is $= \pi (\text{radius})^2$

Ex 8 Find the area bounded by the parabola $x^2 = 4by$, the x -axis and the ordinate at $x = 3$

$$\begin{aligned}
 \text{The required area} &= \int_0^3 y \, dx = \int_0^3 \frac{x^2}{4b} \, dx \\
 &\quad (\text{See Fig 13.4}) \\
 &= \left[\frac{x^3}{12b} \right]_0^3 = \frac{27}{12b} = \frac{9}{4b}
 \end{aligned}$$

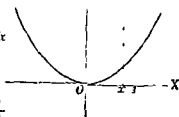


Fig 13.4

Ex 9 Find the area bounded by the parabola $x^2 = 4by$, axis of y and $y = 4$

The required area to be found is $OAB = 2OAC$ (Fig 13.5)

$$\begin{aligned}
 &= \int_0^4 x \, dy = \int_0^4 \sqrt{4by} \, dy \\
 &= 2\sqrt{b} \left[\frac{y^{3/2}}{3/2} \right]_0^4 = \frac{32}{3} \cdot \sqrt{b}
 \end{aligned}$$

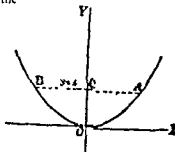


Fig 13.5

13.2. Area Between Two Curves

Let $y_1 = f(x)$ be the curve ACB
and $y_2 = \phi(x)$ be the curve ADB

Suppose they intersect at A and B , where x coordinates are $OM = a$ and $ON = b$ (Fig. 13.6)

We have to find the area $ACBD$, which would be equal to area $MACBN$ minus area $MABDN$, i.e., $ACBD$

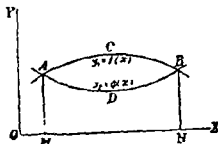


Fig 13.6

$$\begin{aligned} &= \int_a^b f(x) \cdot dx - \int_a^b \phi(x) \cdot dx \\ &= \int_a^b \{f(x) - \phi(x)\} dx \end{aligned}$$

which can be easily simplified for given functions

Ex.10. Find the area included between the two parabolas

$$y^2 = 4x \quad \text{and} \quad x^2 = 4y$$

We are given two curves $y^2 = 4x$ and $x^2 = 4y$

Substituting the value of y in the first equation, we obtain

$$\left(\frac{x^2}{4}\right)^2 = 4x, \quad \text{or} \quad \frac{x^4}{16} - 4x = 0$$

$$\text{or, } x(x^3 - 64) = 0$$

$$\therefore \text{ either } x=0 \quad \text{or} \quad x = \sqrt[3]{64} = 4$$

Hence the limits are $x=0$ and $x=4$, and

$$\text{the required area} = \int_0^4 \left(\sqrt{4x} - \frac{x^2}{4} \right) dx$$

$$= \left[2 \frac{x^{3/2}}{3/2} - \frac{x^3}{12} \right]_0^4 = \frac{4}{3} (4)^{3/2} - \frac{64}{12} = \frac{16}{3} = 5.33$$

Ex. 11. Find the area above the axis of x bounded by

$$x - 2y + 4 = 0, x = 3 \text{ and } x = 6$$

We are given $x - 2y + 4 = 0$

$$y = \frac{x}{2} + 2$$

$$\begin{aligned}
 \text{Required area} &= \int_3^6 \left(\frac{x}{2} + 2 \right) dx \\
 &= \left[\frac{1}{2} \frac{x^2}{2} + 2x \right]_3^6 \\
 &= 21 - \frac{33}{4} = \left(\frac{51}{4} \right) = 12.75
 \end{aligned}$$

13.3. Properties of Definite Integrals

(1) The interchange of the limits of integration changes only the sign of the definite integrals, thus,

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

(2) The limits can be broken. A definite integral can be expressed as a sum of a finite number of definite sub-integrals as follows

$$\int_a^b f(x) dx = \int_a^{c_1} f(x) dx + \int_{c_1}^{c_2} f(x) dx + \int_{c_2}^b f(x) dx$$

(where c_1 and c_2 lie between a and b)

This property of limits can be extended indefinitely

For example

$$\int_{-1}^1 x dx = \int_{-1}^0 x dx + \int_0^1 x dx$$

(3) The value of definite integral is independent of the variable of integration :

$$\text{For example } \int_a^b f(x) dx = \int_a^b f(y) dy$$

$$(4) \quad \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\text{Thus, } \int_0^{\frac{\pi}{2}} \sin x dx = \int_0^{\frac{\pi}{2}} \sin \left(\frac{\pi}{2} - x \right) dx = \int_0^{\frac{\pi}{2}} \cos x dx$$

(5) Substitution necessitates changes in the limits

$$\text{Ex. 12 Integrate } \int_0^{\frac{\pi}{2}} \frac{\cos x}{1 + \sin x} dx$$

$$\text{Suppose, } I = \int_0^{\frac{\pi}{2}} \frac{\cos x}{1 + \sin x} dx$$

$$\text{Let } \sin x = t \quad \therefore \cos x \, dx = dt.$$

With this substitution, limits also change;

$$\text{when } x = 0 \quad t = \sin 0 = 0.$$

$$\text{when } x = \frac{\pi}{2} \quad t = \sin \frac{\pi}{2} = 1$$

$\therefore t$ varies from 0 to 1.

$$\begin{aligned} I &= \int_0^1 \frac{dt}{1+t} = \left[\log_e (1+t) \right]_0^1 \\ &= \log_e 2 - \log_e 1 = \log_e 2. \end{aligned}$$

(6) If $f(x)$ is an even function of x

$$\text{that is } f(-x) = f(x)$$

$$\text{then } \int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$$

Alternatively, if $f(x)$ is an odd function of x

$$\text{that is, } f(-x) = -f(x)$$

$$\text{then } \int_{-a}^a f(x) \, dx = 0$$

Ex. 13. Find $\int_0^a \sqrt{a^2 - x^2} \, dx$

Put $x = a \sin \theta$, $dx = a \cos \theta \, d\theta$ Change the limits :

$$x = 0 \quad \text{to} \quad \theta = 0$$

$$x = a \quad \text{to} \quad \theta = \frac{\pi}{2}$$

Hence,

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sqrt{a^2 - x^2} \, dx &= \int_0^{\frac{\pi}{2}} \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta \, d\theta \\ &= \int_0^{\frac{\pi}{2}} a^2 \cos^2 \theta \, d\theta = a^2 \int_0^{\frac{\pi}{2}} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta \\ &= a^2 \left[\frac{\theta}{2} + \left(\frac{1}{2} \right) \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\ &= a^2 \left[\frac{\pi}{2} \cdot \frac{1}{2} + 0 \right] = \frac{a^2 \pi}{4} \end{aligned}$$

Ex 14 Show that

$$\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$$

We use the property

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

Here, $a = \frac{\pi}{2}$

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad (i)$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)} + \sqrt{\sin\left(\frac{\pi}{2}-x\right)}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad (ii)$$

Adding the two values of I we have

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx + \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \\ &= \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} dx \\ &= \left[x \right]_0^{\pi/2} = \frac{\pi}{2} \end{aligned}$$

$$2I = \frac{\pi}{2} \quad \text{or} \quad I = \frac{\pi}{4}$$

Ex 15 Find (i) $\int_0^{\pi/2} \cos^2 x dx$ (ii) $\int_0^{\pi/2} \cos^4 x dx$

(i) Let $I = \int_0^{\pi/2} \cos^2 x dx$

Integrate by parts

$$I = \int_0^{\pi/2} \cos x \cos x dx = \int_0^{\pi/2} (\cos x) (d \sin x) dx$$

(Take $\cos x$ as the first term)

$$= \left[\cos x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\sin x) \sin x \, dx$$

$$= \left[\cos \frac{\pi}{2} \sin \frac{\pi}{2} - \cos 0 \sin 0 \right] + \int_0^{\frac{\pi}{2}} (1 - \cos^2 x) \, dx$$

$$1 = 0 + \int_0^{\frac{\pi}{2}} dx - \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$$

$$\text{or } 2I = \frac{\pi}{2} \quad \text{or} \quad I = \frac{\pi}{4}$$

$$\begin{aligned} \text{(ii) Let } I &= \int_0^{\frac{\pi}{2}} \cos^4 x \, dx \\ &= \int_0^{\frac{\pi}{2}} \cos^3 x \, d \sin x \\ &= \left[\cos^3 x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \cos^2 x (-\sin x) \sin x \, dx \\ &= 0 + 3 \int_0^{\frac{\pi}{2}} \cos^2 x (1 - \cos^2 x) \, dx \\ &= 3 \int_0^{\frac{\pi}{2}} \cos^2 x \, dx - 3 \int_0^{\frac{\pi}{2}} \cos^4 x \, dx \\ I + 3I &= 4I - 3 \int_0^{\frac{\pi}{2}} \cos^4 x \, dx \\ I &= \left(\frac{3}{4} \right) \left(\frac{\pi}{4} \right) = \frac{3\pi}{16} \end{aligned}$$

EXERCISE 13.1

Evaluate

1. $\int_1^3 (1+x+x^2) \, dx$
2. $\int_{\frac{1}{2}}^1 \log x \, dx$
3. $\int_0^{\frac{\pi}{2}} x^2 \sin x \, dx$
4. $\int_0^{\frac{\pi}{2}} e^x (\sin x + \cos x) \, dx$
5. $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$

6 $\int_{2a}^{3a} \frac{1}{x^2 - a^2} dx$

7 $\int_0^2 x \log x dx$

8 $\int_0^{\pi/2} \tan^2 x dx$

9 $\int_0^1 \sin^{-1} x dx$

10 $\int_1^2 x^2 \log x dx$

11 $\int_0^{\pi/2} \frac{\sin x}{\cos^2 x} dx$

12 $\int_1^{+1} \frac{dx}{1+x^2}$

13 $\int_0^{\infty} \frac{dx}{a^2 + x^2}$

14 $\int_0^3 \frac{x}{\sqrt{4-x}} dx$

15 $\int_0^1 \frac{x dx}{\sqrt{1-x^2}}$

16 $\int_0^1 \frac{dx}{\sqrt{12-4x-x^2}}$

17 From the sum definition of an integral, show that $\int_a^b 1 dx = b - a$.

What is the indefinite integral of 1?

18 Show that

(i) $\int_0^1 x^3 dx = \frac{1}{4}$

(ii) $\int_1^4 \sqrt{x} dx = \frac{14}{3}$

(iii) $\int_0^{+1} \frac{dx}{1+x} = 1$

19 Show that $\int_1^3 (1+3x-x^2) dx = -4$

20 Find the area between the following curves, axis of y and $x=4$,

(i) $x = 7y$ (ii) $y = 3x^2$ (iii) $x = 9y^2$ (iv) $y = 4 + 3x$

21 Find the area between the following curves, axis of x and ordinate at $x=2$ $x=3$

(i) $x = 7y$ (ii) $x^2 = 4y$ (iii) $x^2 + y^2 = 4$

22 Find the area between the following curves at $x=2$ and $x=3$

(i) $y = xe^x$ (ii) $y^2 = x^2 = 16$ (iii) $x^2 - y^2 = 16$

23 Find the area between the following curves axis of y and the coordinate at $y=2$, $y=3$

(i) $x^2 = 3y$ (ii) $y = 4x + 7$ (iii) $y = \sin^{-1} x$

24 Find the area of the parabola $y^2 = 16x$ which is cut off by the line $y = x$

25 Find the area included between curve $y = x^3$ and the line $y = 4x$ lying in the first quadrant.

26. Show that
$$\int_0^{\pi/2} \sin^n x \, dx = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x \, dx$$

Hence show that
$$\int_0^{\pi/2} \sin^2 x \, dx = \frac{\pi}{4}$$

and,
$$\int_0^{\pi/2} \sin^4 x \, dx = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{\pi}{2} = \frac{3\pi}{16}$$

27. Show that
$$\int_0^{\sqrt{3}} \frac{x}{\sqrt{4-x^2}} \, dx = 1$$

28. Show that
$$\int_0^{\pi/4} \tan^3 x \, dx = \frac{1-\log x}{2}$$

29. Evaluate

(i) $\int_0^{\infty} x e^{-x^2} \, dx$ (ii) $\int_1^{\infty} \frac{dx}{1+x^2}$

(iii) $\int_1^{\infty} \frac{dx}{(x^2+1)^2}$ (iv) $\int_0^2 \frac{dx}{(x-1)^2}$

30. Show that
$$\int_8^{15} \frac{dx}{(x-3)\sqrt{x+1}} = \frac{1}{2} \log \frac{5}{3} \quad [\text{Hint: Put } \sqrt{x+1} = z]$$

31. Show that
$$\int_{1/3}^1 \frac{(x-x^3)^{1/3}}{x^4} \, dx = 6 \quad [\text{Hint: Put } x = \frac{1}{t}]$$

32. Show that
$$\int_1^{\infty} \frac{1}{x^2(1+x)} \, dx = \log \frac{e}{2}$$

33. Prove that
$$\int_0^1 x \cdot e^{2x} \, dx = \frac{1}{4} (e^2 + 1)$$

Compute the area bounded by the following curves :

34. $y^2 = 2px$, $x = h$ 35. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

36. $y^2 = 2x + 4$, $x > 0$ 37. $y^2 = x^3$, $y = 8$, $x > 0$

38. $y = x^2$, $y = 2 - x^2$ 39. $y = x^2 + 4x$, $y = x + 1$

40. Show that
$$\int_0^{\frac{\pi}{2}} \sqrt{1 + \sin x} \, dx = 2$$

41. Show that $\int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta}{\sin^4 \theta + \cos^4 \theta} d\theta = \frac{\pi}{4}$

42. Prove that

$$\int_0^1 x^m (1-x)^n dx = \int_0^1 x^n (1-x)^m dx \quad (m > 0, n > 0)$$

Evaluate the following

43. $\int_{-1}^1 \frac{dx}{(1+x^2)^2}$ (Hint Put $x = \tan \theta$)

44. $\int_1^5 \frac{\sqrt{x-1}}{x} dx$ (Hint Put $x-1 = t^2$)

45. $\int_0^{\frac{\pi}{2}} \log (\sin x) dx$ 46. $\int_0^{\pi} \frac{dx}{5+3 \cos x}$

47. $\int_0^1 \frac{x}{\sqrt{1-x^2}} dx$ 48. $\int_0^1 \log x dx$

13.4. Application of Definite Integrals

Ex. 16. Given the marginal cost function $f'(x)$, find the total cost function when fixed cost is 50 units and $f'(x) = 2 + x + x^2$, x being output produced

$$\begin{aligned} \text{Total cost} = f(x) &= \int f'(x) dx \quad (\text{i.e., sum of all marginal costs}) \\ &= \int (2 + x + x^2) dx = 2x + \frac{x^2}{2} + \frac{x^3}{3} + C \end{aligned}$$

But since fixed cost = 50, i.e., total cost when output $x = 0$ is 50

($\because$ Variable cost will be zero)

$$\therefore f(0) = 0 + 0 + 0 + C = 50$$

$$\therefore C = 50$$

Hence the total cost function is $f(x) = 2x + \frac{x^2}{2} + \frac{x^3}{3} + 50$

Ex. 17. If the marginal cost function is $f'(x) = 2 + 3\sqrt{q} + \frac{5}{\sqrt{q}}$, find total cost function $F(q)$ when $F(1) = 21$.

$$F(q) = \int f(q) dq = \int \left(2 + 3\sqrt{q} + \frac{5}{\sqrt{q}} \right) dq$$

$$\text{Total cost } TC = 2q + 3 \frac{q^{3/2}}{3/2} + 5 \frac{q^{1/2}}{1/2} + C = 2q + 2q^{3/2} + 10q^{1/2} + C$$

When $q = 1$, $F(q)$ is given to be 21

$$21 = 2(1) + 2(1)^{3/2} + 10(1)^{1/2} + C$$

$$\text{i.e., } 21 = 14 + C \quad \text{or, } C = 7 \quad (\text{which is fixed cost})$$

$$\text{Total cost function } F(q) = 2q + 2q^{3/2} + 10q^{1/2} + 7$$

Ex 19 If the marginal revenue function is $p_m = \left\{ \frac{ab}{(x+b)^2} - C \right\}$

show that $p = \left\{ \frac{ab}{(x+b)} - C \right\}$, is the demand law

Let R be the total revenue, p the price and x the total output.

$$R = px \quad \text{and} \quad MR = \frac{dR}{dx}$$

$$\text{We are given that } MR = p_m = \frac{dR}{dx} = \left\{ \frac{ab}{(x+b)^2} - C \right\}$$

Hence, by integrating, total revenue is

$$R = \int \frac{dR}{dx} dx = \int \left\{ \frac{ab}{(x+b)^2} - C \right\} dx$$

$$R = ab \frac{(x+b)^{-1}}{-1} - Cx + A \quad (A \text{ is constant})$$

$$R = px = \frac{-ab}{x+b} - Cx + A \quad (1)$$

We know that if output $x = 0$ revenue must also be zero. Therefore, by substituting $R = 0$ when $x = 0$ in (1) we get

$$0 = \frac{-ab}{b} + A, \quad \text{i.e., } A = a$$

Now Eqn. (1) will be $R = px = \frac{-ab}{x+b} - Cx + a$

$$\text{i.e., } px = \left\{ a - \frac{ab}{x+b} \right\} - Cx = \frac{ax}{x+b} - Cx$$

$$\text{Hence } p = \frac{a}{x+b} - C$$

Ex 20 If the MR function for output q is given by

$$MR = \frac{6}{(q+2)^2} - 5$$

Show that the demand function is $p = \frac{3}{q+2} - 5$ (solve yourself)

Ex 21. Let the marginal cost function of a firm be $100 - 10x + 0.1x^2$ where x is the output. Obtain the total cost function of the firm under the assumption that its fixed cost is Rs 500

$$MC = 100 - 10x + 0.1x^2$$

$$\begin{aligned} TC &= \int (100 - 10x + 0.1x^2) dx \\ &= 100x - 5x^2 + \frac{x^3}{30} + C \end{aligned}$$

Fixed cost is given as Rs 500

$$TC = 100x - 5x^2 + \frac{x^3}{30} + 500$$

$$TC = \frac{x^3}{30} - 5x^2 + 100x + 500$$

Ex 22 If the MR function is given by

$$P_m = \frac{a}{(x+b)^2} - C, \text{ where } x \text{ is output and } a, b, \text{ are constants, show that}$$

demand law is $x = \frac{a}{b(b+c)^2} - b$ (solve yourself)

Ex. 23. The marginal cost function for some product is $(1 + x + 6x^2)$ where x is the output. Find the total cost function if the fixed cost is Rs 100 when the output is zero

$$MC = 1 + x + 6x^2$$

$$\begin{aligned} TC &= \int (1 + x + 6x^2) dx \\ &= x + \frac{x^2}{2} + 2x^3 + C \end{aligned}$$

Since fixed cost is given to be 100,

$$\text{Therefore, } TC = 2x^3 + \frac{x^2}{2} + x + 100$$

Ex. 24. The marginal cost function of a firm is $2 + 3e^x$, where x is the output. Find the total average cost functions if the fixed cost is Rs 500

$$\text{Given } MC = 2 + 3e^x$$

$$\begin{aligned} TC &= \int (2 + 3e^x) dx \\ &= 2x + 3e^x + C \end{aligned}$$

since fixed cost = 500

Therefore $TC = 2x + 3e^x + 500$ and $AC = 2 + 3x^{-1}e^x + 500x^{-1}$

Ex 25 If $MR = 16 - x^2$ find the maximum total revenue. Also find the total and average revenue and demand.

Given $MR = 16 - x^2$

We know that TR is maximum when $MR = 0$

$$16 - x^2 = 0 \quad \therefore x = \pm 4$$

Hence total revenue (TR) is maximum when output is 4 units. Our problem now is to find the maximum total revenue which happens when output is 4 units.

$$\begin{aligned} TR &= R = \int_0^4 (MR) dx = \int_0^4 (16 - x^2) dx \\ &= \left[16x - \frac{x^3}{3} \right]_0^4 = \frac{128}{3} \end{aligned}$$

$\frac{128}{3}$ is the maximum total revenue

$$(ii) \text{ Total revenue is } R = \int (16 - x^2) dx = 16x - \frac{x^3}{3} + C$$

when $x = 0$ revenue must also be zero $C = 0$

$$R = \left(16x - \frac{x^3}{3} \right) \text{ which is the required total revenue}$$

$$(iii) \text{ Average revenue} = \frac{TR}{\text{output}} = \frac{R}{x} = \frac{16x - \frac{x^3}{3}}{x}$$

$$\left(16 - \frac{x^2}{3} \right) \text{ is the required average revenue}$$

Since $AR = p$ naturally $p = \left(16 - \frac{x^2}{3} \right)$ is the required demand function

Ex 26 The marginal cost and revenue of a firm are given as

$$MC = 4 + 0.08x, MR = 12$$

Compute the total profit given that total cost at zero output = 0

For profit maximisation, $MC = MR$

$$\text{i.e., } 4 + 0.08x = 12$$

$$\text{or, } 0.08x = 8 \quad \therefore x = 100$$

Since $MR = \text{constant}$, hence there is perfect competition,

$$\therefore p = MR = 12. \text{ Hence total revenue} = 12x.$$

$$\text{Total cost} = \int 4 + 0.08x = 4x + \frac{0.08x^2}{2} + A = 4x + 0.04x^2 + A$$

$$\text{Since total cost} = 0 \text{ when } x = 0 \quad \therefore A = 0 \text{ and } TC = 4x + 0.04x^2$$

$$\text{Hence total profit} = TR - TC$$

$$= 12x - (4x + 0.04x^2)$$

$$= 8x - 0.04x^2 = 400$$

(At profit maximisation output $x = 100$)

Ex. 27. The marginal cost function for some product is $(1 + 2x + 6x^2)$ where x is the output. Find the total cost function if the fixed cost is Rs. 100 when the output is zero.

$$MC = 1 + 2x + 6x^2$$

$$TC = \int (1 + x + 6x^2) dx$$

$$= x + \frac{x^2}{2} + 2x^3 + C$$

Since fixed cost is given to be 100,

$$\therefore TC = 2x^3 + \frac{x^2}{2} + x + 100$$

Ex. 28. The marginal cost function of a firm is $2 + 3e^x$, where x is the output. Find the total average cost functions if the fixed cost is Rs. 500.

$$\text{Given } MC = 2 + 3e^x$$

$$\therefore TC = \int (2 + 3e^x) dx$$

$$= 2x + 3e^x + C$$

since fixed cost = 500,

$$\text{therefore, } TC = 2x + 3e^x + 500, \text{ and } AC = 2 + 3x^{-1}e^x + 500x^{-1}$$

Ex. 29. Marginal propensity to consume function is given as follows.

$$\frac{dC}{dy} = 0.5 - 0.001y$$

where C is consumption and y is disposable income. Find the total consumption function if at income zero C is 0.2

$$\text{Consumption function } C = f(y) = \int \frac{dC}{dy} \cdot dy$$

$$C = \int (0.5 - 0.001y) \, dy = 0.5y - \frac{0.001}{2}y^2 + A \text{ (constant)}$$

We are given that when $y = 0$, $C = 0.2$. Hence $A = 0.2$

Therefore, Total Consumption function $= 0.5y - \frac{0.001y^2}{2} + 0.2$

$$C = 0.2 + 0.5y - 0.0005y^2$$

($C = 0.2$ when $y = 0$ may be termed as subsistence or survival consumption level)

Ex 30 If marginal propensity to save is given to be $0.5 + 0.2y^{-2}$ (y is income) Find consumption function if consumption is Rs. 50.01 when income is 200

Let S depict total Saving, then

$$m.p.s = \frac{dS}{dy} = 0.5 + 0.2y^{-2} \text{ (given)}$$

$$\text{Therefore, } S = \int (0.5 + 0.2y^{-2}) \cdot dy$$

$$S = 0.5y - 0.2y^{-1} + A$$

$$\text{Consumption } C = y - S$$

$$C = y - (0.5y - 0.2y^{-1} + A) = 0.5y + \frac{0.2}{y} - A$$

We are also given that when $y = 200$, the level of consumption = 50.01. Substituting this in consumption function, we obtain A (constant) = 50

$$\text{Hence, the required consumption function is } C = -50 + 0.5y + \frac{0.2}{y}$$

13.5. Capital Formation

Capital formation is the process of addition to a given stock of capital. Rate of capital formation may thus be expressed as $\frac{dk}{dt}$, where k is a function of time t . But rate of capital formation is the same as the rate of net investment flow $I(t)$

$$\text{Since, } \frac{dk}{dt} = I(t)$$

$$\text{Therefore } k(t) = \int \frac{dk}{dt} \, dt = \int I(t) \cdot dt$$

Ex. 31. If $I(t) = 3t^{1/3}$ crores of rupees per year, what will be the capital formation in the time period of 5 years, and during the last year of the plan period, i.e., 5th year? Also find capital growth equation or time path of capital when $k(0) = 25$

$$\begin{aligned} \text{(i) Given } k(t) &= \int_0^5 3t^{1/3} dt \\ &= \left[\frac{3t^{4/3}}{\frac{4}{3}} \right]_0^5 = \frac{9}{4} 5^{4/3} = \text{Rs. } \frac{9}{4} \sqrt[3]{625} \text{ crores} \end{aligned}$$

$$\begin{aligned} \text{In the fifth year, } k(t) &= \int_4^5 3t^{1/3} dt = \frac{9}{4} (t^{4/3})_4^5 \\ &= \frac{9}{4} (\sqrt[3]{625} - \sqrt[3]{256}) \end{aligned}$$

(ii) Capital Growth Equation

$$k(t) = \int 3t^{1/3} dt = \frac{9}{4} t^{4/3} + C$$

It is also given that initial stock of capital is 25

$$k(0) = 25 = 0 + C \quad \text{or, } C = 25$$

$\therefore k(t) = 25 + \frac{9}{4} t^{4/3}$ is the required capital growth equation from

given rate of capital formation of $3t^{1/3}$

13.6. Compound Interest

A sum of Rs 100 is invested and accumulates at compound rate of 4% a year. If interest is added yearly then $100 + 4 = 104 = (1.04)$ is the amount at the end of 1st year.

Again $104 + \frac{4}{100} \times 104 = 104(1.04) = 100(1.04)^2$ will be the amount at the end of second year. Similarly $100(1.04)^3$ will be the amount at the end of third year.

Hence the general formula for finding the amount after t years will be

$$P_t = P_0 (1 + r)$$

where, P_0 is the initial amount invested and r is the rate of interest

Suppose interest is added twice a year instead of once a year. Then with the same rate of interest of 4%; 2% will be added in each half-year and another 2% in each second half of the year

Therefore, amount of Rs 100 at the end of successive half years will be
 $100(1.02)$, $100(1.02)^2$, $100(1.02)^3$... $100(1.02)^{2t}$

And the general formula in this case becomes

$P_t = P_o \left(1 + \frac{r}{2}\right)^{2t}$, i.e., number of periods become twice while rate of interest half

Similarly suppose amount invested is Re 1 or $P_o = 1$ and $r = 100\%$,
 $\left(\text{i.e., } r = \frac{100}{100} = 1\right)$ and that this amount of Re 1 is invested for one year but interest is being added n times a year

$$P_t = 1 \left(1 + \frac{1}{n}\right)^{n \times 1} = \left(1 + \frac{1}{n}\right)^n$$

This expression $\left(1 + \frac{1}{n}\right)^n$ tends to a definite limit, which is the value approximating to 2.718 as n tends to infinity

$$\text{Symbolically, } \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = (\text{nearly}) 2.71828 = e$$

So, we say that if Rupee one is invested for one year with rate of interest = 100% which is being added more and more frequently, then the amount of Rupee one is Rs e .

That is P_o after t years will rate of interest $r\%$ turns out to be
 $P_t = P_o e^{rt}$, when interest rate is compounded continuously or frequently n times

Taking the general formula once again

$$P_t = P_o \left(1 + \frac{r}{n}\right)^{nt}$$

$$P_t = P_o \left\{ \left(1 + \frac{r}{n}\right)^{n/m} \right\}^{rt} \quad \text{Let, } \frac{n}{r} = m \text{ or } \frac{r}{n} = \frac{1}{m}$$

$$P_t = P_o \left\{ \left(1 + \frac{1}{m}\right)^m \right\}^{rt}$$

$$\text{But } \lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^m = e,$$

$$P_t = P_o e^{rt}$$

13.7 Present Value or Discounted Value

From the general formula we know, if P_0 is invested at 100 $r\%$ it will amount to $P_0 (1 + r)^t = P_t$ after t years (if compounded once yearly). To put it in another way, if (P_t) amount is known we may be asked to find the amount (P_0) which would amount to (P_t) after t years.

This can be easily found from the formula $P_t = P_0 (1 + r)^t$

$$\text{that is } P_0 = \frac{P_t}{(1 + r)^t}$$

As such we are able to find the value of P_0 (or the initial amount which ought to be invested) with given P_t , r and t .

This amount, $\left[\frac{P_t}{(1 + r)^t} \right]$ is termed as *present value or discounted value* of P_t .

Similarly, if the interest is added n times a year at 100 $r\%$, we have the

$$\text{present value } P_0 = \frac{P_t}{\left(1 + \frac{r}{n}\right)^{V_n}}$$

If the rate of interest is added more frequently or continuously at r 100% then present value of investment

$$P_0 = P_t e^{-rt} \quad \left(\text{since } P_t = P_0 e^{rt} \text{ or } P_0 = \frac{P_t}{e^{rt}} = P_t e^{-rt} \right)$$

$$\text{Symbolically } \lim_{n \rightarrow \infty} \frac{P_t}{\left(1 + \frac{r}{n}\right)^{V_n}} = P_t e^{-rt}$$

Summarizing, therefore, we find that the present value of any investment depends upon the rate of interest and its frequency of being added to P_0 . *Higher the r , and more frequently it being added, smaller will be the present value*

13.8 Capital Value and Flow Value

Let us suppose we obtain different incomes in different years from a machine or from certain investment

Years	This yr	1st yr	2nd yr	3rd yr	nth yr
Income (Rs)	a_0	a_1	a_2	a_3	a_n

In order to get present value of above income stream, we discount it by the rate of interest,

Rs. a_0 is the income which is received this year, hence it is present value in itself

Rs. a_1 is the income which is received after one year.

Rs. a_2 is the income which is received after two years, and so on.

Rs. a_m is the income which is received after m year.

We need to find the present value of $a_1, a_2, \dots, a_m$ incomes if rate of interest is $r.100\%$ per year

$$\text{Therefore present value of } a_1 = \frac{a_1}{1+r} \quad [P_t = P_0 (1+r)^t]$$

$$\text{present value of } a_2 = \frac{a_2}{(1+r)^2}$$

$$\text{present value of } a_3 = \frac{a_3}{(1+r)^3}$$

$$\text{present value of } a_m = \frac{a_m}{(1+r)^m}$$

Therefore present value of the whole stream of income

$$= a_0 + \frac{a_1}{1+r} + \frac{a_2}{(1+r)^2} + \dots + \frac{a_m}{(1+r)^m}$$

This sum will be known as *capital value* or *capitalised value* of the present investment. Naturally, therefore, if the capital value of the investment is greater than the actual value of the investment presently made, it will be worth while to make the investment, otherwise not.

Thus generalising the whole discussion, we say that :

In a *discrete case*, the present value (or capitalised value or discounted value) of Rs. a_t which is received in the t th period is $= \frac{a_t}{(1+r)^t}$ where r is the rate of interest. The present value in case of stream of incomes given by $a_0, a_1, a_2, \dots, a_m$ in 1st, 2nd, 3rd m th years will be :

$$y = \sum_{t=0}^m \frac{a_t}{(1+r)^t}$$

This result can be generalised for a *continuous income stream* and when interest is also compounded continuously.

If the interest is added frequently (say, n times) at $100.r\%$ then

Present value (PV_0) of income stream is given by

$$y = PV_0 = a_0 + \frac{a_1}{\left(1 + \frac{r}{n}\right)^n} + \frac{a_2}{\left(1 + \frac{r}{n}\right)^{2n}} + \dots + \frac{a_t}{\left(1 + \frac{r}{n}\right)^{nt}}$$

Present value of *continuous equal income stream* for x years $\left. \vphantom{\int_0^x} \right\} y = PV_0 = \int_0^x a e^{-rt} dt$

This is a definite integral, when it is assumed that income every year received is Rs. a per year and x is the number of years for which this income is received, so that,

$$PV_0 = \int_0^x a e^{-rt} dt = \left[-\frac{a}{r} e^{-rt} \right]_0^x$$

$$PV_0 = \frac{a}{r} (1 - e^{-rx})$$

Here again we find that the capitalised value of continuous equal stream depends on the size of income-stream, number of years it flows and the rate of interest

Ex. 32. Suppose a share worth Rs. 100 in a mining company is expected to produce dividends of Rs. 40, Rs. 32, Rs. 24, Rs. 16 and Rs. 8 in the present and in the four following years and to be worth nothing thereafter. If the interest is added once yearly at 5%, find whether this investment is worth making

If the same investment in a new trading company yields the dividends of Rs. 1, 2, 4 and 6 in the present and in the three following years, while in the fourth year it is expected that the share can be sold for Rs. 120, should the investment be made here or in the above one?

It is necessary to find the present value of the given income stream in both the cases.

$$\begin{aligned} \left. \begin{array}{l} \text{1st case Present value or} \\ \text{capital value} \end{array} \right\} &= \text{Rs} \left(40 + \frac{32}{(1 + 0.05)} + \frac{24}{(1 + 0.05)^2} \right. \\ &\quad \left. + \frac{16}{(1 + 0.05)^3} + \frac{8}{(1 + 0.05)^4} \right) \\ &= \text{Rs. } 112.6 \end{aligned}$$

$$\begin{aligned} \left. \begin{array}{l} \text{2nd case Present value} \end{array} \right\} &= \text{Rs} \left(1 + \frac{2}{1.05} + \frac{4}{(1.05)^2} \right. \\ &\quad \left. + \frac{6}{(1.05)^3} + \frac{120}{(1.05)^4} \right) \\ &= \text{Rs. } 110.4 \end{aligned}$$

Thus the investment of Rs. 100 is worth making in both the cases as investor realises more than what he invests. The rate of profit in each case is

% of profit in case $I = 112.6 - 100 = 12.6\%$ in 5 years

% of profit in case $II = 110.4 - 100 = 10.4\%$ in 5 years

and therefore investment in case I should be preferred

Ex 33 A sinking fund is formed by investing Rs. x at the end of each year for m years. Show that the final amount of the fund is $x + x(1+r) + x(1+r)^2 + \dots + (1+r)^m$ when interest is added yearly at $100r\%$ per year. Show that the fund will amount finally to Rs. a if

$$x = \frac{ar}{(1+r)^m - 1}$$

Also show that about Rs. 79.5 must be set aside each year if the sinking fund (5% interest) is to replace a machine costing Rs. 1000 after 10 years.

If amount of Rs. x is invested every year with rate of interest r 100% then the total amount will be more than $(x \times m)$ since the interest acquired is also invested every year.

No. of year	Amt. invested	Total amount at the end of that year
1st year	x	x
2nd year	x	$x + x(1+r)$ $\downarrow$ $\downarrow$ $\left\{ \begin{array}{l} \text{Since the amount of second year} \\ \text{will earn no interest while} \\ \text{amount } x \text{ of first year will earn} \\ \text{interest.} \end{array} \right.$
3rd year	x	$x + x(1+r) + x(1+r)^2$
4th year	x	$x + x(1+r) + x(1+r)^2 + x(1+r)^3$
		$\dots \dots \dots$
		$\dots \dots \dots$
m th year	x	$x + x(1+r) + x(1+r)^2 \dots x(1+r)^{m-1}$

But as the final amount of the fund after m years = a (given)

$$a = x + x(1+r) + x(1+r)^2 + \dots + x(1+r)^{m-1}$$

$$a = x[1 + (1+r) + (1+r)^2 + \dots + (1+r)^{m-1}]$$

$$a = x \left[1 + \frac{1 - (1+r)^m}{1 - (1+r)} \right]$$

$$a = x \left[\frac{1 - (1+r)^m}{-r} \right]$$

$$a = x \left[\frac{(1+r)^m - 1}{r} \right]$$

$$\text{Hence, } x = \left[\frac{ar}{(1+r)^m - 1} \right]$$

In the second part of the question, the total amount required to be collected is Rs 1000 in 10 years period with $r = 5\%$. We are required to find the yearly investment

$$\text{Given } a = 1000, r = \frac{5}{100} \text{ and } m = 10$$

$$\text{Substituting in } x = \frac{ar}{(1+r)^m - 1}, \text{ we obtain}$$

$$x = \frac{1000 \times \frac{5}{100}}{\left(1 + \frac{5}{100}\right)^{10} - 1} = \frac{50}{\left(1 + \frac{1}{20}\right)^{10} - 1} = \frac{50}{(1.05)^{10} - 1}$$

$$x = 79.50$$

In other words, Rs. 79.5 must be invested every year to sink the fund of Rs 1000 in 10 years with $r = 5\%$

Ex. 34 If the interest is reckoned continuously at 4% per year show that the capital value of uniform income stream of Rs 100 per year for 10 years is Rs 824 and for 100 years = Rs 2454 ($e = 2.71828$)

$$\text{First case} \quad \text{Capital value (in continuous case)} \quad y = \frac{a}{r} (1 - e^{-rx})$$

$$\text{We are given } a = 100; \quad r = \frac{4}{100} = 0.04 \text{ and } x = 10 \text{ years}$$

$$\begin{aligned} \text{Hence, } y &= \frac{100}{0.04} (1 - e^{-0.04 \times 10}) = 2500 (1 - e^{-0.4}) \\ &= 2500 \{1 - (2.71828)^{-4}\} = 824 \end{aligned}$$

$$\text{Second case. } a = 100, r = \frac{4}{100} = 0.04 \text{ and } x = 100 \text{ years}$$

$$\text{Hence, } P_0 = 2500 (1 - e^{-4.0}) = 2500 \{1 - (2.71828)^{-4.0}\} = 2454$$

Ex. 35. Given the continuous income stream at the constant rate of Rs 20,000 per year, what will be its present value for the next three years if the discount rate is 0.06 per year

For solving this question, we make use of the series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$P_0 = \frac{a}{r} (1 - e^{-rx})$, and it is given that $a=20,000$; $r=0.06$, and $x=3$ years

$$P_0 = \frac{20,000}{0.06} [1 - e^{-0.06 \times 3}] = \frac{10,00,000}{3} \left[1 - \frac{1}{e^{0.18}} \right]$$

$$P_0 = \frac{10,00,000}{3} \left\{ 1 - \left[1 + (-0.18) + \frac{(-0.18)^2}{2!} \right] \right\}$$

$$P_0 = \frac{10,00,000}{3} [1 - (1 - 0.18 + 0.0162)]$$

$$P_0 = \frac{10,00,000}{3} [1 - (-0.8362)] = \left[\frac{10,00,000}{3} \right] 0.1638$$

$$= \text{Rs } 54,600$$

is the *Present Value* of the given income stream

Ex. 36 If population of a region, whose present population is 10 million, is increasing at 10% per year; obtain the population of the region after four years

Since the population increases continuously, the exponential growth formulae is appropriate

$$P_t = P_0 e^{rt}$$

In this case $r = 0.1$ and $t = 4$

$$P_t = 10e^{0.1 \times 4} = 14.918$$

Ex. 37. The sales of a product is depicted by a function $S(t) = 100e^{-0.5t}$, where t is the number of years since the launching of the product. Find

- the total sales in the first three years,
- the sales in the fourth year, and
- the total sales in the future

$$(i) \quad S(3) = \int_0^3 100e^{-0.5t} dt = 155.40$$

$$(ii) \quad \text{Sales in the fourth year } S_4 = S(4) - S(3)$$

$$S_4 = \int_3^4 100 e^{0.5t} dt = 17.6 \quad (e^{-1.5} = 0.2231)$$

(iii) Total Sales in the future $= S(\infty) - S(0)$

$$S = \int_0^{\infty} 100 e^{-0.05t} dt = 2000$$

Ex. 38 If the sales of a firm are given by a function

$$S(t) = e^{a+bt+\frac{1}{2}ct} \quad \text{and}$$

$S(0)$, $S(1)$ and $S(2)$ are given to be 1, 2.7 and 7.3 respectively find a , b and c . Also find t when yearly sales will be 20 and cumulative sales will be 20.

$$\text{Sol. Let } e^0 = 1, e^1 = 2.7 \text{ and } e^2 = 7.3$$

$$\text{We are given } S(t) = e^{a+bt+\frac{1}{2}ct}$$

$$S(0) = 1 = e^a = e^0$$

$$\text{Hence } a = 0 \quad (i)$$

$$S(1) = 2.7 = e^{a+b+\frac{1}{2}c} \quad (t = 1)$$

$$\text{Hence } b + \frac{c}{2} = 1 \quad (a = 0) \quad (ii)$$

$$S(2) = 7.3 = e^{2b+2c}$$

$$\text{Hence } 2b + 2c = 2 \quad (iii)$$

Sol. (i), (ii) and (iii) we get $b = 1$ and $c = 0$. The given sales function $S(t)$ becomes

$$e^{a+bt+\frac{1}{2}ct} = e^t$$

$\therefore$ Sales 20 units will be at time t when $20 = e^t$

$$\text{or } \log 20 = t \log e \quad (\log \text{ base is } 10)$$

$$t = \log \frac{20}{\log 2.7} = \frac{1.3010}{0.4314} = 3.02$$

Cumulative sales of 20 units shall be when

$$\int_0^t e^t dt = 20$$

$$\text{thus is, } e^t = 21, t \log e = \log 21$$

$$t = \frac{1.3222}{0.4313} = 3.07$$

EXERCISE 13.2

1. MC (marginal cost) of a firm is given by $C'(q) = 3q^2 - 2 + 5$ where C is total cost function $C(q)$ given that fixed cost is Rs. 100

15. An income stream decreases continuously over time for m years the rate being Rs ae^{-bt} per year at t years from now. Find the capital value at 100 $r\%$ continuous compound interest

13.9. Consumer's Surplus

If a consumer with a particular demand function purchases x_0 quantity at p_0 price, the consumer's surplus (CS) can be depicted as shown in Fig 13.7

If we could find out the market demand function having demand x_0 at p_0 then,

CS = Total area of the curve below the demand function from 0 to x_0 minus the area of the rectangle $O_p Cx_0$.

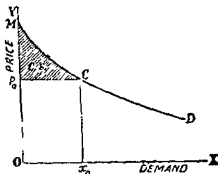


Fig 13.7

This area can easily be found out with the help of definite integral

Ex. 39. If the demand function is $p = 35 - 2x - x^2$, and the demand x_0 is 3, what will be the consumer's surplus?

Given demand function is

$$p = 35 - 2x - x^2$$

For $x = 3$, $p = 20$

Therefore, CS = Area of the curve below the demand curve from 0 to 3 minus Area of rectangle $(20 \times 3) = 60$

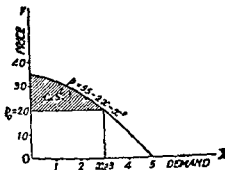


Fig 13.8

$$\begin{aligned} CS &= \int_0^3 (35 - 2x - x^2) dx - 60 \\ &= \left[35x - x^2 - \frac{x^3}{3} \right]_0^3 - 60 \\ &= 105 - 9 - 9 - 60 = 27 \text{ units. (see Fig. 13.8)} \end{aligned}$$

Ex. 40. If the demand law is $p = 85 - 4x - x^2$, what will be the consumer's surplus (a) if $x_0 = 5$, and (b) if $p_0 = 64$.

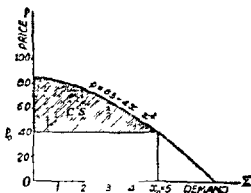


Fig. 13.9

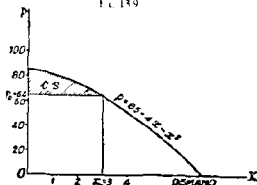


Fig. 13.10

- (a) If $x_0 = 5$ p_0 will be = 40

$$C.S. = \int_0^5 (85 - 4x - x^2) dx - (40 \times 5)$$

$$= \left[85x - 2x^2 - \frac{x^3}{3} \right]_0^5 - 200$$

$$C.S. = \left[333 \frac{1}{3} \right] - 200 = 133 \frac{1}{3} \text{ units (see Fig. 13.9)}$$

- (b) If $p_0 = 64$ then $64 = 85 - 4x - x^2$ i.e., either
 $x_0 = 3$ or $x_0 = -7$ (which has no meaning in demand)

$$C.S. = \int_0^3 (85 - 4x - x^2) dx - (64 \times 3) \text{ [see Fig. 13.10]}$$

$$= \left[85x - 2x^2 - \frac{x^3}{3} \right]_0^3 - 192 = 36$$

Consumer's Surplus under Pure Competition

Under pure competition this price is determined by the equality of demand and supply functions. Thus if supply and demand laws are given we find p_0 by equating the given functions and then find x_0 the market demand.

Ex 41 The price of the quantity demanded under pure competition is determined by the demand and supply laws

$$p_d = \frac{8}{x+1} - 2 \text{ and } p_s = \frac{1}{2}(x+3) \text{ respectively}$$

Determine the corresponding consumer's surplus

The equilibrium quantity and price is found by equating and then solving the given functions

$$\text{Demand } p_d = \left(\frac{8}{x+1} - 2 \right)$$

$$\text{Supply } p_s = \left(\frac{x+3}{2} \right)$$

Equating $D = S$

$$\left(\frac{8}{x+1} - 2 \right) = \left(\frac{x+3}{2} \right)$$

$$\text{So that } x^2 + 8x - 9 = 0$$

Which gives $x = 1$ or $x = -9$

At $x = 1$, p will be 2

$$CS = \int_0^1 \left(\frac{8}{x+1} - 2 \right) dx - (2 \times 1)$$

$$CS = \left[8 \log(x+1) - 2x \right]_0^1 - 2 = 8 \log 2 - 2 - 2$$

$$CS = 8(0.693) - 4 = 1.54 \text{ (see Fig 13.11)}$$

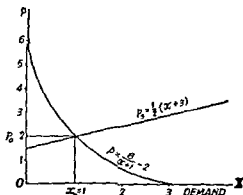


Fig 13.11

Consumer's Surplus under Monopoly

In order to determine price under monopoly we must equate MR and MC

Ex. 42. The demand and supply laws are $p_d = (6 - x)^2$ and $p_s = 14 + x$ respectively. Find the consumer's surplus, if (i) the demand and price are determined under pure competition, and (ii) the demand and price are determined under monopoly (so as to maximise the profit) and the supply function is identified with the marginal cost function.

(i) *C.S. under pure competition* Market equilibrium conditions are found by equating demand and supply, i.e., $(6-x)^2 = 14 + x = p$

$$\text{i.e.,} \quad p = 16 \text{ for } x = 2$$

$$\begin{aligned} C.S. &= \int_0^2 (36 - 12x + x^2) dx - (16 \times 2) \\ &= \left[36x - 6x^2 + \frac{x^3}{3} \right]_0^2 - 32 = \frac{56}{3} \text{ units} \end{aligned}$$

(ii) *To find demand and price under monopoly* we have

$$TR = p_d \times x$$

$$TR = (36 - 12x + x^2) x = 36x - 12x^2 + x^3$$

$$MR = \frac{dR}{dx} = 36 - 24x + 3x^2$$

$$\text{Supply price } p_s = 14 + x$$

But since supply price is to be identified with MC

$$14 + x = MC$$

To maximise profit, $MC = MR$

$$36 - 24x + 3x^2 = 14 + x, \quad (x-1)(3x-22) = 0$$

$$\text{i.e., either } x = 1, \text{ or, } x = \frac{22}{3} \approx 7.33$$

$$\text{Therefore, at } x = 1, \quad p_d = 25$$

$$\text{Hence,} \quad C.S. = \int_0^1 (36 - 12x + x^2) dx - (25 \times 1) = \frac{16}{3} \text{ units}$$

Similarly obtain $C.S.$ at $x = 7.33$

14.10 Producer's Surplus

With the given supply function the producer would have supplied quantities at different prices less than p_0 . At p_0 he supplies all x_0 units. Hence the shaded area becomes producer's surplus as shown in Fig. 14.12.

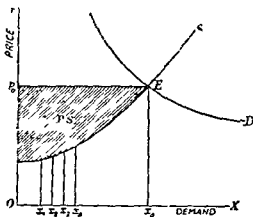


Fig. 14.12

Producer's Surplus = Area of the whole rectangle $p_0 E x_0 O$ —

Area of the curve under the supply curve from O to x_0

Ex. 14.11 Find the producer's surplus when

$$p_1 = 3x^2 - 20x + 5$$

$$p_2 = 15 + 9x \quad (x \text{ is the quantity})$$

in equilibrium $D = S \quad 3x^2 - 20x + 5 = 15 + 9x$

$$x = (10) \text{ or } \left(-\frac{1}{3}\right)$$

$x = 10$ is then, therefore, considered

At $x = 10$ equilibrium price = 105 units

$$\text{Total Revenue} = p \cdot x = 1050$$

$$\text{Total Supply price} = \int_0^{10} (15 + 9x) dx = 600$$

$$\text{Hence } P.S. = 1050 - 600 = 450$$

Ex 44 If the demand and supply functions are given $p_d = 20 - 5x$ and $p_s = 4x + 8$ obtain producer's surplus

In equilibrium $20 - 5x = 4x + 8$

$$x = \frac{4}{3}$$

$$\text{Equilibrium price} = 4 \left(\frac{4}{3} \right) + 8 = \frac{40}{3}$$

$$PS = \frac{40}{3} \times \frac{4}{3} - \left(\frac{128}{9} \right) = \frac{352}{9}$$

Hence required producer's surplus is 39.11 units

EXERCISE 13.3

1 If the demand law is $p = 39 - 3x^2$ find CS

(a) If $x_0 = \frac{5}{2}$ and (b) if commodity is free (i.e. $p_0 = 0$)

2 If the demand law is $p = \sqrt{9-x}$ and the demand is fixed at $x_0 = 9$ find the CS

3 The quantity sold and the corresponding price under monopoly are determined by the demand law $p = 16 - x^2$ and by the $MC = 6 + x$ in such a way as to maximise the profit. Determine corresponding CS

4 In question 3 if demand law is $p = 45 - x^2$ and

$$MC = 6 + \frac{x^2}{4}, \text{ determine } CS$$

5 Demand and supply laws are $p_d = 16 - x^2$ and $p_s = 4 + x$, determine CS under pure competition

6 Demand and supply laws are $p_d = 36 - x^2$ and $p_s = 6 + \frac{x^2}{4}$

Determine CS under pure competition

7 If the supply curve is $p = \sqrt{9-x}$ and the quantity sold in market $x_0 = 7$ find producer's surplus

8 Assume that the demand and average cost curves of steel are

$$p = 2.34 - 1.34 q \text{ and}$$

$$AC = \frac{1}{q} - 0.83 + 0.85 q$$

q is the quantity of steel demanded or produced

Show that the consumer's surplus under monopoly and perfect competition is 0.351 and 0.129 respectively

Show also that C.S. would have been equal to 2.043 if steel were a free good

9 Show that if the demand function of a commodity is given to be

$$p = 100 - 5x - x^2$$

C.S. at prices 0, 11 and 25 would be 469.71, 4661 and 453.18 respectively

Chapter 14

Differential Equations

14.1 Definitions

In the previous chapter we discussed the method of derivation of revenue function from marginal revenue function. We recall that

$$MR = \frac{dR}{dx}$$

$$\text{say } \frac{dR}{dx} = f(x)$$

This equation involves an independent variable x (quantity sold) and derivative of dependent variable R with respect to the independent variable. This equation is termed as a **Differential Equation**.

A *Differential equation* is one which connects an independent variable (x) , a dependent variable y which is a function $y = f(x)$ and its derivatives (differential coefficients) $y, y', y'', \dots, y^{(n)}$.

Symbolically, a differential equation may be written as

$$F(x, y, y', y'', \dots, y^{(n)}) = 0$$

$$\text{or } F\left[x, f(x), \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right] = 0$$

By integrating both sides of the differential equation, we are able to find the relation between the dependent variable and the independent variable.

$$\text{Thus if } \frac{dR}{dx} = 2 - 3x + x^2$$

$$\int dR = \int (2 - 3x + x^2) dx$$

$$\text{so that } R = 2x - \frac{3x^2}{2} + \frac{x^3}{3} + C$$

This equation represents the total revenue function. We now call it as the **solution** or **integral** of the differential equation.

14.2 Kinds of Differential Equations

- (i) *Ordinary type* involves only one independent variable

(ii) *Partial differential equations* involving more than one independent variable and partial derivatives

We shall concern ourselves mainly with ordinary differential equations in this chapter

14.3 Order of Differential Equation

The order of any differential equation is the order of the highest derivative or differential coefficient appearing in it

For example,

$\frac{dy}{dx} = 2x^2 - 3x + 4$ is the *1st Order* Differential Equation

$\frac{d^2y}{dx^2} = 3x - 4$ is the *2nd Order* Differential Equation

$\frac{d^3y}{dy^3} = 4x + 3$ is the *3rd Order* Differential Equation

14.4. Degree of a Differential Equation

The degree of a differential equation is equal to the highest power of the highest order derivative in the equation, after the equation has been simplified by clearing the radicals

For example $y = \sqrt{1 + \frac{d^2y}{dx^2}}$ is second order and first degree equation

Simplifying, we obtain $y^2 = \left\{ 1 + \frac{d^2y}{dx^2} \right\}$

The highest differential (derivative) is $\frac{d^2y}{dx^2}$ Hence the order of equation is 2. Highest degree of this highest differential is 1, hence the degree of equation is 1

In the equation, $\left(\frac{d^3y}{dx^3} \right)^2 + \frac{dy}{dx} + y = 0$

Highest derivative is $\frac{d^3y}{dx^3}$. Order is 3.

Highest power of $\frac{d^3y}{dx^3}$ is 2 $\therefore$ Degree is 2

14.5. Solution of a Differential Equation

A solution of a differential equation is any relation between the variables which is free of derivatives or differentials and which satisfies the equation

identically. It should be stated therefore as an integral of the function given in the differential equation

Suppose we are given the equation

$$\frac{dR}{dx} = 2 - 2x + 3x^2 \quad (1)$$

Integrating we get $R = 2x - \frac{2x^2}{2} + \frac{3x^3}{3} + C$

or $R = 2x - x^2 + x^3 + C \quad (2)$

(2) is the *general solution* of (1) because it contains a constant C which can take any value.

But if total revenue R for output $x = 0$ is zero then substituting $x = 0$ and $R = 0$ in (2) we get $C = 0$ and the solution of equation (1) will be

$$R = 2x - x^2 + x^3 + 0 = 2x - x^2 + x^3$$

which is the *particular solution* of equation (1)

Let us take another equation $\frac{d^2y}{dx^2} = x \quad (3)$

On integrating both sides

$$\frac{dy}{dx} = \frac{x^2}{2} + C$$

and by integrating again $y = \frac{x^3}{6} + Cx + D \quad (4)$

(4) is general solution of (3), (3) is of second order, hence its general solution contains two arbitrary constants C and D . If $C = 1$ and $D = 2$ say, then the solution of the given equation becomes a particular solution

$$y = \frac{x^3}{6} + x + 2$$

Again, these solutions do not contain any derivative, therefore these solutions are called *complete primitives* of the given equations (1) and (3)

Thus, always remember that

(1) A *general solution* of a differential equation must contain as many arbitrary constants as the order of the equation

(2) A *particular solution* of a differential equation is a solution obtained from the general solution by giving particular values to arbitrary constants of the general solution

14.6 Solution of Differential Equations — (Separated and Separable Variables)

Consider the following three forms

$$\frac{dy}{dx} = \varphi(x) \cdot \psi(y) \quad (i)$$

$$\varphi(y) dx + \psi(x) dy = 0 \quad (ii)$$

$$\text{and} \quad M_1(x) N_1(y) dx + M_2(x) N_2(y) dy = 0 \quad (iii)$$

In all these forms the functions are dependent on only one of the variables, either x or y . Equation (ii) is called an equation with separated variables and equations (i) and (iii) can easily be transformed into the form (ii). Hence they are equations with separable variables, i.e.

$$\frac{dy}{\varphi(y)} = \frac{dx}{\psi(x)} \quad (i)$$

$$\frac{M_1(x)}{M_2(x)} dx + \frac{N_2(y)}{N_1(y)} dy = 0 \quad (iii)$$

Solutions of these are obtained by integrating one part with respect to x and the other with respect to y .

We shall now consider three cases

(a) When the dependent variable is absent

$$\text{General form } \frac{dy}{dx} = f(x) \text{ or } dy = f(x) dx$$

$$\text{Solution is (by integrating) } y = \int f(x) dx$$

Ex. 1 Marginal cost = $1 + x + x^2 + x^3$. Find the total cost (C)

$$\text{Here } MC = \frac{dC}{dx} = 1 + x + x^2 + x^3$$

$$C = \int (1 + x + x^2 + x^3) dx = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + A$$

which is the general solution

(b) When the independent variable is absent

$$\text{General form } \frac{dy}{dx} = f(y) \text{ or } dy = f(y) dx$$

$$\text{This can be re-written in the form } \frac{dy}{f(y)} = dx$$

$$\text{Solution : } dx = \int \frac{dy}{f(y)}$$

Ex. 2. Solve the differential equation $\frac{dy}{dx} = ae^x$

By transposing, we get $\frac{dy}{e^x} = a \, dx$

By integrating both sides, $\int \frac{dy}{e^x} = \int a \, dx$

$$ay = \int e^{-x} dy = -e^{-x} + C$$

is the general solution of the given differential equation

(c) *General case*

Ex. 3 Solve the equation $y(1-x) + x \frac{dy}{dx} = 0$

The given equation can be put as $y(1-x) dx = x dy$

On L.H.S. we should have terms involving only x and on R.H.S. we must have terms involving only y

Rearranging we put the equation as $\left(\frac{1-x}{x}\right) dx = \frac{dy}{y}$

$$\text{i.e.} \quad \left(\frac{1}{x} - 1\right) dx = \frac{1}{y} dy$$

$$\text{So that} \quad \int \left(\frac{1}{x} - 1\right) dx = \int \frac{1}{y} dy$$

$$\log x - x = \log y + C$$

$$\log x - \log y = C + x$$

$$\log \frac{x}{y} = C + x \quad \text{or,} \quad \frac{x}{y} = e^{C+x}$$

$$\text{or,} \quad \frac{y}{x} = e^{-(C+x)} \quad \text{or,} \quad y = x e^{-(C+x)}$$

$$y = x \cdot e^{-x} \cdot e^{-C}$$

$$y = A \cdot x e^{-x}$$

(Since e^{-C} is a constant $= A$)

The above is the required general solution of the given differential equation

$$\left(\frac{1-x}{x}\right) dx = \frac{dy}{y}$$

Ex 4 Solve $\left(\frac{1+x^2}{1+y}\right) = xy \frac{dy}{dx}$

Rearranging we have $\frac{(1+x^2)}{x} dx = y(1+y) dy$

$$\int \frac{1+x^2}{x} dx = \int (y+y^2) dy$$

or $\log x + \frac{x^2}{2} = \frac{y^2}{2} + \frac{y^3}{3} + C$

∴ the required general solution of the given equation

Ex 5 Solve $(x^2 - yx^2) dy + (y^2 + xy^2) dx = 0$

Rearranging we have $x^2(1-y) dy + y^2(1+x) dx = 0$

i.e. $x^2(1-y) dy = -y^2(1+x) dx$

$$\int \left(\frac{1-y}{y^2}\right) dy = -\int \left(\frac{1+x}{x^2}\right) dx$$

Integrating and rearranging we have

$$\log x - \log y - \frac{1}{x} - \frac{1}{y} = C$$

or $\log \frac{1}{y} - \left(\frac{x+y}{x}\right) = C$

∴ the required general solution of the given equation

Ex 6 Solve $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

Rearranging $\frac{dy}{dx} = e^x e^{-y} + x^2 e^{-y}$

or $\frac{dy}{dx} = e^{-y}(e^x + x^2)$

$$\int \frac{dy}{e^y} = \int (e^x + x^2) dx$$

i.e., $e^{-y} = e^x + \frac{x^3}{3} = C$ is the required solution

$$20. \quad y = c_1 x + \frac{c_2}{x} + c_3 \qquad \frac{d^3 y}{dx^3} + \frac{3}{x} \frac{d^2 y}{dx^2} = 0$$

$$21. \quad y = \frac{c_1}{x} + c_2 \qquad \frac{d^2 y}{dx^2} + \frac{2}{x} \frac{dy}{dx} = 0$$

14.7. Solution of Linear Homogeneous Equations

We take up a special form of the equation $\frac{dy}{dx} + P = 0$, where P is a function of $\left(\frac{y}{x}\right)$. Such an equation is known as Homogeneous in x and y if it is homogeneous of degree zero in x and y .

This type of equation is solved by putting $\left(\frac{y}{x}\right) = V$, so that $y = Vx$, and $\frac{dy}{dx} = V + x \frac{dV}{dx}$ or, $dy = V dx + x dV$, so that the variables are separated.

We solve some examples to explain this method.

Ex. 7. Solve : $(x+y) + (y-x) \frac{dy}{dx} = 0$

It is homogeneous equation as it is of the form :

$$\frac{dy}{dx} + \frac{x+y}{y-x} = 0 \quad \text{or; } \frac{dy}{dx} + \frac{1 + \left(\frac{y}{x}\right)}{\left(\frac{y}{x}\right) - 1} = 0$$

Substitute : $y = Vx$, and $dy = V dx + x dV$ in the given equation

$\therefore$ The equation now becomes

$$(x+y) dx + (y-x) dy = 0$$

$$(x + Vx) dx + (Vx - x) (V dx + x dV) = 0$$

$$\text{or, } (1 + V) dx - (1 - V) (V dx + x dV) = 0$$

$$\text{or, } (1 + V) dx - (1 - V) V dx - (1 - V) x dV = 0$$

$$\text{or, } dx [1 + V - V + V^2] - (1 - V) x dV = 0$$

$$\text{or, } (1 + V^2) dx = (1 - V) x dV$$

$$\text{or, } \frac{dx}{x} = \left(\frac{1 - V}{1 + V^2} \right) dV \quad [\text{It is separated variables form}]$$

$$\text{or,} \quad \frac{dx}{x} = \left(\frac{1}{1+V^2} - \frac{V}{1+V^2} \right) dV$$

$$\therefore \quad \int \frac{dx}{x} = \int \frac{1}{1+V^2} dV - \int \frac{V}{1+V^2} dV$$

$$\text{or,} \quad \log x = \tan^{-1} V - \frac{1}{2} \log (1+V^2) + C.$$

Substituting back the value of $V = \frac{y}{x}$, we obtain

$$\log x + \frac{1}{2} \log \left(1 + \frac{y^2}{x^2} \right) - \tan^{-1} \frac{y}{x} = C$$

Ex. 8. Solve . $y - x \frac{dy}{dx} = x + y \frac{dy}{dx}$

The given equation can be put as $y - x = (y + x) \frac{dy}{dx}$

$$\text{i.e.,} \quad \frac{dy}{dx} = \frac{y-x}{y+x}$$

$$\text{Put} \quad y = Vx \quad \therefore \quad \frac{dy}{dx} = V + x \cdot \frac{dV}{dx}$$

Substituting value of y and $\frac{dy}{dx}$ in the equation, we have

$$V + x \cdot \frac{dV}{dx} = \frac{Vx - x}{Vx + x} = \frac{V-1}{V+1}$$

$$\text{or,} \quad x \frac{dV}{dx} = \frac{V-1}{V+1} - V = - \left(\frac{V^2 + 1}{V+1} \right)$$

$$\text{or,} \quad \left(\frac{V+1}{V^2+1} \right) dV = - \frac{dx}{x}$$

$$\text{or,} \quad \int \frac{V+1}{V^2+1} dV = - \int \frac{1}{x} dx$$

$$\text{or,} \quad \int \frac{V}{V^2+1} dV + \int \frac{1}{V^2+1} dV = - \int \frac{1}{x} dx$$

$$\text{or,} \quad \frac{1}{2} \log(V^2+1) + \tan^{-1} V = - \log x + C$$

$$\text{or,} \quad \frac{1}{2} \log \left(\frac{y^2}{x^2} + 1 \right) + \tan^{-1} \frac{y}{x} = -\log x + C$$

$$\text{or,} \quad \frac{1}{2} \left[\log (x^2 + y^2) - \log x^2 \right] + \log x + \tan^{-1} \frac{y}{x} = C$$

$$\text{or,} \quad \frac{1}{2} \log (x^2 + y^2) + \tan^{-1} \frac{y}{x} = C \text{ is the solution}$$

14.8. Equations Reducible to Homogeneous Equations

Equations of the form .

$$\frac{dy}{dx} = \frac{ax + by + c}{a_1x + b_1y + c_1}$$

are homogeneous if $c = 0$, $c_1 = 0$

If $c \neq 0$, $c_1 \neq 0$, then try the substitution

$$x = x_1 + h, \text{ and}$$

$$y = y_1 + k,$$

$$\text{So that} \quad \frac{dy}{dx} = \frac{dy_1}{dx_1}$$

$$\text{Ex. 9. Solve } \frac{dy}{dx} = \frac{x+y-3}{x-y-1}$$

$$\text{Put } x = x_1 + h; \quad y = y_1 + k$$

$$\begin{aligned} \text{Hence;} \quad \frac{dy_1}{dx_1} &= \frac{x_1 + h + y_1 + k - 3}{x_1 + h - y_1 - k - 1} \\ &= \frac{x_1 + y_1 + h + k - 3}{x_1 - y_1 + h - k - 1} \end{aligned}$$

$$\text{On putting; } h + k - 3 = 0$$

$$h - k - 1 = 0$$

$$\text{we get, } h = 2 \text{ and, } k = 1$$

so that the equation becomes :

$$\frac{dy_1}{dx_1} = \frac{x_1 + y_1}{x_1 - y_1}$$

This is homogeneous equation, which can now be solved easily

Substitute $y_1 = ux_1$. This leads to

$$u + x_1 \frac{du}{dx_1} = \frac{1+u}{1-u}$$

or, $x_1 \frac{du}{dx_1} = \frac{1+u^2}{1-u}$

or, $\left(\frac{1-u}{1+u^2} \right) du = \frac{dx_1}{x_1}$

or, $\tan^{-1} u - \frac{1}{2} \log(1+u^2) = \log x_1 + C$

$$Cx_1 \sqrt{1+u^2} = e^{\tan^{-1} u}$$

Substitute for u and x_1

EXERCISE 14.2

Solve the following equations

1. $(x+y) dx + x dy = 0$

2. $(x+y) dx - x dy = 0$

3. $y^2 - x^2 \frac{dy}{dx} = 0$

4. $(x-2y) dy + x dx = 0$

5. $(y^2 - xy) dx + x^2 dy = 0$

6. $(x+y) dx + (x-y) dy = 0$

7. $y^3 dx - x^3 dy = 3$

8. $(xy - x^2) \frac{dy}{dx} = y^2$

9. $(x^2 - 2y^2) dx = x y dy$

10. $\frac{y^2}{x} dx + (x-y) dx = 0$

11. $(xy - y^2) dx + \left(y^3 - \frac{x^2}{2} \right) dy = 0$

12. $\frac{dy}{dx} = \frac{xy+y}{x+xy}$

13. $x dy - y dx = \sqrt{x^2 + y^2} dx$

14. $(x^2 + y^2) dx - 2xy dy = 0$

15. $(x^2 - y^2) dx + 2xy dy = 0$

16. $(x^2 + y^2) dx + 2xy dy = 0$

17. $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$

18. $(x^2 + y^2) \frac{dy}{dx} = xy$

$$19 \quad \frac{dy}{dx} = \frac{y^3 + 3x^2y}{x^3 - 3xy^2}$$

$$20 \quad \frac{y^3 - 3x^2y}{x^3 - 3xy^2} = \frac{dy}{dx}$$

$$21 \quad \frac{dy}{dx} = \frac{xy}{x^2 - y^2}$$

$$22 \quad (8y + 10x) dx + (5y + 7x) dy = 0$$

$$23 \quad xy^2 dx = (x^3 + y^3) dx$$

$$24 \quad (2\sqrt{s}t - s) dt + ds = 0$$

$$25 \quad (3y - 7x + 7) dx - (3x - 7y - 3) dy = 0$$

$$26 \quad (x + 2y + 1) dx - (2x + 4y + 3) dy = 0$$

$$27 \quad (x + 2y + 1) dx - (2x + 3) dy = 0$$

$$28 \quad (2x + y - 3) dy = (x + 2y - 3) dx$$

14.9 Exact Differential Equations

General form

$$M(x, y) dx + N(x, y) dy = 0$$

An equation is called an exact differential equation when it is formed from its primitive by simple differentiation

Let there be a function of two variables $F(x, y)$. Total differential of the function is

$$dF = \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial x} dx,$$

$$\text{Thus for } F(x, y) = 0 \quad dF = \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial x} dx = 0 \quad (u)$$

This form of equation is known as exact differential equation since its

$$\text{L.H.S.} \left(= \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial x} dx \right) \text{ is the exact differential of } F(x, y)$$

Exact differential equations are therefore, those which are obtained from their primitive directly by differentiation without any subsequent multiplication, elimination or other manipulation.

Comparing (i) and (u) we find

$$M = \frac{\partial F}{\partial x}, \quad \text{and} \quad N = \frac{\partial F}{\partial y}$$

$$\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial x \partial y} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 F}{\partial y \partial x}$$

Since we know that $\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

As such, if partial derivative of M with respect of y $\left(= \frac{\partial M}{\partial y} \right)$ is equal to the partial derivative of N with respect of x $\left(= \frac{\partial N}{\partial x} \right)$ then the equation $M dx + N dy = 0$ is an exact differential equation. However, an exact equation may involve terms of y^2 , y^3 etc., and therefore may not be linear in y .

Solution of an Exact Differential Equation. We have seen above that

$$dF(x, y) = \frac{\partial F}{\partial x} \cdot dx + \frac{\partial F}{\partial y} \cdot dy$$

$$\text{Integrating both sides, we get } F(x, y) - C = \int \left[\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \right]$$

$\therefore$ Solution of the exact differential equation involves constant and $F(x, y)$, which is the primary function; the differential of which yields the exact differential equation.

The solution involves three steps :

1. Integrate : $\int M \cdot dx$ assuming y as constant.
2. Integrate : $\int N \cdot dy$ assuming x as constant.
3. Add the results, $\left(\text{i.e., } \int M dx + \int N dy \right)$, but terms common to both integrals are written only once.

Ex. 10. Solve the equation : $(x + y) dx + (x + 4y) dy = 0$

Here, $M = x + y$, $N = x + 4y$

$$\text{And } \frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(x + y) = 0 + 1 = 1, \text{ and } \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(x + 4y) = 1 + 0 = 1$$

$$\text{Hence, } \frac{\partial}{\partial y}(x + y) = \frac{\partial}{\partial x}(x + 4y),$$

therefore, the given equation is an exact differential equation. To find its solution, we follow the above-mentioned steps :

$$(1) \int M dx = \int (x + y) dx = \frac{x^2}{2} + yx \quad (\text{by treating } y \text{ as constant})$$

$$(2) \int N dy = \int (x + 4y) dy = xy + \frac{4y^2}{2} \quad (\text{by treating } x \text{ as constant})$$

(3) Add the results (1) and (2), but since xy occurs in both integrals, hence, xy is to be added only once

Therefore, the solution of the given equation is

$$\frac{x^2}{2} + 2y^2 + xy = C \text{ (constant)}$$

$$\text{i.e., } x^2 + 2xy + 4y^2 = C \text{ (a new constant)}$$

Ex. 11. Solve the differential equation $\frac{dy}{dt} + \frac{y(1+2t)}{t(1+t)} = 0$

$$t(1+t) dy + y(1+2t) dt = 0, \quad M = t(1+t), \quad N = y(1+2t)$$

Here, $\frac{\partial}{\partial t}(t + t^2) = (1 + 2t)$ by treating y to be constant, and

$$\frac{\partial}{\partial y}[y(1+2t)] = (1+2t) \text{ by treating } t \text{ to be constant}$$

Therefore, the equation is an exact differential equation

Hence, for solution, we note

$$(1) \int (t + t^2) dy = (t + t^2)y, \text{ (by treating } t \text{ as constant)}$$

$$(2) \int y(1+2t) dt = y \left(t + \frac{2t^2}{2} \right) = y(t + t^2) \text{ (treating } y \text{ as constant)}$$

(3) Add both the results, but since $y(t + t^2)$ occurs in both the results, it will be counted only once.

$$\therefore \text{ The solution of the given equation is } y(t + t^2) = C$$

Transformation of Inexact Differential Equations to Exact Differential Equations Sometimes an inexact differential equation can be made exact differential equation on multiplying every term of the equation by a particular common factor. This factor is called the integrating factor. Often it is not easy to find an integrating factor, but whenever it is easy to find by inspection, we first transform the equation to *exact form*.

$$\text{Let,} \quad M(x, y) dx + N(x, y) dy = 0 \quad \dots(i)$$

not be an exact differential. Let us try to find out the integrating factor of equation (i),

Find $\frac{\partial N}{\partial x}$ and $\frac{\partial M}{\partial y}$ as before, and form,

$$\left\{ \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} \right\}$$

(i) If it is independent of x , then obtain the integrating factor by solving

$$\frac{d \log \mu}{dy} = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

(ii) If it is independent of y , then find the integrating factor by solving

$$\frac{d \log \mu}{dx} = - \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{N}$$

Clearly now the (Integrating Factor) $I \Gamma$ depends only on x .

Ex 12 Solve $x \, dy - y \, dx = x^2 \, dx$

This may be put as $x \, dy + (-y - x^2) \, dx = 0$

Here $N = -$, $M = (-y - x^2)$

Hence, $\frac{\partial N}{\partial x} = 1$, $\frac{\partial M}{\partial y} = -1$

Therefore, $\frac{\partial N}{\partial x} \neq \frac{\partial M}{\partial y}$

This equation is not exact

However,

$$\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) / N \quad \text{and} \quad \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) / M$$

are $\frac{1+1}{x} = \frac{2}{x}$ and, $\frac{1+1}{-(y+x^2)}$ respectively

So that, $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{N}$ is independent of y . Hence $I \Gamma$ is obtained by solving

$$\frac{d \log \mu}{dx} = - \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{N} = - \frac{2}{x}$$

Integrating, we obtain

$$\log \mu = -2 \log x = \log \frac{1}{x^2}$$

$$\text{or, } \mu = \frac{1}{x^2}$$

Multiplying the original equation by the I.F. = $\left(\frac{1}{x^2}\right)$, we get

$$\frac{dy}{x} - \left(1 + \frac{y}{x^2}\right) dx = 0$$

$$\text{Now } \frac{\partial}{\partial x} \left(\frac{1}{x}\right) = -\frac{1}{x^2} \text{ and,}$$

$$\frac{\partial}{\partial y} \left(-1 - \frac{y}{x^2}\right) = -\frac{1}{x^2}$$

Therefore, the equation now becomes exact differential equation and so the solution will be as follows

$$(i) \quad \text{Integral of } \frac{1}{x} \text{ with respect to } y \text{ treating } x \text{ as constant } \left(= \frac{y}{x}\right)$$

$$(ii) \quad \text{Integral of } \left(-\frac{y}{x^2} - 1\right) \text{ with respect to } x \text{ treating } y \text{ as constant}$$

$$\left(= +\frac{y}{x} - x\right)$$

(iii) Hence the required solution is $\frac{y}{x} - x = C$ (since $\frac{y}{x}$ occurs in both the results) which may be put as $y = x(x + C)$

EXERCISE 14.3

Solve the following differential equation (Also test their

$$1. \quad (3x^2 + 6xy) dy + (3x^2 + 3y^2) dx = 0 \quad 2. \quad x$$

$$3. \quad (2x + 3y) dx + (3x - 2y) dy = 0$$

4. $(2x + 3y + 1) dx + (3x - 2y + 1) dy = 0$

5. $(x^2 + 2xy^2) dx + (2x^2 + y^2) dy = 0$

6. $(x^2 + y^2) dx + 2xy dy = 0$

7. $2xy dx + (x^2 + 4y) dy = 0$

8. $2yz^3 dy + 3y^2 z^2 dz = 0$

9. $\frac{dy}{dx} = \frac{y(1+y)}{x(1+2y)} = 0$

10. $(6x^2 - 10xy + 3y^2) dx - (5x^2 - 6xy + 3y^2) dy = 0$

11. $xy dy - (y^2 - x^2) dx = 0$

12. $(x^2 - y^2 - 1) dx + 2xy \cdot dy = 0$

13. $y dx + (2x + y) dy = 0$

14. $(2x + y) dx + (x + 2y) dy = 0$

15. $(y + xy^2) dx - x dy = 0$

16. $(x^2 + y) dx - (x - 2y) dy = 0$

17. $(y^2 - x) \frac{dy}{dx} = y$

18. $\left(\frac{1}{x^2} + \frac{3y^2}{x^4} \right) dx = \frac{2y dy}{x^3}$

19. $\frac{x^2 dy - y^2 dx}{(x-y)^2} = 0$

20. $x \cdot dx + y \cdot dy = \frac{y \cdot dx - x \cdot dy}{x^2 + y^4}$

14.10. General First Order Linear Equations

General Form of Linear Equation $\frac{dy}{dx} + Py = Q$

or, $dy + (Py - Q) dx = 0$, where P and Q are constants or functions of x only.

Its integrating factor is then given by ; I.F. = $e^{\int P dx}$.

Equation : $\frac{dy}{dx} + Py = Q$ or, $dy + (Py - Q) dx = 0$ is not an exact equation .

On multiplying it by I.F. = $e^{\int P dx}$

$$e^{\int P dx} \cdot dy + e^{\int P dx} (Py - Q) dx = 0$$

makes it an exact differential equation.

Let us examine its exactness :

$$\frac{\partial}{\partial x} \cdot e^{\int P dx} = e^{\int P dx} \cdot \frac{\partial}{\partial x} \int P dx = e^{\int P dx} \cdot P$$

$$\frac{\partial}{\partial y} \cdot e^{\int P dx} (Py - Q) = e^{\int P dx} \cdot \frac{\partial}{\partial y} (Py - Q) = e^{\int P dx} \cdot P$$

The test gets satisfied, therefore, the solution should be found as usual.
The general form of the solution shall be :

$$y \cdot e^{\int P dx} = \int Q \cdot e^{\int P dx} dx.$$

Ex. 13. Solve ; $\frac{dy}{dx} + xy = x$

This equation is not exact

Here, $P = x, Q = 2$

Hence I.F. will be $e^{\int P dx} = e^{\int x dx} = e^{x^2/2}$

The required solution on substituting in the general form, shall be

$$y \cdot e^{x^2/2} = \int x \cdot e^{x^2/2} dx$$

But $\int x \cdot e^{x^2/2} dx = \int e^t dt$ (By substituting $x^2/2 = t$)

$$= e^t + C = e^{x^2/2} + C$$

$\therefore$ Solution is : $y e^{x^2/2} = e^{x^2/2} + C$

or, $y = \frac{e^{x^2/2} + C}{e^{x^2/2}} = 1 + C e^{-x^2/2}$

Ex. 14 Solve the equation : $\cos^2 x \frac{dy}{dx} + y = \tan x$.

Changing it into the standard form : $\frac{dy}{dx} + \frac{1}{\cos^2 x} \cdot y = \frac{\tan x}{\cos^2 x}$

$\therefore P = \frac{1}{\cos^2 x} = \sec^2 x$ and $Q = \sec^2 x \tan x$

Hence, I.F. = $e^{\int P dx} = e^{\int \sec^2 x dx} = e^{\tan x}$

$\therefore$ The solution is $y e^{\tan x} = \int Q e^{\int P dx} dx = \int \sec^2 x \cdot \tan x e^{\tan x} dx$

Put $\tan x = t$, $\therefore \sec^2 x dx = dt$.

$y e^{\tan x} = \int \sec^2 x \cdot \tan x e^{\tan x} dx = \int t e^t dt$

But $\int t \cdot e^t dt = t \cdot e^t - \int e^t \cdot dt = t \cdot e^t - e^t + C$ (Integrating by parts)

$$= e^t \cdot (t - 1) + C = e^{\tan x} (\tan x - 1) + C$$

The required solution is $y e^{\tan x} = e^{\tan x} (\tan x - 1) + C$

$$\text{or, } y = (\tan x - 1) + C e^{-\tan x}$$

Ex 15 Prove that $y = 2cx^2 + c^2$ is the solution of the equation

$$\left(\frac{dy}{dx}\right)^2 + 8x^3 \left(\frac{dy}{dx}\right) = 16x^2 y$$

and find the definite solution that satisfies the condition $y(1) = -1$

We are given solution $y = 2cx^2 + c^2$

$$\frac{dy}{dx} = 4cx$$

and this should satisfy the given differential equation

$$(4cx)^2 + 8x^3 (4cx) = 16x^2 (2cx^2 + c^2)$$

$$\text{or } 16c^2 x^2 + 32cx^4 = 16cx^2 (c + 2x^2)$$

$$\text{or, } 16cx^2 (c + 2x^2) = 16cx^2 (c + 2x^2)$$

Hence proved

Definite solution We are given $y = -1$ at $x = 1$

$y = 2cx^2 + c^2$ becomes

$$-1 = 2c + c^2, \quad \text{or } c = -1$$

Therefore definite solution is $y = 1 - 2x^2$

Ex 16 Solve $(x^2 y^3 + xy) dy = dx$

The given equation is, $\frac{dx}{dy} - xy = x^2 y^3$

This is of the form $\frac{dy}{dx} + P y = Q y^n$

where P and Q are functions of x (the independent variable only). This is Bernoulli's equation, which differs from the standard linear form in that the R.H.S. consists of y , the dependent variable. However, such an equation can be converted into standard form by a simple substitution. Divide by y^n throughout,

$$\frac{1}{y^n} \frac{dy}{dx} + \frac{P}{y^n} y = Q$$

or $y^{-n} \frac{dy}{dx} + y^{1-n} P = Q$

Put the coefficient of P (i.e. y^{1-n}) = λ and proceed

In the given equation (note y is the independent variable)

$$\frac{dx}{dy} - xy = x^2 y^3$$

$$\frac{1}{x^2} \frac{dx}{dy} - \frac{x}{x^2} y = y^3$$

Put $\frac{1}{x} = \lambda$ so that $-\frac{1}{x^2} \frac{dx}{dy} = \frac{d\lambda}{dy}$

Substituting

$$-\frac{d\lambda}{dy} - \lambda y = y^3 \quad \text{that is} \quad \frac{d\lambda}{dy} + \lambda y = -y^3$$

which is a linear differential equation

where, $P = y$ and $Q = -y^3$

$$\text{I.F.} = e^{\int P dy} = e^{\int y dy} = e^{y^2/2}$$

Hence the solution is

$$\lambda e^{y^2/2} = \int e^{y^2/2} (-y^3) dy + C$$

$$= -\int 2e^t t dt + C \quad \left[\text{on putting } \frac{y^2}{2} = t \right]$$

$$= -2 \left[t e^t - \int 1 \cdot e^t \right] + C$$

$$= -2 t e^t - 2 e^t + C$$

$$\text{Therefore, } \frac{1}{x} e^{y^2/2} = -2 \frac{y^2}{2} e^{y^2/2} + 2 e^{y^2/2} + C$$

or, $\frac{1}{x} = -y^2 + 2 + C e^{-y^2/2}$

is the required solution

14.11 Linear Differential Equations With Constant Coefficients

In the general form of linear differential equation

$$\frac{dy}{dx} + Py = Q \text{ where } P \text{ and } Q \text{ are given functions of } x$$

Let P and Q be constants, we consider the two cases

Case I (Simple form) When P is a constant ($=a$) and Q is zero the general form of the linear equation

$$\frac{dy}{dx} + ay = 0$$

this is a simple form of differential (first order) equation

Case II (Non simple form) When P is a constant ($=a$) and Q is another constant b , the general form of equation becomes

$$\frac{dy}{dx} + ay = b$$

This is called the non simple form of the first order linear differential equation. Its solution consists of two parts

(i) *Complementary function* y_c

and (ii) *Particular integral* y_p

We take each case separately

(i) Simple form $\frac{dy}{dx} + ay = 0$

$$\frac{dy}{dx} = -a dx$$

$$\int \frac{1}{y} dy = -a \int dx$$

$$\log y = -ax + c$$

$$y = e^{-ax+c} = e^c \cdot e^{-ax} = A \cdot e^{-ax}$$

$$y = A e^{-ax} \text{ (where } A \text{ is constant } = e^c)$$

The solution of simple form of equation $\frac{dy}{dx} + ay = 0$ is given by

$$y = A e^{-ax}$$

(ii) Non simple Form $\frac{dy}{dx} + ay = b$

As explained above the solution of this equation is y_c plus y_p

(i) In order to find y_c , we write the simple or auxiliary form of the given equation, which is $\frac{dy}{dx} + ay = 0$. The solution of this equation, as has already been found, is $y_c = A e^{-ax}$. Thus the complementary function is found

(ii) Particular integral is any particular solution of the complete equation. Let us make it clear by taking a simple example. Suppose we are required to find particular solution of $y = \text{constant}$

$\frac{dy}{dx} = 0$. Hence the complete equation $\frac{dy}{dx} + ay = b$ gives the solution $ay = b$ or $y = \frac{b}{a}$ where b is some constant. Therefore, particular integral $y_p = \frac{b}{a}$

[Note: If $a = 0$, the solution will not be possible. Thus the condition for a solution is valid when $a \neq 0$]

Thus we have found both particular and complementary solutions

Therefore, the complete solution of the given differential equation

$$\frac{dy}{dx} + ay = b \text{ is } y = y_p + y_c$$

or,
$$y = \frac{b}{a} + A e^{-ax}$$

We can go further and find the definite solution of the given equation for certain definite values of x and y

Suppose at $x = 0$, $y = y(0)$, then the solution will be

$$y(0) = \frac{b}{a} + A e^0 = \frac{b}{a} + A$$

So that,
$$A = \left[y(0) - \frac{b}{a} \right]$$

Substituting value of constant A in the complete solution we get the definite solution which will be

$$y = \frac{b}{a} + \left[y(0) - \frac{b}{a} \right] e^{-ax}$$

Ex. 17. Find the general solution of the differential equation

$$2 \frac{dy}{dx} + 4y = 6$$

Also find definite solution for $y(0) = 1$

Reducing the given equation to simple (auxiliary) form to obtain y_c

$$2 \frac{dy}{dx} + 4y = 0, \quad \text{or,} \quad \frac{dy}{dx} + 2y = 0$$

Here, $a = 2$ and $b = 3$

$$\therefore y_c = A e^{-ax} \text{ and } y_p = \frac{b}{a} = \frac{3}{2}$$

$$\text{The complete solution is } y = \frac{3}{2} + A e^{-2x}$$

EXERCISE 14.4

Solve

1. $dy + (4xy - 4x) dx = 0$

2. $dy + (2xy - x) dx = 0$

3. $\frac{dy}{dx} + 3x^2y = e^{-x^2}$

4. $dy = (5y - 10) dx = 0$

5. $dy + (x^2y - 3x^2)dx = 0$

6. $dy + (3x^2y - e^{-x^2}) dx = 0$

7. $2 \frac{dy}{dx} + (12y + 2e^x) = 0$

8. Find the definite solution of the equation

$$x^2 \frac{dy}{dx} + xy = -1, \text{ when } y(0) = 1.$$

Find the complementary function, the particular integral, general solutions of the following equations

9. $\frac{dy}{dx} + y = 0, y(0) = 10$

10. $\frac{dy}{dx} - 7y = 7; y(0) = 7$

11. $2 \frac{dy}{dx} + 10y = 15, y(0) = 0$

12. $3 \frac{dx}{dt} + 6x = 7; y(0) = 0$

13. $\frac{dx}{dy} = 2, x(0) = 0$

14. $\frac{av}{dx} - 4y = 2; y(0) = 0$

15. $\sqrt{2} \cdot \frac{dy}{dx} - \sqrt{6y} = \sqrt{8}, y(0) = 0$

14.12 : Second Order Linear Differential Equations with Constant Coefficients

$$\frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + P_2 y = Q \quad (1)$$

where P_1 and P_2 are constants and Q is a function of x . This can also be written as

$$f(D)y = Q \quad \text{where } D \text{ stands for } \frac{d}{dx}$$

thus, the expression is equivalent to

$$(D^2 + P_1 D + P_2)y = Q$$

Some results for the solution of a linear Difference Equation (D.E.) with constant coefficient

1 A linear combination of solutions is a solution. That is, if $y = y_1, y_2, \dots, y_n$ are n linearly independent solutions of the linear homogeneous Differential Equation (1) then $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ is also the solution. This is also termed as the general solution of the equation. Note that the number of constants is equal to the order of the linear D.E. Hence in a second order equation, there will be two constants in the general solution.

2 The complete solution of a non homogeneous D.E. is composed of the complementary function (C.F.) and the particular integral (P.I.)

The complementary function is the general solution of the homogeneous part of the D.E., i.e., the equation obtained by putting the R.H.S. = 0 that is by putting $f(D)y = 0$.

3 To obtain the complementary function, we find the roots of the auxiliary equation obtained by writing $f(D) = 0$ in the D.E.

$$\frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + P_2 y = 0$$

the auxiliary equation is

$$m^2 + P_1 m + P_2 = 0$$

(This is obtained by putting $y = A_1 y^m$ and inserting it in the D.E.)

This is obviously a polynomial of degree n in m . Hence it will have n roots. In the second order equation, there will be two roots. The roots may be

- Real and different m_1 and m_2 in D.E. of second order
- Real and equal (Repeated roots)
- Complex Conjugate.

(i) Solution of homogeneous equation in the case of real and different roots will be

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

c_1 and c_2 are two arbitrary constants

(ii) Repeated roots For each repeated root, we multiply the constant by x , if $m_1 = m_2 = m$, then

$$y_c = (c_1 + c_2 x) e^{mx}$$

If in a D.E. of order n , three roots are equal, i.e., $m_1 = m_2 = m_3 = m$ then

$$y_c = (c_1 + c_2 x + c_3 x^2) e^{mx} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$$

(iii) Complex conjugate roots If the roots are of the form $\alpha + i\beta$ and $\alpha - i\beta$ (i.e., $\alpha \pm i\beta$), then

$$y_c = e^{\alpha x} [c_1 e^{i\beta x} + c_2 e^{-i\beta x}]$$

[Putting $e^{i\theta} = \cos \theta + i \sin \theta$]

$$\begin{aligned} y_c &= e^{\alpha x} [c_1 (\cos \beta x + i \sin \beta x) + c_2 (\cos \beta x - i \sin \beta x)] \\ &= e^{\alpha x} [(c_1 + c_2) \cos \beta x + i(c_1 - c_2) \sin \beta x] \\ &= e^{\alpha x} (A \cos \beta x + B \sin \beta x) \end{aligned}$$

Here α and β are found from the auxiliary equation and c_1 and c_2 from the initial conditions

The R.H.S. can be written as

$$y = e^{\alpha x} A \sin (\beta x + B)$$

where A and B are obtained from the initial conditions

Rules for Obtaining Particular Integral

Let the D.E. be

$$\frac{d^2 y}{dx^2} + P_1 \frac{dy}{dx} + P_2 y = Q \quad \dots (1)$$

(We can take D.E. of order n) Note that P_1 and P_2 are constants and Q is a function of x .

(i) Let $Q = x$, then try P.I.,

$$y_p = (A_0 x + A_1)$$

substitute in (1) and solve for A_0 and A_1

(ii) Let $Q = x^n$, then write

$$y_p = A_1 x^n + A_2 x^{n-1} + \dots + A_n$$

(iii) Let $Q = e^{\alpha x}$, then try

$$y_p = A e^{\alpha x}$$

(iv) Let $Q = x e^{\alpha x}$, then try

$$y_p = (Ax + B)e^{\alpha x}$$

(v) Let $Q = \cos x$, then try

$$y_p = A \cos x + B \sin x$$

Ex 18 Show that $y = a_1 e^{kx} + a_2 e^{-kx}$

is the solution of equation $\frac{d^2 y}{dx^2} - k^2 y = 0$

The given solution is, $y = a_1 e^{kx} + a_2 e^{-kx}$

$$\frac{dy}{dx} = k a_1 e^{kx} + a_2 (-k) e^{-kx}$$

and,

$$\begin{aligned} \frac{d^2 y}{dx^2} &= k^2 a_1 e^{kx} + a_2 k^2 e^{-kx} = k^2 (a_1 e^{kx} + a_2 e^{-kx}) \\ &= k^2 y \quad [\quad a_1 e^{kx} + a_2 e^{-kx} = y] \end{aligned}$$

$$\frac{d^2 y}{dx^2} = k^2 y \quad \text{or,} \quad \frac{d^2 y}{dx^2} - k^2 y = 0 \text{ is the equation}$$

Ex. 19 Prove that $y = 2cx^2 + c^2$ is the solution of the equation

$$\left(\frac{dy}{dx} \right)^2 + 8x^2 \left(\frac{dy}{dx} \right) = 16x^2 y$$

and find the definite solution that satisfies

$$y(1) = (-1)$$

We are given solution $y = 2cx^2 + c^2$

$$\frac{dy}{dx} = 4cx,$$

this should satisfy the given differential equation

$$(4cx)^2 + 8x^2(4cx) = 16x^2(2cx^2 + c^2) \left\{ \begin{array}{l} \text{since } y = 2cx^2 + c^2 \\ \text{and } \frac{dy}{dx} = 4cx \end{array} \right\}$$

$$\text{or, } 16c^2x^2 + 32cx^4 = 16cx^2(c + 2x^2)$$

$$\text{or } 16cx^2(c + 2x^2) = 16cx^2(c + 2x^2)$$

Hence proved

Definite solution We are given $y = -1$ at $x = 1$

$$y = 2cx^2 + c^2 \text{ becomes } -1 = 2c + c^2$$

$$\text{or, } c^2 + 2c + 1 = 0, \text{ i.e., } c = -1$$

Therefore, the definite solution from general solution

$$y = 2cx^2 + c^2 \text{ becomes } y = 1 - 2x^2$$

$$\text{Ex 20 Solve } \frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 12y = 0$$

The given equation is the particular form of second order equation. We give below the method of finding solution to such an equation without proof

$$\text{In the given equation } \frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 12y = 0$$

$$\text{Let } y = e^{mx}$$

$$\frac{dy}{dx} = me^{mx} \text{ and } \frac{d^2y}{dx^2} = m^2 e^{mx}$$

The given equation will be now

$$m^2 e^{mx} - 7me^{mx} + 12 e^{mx} = 0$$

$$\text{or, } e^{mx}(m^2 - 7m + 12) = 0$$

$$\text{or, } m^2 - 7m + 12 = 0$$

$$\text{or, } (m - 4)(m - 3) = 0$$

(This equation is called Auxiliary equation)

The roots of Auxiliary equation are $m = 4$ or $m = 3$

Therefore, the complete solution (which, in this example, will consist of complementary function only) of the given equation will be

$$y = y_c = c_1 e^{3x} + c_2 e^{4x} \quad (c_1 \text{ and } c_2 \text{ are constants})$$

Ex. 21. Solve $\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 9y = 0$

Substitute $y = e^{mx}$

$$\frac{dy}{dx} = m e^{mx} \quad \text{and} \quad \frac{d^2 y}{dx^2} = m^2 e^{mx}$$

So the given equation will be $e^{mx} (m^2 - 6m + 9) = 0$

The roots are $m = 3, 3$

This is the case of repeated roots. Solution of the equation in such a case will be $y = c_1 e^{3x} + c_2 e^{3x}$

Ex. 22 Solve $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = e^{4x}$

Given equation is no more a homogeneous equation. Hence the complete solution will consist of y_c plus y_p .

To find y_c , we reduce the equation to homogeneous form

$$\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$$

On substituting, $y = e^{mx}$, we get $e^{mx} (m^2 - 5m + 6) = 0$

i.e., $(m-2)(m-3) = 0$ The roots are $m = 2$ and $m = 3$

$$y_c = c_1 e^{2x} + c_2 e^{3x}$$

To find y_p we put $y_p = A e^{4x}$ [From rule (iii) section 4.12]

which when substituted in the D.E.

$$A = \frac{1}{4^2 - 5 \times 4 + 6}$$

$$y_p = \left\{ \frac{1}{(4)^2 - 20 + 6} \right\} e^{4x} = \frac{1}{2} e^{4x}$$

Complete (general) solution of the given equation will be

$$y = y_c + y_p = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{2} e^{4x}$$

Ex. 23 Solve $\frac{d^2 y}{dx^2} - 8 \frac{dy}{dx} + 16 = e^x$

To find y_c $\frac{d^2 y}{dx^2} - 8 \frac{dy}{dx} + 16y = 0$

On putting $y = e^{mx}$, we find $e^{mx} (m^2 - 8m + 16) = 0$

So that, $(m - 4)^2 = 0$ $m = 4$, (repeated root)

$$y_c = c_1 e^{4x} + c_2 x e^{4x}$$

Also, $y_p = A e^{ax} = \left\{ \frac{1}{a^2 - 8a + 16} \right\} e^x = \left\{ \frac{1}{1 - 8 + 16} \right\} e^x = \frac{1}{9} e^x$,

since $a = 1$ in this question,

Thus, $y = y_c + y_p = c_1 e^{4x} + c_2 x e^{4x} + \frac{1}{9} e^x$

is the required general solution of the given equation

Ex. 24. Solve: $\frac{d^2 y}{dx^2} - 7 \cdot \frac{dy}{dx} + 6y = (x - 2) e^x$

To obtain the C.F. write

$$\frac{d^2 y}{dx^2} - 7 \cdot \frac{dy}{dx} + 6y = 0$$

Put, $y = e^{mx}$; the auxiliary equation is

$\therefore m^2 - 7m + 6 = 0$, or $m = 1, m = 6$.

$y_c = c_1 e^x + c_2 e^{6x}$

Now, note that the exponent of e on the R.H.S. is 1, which is one of the roots of the auxiliary equation, hence one root is repeated. Thus try,

$$y_p = x(Ax + B) e^x = (Ax^2 + Bx) e^x$$

Now, $\frac{dy}{dx} = \frac{d(Ax^2 + Bx) e^x}{dx}$

$$= (Ax^2 + Bx) e^x + e^x (2Ax + B)$$

$$\frac{d^2y}{dx^2} = (Ax^2 + Bx) e^x + e^x (2Ax + B) + e^x (c_2 Ax + B) + e^x 2A$$

$$= [(Ax^2 + Bx) + (5Ax + 2B) + 2A] e^x$$

Hence the D.E. becomes

$$e^x [(Ax^2 + Bx + 4Ax + 2B + 2A) - 7(Ax^2 + Bx + 2Ax + B) + 6(Ax^2 + Bx)] = (x - 2) e^x$$

or $[-10Ax - 5B + 2A] = x - 2$

Comparing terms we have

$$-10A = 1 \quad -5B + 2A = -2$$

$$A = -\frac{1}{10} \text{ and } B = \frac{9}{25}$$

Hence $y_p = x \left[-\frac{1}{10}x + \frac{9}{25} \right] e^x$

The complete general solution is

$$y = y_c + y_p = c_1 e^x + c_2 e^{6x} + x \left[-\frac{1}{10}x + \frac{9}{25} \right] e^x$$

Ex 25 Solve $y'' - 2y' + 3y = e^{-x} \cos x$

The homogeneous part is $y'' - 2y' + 3y = 0$

and the auxiliary equation is $m^2 - 2m + 3 = 0$

i.e., $m = \frac{2 \pm \sqrt{4 - 12}}{2} = \frac{2 \pm \sqrt{-8}}{2}$

$$= \left[\frac{2 \pm i\sqrt{8}}{2} \right] = 1 \pm i\sqrt{2}$$

$$\alpha = 1 \quad \text{and} \quad \beta = \sqrt{2}$$

Hence $y_c = e^{\alpha x} [A \cos \beta x + B \sin \beta x]$

$$= e^x [A \cos \sqrt{2}x + B \sin \sqrt{2}x]$$

For y_p try $y = (A \cos x + B \sin x) e^{-x}$

Hence

$$\begin{aligned}\frac{dy}{dx} &= -e^{-x} [A \cos x + B \sin x] + e^{-x} [-A \sin x + B \cos x] \\ &= e^{-x} [-A \sin x + B \cos x - A \cos x - B \sin x] \\ &= e^{-x} [(B - A) \cos x - (A + B) \sin x]\end{aligned}$$

$$\begin{aligned}\frac{d^2y}{dx^2} &= -e^{-x} [(B - A) \cos x - (A + B) \sin x] \\ &\quad + e^{-x} [-(B - A) \sin x - (A + B) \cos x] \\ &= e^{-x} [-(B - A) \sin x - (A + B) \cos x \\ &\quad + (A + B) \sin x - (B - A) \cos x] \\ &= e^{-x} [\sin x (A + B - B + A) + \cos x [-A - B - B + A)] \\ &= e^{-x} [2A \sin x - 2B \cos x]\end{aligned}$$

Putting in the differential equation we have

$$\begin{aligned}e^{-x} [2A \sin x - 2B \cos x] - 2e^{-x} \\ [(B - A) \cos x - (A + B) \sin x] \\ + 3e^{-x} [A \cos x + B \sin x] = e^{-x} \cos x\end{aligned}$$

$$\begin{aligned}\text{or, } [2A \sin x - 2B \cos x - 2B \cos x + 2A \cos x \\ + 2A \sin x + 2B \sin x + 3A \cos x + 3B \sin x] \\ = e^{-x} \cos x\end{aligned}$$

$$\text{or, } (4A + 5B) \sin x + (5A - 4B) \cos x = \cos x$$

$$\text{Hence } 4A + 5B = 0 \quad A = -\frac{5}{4}B$$

$$5A - 4B = 1 \quad -\frac{25}{4}B - 4B = 1$$

$$\therefore B = -\frac{4}{41} \text{ and, } A = \frac{5}{41}$$

$$\text{Hence, } y_p = \left[\frac{5}{41} \cos x - \frac{4}{41} \sin x \right] e^{-x}$$

$$\text{and, } y = y_c + y_p$$

$$= e^x [A \cos \sqrt{2}x + B \sin \sqrt{2}x] + \frac{e^{-x}}{41} [5 \cos x - 4 \sin x]$$

EXERCISE 14.5

1. Show that $y = \frac{x^3}{3} + \frac{A}{x}$ is the solution of $x \frac{dy}{dx} + y = x^2$
2. Verify that $y = e^{-x}(x + c)$ is the solution of $\frac{dy}{dx} + y = e^{-x}$
3. Show that $y^2 = cx + \frac{1}{8}c^3$ is a solution of $y = 2x \frac{dy}{dx} + y^2 \left(\frac{dy}{dx} \right)^3$
4. Solve the following equations
 - (i) $\frac{d^2y}{dx^2} = \frac{1}{x^2}$
 - (ii) $\frac{d^2y}{dx^2} = \frac{1}{x}$
 - (iii) $\frac{d^2y}{dx^2} = x^2$
 - (iv) $\frac{d^2y}{dx^2} = -\frac{y}{x}$
 - (v) $\frac{d^2y}{dt^2} + 3 \frac{dy}{dt} + 2y = 0$
 - (vi) $\frac{d^2y}{dx^2} - 3 \frac{dy}{dx} - 4y = 0$
 - (vii) $\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - 54y = 0$
 - (viii) $2 \frac{d^2y}{dx^2} + 5 \frac{dy}{dx} - 12y = 0$
 - (ix) $9 \frac{d^2y}{dx^2} - 18 \frac{dy}{dx} - 16y = 0$
 - (x) $\frac{d^2y}{dx^2} - 4 \frac{dy}{dx} + 4y = 0$
5. Solve the following equations
 - (i) $\frac{d^2y}{dx^2} = e^{-x}$
 - (ii) $\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + y = 2e^{2x}$
 - (iii) $\frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^{5x}$
 - (iv) $\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = e^{2x}$

Also find definite solution given $y(0) = 0$ in all the above cases

6. Solve the following Differential Equations
 - (i) $y'' + 4y' + 3y = x$
 - (ii) $y + 9y' = (x^2 + 1)e^{3x}$
 - (iii) $y'' + 2y' + 5y = 2\cos x$
 - (iv) $y - y' = 3e^{2x} \cos x$
 - (v) $y'' + 5y + 4 = x^2 + 7x + 9$
 - (vi) $y'' + 9y' = 6e^{3x}$
 - (vii) $y'' - y = 5x + 2$
 - (viii) $y'' + y' - 2y = 8 \sin 2x$
 - (ix) $y'' - a^2y = x + 1$

Chapter 15

Applications of Differential Equations

Differential Equations find wide applications in all branches of economics. We present below a few illustrations from micro economics and macro and growth economics.

15.1 Applications in Micro economics

A Cost Function

Ex 1 The rate of increase of total cost Y as the number of units manufactured x increases proportional to the number of units manufactured plus a constant and inversely proportional to the cost. Find the cost function if $Y = Y_0$ when $x = 0$.

Total cost is denoted by Y , we are then given

$$\frac{dY}{dx} \propto \left[\frac{x+C}{Y} \right] = \frac{a(x+C)}{Y} \quad (1)$$

where C is a constant and a is some assumed measure of proportionality

$$Y.dY = a (x + C) dx$$

Integrating both sides we get the solution of the above equation

$$\frac{Y^2}{2} = a \left(\frac{x^2}{2} + Cx \right) + C_2 \quad (2)$$

when, $x = 0$, $Y = Y_0$

$$\text{Hence } \frac{Y_0^2}{2} = C_2$$

So that the solution of Eqn (2) becomes

$$\frac{Y^2}{2} = a \left(\frac{x^2}{2} + Cx \right) + \frac{Y_0^2}{2}$$

$$\text{or } Y^2 = ax^2 + 2aCx + Y_0^2$$

which is the required cost function and can be put in the form

$$Y = \sqrt{ax^2 + 2aCx + Y_0^2}$$

B. Utility and Demand Analysis.

Ex 2 An individual's preference scale for the goods X and Y is defined by the marginal rate of substitution of Y for $X = \frac{X-a}{Y-b}$

Show that $U = (X-a)^2 + (Y-b)^2$

is the form of the utility function

MRS between Y and X is given by $\frac{dY}{dX} = \frac{X-a}{Y-b}$

$$dY + \left(\frac{X-a}{Y-b}\right) dX = 0$$

$$(Y-b) dY + (X-a) dX = 0$$

Integrating, we have $\left(\frac{Y^2}{2} - bY\right) = -\left(\frac{X^2}{2} - aX\right) + C$

$$Y^2 - 2bY = -(X^2 - 2aX) + 2C$$

$$\text{or, } (Y-b)^2 = - (X-a)^2 + \underbrace{2C - a^2 - b^2}_{\downarrow}$$

$$(Y-b)^2 + (X-a)^2 = U \quad (\text{some constant value})$$

$$\text{i.e., } U = (Y-b)^2 + (X-a)^2$$

is the form of utility function

Ex. 3. It is given that elasticity of demand in the demand curve of the form $x = f(p)$ is $(a-bp)$, where a, b are constants. Find the demand curve

$$\text{Elasticity of demand} = \frac{\frac{dD}{D}}{\frac{dp}{p}} = \frac{dD}{dp} \cdot \frac{p}{D}$$

$$\text{In the given example, } \frac{dD}{dp} \cdot \frac{p}{D} = (a-bp)$$

$$\frac{dD}{D} = \left(\frac{a-bp}{p}\right) dp$$

$$\text{or} \quad \int \frac{dD}{D} = \int \left(\frac{a-bp}{p} \right) dp = \int \left(\frac{a}{p} - b \right) dp$$

$$\text{or,} \quad \log D = a \log p - bp + C$$

$$D = e^{a \log p - bp + C} = e^{\log(p)^a - bp + C}$$

$$D = p^a \cdot e^{-bp+C} = p^a \cdot e^C \cdot e^{-bp}$$

is the required demand curve

C Market equilibrium

Ex 4 Demand and supply functions for tea are given by

$$x_d = \left\{ 120 - 2p + 5 \frac{dp}{dt} \right\} \text{ kg per week, and}$$

$$x_s = \left\{ 3p - 30 + 50 \frac{dp}{dt} \right\} \text{ kg per week,}$$

where p is the price at time t

If the initial price is 36 rupees per kg find the time path of price

At equilibrium $x_d = x_s$

$$120 - 2p + 5 \frac{dp}{dt} = 3p - 30 + 50 \frac{dp}{dt}$$

$$45 \frac{dp}{dt} + 5p - 150 = 0$$

$$9 \frac{dp}{dt} + p = 30$$

$$\text{that is,} \quad \frac{dp}{dt} + \frac{p}{9} = \frac{10}{3} \quad (1)$$

This is the differential equation of the form $\frac{dy}{dx} + ay = b$ of which the solution is $Y = \frac{b}{a} + \left(Y(0) - \frac{b}{a} \right) e^{-ax}$

In this case $a = \frac{1}{9}$ and $b = \frac{10}{3}$ Solution of differential equation (1) shall be

$$p(t) = \frac{10}{3} + 9 [p(0) - 30] e^{-t/9}$$

$$p(t) = 30 + [p(0) - 30] e^{-t/9}$$

Initial price, i.e., at $t = 0$, $p(0) = 36$ (given)

$$p(t) = 30 + (36-30)e^{-t/9} = 30 + 6e^{-t/9}$$

is the time path of price for tea

What will be the price after 10 weeks ?

It will be $p(10) = 30 + 6e^{-10/9}$

Ex 5. The demand and supply functions, when p is the price, Q_d quantity demanded and Q_s the quantity supplied, are given as

$$Q_d = a - bp \quad (a, b > 0) \quad (1)$$

$$Q_s = -c + dp \quad (c, d > 0) \quad (2)$$

$$\text{and } \frac{dp}{dt} = \alpha (Q_d - Q_s) \quad (\alpha > 0) \quad (3)$$

Find the time path of price

[Eqn (3) implies that change in price with respect to time (t) is directly proportional to the excess of demand over supply ($= Q_d - Q_s$)]

Substituting (2) and (1) in Eqn (3), we get

$$\frac{dp}{dt} = \alpha [(a - bp) - (-c + dp)]$$

$$\frac{dp}{dt} = \alpha [(a + c) - (b + d)p] \quad (4)$$

$$\frac{dp}{dt} + \alpha (b + d) P = \alpha (a + c)$$

Hence, $y_c = Ae^{-\alpha(b+d)t}$, and

$$y_p = \frac{\alpha(a+c)}{\alpha(b+d)} = \frac{a+c}{b+d} = \bar{p} \text{ (say)}$$

The complete solution, therefore, is $y_c + y_p$

That is, $P_t = \frac{a+c}{b+d} + Ae^{-\alpha(b+d)t}$

At $t = 0$, $P_t = P_0$, $A = P_0 - \left(\frac{a+c}{b+d}\right) = P_0 - \bar{p}$

Therefore, $P_t = \bar{p} + (P_0 - \bar{p})e^{-\alpha(b+d)t}$ where $\bar{p} = \left(\frac{a+c}{b+d}\right)$

Now as $t \rightarrow \infty$, ($t \rightarrow \infty$, in the long run) $e^{-a(b+p)t} = 0$

$$\lim_{t \rightarrow \infty} L t^p = \bar{p} + 0 = \bar{p}$$

In other words, in the long run, price will converge to the equilibrium price ($\bar{p}$) and in this way the dynamic stability will be obtained

In the above case y_p which depicts the particular integral gives the equilibrium price while y_c the complementary function gives the *deviations* from the equilibrium

15.2 Applications in Macro-Growth Economics

Dynamic Multiplier

The model consists of,

$$(i) \text{ Linear or proportional consumption function } C = cY \quad (1)$$

(c is marginal propensity to consume and Y is national income which is equal to net national product in a closed economy)

$$(ii) \text{ Total demand } Z = C + A \quad (2)$$

where A is autonomous investment

I Dynamic Multiplier When Autonomous Investment Is Constant

Let there be equilibrium initially. Assume that economy is given a push by raising autonomous investment. The economy will adjust by change in income. Suppose change in income is proportional to the gap between aggregate demand and income (also national product). Hence,

$$\frac{dY}{dt} = \lambda (Z - Y), \quad \frac{dY}{dt} + \lambda Y = \lambda Z$$

Hence λ is the speed of response, and $\frac{1}{\lambda} = T$ is the length of lag

$$\frac{dY}{dt} + \lambda Y = \lambda (C + A) = \lambda (cY + A), \quad C = cY$$

$$\text{or, } \frac{dY}{dt} + \lambda Y - \lambda cY = \lambda A \quad \text{or, } \frac{dY}{dt} + \lambda Y (1 - c) = \lambda A$$

$$\text{i.e., } \frac{dY}{dt} + s\lambda Y = \lambda A \quad (3)$$

where $(1 - c) = s =$ marginal propensity to save

If s , λ and A are all constants, the solution of the Eqn (3) will be of the type $Y = Y_p + Y_c$ and in the form

$$Y = \frac{b}{a} + \left(Y_0 - \frac{b}{a} \right) e^{-at}$$

Here, $b = \lambda A$ and $a = s\lambda$

$$\text{the solution is } Y = \frac{\lambda A}{s\lambda} + \left(Y_0 - \frac{\lambda A}{s\lambda} \right) e^{-\lambda s t}$$

$$\text{That is, } Y = \frac{A}{s} + \left(Y_0 - \frac{A}{s} \right) e^{-\lambda s t} \quad (4)$$

This gives the time path of income As $t \rightarrow \infty$, $e^{-\lambda s t} \rightarrow 0$ hence,

$$\lim_{t \rightarrow \infty} Y = \frac{A}{s}$$

Thus the disequilibrium factor is $\left(Y_0 - \frac{A}{s} \right) e^{-\lambda s t}$ but in the long run, equilibrium is at $Y = \frac{A}{s}$

II Dynamic Multiplier . When Autonomous Investment is not Constant. Suppose that A is not constant, but is a function of time, say of the form $A(t) = A_0 e^{nt}$ that is, investment grows at a constant proportional rate of n

In this case our differential Eqn (3) above becomes

$$\frac{dY}{dt} + s\lambda Y + \lambda A_0 e^{nt} \quad (5)$$

and in this case, the integrating factor is $e^{\int P dt} = e^{\int s\lambda dt} = e^{s\lambda t}$

Therefore, the solution of the differential equation (5) will be

$$Y e^{s\lambda t} = \int \lambda A_0 e^{nt} e^{s\lambda t} dt = \lambda A_0 \int e^{(n+s\lambda)t} dt$$

$$= \lambda A_0 \frac{1}{(n+s\lambda)} e^{t(n+s\lambda)} + C$$

$$\text{or, } Y = \frac{\lambda A_0 e^{-s\lambda t}}{(n+s\lambda)} e^{t(n+s\lambda)} + C e^{-s\lambda t}$$

$$= \frac{\lambda A_0}{(n+s\lambda)} e^{nt} + C e^{-s\lambda t}$$

Substituting $T = \frac{1}{\lambda}$, we get $Y = \frac{A_0}{nT+s} e^{nt} + C e^{-t/T}$

If $t = 0$, $Y = Y_0$

then,
$$C = Y_0 - \left(\frac{A_0}{nT + s} \right).$$

Therefore, the required solution of (5) shall be :

$$Y = \frac{A_0}{s + nT} e^{nt} + \left(Y_0 - \frac{A_0}{s} + nT \right) e^{-n/T}$$

15.3. Harrod-Domar Model

(A) Harrod's Model

(a) *Equilibrium on product market is specified by .*

(i) Full capacity use of capital stock (K) i.e., $K = vY$, (1)
(v is capital-output ratio)

and, (ii) Equality of investment and saving (no lags are present)

i.e., $I = S$, or $\frac{dK}{dt} = sY$. (2)

($\therefore$ Investment $= \frac{dK}{dt}$ and s is propensity to save)

Therefore putting $K = vY$ in (2), $\frac{vaY}{dt} = sY$, using (1)

(b) *Equilibrium on labour market, i.e., full employment of labour force*
Suppose labour is growing at constant proportional rate of n per year, so that :

$$\frac{\frac{dL}{dt}}{L} = n$$

or,
$$\frac{dL}{L} = n \cdot dt.$$

$$\therefore \int \frac{dL}{L} = \int n dt, \text{ or } \log L = nt + C$$

At initial period $t = 0$, and $L = L_0$ so that $c = \log L_0$

Hence,
$$\log L = nt + \log L_0 \text{ or, } \log \frac{L}{L_0} = nt.$$

Hence $L_t = L_0 e^{nt}$ where L depicts the labour supply function.

Demand of labour is proportional to production Y (by assuming fixed labour output ratio)

$$L_D = l Y \quad (l \text{ — is the labour output ratio})$$

Labour market will be in equilibrium when $L_D = L_S$

$$\text{i.e., } l Y = L_0 e^{gt}$$

(Remember that here L is measured in man-years)

From Eqns (1) and (2), we have

$$\frac{dK}{dt} = s Y = \frac{sK}{v}$$

$$\frac{dK}{K} = \frac{s}{v} dt$$

On integrating, we get $\log K = \frac{s}{v} t + C$

$$K = C e^{s/v t} = C_1 e^{gt} \text{ where } g = \frac{s}{v}$$

If at $t = 0$, $K = K_0$ then $K_0 = C_1 e^0$ or $C_1 = K_0$

Hence, $K = K_0 e^{gt}$, where $g = \frac{s}{v}$

This shows that capital grows at the rate of $g = \frac{s}{v}$

Similarly, $\frac{dK}{dt} = s Y$

or, $\frac{d(vY)}{dt} = s Y$ (by substituting $K = vY$)

or, $\frac{v dY}{Y} = s dt$

or, $\int \frac{dy}{Y} = \int \frac{s}{v} dt$

or, $Y = B e^{st} \quad (B \text{ is some constant})$

Now, if at $t = 0$, $Y = Y_0$ we have $B = Y_0$

$$Y = Y_0 e^{st}$$

Here g is termed as the warranted rate of growth. Both Y and K therefore grow at the warranted rate of growth.

For full employment, growth of labour supply must be equal to growth in labour demand i.e. $L_t = L_0 e^{nt} = I Y = L_D$

$$L_0 e^{nt} = I Y_0 e^{gt} \quad (\text{By substituting } Y = Y_0 e^{gt})$$

Since in the initial period $L_0 = I Y_0$

$$e^{nt} = e^{gt} \quad \text{or} \quad n = g$$

Hence for the full employment growth, the warranted rate of growth should equal the natural rate of growth $= n$.

Summarising We get the following results which make the complete solution

$$Y = Y_0 e^{nt}, \quad K = K_0 e^{nt}, \quad I = L_0 e^{nt} \quad \text{and} \quad g = \frac{s}{v}$$

Growth path Equation of Investment Instead of the growth path equation for K as derived above, we can also derive growth path equation for investment.

From the capital stock equilibrium equation $K = vY$, we can have

$$\frac{dK}{dt} = v \frac{dY}{dt}$$

$$\text{Also} \quad I = v \frac{dY}{dt} \left(\text{since } \frac{dK}{dt} = I \right), \quad (1)$$

$$\frac{dK}{dt} = sY, \text{ so that } I = sY \quad (2)$$

From Eqn (1) and (2) we obtain $v \frac{dY}{dt} = sY$

$$\frac{dY}{Y} = \frac{s}{v} dt \quad \text{On integrating We obtain}$$

$$\log Y = \frac{s}{v} t + C$$

$$Y = Y_0 e^{s/v t} = Y_0 e^{nt}$$

From Eqn (2), we have $I = sY$ and $I_0 = sY_0$ (given initial condition)

* If technological growth is displacing labour at the rate m , the natural rate of growth will be $n + m$.

$$\frac{I}{s} = \frac{I_0}{s} e^{st}, \text{ so that we have}$$

$$I = I_0 e^{st},$$

which is the required growth path equation of investment

(B) Domar Model

Domar attempts to find the condition for steady growth in a full employment economy initially in equilibrium

Whenever investment is made, it has two effects, it creates production capacity and it generates demand

Let I depict investment and $\frac{dI}{dt}$ change in the investment ρ be the average as well as marginal productivity of capital

ρ is, therefore, reciprocal of capital output ratio The increase in productive capacity is therefore given by $I \rho$

If s = marginal propensity to save, multiplier shall be $\left(\frac{1}{s}\right)$ and the increase in demand = $\frac{dI}{dt} \left(\frac{1}{s}\right)$

$$\text{For equilibrium growth} \quad \frac{dI}{dt} \frac{1}{s} = I \rho$$

$$\frac{dI}{dt} \frac{1}{I} = s \rho$$

Integrating both sides, we obtain $\log I = s \rho t + A$

$$\text{or } I = B e^{s \rho t} \text{ at } t = 0, \quad I = I_0, \quad B = I_0$$

$$I = I_0 e^{s \rho t}$$

which is the basic Domar equation It describes that warranted growth rate is $s \rho = \frac{s}{v}$, same as we obtained under Harrod Model.

15.4. The Basic Neo-classical Model (Solow Model)

The Harrod Domar model assumes fixed output-capital ratio and the production function is simple

$$Y = \frac{K}{v}, \text{ or}$$

$$\text{or} \quad Y = \frac{L}{l}$$

The *Neo-Classical Model* does away with the assumption of fixed output-capital ratio, it takes the output-capital ratio $\left(\frac{1}{v}\right)$ to vary continuously. In the long run, capital and labour inputs are substitutable and the ratio in which the two inputs are used may change. A purely capitalist economy can choose from these infinitely available ratios, only one of which will ensure a steady state growth which is warranted as well as natural rate of growth.

The basic assumptions of *Solow model* include perfect foresight for all individuals, and smooth adjustment in goods, labour and capital markets.

The Solow Model

1. *Production function* $Y = f(L, k)$

$$f_K, f_L > 0$$

$$f_{KK}, f_{LL} < 0$$

The production is subject to diminishing returns to both the factors Labour (L) and Capital (K).

In addition, it is also assumed that the production function is linear homogeneous function. That is if we divide each factor by λ , output is also divided by λ .

$$\frac{Y}{\lambda} = f\left(\frac{L}{\lambda}, \frac{K}{\lambda}\right) \text{ for } \lambda > 0$$

$$\text{If } \lambda = L, \quad \frac{Y}{L} = f\left(1, \frac{K}{L}\right)$$

$$y = \varphi(1, r) \quad \text{where } r = \frac{K}{L} \text{ and } y = \frac{Y}{L}$$

which can be written as,

$$\frac{Y}{L} = y = f(r) \text{ or, } y = L f(r) \quad (1)$$

2 Output market

Like Domar, Solow also assumes an instantaneous multiplier. Time subscript has been, therefore, dropped.

$$S = sY$$

$$I = S$$

$$\text{Hence } I = sY = sL f(r)$$

$$\text{But } I \text{ is also } \frac{dK}{dt}$$

$$\text{Therefore, } \frac{dK}{dt} = sY = sL f(r)$$

$$\text{Now, } r = \frac{K}{L}$$

Hence, $\log r = \log K - \log L$, differentiating w.r.t. 'time'

$$\frac{dr}{r} = \frac{dK}{K} - \frac{dL}{L}$$

$$\frac{1}{r} \frac{dr}{dt} = \frac{sY}{K} - \frac{1}{L} \frac{dL}{dt} = \frac{sL f(r)}{K} - \frac{1}{L} \frac{dL}{dt}$$

$$\frac{dr}{dt} = s f(r) - \frac{dL}{dt} \frac{1}{L} r \quad (2)$$

$$3 \quad \text{Labour force grows at a constant rate } L = L_0 e^{nr}, \quad \frac{1}{L} \frac{dL}{dt} = n$$

$$\text{So that, from (2) } \frac{dr}{dt} = s f(r) - nr$$

$$\text{or, } s f(r) = \frac{dr}{dt} + nr \quad (3)$$

$$\text{Since } I = L s f(r)$$

$$\frac{I}{L} = s f(r) = \frac{dr}{dt} + nr \quad (\text{from (3)})$$

$$i = r + nr, \text{ where } \frac{I}{L} = i, \text{ and } r = \frac{dr}{dt}$$

Now, in the absence of government and foreign trade $Y = C + I$

In per capita terms, $y = c + i$

$$\text{i.e., } y = f(r) = c + r + nr,$$

$$\text{or, } y - nr = c + r \quad (4)$$

Eqs (3) and (4) are the fundamental differential equations of the Solow model

Consider, the following situations

(i) If $r = \frac{K}{L} = \text{constant}$, i.e., $r = 0$

$$\text{Then, } sf(r) - nr = 0 \quad \left[\text{Now, } \frac{f(r)}{r} = \frac{L}{L} \frac{f(r)}{r} = \frac{Y}{K} \right]$$

$$\frac{f(r)}{r} = \frac{n}{s}$$

i.e., output-capital ratio is just equal to $\frac{n}{s}$ (i.e., $\frac{1}{v} = \frac{n}{s}$ or, $n = \frac{s}{v}$)

Harrod's *Warranted* rate of growth is equal to *Natural* rate of growth. This is obtained when capital-output ratio remains constant

(ii) Again, $\frac{dr}{dt} = sf(r) - nr$ implies that $\frac{dr}{dt} > 0$ so far as $sf(r) > nr$, i.e., capital deepening takes place if

$$sf(r) - nr > 0, \text{ i.e., } \frac{f(r)}{r} > \frac{n}{s}$$

and, labour deepening takes place if $\frac{f(r)}{r} < \frac{n}{s}$

(iii) In case when $s = 0$, $\frac{dr}{dt} = -nr$

$$\text{So that, } \frac{\frac{dr}{dt}}{r} = -n$$

Capital labour ratio will, therefore, decline at the labour-supply growth rate

$$(iv) \text{ If } n = 0 \text{ then, } \frac{dr}{dt} = sf(r) = \frac{dK}{L} \left(\therefore \frac{dK}{dt} = s \cdot L \cdot f(r) \right)$$

$$\frac{dr}{dt} = \frac{\frac{dK}{dt}}{L \cdot \frac{K}{K}} = \frac{\frac{dK}{dt}}{K} \cdot \frac{K}{L} = r \cdot \frac{dK}{dt}$$

$$\therefore \frac{\frac{dr}{dt}}{r} = \frac{dK}{K}$$

Hence in the case of no population growth, the rate of change in capital-labour ratio will be equal to the rate of change in capital stock

(v) $\frac{dr}{dt} = sf(r) - nr$ is sum of two components, the first related to change in labour and the second to change in capital

The relationship can be shown graphically (Fig 15.1)

Put $s = 0$, then $r = -nr$ neglecting $(-)$ sign get line OA . Set $n = 0$ and plot $r = sf(r)$, the curved line OB . Vertical difference between the two will give $\frac{dr}{dt} = sf(r) - nr$ when $s \neq 0$ and $n \neq 0$

Example Let $y = \frac{1}{100} r^{1/4}$

and, $s = \frac{1}{5}$ $n = 0.02$, then

$$\frac{dr}{dt} = 0.02r^{1/4} - 0.02r$$

Clearly, therefore, $y = f(r) = \frac{1}{100} r^{1/4}$. It depicts a *Cobb Douglas* type production function, hence, the shape of the curve OP [$s f(r) = 0.02r^{1/4}$] is that of a well behaved production function

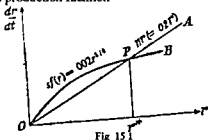


Fig 15.1

At the point P , $r = r^*$ and $r = 0$, $nr = sf(r)$

This gives the equilibrium per capita capital output ratio. Both capital and output will grow at the same rate as labour. This neo-classical model has steady state growth as its unique equilibrium provided the initial capital stock per capita is r^* (at point, P in Fig 15.1)

Dynamics of Disequilibrium

There will be a state of disequilibrium when $r \neq r^*$, i.e., when the capital labour ratio is not equal to the initial capital stock per capita (man)

If $r < r^*$, $sf(r) > nr$ and $r > 0$, hence actual r will increase towards r^* , opposite will be the case for $r > r^*$. Now,

$$\frac{dr}{dt} = sf(r) - nr$$

$$\frac{dr}{dt} = \frac{s f(r)}{r} - n$$

$$\frac{d}{dr} \left(\frac{r}{r} \right) = \frac{d}{dr} \left[\frac{s f(r)}{r} \right] - 0 < 0$$

(Remember $r = \frac{dr}{dt}$ and because there are diminishing returns to factors)

Hence average product of capital falls as the capital labour ratio rises
Thus r and $\dot{r}$ move in opposite directions. A fall in r will be followed by rise in r (then $\dot{r} > 0$)

From our numerical illustration

$$\begin{aligned} \frac{d}{dr} \left(\frac{r}{r} \right) &= \frac{d}{dr} \left(0.02 \frac{r^{3/4}}{r} \right) \\ &= 0.02 \times -\frac{3}{4} r^{-1/4} \text{ which is } < 0 \end{aligned}$$

15.5. Unbalanced Growth in Solow Model

Assume $Y = K^\alpha L^{1-\alpha}$

Then, $\frac{dK}{dt} = I - s Y = s K^\alpha L^{1-\alpha}$

$$\frac{dK}{K^\alpha} = s L_o^{1-\alpha} e^{(1-\alpha)nt} dt \quad (L = L_o e^{nt})$$

Integrating, we have,

$$K(t) = \left[A + s \frac{L_o^{1-\alpha} e^{(1-\alpha)nt}}{n} \right]^{1/(1-\alpha)}$$

when $t = 0$, $K(0) = K_o$, hence $A = K_o^{1-\alpha} - s \left(\frac{L_o^{1-\alpha}}{n} \right)$

$$K(t) = \left[K_o^{1-\alpha} + \frac{s L_o^{1-\alpha}}{n} (e^{(1-\alpha)nt} - 1) \right]^{1/(1-\alpha)}$$

$$y(t) = K_t^\alpha [L_o e^{nt}]^{1-\alpha}$$

Substitute the value of $K(t)$ to obtain the time path of income

Thus we obtain time paths of capital and income

15.6. Kaldorian Modification of Harrod-Domar Model.

Kaldor allows for a varying propensity to save based on the change in distribution of income between wage (W) and profits (P)

$$\begin{aligned} \text{Total saving } S &= s_w + s_p & \left[\begin{array}{l} \text{Where } s_w \text{ Total savings of workers} \\ \text{and } Y = W + P & \text{and } s_p \text{ Total savings of capitalists} \end{array} \right. \end{aligned}$$

But if s_w depicts the propensity to save of workers and s_p depicts the propensity to save of capitalists, then

$$\begin{aligned} S &= s_w W + s_p P \\ S &= s_p P + s_w (Y - P) \\ S &= s_p P + s_w Y - s_w P \\ S + P (s_p - s_w) &= s_w Y \\ \frac{S}{Y} &= \frac{P}{Y} (s_p - s_w) + s_w \\ s &= \frac{P}{Y} (s_p - s_w) + s_w \quad 0 \leq s_w < s_p \leq 1 \end{aligned} \quad (1)$$

There will still be a unique solution in which Y, K and L all grow at the rate of $n = g = \frac{s}{v}$, where a is given by Eq (1), v is $\frac{K}{Y}$

But now s is no more a constant, it depends upon the rate of profit $= \frac{P}{K} = r$ (say)

$$\begin{aligned} \text{Since,} \quad g &= \frac{s}{v} = \frac{1}{v} \left[s_w + (s_p - s_w) \frac{P}{K} \frac{K}{Y} \right] \\ \text{Again,} \quad n &= \frac{s}{v} = g = \frac{1}{v} \left[s_w + (s_p - s_w) \frac{K}{Y} \right] \\ n &= \frac{s_w}{v} + (s_p - s_w) r \frac{K}{Y} \quad \frac{1}{v} = \frac{s_w}{v} + (s_p - s_w) r \\ &= \frac{s_w}{v} + (s_p - s_w) r \end{aligned} \quad (2)$$

But since profit cannot be more than total income earned

$$\begin{aligned} 0 &\leq P \leq Y \\ \frac{0}{K} &\leq \frac{P}{K} \leq \frac{Y}{K} \\ 0 &\leq r \leq \frac{Y}{K} \end{aligned}$$

$$0 \leq r \leq \frac{1}{v}$$

We write Eqn (2) again $n = \frac{s_w}{v} + (s_p - s_w) r$

(i) When $r = 0$, $n = \frac{s_w}{v}$

(ii) When $r = \frac{1}{v}$, $n = \frac{s_w}{v} + \frac{1}{v} (s_p - s_w)$

Therefore, $n = \frac{s_p}{v}$

Hence we can say that n lies between $\frac{s_w}{v}$ and $\frac{s_p}{v}$

that is, $\frac{s_w}{v} \leq n \leq \frac{s_p}{v}$

Whereas n must equal the single value $\frac{s}{v}$ in the Harrod-Domar model, it can now be anywhere between $\frac{s_w}{v}$ and $\frac{s_p}{v}$. This is an improvement if s_p and s_w are wide apart.

Eqn (2) gives the unique value of r

For steady state growth

$$r = \frac{n - \frac{s_w}{v}}{s_p - s_w}$$

where, $\frac{s_w}{v} \leq n \leq \frac{s_p}{v}$ and $0 \leq r \leq \frac{1}{v}$.

Therefore, income distribution should be fixed in such a way that profit rate $\frac{P}{K} = r$ has the rate given by

$$r = \frac{n - \frac{s_w}{v}}{s_p - s_w}$$

If $s_w = 0$, then for balanced growth, $r = \frac{n}{s_p}$

Ex. 6 Suppose that the profit rate is fixed at 10% per year in the Kaldor (Keynesian) model. If $s_p = 0.5$ and $v = 2$ on the annual data, show that for steady state growth s_w and n must be related linearly $s_w = 20n - 1$

Check that labour must grow by as much as 5% even if there is no saving out of wages

We know that for steady growth

$$r = \frac{n - \frac{s_w}{v}}{s_p - s_w}$$

Here, $r = \frac{P}{K} = 10\% = 0.1$, $s_p = 0.5$, and $v = 2$

Therefore, $0.1 = \frac{n - \frac{s_w}{2}}{0.5 - s_w}$

$$0.05 - 0.1 s_w = n - \frac{s_w}{2}$$

i.e., $8 s_w = 20n - 1$, hence proved

If there is to be no saving from wages, $s_w = 0$

Substituting this in the above equation, we get, $0 = 20n - 1$

or, $n = \frac{1}{20} = 0.05 = 5\%$

which suggests that labour force should still grow at 5% per year even if there is no saving out of wages

15.7. Domar Debt Model

If D is national debt and Y is national income, then Domar debt model can be stated as

(a) When national income increases at a constant rate

$$\frac{dY}{dt} = \beta, \quad \beta > 0 \quad (1)$$

(b) When rate of increase of national debt is a fixed proportion of national income

$$\frac{dD}{dt} = \alpha Y(t), \quad \alpha > 0 \quad (2)$$

Initial Conditions

Initial Income $Y(0) = Y_0$

Initial Debt $D(0) = D_0$

Integrating eqn (1), we get $Y = \beta t + C_1$

From initial conditions $Y = Y_0$, $t = t_0$

$$C_1 = Y_0$$

$$Y = \beta t + Y_0$$

$$\begin{aligned} \text{Substituting (3) in Eqn (2) we get } \frac{dD}{dt} &= \alpha (\beta t + Y_0) \\ &= \alpha \beta t + \alpha Y_0 \end{aligned}$$

$$\text{Integrating, we get, } D = \alpha \beta \frac{t^2}{2} + Y_0 \alpha t + C_2$$

$$\text{From the initial conditions when } t = 0, D = D_0$$

$$C_2 = D_0$$

$$\text{The debt equation } D = \alpha \beta \frac{t^2}{2} + Y_0 \alpha t + D_0,$$

so that the solution of the model is given by

$$\left. \begin{aligned} D_t &= \alpha \beta \frac{t^2}{2} + Y_0 \alpha t + D_0 \\ \text{and, } Y_t &= \beta t + Y_0 \end{aligned} \right\} \alpha \text{ and } \beta \text{ both } > 0$$

From two more equations we can find the ratio of national debt to national income. This will be

$$\frac{D_t}{Y_t} = \frac{\frac{1}{2} \alpha \beta t^2 + \alpha Y_0 t + D_0}{\beta t + Y_0}$$

$$\frac{D_t}{Y_t} = \frac{D_0}{\beta t + Y_0} + \frac{\alpha Y_0 t}{\beta t + Y_0} + \frac{\frac{1}{2} \alpha \beta t^2}{\beta t + Y_0}$$

$$\frac{D_t}{Y_t} = \frac{D_0}{\beta t + Y_0} + \frac{\alpha Y_0 t}{t \left(\beta + \frac{Y_0}{t} \right)} + \frac{\frac{1}{2} \alpha \beta t^2}{t \left(\beta + \frac{Y_0}{t} \right)}$$

$$\frac{D_t}{Y_t} = \frac{D_0}{\beta t + Y_0} + \frac{\alpha Y_0}{\beta + \frac{Y_0}{t}} + \frac{\frac{1}{2} \alpha \beta}{\beta + \frac{Y_0}{t}} t$$

$$\text{when } t \rightarrow \infty, \frac{D_t}{Y_t} \rightarrow \left[0 + \frac{\alpha Y_0}{\beta} + \infty \right] \rightarrow \infty$$

Hence in this model the time path of the ratio of the national debt to national income is explosive, it increases over time without limit

Modified Domar Debt Model Suppose it is assumed that national income increases by a constant proportion, i.e.,

$$\frac{dY}{dt} = \text{constant} = \beta \text{ (say)}$$

So that,
$$\frac{dY}{dt} = \beta Y_t \quad (1)$$

and
$$\frac{dD}{dt} = \alpha Y_t \quad (\alpha > 0 \text{ and } \beta > 0) \quad (2)$$

with initial conditions, $D(0) = D_0$, $Y(0) = Y_0$ as before

Integrating Eqn (1),
$$\int \frac{dY}{Y} = \int \beta dt,$$

$$\log Y = \beta t + C_1$$

$$Y = C e^{\beta t}$$

From initial conditions

$$C = Y_0$$

$$Y_t = Y_0 e^{\beta t}, \quad (3)$$

Substituting (3) in Eqn (2) we get
$$\frac{dD}{dt} = \alpha Y_0 e^{\beta t},$$

Integrating it,
$$D = \frac{\alpha Y_0}{\beta} e^{\beta t} + C_2 \quad (4)$$

Again from initial conditions at $t = 0$, $D(0) = D_0$, we have

$$D = \frac{\alpha Y_0}{\beta} + C_2$$

$$C_2 = D_0 - \frac{\alpha Y_0}{\beta}$$

Hence the solution of Eqn (4) becomes

$$D_t = D_0 - \frac{\alpha}{\beta} Y_0 + \frac{\alpha}{\beta} Y_0 e^{\beta t}$$

$$D_t = D_0 + \frac{\alpha}{\beta} Y_0 (e^{\beta t} - 1)$$

i.e., the ratio of national debt to national income (from Eqns 3 and 4) will now be

$$\frac{D_t}{Y_t} = \frac{D_0 + \frac{\alpha}{\beta} Y_0 (e^{\beta t} - 1)}{Y_0 e^{\beta t}}$$

$$\frac{D_t}{Y_t} = \frac{D_0}{Y_0} e^{\alpha t} + \frac{\alpha}{\beta} - \frac{\alpha}{\beta} \frac{\alpha}{Y_0 e^{\beta t}}$$

Therefore, Limit $t \rightarrow \infty \frac{D_t}{Y_t} \rightarrow \left[0 + \frac{\alpha}{\beta} - 0 - \frac{\alpha}{\beta} \right]$

That is $\lim_{t \rightarrow \infty} \frac{D_t}{Y_t} \rightarrow \frac{\alpha}{\beta}$

In other words, the path of debt over time is *convergent* and it approaches a constant term $\frac{\alpha}{\beta}$ over time.

EXERCISE 15.1

1 If C is the total cost of an output x , it is known that marginal cost always equals average cost. Show that C is a fixed multiple of x , i.e., average cost is a constant.

2 Prove that as in the case of perfect competition, a firm's AR curve will coincide with its MR when AR is constant.

3 If MRS of Y for X is $\frac{A}{B} \frac{Y+b}{X+a}$, show that one form of the individual's utility function is $U = (x+a)^A (Y+b)^B$ where a, b, A and B are constants.

4 The MRS of Y for X is $\frac{aX + bY + C}{aX + \beta Y + \gamma}$, where a, b, c, α, β and γ are given constants.

If $\alpha = b$, show that one form of the utility function is quadratic in X and Y .

5. Demand and supply functions for tea are given by

$$x_d = 100 - p + \frac{dp}{dt} \text{ million kg per week}$$

$$\text{and } x_s = -50 + 2p + 10 \frac{dp}{dt} \text{ million kg per week}$$

Find the time-path of p for dynamic equilibrium if initial price is given to be Rs 10 per kg. What will be the price at time $t = 10$?

6 If MRS of Y for X is $\frac{3Y+2}{2X-3}$, find the indifference map.

7. If MRS of Y for X is $\frac{2X}{3Y}$, find the indifference map.

8. What will be the equilibrium rates of investments in the Domar model if :

$$(i) \quad \frac{1}{v} = t^{\frac{1}{2}}, \quad s = 0.4$$

$$(ii) \quad \frac{1}{v} = 2t^{\frac{1}{2}}, \quad s = 3t^{\frac{1}{2}}$$

$$(iii) \quad v = t^{\frac{1}{2}}, \quad s = t^{\frac{1}{2}}$$

9. Suppose that the profit rate is fixed at 8% per year in the Kaldor (Keynesian) model. If $s_p = 0.6$ and $v = 2$ (on an annual data), find the condition for steady-state growth

10. The change in net profit P , as advertising expenditure (x) changes, is given by

$$\frac{dP}{dx} = 3 - 2(P + x).$$

Find P , if $P = 10$ at $x = 0$

11. Manufacturing and marketing costs C are related to the number of items (x) by

$$\frac{dC}{dx} + 2C = 2 + 2x$$

Find C as a function of x , if $C = 0$ when $x = 0$

12. If R is the running cost of a car, S is re-sale value, solve the model :

$$\frac{dR}{dt} = \frac{\alpha}{St} \quad \alpha > 0$$

$$\frac{dS}{dt} = \frac{\lambda}{St} \quad \lambda > 0$$

Initial condition : $R(0) = 0$, $S(0) = S_0$ (original cost)

Discuss whether the time-path will be explosive or not

13. If x is per capita consumption of a product and y is per capita income, find the time-path of x and y . Find also the path of ratio $\frac{x(t)}{y(t)}$ and discuss its nature (whether explosive or not). The model (given) is as under.

$$\left. \begin{aligned} \frac{dx}{dt} &= \frac{2}{\alpha y(t)} + k e^{\beta x} \\ \frac{dy}{dt} &= \beta y(t) \end{aligned} \right\} \begin{aligned} \beta &> 0 \text{ and} \\ \alpha &> \frac{k}{\beta y_0} \end{aligned}$$

and, $x(0) = x_0$, $y(0) = y_0$

14. As the size of the firm increases, the average monthly cost y of office supplies is related to the number x of employees by the equation

$$\frac{dy}{dx} + 2y = y^2 e^{-x}$$

find y as a function of x if $y = 0$ when $x = 0$

15. Find the limits as $t \rightarrow \infty$ of the ratio of national debt (D) to national income (Y) in the following model

$$\frac{dD}{dt} = 0.2 Y_t + 10$$

$$\frac{dY}{dt} = 0.05 Y_t$$

$$\text{and, } Y(0) = 100, D(0) = 0$$

16. The relationship between revenue R and quantity demanded x is such that the rate of increase in revenue as quantity demanded increases is equal to twice the cube of the revenue *minus* the cube of the quantity demanded, all divided by three times the product of the quantity demanded and the square of the revenue. Find the relationship between R and x if $R = 0$ when $x = 10$

$$\left[\text{Hint: } \frac{dR}{dx} = \frac{2R^3 - x^3}{3R^2 x} \right]$$

17. Discuss the neo-classical one sector model of economic growth as presented by Solow. Would you consider the Harrod-Domar model as a special case of the neo-classical model?

18. The demand for a good is $x_d = ap + b$ and the supply $x_s = \alpha p + \beta$, where p is price. If there is excess demand then price rises to reduce this excess, i.e.,

$$\frac{d}{dt} (x_d - x_s) = -k (x_d - x_s)$$

where k is a positive constant. Find time path of price and interpret

Chapter 16

First Order Difference Equations and Applications

Introduction

In many economic studies the focus is not on relating continuous changes of variables but is on discrete changes. In planning models, for example, the comparison is between the initial base year and the terminal year and change in investment over the period is related to change in time over a period. Both the changes are said to be discrete.

Let $Y(t)$ be a function of t . The change in $Y(t)$ due to a change in t by a positive number h is called the first forward difference of Y_t , relative to the increment h , and is denoted by $\Delta Y(t)$.

$$\text{Thus,} \quad \Delta Y(t) = Y(t+h) - Y(t)$$

In economic applications t generally denotes time and h is equal to one period. $Y(t)$ is also written as Y_t . Hence,

$$\Delta Y(t) = Y_{t+1} - Y_t \quad (i)$$

The corresponding forward differences of higher order are then easily defined. Thus,

$$\begin{aligned} \Delta^2 Y_t &= \Delta[\Delta Y_t] \\ &= \Delta[Y_{t+1} - Y_t] \\ &= \Delta Y_{t+1} - \Delta Y_t \\ &= [Y_{t+2} - Y_{t+1}] - [Y_{t+1} - Y_t] \\ &= Y_{t+2} - 2Y_{t+1} + Y_t \end{aligned} \quad (ii)$$

$$\begin{aligned} \Delta^3 Y_t &= \Delta[\Delta^2 Y_t] \\ &= \Delta[Y_{t+2} - 2Y_{t+1} + Y_t] \\ &= \Delta Y_{t+2} - 2\Delta Y_{t+1} + \Delta Y_t \\ &= (Y_{t+3} - Y_{t+2}) - 2(Y_{t+2} - Y_{t+1}) + (Y_{t+1} - Y_t) \\ &= Y_{t+3} - 3Y_{t+2} + 3Y_{t+1} - Y_t \end{aligned} \quad (iii)$$

It should be noted that the expressions in (i), (ii) and (iii) are similar to the binomial expansion of $(1-x)^n$. Thus,

$$\Delta^n Y_t = Y_{t+n} - {}^nC_1 Y_{t+n-1} + {}^nC_2 Y_{t+n-2} \\ + \dots + (-1)^r {}^nC_r Y_{t+n-r} + \dots$$

16.1. Difference Equation

A difference equation relates the independent variable, the dependent variable and its successive differences

Therefore, $F(t, Y_t, \Delta Y_t, \Delta^2 Y_t, \dots) = 0$ is a difference equation

The order of the difference equation is that of the highest difference contained in the equation

$\Delta Y_t = 4$ is a first order difference equation

$\Delta Y_t = aY_t$ is also first order difference equation

$\Delta^2 Y_t + \Delta Y_t = Y_t$ is second order difference equation and so forth

We have already seen that difference can be expressed in a form free from Δ expression

$$\Delta Y_t = 4 \text{ is equivalent to } Y_{t+1} - Y_t = 4$$

The order of the equation is 1 since the highest difference involved is

$$t+1-t=1$$

$\Delta^2 Y_t + \Delta Y_t = Y_t$ is equivalent to

$$Y_{t+2} - 2Y_{t+1} + Y_t + Y_{t+1} - Y_t = Y_t$$

$$\text{or, } Y_{t+2} - Y_{t+1} - Y_t = 0$$

The order of this equation is 2, since the highest difference involved is $t+2-t=2$.

Linear Difference Equations : We are confining our attention to linear difference equations with constant coefficients. A difference equation is said to be linear with constant coefficients if it is of the form

$$a_0 Y_{t+n} + a_1 Y_{t+n-1} + \dots + a_{n-1} Y_{t+1} + a_n Y_t = f(t)$$

where $a_1, a_2, \dots$ are constants and $f(t)$ is any function defined for

$$t = 0, 1, \dots, n$$

16.2. Solution of Difference Equations

A function is a solution of a difference equation if it makes the equation a true statement and it satisfies the equation

If the solution satisfies the equation without reference to initial conditions, it is called *general solution*. If, in addition, it satisfies the initial condition(s) it is termed as a *particular solution*.

Ex. 1 Show that $Y_t = t + C$ is a solution of $Y_{t+1} - Y_t = 1$

Also find a *particular solution* if $Y_0 = 2$ when $t = 0$

Since $Y_t = t + C$ is the solution of the difference equation

$$Y_{t+1} - Y_t = 1$$

the equation should get satisfied on substituting the values of Y_{t+1} and Y_t

$$\text{Given } Y_t = t + C \quad (i)$$

$$Y_{t+1} = (t+1) + C \quad (ii)$$

Substituting (i) and (ii) in the equation $Y_{t+1} - Y_t = 1$, we get

$$(t+1) + C - (t + C) = 1$$

Hence the equation is satisfied therefore, $Y_t = t + C$ is the general solution of the given equation

Initial condition $Y_0 = 2$

Substituting in the solution $2 = 0 + C$, or $C = 2$

So the particular solution becomes $Y_t = t + 2$

Ex. 2. Show that $Y_t = \frac{C}{1+C_t}$ is the solution of $Y_{t+1} = \frac{Y_t}{1+Y_t}$ and find a particular solution if $Y_0 = -10$ when $t = 0$

We are given the solution $Y_t = \frac{C}{1+C_t}$

so that, $Y_{t+1} = \frac{C}{1+C_{t+1}}$

Substituting the values on the left and right hand sides of the given equation

$$Y_{t+1} = \frac{Y_t}{1+Y_t}, \text{ it is found that L.H.S.} = \text{R.H.S.}$$

Hence, it is proved that $Y_t = \frac{C}{1+C_t}$ is the general solution of the given

difference equation $Y_{t+1} = \frac{Y_t}{1+Y_t}$

The solution is $Y_t = \frac{C}{1+C_t}$

$$\text{Substituting for initial condition} \quad -10 = \frac{C}{1 + C(0)} = C$$

$$\text{i.e.,} \quad C = -10$$

So that the particular solution now becomes:

$$Y_t = \frac{-10}{1 - 10t} = \frac{10}{10t - 1}$$

16.3. Solution of First Order Difference Equations (Iterative Method)

We shall illustrate the method by solving one example.

Ex. 3 Suppose we are required to solve equation $\Delta Y_t = 4$, with the given initial condition of $Y_0 = 20$

Let us first put the given equation in the lag form

$$\Delta Y_t = 4$$

$$Y_{t+1} - Y_t = 4$$

$$\text{Similarly,} \quad Y_t - Y_{t-1} = 4$$

$$Y_{t-1} - Y_{t-2} = 4$$

$$\dots \dots \dots$$

$$\dots \dots \dots, \quad \text{and finally}$$

$$Y_1 - Y_0 = 4$$

Adding all $(t+1)$ equations, we get;

$$Y_{t+1} - Y_0 = (t+1) 4$$

$$Y_{t+1} = Y_0 + (t+1) 4$$

$$Y_{t+1} = Y_0 + 4t + 4$$

$$Y_{t+1} = Y_0 + 4t + (Y_{t+1} - Y_t) \quad \therefore Y_{t+1} - Y_t = 4$$

$$Y_t = Y_0 + 4t$$

Alternatively, adding all t equations except the first (i.e., $Y_{t+1} - Y_t = 4$), we directly get $Y_t = Y_0 + 4t$

so that, $Y_t = 20 + 4t$ (since it is given $Y_0 = 20$)

$Y_t = 4t + 20$ is the required solution of the given difference equation $\Delta Y_t = 4$

Note In continuous analysis the equation would have been $\frac{dY}{dt} = 4$ and which on integrating gives $Y = 4t + C$. This is the same result as obtained above ($C = 20$ from the initial condition)

Ex 4 Solve the equation $\Delta Y_t = -0.2 Y_t$

We are given $\Delta Y_t = -0.2 Y_t$

$$Y_{t+1} - Y_t = -0.2 Y_t$$

$$Y_{t+1} - 0.8 Y_t = 0$$

$$Y_{t+1} = 0.8 Y_t$$

when $t = 0$ $Y_1 = 0.8 Y_0$

$$t \approx 1 \quad Y_2 = 0.8 Y_1 = 0.8 \{0.8 Y_0\} = (0.8)^2 Y_0$$

$$t \approx 2 \quad Y_3 = 0.8 Y_2 = 0.8 \{0.8^2 Y_0\} = (0.8)^3 Y_0$$

$$t \approx 3 \quad Y_4 = 0.8 Y_3 = 0.8 \{0.8^3 Y_0\} = (0.8)^4 Y_0$$

Generalising $Y_t = 0.8 Y_{t-1} = 0.8 \{(0.8)^{t-1} Y_0\} = (0.8)^t Y_0$

Thus the solution of the given equation will be

$$Y_t = Y_0 (0.8)^t$$

Ex 5 Solve $Y_{t+1} = aY_t + b$ by the iterative method

Given $Y_t = aY_{t-1} + b$

$$= a(aY_{t-2} + b) + b \quad \text{putting } Y_{t-1} = aY_{t-2} + b$$

$$= a^2 Y_{t-2} + ab + b$$

$$= a^2 (aY_{t-3} + b) + ab + b \quad \text{putting } Y_{t-2} = aY_{t-3} + b$$

$$= a^3 Y_{t-3} + a^2 b + ab + b$$

Generalising $Y_t = a^t Y_0 + b(1 + a + a^2 + \dots + a^t)$

$$Y_t = Y_0 a^t + b \frac{1 - a^{t+1}}{1 - a} \quad (\text{if } a \neq 1)$$

If $a = 1$ then $Y_t = a^t Y_0 + b(1 + 1 + 1 + \dots)$ so that

$$Y_t = Y_0 a^t + bt \quad \text{is the required solution}$$

EXERCISE 16.1

Find the solution of the following difference equations by *iterative method* (or by induction)

1 $\Delta Y_t = 3$

2 $\Delta Y_t = 0.15 Y_{t-1}$

3 $y_{t+1} - 0.3 Y_t = 0$

4 $m Y_{t+1} - n Y_t = 0$

5 $Y_{t+1} - 0.3 Y_t = 1$

6 $Y_{x+1} - 0.2 Y_x = 8 (Y_0 = 1)$

7 $Y_{x+1} - 0.3 Y_x = 3 (Y_0 = 2)$

8 $P_t = 5 P_{t+1} (P_0 = 3)$

9 $y_t = L Y_{t+1} (Y_0 = M)$

10 Show that $\Delta y_x z_x = z_{x+1} \Delta y_x + y_x \Delta z_x$

Hint L.H.S. $= y_{x+1} z_{x+1} - y_x z_x$

Start with R.H.S. and prove it is equal to L.H.S.

11 $Y_x = \alpha Y_t (Y_0 = \beta)$

12 Prove that $\Delta \log x = \log \left(1 + \frac{1}{x} \right)$

13 Prove that $\Delta \sin ax = 2 \sin \frac{a}{2} \cos a \left(x + \frac{1}{2} \right)$

14 If $y_{t+1} - y_t = 2 \sin \frac{a}{2} \cos a \left(t + \frac{1}{2} \right)$ prove that the solution is $y_t = \sin at$

Hint L.H.S. $= \Delta Y_t$, R.H.S. $= \Delta \sin at$ from question No. 13. Now integrate both sides and get the result

15 If Y_x is the sum of squares of the first x integers show that

$$\Delta x = (x+1)^2 \text{ subject to } y_0 = 0$$

16 Show that $y_t = c_1 + c_2 t$ is the solution of $Y_{t+2} - 3y_{t+1} - 2y_t = 0$

Find particular solution if $y_0 = 1$ and $y_1 = 3$

17 Show that $y_x = c_1 + c_2 (-1)^x$ is the solution of $y_{x+2} - y_x = 0$

Find the particular solution if $y_0 = 2$ and $y_1 = 5$

18 Solve by induction $y_{t+1} - y_t = \alpha y_{t+1}$

19 $Y_t = 3Y_{t-1} + 5 (Y_0 = 2)$

16.4 General Method of Solving First Order Difference Equations

We observed that solution of a difference equation is similar to that of a differential equation. The general solution of the difference equation also consists of two components: a particular integral, Y_p , and a complementary function Y_c .

The Y_p component will again represent the equilibrium level of Y and Y_c component will depict deviations of the time path from the equilibrium level.

The sum Y_p and Y_c will give the general solution of any difference equation.

The Basic Theorem The general discrete solution of an ordinary difference equation of n th order gives Y as a function of t ($t = 0, 1, 2, \dots, n$) involving exactly n arbitrary constants $c_1, c_2, \dots, c_n$.

$$Y = f(t, A_1, A_2, \dots, A_n)$$

for difference equation of order n .

For a linear difference equation $n = 1$, the solution will involve only one arbitrary value $Y_1 = Y_0$ at $t = 0$.

Rules for Practical Work. General form of difference equation of n th order is

$$a_0 Y_t + a_1 Y_{t-1} + a_2 Y_{t-2} + \dots + a_n Y_{t-n} = f(t) \quad (1)$$

Its homogeneous form is depicted as

$$a_0 Y_t + a_1 Y_{t-1} + a_2 Y_{t-2} + \dots + a_n Y_{t-n} = 0 \quad (2)$$

Important rules are

(i) If homogeneous Eqn (2) has discrete solutions $Y_1(t)$ and $Y_2(t)$, then $Y_1(t) + Y_2(t)$ is also a discrete solution.

(ii) If homogeneous Eqn (2) has a discrete solution $Y(t)$, then $AY(t)$ is also a discrete solution for any constant A .

(iii) If $\bar{Y}(t)$ is any particular integral of Eqn (1) and if $f(t, A_1, A_2, A_3, \dots, A_n)$ is general discrete solution of the corresponding homogeneous form Eqn (2) which may be termed as complementary solution, then the complete solution of Eqn (1) is

$$\bar{Y}(t) + f(t, A_1, A_2, A_3, \dots, A_n)$$

There are two methods for solving difference equations.

Method I

(i) First find n solutions $Y_1(t), Y_2(t), Y_3(t), \dots, Y_n(t)$ of the equation 2.

(ii) Combine them to get the discrete general solution of homogeneous Eqn (2) involving n arbitrary constants for equation of order n . This gives us

$$a_1 Y_1(t) + a_2 Y_2(t) + a_3 Y_3(t) + \dots + a_n Y_n(t)$$

as a solution. This is a complementary solution denoted by Y_c .

(iii) Get a particular integral of Eqn (1) = $\bar{Y}(t)$

(iv) Thus the general solution of Eqn (1) becomes

$$\underbrace{\bar{Y}(t)}_{\text{Particular}} + \underbrace{a_1 Y_1(t) + a_2 Y_2(t) + \dots + a_n Y_n(t)}_{\text{Complementary}}$$

Method II An alternative method (also called *reverse process*) is of more practical utility. This involves following steps

(i) Attempt a particular solution of Eqn (1) i.e., $\bar{Y}(t)$ which gives trend variation

(ii) Write $y_t = Y_t - \bar{Y}(t)$, then the equation of y_t becomes homogeneous of the form of Eqn (2)

(iii) Find the general solution which is given by $Y_p + Y_c$

Ex. 6 Solve $Y_{t+1} + aY_t = C$

The solution of the given equation will consist of y_p and y_c .

Method I Make $Y_{t+1} + aY_t = 0$,

the complementary function of which must be of the form $Y_t = Ab^t$

If this solution holds for $Y_{t+1} + aY_t = 0$

$$\text{then, } Ab^{t+1} + aAb^t = 0$$

$$\text{so that } b + a = 0 \quad b = -a$$

Therefore, the complementary function $Y_c = A(-a)^t$

To find Y_p , we may choose any solution of $Y_{t+1} + aY_t = C$

We try a solution of simplest form $Y_t = \bar{Y}$ (some constant)

for all values of t so that our equation now reduces to $\bar{Y} + a\bar{Y} = C$

$$\text{i.e., } \bar{Y} = \frac{C}{1+a}$$

Since $Y_t = \bar{Y} = \frac{C}{1+a}$ satisfies the equation $Y_{t+1} + aY_t = C$,

the particular solution will be $Y_p = \frac{C}{1+a}$

(When $a = -1$, Y_p will be indeterminate and hence the solution is possible only when $a \neq -1$)

Thus the complete solution of the given equation $Y_{t+1} + aY_t = C$ is given by

$$Y_t = Y_p + Y_c$$

$$Y_t = \bar{P} + A(-a)^t \quad \left(\text{where } \bar{P} = \frac{C}{1+a} \right)$$

The above solution is general solution since there is one arbitrary constant A in the solution. The value of A is found by using (given) initial conditions.

Suppose, the initial conditions are given to be $Y_t = Y_0$ when $t = 0$

The solution will be $Y_0 = \bar{P} + A$ or $A = Y_0 - \bar{P}$

And, therefore, the solution will be $Y_t = \bar{P} + (Y_0 - \bar{P})(-a)^t$

where, $\bar{P} = \frac{C}{1+a}$ and $a \neq -1$

Method II First we find Y_p . Suppose $Y_t = \bar{P}$ for all values of t

$$\bar{P} + a\bar{P} = C$$

$$Y_p = \bar{P} = \frac{C}{1+a} \quad a \neq -1$$

To find Y_c we write the equation in the following form

$$Y_{t+1} + aY_t = C$$

with, $\bar{P} + a\bar{P} = C$

Subtracting, $Y_{t+1} - \bar{P} + a(Y_t - \bar{P}) = 0$

$$Y_{t+1} - \bar{P} = -a(Y_t - \bar{P})$$

$$y_{t+1} = -ay_t \quad [y_t = Y_t - \bar{P}]$$

or, $y_{t+1} + ay_t = 0$,

the solution of which will be $y_t = A(-a)^t$

and, if initial condition $y_t = y_0$ (when $t = 0$) is given,

then $A = y_0$ and the solution is $y_t = y_0(-a)^t$

$$i.e., \quad Y_t - \bar{Y} = (Y_0 - \bar{Y})(-a)^t$$

The complete solution is $Y_t = \bar{Y} + (Y_0 - \bar{Y})(-a)^t$, where $\bar{Y} = \frac{c}{1+a}$

Note. In the difference equation $Y_{t+1} + aY_t = b$, the coefficient of Y_{t+1} is unity. If, however, a given difference equation has a non unity coefficient of Y_{t+1} , it must be normalized. In other words, in such cases all the terms must be divided by this coefficient before proceeding to find the solution.

How to check the solution To check whether our solution is correct, we ensure that it satisfies the following conditions,

- (i) Solution must give the correct initial values, in other words, the solution must satisfy the given initial conditions and
- (ii) Solution must satisfy the requirements of the given difference equation

Ex 7 Solve $Y_t = 2Y_{t-1} + t$

The given equation may be written as $Y_t - 2Y_{t-1} = t$ (1)

On comparing with $Y_t + aY_{t-1} = b$

$$a = -2 \quad \text{and} \quad b = t$$

The simple solution of the previous example cannot be applied here because b is not a constant. We proceed as follows

To find particular integral, Y_p we try $Y_t = \mu_1 t + \mu_2$

$$\text{Eqn (1) becomes} \quad (\mu_1 t + \mu_2) - 2[\mu_1(t-1) + \mu_2] = t$$

$$\mu_1 t + \mu_2 - 2\mu_1 t + 2\mu_1 - 2\mu_2 - 1 = 0$$

$$-\mu_1 t + 2\mu_1 - \mu_2 - 1 = 0 \quad (2)$$

$$\text{On substituting } t = 0 \text{ in (2),} \quad 2\mu_1 - \mu_2 - 1 = 0 \quad (3)$$

$$\text{On substituting } t = 1 \text{ in (2),} \quad \mu_1 - \mu_2 - 1 = 0 \quad (4)$$

Solving (3) and (4) simultaneously, we obtain

$$\mu_1 = -1, \text{ and } \mu_2 = -2$$

Therefore, $Y_p = \mu_1 t + \mu_2$ should be written as

$$Y_p = -t - 2 = \bar{Y}$$

To find complementary function, Y_c , we write the equation in deviation form as

$$Y_t = 2Y_{t-1} + t \quad [\text{given equation}]$$

$$\bar{Y} = 2\bar{Y} + t \quad (\text{by substituting } \bar{Y} \text{ for all values of } t)$$

Subtracting from eq (1) we get $Y_t - \bar{Y} = 2(Y_{t-1} - \bar{Y})$ which in deviation form is $y_t = 2y_{t-1}$

Hence, the complementary function will be $y_c = A \cdot 2^t$

Therefore, the complete solution will be $y_c + y_p$ $Y_t = (-t-2) + A \cdot 2^t$

16.5. Geometrical Representation of the Solution of First Order Difference Equations Dynamics of the Equilibrium

The general form of the first order difference equation is $Y_{t+1} - aY_t = f(t)$

which may also be written as $Y_{t+1} - aY_t = f(t)$

Solution of this equation involves particular integral $Y_p = \bar{Y}$ and the complementary function $Y_c = A_0 a^t$ or $Y_0 a^t$

The complementary function determines the dynamic stability of the equilibrium. If the complementary function $A_0 a^t$ tends to zero as $t \rightarrow \infty$, the equilibrium will be stable. If it does not tend to zero as $t \rightarrow \infty$ then there will be dynamic instability. The convergence or divergence, as we shall find below, depends upon the value and sign of the term a — which is the base of the exponential term a^t (we must remember here that t is always positive)

Table 16.1 : Values of a^t with different values of ' a '.

Value of 'a'	a^t	Values of a^t when				
		$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$
1 $a > 1$, say $a = 2$	2^t	1	2	4	8	16
2 $a = 1$	1^t	1	1	1	1	1
3 $0 < a < 1$, say $a = \frac{1}{2}$	$(\frac{1}{2})^t$	1	1/2	1/4	1/8	1/16
4 $a = 0$	$(0)^t$	0	0	0	0	0
5 $-1 < 0 < 1$, say $a = -\frac{1}{2}$	$(-\frac{1}{2})^t$	1	-1/2	1/4	-1/8	1/16
6 $a = -1$	$(-1)^t$	1	-1	1	-1	1
7 $a < -1$, say $a = -2$	$(-2)^t$	1	-2	4	-8	16

If we plot the value a^t on the y-axis with value of t on the x-axis, we shall get the following graphs depicting dynamic stability or instability.

(1) $a > 1$, a^t increase steadily without limit. Graphical form will be as shown in Fig. 16.1

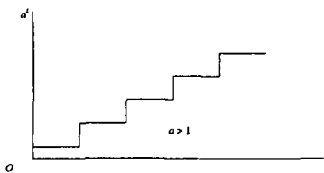


Fig. 16.1

This will be the case of increasing divergence from equilibrium with explosive fluctuations.

(2) If $a = 1$, graphical form will be as shown in Fig. 16.2. This will be the case of constant divergence from the equilibrium.

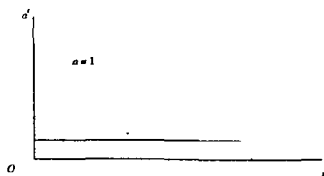
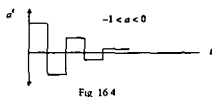
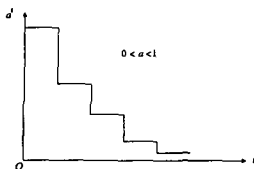


Fig. 16.2.

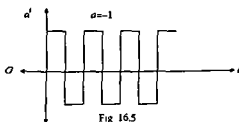
(3) If $0 < a < 1$, its form will be as shown in Fig. 16.3. Value of a^t will go on decreasing with rise in the value of t .

(4) If $a = 0$, there will be no disturbance from the equilibrium at all.

(5) If $-1 < a < 0$, its graphical form will be as shown in Fig. 16.4. Values of a^t will alternate between + and - but absolute values go on decreasing — ultimately converge to the equilibrium. Thus, this will be the case of dynamic stability with damped fluctuations.

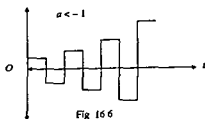


(6) If $a = -1$, fluctuations are equal and hence there is no dynamic stability (Fig 16.5)



(7) If $a < -1$, the values of a' alternate and the fluctuations deviate farther and farther from the equilibrium

This is the case of explosive fluctuations (Fig 16.6)



Constant A_0 of solution $A_0 a'$ So far we have considered the values of a' with the change in t . Let us now examine the effect of constant A_0 on the path a' .

Constant A_0 when multiplied with a' will only change the scale of the path but will not affect the basic change in (or the shape of) the path. In other words, it will only produce the scale effect and thus will only blow up (if $A_0 > 1$) or pull down (if $0 < A_0 < 1$) the values of a' . What will happen if $A_0 < 0$?

Negative sign of A_0 definitely affects the shape of the path. If a' is multiplied by $A_0 = -1$, the time path gets reversed. In other words, the negative values of A_0 produce a mirror effect as well as a scale effect.

Thus the above geometrical representation of the solution of the first order difference equation gives us an interesting trend phenomenon which could not arise while dealing with differential equations.

Ex 8 $y_x + 2y_{x-1} = 2^x(3x + 4)$

We try particular integral

$$y_x = 2^x(\mu_1 + \mu_2 x)$$

$$\text{Hence } 2^x(\mu_1 + \mu_2 x) + 2 \cdot 2^{x-1}[\mu_1 + \mu_2(x-1)] = 2^x(3x + 4)$$

$$\mu_1 + \mu_2 x + \mu_1 + \mu_2 x - \mu_2(3x + 4)$$

$$2\mu_1 + 2\mu_2 x - \mu_2 = 3x + 4$$

Comparing the corresponding terms,

$$2\mu_1 - \mu_2 = 4$$

$$\text{Also, } 2\mu_2 = 3, \text{ or } \mu_2 = \frac{3}{2}$$

$$\mu_1 = \frac{1}{2}(\mu_2 + 4)$$

$$\mu_1 = \frac{1}{2}\left(\frac{3}{2} + 4\right)$$

$$\text{Hence, the particular solution is } y_p = 2^x\left(\frac{11}{4} + \frac{3}{2}x\right)$$

The complementary function is found by solving $y_x + 2y_{x-1} = 0$ which gives

$$y_c = y_0(-2)^x$$

Hence, the complete solution is

$$y_x = y_p + y_c$$

$$y_x = y_0(-2)^x + 2^x\left(\frac{11}{4} + \frac{3}{2}x\right)$$

EXERCISE 16.2

Solve the following difference equations

1 $y_t - 2y_{t-1} = 3$

2 $y_t - ay_{t-1} = \alpha^t$ (Try $y_t = \mu \alpha^t$ for particular solution)

3 $y_x - 3y_{x-1} = x + 2$

4 $y_{t+1} - 2y_t = 0$

5 $y_x = \frac{1}{2}y_{x-1} - 1$

6 $y_{x+1} - y_x = \alpha(y_x + 1) + \beta$

Solve the following equations and also find the nature of the time path

7 $y_{t+1} - \frac{1}{3}y_t = 6$ ($y_0 = 1$)

8 $y_t + 3y_{t-1} = t - 1$, ($y_0 = 3$)

9 $y_{t+1} + \frac{1}{3}y_t = 3$ ($y_0 = 0$)

10 $4y_{t+1} - 5t - 1$ ($y_0 = 3$)

11 $y_{x+1} = 3y_{x-1}$ ($y_0 = \frac{1}{2}$)

12 $y_{t+1} = \alpha y_t - \beta$, ($y_t = y_0$ when $t = 0$)

13 $y_t - L y_{t-1}$ ($y_0 = M$)

14 $P_t = 5P_{t-1}$ ($P_0 = 3$)

15 $y_{t+1} = 0.3 y_{t-3}$ ($y_0 = 2$)

Solve the equations and discuss the growth sequence. Calculate first three values of the solution sequence

16 $8y_{t+1} + 4y_t = 3$ ($y_0 = \frac{1}{2}$)

17 $2y_{t+1} = y_t + 2$, ($y_2 = 4$)

18 $y_{t+1} - 10y_t = 3$ ($y_0 = 1$)

19 $3y_{t+1} - 2y_t - 6 = 0$, ($y_2 = 4$)

20 $y_{t+1} + 4y_t + 12 = 0$, ($y_0 = 6$)

21 Show that $y_x = c_1 + c_2 2^x - x$ is a solution of $y_{x+2} - 3y_{x+1} + 2y_x = 1$ and find particular solution if $y_0 = 3$ when $y_1 = 3$

16.6 Application in Economics

The Cobweb model : The basic Cobweb model assumes that today's demand for any commodity is a function of the present price (p_t) while today's supply depends upon yesterday's decisions about the output. Hence, output is naturally influenced by yesterday's price (p_{t+1})

To put this in symbolic form

Demand function $\lambda_{Dt} = a + \alpha P_t$

Supply function $\lambda_{St} = b + \beta P_{t-1}$

Thus both demand and supply functions are assumed to be linear

$$\begin{aligned} \text{For equilibrium } X_{D_t} &= X_{S_t} \\ a + \alpha P_t &= b + \beta P_{t-1} \end{aligned} \quad (1)$$

Let $\bar{P}$ be the equilibrium price, so that $P_t = \bar{P}$ for all t 's

$$\begin{aligned} \text{Thus, } a + \alpha \bar{P} &= b + \beta \bar{P} \\ \text{and, } \bar{P} &= \frac{b-a}{\alpha-\beta} \end{aligned} \quad (2)$$

Subtracting (2) from (1), we get $\alpha(P_t - \bar{P}) = \beta(P_{t-1} - \bar{P})$

$$\begin{aligned} \alpha p_t &= \beta(p_{t-1}) \\ p_t &= \frac{\beta}{\alpha} (p_{t-1}) \end{aligned}$$

which represents the first order difference equation

The complementary function of this equation will be

$$P_t = P_0 \left(\frac{\beta}{\alpha} \right)^t, \text{ where } p_t = P_t - \bar{P}, p_0 = P_0 - \bar{P}$$

$$\text{or, } P_t - \bar{P} = (P_0 - \bar{P}) \left(\frac{\beta}{\alpha} \right)^t$$

$$\text{or, } P_t = \bar{P} + (P_0 - \bar{P}) \left(\frac{\beta}{\alpha} \right)^t \quad P_0 \text{ is the initial price}$$

The result can also be obtained directly by letting $p_t = \lambda \lambda^t$ to be the solution of the difference equation. Then $\lambda \lambda^t = \frac{\beta}{\alpha} \lambda \lambda^{t-1} \quad \lambda = \frac{\beta}{\alpha}$

$$\text{And the solution is } p_t = \lambda \left(\frac{\beta}{\alpha} \right)^t \text{ From the initial condition } \lambda = p_0$$

We have the following conclusions

(i) $\bar{P} = \frac{b-a}{\alpha-\beta}$ which is the equilibrium price of the model. This being a constant term, it gives a stationary equilibrium price

(ii) The term $(P_0 - \bar{P})$ depicts the scale effect. If $(P_0 - \bar{P}) > 0$ the time path, as said above, will blow up. If $(P_0 - \bar{P}) < 0$, the time path will start from below the equilibrium price

The magnitude of $(\bar{P} - P_0)$ determines how far above or below P_0 the time path starts

(iii) The crucial factor is $\left(\frac{\beta}{\alpha}\right)'$ where β and α are the slopes of the supply and demand curves. Since the demand curve is generally negatively sloped, $\alpha < 0$, and since supply curve is positively sloped, therefore, $\beta > 0$, and

$$\frac{\beta}{\alpha} = \frac{(+)\text{sign}}{(-)\text{sign}} = (-)\text{sign}$$

Hence the time-path must always be oscillatory

Three possibilities arise

(1) If $\beta > |\alpha|$, then $\frac{\beta}{|\alpha|} > 1$, then there will be explosive oscillations (See Fig 16.8)

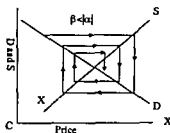


Fig 16.7

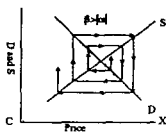


Fig 16.8

(2) If $\beta = |\alpha|$ then $\frac{\beta}{|\alpha|} = 1$, then there will be regular oscillations (See Fig 16.9)

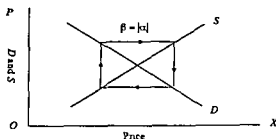


Fig 16.9

(3) If $\beta < |\alpha|$ then $\frac{\beta}{|\alpha|} < 1$ then there will be damped oscillations as shown in Fig 16.7

Thus the dynamic equilibrium can only be obtained in the third case when $\beta < |\alpha|$ or when the demand curve is steeper than the supply curve

The disequilibrium price P , therefore, oscillates over successive periods around the equilibrium price $\bar{P}$ and converges to $\bar{P}$ if and only if $|\beta| < |\alpha|$, or if and only if D is steeper than S around the point of intersection

Ex. 9. Examine the path represented by $Y_t = 5\left(\frac{-1}{10}\right)^t + 3$.

Here we find that $a = -\frac{1}{10}$ which is < 0 , the time-path will be *oscillatory* but since $|a| = \frac{1}{10} < 1$, the oscillation is *damped*. Therefore the time-path converges to the equilibrium level 3 (which is the particular integral of the given equation).

Ex. 10. How do you characterise the time-path : $Y_t = 3^t + 17$

Here, since $a = 3 > 0$, no oscillations will occur.

Again since $|a| = 3 > 1$,

the time-path will *explode* and will diverge from the equilibrium level which is 1

Ex. 11. Given the following demand and supply functions, find equilibrium price and determine whether the equilibrium is stable

- | | | | |
|-----|-----------------------|---|-----------------------|
| (a) | Demand function | : | $X_d = 18 - 3p_t$ |
| | Supply function | : | $X_s = -3 + 4p_{t-1}$ |
| (b) | Demand function | : | $X_d = 19 - 6p_t$ |
| | Supply function | : | $X_s = 6p_{t-1} - 5$ |
| (a) | Slope of demand curve | : | $ \alpha = 3$ |
| | Slope of supply curve | : | $ \beta = 4$ |

Since $|\beta| > |\alpha|$, and $\frac{\beta}{\alpha} < 0$.

There will be *explosive oscillations* and equilibrium will be *unstable*

Equilibrium price : $\bar{p} = 3$. [From $18 - 3\bar{p} = -3 + 4\bar{p}$]

- | | | |
|-----|-------------------------|----------------|
| (b) | Slope of demand curve : | $ \alpha = 6$ |
| | Slope of supply curve : | $ \beta = 6$ |

Since, $|\beta| = |\alpha|$, and $\frac{\beta}{\alpha} < 0$

There will be *regular oscillations* and the equilibrium will be *unstable*

16.7 Market Model With Inventory (Stocks)

The above Cobweb model was built on the assumption that current output is cleared by the current demand. However, the sellers may keep inventory of the commodity. Let Q_t be the stock at the end of period t .

We assume that (i) Demand and Supply are both unlagged functions of price, and (ii) that the prices are set according to stock changes, i.e., price changes are inversely proportional to the observed changes in the stock of the commodity (ΔQ_t , $t \in \mathbb{Z}$, $Q_t - Q_{t-1}$). Of course, change in stock in any period is due to excess (or deficit) or supply over demand.

Thus our model becomes

$$\begin{aligned} X_d &= a + \alpha p_t & \begin{cases} a > 0 \text{ and } \alpha < 0 \\ b > 0 \text{ and } \beta > 0 \end{cases} \\ X_s &= b + \beta p_t \end{aligned}$$

$$\begin{aligned} \text{Assume } p_t - p_{t-1} &= -\sigma(\Delta Q_t) = (\Delta Q_t = X_{s,t-1} - X_{d,t-1}) \\ p_t &= p_{t-1} - \sigma(X_{s,t-1} - X_{d,t-1}) \end{aligned}$$

Here $\sigma > 0$ and represents the *Stock Induced Price Adjustment Coefficient*.

$$\begin{aligned} p_t &= p_{t-1} - \sigma(b + \beta p_{t-1} - a - \alpha p_{t-1}) \\ p_t &= \sigma(a - b) + p_{t-1} - \sigma(\beta - \alpha)p_{t-1} \\ p_t &= \sigma(a - b) + [1 - \sigma(\beta - \alpha)]p_{t-1} \end{aligned} \quad (1)$$

$$\text{Hence the time path will be } p_t = \bar{p} + (p_0 - \bar{p})[1 - \sigma(\beta - \alpha)]^t \quad (2)$$

where $\bar{p}$ (as before) is $\frac{\beta - a}{\alpha - \beta}$ which gives the equilibrium price.

Thus the dynamic stability of the model hinges on the term

$$[1 - \sigma(\beta - \alpha)] = C \text{ (some constant)}$$

We know that generally $\alpha < 0$ and $\beta > 0$

$(\beta - \alpha)$ will always be positive

It may be seen from the table below that the dynamic stability will be attained only if $0 < C < 1$, i.e. when $0 < \sigma < \frac{1}{(\beta - \alpha)}$ [which means when stock induced price adjusted coefficient is less than $\left(\frac{1}{\beta - \alpha}\right)$ and greater than zero] time path will be non-oscillatory convergent.

$$\text{Also when } \frac{1}{\beta - \alpha} < \sigma < \frac{2}{\beta - \alpha}$$

there will be damped oscillations. Equilibrium will be restored at $p_t = \bar{p}$ in both these cases.

Let us now tabulate the nature of time path with different values of

$$C = [1 - \alpha(\beta - \alpha)]$$

Values of $C = [1 - \alpha(\beta - \alpha)]$	Value of α	Nature of the time path of p_t
1 $0 < C < 1$	$0 < \alpha < \frac{1}{(\beta - \alpha)}$	No oscillations — Damped and Convergent
2 $C = 0$	$\alpha = \frac{1}{(\beta - \alpha)}$	In equilibrium
3 $-1 < C < 0$	$\frac{1}{\beta - \alpha} < \alpha < \frac{2}{\beta - \alpha}$	Oscillations — Damped
4 $C = -1$	$\alpha = \frac{2}{\beta - \alpha}$	Oscillations — Regular
5 $C < -1$	$\alpha > \frac{2}{\beta - \alpha}$	Oscillations — Explosive

16.7 Dynamic Multiplier (with Constant Autonomous Investment)

We know that

$$Y_t = C_t + I_t + A_t$$

where

C_t = Consumption in period t .

I_t = Induced investment in period t and

A_t = Autonomous investment including autonomous inventory consumption

Now if $I_t = 0$ and $A_t = A$, then the above equation turns into $Y_t = C_t + A$. In case consumption is a lagged function of income, i.e., today's consumption depends upon yesterday's income then

$$C_t = cY_{t-1} \text{ where } c \text{ is MPC}$$

$$Y_t - cY_{t-1} = A \quad (i)$$

The solution of (i) will be $Y_t = \bar{Y} + (Y_0 - \bar{Y})c^t$,

where the equilibrium income $\bar{Y}$ is obtained by putting

$$Y_t = \bar{Y} \text{ and } Y_{t-1} = \bar{Y} \text{ in Eqn (i)}$$

Therefore, $\bar{Y} = \frac{A}{1-c}$

Since MPC is usually less than 1 (but greater than zero) c^t will tend to zero as $t \rightarrow \infty$, therefore in the long run Y_t converges steadily to $\bar{Y}$. There is stable equilibrium. The speed of response depends upon (1) time-lag, and (2) MPC. Shorter the lag the more will be the speed of response and smaller the MPC, the smaller will be the change in Y_t .

16.8. Dynamic Multiplier with change in Autonomous Investment

Now suppose autonomous investment grows according to the geometrical law

$A_t = A_0(1+r)^t$ where A_0 is the initial autonomous investment and r is the rate of growth of autonomous investment

Hence $Y_t = cY_{t-1} + A_0(1+r)^t$

$$Y_t - cY_{t-1} = A_0(1+r)^t$$

which still has the homogeneous equation of the form

$$Y_t - cY_{t-1} = 0$$

of which the complementary function (Y_c) will be

$$Y_c = E c^t \text{ where } E \text{ is an arbitrary constant}$$

Try, $Y_t = F(1+r)^t$ as a particular solution, then

$$F(1+r)^t - c[F(1+r)^{t-1}] = A_0(1+r)^t$$

$$F(1+r) - cF = A_0(1+r)$$

$$F = \frac{A_0(1+r)}{(1+r-c)}$$

Therefore, the particular solution (Y_p) is

$$Y_p = \frac{A_0(1+r)}{(1+r-c)} (1+r)^t = \left[\frac{A_0}{1 - \frac{c}{1+r}} \right] (1+r)^t$$

$$\text{If, } t=0, Y_0 = \left[\frac{A_0}{1 - \frac{c}{1+r}} \right]$$

$$A_0 = Y_0 \left[1 - \frac{c}{1+r} \right]$$

and

$$Y_p = Y_0(1+r)^t$$

Thus, if the initial income is $\frac{A_0}{1 - \frac{c}{1+r}}$, then income will expand according to $Y_0(1+r)^t$

It should also be noted that Y_t in the general solution is composed of two terms

(i) $Y_0(1+r)^t$ which gives the steadily expanding time-path due to the assumed expansion of the autonomous investment, and

(ii) $(Y_0 - \bar{Y})c'$ which depicts the damped nature of oscillations tending steadily to zero

Therefore, the total effect of the two is that Y_t steadily approaches not to a given equilibrium level $\bar{Y}$ but rather to an expanding level $Y_0(1+r)^t$

In case autonomous investment stops at any point or stops increasing at any point of time, then Y_t will converge to an appropriate value

Usually autonomous investment is *Schumpeterian*, and in the sense that it is in spurts, the time path A_t is not mathematical but an empirical time series. In that case great care will have to be taken in interpreting the time path of income

16.9. Multiplier-Accelerator Model : The Harrod Model

1 Planned induced investment (*ex-ante* Investment) is fixed by the change in output from previous period, hence

$$I_t = v(Y_t - Y_{t-1})$$

where v is the acceleration coefficient

2 Ex-post relationship : $Y_t = C_t + I_t + A_t$,

So that $S_t = I_t + A_t$ (i.e., Ex-post saving = Ex post investment)

3 Ex-ante saving is $S_t = s Y_{t-1}$ or $C_t = c Y_{t-1}$.

We have introduced one period lag in consumption and saving functions but not in the investment function

4 Labour market equation $L_t = L_0(1+n)^t$

Now if the entrepreneurs are to be satisfied, the investment plans must be realised. Harrod assumes that saving plans are realised,

i.e., ex-post investment = ex-ante investment = ex-ante saving

(ex-ante savings are equal to ex post savings)

Hence, in equilibrium, $v(Y_t - Y_{t-1}) = s Y_{t-1}$

$$vY_t = (v + s)Y_{t-1}$$

$$Y_t = \left[1 + \frac{s}{v}\right] Y_{t-1},$$

the solution of which will be $Y_t = Y_0 \left[1 + \frac{s}{v}\right]^t$, if Y_0 is the initial income in period zero

If there is to be full employment, labour force should also grow at this rate of growth. So for full employment equilibrium growth $n = \frac{s}{v}$, where n is the rate of growth of labour force termed as natural rate of growth by Harrod.

Development of Domar model is similar to that in 15.3 leading to $\frac{\Delta I}{s} = I - p$ leading to $g = p - s$.

EXERCISE 16.3

Given the demand and supply functions as follows, find the equilibrium price, the time path and determine whether or not the equilibrium is stable.

$$1. \quad X_d = 22 - 3p_t, \quad X_s = -2 + 3p_{t-1}$$

$$2. \quad 3X_d = 20 - 5p_t, \quad 3X_s = -20 + 7p_{t-1}$$

$$3. \quad D_t = m_d P_t + b_d, \quad S_{t+1} = m_s p_t + b_s,$$

$$m_d = 1, \quad b_d = 5, \quad m_s = 1$$

$$4. \quad D_t = m_d p_t + b_d, \quad S_{t+1} = m_s p_t + b_s,$$

$$m_s = 2, \quad b_s = 1, \quad m_d = 1 \text{ and } b_d = 7$$

$$5. \quad X_t^d = a + bP_t,$$

$$X_t^s = c + dP_t,$$

$$X_t^d = X_t^s$$

$$6. \quad q_t^d = 3.0 - 0.75p_t,$$

$$q_t^s = 0.5 + 1.1p_{t-1},$$

$$p_0 = 1.33$$

$$q_t^d = p_t^s$$

$$7. \quad D_t = 100 - 0.5p_t,$$

$$S_{t+1} = 8 + 0.75p_t$$

8. $X^d = 10 - 3p_t$, $X^s = 2p_{t-1}$. Deduce Cobweb model. Find p_t for $t = 1, 2, 3$, and 4, given $p_0 = 3$.

9. State the conditions of equilibrium if in the Cobweb model the supply curve is assumed to have negative slope.

10. If the seller in our market model always increases or decreases the price by 10% of the amount of the decrease or increase of stocks, the slopes of

demand and supply curves being -1 and 15 respectively, what types of the path shall we get

(Hint $\sigma = 0.1$, $\alpha = -1$ and $\beta = 15$, hence $c = -6$ the time-path will be characterised by damped oscillations)

11. Determine the result (in the above exercise 10) when changes in price amount to 12% and slope of $D = -2$ and that of $S = 10$

12. Construct a dynamic model of Cobweb type with one-period lag on demand side and no supply lag. Would it lead to different mathematical solution if you had taken one period time-lag on the supply side instead of the demand side?

Are there any economic considerations for choosing one model in preference to the other?

13. Show that the multiplier-accelerator macro-model can be reduced to linear difference equation with constant coefficients

$$\begin{aligned}\text{Solution.} \quad Y_t &= C_t + I_t \\ C_t &= C_o + cY_{t-1} \quad (0 < c < 1) \quad c = \text{MPC}\end{aligned}$$

According to accelerator principle, the amount of capital is proportional to the amount of consumption (c , demand)

Or $K_t = \beta(C_t)$, where $\beta > 0$ and represents the acceleration coefficient

$$\begin{aligned}\text{Also,} \quad I_t &= K_t - K_{t-1} \\ I_t &= \beta(C_t - C_{t-2})\end{aligned}$$

$$\text{Since,} \quad Y_t = C_t + I_t$$

$$\begin{aligned}\text{therefore,} \quad Y_t &= C_o + cY_{t-1} + \beta[(C_o + cY_{t-1}) - (C_o + cY_{t-2})] \\ Y_t &= C_o + cY_{t-1} + c\beta(Y_{t-1} - Y_{t-2}) \\ Y_{t+2} &= C_o + \beta(1+c)Y_{t+1} - c\beta Y_t \quad (\text{By shifting the lag}) \\ Y_{t+2} - c(1+\beta)Y_{t+1} + c\beta Y_t &= C_o\end{aligned}$$

is the second order difference equation, the solution of which would generate the cycles depending on the different values of β and c .

14. If the market model with inventories has the following form, find the time-path and difference its nature

$$(i) \quad X_d = 20 - pt, \quad X_s = -5 + 4p_t, \quad \text{and} \quad P_{t+1} = p_t - 0.2(\Delta Q_t)$$

15. In the Harrod model, find the paths of Y_t for the following problems:

$$(i) \quad c = \frac{1}{5}, b = 2, Y = 100 \qquad (ii) \quad c = -\frac{1}{5}, \beta = 2, Y_t = 100$$

$$(iii) \quad c = 4, b = 2, Y_o = 100 \qquad (iv) \quad c = \frac{1}{5}, \beta = 2, Y_o = -100.$$

Chapter 17

Second Order Difference Equations and Applications

Introduction Thus far we considered only one period lag, but it may be that the consumption of a commodity depends not only on preceding period's income but also on income earned in the period preceding it, i.e., C_t depends on Y_{t-1} and Y_{t-2} . Such a situation will give rise to a difference equation of second order. If the equation involves only Y_{t-1} and Y_{t-2} and no terms in Y_{t-1} and Y_{t-2} of higher power, then the equation will also be linear.

Strictly speaking, a second order difference equation involves expression $\Delta^2 Y_t$ called the second order difference of Y_t . But as we have already shown, a second order difference can be transformed into a sum of terms involving a two-period lag as such

$$\begin{aligned}\Delta^2 Y_t &= \Delta (\Delta Y_t) \\ &= \Delta (Y_{t+1} - Y_t) \\ &= \Delta Y_{t+1} - \Delta Y_t \\ &= (Y_{t+2} - Y_{t+1}) - (Y_{t+1} - Y_t) \\ &= Y_{t+2} - 2Y_{t+1} + Y_t\end{aligned}$$

Since expressions like $\Delta^2 Y_t$ are cumbersome to operate with we use the equation in the lag form. Here we shall deal with only linear equations.

17.1. Solution of Second Order Difference Equations

$$Y_t + aY_{t-1} + bY_{t-2} = f(t)$$

is a second order linear non homogeneous difference equation, a and b are structural constants ($b \neq 0$). The method of finding solution to this equation involves two steps

- (i) to find a particular integral $Y_t = \hat{Y}$, and
 - (ii) to find the homogeneous solution or the complementary function.
- First, let us take $f(t) = \text{a constant} = c$

(1) *Particular integral* Try the solution $Y_t = \hat{Y}$, for all values of t then

$$\bar{Y} + a\bar{Y} + b\bar{Y} = c$$

$$\bar{Y} = \frac{c}{1+a+b} \text{ for all values if } a+b \neq -1$$

(2) *Complementary function.*

Our auxiliary equation is $Y_t + aY_{t-1} + bY_{t-2} = 0$

Try the solution $Y_t = A\lambda^t$, $\lambda \neq 0$, then

$$A\lambda^t + Aa\lambda^{t-1} + Ab\lambda^{t-2} = 0$$

$$\lambda^2 + a\lambda + b = 0 \quad [\lambda \neq 0]$$

This quadratic equation ($\lambda^2 + a\lambda + b = 0$) is termed as the characteristic equation of the difference equation. This possesses two roots λ_1 and λ_2 which are called the *characteristic roots*

$$\lambda_1 = \frac{1}{2} [-a + \sqrt{a^2 - 4b}]$$

$$\lambda_2 = \frac{1}{2} [-a - \sqrt{a^2 - 4b}]$$

Hence the homogeneous solution is $Y_t = A_1\lambda_1^t + A_2\lambda_2^t$

Three types of situations may arise with.

(i) Real but unequal (distinct) roots, if $a^2 - 4b > 0$

(ii) Real and equal (repeated) roots, if $a^2 - 4b = 0$

(iii) Complex conjugate roots, if $a^2 - 4b < 0$

We discuss each of them separately

L. Real Different Roots : The general solution will be

$$Y_t = P + A_1\lambda_1^t + A_2\lambda_2^t$$

$$\text{Here, } P = \frac{c}{1+a+b},$$

and λ_1 and λ_2 are as given above. The bigger of the two roots in absolute value is termed as the *dominant root*, let it be λ_1 , then $|\lambda_1| > |\lambda_2|$. The root is termed as dominant root because it will swamp the effect of the smaller one. The smaller root will have effect only in the initial stages. Four situations can arise

(i) If $\lambda_1 > 1$, λ_1^t will increase as time passes, hence the time-path will be explosive

(ii) If $0 < \lambda_1 < 1$, λ_1^t will decline as time passes, hence the time-path will be damped

(iii) If $-1 < \lambda_1 < 0$, damped oscillations will take place

(iv) $\lambda_1 < -1$, explosive oscillations will be observed

Thus the time-path will be convergent or damped only if the dominant root lies between -1 and 1 , i.e., if $-1 < \lambda_1 < 1$

Ex. 1. Solve $Y_{t+2} - 6Y_{t+1} + 8Y_t = 2$.

Particular integral Y_p We try $Y_t = \bar{Y}_t$ then, $\bar{Y}_t = \frac{2}{1-6+8} = \frac{2}{3}$

To obtain complementary function Y_c

our auxiliary equation is $Y_{t+2} - 6Y_{t+1} + 8Y_t = 0$

We try $Y_t = A\lambda^t$

$$\lambda^2 - 6\lambda + 8 = 0$$

which gives, $\lambda_1 = 4$, and $\lambda^2 = 2$

Hence the complete solution is $Y_t = \frac{2}{3} + A_1 4^t + A_2 2^t$ (1)

Suppose the initial conditions are

$$Y = 1 \text{ when } t = 0 \text{ and, } Y = 4 \text{ when } t = 1$$

Thus when $Y_0 = 1$, we have $\frac{2}{3} + A_1 + A_2 = 1$ (2)

and, when $Y_1 = 4$, we have $\frac{2}{3} + 4A_1 + 2A_2 = 4$ (3)

Solving Eqns (2) and (3), we have

$$A_1 = \frac{4}{3} \text{ and } A_2 = -1$$

Hence the solution is $Y_t = \frac{2}{3} + \frac{4}{3} 4^t - 2^t$

Since the dominant root $\lambda_1 = 4 > 1$, the time-path will be explosive and there will be no dynamic equilibrium

Ex. 2. Solve $5Y_t - 3Y_{t-1} - 2Y_{t-2} = 3$

Normalising the equation, i.e., putting it in the standard form, we have

$$Y_t - \frac{3}{5} Y_{t-1} - \frac{2}{5} Y_{t-2} = \frac{3}{5}$$

Here, $a + b = -1$, $\left[-\frac{3}{5} + \left(\frac{-2}{5} \right) = -1 \right] = P$

Hence $1 - a - b = 0$ and the particular integral cannot be found by trying

$Y_t = P$

In such cases we should try: $Y_t = \bar{Y}(t)$

$$Y_{t+1} = \bar{Y}(t-1) \text{ and } Y_{t+2} = \bar{Y}(t-2)$$

From the given eqn, we have $5\bar{Y}(t) - 3\bar{Y}(t-1) - 2\bar{Y}(t-2) = 3$, which gives

$$\bar{Y} = \frac{3}{7}t$$

Hence, the particular solution is $Y_t = \frac{3}{7}t$

The particular solution represents a *moving equilibrium*

For finding the complementary function, we try $Y_t = A\lambda^t$, then the characteristic equation is,

$$\lambda^2 - \frac{3}{5}\lambda - \frac{2}{5} = 0 \text{ which gives two roots,}$$

$$\lambda_1 = 1, \text{ and } \lambda_2 = -\frac{2}{5}$$

$$|\lambda_1| > |\lambda_2|$$

$\lambda_1 = 1$ is the dominant root

The complete solution is $Y_t = \frac{3}{7}t + A_1(1) + A_2\left(-\frac{2}{5}\right)^t$

$$Y_t = \frac{3}{7}t + A_1 + A_2\left(-\frac{2}{5}\right)^t$$

λ_1 is the dominant root but is ineffective since its value is 1. It at least points out that the time-path is non-explosive and damped. Since $\lambda_2 = -\frac{2}{5}$, there will be oscillations. Hence around the moving equilibrium given by the particular integral $3t/7$, there will be damped oscillations.

II. Equal or Repeated Roots: $\lambda_1 = \lambda_2 = \lambda$. Equal roots can occur only if the roots are real in the case of second order linear difference equations. The complementary function is

$$Y_c = A_1\lambda^t + A_2 \cdot t \cdot \lambda^t$$

One t is introduced for each repeated root.

Ex. 3. Solve: $Y_{t+1} - 4Y_t + 4Y_{t-1} = 3$.

(i) Particular solution is obtained by putting $Y_t = \bar{Y}$,

$$\bar{Y} - 4\bar{Y} + 4\bar{Y} = 3, \text{ hence } \bar{Y} = 3$$

(ii) Complementary function is $Y_{t+1} - 4y_t + 4Y_{t-1} = 0$

and the characteristic (or auxiliary) equation is $\lambda^2 - 4\lambda + 4 = 0$

Which gives $\lambda_1 = \lambda_2 = 2$

Hence the complete solution is $cY' = 3 + A_1 2^t + A_2 t 2^t$

when $Y_0 = 1$ then $1 = 3 + (A_1 + 0)$, and

when $Y_1 = 6$ then $6 = 3 + 2(A_1 + A_2)$

Thus, $A_1 = -2$ and, $A_2 = \frac{7}{2}$

The solution is, therefore $Y_t = 3 - 2 \cdot 2^t + \frac{7}{2} t 2^t$, hence the time path is explosive

III. Complex Roots : The roots as described above are

$$\lambda_1 = \frac{-a + \sqrt{a^2 - 4b}}{2}, \quad \text{and}$$

$$\lambda_2 = \frac{-a - \sqrt{a^2 - 4b}}{2}$$

If $(a^2 - 4b)$ is negative, i.e., $(a^2 < 4b)$, then it will not be possible to find out the square root of $a^2 - 4b$, but we shall be able to find the square root of $(4b - a^2)$. Hence writing $a^2 - 4b = -1(4b - a^2)$, we have the two roots as,

$$\lambda_1 = \frac{-a + \sqrt{(-1)(4b - a^2)}}{2},$$

$$\lambda_2 = \frac{-a - \sqrt{-1} \sqrt{4b - a^2}}{2}$$

$$\lambda_1 = \frac{-a + i \sqrt{4b - a^2}}{2},$$

$$\lambda_2 = \frac{-a - i \sqrt{4b - a^2}}{2} \quad [\text{where } \sqrt{-1} = i \text{ (iota)}]$$

λ_1 and λ_2 are called *conjugate complex numbers*, each consisting of a real part $\left(-\frac{a}{2}\right)$ and an imaginary part $\left(\frac{i \sqrt{4b - a^2}}{2}\right)$ the only difference being of the sign between the real and imaginary parts.

Let, $u = -\frac{a}{2}$ (real part),

and $v = \frac{\sqrt{4b-a^2}}{2}$ (imaginary part)

$\therefore \lambda_1 = u + iv$ and, $\lambda_2 = u - iv$

Then the homogeneous solution is $Y_h = A_1 (u + iv)^t + A_2 (u - iv)^t$.

But it is difficult to interpret this result. It is convenient to transform it in polar coordinates.

Put $u = r \cos \theta$ and $v = r \sin \theta$

Hence, $u^2 + v^2 = r^2$ and $\tan \theta = \frac{v}{u}$

We can then put $\lambda_1 = r (\cos \theta + i \sin \theta)$ and $\lambda_2 = r (\cos \theta - i \sin \theta)$

and, $Y = r [(\cos \theta + i \sin \theta)^t + Y_2 (r (\cos \theta - i \sin \theta))^t]$
 $= A_1 r^t (\cos \theta + i \sin \theta)^t + A_2 r^t (\cos \theta - i \sin \theta)^t$

Let us apply DeMoivre's theorem* which states that

$$(\cos \theta \pm i \sin \theta)^t = (\cos t\theta \pm i \sin t\theta)$$

$$Y_t = A_1 r^t (\cos t\theta + i \sin t\theta) + A_2 r^t (\cos t\theta - i \sin t\theta)$$

$$= r^t (A_1 + A_2) \cos t\theta + r^t (A_1 - A_2) i \sin t\theta$$

(collecting terms of $\cos t\theta$ and $\sin t\theta$)

$\therefore Y_t = r^t [k_1 \cos t\theta + k_2 \sin t\theta],$

Putting $A_1 + A_2 = k_1$ and $i (A_1 - A_2) = k_2$

Put $k_1 = k \cos \epsilon$ and $k_2 = k \sin \epsilon$,

where k and ϵ are another set of constants.

* DeMoivre's theorem can be derived as follows :

$$(\cos \theta + i \sin \theta)^2 = \cos^2 \theta - r^2 \sin^2 \theta + 2i \cos \theta \sin \theta$$

$$= \cos^2 \theta - \sin^2 \theta + i 2 \sin \theta \cos \theta$$

$$= \cos 2\theta + i \sin 2\theta$$

[Because, $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, and $2 \sin \theta \cos \theta = \sin 2\theta$]

Hence, $(\cos \theta + i \sin \theta)^3 = (\cos 2\theta + i \sin 2\theta) (\cos \theta + i \sin \theta)$

$$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta +$$

$$i [\sin 2\theta \cos \theta + \cos 2\theta \sin \theta]$$

$$= \cos (2\theta + \theta) + i \sin (2\theta + \theta)$$

$$= \cos 3\theta + i \sin 3\theta$$

(Because, $\cos (A + B) = \cos A \cos B - \sin A \sin B$, and
 $\sin (A + B) = \sin A \cos B + \cos A \sin B$).

$$\begin{aligned}\text{Therefore, } Y_t &= r^t [k \cos \varepsilon \cos t \theta + k \sin \varepsilon \sin t \theta] \\ &= r^t k \cos t (\theta - \varepsilon)\end{aligned}$$

Therefore, the time path of Y_t when roots are complex is given by

$$Y_t = \bar{Y} + r^t k \cos (\theta t - \varepsilon)$$

where,

$$\begin{aligned}r &= \sqrt{u^2 + v^2} \\ &= \sqrt{\frac{a^2}{4} + \frac{4b-a^2}{4}} = \sqrt{b}\end{aligned}$$

determines the amplitude of the cycle. The period of the cycle is determined by $\frac{2\pi}{\theta}$, and phase of the cycle by ε , k and ε are constants depending upon the initial conditions

If initial values y_0 and y_1 are given the values of k (*initial amplitude*) and ε (*phase*) are determined by the values of r and θ , or a and b

Cycle is generated because of the trigonometric form of the function. Amplitude depends upon b alone but the period depends upon θ , i.e., a and b

17.2 Solution of Third-order Difference Equation

To solve third-order difference equation it is necessary to solve a cubic equation and then proceed as above. But the solution of the cubic is likely to become complex and difficult to handle

Ex 4 For what values of a and b does the auto-regressive linear set U_t oscillate, when $U_t = aU_{t-1} + b(U_{t-1} - U_{t-2}) + V_t$

The auxiliary equation becomes $\lambda^2 - a\lambda - b(\lambda - 1) = 0$
or, $\lambda^2 - \lambda(a+b) + b = 0$

$$\text{The roots are } \lambda_1, \lambda_2 = \frac{(a+b)}{2} \pm \frac{1}{2} \sqrt{(a+b)^2 - 4b}$$

There will be oscillations if the roots are conjugate complex roots i.e., $(a+b)^2 - 4b$ is less than zero

$$\text{i.e., if } a^2 + b^2 + 2ab - 4b < 0$$

For values of a and b satisfying this inequality, the value of U_t will oscillate. Now

$$\begin{aligned}(a+b)^2 - 4b &\leq 0 \\ (a+b)^2 &< 4b\end{aligned}$$

$$a + b < \pm 2\sqrt{b}$$

$$a < -b \pm 2\sqrt{b}$$

Hence value of a should be between $-b - 2\sqrt{b}$ and $-b + 2\sqrt{b}$

If $b = 4$, then $a < -4 \pm 4$, i.e., $a < 0$ or < -8

a should lie between -8 and 0 (when $b = 4$)

Ex 5. Solve the difference equation $y_{n+2} = y_{n+1} - y_n$

Given $y_{n+2} - y_{n+1} + y_n = 0$

The auxiliary equation becomes $\lambda^2 - \lambda + 1 = 0$

$$\lambda_1 = \frac{1}{2} [1 + \sqrt{-3}] \text{ and}$$

$$\lambda_2 = \frac{1}{2} [1 - \sqrt{-3}]$$

the roots are conjugate complex

Let, $u = \frac{1}{2}$ and $v = \frac{\sqrt{3}}{2}$,

then, $\lambda_1 = u + iv$ and $\lambda_2 = u - iv$

If, $u = r \cos \theta$ and $v = r \sin \theta$

the solution is $Y_n = r^n [k_1 \cos n\theta + k_2 \sin n\theta]$

We have here, $r = \sqrt{u^2 + v^2}$

$r = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$ Therefore, the amplitude of the cycle is constant

$$\text{Now, } \tan \theta = \frac{v}{u} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

Therefore, $\theta = \frac{\pi}{3}$ (Because $\tan \frac{\pi}{3} = \sqrt{3}$)

The solution of the given equation is, therefore, given by

$$y = k_1 \cos \frac{\pi n}{3} + k_2 \sin \frac{\pi n}{3}, \text{ with Amplitude} = 1, \text{ and}$$

$$\text{period} = \frac{2\pi}{\pi/3} = \frac{6\pi}{\pi} = 6$$

Ex 6 Solve the difference equation $y_{n+2} - \sqrt{3} y_{n+1} + y_n = 0$

given that $y_0 = y_1 = 1$

The auxiliary equation becomes $\lambda^2 - \sqrt{3}\lambda + 1 = 0$

$$\text{Therefore, } \lambda_1, \lambda_2 = \frac{1}{2} [\sqrt{3} \pm \sqrt{3-4}]$$

$$= \frac{\sqrt{3}}{2} \pm i \left(\frac{1}{2} \right)$$

$$\text{giving, } u = \frac{\sqrt{3}}{2}, v = \frac{1}{2}$$

The solution being $y_n = r^n [k_1 \cos n\theta + k_2 \sin n\theta]$

$$\text{where, } r = \sqrt{u^2 + v^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$$

$$\tan \theta = \frac{v}{u} = \frac{\frac{1}{2}}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$$

$$\text{Therefore, } \theta = \frac{\pi}{6} \quad \left(\tan \frac{\pi}{6} = \tan 30^\circ = \frac{1}{\sqrt{3}} \right)$$

$$\text{The solution is } y_n = k_1 \cos \frac{\pi n}{6} + k_2 \sin \frac{\pi n}{6} \quad (1)$$

Now from the given condition ($y_0 = y_1 = 1$), we have

$$y_0 = 1 = k_1 \cos 0 + k_2 \sin 0$$

$$1 = k_1 + 0, k_1 = 1$$

$$\text{Again, } y_1 = 1 = k_1 \cos \frac{\pi}{6} + k_2 \sin \frac{\pi}{6}$$

$$1 = k_1 \left(\frac{\sqrt{3}}{2} \right) + k_2 \left(\frac{1}{2} \right)$$

$$\text{Therefore, } k_2 = 2 - \sqrt{3} \quad [k_1 = 1]$$

$$\text{The complete solution is given by, } y_n = \cos \frac{\pi n}{6} + (2 - \sqrt{3}) \sin \frac{\pi n}{6}$$

EXERCISE 17.1

Find the general solution of each of the following difference equations

1. $y_{t+2} - 25y_t = 0$

2. $y_{t+2} - 6y_{t+1} + 9y_t = 0$

3. $y_{t+2} - 5y_{t+1} - 66y_t = 0$

4. $y_{t+3} - 7y_{t+2} + 11y_{t+1} - 6y_t = 0$

5. $y_t - \frac{1}{4}y_{t-2} = 0$

Find the solution of the following equations satisfying the given conditions

6. $y_t - \frac{1}{4}y_{t-2} = 0$, when $y_0 = 1$, $y_1 = 0$

7. $y_{n+2} - 4y_{n+1} + 3y_n = 0$, when $y_0 = 1$ and $y_1 = 1$

8. $y_{x+2} + y_{x+1} + 3y_x = 0$ when $y_0 = 2$, $y_1 = 0$

Find the complete solutions of the following difference equations

9. $y_{x+2} - 4y_{x+1} + 3y_x = 5^x$ [Try $y_x = C \cdot 5^x$]

10. $y_{x+1} - 5y_{x+1} - 6y_x = 6^x$ [Try $y_x = C \cdot 6^x$]

11. $y_{x+2} - 4y_{x+1} + 4y_x = 2^x$ [Equal roots try $y_x = ux^2 2^x$]

12. $y_{n+2} - 5y_{n+1} + 6y_n = n^2$ [Try $y_n = A_0 + A_1n + A_2n^2$]

13. $y_{n+1} - 4y_{n+1} - 5y_n = 0$

14. $y_{t+2} - 3y_{t+1} + 10y_t = 0$, when $y_0 = 1$, $y_1 = 2$

17.3. Applications in Economics

Second order difference equations are used very widely in economics, specially in macro-economics. These are used in extended Cobweb and Harrod-Domar models. But the most popular use is in the theory of economic cycles.

The Multiplier-Accelerator Interaction Model of Samuelson

Samuelson's economic cycle model is based on the interaction of multiplier and accelerator. He considers a one period lag in consumption function and investment as a function of change in consumption during current period, that is :

$$C_t = cY_{t-1} \quad (1)$$

$$I_t = \beta (C_t - C_{t-1}) \quad (2)$$

$$\text{and } Y_t = I_t + C_t + A_t \quad (3)$$

with the usual symbols, c and β are propensity to consume and accelerator respectively

Putting values of C_t and I_t from Eqns (1) and (2) in Eqn (3) we have

$$\begin{aligned} Y_t &= cY_{t-1} + \beta (C_t - C_{t-1}) + A_t \\ &= cY_{t-1} + c\beta (Y_{t-1} - Y_{t-2}) + A_t \\ Y_t &= Y_{t-1} (c + c\beta) - c\beta Y_{t-2} + A_t \end{aligned} \quad (4)$$

A_t is supposed to be constant ($=A$)

Eqn (4) is a second order difference equation

(a) Particular solution is obtained by putting $Y_t = \bar{Y}$ which gives

$$\begin{aligned} \bar{Y} - c(1 + \beta)\bar{Y} + c\beta\bar{Y} &= A_t \\ \bar{Y} &= \frac{A}{1 - c - c\beta + c\beta} = \frac{A}{1 - c} \end{aligned}$$

which is the solution obtained when only multiplier is operating

(b) Homogeneous solution is $A_1\lambda_1^t + A_2\lambda_2^t$

where λ_1 and λ_2 are obtained from the auxiliary equation

$$\begin{aligned} \lambda^2 - c(1 + \beta)\lambda + c\beta &= 0 \\ \lambda_1, \lambda_2 &= \frac{c(1 + \beta) \pm \sqrt{c^2(1 + \beta)^2 - 4c\beta}}{2} \end{aligned}$$

The roots will be real if $c^2(1 + \beta)^2 - 4c\beta > 0$ (It should be noted that since the coefficient of Y_{t-1} in the difference equation is negative the term outside the root sign will always be positive)

Also, $c(1 + \beta) > \sqrt{c^2(1 + \beta)^2 - 4c\beta}$ Hence, in this case, the roots λ_1, λ_2 will always be greater than zero and the time path will be non-cyclic.

$$C \geq \frac{4\beta}{(1 + \beta)^2}$$

Both the roots are less than one

Hence the complete solution is

$$Y_t = \frac{A}{1-0.9} + A_1 (0.75)^t + A_2 (0.60)^t$$

as $t \rightarrow \infty$ both $(0.75)^t$ and $(0.60)^t$ will tend to zero. Hence the time-path will converge to $\frac{A}{1-c}$ as given by the multiplier alone (Region A in Fig. 17.1)

$$(2) \quad \text{Real roots and } \beta = 1, \text{ therefore } C \geq \frac{4\beta^2}{(1+\beta)} \geq 1$$

Since ordinarily $c \leq 1$, hence for real roots with $\beta = 1$, the only permissible value of c is 1

Then $\lambda = \pm 1$ (the auxiliary becomes $\lambda^2 - 2\lambda + 1 = 0$)

Solution when the roots are equal is as follows

The homogeneous solution is given by

$$Y_t = A_1 \lambda^t + A_2 t \lambda^t \quad (\lambda_1 = \lambda_2)$$

(Note that for each repeated root the constant A will be multiplied by t)

$$Y_t = a_1 + A_2 t$$

For particular solution

$$\text{try } Y_t = \bar{A} t^2$$

(because the root 1 is repeated second time)

$$Y_t - c(1+\beta)Y_{t-1} + c\beta Y_{t-2} = A$$

$$\bar{A} t^2 - \bar{A}(1+1)(t-1)^2 + \bar{A}(t-2)^2 = A$$

$$\bar{A}(t^2 - 2t^2 - 2 + 4t - t^2 - 4t + 4) = A$$

$$\text{or, } 2\bar{A} = A \quad \text{or, } \bar{A} = \frac{A}{2}$$

$$\text{Particular solution is } \frac{t^2 A}{2} \quad \text{and}$$

the complete solution is $Y_t = \frac{\bar{A}}{2} t^2 + A_1 + A_2 t$

Obviously, $Y_t \rightarrow \infty$ as $t \rightarrow \infty$

Hence, the path is explosive and non-cyclic.

(3) If $\beta > 1$ and *real roots*

Let $\beta = 2$, then for real roots

$$c \geq \frac{4 \times 2}{(1+2)^2} \geq \frac{8}{9}$$

If $c = \frac{8}{9}$ then the roots will be equal ($\lambda_2 = \lambda_1 = \lambda$) and the homogeneous solution will be $P_t = A_1 \lambda^t + A_2 t \lambda^t$, the roots will be $\frac{c(1+\beta)}{2} = \frac{8}{9} \cdot \frac{(1+2)}{2} = \frac{4}{3}$ which is greater than 1. Hence the time path will be explosive non-cyclic.

In fact, it can be shown that when real roots are equal and $\beta > 1$, then $\frac{c(1+\beta)}{2} > 1$ and both the roots will be inevitably greater than 1, hence they will give explosive time-path

$$\begin{aligned} \text{Since, } c &= \frac{4\beta}{(1+\beta)^2} \\ \frac{c(1+\beta)}{2} &= \frac{2\beta}{1+\beta} \end{aligned}$$

$\frac{2\beta}{1+\beta}$ will be greater than 1 when $\frac{2\beta}{1+\beta}$ is greater than 1 or

$\frac{2\beta}{1+\beta} - 1 = \frac{\beta-1}{1+\beta}$ is greater than zero. For every value of $\beta > 1$, $\frac{\beta-1}{1+\beta}$ will be positive and greater than zero.

Hence when $\beta > 1$, equal real roots will give an explosive time-path. When $\beta < 1$, equal roots will lead to convergent time path.

If the roots are not equal, but still if $\beta > 1$, then the result $\frac{c(1+\beta)}{2} > 1$ will still lead to explosive time-path.

$$\begin{aligned} \text{The solution is } \lambda_1 &= \frac{c(1+\beta)}{2} + \frac{\sqrt{c^2(1+\beta)^2 - 4c\beta}}{2}, \text{ and} \\ \lambda_2 &= \frac{c(1+\beta)}{2} - \frac{\sqrt{c^2(1+\beta)^2 - 4c\beta}}{2} \end{aligned}$$

Since $\frac{c(1+\beta)}{2} > 1$, if anything is added to it, it will be still greater hence one of the roots (termed as the dominant root) λ_1 will be greater than 1. This will give explosive time-path (Region D in Fig. 17.1).

(4) *Complex roots* For roots to be complex conjugate $c < \frac{4\beta}{(1+\beta)^2}$

Let $u = \frac{c(1+\beta)}{2} = r \cos \theta$

$$v = \frac{\sqrt{4c\beta - c^2(1+\beta)^2}}{2} = r \sin \theta$$

$$r^2 = u^2 + v^2 = \frac{c^2(1+\beta)^2}{4} + \frac{4c\beta - c^2(1+\beta)^2}{4} = c\beta$$

or $r = \sqrt{c\beta}$

and $\tan \theta = \frac{v}{u} = \frac{\sqrt{4c\beta - c^2(1+\beta)^2}}{c(1+\beta)}$

$$= \sqrt{\frac{4\beta}{(1+\beta)^2} - 1}$$

and the time-path is $Y_t = r^2 (k_1 \cos t\theta + k_2 \sin t\theta)$

or, $Y_t = r' k \cos (\theta t - \epsilon)$

that is, $Y_t = (c\beta)^{1/2} (k_1 \cos t\theta + k_2 \sin t\theta)$

The time-path will be cyclic explosive if $c\beta > 1$, damped if $c\beta < 1$ and regular cyclic if $c\beta = 1$

$$\tan \theta = \sqrt{\frac{4\beta}{(1+\beta)^2} - 1}$$

will give the period of the cycle (note that the period is independent of the value of marginal propensity to consume c)

In Fig 17.1, within the parabola giving the cyclic time-path the rectangular hyperbola $c\beta = 1$ demarcates the regions of damped and explosive cycles, regions B and C respectively

Ex 7. If $\beta = 4$, and $c = 0.8$, find the time-path of Y in Samuelson's trade cycle model

The second order difference equation in this case will be

$$Y_t - 0.8(1+4)Y_{t-1} + 0.8 \times 4 Y_{t-2} = A$$

Now, since, $\frac{4\beta}{(1+\beta)^2} = \frac{16}{25}$ and, $c = 0.8$

$$c > \frac{4\beta}{(1+\beta)^2}$$

hence, the roots are real

$$\begin{aligned}\lambda_1, \lambda_2 &= \frac{0.8 \times (5) \pm \sqrt{(0.8 \times 5)^2 - 4 \times 0.8 \times 4}}{2} \\ &= \frac{4 \pm \sqrt{16 - 12.8}}{2} = \frac{4 \pm 1.8}{2} = 2.9 \text{ or } 1.1\end{aligned}$$

Dominant root, $\lambda_1 = 2.9 > 1$, hence the time path is explosive non cyclic.

$$Y_t = \frac{A}{1-0.8} + A_1 (2.9)^t + A_2 (1.1)^t = 5A + A_1 2.9^t + A_2 1.1^t$$

Ex 8 If $\beta = 4$ and $c = 0.5$ in the above illustration, what will be the time path of Y ?

$$\text{Here } \lambda_1, \lambda_2 = \frac{0.5 \times 5 \pm \sqrt{(0.5 \times 5)^2 - 4 \times 0.5 \times 4}}{2} = \frac{2.5 \pm \sqrt{6.25 - 8}}{2}$$

Hence the roots are complex conjugate, they are

$$\lambda_1, \lambda_2 = \frac{2.5}{2} \pm \sqrt{-1} \left(\frac{\sqrt{8-6.25}}{2} \right) = 1.25 \pm i(0.616)$$

Now let, $u = 1.25 = r \cos \theta$ and $v = 0.616 = r \sin \theta$

$$\begin{aligned}\text{So that, } r^2 &= u^2 + v^2 \\ &= (1.25)^2 + (0.616)^2 = 1.5625 + 0.3795 = 1.9420\end{aligned}$$

That is, $r = \sqrt{1.9420} > 1$

Hence the time path is explosive and cyclic. It is given by

$$Y_t = (1.9420)^{t/2} [A_1 \cos \theta t + A_2 \sin \theta t]$$

$$\text{Here, } \tan \theta = \frac{v}{u} = \frac{0.616}{1.25} = 0.4928, \text{ i.e., } \theta = \tan^{-1} 0.4928$$

EXERCISE 17.2

1. If consumption is a constant proportion of last period's income and investment is function of change in the income lagged one period, then find the time path of income (*Hicks model*)

$$\text{Hint} \quad C_t = cY_{t-1}$$

$$I_t = v(Y_{t-1} - Y_{t-2})$$

$$Y_t = C_t + I_t$$

2. If in the above example average propensity to consume is not equal to marginal propensity to consume, what will be the time path ?

Hint. $C_t = C_0 + cY_{t-1}$

3 If consumption depends upon past two years' income and investment depends upon

(a) last year's change in income, or

(b) last year's level as well as change in income, find the time path of income

Hint (a) $C_t = c_0 + c_1 Y_{t-1} + c_2 Y_{t-2}$

4 If $\frac{C_t - cY_{t-1}}{I_t - \beta(C_t - C_{t-1})}$ and Autonomous Investment = 100

Find the time path of Y , given that

(a) $c = 0.4, \beta = 1$ (b) $c = 0.9, \beta = 4$

(c) $c = 0.6, \beta = 3$ (d) $c = 0.75, \beta = 3$

5 In the modified Cobweb model (Goodwin model) supply depends upon the expected price. If it is assumed that price is expected to move in opposite direction to that of the previous period, find the time path of price.

Hint. Demand function $X_t = \alpha + P_t$

Supply function $X_t = \beta + bP_t$

where P_{et} = expected price.

Let $P_{et} = P_{t-1} (P_{t-1} - P_{t-2})$ and 0 (opposite direction expectations)

Form the difference equation by equating supply and demand. Take the usual case $a < 0, b > 0$

6 Find the time path of P in Exercise 5 if $b = 2, a = -3$ and $p = 0.5$

Hint Auxiliary equation in this case will be $\lambda^2 - \frac{1}{3}\lambda + \frac{1}{3} = 0$. Now proceed.

7. In a market price determination model price is determined by stock changes in the previous period, find the time path of price

Hint. $D_t = \alpha + aP_t$

$S_t = \alpha + aP_t$, and

$P_t = P_{t-1} - \lambda \Delta Q_{t-1}$ where $\Delta Q_t = S_t - D_t$

8 If in the Harrod Domar model, there is disturbance in the equilibrium path such that investment plans are not realised, find the time path of y given that income increases in period t by just enough to make good the deficiency in the preceding period

Hint. Deficiency in period t is given by

$$U_t = \text{Planned investment} - \text{Realised investment}$$

Therefore, $U_t = v(y_t + y_{t-1}) - sy_{t-1}$, so that we have

$$y_t = y_{t-1} + U_{t-1}$$

Now show that the model can generate oscillatory motion

9 In the above exercise, find the time path if

(a) $v = 2$ and $m.p.c. = 0.8$ and

(b) $v = 2$ and $m.p.c. = 0.2$

10 In Exercise 8 in the situation of disturbance if entrepreneurs behave in a way such that output increases in period t by the warranted rate of growth with an adjustment for the preceding period deficiency, then what will be the time path? Show that there cannot be oscillatory motion in such a situation. What is the nature of the path?

$$\text{Hint. } Y_t = \left(1 + \frac{s}{v}\right) Y_{t-1} + U_t$$

11. Consider the model

$$C = \alpha + \beta Y_{t-1}$$

$$I = r + \delta (Y_{t-1} + Y_{t-2})$$

$$Y = C + I$$

where C , Y and I denote consumption, income and investment. $\alpha, \beta, \gamma, \delta$ are fixed constants. Discuss the equilibrium path of Y over time. Do you consider the model to be realistic enough to describe the observed business cycles?

12. If in question 11 autonomous investment (A_t) grows at 100 g per cent, find the time-path of income.

Hint. Time-path of A_t is $I_0 r^t$ (where $r = 1+g$)

$$\text{Then } Y_t - (c + v) Y_{t-1} + v Y_{t-2} = I_0 r^t$$

For particular solution take

$$Y_t = A_0 r^t$$

Then
$$A_0 = \frac{I_0 r^2}{r - (c+v)r + v}$$

13. In the above example, if $s = 0.2$, and $U = 2$ or 1 or 0.02 , find the time-path of income

Hint. Particular solution in this situation shall be

$$\frac{I_0 (1.02)^2}{(1.02)^2 - (0.8 + 2)(1.02) + 2} \quad \text{The roots will be}$$

$$\lambda_1, \lambda_2 = \frac{(0.8+2) \pm \sqrt{(2.8)^2 - 8}}{2}, \quad \text{which show that there will be}$$

explosive cyclic growth

Chapter 18

Linear Programming

18.1. Introduction

Programming is a technique which is applied for finding maximum or minimum values in problems confronting the decision-making authorities, subject to certain constraints or side-conditions which limit decisions. In fact, such problems which require determination of maxima or minima could also be solved with the help of Calculus. But it is found that in the presence of certain side-conditions which are not exact but limit the requirements, the application of Calculus breaks down. Programming, therefore, helps to find out maximum or minimum values when side-conditions are inequalities and not equations.

In case the expression to be maximised or minimised and the inequalities involving side-conditions are of 1st degree (i.e., linear), the method of solving the problem is known as linear programming.

With given cost-function and price we can locate the point of maximum profit by determining a point at which marginal profitability is zero. For profit (net revenue) maximisation the conditions are

$$\frac{\partial \pi}{\partial Q} = 0 \text{ (1st-order condition)}$$

$$\text{and } \frac{\partial^2 \pi}{\partial Q^2} < 0 \text{ (2nd-order condition)}$$

Supposing that such point occurs at R in Fig. 18.1, at which output must be OQ_M . But this level of output may not be attained by the firm in question, due to certain constraints at different stages of production. Suppose that the firm cannot increase its output beyond OQ_C . In the presence of the constraints, OQ_M level of output in the context of maximising the total

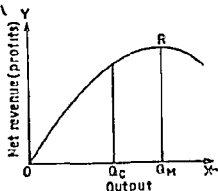


Fig. 18.1

profit becomes irrelevant because the firm cannot attain it even though it might like to do so. As such our problem is to find out whether the point of maximum attainable profit is really OQ_C , the level to which the output could be increased, or is less than this output OQ_C and is represented by some other point to its left.

Also all the points to the left of R will not satisfy the maximisation conditions, and the application of Order conditions of Calculus break down

Problems of such nature as are concerned with the efficient use of limited resources which serve as side-conditions as to achieve the desired objective (of minimisation or maximisation) can be solved with the help of Programming Technique

The term 'Linear' implies that the relationships involved in the problems must be linear, i.e., of 1st degree (sum of constant multiples of variables) while the term 'Programming' refers to determination of particular programme which gives the line of action which would achieve the desired objective.

Example Let us assume that a firm produces two different products I and II. Each product has to undergo three operations before it takes the final shape. As usual, the main object of this firm is to find how much of products I and II be produced with the maximum capacity available for each operation so as to maximise the total net revenue. The revenue per unit and the time required per unit of each product in each operation is tabulated below

TABLE 18.1

Deptt	Time taken by each unit of products I and II in hours		Max. capacity available in each operation
	I	II	
Cutting	1	4	24 hours
Mixing	3	1	21 hours
Packaging	1	1	9 hours
Net revenue per Unit of product	Rs 2	Rs 5	—

This shows that maximum capacity available in cutting, mixing and packaging departments of the firm are 24 hrs, 21 hours and 9 hours respectively, and these capacities are assumed to remain constant, during a given time period. Each unit of product I takes 1 hour in cutting, 3 hours in mixing and 1 hour in packaging operations before it turns out in a final shape. Against this, one unit of product II takes 4 hours, 1 hour and 1 hour in each of the above operations respectively. Also the net revenue contribution per unit of product I is Rs 2.00 and that of product II is Rs 5.00.

Basic Assumptions. The application of LP (Linear Programming) rests on several basic assumptions and requirements

1. *A well defined objective function.* The decision maker wants to achieve certain goal, this may be maximisation of net income, profit, sales or

minimisation of cost. The objective function should be expressed as a linear function of decision variables. In the example given above, the objective is to maximise net revenue by choosing appropriate values of the decision variables — output of the products I and II.

2. Presence of constraints in the activities The L.P. technique can be applied to the problems which involve certain kinds of constraints in their activities. Such constraints usually are because of technological limitations or capacity and resource limitations. They exist primarily because resources are limited.

In the above example, the total capacity of the cutting department does not exceed 24 hours; that of mixing and packaging departments 21 hours and 9 hours. Therefore, number of units of each product which could be produced would depend on (a) the capacity available in each department and (b) the time consumed by each unit of product in each operation.

Let us assume that the firm can maximise the total net revenue by producing X and Y amounts of product I and II respectively. The above table can now be transformed in the form of inequalities describing constraints—generally known as structural constraints.

Since one unit of product I requires one hour of cutting facility, the total time required for X units in this department will be $1 \times (X) = X$ hr. In the same way Y units will require $4Y$ hours of cutting facility. Hence the production of $(X + Y)$ units of two products will require $(X + 4Y)$ hours. The total capacity in the cutting department at the maximum is 24 hours, therefore the production of X and Y amounts of two products should be such that the demand on cutting facility does not exceed 24 hours. In mathematical symbols, since $X + 4Y$ cannot be greater than 24 hours, therefore the constraints in the cutting department would be described in terms of inequality as $X + 4Y \leq 24$. Exactly with the same reasoning we can form the constraints in terms of inequalities for mixing and packaging departments.

Thus we have three structural constraints in our problem:

- (i) $X + 4Y \leq 24$ — for cutting department
- (ii) $3X + Y \leq 21$ — for mixing department
- (iii) $X + Y \leq 9$ — for packaging department

3. Non negativity of decision variables Along with these structural constraints we have "Non negativity" constraints which assume that negative value of the variables is not possible in the solution of L.P. problems.

In our problem, this would mean that output X and Y can at most be zero but can never be below zero; that is, non-negative constraints will be $X \geq 0$ and $Y \geq 0$.

4 Linear relations in the problem LP problems are always stated in linear relations. In fact, functional relationship among variables may not be actually linear in many cases, but such problems cannot be solved with the help of this technique.

In our example, the revenue per unit of product is Rs. 2 and Rs. 5, hence total revenue is $2X + 5Y$ which is a linear relation of X and Y . This function is to be maximised under the above three constraints which also involve linear relations. (The function to be maximised or minimised is termed as the Objective function.)

5 Constant prices In order that we deal with linear relations only, prices of inputs and outputs are to be assumed given and constant during a given time period. Hence prices are not considered as variables throughout the programming exercise.

6 Divisibility Activity levels are permitted to take fractional or integer values.

Besides the above six assumptions, there are some implicit assumptions or requirements of this technique. These are merely mathematical requirements. The number of activities and constraints should be finite, otherwise, solution will not exist. Simultaneously, it is also assumed that each coefficient used in the relations is known with certainty. This is often not a realistic assumption.

The Problem Restated

As seen above, every LP problem in its standard form involves three parts:

- (1) *Objective function* which is required to be maximised or minimised,
- (2) *Structural constraints* which state the side-conditions on the different activities of the problem and
- (3) *Non negativity constraints* which assume that there can be no negative values of variables involved in the problem.

Thus we state our above problem (given in the form of table) in standard form as follows:

$$1 \quad \text{Objective function} \quad \text{Max } f = 2X + 5Y$$

(which states that if products I and II bring revenue of Rs. 2 and Rs. 5 per unit, what amount of these products be produced so that the firm earns maximum revenue.)

$$2 \quad \text{Structural constraints} \quad X + 4Y \leq 24$$

$$3X + Y \leq 21$$

$$X + Y \leq 9$$

(which state that capacity in the three departments is restricted to a given extent.)

3. *Non-negativity constraints* · $X \geq 0$ and $Y \geq 0$

(which state that the production of the products in question can at most be zero but can never be negative.)

Summarising, therefore, we may say L.P. technique is applicable to problems that exist in the situation which

(i) can be quantified, (ii) permit alternative courses of action, (iii) involve some restrictions on feasible choices, and (iv) require optimisation in respect of certain appropriate criterion

The linear objective function implies that the contribution per unit of each product is constant. This is a very restrictive assumption implying pure competition and constant returns. If these assumptions do not hold linear programming shall not be applicable

Linearity of the constraints implies that (i) each resource is homogeneous, (ii) the relation of the resource to output is characterised by constant returns. Hence constant marginal rates of substitution are assumed

Linearity assumption, therefore, can be used as first approximation to reality (as found in empirical structures). While interpreting the results this should be kept in mind.

18.2. The General Linear Programming Problem (L.P.P.)

We shall now generalize the L.P. problem on the basis of the problem explained above

Let us consider a profit maximising firm which possess fixed quantities of capacity (or inputs) in m -operations. The firm produces $x_1, x_2, x_3, \dots, x_n$ commodities which bring $p_1, p_2, p_3, \dots, p_n$ of revenue per unit of commodity $X_1, X_2, \dots, X_n$ respectively. The capacity constraint in each of the m -operations is given as under :

Assume that the firm can maximise the total net revenue by producing $x_1, x_2, x_3, \dots, x_n$ respectively.

Since the price per unit of X_1 product is p_1 , revenue earned by the firm by selling x_1 amount of this product = $x_1 p_1$.

Similarly, the revenue earned by the firm by selling $x_2, x_3, x_4, \dots, x_n$ amounts of products will be $x_2 p_2, x_3 p_3, \dots, x_n p_n$.

That is, the total net revenue of the firm which needs to be maximised will be

$$R = x_1 p_1 + x_2 p_2 + x_3 p_3 + \dots + x_n p_n$$

Table 18.2

Operations	Time taken by products					Max. capacity available in the operations	
	X_1	X_2	X_3	X_4	X_n		
1st	a_{11}	a_{12}	a_{13}	a_{14}	a_{1n}	b_1	units
2nd	a_{21}	a_{22}	a_{23}	a_{24}	a_{2n}	b_2	units
3rd	a_{31}	a_{32}	a_{33}	a_{34}	a_{3n}	b_3	units
4th	a_{41}	a_{42}	a_{43}	a_{44}	a_{4n}	b_4	units
m th	a_{m1}	a_{m2}	a_{m3}	a_{m4}	a_{mn}	b_m	units
Revenue per unit of product	p_1	p_2	p_3	p_4	p_n		

Again since each unit of product X_1 consumes a_{11} unit of 1st operation, total capacity of operation I consumed by x_1 units = $a_{11}x_1$

Similarly, " I x_2 .. = $a_{12}x_2$

and, " I x_n .. = $a_{1n}x_n$

But the total capacity available in operation I is at most b_1 units

$(a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n)$ must not be more than b_1

that is, $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n \leq b_1$

In the same way we shall have

$$\left\{ \begin{array}{l} a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n \leq b_2 \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n \leq b_m \end{array} \right\}$$

Let us now put the problem in its standard form

(1) *Objective function* Maximise $f = p_1x_1 + p_2x_2 + \dots + p_nx_n$

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$$

(2) *Structural constraints*

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m$$

(3) *Non negativity constraints* $x_j \geq 0$ ($j = 1, 2, 3, \dots, n$)

We can also write down this general L.P. problem in the abbreviated form with the help of summation sign and matrix algebra

		<i>In matrix notation</i>
Objective function	Maximise $f = \sum_{j=1}^n p_j x_j$	Maximise $P'X$
Subject to	$\sum_{j=1}^n a_{ij} x_j \leq b_i, (i = 1, 2, \dots, m)$	Subject to $AX \leq B$
and,	$x_j \geq 0 (j = 1, 2, \dots, n)$	and, $X \geq 0$

18.3. Transformation of Linear Inequalities Into Linear Equations—Slack Variables

In the application of L.P. technique the constraints are first of all changed in terms of related system of linear equalities. Let us, therefore, change each of the equalities of structural constraints in terms of equations. This is done by introducing 'Slack' variables in the inequalities.

For example, consider the first inequality in the system which is

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n \leq b_1$$

This inequality states that the sum of the terms on the left of the inequality sign is less than or equal to b_1 . Stated in another way this equation also means that there is some number say $S_1 \geq 0$ which when added to the terms on the left will convert it into an equality. This additional number is known as Slack variable which in these examples cannot take value less than zero.

Similar operation of addition of different slack variables will convert the inequalities into equalities. Since the number of inequalities in the system is equal to m (there are m different operations of production), m number of slack variables are introduced in the whole system, and the resulting linear equation system would be

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n + S_1 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n + S_2 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n + S_3 = b_3$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n + S_m = b_m$$

Now each equation of the above system contains $(n + 1)$ variables (Slack variable) = $(n+1)$ variables, so that the structural constraints can now be represented as a system of m non-homogeneous linear equations in $(n+m)$ unknowns.

And now the L.P. problem involving equalities can be rewritten as follows

$$\text{Maximise } f = p_1x_1 + p_2x_2 + \dots + p_nx_n + 0S_1 + 0S_2 + \dots + 0S_m$$

$$\text{Subject to (a) } a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n + S_1 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n + S_2 = b_2$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n + S_m = b_m$$

$$(b) \ x_j \geq 0 \ (j = 1, \dots, n)$$

In matrix notation

$$\text{Maximise } P.X \text{ subject to (a) } AX = B$$

$$(b) \ X \geq 0$$

18.4 Geometry of Linear Programming Problem (Graphical Solution)

Consider the problem stated above

$$\text{Maximise } f = 2X + 5Y$$

$$\text{Subject to (a) } X + 4Y \leq 24$$

$$3X + Y \leq 21$$

$$X + Y \leq 9$$

$$(b) \ X \text{ and } Y \geq 0$$

We want to represent each inequality graphically

Consider the inequality $X + 4Y \leq 24$. Imagine that had only 1st commodity been produced the whole cutting capacity would have been allowed for it (since now the whole capacity would be utilised for producing commodity 1 only)

$$X + (0)Y = 24 \text{ (at the maximum)}$$

$$\text{or } X = 24 \text{ (when } Y = 0)$$

Again if had only Y amount of commodity been produced the whole cutting capacity of the firm would have been allowed for it

$$(0)X + 4Y = 24$$

$$Y = 6 \text{ (when } X = 0)$$

Plotting these points in Fig. 18.2, we get a straight line MN which represents the inequality $X + 4Y \leq 24$. However, since our purpose is only to graph the inequality $X + 4Y \leq 24$, the region

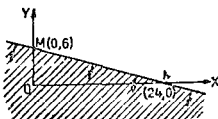


Fig. 18.2

of our interest is only the shaded area—which is below the line MN as indicated by arrows. This shaded area also includes negative production of X and Y (since the shaded area includes all the values in the four quadrants). By introducing the non-negativity constraints (i.e., $X \geq 0$ and $Y \geq 0$), our area remains restricted to quadrant I only.

Any point such as P on the line MN would satisfy the equation $X + 4Y = 24$, while any point such as L , which is below and to the left of MN , would satisfy the inequality $X + Y < 24$.

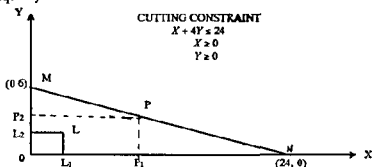


Fig 18.3

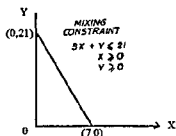


Fig 18.4

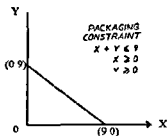


Fig 18.5

In other words, we may say that point P involves such amounts of production of 2 commodities ($X = OP_1$ and $Y = OP_2$) that the full cutting capacity gets utilised, while point L involves such amounts of production of 2 commodities ($X = OL_1$ and $Y = OL_2$) that the cutting capacity of the firm remains underutilised.

Similarly, we can plot the inequalities of mixing ($3X + Y \leq 21$) and packaging ($X + Y \leq 9$) capacities as represented by Fig. 18.4 and 18.5.

Our purpose is to find such amounts of production of two commodities which not only should maximise total revenue but also should satisfy all these constraint lines given in Figs. 18.3, 18.4, and 18.5. Let us, therefore, combine these figures into Fig. 18.6. Fig. 18.6 shows that all the points in the shaded area bounded by three constraint lines would satisfy the three inequalities. For example, point P_1 represents such amounts of two commodities which do not violate only one of the constraints. In other words, such output combination as

given by point P_1 is only within the capacity available in the mixing department of the firm. P_2 represents such amounts of X and Y which are within the available cutting and mixing capacities, but not within the packaging capacity of the firm. P_2 output combination also violates the mixing capacity of the firm.

Hence all the points outside the shaded region represent such amounts of commodities which are not possible to be produced with the existing available capacities of the firm. Therefore the shaded area given by Fig. 18.6 is termed as feasible region, any point such as Q_1 which lies within it or on its boundary Q_2 represents such a combination of output levels of the two commodities as does not violate the constraints. In other words any such output level of two commodities will be within the firm's capacities in all the three operations. For this reason every point within the region or on the boundary line of the region represents what is called a feasible solution of the L.P. problem.

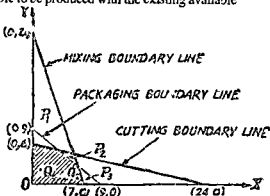


Fig 18.6

So far, we have been able to locate feasible region which can give us the feasible solution of our problem, but our purpose is also to find such a point in this region which would fetch maximum revenue to the firm, i.e., which would account for the maximum value of the function $2X + 5Y$.

Let us assume an arbitrary value, say = 10 units, which is the maximum revenue that the firm can earn. Therefore, $2X + 5Y = 10$. Let us plot this straight line AB as given in Fig. 18.7 which represents the revenue level of 10 units. Now all the lines parallel and above line AB would represent a set of iso-revenue lines of higher levels. Our objective is to pick the optimal among the feasible output combinations (i.e., that highest iso-revenue line which lies in the feasible region). Clearly, it can only be iso-line AB being tangent at point F , because any iso-revenue curve above this would violate the feasible region (of capacities) and any line below it would not fetch maximum revenue to the firm.

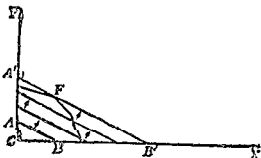


Fig 18.7

* To simplify calculations this is taken to be equal to the product of the co-efficients of X ($= 2$) and Y ($= 5$). To draw the line put $X = 0$, then $Y = 2$ and put $Y = 0$ then $X = 5$.

How to get the point I algebraically? Point F is the interaction of two constraints (lines) of cutting and packaging capacities. Therefore, solving the equations $X + Y = 9$ and $X + 4Y = 24$ we get $(X, Y) = (4, 5)$. Thus the basic feasible solution (these concepts have been explained below) to our problem is $X = 4$ and $Y = 5$. Thus output mix will not only earn maximum revenue = $2X + 5Y = 33$ units to the firm but will also be possible under the given capacity constraints in the 3 production operations.

18.5 Feasible and Basic Solutions

In order to understand these two concepts, let us consider the simplest case of single constraint in a two-variable problem

$$\begin{aligned} \text{Max} \quad & f = AX + BY \\ \text{Subject to (1)} \quad & aX + bY < k \text{ and,} \\ & (2) X > 0 \text{ and } Y > 0 \end{aligned}$$

Introducing the slack variable, the problem can be written as

$$\begin{aligned} \text{Maximise } & f = AX + BY + 0S_1 \\ \text{Subject to (1)} \quad & aX + bY + S_1 = k \\ & (2) X > 0 \text{ and } Y > 0 \end{aligned}$$

Here, A, B, a, b and k are the constants
 X and Y are structural variables (to be determined)
 and S_1 is the slack variable.

We graph the constraint equation (as explained above) $aX + bY + S_1 = k$. In Fig. 18.8 MM' represents the equation, $aX + bY = k$, that is at any point on MM' the available capacity k is fully utilised so that value of slack variable $S_1 = 0$.

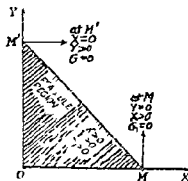


Fig. 18.8

But all the points below MM' give combinations of X and Y such that $aX + bY$ remains $< k$, that is some of M (capacity) available remains unused and, therefore, the slack variable S_1 has a value greater than 0. Also, further we go below MM' , the more unused k we will

have and hence the greater must be the value of S_1 . Consequently (1) the slack variable S_1 will have zero value on all the points lying on MM' and (2) it will have positive values on all the points in the feasible region.

We have learnt already that any point in the feasible region represents a feasible solution. Presently, we have also learnt that such point (feasible solution point) would specify values for all the variables X, Y and S — any of which can

have zero value but not less than zero value in order to satisfy the non negativity constraint

We can thus define feasible solution as a point which specifies such values to all the variables involved in the objective function of the problem which would satisfy both types of constraints structural and non negativity

While *Basic Solution* is a solution wherein the number of variables (including slack variables) getting the non zero value is equal to the number of structural constraints in the given L.P problem

Consider the above problem once again to clarify the difference between the two concepts

$$\text{Maximise} \quad f = 2X + 5Y + 0S_1 + 0S_2 + 0S_3$$

$$\text{Subject to (1) } X + 4Y + S_1 = 24$$

$$3X + Y + S_2 = 21$$

$$X + Y + S_3 = 9$$

$$\text{and, (2) } X \geq 0 \quad Y \geq 0$$

Let us assume the following possible solutions to the problem

Solution 1	$X = 4$	<i>This solution is a basic and feasible solution.</i>
	$Y = 5$	<i>It is basic (because No. of constraints in the</i>
	$S_1 = 0$	<i>problem is 3, No. of variables having value</i>
	$S_2 = 4$	<i>> 0 is 3)</i>
	$S_3 = 0$	<i>It is feasible (because all variables > 0 and</i>
		<i>substituting these values in the equalities of</i>
		<i>the constraints, these are not violated)</i>
Solution 2	$X = 6$	<i>This solution is feasible but not basic</i>
	$Y = 0$	
	$S_1 = 18$	
	$S_2 = 3$	
	$S_3 = 3$	
Solution 3	$X = 9$	<i>This solution is basic but not feasible.</i>
	$Y = 0$	
	$S_1 = 15$	
	$S_2 = -6$	
	$S_3 = 0$	

From the above it may be concluded that basic solution may or may not be a feasible solution and similarly feasible solution may or may not be a basic solution. But what we need is a solution which must be both feasible and basic. Why? Because the optimal points are always extreme or corner points of the feasible region and since it can be shown that all basic feasible solutions are

All the 10 solutions are Basic Solutions, only 5 of these solutions (i.e., No 1, 2, 6, 9 and 10) are Basic Feasible Solutions. One of these five solutions, only one, which maximises the given function shall be termed as Optimum Basic Feasible Solution.

18.6. Concept of Degeneracy

The above-mentioned theorem does not apply in case of some typical L.P. problems. Such problems are known as problems of degeneracy. Degeneracy in problems occurs when more than two constraint lines (when graphed) bounding the feasible region happen to pass from some corner point.

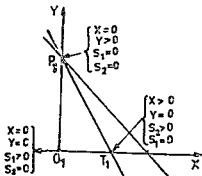


Fig 18.9

Ex.1. Assume that a problem involves two structural constraints as represented by Fig 18.9. Both the constraint lines pass through P_1 .

Feasible region is $P_1 O_1 T_1$, corner points are P_1 , T_1 and O_1 . At P_1 number of non-zero valued variables is 1 while the problem involves two structural constraints. Hence this corner point does not follow the theorem given above. Point O_1 and T_1 follow the theorem.

Point P_1 is, therefore, known as a degenerate extreme point.

Ex. 2. Assume that it involves 3 structural constraints as represented by Fig 18.10. All the constraint lines pass through point Q_2 .

Feasible region is $P_2 Q_2 T_2 O_2$ —corner points are P_2 , Q_2 , T_2 and O_2 . At Q_2 , the number of non-zero valued variables is 2 while the problem involves three structural constraints. Hence this corner point does not follow the theorem given above, while corner points Q_2 , T_2 and P_2 follow the theorem.

Point Q_2 is also the degenerate extreme point.

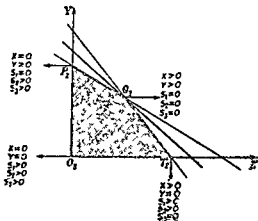


Fig 18.10

In practice, such degeneracy problems of linear programming are dealt with by introducing a tiny artificial variable into one of the structural constraints so that the constraint line does not exactly pass through the intersection of others.

Some other problems also arise in certain L.P. problems

(i) Multiple Optimal Solutions

The L.P. problem will give rise to multiple number of optimal solutions in case the slope of the objective function is same as that of one of the constraints of the problem. If the optimal point falls on this constraint, then all the points on the constraint equation are optimal basic feasible solutions. There will be no unique solution.

(ii) Case of No Feasible Solution

If the given constraints fail to define a feasible region, then no solution to the given L.P. problem will exist. This happens when the constraints are not compatible. For instance, in problem given under

$$\text{Maximise } f = 20x + 50y$$

$$\text{Subject to } 2x + 3y \geq 120$$

$$x + y \leq 40$$

$$2x + 16y \geq 90$$

$$\text{and, } x, y \geq 0$$

involve no feasible solution.

18.7 Simplex Method

We have solved one simple L.P. problem involving two variables and three constraints by graphical method. In case a problem involves more than three variables in objective function and structural constraints, then the problem would not lend itself easily to graphical solution. Such linear programming problems involving more variables and more constraints are dealt by using the Simplex Method which is the most general and powerful method of solving any linear programming problem. In actual practice L.P. problems are usually solved by the application of simplex method.

By the simplex procedure, we can obtain a maximum feasible solution in a finite number of steps starting with any basic (extreme point) feasible solution. The iterations (or different steps) consist in finding a new feasible solution whose corresponding value of the objective function is more than the value of the objective function for the preceding solution. The process is continued until an optimal solution is reached.

We shall consider the same example which we solved by graphical method in order to verify the solution

$$\begin{aligned}
 \text{Ex 1 Maximise} \quad & f = 2X + 5Y \\
 \text{Subject to} \quad & X + 4Y \leq 24 \\
 & 3X + Y \leq 21 \\
 & X + Y \leq 9 \text{ and} \\
 & X, Y \geq 0
 \end{aligned}$$

Introducing the slack variables, we obtain following equations

$$X + 4Y + S_1 = 24$$

$$3X + Y + S_2 = 21$$

$$X + Y + S_3 = 9$$

which can be written in *Vector equation form*

$$\begin{array}{ccc}
 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} X + \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} Y + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} S_1 + \\
 \downarrow \quad \quad \downarrow \quad \quad \downarrow \\
 P_1 \quad \quad P_2 \quad \quad P_3 \\
 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} S_2 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} S_3 = \begin{bmatrix} 24 \\ 21 \\ 9 \end{bmatrix} \\
 \downarrow \quad \quad \downarrow \quad \quad \downarrow \\
 P_4 \quad \quad P_5 \quad \quad P_0 \\
 P_1 X + P_2 Y + P_3 S_1 + P_4 S_2 + P_5 S_3 = P_0
 \end{array}$$

Thus the whole problem reduces to

$$\text{Max} \quad f = 2X + 5Y + 0S_1 + 0S_2 + 0S_3$$

$$\text{Subject to} \quad P_1 X + P_2 Y + P_3 S_1 + P_4 S_2 + P_5 S_3 = P_0$$

$$\text{and} \quad X \geq 0, Y \geq 0, S \geq 0$$

Simplex Tableau is formed in a particular way as explained below*

* For theorems concerning the simplex method see Cass S.I. *Linear Programming*, McGraw Hill Ch 4 and M. H. H. *Mathematical Economics Microeconomic Models*, Sultan Chand, Ch 13.

(1) All the vectors appear on the top of the table, but their order or appearance is changed.

Simplex Tableau (Example 1)

	$C_j \rightarrow$		0	0	0	0	2	5	Replacement Ratios
Stages		Basic	P_0	P_1	P_2	P_3	P_4	P_5	
Stage I (Initial Solution)	$\leftarrow 0$	P_2	24	1	0	0	1	4	$\frac{a_{30}}{a_{22}} = \frac{24}{4} = 6$
	0	P_4	21	0	1	0	3	1	$\frac{a_{42}}{a_{40}} = \frac{21}{1} = 21$
	0	P_5	9	0	0	1	1	1	$\frac{a_{50}}{a_{52}} = \frac{9}{1} = 9$
	Z_j		0	0	0	0	0	0	
	$Z_j - C_j$		0	0	0	0	-2	-5	
Iteration I (Stage 2)	$\rightarrow 5$	P_2	6	1/4	0	0	1/4	1	$\frac{a_{70}}{a_{71}} = \frac{6}{1/4} = 24$
	0	P_4	15	-1/4	1	0	11/4	0	$\frac{a_{80}}{a_{81}} = \frac{15}{1/4} = \frac{60}{11}$
	$\leftarrow 0$	P_5	3	-1/4	0	1	3/4	0	$\frac{a_{90}}{a_{91}} = \frac{3}{3/4} = 4$
	Z_j		30	5/4	0	0	5/4	5	
	$Z_j - C_j$		30	5/4	0	0	-3/4	0	
Iteration II (Stage 3)	3	P_5	5	-1/3	0	-1/3	0	1	
	0	P_4	4	8/12	1	-11/4	0	0	
	$\rightarrow 2$	P_1	4	-1/3	0	4/3	1	0	
	Z_j		33	1	0	1	2	5	
	$Z_j - C_j$		33	1	0	1	0	0	

P_0 vector appears first followed by the Basis (Identity) vectors, viz., P_3 , P_4 and P_5 followed by structural vectors, viz., P_1 and P_2 *

(2) In the first row of the table, (C_j), we write the coefficients of the vector of the objective function (1), which is required to be maximised, following the order described in (1) above

(3) In the first column of table (C_j), we write the coefficients of the basis vectors at the first stage but subsequently these coefficients are replaced by the coefficients of the incoming structural vectors

(4) Formulation of Z_j row is the summation of products of elements of each column vector with corresponding element of C_j column

For example, elements of P_5 column are (0,0,1), while corresponding elements of C_j column are (0,0,0) hence Z_j for P_5 column

$$= (0 \times 0) + (0 \times 0) + (1 \times 0) = 0$$

Since C_j column possesses all zeros in first stage, Z_j for all vectors will be zero (always)

(5) *Formulation of ($Z_j - C_j$) row* Subtract from Z_j value of each column the C_j value of the vector given in the 1st row of the table. Except in case of column P_1 and P_2 all the elements of ($Z_j - C_j$) row will be zero in the first stage. For the vectors P_1 and P_2 the value of $Z_j - C_j$ will be (-2) and (-5) respectively, because for the vector P_1 , $C_j = 2$ and $Z_j = 0$ and for vector P_2 , $C_j = 5$ and $Z_j = 0$

Before going further to stage II (or Iteration I), following test is used to determine whether the solution of the given L.P. problem is an optimal feasible solution, or whether it is necessary to make further manipulations (iterations) or whether there can be no finite solution at all (of the given problem)

Tests

- 1 If all $Z_j - C_j \geq 0$, an optimal solution has been obtained, hence no further iterations are necessary
- 2 $Z_j - C_j < 0$ for some columns, then

(a) if all the elements of those columns for which $Z_j - C_j < 0$ possess negative values, the solutions will be infinite

(b) if some of the elements of those columns (for which $Z_j - C_j < 0$) possess positive values, further iterations are necessary to achieve the optimal solution

* It makes no difference in computations if the vectors are arranged in ascending order P_1

P_2 P_0 . In fact vector P_0 can be the first or the last vector

If we apply the above tests to our problem it is found that in the first stage ($Z_j - C_j < 0$) for vectors P_1 and P_2 . Also all the elements of these 2 vector columns possess positive values hence further iteration is needed to arrive at the optimal solution.

We proceed as follows for further iteration

(A) Structural vectors (P_1 and P_2) are used to replace the basis vectors (P_3 , P_4 and P_5) in turn.

This raises two problems. First we must decide which of the structural vectors should be the 'replacing' vector and second, which of the basis vectors be 'replaced' vector.

For this we apply the following two rules at all stages of the table

1 For deciding replacing structural vector

Replacing vector will be that structural vector which has highest negative $Z_j - C_j$ value amongst them. In the first stage of the tableau, for example, we could select P_2 to be the replacing vector since for P_2 , we have $Z_j - C_j = -5$.

2 For deciding basis vector which is to be replaced

The replaced vector is determined by means of finding the ratio of each element in P_0 vector to the corresponding elements of the replacing vector P_2 . The basis vector associated with the smallest positive ratio would be the vector to be replaced.

Let us use some symbols

Let a_{30} denote the element of the row labelled P_3 and column labelled P_0 . It is 24 in our table.

a_{51} denote the element of the row labelled P_5 and column labelled P_1 . It is 1 in our table and so on.

In the first stage of the table we have three ratios

$$\left. \begin{aligned} \frac{a_{30}}{a_{32}} \text{ (associated with } P_3 \text{ vector)} &= 6 \\ \frac{a_{40}}{a_{42}} \text{ (associated with } P_4 \text{ vector)} &= 21 \\ \frac{a_{50}}{a_{52}} \text{ (associated with } P_5 \text{ vector)} &= 9 \end{aligned} \right\} \rightarrow \begin{array}{l} \text{Of these ratios, 6 being the smallest + } P_3 \text{ vector is selected} \\ \text{as the basis vector to be replaced. Division only by the} \\ \text{positive even tho' the replacing vector is negative.} \end{array}$$

The replacement basis vector is indicated by an arrow pointing away from the table in C_j column.

(B) Formulation of Iteration I (or stage II)

(i) We first write new vector (introduced) P_2 in place of the basic vector replaced (P_3) in the 2nd column of the table

(ii) The elements in the row of this new vector P_2 (introduced) are obtained by dividing each element of P_3 row by corresponding the element of vector P_2

Therefore elements in P_2 row will be

$$\begin{aligned}\frac{a_{30}}{a_{32}} &= \frac{24}{4} = 6 & \frac{a_{33}}{a_{32}} &= \frac{1}{4} \\ \frac{a_{34}}{a_{32}} &= \frac{0}{4} & \frac{a_{35}}{a_{32}} &= \frac{0}{4} = 0 \\ \frac{a_{31}}{a_{32}} &= \frac{1}{4} & \text{and } \frac{a_{32}}{a_{32}} &= \frac{4}{4} = 1\end{aligned}$$

(iii) The elements in the remaining rows are determined by the following rule

Elements of row P_4 are

$$\left\{ \begin{aligned} a_{40} - \left(\frac{a_{42}}{a_{32}} \times a_{30} \right) \\ a_{43} - \left(\frac{a_{42}}{a_{32}} \times a_{33} \right) \\ a_{44} - \left(\frac{a_{42}}{a_{32}} \times a_{34} \right) \\ a_{45} - \left(\frac{a_{42}}{a_{32}} \times a_{35} \right) \\ a_4 - \left(\frac{a_{42}}{a_{32}} \times a_{31} \right) \\ a_{42} - \left(\frac{a_{42}}{1} \times a_{32} \right) \end{aligned} \right\} \quad \text{and}$$

This can be explained as follows, following example from the simplex table

Three Rows of Stage I are

	I	P_4	I_5	P_1	P		
P_3	4	1	0	0	1	4	
	21	0	1	0	3	1	$\frac{0}{4} = 0$
I_3	9	0	0	1	1	1	$\frac{0}{4} = 0$

Three rows of Iteration I (or Stage II)

	P_0	P_3	P_4	P_5	P_1	P_2
P_2	$\frac{24}{4}$	$\frac{1}{4}$	$\frac{0}{4}$	$\frac{0}{4}$	$\frac{1}{4}$	$\frac{4}{4}$
P_4	$\frac{(21 - \frac{1}{4} \times 24)}{15}$	$\frac{(0 - \frac{1}{4} \times 0)}{-1/4}$	$\frac{(1 - \frac{1}{4} \times 0)}{1}$	$\frac{(0 - \frac{1}{4} \times 0)}{0}$	$\frac{(3 - \frac{1}{4} \times 1)}{1/4}$	$\frac{(1 - \frac{1}{4} \times 4)}{0}$
P_5	$\frac{(9 - \frac{1}{4} \times 24)}{3}$	$\frac{(0 - \frac{1}{4} \times 0)}{-1/4}$	$\frac{(0 - \frac{1}{4} \times 0)}{0}$	$\frac{(1 - \frac{1}{4} \times 0)}{1}$	$\frac{(1 - \frac{1}{4} \times 1)}{3/4}$	$\frac{(1 - \frac{1}{4} \times 4)}{0}$

(C) Formulation of Z_j and $(Z_j - C_j)$

Again we determine Z_j row by the same procedure given in stage I, that is, multiply each element in each column by the corresponding element in the C_j column and then add these products. In stage II, elements in the P_0 are 6, 15 and 3. Multiplying each of these by the corresponding elements in C_j column and then adding them, we get

$$(6 \times 5 + 15 \times 0 + 3 \times 0) = 30$$

Value of C_j (given in the first row) is zero for this column

$$Z_j - C_j = 30 - 0 = 30$$

Value of Z_j and $(Z_j - C_j)$ for other columns are determined in the same way

As explained under 'test', this iteration process is carried on until all $(Z_j - C_j)$ values are either $= 0$ or more than zero, i.e., positive. In our example, we stop after iteration II, when all the values of $(Z_j - C_j)$ are either zero or positive, hence an optimal solution has been achieved

This solution is given by P_0 column $P_0 = 5P_2 + 4P_4 + 4P_1$

But the coeff. of vector P_2 is Y $Y = 5$

Coefficient of vector P_1 is X $X = 4$

Coefficient of vector P_4 is S_2 $S_2 = 4$

That is, the given function $2X + 5Y + 0S_1 + 0S_2 + 0S_3$ will be maximum, when $X = 4$ and $Y = 5$ and the maximum value of the function will be $2(4) + 5(5) + 4(0) = 33$, which is also given by the 1st element of Z_j row (Verify that this is the *basic feasible solution*. What does $S_2 = 4$ signify?)

**Eliminating Redundant Constraints*

As already pointed out, solution of an L.P. problem becomes more time consuming with increasing number of constraints. But many times, some of the

constraints have no effect on the solution of the problem. They should be left out of the computations to save time. For example, consider the constraints

$$\begin{aligned}x + y &\geq 5 \\x + y &\geq 10\end{aligned}$$

If the acceptable solution requires $x + y \geq 10$, then $x + y \geq 5$ can have no effect on the solution, $x + y$ cannot have value > 5 and < 10 . Hence the constraint $x + y \geq 5$ should be dropped. However, if $x + y \leq 5$, $x + y \leq 10$, the second constraint $x + y \leq 10$ will not operate.

Suppose the constraints are

$$\begin{aligned}4x + 3y + 6z &\leq 120 & (1) \\10x + 7y + 16z &\leq 250 & (2)\end{aligned}$$

Compute the intercepts on the axes

			Constraints	
			(I)	(II)
$x = 0, y = 0$	z	$=$	20	$15\frac{5}{8}$
$x = 0, z = 0$	y	$=$	40	$35\frac{5}{7}$
$y = 0, z = 0$	x	$=$	30	25

Hence every intercept on axes is smaller in the case of constraint (II) as compared to that in (I). The two planes do not intersect. Plane (2) lies below plane (1). Hence the constraint (II) is redundant, i.e., it should be left out.

Had the constraints been

$$\begin{aligned}4x + 3y + 6z &\geq 120 & (3) \\10x + 7y + 16z &\geq 250 & (4)\end{aligned}$$

then constraint (4) will be *redundant*.

Degeneracy in Simplex Table

If at any stage in carrying out the simplex operation, it is discovered that structural vectors replace more than one basis vector, then L.P. problem is said to degenerate. In other words, this means that in case we get two or more than two minimum ratios identical, then structural vector would be replacing two or more than two basis vectors. This will be the case of degeneration.

Ex 2. Consider the following L.P. problem

$$\begin{aligned}\text{Maximise} \quad & f = X + 4Y \\ \text{Subject to} \quad & 2X + 3Y \leq 4 \\ & 3X + Y \leq 3 \text{ and,} \\ & X \geq 0, Y \geq 0\end{aligned}$$

Introducing the slack variables, we get the equation system

$$2X + 3Y + S_1 = 4$$

$$3X + Y + S_2 = 3$$

which may be written in vector form

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix}_{P_1} X + \begin{pmatrix} 3 \\ 1 \end{pmatrix}_{P_2} Y + \begin{pmatrix} 1 \\ 0 \end{pmatrix}_{P_3} S_1 + \begin{pmatrix} 0 \\ 1 \end{pmatrix}_{P_4} S_2 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}_{P_0}$$

$$P_1 X + P_2 Y + P_3 S_1 + P_4 S_2 = P_0$$

Our problem now reduces to

$$\text{Maximise } f = X + 4Y + 0S_1 + 0S_2 \quad (I)$$

$$\text{Subject to } P_1 X + P_2 Y + P_3 S_1 + P_4 S_2 = P_0 \quad (II)$$

$$\text{and } X \geq 0, Y \geq 0, S_i \geq 0$$

Forming the simplex tableau as explained in Example 1 we find that after iteration I (or stage II) all the values in $(Z_j - C_j)$ row turn out to be ≥ 0 . Hence no further iteration is needed.

Simplex Tableau (Example 2)

Stage	$C_j \rightarrow$	Basis	0 P_0	0 P_2	0 P_4	0 P_1	0 P_3	Ratios
I	$\rightarrow 0$	P_2	4	1	0	2	3	$\frac{a_{20}}{a_{22}} = \frac{4}{3}$
	0	P_4	3	0	1	3	1	$\frac{a_{40}}{a_{42}} = \frac{3}{1}$
	Z_j		0	0	0	0	0	
	$Z_j - C_j$		0	0	0	-1	-4	
II	$\rightarrow 4$	P_2	4/3	1/3	0	2/3	1	
	0	P_4	5/3	-1/3	1	7/3	0	
	Z_j $Z_j - C_j$		16/3 16/3	4/3 4/3	0 0	8/3 5/3	4 0	

Note. See that $(Z_j - C_j) \geq 0$

Optimal solution is in col. P_0 which is equal to $\frac{4}{3}P_2 + \frac{5}{3}P_4$. Hence to maximise the function $f = X + 4Y$, $X = 0$, $Y = \frac{4}{3}$ and $S_3 = \frac{5}{2}$. The maximum value is $\frac{5}{2}(0) + 4\left(\frac{4}{3}\right) = \frac{16}{3}$.

Ex. 3 Maximise $f = 2x_1 + 3x_2 + x_3 + 7x_4$
 Subject to $8x_1 + 3x_2 + 4x_3 + x_4 \leq 6$
 $2x_1 + 6x_2 + x_3 + 5x_4 \leq 3$
 $x_1 + 4x_2 + 5x_3 + 2x_4 \leq 7$, and
 $x_j \geq 0$ ($j = 1, 4$)

After introducing the slack variables, we obtain

$$8x_1 + 3x_2 + 4x_3 + x_4 + S_1 = 6$$

$$2x_1 + 6x_2 + x_3 + 5x_4 + S_2 = 3$$

$$x_1 + 4x_2 + 5x_3 + 2x_4 + S_3 = 7$$

In the vector form

$$\begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix} x_1 + \begin{bmatrix} 3 \\ 6 \\ 4 \end{bmatrix} x_2 + \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix} x_3 + \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} x_4 + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} S_1 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} S_2 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} S_3 = \begin{pmatrix} 6 \\ 3 \\ 7 \end{pmatrix}$$

$$\text{i.e. } P_1 x_1 + P_2 x_2 + P_3 x_3 + P_4 x_4 + P_5 S_1 + P_6 S_2 + P_7 S_3 = P_0$$

Our problem is now

Maximise $f = 2x_1 + 3x_2 + x_3 + 7x_4 + 0S_1 + 0S_2 + 0S_3$

Subject to $P_1 x_1 + P_2 x_2 + P_3 x_3 + P_4 x_4 + P_5 S_1 + P_6 S_2 + P_7 S_3 = P_0$
 and $x_1, x_2, x_3, x_4 \geq 0$

The simplex table tells us that after iteration 1, all the values of $(Z_j - C_j) \geq 0$ hence optimal solution has been achieved

Simplex Tableau (Example 3)

Stages	C_j	Basis	0 P_0	0 P_5	0 P_6	0 P_7	2 P_1	3 P_2	1 P_3	7 P_4	Ratios
Stage I	0	P_5	6	1	0	0	8	3	4	1	$\frac{a_{50}}{a_{54}} = \frac{6}{1}$
	0	P_4	3	0	1	0	2	6	1	5	$\frac{a_{50}}{a_{51}} = \frac{3}{5}$
	0	P_1	7	0	0	1	1	4	5	2	$\frac{a_{70}}{a_{71}} = \frac{7}{2}$
		Z_j	0	0	0	0	0	0	0	0	
		$Z_j - C_j$	0	0	0	0	2	-3	-1	-7	
Stage II	0	P_5	27/5	1	-1/5	0	38/5	9/5	19/5	0	
	7	P_4	3/5	0	1/5	0	2/5	6/5	1/5	1	
	0	P_7	29/5	0	-2/5	1	1/5	8/5	23/5	0	
		Z_j	21/5	0	7/5	0	14/5	42/5	7/5	7	
		$Z_j - C_j$	21/5	0	7/5	0	4/5	27/5	2/5	0	

Optimal solution is $\frac{27}{5}P_3 + \frac{3}{5}P_4 + \frac{29}{5}P_7$

that is, $S_1 = \frac{27}{5}$, $X_4 = \frac{3}{5}$ and $S_3 = \frac{29}{5}$

Thus coordinates of optimal point will be $0 \ 0 \ 0, \frac{3}{5}$ and the value of function (at maximum)

$$= 2(0) + 3(0) + 1(0) + 7\left(\frac{3}{5}\right) = \frac{21}{5} = 4.20$$

EXERCISE 18.1

Consider the following L.P. problems (Use graphic method where possible)

- 1 Maximise $f(L, M) = 2L + 5M$
 Subject to $L < 4$
 $M < 6$

and $L > 0, M > 0$

- 2 Maximise $f(A, B) = 5A + 8B$
 Subject to $5A + 6B \leq 74$
 $A + 2B \leq 22$
 $4A + B \leq 14$
 A and $B \geq 0$

(a) Graph the inequalities given above and determine one optimal feasible solution by graphical method

(b) Introduce slack variables to inequalities and write down converted linear form of the problem

(c) Rewrite (b) in vector form and give geometrical interpretation of the non negativity constraints

(d) Use simplex method to solve the problem and verify that the graphical solution is correct

- 3 Maximise $f(X, Y) = 4x + 6Y$
 Subject to $\frac{1}{2}X + Y \leq 4$
 $2X + Y \leq 8$
 $4X - 2Y \leq 2$
 X and $Y \geq 0$

4. Maximise $g = X_1 + 4X_2 + 5X_3$
 Subject to $3X_1 + 6X_2 + 3X_3 \leq 22$
 $X_1 + 2X_2 + 3X_3 \leq 14$
 $3X_1 + 2X_3 \leq 14$
 $X_j \geq 0 \ (j = 1, 2, 3)$
5. Maximise $f(X, Y) = 6X + 4Y$
 Subject to $2X + 4Y \leq 48$
 $4X + 2Y \leq 60$
 $3Y \leq 36$, and
 $X \geq 0, Y \geq 0$
6. Maximise $f(A, B) = 3A + 3B$
 Subject to $A + 7B \leq 9$
 $2A + B \leq 8$
 $A + B \leq 6$
 A and $B \geq 0$

7. Discuss the role of linear programming in the theory of firm. A firm has only one fixed factor ($e.g.$, labour) and there are several processes involving products and variable factors available at given market prices. Show only one process is used that with the least use of the fixed factor.

8. Prove that the solution of a linear programming problem must be at extreme point of the feasible zone.

9. Maximise $f = 2.5X_1 + 2X_2$
 Subject to $X_1 + 2X_2 \leq 8000$
 $3X_1 + 2X_2 \leq 9000$
 $X_1 \geq 0$
 $X_2 \geq 0$
10. Maximise $f = 3X_1 + 4X_2$
 Subject to $X_1 + X_2 \leq 6$
 $2X_1 + 4X_2 \leq 21$
11. Maximise $f = 2Y + Z$
 Subject to $Y \leq 4$
 $Z \leq 3$
 $Y + 2Z \leq 8$

12. Show that there is no solution to the linear programming problem,

$$\begin{array}{ll}\text{Maximise} & Z = 4X + Y \\ \text{Subject to} & 3X + 5Y \leq 15 \\ & 5X + Y \leq 15 \\ & -X + Y \leq 2 \\ & 4X + 5Y \leq 20 \text{ and } X, Y \geq 0\end{array}$$

13. Maximise $g = 6X + Y + 5Z$
 Subject to $3X + 6Y + 4Z \leq 48$
 $11X + 6Y + 12Z \leq 136$
 $3X + 10Y + 6Z \leq 70$
 $7X + 14Y + 4Z \leq 55$
 $4X + 4Y + 5Z \leq 50$, and $X, Y, Z \geq 0$

14. Show that the constraints

$$\begin{aligned}3X + 2Y - Z &\leq 14 \\ X - 3Y - 4Z &= 19 \\ 2X + 4Y - Z &= 10 \\ 3X + 4Y + 7Z &\leq 30\end{aligned}$$

are inconsistent, and hence that any linear programming problem which includes these constraints will be infeasible

15. Maximise $Z = 8x + 6y$
 Subject to $6x + 3y \leq 126$
 $2x + 4y \leq 96$
 $x, y \geq 0$

18.8 Minimisation Examples of Linear Programming

We have mentioned above that L.P. technique enables us to deal with problems involving objective functions which are either required to be maximised or minimised under certain given constraints. We now consider a problem in which objective function is to be minimised under given conditions.

Ex. 4. Let us assume a hypothetical problem in which a person requires a certain minimum amount of each of 3 types of vitamins per day in order to maintain his health. We also assume that these 3 types of vitamins are found in specific quantities in the 2 food items. If the price per unit of each food item is known, our problem is to find the combination of food items which must be purchased so that the daily vitamin requirements are met and simultaneously the cost of the food items purchased is minimum with the following given data

Vitamins (Types)	Food items containing Vitamins per kg		Minimum daily Vitamin requirement
	I	II	
A_1	10	4	20
A_2	5	5	20
A_3	2	6	12
Price per kg of food item	Rs 0.60	Rs 1.00	—

The table indicates one unit of Food I contains 10 units of vitamin A_1 , 5 units of vitamin A_2 and 2 units of vitamin A_3 . Similarly food II contains 4 units of A_1 , 5 units of A_2 and 6 units of A_3 . The daily minimum requirement for a person to maintain his health is 20 units of A_1 , 20 units of A_2 and 12 units of A_3 .

With the prices of both the food items given, and assuming that x_1 quantity of food I and x_2 quantity of food II will satisfy the minimum requirement of the vitamins, our problem can be put in algebraic form

$$\begin{aligned}
 &\text{Minimise} && f = 0.6x_1 + x_2 \\
 &&& 10x_1 + 4x_2 \geq 20 \\
 &&& 5x_1 + 5x_2 \geq 20 \\
 &&& 2x_1 + 6x_2 \geq 12, \text{ and} \\
 &&& x_1, x_2 \geq 0
 \end{aligned}$$

Note that each structural constraint is of $\geq$ type. This is because to maintain his health the man in question must consume minimum of 20 units of A_1 and not less than that amount. Same is applicable to the case of vitamins A_2 and A_3 .

Graphical method. We plot all the 3 constraint inequalities as explained under the maximisation problem. Here again because of non-negativity restrictions, we need only to consider the positive quadrant as specified in Fig 18.11.

Since each constraint is of $\geq$ type, solution to our problem would lie above these lines of constraints. For this reason, now the shaded area above the lines will be our feasible region and each point in this region will contain feasible solution.

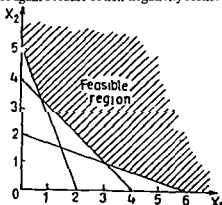


Fig 18.11

Again to minimise cost (of that feasible solution) we must select the lowest possible iso-cost line while still staying in the feasible region. Fig 18.12 gives the boundaries of feasible region. In order that objective function should be minimum, the movement of iso-cost should be as indicated by the arrows in Fig. 18.12.

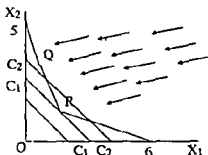


Fig 18.12

CC_1 is the only iso-cost line (tangent at R) satisfying our both conditions

(i) It gives the solution which is feasible (since R is the extreme point of feasible region), and

(ii) It involves least cost (since any iso-cost line below it would violate all the constraints)

How to find point R ? Since this point is the interaction of constraint lines of vitamin A_2 and vitamin A_3 , we solve the equations:

$$\begin{aligned} 5X_1 + 5X_2 &= 20 & X_1 &= 3 \\ 2X_1 + 6X_2 &= 12 & X_2 &= 1 \end{aligned}$$

Hence solution to our problem is that a person should purchase 3 units of food I and one unit of food II. These amounts of food will not only provide the required intake of vitamins but will also involve minimum cost to the consumer in question.

Ex. 5 The Organising Committee of some conference is to decide breakfast menu. Two alternatives can be offered; Idli-Sambhar and Omelette, the cost per unit being Rs. 2 and 3 respectively. It is also decided that on an average the breakfast should provide at least 20 mgs. and 30 mgs. of vitamin A and B respectively. If each unit of idli provides 5 mgs. of vitamin A as well as 5 mgs. of vitamin B and if each unit of omelette provides 5 mgs. and 10 mgs of vitamin A and B respectively; find the least-cost menu subject to nutrition requirements.

Statement of the problem :

Objective function. $\text{Min} : f = 2x + 3y$

(Here x and y depict the number of idlis and omelettes to be served to the participants to minimise the total cost.)

Constraints : Vit. A : $5x + 5y \geq 20$

Vit. B : $5x + 10y \geq 30$, and

$x, y \geq 0$

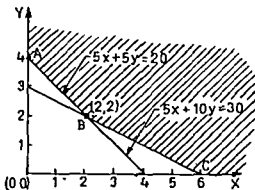


Fig 18 13

The feasible region is the shaded space in Fig 18 13. The extreme points of this region are A, B and C. The coordinates of these points are A (0, 4), B (2, 2) and C (6, 0). They are basic solutions.

The point which minimizes the objective function ($C = 2x + 3y$) gives the optimal basic feasible solution.

$$\text{At } A, \quad C = 2(0) + 3(4) = 12$$

$$\text{At } B, \quad C = 2(2) + 3(2) = 10$$

$$\text{At } C, \quad C = 2(6) + 3(0) = 12$$

Hence point B gives the least cost combination. Two idlis and 2 omlettes should be served in the breakfast to minimise the total cost.

Simplex Method for Solving Minimisation Problem

Though there is a different simplex method for finding the solution of a minimisation linear programming problem, the best way of solving such type of problems is to first find out the dual of given minimisation linear programming problem and then solve this dual maximisation L.P. problem by simplex method as explained above. However, the simplex method of minimisation of linear programming problem has also been given.

We shall be considering minimisation linear programming problems after explaining "Duality".

18.9 Concept of Duality

We have discussed maximisation and minimisation L.P. problems as two separate types of problems but, actually, every L.P. problem has a counterpart called its dual.

The original problem (whether it is in the form of maximisation or minimisation function) is referred to as a primal problem. If the primal problem requires maximisation, the dual problem is one of minimisation and if the primal is a minimisation problem, the dual is a maximisation problem. In this way, minimisation and maximisation problems are really not so distinct as they appear to be. In fact, since the optimal values of the objective functions in the primal and in the dual are always identical, and also that primal can be translated into its dual and *vice versa*, we have always an option of picking either of the two to work on. Nevertheless, the choice will ultimately depend upon

- (1) the formulation that yields more directly the desired result, and
- (2) the formulation that can be more easily solved (this has been explained in the following pages)

Dual of general linear programming problem (or general linear programming problem of minimisation case)

A very good illustration of relationship between optimal problem and its dual is provided in the theory of production and costs. Suppose the primal problem was that of maximisation of the total net revenue with given cost outlay. The dual would be that of minimisation of costs for the given output.

Suppose a firm produces 2 products with 2 inputs, there are capacity constraints of the inputs.

If the prices of two products are p_1 and p_2 then the revenue which the firm will try to maximize will be

$$R = p_1 x_1 + p_2 x_2$$

Subject to capacity constraints of the two inputs which may be written as

$$a_{11} x_1 + a_{12} x_2 \leq b_1$$

$$a_{21} x_1 + a_{22} x_2 \leq b_2$$

for input 1 and 2 respectively.

Obviously, this presentation can be generalised if the number of products are n and number of inputs m . Then the problem will be to

$$\text{Maximise } R = \sum_{i=1}^n p_i x_i$$

$$\text{Subject to } \left[\begin{array}{l} \sum_{i=1}^n a_{1i} x_i \leq b_1 \\ \vdots \\ \sum_{i=1}^n a_{ji} x_i \leq b_j \\ \vdots \\ \sum_{i=1}^n a_{mi} x_i \leq b_m \end{array} \right]$$

$$\text{Or, } \sum_{i=1}^n a_{ji} x_i \leq b_j \quad (j = 1, 2, \dots, m)$$

$$\text{and, } x_i \geq 0$$

In matrix notations maximise $R = P'X$

Subject to $AX \leq B$

and $X \geq 0$

Now consider the *Dual*. Suppose that the firm decides to determine the portion of total revenue from each of its products it owes to each of the inputs (or capacities spent). This can be done if we consider the imputed prices (or opportunity costs or shadow prices) of all the inputs (capacities) spent on each of the products.

We know that one unit of product 1 uses a_{11} and a_{21} amounts of inputs 1 and 2 respectively hence if imputed prices of these inputs are C_1 and C_2 the total cost of producing one unit of product 1 will be $a_{11}C_1 + a_{21}C_2$. This should be at least as much as the price of the product in the market (p_1). Similarly for the other product. Hence

$$\left. \begin{aligned} a_{11}C_1 + a_{21}C_2 &\geq p_1 \\ a_{12}C_1 + a_{22}C_2 &\geq p_2 \end{aligned} \right\} \quad (1)$$

The total imputed cost of inputs available will be

$$b_1C_1 + b_2C_2 \quad (2)$$

Hence the firm will minimize the total imputed cost (2) subject to the constraints (1). This may be generalised as follows

(1) Objective function

$$\text{Minimise } f = b_1C_1 + b_2C_2 + \dots + b_mC_m = \sum_{j=1}^m b_j c_j$$

(2) Structural constraints

$$a_{11}C_1 + a_{21}C_2 + \dots + a_{m1}C_m \geq p_1 \quad \text{or} \quad \sum_{j=1}^m a_{j1}c_j \geq p_1$$

$$a_{12}C_1 + a_{22}C_2 + \dots + a_{m2}C_m \geq p_2 \quad \text{or} \quad \sum_{j=1}^m a_{j2}c_j \geq p_2$$

$$a_{1n}C_1 + a_{2n}C_2 + \dots + a_{mn}C_m \geq p_n \quad \text{or} \quad \sum_{j=1}^m a_{jn}c_j \geq p_n$$

(3) Non negativity constraints $C_j \geq 0$ ($j = 1, 2, \dots, m$)

Minimise $\sum_{j=1}^m b_j c_j$

Subject to $\sum_{j=1}^m a_{ji} C_j \geq p_i$ ($i = 1, 2, \dots, n$)

and, $C_j \geq 0$ ($j = 1, 2, \dots, m$)

In matrix notation,

Minimise $f = B C$

Subject to $A C \geq P$
 $C \geq 0$

where P is the column vector of prices,
 X is the column vector of outputs
 $A = m \times n$ coefficient matrix,
 B = capacity constraint vector

18.10 Symmetry Between Primal and Dual

From the above two general L.P. problems we can easily pinpoint following characteristics of the primal and dual programmes which give them remarkable symmetry

1 *Regarding objective function.* (i) If the primal involves maximisation, the dual involves minimisation and *vice versa*. (ii) The profit constraints in the primal problem replace capacity constraints and *vice versa*.

2 *Regarding structural constraints* (i) If the primal involves $\geq$ signs, the dual involves $\leq$ signs and *vice versa*

(ii) A new set of variables appears in the dual.

(iii) If in the primal the coefficients in the constraints are found by moving from left to right, coefficients are positioned in the dual from top to bottom and *vice versa*

3 *Regarding non negativity constraints* The constraints remain unchanged

4 *Regarding variables* Neglecting the number of non negativity constraints, if there are ' n ' variables and ' m ' inequalities in the primal problem, in its dual there will be ' m ' variables and ' n ' inequalities

These symmetrical characteristics between primal and its dual help us to formulate certain rules for translating primal into its dual or *vice versa*. Let us first illustrate with some examples in the primal and its dual form from which we shall frame rules

Ex. 6.

*Primal*Maximise $f = 2P_1 + 6P_2$

Subject to ,

$$4P_1 + P_2 \leq 5$$

$$3P_1 + 2P_2 \leq 7$$

$$P_1 + P_2 \leq 2$$

and, $P_1 \geq 0, P_2 \geq 0$ *Dual*Minimise $f = 5C_1 + 7C_2 + 2C_3$

Subject to

$$4C_1 + 3C_2 + C_3 \geq 2$$

$$C_1 + 2C_2 + C_3 \geq 6$$

and, $C_1 \geq 0, C_2 \geq 0, C_3 \geq 0$

The structural constraints may be put in matrix form

$$\begin{pmatrix} 4 & 1 \\ 3 & 2 \\ 1 & 1 \end{pmatrix} \begin{bmatrix} P_1 \\ P_2 \end{bmatrix} \leq \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix}, \quad \begin{bmatrix} 4 & 3 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} \geq \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

Ex 7

Minimise $f = 4X + 5Y$

Subject to

$$X \geq 4$$

$$Y \geq 3$$

$$X + Y \geq 8$$

and, $X \geq 0, Y \geq 0$ Maximise $f = 4A + 3B + 8C$

Subject to

$$A + C \leq 4$$

$$B + C \leq 5$$

and, $A \geq 0, B \geq 0, C \geq 0$

Structural constraints may be put in matrix form

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} \geq \begin{pmatrix} 4 \\ 3 \\ 8 \end{pmatrix}, \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} \leq \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

We now formulate the basic rules of dual transformation

(1) The row vector of the coefficients in the primal objective function gives us the column vector of constraints in the dual constraints. Similarly the column vector of constraints in the primal constraints becomes the row vector of the coefficients in the dual objective function

(2) Transpose of the coefficient matrix of the primal constraints gives us the coefficients of the constraints in the dual and *vice versa*

(3) The inequality sign in the dual constraints is reversed, but inequalities of non negativity conditions retain their direction

Let us take an illustration and translate it into its dual with the help of these rules of transformation.

Ex. 8 Write the dual of the programme

$$\text{Minimise } f = x_1 + x_2 + 3x_3 + 2x_5$$

$$\text{Subject to } x_1 + 3x_2 - x_3 + 2x_5 \geq 7$$

$$-2x_2 + 4x_3 + x_4 \geq 12$$

$$-4x_2 + 3x_3 + 9x_5 + x_6 \geq 10, \text{ and}$$

$$x_j \geq 0 \quad (j = 1 \quad 6)$$

(1) The row vector of objective function is $= [1, 1, 3, 0, 2, 0]$

This becomes the column vector of the constraints of the dual—

$$= \begin{pmatrix} 1 \\ 1 \\ 3 \\ 0 \\ 2 \\ 0 \end{pmatrix}$$

(2) The col vector of the constraints is $= \begin{bmatrix} 7 \\ 12 \\ 10 \end{bmatrix}$ This becomes row vector

of the coefficients of the objective function with new set of variables (X, Y, Z), objective function is described as

$$f = 7X + 12Y + 10Z$$

(3) The coefficients of constraints of primal are given by matrix

$$A = \begin{pmatrix} 1 & 3 & -1 & 0 & 2 & 0 \\ 0 & -2 & 4 & 1 & 0 & 0 \\ 0 & -4 & 3 & 0 & 8 & 1 \end{pmatrix}$$

$$A' = \begin{pmatrix} 1 & 0 & 0 \\ 3 & -2 & -4 \\ -1 & 4 & 3 \\ 0 & 1 & 0 \\ 2 & 0 & 8 \\ 0 & 0 & 1 \end{pmatrix} \text{ and, } \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \leq \begin{pmatrix} 1 \\ 1 \\ 3 \\ 0 \\ 2 \\ 0 \end{pmatrix}$$

Since we have introduced a new set of variables (X, Y, Z), therefore, the required constraints in the dual will now be :

$$X \leq 1 \quad (1)$$

$$3X - 2Y - 4Z \leq 1 \quad (2)$$

$$-X + 4Y + 3Z \geq 3 \quad (3)$$

$$Y \leq 0 \quad (4)$$

$$2X + 8Z \leq 2 \quad (5)$$

$$Z \leq 0 \quad (6)$$

$$\text{and, } X, Y, Z \geq 0 \quad (7)$$

with the objective function Maximise $f = 7X + 12Y + 10Z$

Ex. 9 Obtain dual of the following L P P

$$\text{Maximise } f = 2x_1 + 3x_2$$

$$\text{Subject to } x_1 + 3x_2 \leq 12$$

$$2x_1 + x_2 \geq 6$$

$$x_1 + 5x_2 = 10$$

$$\text{and, } x_1, x_2 \geq 0$$

This primal is of mixed constraints, that is, constraints are of $\leq$, $\geq$ and equality types. To find the dual, all the constraints must first be put or stated in the *standard form*

The second and the third constraints are to be stated in the form of $\leq$ in the primal in the following way

$$2x_1 + x_2 \geq 6 \text{ can be written as}$$

$$-2x_1 - x_2 \leq -6 \text{ while the third constraint } x_1 + 5x_2 = 10$$

$$\text{can be written as}$$

$$x_1 + 5x_2 \leq 10 \text{ and}$$

$$x_1 + 5x_2 \geq 10$$

$$\text{Again, } x_1 + 5x_2 \geq 10 \text{ can be stated as}$$

$$-x_1 - 5x_2 \leq -10$$

Hence the given primal may now be described as

$$\text{Maximise } f = 2x_1 + 3x_2$$

$$\text{Subject to } x_1 + 3x_2 \leq 12$$

$$-2x_1 - x_2 \leq -6$$

$$x_1 + 5x_2 \leq 10$$

$$-x_1 - 5x_2 \leq -10$$

The dual of this primal will be as under

$$\text{Minimise .} \quad 12y_1 - 6y_2 + 10y_3 - 10y_4$$

$$\text{Subject to .} \quad y_1 - 2y_2 + y_3 - y_4 \geq 2$$

$$3y_1 - y_2 + 5y_3 - 5y_4 \geq 3$$

$$\text{and,} \quad y_1, y_2, y_3, y_4 \geq 0$$

We always select the problem in the form which involves lesser number of constraints. But in case the primal and its dual have an equal or nearly equal number of constraints, preference should be given to the problem in its maximisation form, because there is no need to introduce artificial variables along with the slack variables as would be the case in the minimisation form.

Relation between primal and dual in matrix notation. More generally on the basis of above illustration and rules we can relate the primal and dual programme in the matrix notations

<i>Primal</i>		<i>Dual</i>	
Maximise	$f = CX$	Minimise	$f = B'U$
Subject to	$AX \leq B$	Subject to	$A'U \geq C'$
and,	$X \geq 0$	and,	$U \geq 0$

where matrix A is the matrix of coefficients of constraints of the primal of order $m \times n$ and A' is the transpose of A

B is the $(m \times 1)$ column vector of constraints

B' is the transpose of B , i.e. B' is the $(1 \times m)$ row vector

C is the $(1 \times n)$ row vector of coefficients of the objective function of the primal.

C' is the transpose of C , i.e., C' is the column vector

X is the column vector of order $n \times 1$

U is the column vector of new variables having number of elements equal to the number of constraints in the primal.

18.11. An Economic Interpretation of Duality

Suppose a firm has

(1) m -inputs and n outputs

(2) x_j is the quantity of product X_j produced by the firm

(3) r_j is the revenue earned per unit of product X_j

(4) a_{ij} is the quantity of input ' i ' required to produce one unit of product X_j

(5) w_i is the per unit cost of input ' i '

(6) b_i is the maximum amount of input 'i' available to the firm

(A) Maximisation problem for the firm would then be

$$\text{To maximise total revenue } \sum_{j=1}^n r_j x_j$$

$$\text{Subject to } \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1 \text{ to } m) \text{ and } x_j \geq 0 \quad (j = 1 \text{ to } n)$$

Dual of this problem will be to minimise the total opportunity cost

$$\text{i.e., to minimise cost} = \sum_{i=1}^m b_i w_i$$

$$\text{Subject to } \sum_{j=1}^n a_{ij} w_j \geq r_j \quad (j = 1 \text{ to } n) \text{ and } w_i \geq 0 \quad (i = 1 \text{ to } m)$$

Now, in fact, each firm would see that total net revenue earned should be greater than the total cost, or at the most they should be equal

In mathematical form for the feasible solution of the two problems

$$\sum_{j=1}^n r_j x_j \leq \sum_{i=1}^m b_i w_i$$

$$\text{hence, } \quad \text{Max } \sum_{j=1}^n r_j x_j = \text{Min } \sum_{i=1}^m b_i w_i$$

Note. Since x_j and b_i are physical measurements and since r_j 's are expressed in money terms, the w_i must also be expressed in money terms if the above equations are to make any sense

Thus the primal and the dual can be interpreted as follows

The Primal Problem With a given revenue per unit of output (r) and a given upper limit for the availability of each input (b) how much of each output (x) to produce in order to maximise total revenue?

The Dual Problem With a given availability of each input (b) and given lower limit of revenue per unit for each output (r), what unit value (prices) should be assigned to each input (w) in order to minimise the value (cost) of total inputs?

These prices (w) are generally termed as *accounting or shadow prices*

EXERCISE 18 2

Given following primals, formulate the duals

- Maximise $f = 10X + 15Y + 20Z$
 Subject to $107X + 5Y + 2Z \leq 2705$
 $54X + 10Y + 4Z \leq 2210$
 $7X + Y \leq 445$
 $X, Y, Z \geq 0$

2. Maximise $f = 6X_1 + 4X_2 + 7X_3 + 5X_4$
 Subject to $X_1 + 2X_2 + 3X_3 + 2X_4 \leq 20$
 $6X_1 + 5X_2 + 3X_3 + 2X_4 \leq 100$
 $3X_1 + 5X_2 + 4X_3 + 21X_4 \leq 75$, and
 $X_j \geq 0 \quad (j = 1 \text{ to } 4)$
3. Minimise $f = 7A + 3B + 8C$
 Subject to $8A + 2B + C \geq 3$
 $3A + 6B + 4C \geq 4$
 $4A + B + 5C \geq 1$
 $A + 5B + 2C \geq 7$, and
 $A, B, C \geq 0$
4. Minimise $f = X_1 - 3X_2 + 2X_3$
 Subject to $3X_1 - X_2 + 2X_3 + X_4 \geq 7$
 $4X_2 - 2X_3 + X_4 \geq 12$
 $3X_2 - 4X_1 + 8X_3 + X_4 \geq 10$, and
 $X_j \geq 0 \quad (j = 1 \text{ to } 6)$

5. Prove in case of question 3 and 4 dual of their dual is the primal

18.12. Solution of Minimisation Problem by Simplex Method

Ex. 10. Minimise $f = 9X + 12Y + 15Z$

Subject to $2X + 2Y + Z \geq 10$

$2X + 3Y + 7Z \geq 12$

$X + Y + 5Z \geq 14$

$X, Y, Z \geq 0$

As said above, we shall first find the dual of given problem so that we have to

Maximise $f = 10A + 12B + 14C$

Subject to $2A + 2B + C \leq 9$

$A + B + 5C \leq 15$ and,

$A, B, C \geq 0$

Simplex Table Example 10

$C_j \rightarrow$		0	0	0	0	10	12	14	
$\downarrow$	Basis	P_0	P_4	P_5	P_6	P_1	P_2	P_3	
0	P_4	9	1	0	0	2	2	1	-14 is the most negative number in $Z_j - C_j$ replacing vector P_3
0	P_5	12	0	1	0	2	3	1	
0 $\leftarrow$	P_6	15	0	0	1	1	1	5	
	Z_j	0	0	0	0	0	0	0	15/5 is the smallest positive ratio replaced vector P_6
	$Z_j - C_j$	0	0	0	0	-10	-12	-14	
0	P_4	6	1	0	-1/5	9/5	9/5	0	Replacing vector P_2 Replaced vector P_5
0 $\leftarrow$	P_5	9	0	1	1/5	9/5	14/5	0	
14 $\rightarrow$	P_3	3	0	0	1/5	1/5	1/5	1	
	Z_j	42	0	0	14/5	14/5	14/5	14	
	$Z_j - C_j$	42	0	0	14/5	-36/5	-46/5	0	
0 $\leftarrow$	P_4	3/14	1	-9/14	-1/14	9/14	0	0	Replacing vector P_1 Replaced vector P_1
12 $\rightarrow$	P_5	45/14	0	5/14	-1/14	9/14	1	0	
14	P_6	33/14	0	-1/14	3/14	1/14	0	1	
	Z_j	501/7	0	23/7	15/7	122/4	12	14	
	$Z_j - C_j$	501/7	0	23/7	15/7	-9/7	0	0	
10 $\rightarrow$	P_1	1/3	14/9	-1	-1/9	1	0	0	
12	P_2	3	-1	1	0	0	1	0	
14	P_3	7/3	-1/9	0	2/9	0	0	1	
	Z_j	72	2	2	2	10	12	14	
	$Z_j - C_j$	72	2	2	2	0	0	0	

The solution of maximisation problem as per above table is $\left(\frac{1}{3}, 3, \frac{7}{3}\right)$, and that of minimisation dual (which in fact is the solution of the given problem) is $(2, 2, 2)$. Therefore, the minimum value of the given function $(9X + 12Y + 15Z)$ will be $(9 \times 2 + 12 \times 2 + 15 \times 2) = 72$.

We obtain the same solution by solving the *minimisation problem by Simplex method directly as follows*

Ex. 11. Minimise $f = 9X + 12Y + 15Z$

$$\begin{aligned}
 \text{Subject to } & 2X + 2Y + Z \geq 10 \\
 & 2X + 3Y + Z \geq 12 \\
 & X + Y + 5Z \geq 14 \text{ and} \\
 & X, Y, Z \geq 0
 \end{aligned}$$

Introducing the slack variables in the constraints we obtain

$$2X + 2Y + Z - S_1 = 10$$

$$2X + 3Y + Z - S_2 = 12$$

$$X + Y + 5Z - S_3 = 14$$

and X, Y, Z, S_1, S_2 and $S_3 \geq 0$

We note here that the (non negative) S_1, S_2 and S_3 variables, representing surpluses, are to be subtracted from the left side of the constraints. Now if we try to set $X = Y = Z = 0$, the three constraints will yield the solution $(S_1, S_2, S_3) = (-10, -12, -14)$, which is clearly not feasible. Therefore, we must search for some other initial Basic Feasible solution. For this we add a non negative artificial variable to every constraint. Let that be (A_1, A_2, A_3) . It should be noted that M , the coefficient of artificial vectors, is taken to be an unspecified very large positive number in minimization problem and large negative number in maximization problem. Artificial variables are used for obtaining a unit initial basis matrix.

$$\text{Minimise} \quad 9X + 12Y + 15Z + M(A_1 + A_2 + A_3) + 0S_1 + 0S_2 + 0S_3$$

$$\text{Subject to} \quad 2X + 2Y + Z - S_1 + A_1 = 10$$

$$2X + 3Y + Z - S_2 + A_2 = 12$$

$$\sim \quad X + Y + 5Z - S_3 + A_3 = 14$$

$$\text{and,} \quad X, Y, Z, S_1, S_2, S_3, A_1, A_2, A_3 \geq 0$$

The simplex method can now be applied in substantially the same way, except for a few simple modifications. (See Simplex Tableau of Ex. 11)

18.13 Equality and Mixed Constraints

Up to now we have discussed only $\leq$ constraints in maximisation and $\geq$ in minimisation problems. Recollect that the simplex tableau is formed by placing the basis matrix (Identity matrix) by the side of the structural matrix. In the minimisation problems with only $\geq$ constraints we introduce artificial vectors in the basis for this purpose. In the pure $\leq$ constraint case the basis is formed by the coefficients of the slack variables. But in the case of equality or mixed constraints, the initial basis might not be formed, then artificial vectors may have to be introduced to form the basis. For instance, in the Ex. 12 there are equality constraints.

$$\text{Ex. 12. Maximise} \quad x_2 + 3x_3 + 2x_5, \text{ or}$$

$$\text{Minimise} \quad -x_2 - 3x_3 - 2x_5$$

$$\text{Subject to} \quad x_1 + 2x_2 + 4x_3 + 2x_5 = 20$$

$$x_2 + 3x_3 + x_4 = 20$$

$$2x_2 + x_3 + 5x_5 + x_6 = 10, x_i \geq 0 \ (i = 1 \text{ to } 6)$$

Simplex Tableau Ex. II
 (Minimisation Problem)

C_j $\rightarrow$ $Basis$ $\rightarrow$ P_0	O P_0	O P_4	O P_5	O P_6	M A_1	M A_2	M A_3	M A_3	9 P_1	12 P_2	15 P_3	$Ratios$
M A_1	10	-1	0	0	1	0	0	0	2	2	1	$\frac{10}{1} = 10$
M A_2	12	0	-1	0	0	1	0	0	2	3	1	$\frac{12}{1} = 12$
M A_3	14	0	0	-1	0	0	1	1	1	1	5	$\frac{14}{5} = 2.8$
Z_j $Z_j - C_j$	$36M$ $36M$	-M	-M	-M	M	O	M	O	$5M$ $(5M - 9)$	$6M$ $(6M - 12)$	$7M$ $(7M - 15)$	Biggest +ive M value is possessed by P_3 vector
M A_1	7.2	-1	0	$\frac{1}{5}$	1	0	$-\frac{1}{5}$	$-\frac{1}{5}$	$\frac{9}{5}$	$\frac{9}{5}$	0	$7.2 + \frac{9}{5} = 4$
M A_2	9.2	0	-1	$\frac{1}{5}$	0	1	$-\frac{1}{5}$	$-\frac{1}{5}$	$\frac{9}{5}$	$\frac{14}{5}$	0	$9.2 + \frac{14}{5} = 3.9$
15 P_3	2.8	0	0	$-\frac{1}{5}$	0	0	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	1	$2.8 + \frac{1}{5} = 14$
Z_j	$(16.4M + 42)$	-M	-M	$\left(\frac{2}{5}M - 3\right)$	M	M	$\left(3 - \frac{2}{5}M\right)$	$\left(\frac{18}{5}M + 3\right)$	$\left(\frac{23}{5}M + 3\right)$	$\left(\frac{23}{5}M + 3\right)$	15	Biggest +ive M value is possessed by P_2 vector
$Z_j - C_j$	$(16.4M + 42)$	-M	-M	$\left(\frac{2}{5}M - 3\right)$	0	0	$\left(3 - \frac{7}{5}M\right)$	$\left(\frac{18}{5}M - 6\right)$	$\left(\frac{23}{5}M - 9\right)$	$\left(\frac{23}{5}M - 9\right)$	0	

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Putting this in simplex format, we have

C_j	Basis	P_0	0 P_1	1 P_2	3 P_3	0 P_4	2 P_5	0 P_6
0	P_1	20	1	2	4	0	2	0
0	P_4	20	0	1	3	1	0	0
0	P_5	10	0	2	1	0	5	1
Z_j		0	0	0	0	0	0	0
$Z_j - C_j$		0	0	-1	-3	0	-2	0

and proceed as usual here P_1, P_4, P_5 form the initial basis. No artificial basis is required

Ex. 13. Minimise $2x_1 - 2x_2 + 3x_3 - x_4$

Subject to $x_1 + 2x_2 + 3x_3 = 14$

$2x_1 + 3x_2 + x_3 = 20$

$-x_1 + 2x_2 + x_3 + x_4 = 10, x_i \geq 0$

This minimisation problem is equivalent to

Maximise $-2x_1 + 2x_2 - 3x_3 + x_4$

Subject to $x_1 + 2x_2 + 3x_3 = 14$

$2x_1 + 3x_2 + x_3 = 20$

$-x_1 + 2x_2 + x_3 + x_4 = 10$

Putting this in simplex format we have (note that the order of vectors in the basis should be as in the identity matrix)

C_j	Basis vector	P_0	2 P_1	2 P_2	3 P_3	-1 P_4	0 P_5	M A_1
M	A_1	14	1	2	3	0	0	1
0	P_5	20	2	3	0	0	1	0
-1	P_4	10	-1	2	1	1	0	0

Since the given system contains two unit vectors P_4 and P_5 we need only one artificial vector $A_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, which we have introduced in the last column, hence basis vectors will be A_1, P_5, P_4 in that order so that they form an identity matrix. The coefficient of the new vector A_1 is taken to be M . Proceed as usual

Ex 14 Maximise $3x_1 - x_2$

Subject to $2x_1 + x_2 \geq 2$

$x_1 + 3x_2 \leq 3$

$x_2 \leq 4$ and $x_i \geq 0$ ($i = 1$ to 2)

Introducing positive slack variables, we have

Maximise $3x_1 - x_2$

Subject to $2x_1 + x_2 - S_1 = 2$, $2x_1 + x_2 - S_1 + A_1 = 2$

$x_1 + 3x_2 + S_2 = 3$, $x_1 + 3x_2 + S_2 = 3$

$x_2 + S_3 = 4$, $x_2 + S_3 = 4$

Putting this in simplex format, we have

C_j	Vector	P_0	3 P_1	-1 P_2	0 P_3	0 P_4	0 P_5	M A_1
M	A_1	2	3	1	-1	0	0	1
0	P_4	3	1	3	0	1	0	0
0	P_5	4	0	1	0	0	1	0

Since the original problem contains two unit vectors, we need introduce only 1 vector in the basis, this is $A_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. In the basis the vectors will be in the order A_1, P_4, P_5 . Now proceed as usual.

Ex 15 Minimise $-x_1 - 2x_2 - 3x_3 + x_4$

Subject to $x_1 + 2x_2 + 3x_3 = 15$

$2x_1 + x_2 + 5x_3 = 20$

$x_1 + 2x_2 + x_3 + x_4 = 10$

$x_j \geq 0$ ($j = 1$ to 4)

We shall solve it as a minimization problem, since the solution is very similar to that given in example 10.

The following points should be noted

(i) Since there are three constraints not more than 3 variables can take positive values

(ii) We shall have to introduce two artificial vectors with M as the coefficient of the new variables in the objective function

(iii) As long as there is an artificial vector in the basis, Z will be a linear function of M , with M and non M terms

(iv) The replacing vector is associated with the largest M value in $Z_j - C_j$

(v) Then proceed as usual. Even after all the artificial vectors have been removed from the basis, further iteration might be necessary. If there are m vectors in the basis in general approximately $2m$ iterations will be required

(vi) Since we do not require the inverse of the matrix, the eliminated vector is never selected to re-enter the basis

The solution is $x_1 = x_2 = x_3 = 5/2$ and the minimum value of the function is $(-)$ 15

Note Had the problem been to maximize $x_1 + 2x_2 + 3x_3 - x_4$ subject to the same conditions, the solution procedure would have been exactly same. But the maximum value of the function will turn out to be $(+)$ 15. See Ex. 15 below for such case

Simplex Tableau Ex. 15

C_j		0	-1	-2	-3	1	M	M	Remarks
	Vec tor	P_0	P_1	P_2	P_3	P_4	A_1	A_2	
M	A_1	15	1	2	3	0	1	0	The largest M Coefficient = $8M$. Hence P_3 is the replacing vector. $20/5$ is the smallest
M	A_2	20	2	1	5	0	0	1	
1	P_4	10	1	2	1	1	0	0	
Z_j $Z_j - C_j$		$10 + 35M$ $10 + 35M$	$1 + 3M$ $2 + 3M$	$2 + 3M$ $4 + 3M$	$1 + 8M$ $4 + 8M$	1 0	M 0	M 1	
M ←	A_1	3	$-\frac{1}{5}$	$\frac{7}{5}$	0	0	1		Vector A_1 is eliminated. Largest positive M Coefficient $7/5$. Vector A_1 is replaced by P_2
-3	P_3	4	$\frac{2}{5}$	$\frac{1}{5}$	1	0	0		
1	P_4	6	$\frac{3}{5}$	$\frac{9}{5}$	0	1	0		
Z_j $Z_j - C_j$		$-6 + 3M$ $-6 + 3M$	$-\frac{3}{5} - \frac{1}{5}M$ $\frac{2}{5} - \frac{1}{5}M$	$\frac{6}{5} + \frac{7}{5}M$ $\frac{16}{5} + \frac{7}{5}M$	-3 0	1 0	M 0		

-2 →	P_2	$\frac{15}{7}$	$-\frac{1}{7}$	1	0	0	Vector P_4 is replaced by P_1
-3	P_3	$\frac{25}{7}$	$\frac{3}{7}$	0	1	0	
1 ←	P_4	$\frac{15}{7}$	$\frac{6}{7}$	0	0	1	
Z_j $Z_j - C_j$		$-\frac{90}{7}$	$\frac{6}{7}$	0	0	0	
-2	P_2	$\frac{5}{2}$	0	1	0	$\frac{1}{6}$	P_1 is introduced which then provides the final solution
-3	P_3	$\frac{5}{2}$	0	0	1	$-\frac{3}{6}$	
-1 →	P_1	$\frac{5}{2}$	1	0	0	$\frac{7}{6}$	
Z_j $Z_j - C_j$		-15	0	0	0	-1	

Ex 16. Maximise $Z = x_1 + 2x_2 + 3x_3 - x_4$

Subject to $x_1 + 2x_2 + 3x_3 = 15$

$2x_1 + x_2 + 5x_3 = 20$

$x_1 + 2x_2 + x_3 + x_4 = 10$

and, $x_j \geq 0$ ($j = 1, 2, 3, 4$)

Simplex Tableau Ex. 16

$C_j \rightarrow$		1	2	3	-1	M	M		Replace ment Ratios
	Basic	P_1	P_2	P_3	P_4	A_1	A_2	P_0	
M	x_1	1	2	3	0	1	0	15	5
M	x_2	2	1	5	0	0	1	20	4
-1	x_4	1	2	1	1	0	0	10	10
	Z_j	$3M-1$	$3M-2$	$8M-1$	-1	M	M	$35M-10$	
	$Z_j - C_j$	$3M-2$	$3M-4$	$8M-4$	0	0	0		

Since M is largest negative value in this case, the replacing variable is the one which has largest M coefficient in $Z_j - C_j$, it is P_3 . Dividing elements of P_0 by those of P_3 , we find that A_2 is leaving variable

Iteration 1

C_j	Basis	P_1	P_2	P_3	P_4	A_1	A_2	P_0	Ratios
M	A_1	$-\frac{1}{5}$	$\boxed{\frac{7}{5}}$	0	0	1	$\frac{3}{5}$	3	$15/7 \rightarrow$
3	x_3	$\frac{2}{5}$	$\frac{1}{5}$	1	0	0	$\frac{1}{5}$	4	20
-1	x_4	$\frac{3}{5}$	$\frac{9}{5}$	0	1	0	$-\frac{1}{5}$	6	$\frac{30}{9}$
Z_j		$\left(\frac{3}{5} \quad M/5\right)$	$\left(\frac{7}{5}M \quad \frac{6}{5}\right)$	3	1	M	$\left(\frac{4}{5} \quad \frac{3M}{5}\right)$	$(6+3M)$	
$Z_j - C_j$		$\left(\frac{2}{5} \quad M/5\right)$	$\left(\frac{16}{5} + \frac{7M}{5}\right)$	0	0	0	$\left(\frac{4}{5} - \frac{3M}{5}\right)$		

Iteration 2

C_j	Basis	P_1	P_2	P_3	P_4	A_1	A_2	P_0	Ratios
2	x_1	$-\frac{1}{7}$	1	0	0	$\frac{5}{7}$	$-\frac{3}{7}$	$\frac{15}{7}$	-15
3	x_3	$\frac{3}{7}$	0	1	0	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{25}{7}$	$\frac{25}{3}$
-1	x_4	$\boxed{\frac{16}{7}}$	0	0	1	$-\frac{9}{7}$	$\frac{4}{7}$	$\frac{15}{7}$	$\frac{15}{6} \rightarrow$
Z_j		$\frac{1}{7}$	2	3	1	$\frac{16}{7}$	$-\frac{4}{7}$	$\frac{90}{7}$	
$Z_j - C_j$		$-\frac{6}{7}$	0	0	0	$\left(\frac{16}{7} \quad M\right)$	$\left(\frac{4}{7} - M\right)$		

Iteration 3

C_j	Basis	P_1	P_2	P_3	P_4	A_1	A_2	P_0	Ratios
2	x_2	0	1	0	$\frac{1}{6}$	$-\frac{1}{2}$	$-\frac{1}{3}$	$\frac{5}{2}$	
3	x_3	0	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{5}{2}$	
1	x_4	1	0	0	$\frac{7}{6}$	$-\frac{3}{2}$	$\frac{2}{3}$	$\frac{5}{2}$	
Z		1	2	3	0	1	0	15	
$Z_j - C_j$		0	0	0	1	$1 - M$	$-M$		

Since all $Z_j - C_j \geq 0$ the optimal solution has been achieved, which is $x_1 = \frac{5}{2}$, $x_2 = \frac{5}{2}$, $x_3 = \frac{5}{2}$, $x_4 = 0$, $A_1 = 0$, $A_2 = 0$ and $Z = 15$

18.14 Some Special Cases

(I) Infeasibility When no solution satisfies all the given constraints, it is a situation of infeasibility in the L.P.P.

In the simplex method, an infeasible solution is indicated by looking at the final simplex tableau. If all the $Z_j - C_j$ row entries are of the proper sign to imply optimality, yet an artificial variable still exists in the solution mix, it means there is no solution which could satisfy all the constraints.

(II) Unbounded Solutions Unbounded feasible regions do not have finite solutions. In any simplex tableau, if the replacement ratios (in order to decide which variable to leave the solution mix) turn out to be negative or undefined, it indicates that the given L.P.P. is unbounded.

(III) Tie for Smallest Ratio Whenever there is tie for smallest replacement ratios (that is when there is more than one smallest ratios to decide which variable to leave), it is said to be the condition of degeneracy in the L.P.P. In other words, tie for the smallest ratio indicates degeneracy.

EXERCISE 18.3

Solve the following problems by Graphical and (Ex. 1-5) Simplex methods

1. Minimise $f = 2P + 3Q$
 Subject to $P + Q \geq 6$
 $2P + Q \geq 7$
 $P + 4Q \geq 8$
 P and $Q \geq 0$
2. Minimise $f = 6A + 24B$
 Subject to $A + 2B \geq 3$
 $A + 4B \geq 4$
 $A + B \geq 0$
 A and $B \geq 0$

Obtain the solution of the following by applying minimisation Simplex technique. Also verify your solution by applying the Simplex method to the duals of the problems.

$$\begin{array}{ll} 3. & \text{Minimise} \quad f = 4x + 6y + z \\ & \text{Subject to} \quad x + y \geq 0 \\ & \quad y + 4z \geq 20 \\ & \quad 3x + z \geq 40 \\ & \quad x, y, z \geq 0 \end{array}$$

$$\begin{array}{ll} 4. & \text{Minimise} \quad f = 12X_1 + 42X_2 \\ & \text{Subject to} \quad X_1 + 2X_2 \geq 3 \\ & \quad X_1 - 4X_2 \geq 4 \\ & \quad 3X_1 + X_2 \geq 3 \\ & \quad X_1 \text{ and } X_2 \geq 0 \end{array}$$

$$\begin{array}{ll} 5 & \text{Minimise} \quad f = X + Y \\ & \text{Subject to} \quad X + 8Y \geq 1 \\ & \quad 8X - 3Y \geq -1 \\ & \quad X \text{ and } Y \geq 0 \end{array}$$

$$\begin{array}{ll} 6 & \text{Minimise} \quad f = 8U + 10V + 6t \\ & \text{Subject to} \quad U + 4V + 6t \geq 12 \\ & \quad 3U + V + t \geq 20 \\ & \quad U, V, t \geq 0 \end{array}$$

$$\begin{array}{ll} 7 & \text{Maximise} \quad f = 60x_1 + 60x_2 + 90x_3 + 90x_4 \\ & \text{Subject to} \quad 100x_1 + 100x_2 + 100x_3 + 100x_4 \geq 1500 \\ & \quad 7x_1 + 5x_2 + 3x_3 + 2x_4 \leq 100 \\ & \quad 8x_1 + 5x_2 + 10x_3 + 15x_4 \leq 100 \\ & \quad x_1, x_2, x_3, x_4 \geq 0 \end{array}$$

$$\begin{array}{ll} 8. & \text{Maximise} \quad f = x_1 - x_2 \\ & \text{Subject to} \quad 2x_1 + x_4 \leq 2 \\ & \quad x_1 + x_2 \leq 5 \\ & \quad x_1 \text{ and } x_2 \geq 0 \end{array}$$

Show that $x_1 = 4$ and $x_2 = 1$ is an optimal solution

$$\begin{array}{ll} 9. & \text{Minimise} \quad g = 5x + 4y \\ & \text{Subject to} \quad 2x + y \geq 1 \\ & \quad x + 3y \geq 2 \\ & \quad -x + y \geq 2 \end{array}$$

where x and y are not restricted to be non negative (Tackle the problem graphically)

10. Find graphically the solution of the problem maximising

$$\begin{aligned}\text{Subject to : } & 5x - 2y + 3z \\ & 3x - 4y + 2z \leq 9 \\ & 2x + 7y + 5z \leq 11\end{aligned}$$

(Hint. The dual will contain only 2 variables in the objective function and 3 constraints which can be handled graphically)

11. Minimise $g = x_1 - 3x_2 + 2x_3$
 Subject to $3x_1 - x_2 + 2x_3 \geq 7$
 $2x_1 + 4x_2 \geq 12$
 $-4x_1 + 3x_2 + 8x_3 \geq 10$
 $x_1, x_2, x_3 \geq 0$
 $[x_1 = 4, x_2 = 5, x_3 = 0, g_2 = -11]$

12. Show that in the problem minimise : $3x + 4y$ subject to the constraints

$$\begin{aligned}x + 2y &\geq 6, & 2x + 3y &\geq 9; & 3x - y &\geq 7; \\ x - 2y &\geq 1 \text{ and } 3x - 4y &\geq 0\end{aligned}$$

three of the constraints are redundant, Solve this problem

13. Maximise $f = 2a_1 + 4a_2 + a_3 + a_4$
 Subject to : $a_1 + 3a_2 + a_4 \leq 4$
 $2a_1 + a_4 \leq 3$
 $a_2 + 4a_3 + a_4 \leq 3$
 $a_1, a_2, a_3, a_4 \geq 0$

14. Maximise : $f = -x_1 + 2x_2$
 Subject to : $5x_1 - 2x_2 \leq 3$
 $x_1 + x_2 \leq 1$
 $-3x_1 + x_2 \leq 3$
 $-3x_1 - 3x_2 \leq 2$
 $x_1, x_2 \geq 0.$

15. A farmer has the option of producing one or more of four different products : Barley, Rye, Oats and Wheat, whose unit profits are Rs. 3, 4, 1 and 7, respectively. The farmer has 7 man-months of labour, 3 acres of land and 8 tractor-months available for the whole job

Barley requires 8 man-months of labour, 2 acres of land 1 tractor-month

Rye requires 3 man-months of labour, 1 acre of land and 5 tractor-months.

Wheat requires 8 man months of labour, 5 acres of land and 2 tractor-months

Write out the complete linear programming formulation of the problem in terms of profit maximisation and find out the optimal solution of the problem

16 Product x_1 requires 1 minute of work per unit on machine A and 16 minutes of work per unit on machine B. Product x_2 requires 2 minutes of work per unit on machine A and 8 minutes of work per unit on machine B.

During a certain period, 100 minutes are available on machine A and 640 minutes on machine B. Each unit of x_1 has a profit margin of 90 paise and each unit of x_2 has a profit margin of 60 paise.

What amounts of x_1 and x_2 should be scheduled to maximise profits? Set up a Linear Programming model for this problem and illustrate with a diagram.

17 "Hidden in every competitive general equilibrium system is a maximum problem for the value of output and a minimum problem for factor returns."

Demonstrate this by the use of Linear Programming model.

18 Two products A and B when manufactured must pass through machine operations I, II and III. The machine hours required per unit produced are as follows

	I	II	III
A	2	4	3
B	1/2	2	1

The total machine hours available on machines I, II and III are 46, 100 and 300 hours respectively. Product A sells for Rs. 6 per unit and product B for Rs. 4 per unit. What combination of A and B should be manufactured in order to maximise profit?

19. A profit maximising firm can produce one or more of the four products W, X, Y, Z whose unit profits are Rs. 6, Rs. 5 1/2, Rs. 9 and Rs. 8 respectively. Each product utilises two different types of raw materials and the total available raw materials of the first type is 120 kg and that of the second type is 100 kg. Each unit of W, X, Y, Z requires 7, 5, 3 and 2 kg of raw material of the first type and 3, 5, 10 and 15 kg of raw material of the second type respectively. Each product requires one machine-hour and total available machine-hours are 150. The labour hour requirements of the products are 0.5, 2, 1.9 and 3.1 respectively and the labour hours available are 320.

Write out the complete Linear programming formulation of the problem in terms of profit maximisation and find out the optimum feasible solution.

[Hint The above problem can be put in the following form]

Products	W	X	Y	Z	
Raw material I	7	5	3	2	≤ 120
Raw material II	3	5	10	15	≤ 100
Machine Hrs	1	1	1	1	≤ 150
Labour Hrs	0.5	2	1.9	3.1	≤ 320
Unit profits	6	5 1/2	9	8	

Apply Simplex technique now]

20 A firm can produce one or more of five different products W, X, Y, Z and V with unit profit of Rs. 0.40, 0.50, 1.60, 1.40 and 1.30 respectively. These products require four different types of raw materials and total available quantities of raw materials of types I, II, III and IV are 100 kg, 60 kg, 20 kg, and 80 kg, respectively. The raw material requirements per unit of B, C, D, E and F are

Raw materials	W	X	Y	Z	V
Type I	2.1	2	5.1	5	5
II	3	0	8.1	0	0
III	0	3	0	8.2	0
IV	0	0	0	0	10.3

Write out a complete linear programming formulation of the problem in terms of profit maximisation, and find out the optimum feasible solution.

[Hint. The above problem can be put as follows]

Maximise	$f = 0.40W + 0.50X + 1.60Y + 1.40Z + 1.30V$
Subject to	$2.1W + 2X + 5.1Y + 5Z + 5V \leq 100$
	$3W + 8.1Y \leq 60$
	$3X + 8.2Z \leq 20$
	$10.3V \leq 80$

Apply Simplex technique now]

21 A manufacturer produces three products A, B and C whose profits per unit are Rs. 5, Rs. 7 and Rs. 10 respectively. These products require processing through four machine centres. The available machine hours in each centre and the input coefficients are shown below.

	Machine Centres				Machine hours Available	
	I	II	III	IV		
A	0.2	0.6	0.2	0.4	Centre I	320
B	0.3	0.5	0.5	0.2	Centre II	400
C	0.4	0.4	0.3	0.8	Centre III	160
					Centre IV	225

Carefully set up the problem for Simplex Solution and state the optimum feasible solution.

[Hint The above problem can be set up in the following way

Maximise $f \ 5a + 7b + 10c$

Subject to $0.2a + 0.3b + 0.4c \leq 320$

$0.6a + 0.5b + 0.4c \leq 400$

$0.2a + 0.5b + 0.3c \leq 160$

$0.4a + 0.2b + 0.8c \leq 225$

Apply Simplex technique to get the optimum feasible solution]

22. A firm manufactures three types of products A, B, and C. These products must pass through at least two of the three processing centres. Each processing centre has a limited amount of production time per month, as given below

Centre I 320 hrs Centre II 400 hrs and Centre III 160 hrs

Product A requires 0.2 hrs in centre I, 0.6 hr in centre II and zero in III centre

Product B requires 0.3 hrs in centre I, 0.6 hrs in centre II and zero in centre III

Product C requires 0.4 hrs in centre I, 0.4 hr in centre II and 0.8 in centre III

Profit per unit is Rs. 6 for A, Rs. 7 for B and Rs. 10 for C

Set up the problem for simplex solution. State the optimum feasible solution and the maximum profit

23. A farmer produces 2 products A and B with unit profits of Rs. 6 and Rs. 3 respectively. These products require two types of fertilisers and the total available quantities of fertiliser are 60 kg (Type I) and 200 kg (Type II). The total man hours and tractor hours available for the whole job are 300 and 80 respectively. The input requirements per unit of output are as follows

	A	B
Fertiliser I	2	1/4
Fertiliser II	4	2
Man hours	3	1
Tractor hours	1	4

Write out a complete linear programming formulation of the problem and find the optimum feasible solution

[Hint. The formulation of the above problem would be as follows

Let X units of product A be produced

and Y " " B "

$$\begin{aligned}
 &\text{Maximise} && f \quad 6X + 3Y \\
 &\text{Subject to} && 2X + \frac{1}{4}Y \leq 60 \\
 &&& \begin{cases} 4X + 2Y \leq 200 \\ 3X + Y \leq 300 \end{cases} \\
 &&& (0) X + 4Y \leq 80]
 \end{aligned}$$

24 A firm produces two products *A* and *B* with unit profits Rs. 30 and Rs. 50 respectively

Product *A* requires 2 units of labour hrs

1 unit of machine hr and raw material of Rs. 3

Product *B* requires 5 units of labour hrs

4 unit of machine hr and raw material of Rs. 2

Total available resources are 90 units of labour hours 75 units of machine hours and raw material of Rs. 120

$$\begin{aligned}
 &\text{25 Minimise} && 3x + 4y \\
 &\text{Subject to} && 5x + 10y \geq 800 \\
 &&& 15x + 10y \geq 1200 \\
 &&& x + y \geq 100 \\
 &&& x, y \geq 0
 \end{aligned}$$

Drop the constraint $x + y \geq 100$ and solve again Compare

Hence show (logically or mathematically) that increase in the number of constraints reduces the maximum value of the objective function and increases the minimum value of the objective function

$$\begin{aligned}
 &\text{26 Minimise the cost function} && 2x + y \\
 &\text{Subject to} && 0.10x + 0.50y \geq 5 \\
 &&& 0.45x + 0.15y \geq 7 \\
 &&& 0.25x + 0.15y \geq 6 \\
 &&& 0.20x + 0.20y \geq 6 \\
 &&& x, y \geq 0
 \end{aligned}$$

27 Maximise

$$5Q + 7R + 6S + 6T$$

Subject to

$$0.10Q + 0.45R + 0.25S + 0.20T \leq 2$$

$$0.50Q + 0.15R + 0.15S + 0.20T \leq 1$$

(not more than 2 variables can have positive values)

28. The following information is available regarding furniture manufacturing

	Departments			Profit per unit
	I (Roughing)	II (Finishing)	III (Painting)	
Table (Hours per unit)	10	7	2	Rs. 12
Chair (Hours per unit)	2	3	4	Rs. 3
Stools (Hours per unit)	1	2	1	Rs. 1
Total hours available	100	77	80	

Find the product mix to maximise profit. What time will be unused in each department?

Change the objective to maximisation of output (T+C+S). Find the optimal product mix.

29. A firm produces three products each of which uses two inputs, product and input prices are also given

Product	Inputs		Price per unit
	1	2	
I	1	1	6
II	6/5	6/5	6
III	3	1	13
Price per unit	3	2	

The maximum amount of inputs available are 18 and 14 respectively. The fixed cost is C. Find the profit maximising product mix. What is the value of the slack variables? Interpret.

Form the dual of the above example and solve.

Chapter 19

Input-Output Analysis

19.1. Introduction

Input-output analysis is a technique which was invented by Prof Wassily W Leontief in the year 1951. This technique deals with the type of problems, one of which may be described in the following words:

"What should be the level of output of each industry with the existing technology so that the total output goal for consumer and industrial use of its product gets fully satisfied, or alternatively, what level of output of each producing sector in an economy can bring about equilibrium for its product in the economy as a whole."

The basic idea behind this problem is quite simple to understand. Since inputs of one industry are the outputs of another industry and vice versa, ultimately their mutual relationship must lead to equilibrium between supply and demand in the economy consisting of n industries. For example, the output of industry 1 is needed as an input in many other industries and perhaps for that industry itself, naturally, therefore, the total output level of industry 1 must take account of the input requirements of all the industries in the economy. Exactly in the same way since the output of industry n enters into other industries as their "input requirements" the total output of n th industry must be one that is consistent with all input requirements so as to avoid any bottlenecks anywhere in the economy.

Thus, the essence of input-output analysis is that, given certain technological coefficients and final demand, each endogenous sector would find its output uniquely determined as a linear combination of multi sector demand.

Let us suppose that an economic system consists of n producing sectors. In order to avoid bottlenecks in the economy, the total output of each producing sector must satisfy the total demand for its product which, in fact, would arise because

- (1) Its product is being used as an intermediate product (i.e., input) elsewhere in the industrial structure of production, and
- (2) Its product is used for household consumption, capital formation, Government consumption or for export.

For example, total output of agricultural sector may be used (or demanded)

(1) as an input in food or other manufacturing sector (grain for producing bread or cotton for producing cloth), and

(2) as a final consumption, Government or household (vegetables or grain) and/or as an export demand

Let us assume that an economy consists of 4 producing sectors only, and that the production of each sector is being used as an input in all the sectors and is used for final consumption

Suppose (i) X_1, X_2, X_3 and X_4 are total outputs of the 4 sectors ,

(ii) F_1, F_2, F_3 and F_4 are the amounts of final demand, consumption, capital formation and exports for output of these sectors

We can now translate the distribution of total product of 4 producing sectors in the following way

Input-Output Transaction Table

Producing sector No	Total output of the sector	Input requirements of Producing sectors				Requirements for final uses
		X_1	X_2	X_3	X_4	
1	2	3	4	5	6	7
1	X_1	X_{11}	X_{12}	X_{13}	X_{14}	F_1
2	X_2	X_{21}	X_{22}	X_{23}	X_{24}	F_2
3	X_3	X_{31}	X_{32}	X_{33}	X_{34}	F_3
4	X_4	X_{41}	X_{42}	X_{43}	X_{44}	F_4
Primary Input (Labour)	Total Primary Input = L	L_1	L_2	L_3	L_4	—

We derive two important equations from the above table

(1) Columns 3, 4, 5 and 6 of the above table give us total inputs (from all sectors) utilised by each sector for its production. In other words, col 3 gives the production function of sector 1 and col 6 represents the production function of sector 4

$$X_1 = f_1(X_{11}, X_{21}, X_{31}, X_{41}, L_1)$$

$$X_2 = f_2(X_{12}, X_{22}, X_{32}, X_{42}, L_2)$$

$$X_3 = f_3(X_{13}, X_{23}, X_{33}, X_{43}, L_3)$$

$$X_4 = f_4(X_{14}, X_{24}, X_{34}, X_{44}, L_4)$$

In general terms, if there are 'n' number of producing sectors then the production function of sector n will be represented by

$$X_n = f_n(X_{1n}, X_{2n}, X_{3n}, \dots, X_{nn}, L_n)$$

(2) Rows of the table give us the equality between the demand and supply of each product

$$X_1 = X_{11} + X_{12} + X_{13} + X_{14} + F_1$$

$$X_2 = X_{21} + X_{22} + X_{23} + X_{24} + F_2$$

$$X_3 = X_{31} + X_{32} + X_{33} + X_{34} + F_3$$

$$X_4 = X_{41} + X_{42} + X_{43} + X_{44} + F_4$$

$$L = L_1 + L_2 + L_3 + L_4$$

In general terms, if there are n producing sectors

$$X_1 = X_{11} + X_{12} + X_{13} + \dots + X_{1n} + F_1$$

$$X_2 = X_{21} + X_{22} + X_{23} + \dots + X_{2n} + F_2$$

$$\dots \dots \dots \dots \dots$$

$$X_n = X_{n1} + X_{n2} + X_{n3} + \dots + X_{nn} + F_n$$

$$L = L_1 + L_2 + L_3 + L_4 + \dots + L_n$$

That is
$$X_i = \sum_{j=1}^n X_{ij} + F_i, \text{ and } L = \sum_{i=1}^n L_i$$

Here X_i = Total output of i th sector

X_{ij} = Output of i th sector used as input in j th sector, and

F_i = Final demand for i th sector

The above identity states that all the output of a particular sector could be utilised either as an input in one of the producing sectors of the economy and/or as a final demand. Basically, therefore, input-output analysis is nothing more than finding the solution of these simultaneous equations.

19.2 Assumptions

The economy can be meaningfully divided into a finite number of sectors (industries)

1 Each industry produces only one homogeneous output. No two products are produced jointly, but if at all there is such a case then it is assumed that they (products) are produced in fixed proportions.

2 Each producing sector satisfies the properties of linear homogeneous production function—in other words, production of each sector is subject to constant returns to scale so that a fold change in every input will result in an exactly k fold change in the output.

3 A far stronger assumption is that each industry uses a fixed input ratio for the production of its output, in other words, input requirements per unit of output in each sector remain fixed and constant. The level of output in each sector (industry) uniquely determines the quantity of each input which is purchased.

19.3. The Technological Coefficient Matrix

From the assumption of fixed input requirements we see that in order to produce one unit of j th commodity, the input used of i th commodity must be a fixed amount, which we denote by a_{ij} , thus $a_{ij} = (X_{ij})/X_j$. If X_j represents the total output of the j th commodity (or j th producing sector) the input requirements of i th commodity will be equal to $a_{ij}X_j$ or, $X_{ij} = a_{ij}X_j$.

As such we can now put the input output transaction table in terms of technical coefficients as follows

Producing sector No.	Total output of the sector	Purchases →	Input requirements of Producing sectors				Requirements for final uses
			X ₁	X ₂	X ₃	X ₄	
1	2		3	4	5	6	7
	Sales →						
1	X ₁		a ₁₁ X ₁₁	a ₁₂ X ₁₂	a ₁₃ X ₁₃	a ₁₄ X ₁₄	F ₁
2	X ₂		a ₂₁ X ₂₁	a ₂₂ X ₂₂	a ₂₃ X ₂₃	a ₂₄ X ₂₄	F ₂
3	X ₃		a ₃₁ X ₃₁	a ₃₂ X ₃₂	a ₃₃ X ₃₃	a ₃₄ X ₃₄	F ₃
4	X ₄		a ₄₁ X ₄₁	a ₄₂ X ₄₂	a ₄₃ X ₄₃	a ₄₄ X ₄₄	F ₄
Primary Input (Labour)	Total Primary Input L →		l ₁ X ₁	l ₂ X ₂	l ₃ X ₃	l ₄ X ₄	

It should be noted that all these coefficients are non negative (>0)

The above table gives us the total output of each sector in terms of technical coefficients, and if there are "n" producing sectors

$$X_1 = a_{11}X_1 + a_{12}X_2 + a_{13}X_3 + \dots + a_{1n}X_n + F_1$$

$$X_2 = a_{21}X_1 + a_{22}X_2 + a_{23}X_3 + \dots + a_{2n}X_n + F_2$$

$$X_n = c_n X_1 + a_{n2} X_2 + a_{n3} X_3 + \dots + a_{nm} X_m + F_n$$

$$L = l_1 X_1 + l_2 X_2 + l_3 X_3 + l_4 X_4$$

$$X_i = \sum_{j=1}^n a_{ij} X_j + F_i, \quad (i = 1, 2, \dots, n)$$

$$\text{and, } L = \sum l_i X_i$$

The equations may be put in matrix notations

$$\begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} + \begin{pmatrix} F_1 \\ F_2 \\ \vdots \\ F_n \end{pmatrix}$$

$$X = AX + F \quad \text{and}$$

$$L = \sum l_i X_i$$

19.4 Closed and Open Input Output Models

In the above example besides n industries our model contains exogenous sector of final demand which supplies primary input factors (labour services — which are not produced by n industries) and consumes the outputs of the n producing industries (not as input). Such an input output model is known as **open model**. It includes exogenous sectors in terms of "final demand bill" — along with the endogenous sectors in terms of n producing sectors. Input output model which has endogenous final demand vector is known as **closed input-output model**.

19.5 Coefficient Matrix and Open Model

Our open model in matrix notation is given by

$$X = AX + F$$

where A is the input coefficient matrix, F is the final demand vector and X is the total output matrix.

The input coefficient matrix represented by $[a_{ij}]$

$$= \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix}$$

is of great importance. Each column of this matrix specifies the input requirements for the production of one unit of a particular commodity. The second column, for example, states that to produce a unit of commodity 2, the inputs needed are a_{12} units of commodity 1, a_{22} units of commodity 2, a_{32} units of commodity 3, and a_{n2} units of commodity n .

If no industry uses its own product as an input then,

$$a_{ij} = \begin{pmatrix} 0 & a_{12} & a_{13} & a_{1n} \\ a_{21} & 0 & a_{23} & a_{2n} \\ a_{31} & a_{32} & 0 & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & 0 \end{pmatrix}$$

(Note Elements of matrix will all be zero whenever the sectors do not trade with each other. It should be noted that no coefficient can be negative, there can be no negative inputs.)

Coefficient Matrix in Value Terms

Again, if we assume prices of all the outputs to be given, the ij th elements of $[a_{ij}]$ matrix will represent the amount of i th commodity in money terms needed for producing "a rupee worth" of j th commodity. For example, if $a_{ij} = 0.35$, it means that 35 paise worth of i th commodity is required as an input for producing a rupee worth of j th commodity.

Also, in view of the presence of exogenous sector (which supplies primary inputs) the sum of the elements of each input coefficient column $[a_{ij}]$ must be less than 1. Each column sum represents the partial input cost (excluding the cost of primary input) incurred in producing a rupee worth of some commodity, if this sum is greater than or equal to one rupee, the production will not be economically justifiable. Symbolically this fact may be stated as

$$\sum_{i=1}^n a_{ij} < 1 \quad (j = 1, 2, \dots, n) \quad \text{and each } a_{ij} \text{ is non negative, i.e., either zero or greater than zero}$$

The cost of the primary inputs (which is also termed as value added) needed in producing a unit of j th commodity should be

$$\left(1 - \sum_{i=1}^n a_{ij} \right)$$

Note here that a_{ij} 's are in value terms

Solution of Open Model

Let us consider an economy with n industries. If producing sector 1 is to produce an output just sufficient to meet the input requirements of the n industries as well as the final demand of the exogenous sector, its output level X_1 must satisfy the following equations

$$X_1 = a_{11}X_1 + a_{12}X_2 + a_{13}X_3 + a_{1n}X_n + F_1$$

$$\text{or, } (1 - a_{11})X_1 - a_{12}X_2 - a_{13}X_3 - a_{1n}X_n = F_1$$

For the entire set of n industries, the correct output levels, therefore, can be symbolised by the following set of n linear equations

$$\begin{aligned} (1 - a_{11})X_1 - a_{12}X_2 - a_{13}X_3 - a_{1n}X_n &= F_1 \\ -a_{21}X_1 + (1 - a_{22})X_2 - a_{23}X_3 - a_{2n}X_n &= F_2 \end{aligned}$$

$$-a_{n1}X_1 - a_{n2}X_2 - a_{n3}X_3 - a_{nn}X_n = F_n$$

In the matrix notation this may be written as

$$\begin{pmatrix} (1 - a_{11}) & -a_{12} & -a_{13} & -a_{1n} \\ -a_{21} & (1 - a_{22}) & -a_{23} & -a_{2n} \\ -a_{31} & -a_{32} & (1 - a_{33}) & -a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ -a_{n1} & -a_{n2} & -a_{n3} & (1 - a_{nn}) \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \\ X_n \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ \vdots \\ F_n \end{pmatrix}$$

$$[I - A] X = F$$

$$\text{that is, } X = [I - A]^{-1} F$$

We arrive at the same result if we take the equation derived under section 19.3

$$X = AX + F$$

$$\text{or, } X - AX = F$$

$$\text{or, } [I - A] X = F$$

$$X = [I - A]^{-1} F$$

where A is the given matrix of input coefficients, while X and F are the vectors of output and final demand of each producing sector. If $|I - A| \neq 0$, then $[I - A]^{-1}$ exists, we can then estimate for either of the two matrices X and F by assuming one of them to be given exogenously.

It is here that we observe that assumptions made in input-output analysis go a long way in making the problem simplified.

For example, with the assumption of linear homogeneous function, it is possible to write a linear equation of each producing sector which then can be easily transformed into matrix notation.

On the other hand, as long as the input coefficients remain fixed (as assumed) the matrix A will not change or $[I - A]$ will not vary. Therefore, in finding the solution of $X = [I - A]^{-1}$ only one matrix inversion need be performed even if we are to consider thousands of different final demand vectors according to alternative development targets. Hence such an assumption of fixed technical coefficients has meant considerable saving in computational effort.

Ex. 1. Suppose there are only three industries in an economy and we have to estimate the output of each (sector) industry with the given input coefficient matrix and final demand as follows (the coefficient matrix is in value terms) :

$$A = \begin{bmatrix} 0.3 & 0.5 & 0.2 \\ 0.2 & 0 & 0.5 \\ 0.1 & 0.3 & 0.1 \end{bmatrix} \text{ and } F = \begin{bmatrix} 100 \\ 40 \\ 50 \end{bmatrix} \text{ million rupees}$$

Here we note that 3-column sums of A are $(0.3 + 0.2 + 0.1) = 0.6$, $(0.4 + 0 + 0.3) = 0.7$ and $(0.2 + 0.5 + 0.1) = 0.8$, which are less than 1 in each case. In other words, $(1 - 0.6) = 0.4$, $(1 - 0.7) = 0.3$ and $(1 - 0.8) = 0.2$ is the maximum amount of primary input which can be used for producing "a rupee worth" of the three commodities (P , Q and R) respectively.

$$\text{Since } A = \begin{bmatrix} 0.3 & 0.4 & 0.2 \\ 0.2 & 0 & 0.5 \\ 0.1 & 0.3 & 0.1 \end{bmatrix}, [I - A] = \begin{bmatrix} +0.7 & -0.4 & -0.2 \\ -0.2 & +1 & -0.5 \\ -0.1 & -0.3 & +0.9 \end{bmatrix}$$

Substituting these values in $X = [I - A]^{-1}F$, we get

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 0.7 & -0.4 & -0.2 \\ -0.2 & 1 & -0.5 \\ -0.1 & -0.3 & 0.9 \end{bmatrix}^{-1} \begin{bmatrix} 100 \\ 40 \\ 50 \end{bmatrix}$$

$$\text{But } \begin{bmatrix} 0.7 & -0.4 & -0.2 \\ -0.2 & 1 & -0.5 \\ -0.1 & -0.3 & 0.9 \end{bmatrix}^{-1} = \frac{1}{0.401} \begin{bmatrix} 0.75 & 0.42 & 0.40 \\ 0.23 & 0.61 & 0.39 \\ 0.16 & 0.25 & 0.62 \end{bmatrix}$$

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \frac{1}{0.401} \begin{bmatrix} 0.75 & 0.42 & 0.40 \\ 0.23 & 0.61 & 0.39 \\ 0.16 & 0.25 & 0.62 \end{bmatrix} \begin{bmatrix} 100 \\ 40 \\ 50 \end{bmatrix}$$

$$X_1 = \frac{1}{0.401} \{0.75(100) + 0.42(40) + 0.40(50)\}$$

$$= \text{Rs } 270 \text{ million (approx)}$$

$$X_2 = \frac{1}{0.401} \{0.23(100) + 0.61(40) + 0.39(50)\}$$

= Rs. 167 million (approx), and

$$X_3 = \frac{1}{0.401} \{0.16(100) + 0.25(40) + 0.62(50)\}$$

= Rs. 142 million (approx)

We may ask another question. What amount of primary input would be needed to get the above amount of output-mix (of Rs. 279, Rs. 161 and Rs. 142 million) in three producing sectors? Would the amount so required be consistent with what is available in the economy?

We have already mentioned above that the maximum amounts of primary inputs (in money terms) required for producing 'a rupee worth' of commodities in the three sectors are : 0.4, 0.3 and 0.2. Therefore, the total primary input requirements (of the three sectors) will be equal to

$$0.4(279) + 0.3(167) + 0.2(142) = \text{Rs. } 190.1 \text{ million}$$

We may also, therefore, say that the specific final demand,

$$\begin{bmatrix} 100 \\ 40 \\ 50 \end{bmatrix}$$

will be feasible if and only if the available amount of primary input is at least Rs. 190 million. If this amount falls short, the calculated production targets will have to be revised downwards accordingly.

19.6. The Hawkins-Simon conditions (The viability of the system)

Many a time input-output matrix solution may give outputs expressed by negative numbers. If our solution gives negative outputs, it means that more than one tonne (or any unit) of that product is used up in the production of every one tonne of that product; which is definitely an unrealistic situation. Such a system is not a viable system.

Hawkins-Simon conditions guard against such eventualities.

Our basic equation is $X = [I - A]^{-1} F$, in order that this does not give negative numbers as a solution, the matrix $[I - A]$, which in fact is :

$$\begin{pmatrix} (1 - a_{11}) & -a_{12} & -a_{13} & \dots & -a_{1n} \\ -a_{21} & (1 - a_{22}) & -a_{23} & \dots & -a_{2n} \\ -a_{31} & -a_{32} & (1 - a_{33}) & \dots & -a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ -a_{n1} & -a_{n2} & -a_{n3} & \dots & (1 - a_{nn}) \end{pmatrix}$$

should be such that

(1) the determinant of the matrix must always be positive, and

(2) the diagonal elements $(1 - a_{11}), (1 - a_{22}), (1 - a_{33}), \dots, (1 - a_{nn})$ should all be positive or, in other words, elements $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ should all be less than one. Thus, one unit of output of any sector should use not more than 1 unit of its own output.

These are called *Hawkins-Simon conditions*.

Ex. 2. Suppose $[A] = \begin{bmatrix} 0.8 & 0.2 \\ 0.9 & 0.7 \end{bmatrix}$

$$[I - A] = \begin{bmatrix} 0.2 & -0.2 \\ -0.9 & 0.3 \end{bmatrix}$$

and the value of determinant of $[I - A]$ will be $0.06 - 0.18 = (-) 0.12$ which is less than zero.

As such Hawkins-Simon conditions are not satisfied.

No solution will be possible in this case.

Ex. 3. The following inter-industry transactions table was constructed for an economy for the year 1990.

Industry	1	2	Final Consumption	Total
1	500	1 600	400	2,500
2	1,750	1 600	4 650	8 000
Labour	250	4 800	—	5 050
Total	2,500	8 000	5 050	15,550

Construct technology coefficient matrix showing direct requirements.

Does a solution exist for this system?

Technology matrix showing direct requirements per rupee of output is obtained by dividing each input by the total output of the sector.

That is,

$$a_{11} = \frac{500}{2,500} = 0.20 \left(= \frac{X_{11}}{X_1} \right)$$

$$a_{12} = \frac{1,600}{8,000} = 0.20 \left(= \frac{X_{12}}{X_2} \right)$$

$$a_{21} = \frac{1,750}{2,500} = 0.70 \left(= \frac{X_{21}}{X_1} \right)$$

$$a_{22} = \frac{1,600}{8,000} = 0.20 \left(= \frac{X_{22}}{X_2} \right)$$

Hence technology matrix is given by

	Industry	1	2
A =	1	0.20	0.20
	2	0.70	0.20
	Labour	0.10	0.60

$$\text{and } [I - A] = \begin{pmatrix} 1 - 0.20 & -0.20 \\ -0.70 & 1 - 0.20 \end{pmatrix} = \begin{pmatrix} 0.80 & -0.20 \\ -0.70 & 0.80 \end{pmatrix}$$

$$|I - A| = \begin{vmatrix} 0.80 & -0.20 \\ -0.70 & 0.80 \end{vmatrix}$$

$$= 0.80 \times 0.80 - 0.20 \times 0.70 = 0.50$$

Since $|I - A|$ is positive and all elements of the principal diagonal of $(I - A)$ are also positive the Hawkins Simon conditions are satisfied. Hence, the empirical system has a solution.

Ex 4 Given

$$A = \begin{bmatrix} 0.1 & 0.3 & 0.1 \\ 0 & 0.2 & 0.2 \\ 0 & 0 & 0.3 \end{bmatrix}$$

and final demands are F_1 , F_2 and F_3 . Find the output levels consistent with the model. What will be the output levels if $F_1 = 20$, $F_2 = 0$ and $F_3 = 100$?

We know that

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = (I - A)^{-1} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

$$\text{Now } [I - A] = \begin{bmatrix} 0.9 & -0.3 & -0.1 \\ 0 & 0.8 & -0.2 \\ 0 & 0 & 0.7 \end{bmatrix}$$

Co-factors are as follows

$$A_{11} = \begin{vmatrix} 0.8 & -0.2 \\ 0 & 0.7 \end{vmatrix} = 0.56$$

$$A_{12} = - \begin{vmatrix} 0 & -0.2 \\ 0 & 0.7 \end{vmatrix} = 0$$

$$A_{13} = + \begin{vmatrix} 0 & 0.8 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{21} = - \begin{vmatrix} -0.3 & -0.1 \\ 0 & 0.7 \end{vmatrix} = 0.21$$

$$A_{22} = \frac{1}{0.9} \begin{vmatrix} 0.9 & -0.1 \\ 0 & 0.7 \end{vmatrix} = 0.63$$

$$A_{23} = \frac{1}{0.9} \begin{vmatrix} 0.9 & -0.3 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{31} = \frac{1}{0.8} \begin{vmatrix} -0.3 & -0.1 \\ 0.8 & 0.2 \end{vmatrix} = 0.14$$

$$A_{33} = \frac{1}{0.9} \begin{vmatrix} 0.9 & -0.1 \\ 0 & -0.2 \end{vmatrix} = 0.18$$

$$A_{33} = \frac{1}{0.9} \begin{vmatrix} 0.9 & -0.3 \\ 0 & 0.5 \end{vmatrix} = 0.72$$

Hence the value of the determinant is $0.9 \times 0.56 = 0.504$

$$\begin{aligned} \text{Hence, } (I-A)^{-1} &= \frac{1}{0.504} \begin{bmatrix} 0.56 & 0.21 & 0.14 \\ 0 & 0.63 & 0.18 \\ 0 & 0 & 0.72 \end{bmatrix} \\ &= \begin{bmatrix} 1.11 & 0.42 & 0.28 \\ 0 & 1.25 & 0.36 \\ 0 & 0 & 1.43 \end{bmatrix} \end{aligned}$$

$$\text{Hence, } \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 1.11 & 0.42 & 0.28 \\ 0 & 1.25 & 0.36 \\ 0 & 0 & 1.43 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

$$\text{Hence, } \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 1.11F_1 + 0.42F_2 + 0.28F_3 \\ 0 + 1.25F_2 + 0.36F_3 \\ 0 + 0 + 1.43F_3 \end{bmatrix}$$

From the given values of F_1 , F_2 and F_3 , we have

$$\begin{aligned} X_1 &= 1.11F_1 + 0.42F_2 + 0.28F_3 \\ &= 1.11 \times 20 + 0 + 0.28 \times 100 \\ &= 50.2 \\ X_2 &= 1.25F_2 + 0.36F_3 \\ &= 0 + 0.36 \times 100 = 36 \text{ and,} \\ X_3 &= 1.43F_3 = 143 \end{aligned}$$

Note It is to be noted that if the technology matrix is upper triangular, i.e., if all elements below the main diagonal are zero or nearly zero, then $(I-A)^{-1}$

matrix will also be triangular. In such cases, output X_3 depends on only the final demand of sector 3 and X_2 on final demands of sector 2 and 3.

Ex. 5. In the above example, if final demands change by 10, 10, 10, then what will be the change in sector outputs?

We have $X = (I - A)^{-1} F$

$\therefore \Delta X = (I - A)^{-1} \Delta F$

where ΔX and ΔF are vectors of small changes in outputs and final demands respectively. Hence

$$\Delta X = \begin{bmatrix} 1.11 & 0.42 & 0.28 \\ 0 & 1.25 & 0.36 \\ 0 & 0 & 1.43 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$

$$\Delta X_1 = 1.11 \times 10 + 0.42 \times 28 \times 10 = 18.1$$

$$\Delta X_2 = 0 + 1.25 \times 10 + 0.36 \times 10 = 16.1$$

$$\Delta X_3 = 14.3$$

Ex. 6 An economy produces only coal and steel. The two commodities serve as intermediate inputs in each other's production. 0.4 tonne of steel and 0.7 tonne of coal are needed to produce a tonne of steel. Similarly 0.1 tonne of steel and 0.6 tonne of coal are required to produce a tonne of coal. No capital inputs are needed. Do you think that the system is viable?

2 and 5 labour days are required to produce a tonne of coal and steel respectively. If the economy needs 100 tonnes of coal and 50 tonnes of steel, calculate the gross output of the two commodities and the total labour required.

Determine the equilibrium prices, if the wage rate is Rs. 10 per man-day.

Here the technology matrix is given as under:

	Steel	Coal	Final demand
Steel	0.4	0.1	50
Coal	0.7	0.6	100
Labour	5	2	

$$\therefore I - A = \begin{pmatrix} 0.6 & -0.1 \\ -0.7 & 0.4 \end{pmatrix}$$

The system will be variable if Hawkins-Simon conditions are satisfied

(i) The diagonal elements of $(I - A)$ are all positive

(ii) The determinant should be positive.

In present problem it is

$$0.6 \times 0.4 - 0.7 \times 0.1 = 0.24 - 0.07 = 0.17 > 0$$

Hence the system is *viable*

The required gross output is given by

$$\begin{aligned} X &= (I - A)^{-1} F \\ &= \begin{pmatrix} 0.6 & -0.1 \\ -0.7 & 0.4 \end{pmatrix}^{-1} \begin{pmatrix} 50 \\ 100 \end{pmatrix} \\ &= \frac{1}{0.17} \begin{pmatrix} 0.4 & 0.1 \\ 0.7 & 0.6 \end{pmatrix} \begin{pmatrix} 50 \\ 100 \end{pmatrix} = \frac{1}{0.17} \begin{pmatrix} 20 + 10 \\ 35 + 60 \end{pmatrix} = \begin{pmatrix} 176.5 \\ 558.8 \end{pmatrix} \end{aligned}$$

Total labour days required are $5 \times \text{steel output} + 2 \times \text{coal output}$

$$= 5 \times 176.5 + 2 \times 558.8 = 882.5 + 1117.6 = 2,000 \text{ labour days}$$

(For last part see Ex 7)

19.7. Solution for Two Industries

Assume that the following information is given to you

Sector of origin	Sector of destination		Final demand
	X_1	X_2	
X_1	a_{11}	a_{12}	F_1
X_2	a_{21}	a_{22}	F_2
Primary input	l_1	l_2	

where l_1 and l_2 are labour (primary input) requirement for producing one unit of commodity 1 and 2 respectively

Then
$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Hence,
$$X = (I - A)^{-1} F$$

$$\begin{aligned} &= \begin{pmatrix} 1 - a_{11} & -a_{12} \\ -a_{21} & 1 - a_{22} \end{pmatrix}^{-1} \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \\ &= \frac{1}{D} \begin{pmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{pmatrix} \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \frac{1}{D} \begin{pmatrix} A_{11}F_1 + A_{21}F_2 \\ A_{12}F_1 + A_{22}F_2 \end{pmatrix} \end{aligned}$$

$$X_1 = \frac{A_{11}F_1 + A_{21}F_2}{D}$$

$$X_2 = \frac{A_{12}F_1 + A_{22}F_2}{D}$$

where D is the determinant of $(I - A)$ and A_{11}, A_{21} , the co-factors of the matrix $I - A$

Here, of course

$$A_{11} = 1 - a_{22}, A_{12} = a_{21}, A_{21} = a_{12}, A_{22} = 1 - a_{11}, \text{ and}$$

$$D = (1 - a_{22})(1 - a_{11}) - a_{21}a_{12}$$

Total labour requirement will be

$$l_1 X_1 + l_2 X_2 = (l_1 \ l_2) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

$$= l X, \text{ where } l \text{ is the row vector of labour coefficients}$$

If there are two primary inputs then the primary input coefficients can be arranged as follows

	Sector 1	Sector 2
Input 1 (Say labour, L)	l_{11}	l_{12}
Input 2 (Say capital, K)	l_{21}	l_{22}

$$R = \begin{pmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \end{pmatrix}$$

Hence factor requirements will be given by

$$\begin{pmatrix} L \\ K \end{pmatrix} = \begin{pmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \end{pmatrix} (X) = RX = R(I - A)^{-1}F$$

$$\text{Factor demand vector} = R(1 - A)^{-1}F$$

Solution for 3 industries It can be seen easily that in the case of 3 industries, the solution will be

$$X_1 = \frac{1}{D} [A_{11}F_1 + A_{21}F_2 + A_{31}F_3]$$

$$X_2 = \frac{1}{D} [A_{12}F_1 + A_{22}F_2 + A_{32}F_3]$$

$$X_3 = \frac{1}{D} [A_{13}F_1 + A_{23}F_2 + A_{33}F_3]$$

and total labour requirements will be

$$L = l_1 X_1 + l_2 X_2 + l_3 X_3$$

This is equilibrium or consistency condition for 3 industries

What is the effect if final demand for any one commodity changes? Differentiating the expression for X_1, X_2, X_3 and L with respect to F_1, F_2 and F_3 we have

$$\frac{\partial X_1}{\partial F_1} = \frac{A_{11}}{D}, \quad \frac{\partial X_2}{\partial F_1} = \frac{A_{12}}{D}, \quad \frac{\partial X_3}{\partial F_1} = \frac{A_{13}}{D}$$

$$\begin{aligned} \frac{\partial L}{\partial F_1} &= l_1 \frac{\partial X_1}{\partial F_1} + l_2 \frac{\partial X_2}{\partial F_1} + l_3 \frac{\partial X_3}{\partial F_1} \\ &= \frac{1}{D} \{ l_1 A_{11} + l_2 A_{12} + l_3 A_{13} \} \end{aligned}$$

Thus all these multipliers are constant, given the technology matrix

Let us watch our results closely. They tell us that a unit change in final demand of commodity will necessitate a change in output of commodity 1, 2, and 3 by $\frac{A_{11}}{D}$, $\frac{A_{12}}{D}$ and $\frac{A_{13}}{D}$ respectively. If all the information is in value terms, this will mean that total output will change by

$$\frac{A_{11}}{D} + \frac{A_{12}}{D} + \frac{A_{13}}{D}$$

and total demand for labour will change by

$$\frac{l_1 A_{11} + l_2 A_{12} + l_3 A_{13}}{D}$$

Now $\left[\frac{A_{11} + A_{12} + A_{13}}{D} \right]$ is nothing but the sum of the first column of the inverse matrix A . In general, therefore, any column j of the inverse matrix shows how much output in each industry i will rise if the final demand for the product of industry j rises by a rupee or a unit of production. *These are thus matrix multipliers.*

19.8 Determination of Equilibrium Prices

Let the prices of commodities 1, 2, be p_1, p_2 respectively and the price of the primary factor inputs be w , then the technology matrix or

* The cost of primary inputs includes labour, interest, import, profit and depreciation. It includes all costs other than those on the inter-industrial purchases.

transaction matrix in quantity may be converted into that in value terms. The problem can be posed as follows

Sector	1	2	Final demand
1	$a_{11} X_1 p_1$	$a_{12} X_2 p_1$	$F_1 p_1$
2	$a_{21} X_1 p_2$	$a_{22} X_2 p_2$	$F_2 p_2$
Primary input	$l_1 X_1 w$	$l_2 X_2 w$	
Total cost	$a_{11} X_1 p_1 + a_{21} X_1 p_2 + l_1 X_1 w$	$a_{12} X_2 p_1 + a_{22} X_2 p_2 + l_2 X_2 w$	

With pure competition and free entry profit in each industry must be zero, i.e., receipts equal costs. Hence for the 1st industry receipts are output $\times$ price and cost is $a_{11} X_1 p_1 + a_{21} X_1 p_2 + l_1 X_1 w$. Similarly for the second industry

Hence, for equilibrium,

$$p_1 X_1 = a_{11} X_1 p_1 + a_{21} X_1 p_2 + l_1 X_1 w$$

$$p_2 X_2 = a_{12} X_2 p_1 + a_{22} X_2 p_2 + l_2 X_2 w$$

which simplify to

$$p_1 - a_{11} p_1 - a_{21} p_2 = l_1 w \quad \text{and,}$$

$$-a_{12} p_1 + p_2 - a_{22} p_2 = l_2 w$$

which can be put in matrix form as under

$$\begin{bmatrix} (1 - a_{11}) & -a_{21} \\ -a_{12} & (1 - a_{22}) \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} l_1 w \\ l_2 w \end{bmatrix}$$

Notice that the set of coefficients here are transposed, this matrix is transpose of $(I - A)$

$$\begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} (1 - a_{11}) & -a_{21} \\ -a_{12} & (1 - a_{22}) \end{bmatrix}^{-1} \begin{bmatrix} l_1 w \\ l_2 w \end{bmatrix} = [I - A']^{-1} V$$

where V is the vector of primary cost or value added coefficients

$$\therefore p_1 = \frac{1}{D} (A_{11} l_1 + A_{12} l_2) w$$

$$p_2 = \frac{1}{D} (A_{21} l_1 + A_{22} l_2) w$$

where A_{11}, A_{12} , etc are the co factors of the matrix $(I - A)$ as in the preceding cases

For the three industry case

$$P_1 = \frac{1}{D} [A_{11} l_1 + A_{12} l_2 + A_{13} l_3] w$$

$$P_2 = \frac{1}{D} [A_{21} l_1 + A_{22} l_2 + A_{23} l_3] w$$

$$P_3 = \frac{1}{D} [A_{31} l_1 + A_{32} l_2 + A_{33} l_3] w$$

We already know that total employment is given by

$$L = l_1 X_1 + l_2 X_2 + l_3 X_3$$

which gives, $\frac{\partial L}{\partial F_1} = \frac{1}{D} (l_1 A_{11} + l_2 A_{12} + l_3 A_{13})$

so that, $\frac{w \partial L}{\partial F_1} = \frac{1}{D} (l_1 w A_{11} + l_2 w A_{12} + l_3 w A_{13})$

$$\frac{1}{D} (w_1 A_{11} + w_2 A_{12} + w_3 A_{13})$$

Here, $w_1 = w l_1$, $w_2 = w l_2$, $w_3 = w l_3$ are wage costs per unit of output

Also, $P_1 = \frac{1}{D} [A_{11} w_1 + A_{12} w_2 + A_{13} w_3]$

or, $P_1 = w \frac{\partial L}{\partial F_1}$
 = wage rate $\times$ marginal employment
 = marginal labour cost

Ex. 7 In Ex. 6 determine the equilibrium prices if the wage rate is Rs 10 per labour day

Given $A = \begin{pmatrix} 0.4 & 0.1 \\ 0.7 & 0.6 \end{pmatrix}$
 $I - A = \begin{pmatrix} 0.6 & -0.1 \\ -0.7 & 0.4 \end{pmatrix}$
 $(I - A) = \begin{pmatrix} 0.6 & -0.7 \\ -0.1 & 0.4 \end{pmatrix}$
 $\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \begin{pmatrix} 0.6 & -0.7 \\ -0.1 & 0.4 \end{pmatrix}^{-1} \begin{pmatrix} 5 \times 10 \\ 2 \times 10 \end{pmatrix}$

$$= \frac{1}{0.17} \begin{bmatrix} 0.4 & 0.7 \\ 0.1 & 0.6 \end{bmatrix} = \begin{pmatrix} 50 \\ 20 \end{pmatrix}$$

$$= \frac{1}{0.17} \begin{bmatrix} 20 + 14 \\ 5 + 12 \end{bmatrix} = \begin{pmatrix} 200 \\ 100 \end{pmatrix}$$

$p_1 = \text{Rs } 200 \text{ per tonne}, p_2 = \text{Rs } 100 \text{ per tonne}$

19.8. Determination of Value Added

We already know that any row (i) shows the sales of j th industry and any column (j) shows the purchases of industry j

Now let all the transactions be given in value terms. Then rows give receipts and columns costs

Value added is defined in two ways

The gross output of each industry minus its inter industry purchases equals the value added by the industry. This value added equals the wage, interest and rental payments and the profits of the industry—i.e., the payments to primary inputs

Ex 8 Let us consider a hypothetical input-output table

Inter-industry purchases	Inter industry sales		Final demand	Gross output
	1	2		
1	500	1 600	400	2,500
2	1 750	1 600	4 650	8 000
Cost of primary input (value added)	250	4 800		5 050
Total cost/ Total receipts/ Gross output	2,500	8 000	5 050	15,550

For Industry 1, output = 2,500 of which

Inter industry purchases = 500 + 1750 = 2250

Hence value added = 2,500 - 2250 = 250

which is also cost of primary inputs

Similarly for industry 2,

Value added = 8000 - (1600 + 1600) = 4,800

Hence gross value added is 250 + 4,800 = 5,050

Now looking from the sales or disposal point of view,

Sales to final consumers by Industry 1

= Total sales - Inter industry sales

$$= 2,500 - (500 + 1,600) = 400$$

$$\text{By Industry 2} = 8,000 - (1,750 + 1,600) = 4,650$$

Hence total disposal to final demand is $= 400 + 4,650 = 5,050$ which is also equal to total value added

Payments to primary inputs = Factor cost = Value added = National income = sum of value added in each sector

Also sales to final demand = Final demand = Goods available for final demand = $NNP = NI$

$$\begin{aligned} &= (X_1 - x_{11} - x_{12}) + (X_2 - x_{21} - x_{22}) + \\ &= (X_1 + X_2 + \dots) - (x_{11} + x_{12} + x_{21} + \dots) \\ &= \text{Gross output} - \text{Inter industry transactions} \end{aligned}$$

Hence, aggregate final demand must equal aggregate gross value added (Note that in this simplified model we have neglected depreciation, taxation and foreign sector)

Thus, value added (V) is also the total factor cost. It is, therefore, equal to $l_1 w X_1 + l_2 w X_2 + l_3 w X_3 + \dots$

$$\begin{aligned} V &= (X_1 \quad X_2 \quad X_3) \begin{pmatrix} l_1 w \\ l_2 w \\ l_3 w \end{pmatrix} \\ &= X' \begin{pmatrix} l_1 w \\ l_2 w \\ l_3 w \end{pmatrix} = (l_1 w \quad l_2 w \quad l_3 w)' X \end{aligned}$$

Ex. 9. In our two-sector model of Ex. 6, we have

	1	2	F
1	0.4	0.1	50
2	0.7	0.6	100
l	5	2	

We had found out that $X_1 = 176.5$, $X_2 = 558.8$

Hence, given $w = 10$ and $l_1 w = 50$, $l_2 w = 20$, we obtain

$$\begin{aligned} V &= (X_1 \quad X_2) \begin{pmatrix} l_1 w \\ l_2 w \end{pmatrix} = (176.5 \quad 558.8) \begin{pmatrix} 50 \\ 20 \end{pmatrix} \\ &= 176.5 \times 50 + 558.8 \times 20 = 20,000 \text{ approximately} \end{aligned}$$

Alternatively, we had calculated total employment or factor demand at 2,000 days, hence $V = 2,000 \times 10$ (wage rate) = 20 000

Or V can be found from the value of the final demand, From Ex 7 we know that,

$$p_1 = 200, \quad p_2 = 100, \quad F_1 = 50 \text{ and } F_2 = 100$$

$$\text{So that,} \quad V = p_1 F_1 + p_2 F_2 = 20,000$$

Note We have taken final demand as one sector's (household sector's) demand only, but more generally final demand is composed of (i) Household consumption demand (ii) Investment demand, (iii) Government purchase of goods, (iv) Net export demand, and (v) Change in inventories

Similarly, we have taken labour as the only primary input, but land and capital are other primary inputs and labour can also be divided into many types

19.9 Coefficient Matrix and Closed Model

We shall now examine whether we will be able to estimate F or X if the model is changed into closed one. If the exogenous sector (final demand bill) of the open input output model is absorbed into the system of endogenous sectors, the model would turn into a closed one. In such a model final demand bill and primary inputs will not appear any more, rather in their place, we shall have the input requirements and output of this newly conceived industry, the 'household industry' producing the primary input labour. Final demand sector would now be considered as one of endogenous sectors. As such now we shall have $(n + 1)$ industries in place of n industries and all producing for the sake of satisfying the input requirements.

This newly conceived industry (of final demand bill) will also be assumed to have a fixed input ratio as any other industry. In other words, the supply of primary input must now bear a fixed proportion to final demand (i.e., consumption of this newly conceived industry). This will mean, for example, that households will consume each commodity in fixed proportion to the labour services they supply.

Looking at the problem in this particular way, it appears that the conversion of open model into a closed one should not create any significant change in our analysis and solution because disappearance of final demand means only an addition of one more homogeneous equation to already existing set of n homogeneous equations. Is it really so? Let us examine.

Let us assume that there are four industries only — including the new one (of final demand) designated by subscript 0. We shall, therefore, have the following set of equations

$$X_0 = a_{00}X_0 + a_{01}X_1 + a_{02}X_2 + a_{03}X_3$$

$$X_1 = a_{10}X_0 + a_{11}X_1 + a_{12}X_2 + a_{13}X_3$$

$$X_2 = a_{20}X_0 + a_{21}X_1 + a_{22}X_2 + a_{23}X_3$$

$$X_3 = a_{30}X_0 + a_{31}X_1 + a_{32}X_2 + a_{33}X_3$$

This gives us a homogeneous equation system

$$\begin{pmatrix} (1 - a_{00}) & -a_{01} & -a_{02} & -a_{03} \\ -a_{10} & (1 - a_{11}) & -a_{12} & -a_{13} \\ -a_{20} & a_{21} & (1 - a_{22}) & -a_{23} \\ -a_{30} & -a_{31} & -a_{32} & (1 - a_{33}) \end{pmatrix} \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$[I - A]X = 0$$

Since the 4 rows of the input coefficient matrix happen to be linearly dependent, $|I - A|$ will turn out to be zero. Hence the solution is indeterminate.

This means that in a closed model no unique output mix of each sector exists. We can at most determine the output levels of endogenous sectors in proportion to one another, but cannot fix their absolute levels unless additional information is made available exogenously.

We reach the same conclusion by analysing the model in a slightly different way.

Let us now assume a competitive economy of three producing sectors. We transform the output and inputs of each sector into its receipts and payments by taking into consideration the prices which are endogenously determined.

$$\text{Receipts} \quad \text{S I } P_1 X_1 = P_1 a_{11} X_1 + P_1 a_{12} X_2 + P_1 a_{13} X_3 + P_1 F_1 \quad (1)$$

$$\text{S II } P_2 X_2 = P_2 a_{21} X_1 + P_2 a_{22} X_2 + P_2 a_{23} X_3 + P_2 F_2 \quad (2)$$

$$\text{S III } P_3 X_3 = P_3 a_{31} X_1 + P_3 a_{32} X_2 + P_3 a_{33} X_3 + P_3 F_3 \quad (3)$$

$$\text{Payments} \quad \text{S I } P_1 X_1 = P_1 a_{11} X_1 + P_2 a_{21} X_1 + P_3 a_{31} X_1 + W L_1 + r K_1 \quad (4)$$

$$\text{S II } P_2 X_2 = P_1 a_{12} X_2 + P_2 a_{22} X_2 + P_3 a_{32} X_2 + W L_2 + r K_2 \quad (5)$$

$$\text{S III } P_3 X_3 = P_1 a_{13} X_3 + P_2 a_{23} X_3 + P_3 a_{33} X_3 + W L_3 + r K_3 \quad (6)$$

where W is the wage rate and r is the capital return, L_1 , L_2 , K_1 , K_2 and K_3 stand for labour and capital employed in each sector.

Therefore,

$(P_1a_{11}X_1 + P_2a_{21}X_1 + P_3a_{31}X_1)$, $(P_1a_{12}X_2 + P_2a_{22}X_2 + P_3a_{32}X_2)$ and $(P_1a_{13}X_3 + P_2a_{23}X_3 + P_3a_{33}X_3)$ are the payments to intermediate inputs and $(WL_1 + rK_1)$, $(WL_2 + rK_2)$ and $WL_3 + rK_3)$ are the payments to capital and labour in each sector

$$\text{Also } NNP = Y = P_1F_1 + P_2F_2 + P_3F_3 = W(L_1 + L_2 + L_3) + r(K_1 + K_2 + K_3)$$

(This also represents the net value added.)

Thus, in the whole system we have only 7 equations while these involve 18 variables (X_1, X_2, X_3) , (F_1, F_2, F_3) , (L_1, L_2, L_3) , (K_1, K_2, K_3) , (P_1, P_2, P_3) , (W, r) and Y . Therefore, the system is indeterminate. Now, if we transform the model into a closed one by assuming that labour and capital are used in fixed proportions to output

$$\begin{aligned} \text{i.e., } L_1 &= U_1 X_1 & K_1 &= V_1 X_1 \\ L_2 &= U_2 X_2 & K_2 &= V_2 X_2 \\ L_3 &= U_3 X_3 & K_3 &= V_3 X_3 \end{aligned} \quad \text{and,}$$

Still the number of equations is less than the variables involved in the closed model. Hence the closed model does not give us the required solution until we have some variables exogenously given.

19.10 Consumption Function Analysis (Closed Model with respect to Consumption)

Our purpose here is to incorporate induced changes in final demand into the model along with induced changes in inter-industry output that we have already included.

Ex. 10. Let us go back to the two-industry input-output model

	1	2	Final demand
1	0.1	0.3	F_1
2	0	0.2	F_2
Labour	0.9	0.5	

Let the final demand for each sector's output be split into an induced component and an autonomous component. Let us take the Keynesian consumption function for each sector, then

$$F_1 = 0.5y + \bar{c}_1$$

$$F_2 = 0.3y + \bar{c}_2$$

where $\bar{c}_1$ and $\bar{c}_2$ are the autonomous components of final demand. This autonomous component may also include investment and government consumption purchases. Clearly 0.5 and 0.3 are the marginal propensities to

consume the outputs of industries 1 and 2 respectively. It is clear that since the sum of these coefficients is 0.8 the aggregate marginal propensity to consume is 0.8 hence the multiplier ought to work out to a value

$$\frac{1}{1-c} = \frac{1}{0.2} = 5.0$$

Now, our input-output system becomes

$$X_1 = .1X_1 + .3X_2 + .5y + \bar{c}_1$$

$$X_2 = .0X_1 + .2X_2 + .3y + \bar{c}_2$$

where national income or total final demand is now a variable to be determined by the model rather than assumed exogenously. Such models are termed as closed loop models. Of course, $\bar{c}_1$ and $\bar{c}_2$ are given exogenously.

Here there are two equations but 3 variables. To obtain the third equation we note that national income is also equal to total value added or total primary factor cost.

Since labour cost per unit of output for sector 1 and 2 are Re. 0.9 and 0.5 respectively, total value added is $0.9X_1 + 0.5X_2 = y$. Hence the complete system is now

$$X_1(1 - .1) - .3X_2 - .5y = \bar{c}_1$$

$$-.0X_1 + X_2[1 - .02] - .3y = \bar{c}_2 \quad \text{and}$$

$$-.09X_1 - .05X_2 + y = 0$$

$$\text{i.e.} \begin{bmatrix} 0.9 & -0.3 & -0.5 \\ 0 & 0.8 & -0.3 \\ -0.9 & -0.5 & +1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ y \end{bmatrix} = \begin{bmatrix} \bar{c}_1 \\ \bar{c}_2 \\ 0 \end{bmatrix}$$

Generalising, we note that last column is the vector of m.p.c.s and the last row is the vector of labour cost per unit of output, hence the partitioned Leontief matrix is

$$\begin{bmatrix} I-A & -\bar{c} \\ -L & +1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} I-A & -\bar{c} \\ +L & -1 \end{bmatrix}$$

In partitioned matrix notation, where $(I-A)$ is the Leontief matrix in the open model, $\bar{c}$ and L are vectors of m.p.c. and labour-cost coefficients.

Now reverting back to our problem, we have the inverse of the new Leontief close loop matrix as

$$\frac{1}{0.144} \begin{bmatrix} 0.65 & 0.55 & 0.49 \\ 0.27 & 0.45 & 0.27 \\ 0.72 & 0.72 & 0.72 \end{bmatrix}$$

$$\begin{pmatrix} X_1 \\ X_2 \\ y \end{pmatrix} = \begin{bmatrix} 4.514 & 3.819 & 3.403 \\ 1.875 & 3.125 & 1.875 \\ 5.000 & 5.000 & 5.000 \end{bmatrix} \begin{bmatrix} \bar{c}_1 \\ \bar{c}_2 \\ 0 \end{bmatrix} \quad (1)$$

It is to be noted that in the open Leontief model the result would have been

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0.9 & -0.3 \\ 0 & 0.8 \end{pmatrix}^{-1} \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$$

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 1.111 & 0.417 \\ 0 & 1.250 \end{pmatrix}^{-1} \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \quad (2)$$

Several differences between (1) and (2), i.e., between the inverse of the closed and open models are worth noting here

(1) Each element of the last row is the multiplier (=5.0). Hence

$$5\bar{c}_1 + 5\bar{c}_2 + 0 = y = 5(\bar{c}_1 + \bar{c}_2)$$

which gives the equilibrium level of income

$$= \text{Multiplier} \times \text{change in autonomous spending,}$$

$$\text{Again, } \Delta y = 5 \times (\Delta\bar{c}_1 + \Delta\bar{c}_2)$$

(2) None of the elements of the inverse of the closed loop model is zero, though the original transaction matrix was triangular. This is because, though industry 1 does not require input from industry 2 and thus an increase in final demand F_1 alone will not necessitate change in output of industry 2 in the open model, this will nevertheless raise the level of income, which in turn will induce an increase in consumption of the output of industry 2.

(3) It is to be noted that elements of the inverse in closed model are much larger than the corresponding elements of the inverse in the open model. This is because in the closed model the effects of changes in the final demand result in income changes which induce further changes in production and income.

(4) In the closed system, the technology matrix is given in value terms

Ex. 11. Solve the above example, if $F_1 = 0.5y + 100$ and,

$$F_2 = 0.3y + 200$$

$$\text{Here, } \begin{bmatrix} X_1 \\ X_2 \\ y \end{bmatrix} = \begin{bmatrix} 4\,514 & 3\,819 & 3\,403 \\ 1\,875 & 3\,125 & 1\,875 \\ 5\,000 & 5\,000 & 5\,000 \end{bmatrix} \begin{bmatrix} 100 \\ 200 \\ 0 \end{bmatrix}$$

$$X_1 = 1215.2, X_2 = 782.5, y = 1500$$

If $F_1 = 100$ and $F_2 = 200$ then from (2) above, we would have

$$X_1 = 194.5, X_2 = 250, y = 300$$

19.11. The Leontief Production Function

We have already stated that Leontief assumes fixed coefficients of production with only one technique of producing each output. Leontief's production function can, therefore, be written as

$$X_1 = f(X_{11}, X_{21}, L_1, K_1) \\ = \min \left(\frac{X_{11}}{a_{11}}, \frac{X_{21}}{a_{21}}, \frac{L_1}{l_1}, \frac{K_1}{b_1} \right)$$

and similar relation for X_2 considering that there are only 2 sectors and Labour (L) and capital (K). l_1 and b_1 are Labour-output and capital-output coefficients. Thus the production function requires a fixed minimum amount of factors. This formulation leads to wastage of resources and bottlenecks.

If there is to be *no wastage* then the production function will be

$$X_1 = \left[\frac{X_{11}}{a_{11}}, \frac{X_{21}}{a_{21}}, \frac{L_1}{l_1}, \frac{K_1}{b_1} \right]$$

Such a production function is L-shaped.

To represent the production function graphically we consider only 2 inputs: say from sector X_2 and the other is labour; then

$$X_1 = \min \left[\frac{X_{21}}{a_{21}}, \frac{L_1}{l_1} \right]$$

This implies,

$$\lambda_1 \leq \frac{X_{21}}{a_{21}}, \quad Y_1 \leq \frac{L_1}{l_1}$$

$$\text{or } Y_1 \geq a_{21}\lambda_1, \quad L_1 \geq l_1\lambda_1$$

with at least one equality holding, the other input being wasted to some extent.

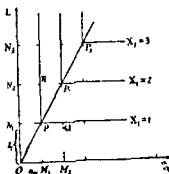


Fig 19.1

In Fig. 19.1 $X_1 = 1$ is the isoquant when output of $X_1 = 1$; $X_1 = 1$ is the isoquant when the output of $X_1 = 2$ is the next isoquant when the output of $X_1 = 1$. The isoquants are L-shaped. Obviously $OP_1 = 1$ and $OP_2 = 2$ and so on $OP_1 = P_1P_2$. At point P_1 , the input from sector 2 :

$$X_{21} = l \cdot (a_{21}) = a_{21} (= OM_1)$$

while Labour input, $L_1 = l_1 \cdot (1) = l_1 (= ON_1)$

$$\text{The slope of } OP_1 = \frac{P_1 M_1}{OM_1} = \frac{l_1}{a_{21}}$$

Since the input coefficients remain constant, hence the slope of the ray OP_1 remains constant. O, P_1, P_2, P_3 are on the same ray. All along this ray, no input is wasted. Obviously, $OM_2 = 2a_{21}$ or, $ON_2 = 2l_1$ and slope of $OP_2 = \frac{l_1}{a_{21}}$ hence in order to double the output of X_1 , both the inputs should be exactly doubled, if there is to be no wastage. Consider, however, the point Q , on the R.H.S. of the isoquant $X_1 = 1$ output, but input X is as much as OM ; thus input X will be wasted up to $M_1 M_2$. Similarly labour input will be wasted at R and all points on PR . Of course both inputs cannot be wasted. If L and X_2 are used in larger quantities, then we shall be on second isoquant above $X_1 = 1$.

19.12. Dynamic Input-Output Model

The Leontief input-output model has appeared in a number of modified forms.

We have discussed above how the model can be changed in an open or closed one by including or ignoring the exogenous sectors. Here we shall discuss a dynamic form of input-output model in which specific account is taken of inter-relationship of the current and past outputs

In our static open model we assumed that current output can be used for current consumption and/or as current input in production of some other outputs. But in dynamic model it is assumed that the current output can also be used (along with these two purposes) as an addition to industry's inventory stock and fixed capital formation

The balance equation will, therefore, be

$$X_i = X_{i1} + X_{i2} + X_{i3} + \dots + X_{in} + \Delta \sum_{j=1}^n S_{ij} + F_i$$

$$(X_i)_t = (X_{i1})_t + (X_{i2})_t + (X_{i3})_t + \dots + (X_{in})_t + \Delta \left(\sum_{j=1}^n S_{ij} \right) + (F_i)_t$$

Here X_i is the total output of i th (endogenous) producing sector in period t , which is used for three purposes

(i) as an input for production in economy's n industries

$$= (X_{i1}) + (X_{i2}) + (X_{i3}) + \dots (X_{in})_t$$

(ii) as remaining final demand $= (F)_i$ (which now includes purchases by government sector, households and foreign sector

(iii) as an addition to the stock of n industries

$$= \left(\sum_{j=1}^n \Delta S_{ij} \right)_t$$

If depreciation is ignored, the net addition to stock of i th output in the n producing sectors will be equal to the difference between the current year's and last year's accumulated capital

$$\left[\Delta \left(\sum_{j=1}^n S_{ij} \right) \right]_t = \left(\sum_{j=1}^n S_{ij} \right)_t - \left(\sum_{j=1}^n S_{ij} \right)_{t-1}$$

But since we assume that there is fixed relation between capital and output

$$b_{ij} = \frac{S_{ij}}{X_j}, \text{ so that}$$

$$b_{ij} (X_j)_t = (S_{ij})_t$$

$$b_{ij} (X_j)_{t-1} = (S_{ij})_{t-1}$$

Therefore, the balance equation of dynamic model turns out to be

$$(X_i)_t = \sum_{j=1}^n a_{ij} (X_j)_t + (F_i)_t + \sum_{j=1}^n b_{ij} [(X_j)_t - (X_j)_{t-1}]$$

This is a basic equation of dynamic model representing a set of first order linear difference equation with constant coefficients a_{ij} 's and b_{ij} 's

Another way of formulating the dynamic model is as follows

Let the Investment demand by sector j be $I_j = b_j (X_j - X_{j,t-1})$

where b_j is the accumulation coefficient

In matrix notation $I = b' (X_t - X_{t-1})$

In the continuous case the above relation will be

$$I = b' \left(\frac{dX}{dt} \right)$$

Now, in order to transform these aggregate figures of capital formation (C F) by sector of use to sectors of origin or to demand for sectoral outputs, a capital formation matrix has to be defined, let it be

$$B = \begin{pmatrix} b_{11} & b_{12} & b_{1n} \\ b_{21} & b_{22} & b_{2n} \\ b_{n1} & b_{n2} & b_{nn} \end{pmatrix}$$

$b_{11}, b_{21}, \dots, b_{n1}$ are the coefficients defined as

$$b_{i1} = \frac{\text{demand for } i\text{th sector's output by sector 1 for C F}}{\text{C F in sector 1}}$$

$$\text{Similarly } b_{j1} = \frac{\text{demand for } j\text{th sector's output by } i\text{th sector 1 for C F}}{\text{C F in sector 1}}$$

Hence the vector of investment by origin is

$$= Bb (X_t - X_{t-1}),$$

$$= Bb \frac{dX}{dt} \quad (\text{in continuous case})$$

Hence the Leontief consistency conditions are

$$X_t + AX_t + F_t + Bb (X_t - X_{t-1}) \quad (\text{This will be discrete case})$$

$$X = AX + F + Bb' \frac{dX}{dt} \quad (\text{This will be continuous case})$$

Thus the system may be described in first order difference or differential equations

However, in a dynamic (model) situation it is better to use the inequalities to account for overproduction and excess capacity. Even for the given set of consumer goods which the economy wishes to turn out, different production patterns will arise depending on the quantities of the various goods which it is decided to put in the capital investment. It gives us a unique set of output requirements for any set of final output goals. Production goals can be achieved by a variety of means and somehow society must make up its mind among them, presumably on the basis of some sort of optimality computation. Planning for long term cannot be reduced to a simple matter of solution of a system of simultaneous equations as in the input-output case.

19.13. Limitations of Input-Output Analysis

- (1) Errors in forecasting final demand will have grave consequences
- (2) Current relative prices of inputs may not be same as the ones implied in the table
- (3) The assumption of linear homogeneous production function may not be valid. The technical coefficients will not remain constant even if input price ratios are held constant in such circumstances
- (4) The constant coefficient formulation also ignores the possibility of industry outputs reaching capacity, changing prices and input proportions in the table
- (5) The assumption of constant technical coefficients goes counter to the possibility of substitution of inputs and factors
- (6) Sectoral division is, for practical purposes, limited. Such a sectorisation is not good enough for many forecasting purposes
- (7) Sectorisation (grouping of commodities in sectors) is often arbitrary. The intra sectoral heterogeneity with respect to technologies, efficiency and demand is not invariant over time.
- (8) Input-output model building is highly costly in terms of time and money
- (9) Regional input-output analysis involves many assumptions and difficulties in construction of such tables

EXERCISE 19.1

1. The following table gives the input-output coefficients for a two-sector economy consisting of agriculture and manufacturing industry

Input Industry	A	M
	0.10	0.50
A	0.20	0.25
M		

The final demand for the two industries is 300 and 100 units respectively. Find the gross outputs of the two industries.

If the input coefficients for labour for the two industries are respectively 0.5 and 0.6, find the total units of labour required.

2. Let $A = \begin{pmatrix} \frac{1}{8} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{6} & \frac{1}{4} \end{pmatrix}$

and the final demand vector be $\begin{pmatrix} 10 \\ 28 \\ 14 \end{pmatrix}$

Find the total output of the 3 sectors. What will be the total outputs if the final demand for sector 1 has increased by 1?

3. Consider a triangular (or upper triangular) matrix

$$A = \begin{bmatrix} 0.4 & 0.2 & 0.3 \\ 0 & 0.1 & 0.3 \\ 0 & 0 & 0.2 \end{bmatrix}$$

(a) Calculate $(I - A)^{-1}$

(b) Take first element (a_{11}) to be 0.3 and calculate $(I - A)^{-1}$

(c) Formulate a general rule for the inverse of $(I - A)$ if A is triangular

(d) In this case what is the special nature of the relationship between X and F or ΔX and ΔF ?

4. Find the demand vector D which is consistent with the output vector

$$X = \begin{bmatrix} 25 \\ 21 \\ 18 \end{bmatrix}, \text{ when the input-output coefficient matrix is}$$

$$A = \begin{pmatrix} 0.2 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.2 \\ 0.1 & 0.3 & 0.3 \end{pmatrix}$$

Also compute the gross value added in each sector and show that total value added in each sector and total final demand are both equal to 1

5. If $F = \begin{pmatrix} 7611 \\ 2246 \end{pmatrix}$ and,

$$A = \begin{bmatrix} 0.204 & 0.089 \\ 0.037 & 0.441 \end{bmatrix}$$

find X and value added in each sector.

(Hint: Value added in each sector is given by:

$$V_1 = X_1 - a_{11}X_1 - a_{21}X_2$$

$$V_2 = X_2 - a_{12}X_1 - a_{22}X_2)$$

6. In exercise 5 if final demand for sectors 1 and 2 increases by 4.25 and 7.16 per cent respectively, calculate the required increase in output of each sector after 1 year.

$$[\text{Hint}] \quad \Delta X = (I - A)^{-1} \Delta F$$

$$\Delta F = \begin{pmatrix} 323 & 47 \\ 150 & 81 \end{pmatrix}$$

$$7 \quad \text{If} \quad A = \begin{pmatrix} 0.2 & 0.4 \\ 0.1 & 0.5 \end{pmatrix}, \text{ and}$$

$$F = \begin{pmatrix} 60 \\ 40 \end{pmatrix}, \text{ find } X \text{ and } V$$

8. In the above exercise, assume gross value added increased by 5% and income elasticities of demand are 0.5 and 1.5 for sectors 1 and 2 respectively. Calculate the balanced growth targets.

(Hint: If population growth rate is zero, growth in annual demand is equal to the product of growth of income and income elasticity of demand, hence

$$g_1 = 0.5 \times 0.5 = 0.025$$

$$g_2 = 1.5 \times 0.05 = 0.075$$

$$\Delta F_1 = 60 \times 0.025 = 1.5$$

$$\Delta F_2 = 40 \times 0.075 = 3.0 \text{ and proceed}$$

9. In the above exercises if the marginal capital output ratios are 2 and 3 respectively, calculate the total investment required for balanced growth.

10. From the following co-eff matrix find out the final output goals of each industry assuming that consumer output targets are Rs. 80 million in steel, Rs. 300 million in coal and Rs. 50 million in railway transport.

	Steel	Coal	Railway transport
Steel	0.3	0.2	0.2
Coal	0.2	0.1	0.5
Railway transport	0.2	0.4	0.2
Labour	0.3	0.3	0.1

What would be the labour requirements in final outputs of three industries?

(Hint: In the above question

$$A = \begin{bmatrix} 0.3 & 0.2 & 0.2 \\ 0.2 & 0.1 & 0.5 \\ 0.2 & 0.4 & 0.2 \end{bmatrix} \text{ and } F = \begin{bmatrix} 80 \\ 30 \\ 50 \end{bmatrix}$$

$$[I - A] = \begin{bmatrix} +0.7 & -0.2 & -0.2 \\ -0.2 & +0.9 & -0.5 \\ -0.2 & -0.4 & +0.8 \end{bmatrix}$$

Substituting the matrix, we get the required result

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 0.7 & -0.2 & -0.2 \\ -0.2 & 0.9 & -0.5 \\ -0.2 & -0.4 & 0.8 \end{bmatrix}^{-1} \begin{bmatrix} 80 \\ 30 \\ 50 \end{bmatrix}$$

After inverting the matrix, we get the required result)

11. From the following matrix find the final output goals of each industry assuming that the consumer output targets are Rs 85 million in steel, Rs 25 million in coal and Rs 55 million in railway transport '

	<i>Steel</i>	<i>Coal</i>	<i>Railway transport</i>
<i>Steel</i>	0.4	0.1	0.2
<i>Coal</i>	0.2	0.2	0.4
<i>Railway transport</i>	0.2	0.3	0.3
<i>Labour</i>	0.2	0.4	0.1

What would be the labour requirements of the final outputs of three industries?

12. If the consumer output requirements are Rs 80 million in steel, Rs 20 million in coal and Rs 55 million in railway transport, determine from the following matrix the final output goals of each industry

	<i>Steel</i>	<i>Coal</i>	<i>Railway transport</i>
<i>Steel</i>	0.2	0.2	0.3
<i>Coal</i>	0.4	0.1	0.3
<i>Railway transport</i>	0.1	0.3	0.3
<i>Labour</i>	0.3	0.4	0.1

Estimate the labour requirements for final outputs of three industries

13. From the following matrix determine the gross level of output and labour requirement of industry A, B and C.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>Consumer's requirements</i>
<i>A</i>	0.1	0.2	0.4	Rs 40 million
<i>B</i>	0.3	0.2	0.1	Rs 50 million
<i>C</i>	0.2	0.4	0.3	Rs 80 million
<i>Labour</i>	0.2	0.3	0.3	

14. From the following technological matrix and final consumer requirements determine gross levels of output of 3 industries A, B and C

	<i>A</i>	<i>B</i>	<i>C</i>	<i>Consumer requirements</i>
<i>A</i>	0.2	0.4	0.5	Rs 70 million
<i>B</i>	0.5	0.1	0.1	Rs 60 million
<i>C</i>	0.1	0.2	0.1	Rs 90 million
<i>Labour</i>	0.2	0.3	0.3	

What will be total labour requirements ?

Substituting the matrix, we get the required result.

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 0.7 & -0.2 & -0.2 \\ -0.2 & 0.9 & -0.5 \\ -0.2 & 0.8 & 0.1 \end{bmatrix}^{-1} \begin{bmatrix} 80 \\ 30 \\ 50 \end{bmatrix}$$

After inverting the matrix, we get the required result.)

11. From the following matrix find the final output goals of each industry assuming that the consumer output targets are Rs 85 million in steel, Rs. 25 million in coal and Rs 55 million in railway transport ;

Steel	0.4	0.1	0.2
Coal	0.2	0.2	0.4
Railway transport	0.2	0.3	0.1
Labour	0.2	0.3	0.4

What would be the labour requirements of the final outputs of three industries?

12. If the consumer output requirements are Rs. 80 million in steel, Rs. 20 million in coal and Rs. 55 million in railway transport, determine from the following matrix the final output goals of each industry :

Steel	0.2	0.1	0.3
Coal	0.4	0.1	0.3
Railway transport	0.1	0.3	0.1
Labour	0.3	0.4	0.3

Estimate the labour requirements for final outputs of three industries.

13. From the following matrix determine the gross level of output and labour requirement of industry A, B and C :

A	0.1	0.2	0.4
B	0.3	0.2	0.1
C	0.2	0.4	0.3
Labour	0.2	0.3	0.3

14. From the following technological matrix and final consumer requirements determine gross levels of output of 3 industries A, B and C :

A	0.2	0.4	0.5
B	0.5	0.1	0.1
C	0.1	0.2	0.1
Labour	0.2	0.3	0.3

What will be total labour requirements ?

15. What are the possible explanations of fixed input coefficient in an input-output system?

Given the following technological matrix and the consumer requirements, determine gross levels of output and labour requirements of the 3 industries A, B and C

	A	B	C	Consumer requirements
80 (Thousand Rs)	0.2	0.3	0.2	
	0.5	0.4	0.3	120
	0.1	0.2	0.2	90
Labour	0.2	0.1	0.3	

16. A three-sector input-output matrix $[I - A]$ is given as

$$\begin{bmatrix} 1 & -0.5 & 0 \\ -0.2 & 1 & -0.5 \\ -0.4 & 0 & 1 \end{bmatrix}$$

With labour coefficients (per unit of output) as 0.4, 0.7, 1.2, if the household demand for the outputs of the 3 sectors is 1,000, 5,000 and 4,000, determine the level of output and employment

If the wage rate is Rs. 10 per labour day, find the equilibrium prices and the total value added

17. Let the technology matrix be

Sector	1	2	3
1	0.1	0.3	0.1
2	0	0.2	0.2
3	0	0	0.3

If the final demand functions are

$$F_1 = 0.1y + \bar{c}_1$$

$$F_2 = 0.3y + \bar{c}_2$$

$$F_3 = 0.4y + \bar{c}_3$$

where y is the national income, then obtain expressions for X_1 , X_2 and y

$$Hint \quad \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ y \end{pmatrix} = \begin{pmatrix} 2.847 & 2.589 & 3.839 & 2.153 \\ 2.857 & 2.857 & 2.857 & 5.000 \\ 2.857 & 2.857 & 2.857 & 5.000 \\ 5.000 & 5.000 & 5.000 & 0 \end{pmatrix} \begin{pmatrix} \bar{c}_1 \\ \bar{c}_2 \\ \bar{c}_3 \\ 0 \end{pmatrix}$$

15. What are the possible explanations of fixed input coefficient in an input-output system ?

Given the following technological matrix and the consumer requirements, determine gross levels of output and labour requirements of the 3 industries A, B and C

	A	B	C	Consumer requirements
A	0.2	0.3	0.2	80 (Thousand Rs)
B	0.5	0.4	0.3	120 "
C	0.1	0.2	0.2	90 "
Labour	0.2	0.1	0.3	

16. A three-sector input-output matrix $[1 - A]$ is given as

$$\begin{bmatrix} 1 & -0.5 & 0 \\ -0.2 & 1 & -0.5 \\ -0.4 & 0 & 1 \end{bmatrix}$$

With labour coefficients (per unit of output) as 0.4, 0.7, 1.2, if the household demand for the outputs of the 3 sectors is 1,000, 5,000 and 4,000, determine the level of output and employment

If the wage rate is Rs. 10 per labour day, find the equilibrium prices and the total value added

17. Let the technology matrix be

Sector	1	2	3
1	0.1	0.3	0.1
2	0	0.2	0.2
3	0	0	0.3

If the final demand functions are

$$F_1 = 0.1y + \bar{c}_1$$

$$F_2 = 0.3y + \bar{c}_2$$

$$F_3 = 0.4y + \bar{c}_3$$

where y is the national income, then obtain expressions for X_1 , X_2 and y

$$\text{Hint: } \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ y \end{pmatrix} = \begin{pmatrix} 2.847 & 2.153 & 2.014 & 1.736 \\ 2.589 & 3.839 & 2.946 & 2.589 \\ 2.857 & 2.857 & 4.286 & 2.857 \\ 5.000 & 5.000 & 5.000 & 5.000 \end{pmatrix} \begin{pmatrix} \bar{c}_1 \\ \bar{c}_2 \\ \bar{c}_3 \\ 0 \end{pmatrix}$$

18. Appraise some of the following critical assumptions of the input model of production

(a) Linearity, (b) fixed factor proportions, and (c) fixed product mix (that is, fixed proportions among the joint output components)

19. Discuss how the technique of inter-industry analysis could be used in constructing planning models. Discuss implications of assuming fixed technological coefficients. Mention and briefly discuss any planning model you know of which is based upon the framework of inter-industry analysis

20. Outline Leontief's analysis of inter-industry relations distinguishing the assumptions relevant to the 'Open' and 'Closed' systems

21. Prove that in a Closed Leontief system, if there is any economic activity at all, the absolute levels of activity are indeterminate

22. Outline mathematically the Leontief dynamic model. Prove that a normal dynamic Leontief system has only a special initial configuration leading to balanced growth. Further prove that if such a configuration exists then Leontief trajectory is an efficient path

23. Explain the main features and basic assumptions of input-output analysis, and comment on the statement that basically the input-output analysis *constant is nothing more complicated than the solution of a set of n simultaneous linear-equations in n -variables*

24. How is input-output model used in checking the consistency of targets and in the assessment of resources? Can it be used to assess the disaggregated multiplier effect?

25. Explain the significance and uses of the simple input-output system $X = (I - A)^{-1} D$, where A is the coefficient matrix, D is the final demand vector and X is the gross output vector. Show that the aggregate final demand must equal aggregate gross value added

Chapter 20

Game Theory

The problems in Linear Programming deal with those optimisation problems which possess the following three characteristics

- (i) A single decision maker is involved
- (ii) Constraints are in the form of inequalities.
- (iii) The nature of Optimising process is either of maximisation or of minimising but never any mixture of the two

This chapter introduces the possibility of more than one decision maker, wherein the Optimum Value of the (objective) function for any one decision maker depends not only on his own choices but also on the choices of the others. Game theory is the study of such situations, situations in which the very framework of optimisation is changed to that of finding either the maximum amongst the set of minima (*maximum*) or the minimum among the set of maxima (*minimax*)

20.1. Basic Concepts of Game Theory

Situations involving more than one decision maker are studied under “*game theory*” — where each *strategy* is consciously adopted by the players

The decision makers are called *players*. The players may be individuals, group of individuals, firms, nations, etc.

A *game* may be defined as the collection of rules, regulations and conventions known to planners and observed by the players. A game is then a situation in which two or more participants, the players, confront one another in pursuit of certain conflicting objectives. Some players may win and receive a positive *payoff*, whereas others may lose and get a negative payoff.

A *strategy* is a plan of action undertaken in the light of own belief about the reaction of the rival. Each player takes into consideration all possible alternative moves in selecting strategy. The final outcome of the game depends jointly on the strategies chosen by the players in the game.

20.2. Classification and Description of Games

There are basically two major categories of games — games of chance and games of strategy. No skill is involved in former type of games. In games of strategy, on the other hand, the outcome depends on the deliberate strategy

undertaken by each player. It is here in selection of that strategy some skill is called for.

There are several ways of classifying the games of strategy by the number of players, the number of strategies, the nature of the payoff function and the nature of preplay negotiations.

(i) Depending on the number of players, we may have two person games, three person games, n person games. 'Person', as pointed out earlier, does not mean a single individual. It refers to 'participating party' in the game.

(ii) Another way of classifying games is by the number of strategies, as finite games or infinite games. Infinite games are those in which there are infinite number of strategies available for one or more players. In the present chapter we are concerned only with finite games in which the number of strategies available to each of the players is finite.

(iii) Depending on the payoff situation, a game can be classified as either constant sum or non-constant sum. In the former type, the payoffs of all players at the end of the game will always add up to a fixed constant, whatever the strategies chosen by the players. In contrast, the game would become a non-constant one if the payoffs to the players in the end do not sum to same constant every time. In a constant sum game, the constant can be any number, in the special case where it is zero, the game becomes a zero sum game.

(iv) A final classification of games is by the nature of preplay negotiation. The game is a co-operative game if the players can form a coalition to discuss their strategies before the game is played and make binding agreements of strategies. The game is non-co-operative if players do not agree to any such coalition and co-operation.

20.3. Payoff Matrix

The two-person constant sum game can be completely described by means of a payoff matrix derived from a hypothetical example given in the form of following table.

Strategies of Player I	Strategies of Player 2		
	I	II	III
I	7	8	4
II	4	7	2

In this table, we assume that player 1 has only two strategies available, but player 2 has three. The total number of possible outcomes is, therefore, $2 \times 3 = 6$.

For each possible outcome, there is a corresponding payoff to each player. The present table gives the payoffs to player 1. Thus the entry 7 represents what

player 1 will receive if both players adopt their respective first strategies. To facilitate reference, let us consider that entry as the a_{11} element of the payoff matrix $A = [a_{ij}]$. Thus, the payoff matrix A is defined as

$$A = \begin{bmatrix} 7 & 8 & 4 \\ 4 & 7 & 2 \end{bmatrix}$$

Here, the first subscript of a_{ij} always refers to the strategy adopted by player 1 and the second to the strategy of player 2. This convention requires the listing of strategies in the particular form illustrated in the above table, i.e., those of players in a vertical sequence, and those of player 2 horizontally. As such a_{23} would mean payoff to player 1 when player 1 adopts strategy two while player 2 strategy three. From the payoff matrix, $a_{23} = 2$.

The payoff matrix for player 2 can also be derived easily. Since in a constant sum game, the payoff to player 2 under any given outcome of the game is necessarily equal to that specific constant minus the payoff to player 1, matrix b is readily determined from matrix A . Assuming that the constant sum in the present case is 10, then the matrix B can be defined as

$$B = \begin{bmatrix} (10-7) & (10-8) & (10-4) \\ (10-4) & (10-7) & (10-2) \end{bmatrix} = \begin{bmatrix} 3 & 2 & 6 \\ 6 & 3 & 8 \end{bmatrix}$$

In zero-sum game one player's payoff must be the negative of the payoff to the other, because a player can only win what his opponent loses. As an economic example of this, consider two firms in a duopolistic market that are striving to lure away each other's established customers. Since such a contest involves existing customers only, the number of customers won over by one firm must be identical to the number of customers lost by the other. Assume that there are 10 customers in all and that original distribution of customers between two firms is, say, 5 to each. Matrices A and B are in terms of number of customers. We can, now, obtain two new payoff matrices A^* and B^* in the present zero-sum game by subtracting the original distribution (5) from every element

$$A^* = \begin{bmatrix} +2 & +3 & -1 \\ -1 & +2 & -3 \end{bmatrix} \quad B^* = \begin{bmatrix} -2 & -3 & +1 \\ +1 & -2 & +3 \end{bmatrix}$$

Since we now have

$$a_{ij}^* + b_{ij}^* = 0$$

A^* and B^* will add up to a ' 2×3 zero matrix'. In this way we have transformed the constant sum game into zero-sum game.

Summarising, *constant sum* games are characterised by

$$a_{ij} + b_{ij} = C = \text{constant}$$

while *zero-sum* games are depicted by

$$a_{ij} + b_{ij} = 0$$

20.4. Saddle-point Solutions (Strictly Determined Games)

While the structure of a two-person, constant-sum game is completely summarised in a single payoff matrix A , this matrix alone does not enable us to tell the final outcome of the game. We make two important assumptions for obtaining the solution

(i) All players involved in the game theory minimise their risks. In other words, they avoid the riskier course of action. Each player seeks to guarantee himself the maximum possible payoff regardless of what the opponent does

(ii) Each player has knowledge of the payoff matrix A , but he is ignorant about the exact strategy the opponent plans to adopt

With these assumptions in mind, a conservative player would always proceed thus

(1) Determine the least payoff he can receive under each of his strategies (i.e., the minimum in each row of A), and

(2) Choose the strategy that has the largest minimum

In this way, he is sure that whatever the opponent does, he will not end up with the worst of all worlds, because he specifically avoids some less favourable outcomes by following the above course of action

Optimal Strategy for Player 1 Applying the above procedure to the payoff matrix A , player 1 will find the minima of the two rows to be 4 and 2 respectively, the maximum among these is 4 which occurs in the first row. This is called the maximin. The optimal strategy for player 1, therefore, is his strategy 1

	Row minimum	Maximin
$A = \begin{bmatrix} 7 & 8 & 4 \\ 4 & 7 & 2 \end{bmatrix}$	$\longrightarrow 4$	$\longrightarrow 4$
	$\longrightarrow 2$	

Optimal Strategy for Player 2 Player 2 shall also seek his own maximin from his own payoff matrix B . If we assume the present game to be 10 Sum, then the payoff matrix is B and the maximin is 6, as explained under

	$B = \begin{bmatrix} 3 & 2 & 6 \\ 6 & 3 & 8 \end{bmatrix}$
Col minimum	$\downarrow \downarrow \downarrow$ 3 2 6
Maximin	$\downarrow$ 6

(Note. Since the second player's strategies are depicted columnwise, he seeks his maximin via columns.)

However, in view of the constant-sum nature of the game, the choice of the maximum among the column minima in matrix B must yield the same strategy that gives the minimum among the column maxima in matrix A. We can, therefore, work with matrix A instead. In this case, player 2 will seek to ensure the lowest payoff to his opponent (i.e., the highest payoff to himself) regardless of the strategy chosen by the opponent. Thus, player 2 will discard all entries in the payoff matrix A except for the maximum payoff in each column and then select as his optimal strategy the column with the smallest maximum payoff. This is shown as under:

$$\begin{array}{rcccl}
 & & \mathbf{A} = \begin{bmatrix} 7 & 8 & 4 \\ 4 & 7 & 2 \end{bmatrix} & & \\
 \text{Col. maximum} & & \begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 7 & 8 & 4 \end{array} & & \\
 \text{Minimax} & & \begin{array}{ccc} & & \downarrow \\ & & 4 \end{array} & &
 \end{array}$$

In sum, therefore, the procedure postulated above essentially amounts to the search for a maximin or a minimax.

In the above example, we observe that the optimal strategy for player 1 is first one where he obtains a payoff of at least $a_{12}=4$, whereas player 2 will pick strategy 3 so that player 1 gets a payoff of at most $a_{13}=4$. Thus, the two players have agreed upon a solution to the game. This result is arrived at because matrix A satisfies the equation

$$\max_i \min_j (a_{ij}) = \min_j \max_i (a_{ij})$$

These symbols can be explained through the following table considered originally

	$j = 1$	$j = 2$	$j = 3$
$i = 1$	7	8	4
$i = 2$	4	7	2

Now, for $i = 1$, $\min (a_{ij}) = 4$ and
for, $i = 2$, $\min (a_{ij}) = 2$.

Combining both results $\min (a_{ij}) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ which is just a listing of row minima. When the above expression is preceded by the expression $\left\{ \max_i \right\}$ it only means to maximise row-wise among the i 's

In other words,

$$\max_i \min_j (a_{ij}) = \max_i \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 4$$

As such the left hand expression of the above equation is first a concise description of the procedure for determining the maximin described under 'Optimal strategy for player 1'. Similarly, the right-hand side of the above equation is the mathematical expression of the minimax procedure outlined under 'Optimal strategy for player 2'.

When an element of a payoff matrix serves both as a maximin and as a minimax, it is called a saddle point. In case a saddle point is present in a payoff matrix, it represents the solution of the game and such game is called a strictly determined game.

Mathematically speaking

$$\max_i \min_j (a_{ij}) = \min_j \max_i (a_{ij}) = a^*_{ij},$$

where a^*_{ij} is the saddle point and solution of the game.

Let us now take an example of zero sum game. Here also we assume that player 1 has a choice of two strategies and player 2 has a choice of 3 strategies. Player 1 figures that if he chooses row 1, then the opponent might choose column 2 resulting in a payoff of 1. Similarly, if he chooses row 2, then he figures that the opponent might choose column 1, resulting in a payoff of -1. These are the row minima as shown below.

		<i>Row minima Maximin</i>	
	$A = \begin{bmatrix} 2 & 1 & 4 \\ -1 & 0 & 6 \end{bmatrix}$	1	
		-1	1
<i>Col maxima</i>	$\begin{matrix} \downarrow & \downarrow & \downarrow \\ 2 & 1 & 6 \end{matrix}$		
<i>Minimax</i>	$\downarrow$ 1		

Maximising over these row minima, player 1 selects his first strategy guaranteeing a payoff of 1 or more (more if player 2 selects column 1 or 3). Similarly, player 2 assumes the worst, figures that the opponent might select the first row if he chooses column 1 or 2 and the second row if he chooses column 3, leading to the column maxima 2, 1 and 6, as shown above. Minimizing over these column maxima, player 2 chooses his second strategy, guaranteeing a loss of not more than 1. Thus, in this game the choices are consistent.

$$\max_i \min_j a_{ij} = \min_j \max_i a_{ij} = 1$$

and the saddle point entry, 1, is the value of game (to player 1).

20.5. Mixed Strategy The Case of No Saddle Point

Not all two-person zero sum (or constant sum) game are strictly determined. Let us now explore the case where the payoff matrix contains no saddle point. Consider a game with the following payoff matrix

$$\begin{array}{rcccl}
 & & & & \text{Row minima} \\
 & & & & -2 \\
 & & & & -4 \\
 A = \begin{bmatrix} 6 & -2 & 3 \\ -4 & 5 & 4 \end{bmatrix} & & & & \\
 & \downarrow & \downarrow & \downarrow & \\
 \text{Col maxima} & 6 & 5 & 4 &
 \end{array}$$

In this case, $\max \min a_{ij} = -2 < 4 = \min \max a_{ij}$. If the players follow the rules developed thus far, player 1 selects strategy 1 and expects player 2 to select strategy 2 and a payoff of -2 , while player 2 selects strategy 3 and expects player 1 to select strategy two and a payoff of 4 . The outcome is $a_{ij} = 3$, which neither player expected.

Furthermore, if player 2 does select his third strategy, then player 1 would do better selecting his second, not his first strategy. Similarly, if player 1 does select his first strategy, then player 2 would do better selecting his second, not his third strategy. The solution concept as outlined so far seems to fail in such games.

The solution concept is still valid, however, if the concept of strategy is broadened to allow for *mixed strategies* (or *random strategies*) which are probability combinations of pure strategies discussed thus far.

For any given zero-sum two-person game where there is no saddle point, the game is solved through mixed strategies in the following way.

Assume the given payoff matrix as

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Suppose A 's strategies are represented by probabilities (p_1, p_2) and that of B 's by $\begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$. Our problem is to calculate the values of p_1, p_2, q_1, q_2 that will cause the expected payoff to A to be maximum.

The values of p_i and q_j that determine the optimal strategies for each player are given by

$$\begin{aligned}
 1. \text{ For player } A \quad p_1 &= \frac{a_{22} - a_{21}}{a_{11} + a_{22} - a_{21} - a_{12}} \\
 p_2 &= \frac{a_{11} - a_{12}}{a_{11} + a_{22} - a_{21} - a_{12}}
 \end{aligned}$$

$$\begin{aligned} 2 \text{ For player B } \quad q_1 &= \frac{a_{22} - a_{12}}{a_{11} + a_{22} - a_{21} - a_{12}} \\ q_2 &= \frac{a_{11} - a_{21}}{a_{11} + a_{22} - a_{21} - a_{12}} \end{aligned}$$

The value of the game of player A is given by

$$A = \frac{a_{11}a_{22} - a_{21}a_{12}}{a_{11} + a_{22} - a_{21} - a_{12}}$$

20.6 Dominated Strategies

Consider the following payoff matrix A

$$\begin{array}{ccccc} & & & \text{Row min} & \\ & & & \longrightarrow & \\ A = \begin{bmatrix} 8 & 7 & 10 \\ 10 & 9 & 12 \\ 11 & 5 & 9 \end{bmatrix} & \longrightarrow & 7 & \longleftarrow & \text{maximum} \\ & \longrightarrow & 9 & & \\ & \longrightarrow & 5 & & \\ & \downarrow & \downarrow & \downarrow & \\ \text{Col. max.} & & 11 & 9 & 12 \\ & & \downarrow & & \\ & & \text{minimax} & & \end{array}$$

If an element-by-element comparison is made of the first two rows in the above payoff matrix, row 1 reveals an inferior payoff to player 1, whichever strategy player 2 happens to adopt. There is naturally no reason, therefore, for player 1 ever to consider adopting strategy 1 when strategy 2 is available to him. Strategy 1 is thus said to be *dominated* by strategy 2 and can be removed forthright from the payoff matrix. Domination can also occur among columns. Column 2 dominates column 3 from the standpoint of player 2, because the former involves a lower payoff to his opponent (player 1) in every row.

A dominated strategy does not have to be inferior in every element. A strategy is considered to be dominated by the other so long as it is no better anywhere than another strategy. For example, a strategy involving payoffs [10, 15, 30] is regarded as dominated by another one with payoffs [10, 16, 30]. However, when a strategy has some other payoffs larger but some payoffs less than the corresponding elements in another strategy, neither of them dominates the other. As such in the above matrix A, row 2 and row 3 do not dominate each other.

After removing these dominated strategies, the payoff matrix reduces to

$$A = \begin{bmatrix} - & - & - \\ 10 & 9 & - \\ 11 & 5 & - \end{bmatrix}$$

The removal of dominated strategies simplifies the payoff matrix and also does not produce any effect on the final outcome of the game. It can be observed that the solution point $a_{22} = 9$ is still retained in the reduced matrix.

Ex 1 Solve the game whose payoff matrix is given by

	B_1	B_2	B_3
A_1	1	3	1
A_2	0	-4	-3
A_3	1	5	-1

To determine the saddle point, we must select the minimum element of each row and the greatest element of each column of the given payoff matrix

$$\begin{array}{ccccc}
 & & & \text{Row min.} & \\
 \left[\begin{array}{ccc} 1 & 3 & 1 \\ 0 & -4 & -3 \\ 1 & 5 & -1 \end{array} \right] & & & \begin{array}{c} 1 \\ 4 \\ -1 \end{array} & \leftarrow \text{Maximum element} \\
 \text{Col max.} & \begin{array}{ccc} 1 & 5 & 1 \\ \uparrow & & \uparrow \end{array} & & \\
 & \text{Minimax element(s)} & & &
 \end{array}$$

Obviously the given payoff matrix has two saddle points a_{11} and a_{13} . The optimum strategy for player A is A_1 while the optimum strategy for B is B_1 or B_3 . The value of game is 1 for A and -1 for B.

Ex 2 Solve the following game

$$A = \begin{bmatrix} -2 & 0 & 0 & 5 & 3 \\ 4 & 2 & 1 & 2 & 5 \\ -4 & -3 & 0 & -3 & 6 \\ 5 & 1 & -5 & -2 & -6 \end{bmatrix}$$

To obtain the saddle point, we obtain row minima and column maxima

$$\begin{array}{ccccc}
 & & & \text{Row min.} & \\
 \left[\begin{array}{ccccc} -2 & 0 & 0 & 5 & 3 \\ 4 & 2 & 1 & 2 & 5 \\ -4 & -3 & 0 & -3 & 6 \\ 5 & 1 & -5 & -2 & -6 \end{array} \right] & & & \begin{array}{c} -2 \\ 1 \\ -1 \\ -6 \end{array} & \leftarrow \text{Maximum element} \\
 \text{Col max.} & \begin{array}{ccccc} 5 & 3 & 1 & 5 & 6 \\ & \uparrow & & & \end{array} & & & \\
 & \text{Maximax element} & & &
 \end{array}$$

Hence the solution to the given game is

- (i) the best strategy for player A is second
- (ii) the best strategy for player B is third, and
- (iii) the solution of the game is 1 to A and -1 to B

Ex 3 Solve the following game

$$A = \begin{bmatrix} 3 & -5 & -4 \\ -2 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}$$

First of all, we make use of the property of dominance to reduce the given pay-off matrix to 2×2

(i) Row 3 dominates row 2, hence the reduced matrix is

$$A = \begin{bmatrix} 3 & -5 & -4 \\ -1 & 1 & 2 \end{bmatrix}$$

(ii) Column two dominates column three, hence the reduced matrix is

$$A = \begin{bmatrix} 3 & -5 \\ -1 & 1 \end{bmatrix}$$

Applying mixed strategies formula

$$p_1 = \frac{1 - (-1)}{3 + 1 - (-1) - (-5)} = \frac{2}{10}, \quad p_2 = \frac{3 - (-5)}{10} = \frac{8}{10}$$

$$q_1 = \frac{1 - (-5)}{10} = \frac{6}{10}, \quad q_2 = \frac{3 - (-1)}{10} = \frac{4}{10}$$

$$\text{Value of game is } v = (-) \frac{20}{100} = -\frac{1}{5}$$

Ex 4 Mohan is paid Rs 10 if two coins turn both heads and Re. 1 if two coins turn both tails. Ram is paid Rs 3 when the two coins do not match. Whom do you consider in the better situation?

To above example may be transformed into the following payoff matrix

		Ram	
		$B_1(H)$	$B_2(T)$
Mohan	$A_1(H)$	8	-3
	$A_2(T)$	-3	1

Since the maximin is not equal to minimax, payoff matrix does not possess any saddle point. The players shall, therefore, use mixed strategies. Let Mohan assign probabilities p to A_1 and $(1-p)$ to A_2 and Ram assign the probabilities q to B_1 and $(1-q)$ to B_2 .

The expected payoff of Ram is then given by

$$8p + (-3)(1-p) = (-3)p + 1(1-p)$$

$$p = \frac{4}{15}, \quad (1-p) = \frac{11}{15}$$

The expected payoff of Mohan is given by

$$8q + (-3)(1 - q) = -3q + 1(1 - q)$$

$$q = \frac{4}{15} \text{ and } (1 - q) = \frac{11}{15}$$

The expected value of the game is

$$[8p - 3(1 - p)]q + [-3p + (1 - p)](1 - q)$$

$$\left(8 \frac{4}{15} - 3 \frac{11}{15}\right) \frac{4}{15} + \left\{-3 \frac{4}{15} + \frac{11}{15}\right\} \frac{11}{15} = -\frac{1}{15}$$

Ex 5 Solve the following game

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 6 & 2 & 7 \\ 5 & 1 & 6 \end{bmatrix}$$

First apply the principle of dominance and obtain

$$A = \begin{bmatrix} 1 & 7 \\ 6 & 2 \end{bmatrix}$$

Minimax $\neq$ Maximin, hence the game has no saddle point

$$A\text{'s expected payoff } p + 6(1 - p) = 7p + 2(1 - p), \quad p = \frac{2}{5}$$

$$B\text{'s expected payoff } q + 7(1 - q) = 6q + 2(1 - q), \quad q = \frac{1}{2}$$

$$\text{Expected value of the game } [p + 6(1 - p)]q + [7p + 2(1 - p)](1 - q) = 4$$

EXERCISE 20 I

1. Given the following sets of payoff matrices A and B , indicate whether the games are zero-sum or constant-sum

$$(i) \quad A = \begin{bmatrix} 3 & 7 & 4 \\ 5 & 1 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 9 & 5 & 8 \\ 7 & 11 & 3 \end{bmatrix}$$

$$(ii) \quad A = \begin{bmatrix} 5 & 10 \\ 7 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -3 \\ 0 & 6 \end{bmatrix}$$

$$(iii) \quad A = \begin{bmatrix} 6 & 8 & 3 \\ 3 & 2 & -2 \\ -1 & 0 & -3 \end{bmatrix} \quad B = \begin{bmatrix} -6 & 8 & -3 \\ -3 & -2 & 2 \\ 1 & 0 & -3 \end{bmatrix}$$

2. If the total payoff for the two players in a constant-sum game is Rs 500, write out the matrix B in the following cases .

$$(i) \quad A = \begin{bmatrix} 240 & 125 \\ 360 & 315 \end{bmatrix} \quad (ii) \quad A = \begin{bmatrix} 165 & 200 & 350 \\ 480 & 310 & 215 \end{bmatrix}$$

3. If player 1 has 30 strategies and if player 2 has 25, how many possible outcomes of the game will there be ?

$$4. \text{ Given : } A = \begin{bmatrix} 5 & 8 & 6 \\ 12 & 13 & 6 \\ 11 & 10 & 9 \end{bmatrix}$$

- (i) Find maximin and minimax.
- (ii) Is there a saddle point ? Which element is it ?
- (iii) What are the optimal strategies for the two players ?
- (iv) What is the payoff to player 1 ? Can you obtain payoff to player 2 ? Why ?

5. What are the dominated rows and columns in *question 4* ?

$$6. \text{ Given : } A = \begin{bmatrix} 4 & 0 & -5 \\ 3 & -2 & -3 \\ 1 & 2 & 5 \end{bmatrix}, \text{ find}$$

- (i) $\min_j (a_{1j})$ (ii) $\min_j (a_{2j})$ (iii) $\max_i (a_{i2})$
- (iv) $\max_i \min_j (a_{ij})$ (v) $\min_j \max_i (a_{ij})$

7. Solve the following two-person games :

$$(i) \quad \begin{bmatrix} 4 & 0 \\ 6 & 3 \end{bmatrix} \quad (ii) \quad \begin{bmatrix} 15 & 0 & -2 \\ 0 & -15 & -1 \\ 1 & 2 & 0 \end{bmatrix}$$

$$(iii) \quad \begin{bmatrix} 1 & 2 & 3 \\ 3 & 0 & 2 \end{bmatrix} \quad (iv) \quad \begin{bmatrix} 4 & 0 & 1 \\ 5 & 3 & 8 \end{bmatrix}$$

$$(v) \quad \begin{bmatrix} 6 & 0 & 7 \\ 3 & 1 & 6 \\ 5 & 4 & 2 \end{bmatrix} \quad (vi) \quad \begin{bmatrix} 5 & 3 & 2 \\ 3 & 4 & 0 \end{bmatrix}$$

$$(vii) \quad \begin{bmatrix} -3 & 6 \\ 8 & 2 \\ 6 & 3 \end{bmatrix}$$

ANSWERS

(1.1)

- 1 (i) $A = \{x \mid x \text{ odd integers}\}$
 (ii) $A = \{x \mid x \text{ is integer divisible by 4}\}$
 (iii) $A = \{x \mid x \text{ is woman president of India}\} = \phi$
 (iv) $A = \{x^2 \mid 1 \leq x \leq 5\}$
 (v) $A = \{x \mid x = -1\} = \phi$
- 2 (i) Set of cubes of positive integers from 1 to 4
 (ii) Set of numbers in geometric series with first term 2 and common ratio 2.
 (iii) Set of all persons aged from 15 years to less than 60 years.
 (iv) Set of root of $x^2 - 7x + 12 = 0$
 (v) Set of perfect squares.
- 3 (i) Infinite Set (ii) Finite Set (iii) Finite Set.
 (iv) Infinite Set (v) Finite (vi) Finite
- 4 (i) (ii) and (v) are correct.
- 5 (i) (ii) and (iv) are defined by inclusion

(1.2)

- 1 ACB ACC
 BCC, DCA DCB DCC
 DCE, DCF, ECC, FCE, FCB
- 2 No, ϕ_1 will be a sub-set of ϕ_2
- 3 (i) Wrong (ii) Wrong (iii) Correct. (iv) Wrong (v) Correct.
- 4 (i) The set of all parents. (ii) The set of all mothers who are widows.
 (iii) The set of all fathers who are widowers
 (iv) The set of all ladies who are not mothers but are widows.
 (v) The set of all mothers who are not widows.
- 6 (i) Set of persons who are fathers as well as mothers — a null set.
 (ii) A null set.
 (iii) Set of widows is a sub-set of the set of mothers. This means widows are mothers.
 (iv) All mothers are widows
- 10 Read, Find in solved examples 34 to 36
- 11 $A \cap B$ is (i) ϕ (iii) B (iv) A.
- 13 (i) $\{1, 2, 3, 4, 5, 6, 7\}$ (ii) $\{3, 4, 5\}$ (iii) ϕ (iv) Ω
 (v) $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ (vi) $\{1, 2, 3, 4, 5\}$ (vii) $\{1, 2\}$
 (viii) $\{3, 4, 5\}$ (ix) ϕ (x) ϕ .
- 15 $P(A) = \{\phi, \{0\}, \{1\}, \{2\}, \{0, 1, 2\}, \{0, 1\}, \{1, 2\}, \{0, 2\}\}$
- 17 (ii) and (vi) are correct.
- 20 (i) and (iii) are correct.
- 22 $A \cup B = \{4, 2, 1\}$, $A \cap B = \{4\}$ and $A - B = \{2\}$

(1.3)

1. $(A \cap B)' = \{1, 3, 4\}$, $(A \cup B)' = \{4\}$.
(i) Ω (ii) Φ (iii) A (iv) A' (v) $\{1, 2, 3\}$ (vi) $\{1, 4\}$
2. (i) and (ii) incorrect, (iii) correct.
5. (i) $\{1, 2, 3, 4, 5, 6, 7\}$ (ii) $\{5\}$ (iii) A .
(iv) $\{1, 2, 4, 5, 6\}$ (v) $\{1, 2, 4, 5, 6\}$
(vi) $\{2, 4, 5\}$ (vii) $\{2, 4, 5\}$
8. $\{1, 2, a, b, c\}$, $\{a, b, c, e, f\}$, $\{c\}$, Φ , Φ , $\{c\}$
9. (i) $\Omega - (A \cap C) - (B \cap D)$ (ii) $(C - B) \cup (D - A)$.
(iii) $A \cap C'$ (iv) $B \cap D'$
10. $n(A' \cap B' \cap C') = 38$, consistent
13. Inconsistent, the returns on pipe smoking are inconsistent.
16. 25, 18. 25, 40, 20, 25, 21. 4, 38
22. (i) 120 (ii) Cricket 108, hockey 80, football 64, Total 252

(1.4)

1. $\{(0, -3), (0, -2), (1, -3), (1, -2)\}$
2. $\{(1, 3), (2, 3), (1, 4), (2, 4)\}$
6. $(A \cap B) \cap C = \left\{ \begin{array}{l} (0, 2, 2), (0, 2, 3), (0, 2, 4) \\ (0, 3, 2), (0, 3, 3), (0, 3, 4) \\ (1, 2, 2), (1, 2, 3), (1, 2, 4) \\ (1, 3, 2), (1, 3, 3), (1, 3, 4) \end{array} \right\}$
 $A \cap (B \cap C) = \{(0, 2), (0, 3), (1, 2), (1, 3)\}$
 $(A \cup B) \cap (B \cup C) = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5)\}$
10. 36 cases $\{(1, 1), (1, 2), \dots, (1, 6), (2, 1), \dots, (6, 6)\}$
16 cases $\{(1, 1), (1, 2), \dots, (1, 4), (2, 1), \dots, (4, 4)\}$
11. Domain $\{2, 1\}$, Range $\{-1, 0, 1\}$
12. (i) $R = \{(1, 0), (2, 0), (3, 0), (2, 1), (3, 1), (3, 2)\}$
(ii) $R = \{(0, 0), (1, 1), (2, 2), (3, 3)\}$
(iii) $R = \{(2, 1)\}$
16. (a), (c) (d) and (e) are functional relations
17. (ii) and (iii) are functional relations
18. $R = \{(4, 3), (6, 3), (6, 5), (8, 3), (8, 5), (8, 7)\}$
Domain : $\{4, 6, 8\}$, Range : $\{3, 5, 7\}$
21. $R = \{(2, 6), (2, 8), (2, 10), (3, 3), (3, 6), (3, 9), (4, 8), (5, 10)\}$
Domain = A, Range = B.
22. $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (4, 1), (5, 1)\}$
24. $R = \{(1, 1), (2, 4)\}$

(2.3)

4. (i), (ii), (iii), (iv), (vi) are false
5. (i) $0 \leq x \leq 3$, (ii) no solution

$$(iii) \begin{cases} x \geq 13 \\ x \leq -7 \end{cases}$$

$$(iv) \begin{cases} x \geq 4 \\ x \leq 4 \end{cases}$$

$x = 4$ when both bounds are same the interval is termed as degenerate interval

(3.1)

1	$x = 1$	2	2.	3	$x = 7$	4	3	5	8	6	6
7	21			8	48			9	14	10	2
11	12	12	2.			13	1			14	(-21).
15	abc	16	0.12			17	$13\frac{6}{7}$			18	$\frac{1}{9}$
19	$(\frac{2}{5})$	20	1			21	$1\frac{2}{3}$			22	$2\frac{-1}{3}$
26	1.5	27	$5\frac{20}{3}$							28	3.2
33	$\frac{5}{3}, \frac{5}{3}$	34	4.1			35	4.16				

(3.2)

1	$x = 3, y = 2$	3	$x = 2, y = 1$	4	2.3	5	1.4
6	-2.1	8	16.-4	9	12.-2	10	$\frac{1}{4}, \frac{1}{5}$
12	40.30	13	13.17	14	$\frac{25}{21}, \frac{15}{10}$	15	$2\frac{2}{3}, 3\frac{2}{3}$
17	1.0	18	$x = \frac{a^2+ab+b^2}{ab}, y = \frac{ab}{a+b}$				
19	4.10	20	2.3	21	$q = 104, p = 28$	22	$4\frac{1}{3}$
23	-3.5.6						

(3.3)

- 1 (i) 1.7543 () 3.8973 (v) 2.9488 (v) 1.516
 2 (ii) Ant log (4.9608) = 0009.37
 4 () 45380

(3.4.1)

- 1 (i) (75.1440) (ii) $n \frac{n(n+1)}{2}$
 (iii) $(3n-2)x \frac{n(3n-1)x}{2}$ (v) $\frac{13n-4}{4}, \frac{n(13n+5)}{8}$
 (v) $\frac{3a}{14}(n+1); \frac{3an(n+3)}{28}$
 2 (i) AM between $\frac{1}{x}$ and $\frac{1}{y}$ () $\frac{x+y}{2xy}$ (ii) $\frac{a+b}{2}$
 3 18.4 17.6 16.8, -6.4

$$4 \quad -12 \ -9 \ -6, \quad 6$$

$$5 \quad 23 \text{ terms}$$

$$6 \quad (i) (-8 \ -16, -32) \quad (ii) -240 \quad 7 \quad (i) 960, \quad (ii) 240$$

$$8 \quad 4, 6, 8 \quad 10, 5, 7, 9 \quad 11 \quad \frac{9}{2}, \frac{7}{2}, \frac{5}{2} \quad 12 \quad 9, 1$$

(3.4.2)

$$1 \quad (i) \frac{1216}{343} \quad (ii) \text{Fifth term}$$

$$(iii) S_n = 7(7 + \sqrt{42}), S_4 = \frac{13}{\sqrt{7}(\sqrt{7} - \sqrt{6})}$$

$$2 \quad \frac{3}{2}, 1, \frac{2}{3} \quad 3 \quad \frac{5}{28}, \frac{45}{4}$$

$$4 \quad 12, 3, \frac{3}{4} \quad 5 \text{ Rs } 2,500 \text{ crores} \quad 6 \quad (i) \frac{116}{475} \quad (ii) \frac{97}{99}$$

$$8 \quad 4, 6, 9 \quad 9 \quad 2, 6, 18$$

$$12 \quad (i) \frac{4}{81}(10^{n+1} - 9n - 10) \quad (ii) \frac{1}{3} \left[n - \frac{1}{9} \left(1 - \frac{1}{10^n} \right) \right]$$

$$(iii) \frac{2}{81}(10^{n+1} - 9n - 10) \quad (iv) \sqrt{101}$$

(3.4.3)

$$1 \quad (i) \frac{1}{11}, \frac{1}{15} \quad (ii) \frac{1}{5}, \frac{1}{6} \quad (iii) 10, -10$$

$$2 \quad 6\sqrt{2} - 4\sqrt{3} \quad 3 \quad \frac{2}{5}, \frac{2}{7}, \frac{2}{9}, \frac{2}{11}$$

$$5 \quad 2b-a, \frac{b^2}{a}, \frac{ab}{2a-b}$$

(3.4.4)

$$1 \quad (i) x^5 + {}^5C_1 x^4 y + {}^5C_2 x^3 y^2 + \dots + y^5 \quad (ii) 32x^3 + 240x^4 + 720x^3 + \dots$$

$$2 \quad (i) -\frac{28}{27}x^3 \quad (ii) {}^{11}C_3 \frac{3^6}{2^3}x^3$$

$$3 \quad {}^{12}C_4 (2x)^8 \left(\frac{y}{2}\right)^4, {}^{12}C_7 (2x)^5 \left(\frac{y}{2}\right)^7$$

$$5 \quad (4 \cdot 12)^{1/2} = (4 \cdot 0 \cdot 12)^{1/2} = 4^{1/2} \left(1 + \frac{0 \cdot 12}{4}\right)^{1/2} = 2(1 + 0 \cdot 3)^{1/2} \\ = 2 \left(1 + \frac{0 \cdot 3}{2} + \text{all other terms neglected}\right) = 2 \left(1 + \frac{0 \cdot 3}{2}\right) = 2 \cdot 0 \cdot 3$$

(4.3)

$$1. \quad \tan 105^\circ = \frac{1+\sqrt{3}}{1-\sqrt{3}}, \tan 165^\circ = \frac{1-\sqrt{3}}{1+\sqrt{3}}, \tan 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$\begin{array}{ll}
 2 & \sin(A+B) = \frac{56}{65} \quad \cos(A-B) = \frac{63}{65} \\
 3 & \frac{77}{36} \quad 4 \quad -\frac{13}{84} \quad 5 \quad \frac{33}{65}, \frac{-63}{65} \quad \frac{56}{65}, \frac{16}{65} \\
 6 & \frac{120}{119} \quad -5 \quad 7 \quad \frac{240}{161} \quad \frac{-3}{5}
 \end{array}$$

(4.4)

$$\begin{array}{llll}
 1 & (i) 0.3491 & (ii) \frac{2\pi}{15} & (iii) 0.6981 \quad (iv) \frac{20}{5} \quad (v) \frac{7\pi}{15} \\
 & (vi) 1.0716 & (vii) 1.9835 & (viii) \frac{7\pi}{6} \quad (ix) \frac{7\pi}{2} \quad (x) \frac{5\pi}{2} \\
 2 & (i) 89^\circ 42' & (ii) 80^\circ & (iii) 420^\circ \quad (iv) 73^\circ 42' \quad (v) 150^\circ \\
 & (vi) 70^\circ 18' & (vii) 22^\circ 30' & (viii) 54^\circ 18'
 \end{array}$$

(4.5)

$$\begin{array}{llll}
 4 & 5.2 \text{ units} & 5 & 4.12 \text{ units} \\
 6 & (i) 3.61 & (ii) 5 & (iii) 6.69 \quad (iv) 4.78 \quad (v) \sqrt{101} \\
 7 & (i) (2, 3) & (ii) (-2, 3) & (iii) (-1, 1.5) \quad (iv) (1.5, 1.9) \quad (v) (0, 0) \\
 8 & (2, 1) \sqrt{5} & 9 & (2, -1)
 \end{array}$$

(4.6)

$$\begin{array}{ll}
 1 & x = 4y \quad x + y = 4 \quad x - y + 2 = 0 \quad 2x + y = 0 \\
 & 3x + 2y + 18 = 0 \quad 2x - 3y = 5 \quad x - 8y - 15 = 0 \quad x + y = 1 \\
 & (2b-a)x + by = b^2 + 2ab - a^2 \\
 2 & -1.9 \quad -4.8 \quad 0.8 \quad 0.8 \quad -\frac{b}{a} \\
 3 & 2x - 3y + 12 = 0 \quad 8x + 5y - 15 = 0 \quad 5y + 4x + 10 = 0 \\
 & 17x - 10y = 0 \quad 3x + 2y + 5 = 0 \\
 4 & (i) \frac{2}{3}, \frac{4}{5} \quad (ii) \frac{-8}{5}, 3 \quad (iii) -0.8, -2 \quad (iv) \frac{2}{\sqrt{3}-1}, \frac{-7}{\sqrt{3}-1} \\
 & (v) \frac{-(2b-3a)}{b}, \frac{b^2 + 2ab - a^2}{b} \\
 6 & 4x - 5y = 23 \quad 7 \quad 1x - 5y = 2 \\
 8 & (i) 3x - 5y = -15 \quad (ii) 3x - 4y = -12 \quad (iii) 8x + 3y = 24 \\
 & (iv) -9x - 6y = 54 \quad (v) x_1y_1 - x_2y_2 = -x_1y_1 \\
 9 & (i) \frac{9}{4}, \frac{9}{2} \quad (ii) \frac{-3}{5}, \frac{1}{2} \quad (iii) \frac{5}{2}, \frac{-5}{3} \\
 & (iv) 2, -4 \\
 10 & (i) x + y = 12 \quad (ii) x - y + 5 = 0 \quad (iii) x + 4y = 7 \\
 11 & 2x + 3y = 1 \quad 12 \quad q = -\frac{1}{50p+3}
 \end{array}$$

(4 7)

$$1 \quad (i) \text{ circle with centre at } (-4, 9) \quad (i) \text{ Rad as } \frac{5}{2} \text{ Centre at } (5, -2)$$

$$(iii) \text{ circle with centre at } (-4, -6)$$

$$(iv) \text{ Radius} = \frac{\sqrt{51}}{2} \text{ Centre } (-5, -3)$$

$$(i) \text{ An imaginary circle centre at } \left(-\frac{15}{4}, -\frac{6}{9}\right)$$

$$2 \quad x^2 + y^2 = 1 \quad x^2 + y^2 = 49 \quad x^2 + y^2 = 100$$

$$3 \quad 2 \text{ units}$$

$$4 \quad x^2 + y^2 - 10x - 6y + 18 = 0$$

$$5 \quad (i) x^2 + y^2 + 8x + 6y = 0$$

$$(ii) x^2 + y^2 - 8x + 4y + 11 = 0$$

$$(i) x^2 + y^2 + 12x - 10y + 57 = 0$$

$$(iv) x^2 + y^2 - 6y - 16 = 0$$

$$(i) x^2 + y^2 + 4x - 12 = 0$$

$$6 \quad (i) (0, 0), 5$$

$$(i) (1, 3), 2$$

$$(i) (-3, -4), 4$$

$$(iv) (-2, 4), 3$$

$$(v) (5, 12), 13$$

$$(iv) (3, -4), 13$$

$$(iii) (1, 0), \frac{5}{2}$$

$$(iii) \left(\frac{4}{3}, -\frac{5}{3}\right), \sqrt{\frac{127}{18}}$$

$$(iv) \left(\frac{2}{3}, -\frac{5}{3}\right), \sqrt{\frac{101}{4}}$$

$$(i) (a, b), \sqrt{a^2 + b^2}$$

$$7 \quad x^2 + y^2 + 6x - 8y = 0$$

$$9 \quad (i) 4x^2 + 4y^2 + 42x + 47y + 138 = 0$$

$$(ii) 3x^2 + 3y^2 - 29x - 19y + 50 = 0$$

$$(i) 2x^2 + 2y^2 - 11y - 1 = 0$$

(4 8)

$$1 \quad V(0, 0), F(0, -4), D: y = 0$$

$$2 \quad V(3, 2), F(3, 0)$$

$$3 \quad V(1, 2), F(0, 2), D: x = 0$$

$$4 \quad V\left(4, \frac{9}{2}\right), F(4, 4)$$

$$5 \quad V(-2, 2), F\left(-2, \frac{3}{2}\right)$$

$$6 \quad \left(\frac{1}{3}, -\frac{1}{2}\right), D: 2y - 1 = 0$$

$$7 \quad V\left(4, \frac{9}{2}\right), F(4, 4), D: y - 5 = 0$$

$$8 \quad V(-1, -2), F\left(-1, -\frac{5}{3}\right), D: 3y + 7 = 0$$

$$9 \quad V\left(-2, \frac{4}{3}\right), F\left(-2, \frac{7}{12}\right), D: 12y - 25 = 0$$

$$10 \quad V\left(-\frac{2}{2}, -\frac{3}{4}\right), F\left(-\frac{28}{15}, -\frac{3}{4}\right), D: 15x - 8 = 0$$

(4 9)

$$1 \quad xy - 3x - 2y + 1 = 0$$

$$2 \quad c = \frac{\sqrt{5}}{3}$$

$$\{\infty\} = \left(\pm \frac{5}{\sqrt{2}}, 0\right)$$

(4 11)

1 Not continuous.

2 Continuous (regular)

Continuity except that point (1, 0) which is regarded part of the curve but completely separate from all other points of the curve

(4 12)

1 Not convergent

2 Convergent

3 Not convergent

4 Conditionally convergent

5 Convergent

6 Conditionally convergent

7 Convergent

8 Convergent if $|c| < 1$ and divergent otherwise

9 Convergent

10 Convergent

11 Convergent

12 Conditionally convergent

13 Not convergent if $0 < p < 1$ convergent if $p > 1$ 14 Convergent if $0 < n < 1$ Divergent if $n \geq 1$

15 Convergent

(5 1)

$$1 \quad (c) \begin{bmatrix} 10 & -13 & 1 & 7 \\ -1 & 2 & 0 & -1 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

$$2 \quad (b) \begin{bmatrix} 12 \\ 34 \end{bmatrix}$$

$$(c) \begin{bmatrix} 18 & -1 \\ 34 & 1 \end{bmatrix}$$

$$(e) \begin{bmatrix} 22 & 31 & 40 \\ 49 & 67 & 85 \\ 48 & 79 & 100 \end{bmatrix}$$

$$(f) \begin{bmatrix} 28 & 37 & 46 \\ 53 & 71 & 89 \end{bmatrix}$$

$$(g) [23 \ 41 \ 46]$$

$$(h) [14]$$

$$(i) [ad+bc+ef]$$

$$(j) \begin{bmatrix} 2 \\ 6 \\ 9 \end{bmatrix}$$

$$(k) [1 \ 2 \ 3]$$

$$(l) [4 \ 5 \ 6]$$

$$3 \quad (i) AB = \begin{bmatrix} 81 & 48 \\ 88 & 88 \\ 12 & 14 \end{bmatrix} \text{ BA not defined}$$

$$(ii) AB = \begin{bmatrix} 19 & -7 & -12 \\ 52 & -4 & -30 \\ 30 & 6 & 30 \end{bmatrix}$$

$$BA = \begin{bmatrix} 24 & 24 & 36 \\ -6 & 48 & 30 \\ -15 & -27 & -27 \end{bmatrix}$$

$$(iii) AB = [40]$$

$$BA = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 4 & 6 & 8 & 10 \\ 6 & 9 & 12 & 15 \\ 8 & 12 & 16 & 20 \end{bmatrix}$$

$$(ii) AB = [5]$$

$$BA = \begin{bmatrix} 0 & 12 & 5 & 20 \\ 0 & 6 & 2 & 10 \\ 0 & -3 & -1 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(vii) AB = \begin{bmatrix} 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 3 & 1 & 0 \\ 0 & 2 & 0 & 3 & 1 \\ 1 & 0 & 2 & 0 & 3 \end{bmatrix}$$

$$(viii) AB = \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix} = 10I$$

$$4 \quad (i) \begin{bmatrix} 82 \\ 58 \end{bmatrix} \quad (iii) \begin{bmatrix} -3 \\ -8 \\ -3 \end{bmatrix} \quad (v) \begin{bmatrix} -2 \\ -13 \end{bmatrix}$$

$$5 \quad (a) \begin{bmatrix} -5 & -1 & 0 & -5 \\ -1 & 7 & -7 & -6 \end{bmatrix} \quad (c) \begin{bmatrix} -16 & -14 \\ -\sqrt{2} & -\sqrt{2} \end{bmatrix}$$

$$7 \quad (i) AB = \begin{bmatrix} 6 & 7 \\ 16 & 12 \end{bmatrix} \quad BA = \begin{bmatrix} 2 & 9 \\ 8 & 16 \end{bmatrix}$$

$$(ii) AB = \begin{bmatrix} 4 & 11 \\ 8 & 12 \end{bmatrix} \quad BA = \begin{bmatrix} 2 & 8 \\ 11 & 12 \end{bmatrix}$$

$$(iii) AB = [19] \quad BA = \begin{bmatrix} 2 & 4 & 3 \\ 4 & 8 & 6 \\ 6 & 12 & 9 \end{bmatrix}$$

AB and BA are not defined

$$9 \quad (a) \begin{bmatrix} 12 & 17 \\ 22 & 29 \end{bmatrix} \quad (b) \begin{bmatrix} 9 & 19 \\ 17 & 13 \end{bmatrix} \quad (c) \begin{bmatrix} 10 & 24 \\ 14 & 31 \end{bmatrix}$$

$$(d) \begin{bmatrix} 12 & 22 \\ 17 & 29 \end{bmatrix} \quad (e) \begin{bmatrix} 5 & 6 \\ 4 & 10 \end{bmatrix}$$

$$10 \quad (a) \begin{bmatrix} 12 & 17 \\ 22 & 29 \end{bmatrix} \quad (b) \begin{bmatrix} 12 & 22 \\ 17 & 29 \end{bmatrix} \quad (c) \begin{bmatrix} 12 & 22 \\ 17 & 29 \end{bmatrix}$$

$$17 \quad (i) \begin{bmatrix} 60 & 44 \\ 27 & 32 \end{bmatrix} \quad (ii) \begin{bmatrix} 58 & 40 \\ 31 & 34 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 44 & 6 \\ -5 & 10 \end{bmatrix} \quad (iv) \begin{bmatrix} 32 & 19 \\ 0 & 18 \end{bmatrix}$$

(v) when $AB = BA$

(5.3)

$$1 \quad 10$$

$$2 \quad 22$$

$$3. \quad -x^2 - 1$$

$$4 \quad 2$$

$$5 \quad \left(-\frac{1}{x^2}\right) \quad 6 \quad (-4ab) \quad 7 \quad 14 \quad 8 \quad ab-3a^2$$

$$9 \quad 40 \quad 10 \quad x^2-10x+10 \quad 11 \quad -40$$

$$12 \quad abc-af^2-bg^2-ch^2+2fgh$$

$$13 \quad 2 \quad 14 \quad -4$$

(5.5)

$$2 \quad (b) \ 2 \quad (d) \ 3 \quad (f) \ 2$$

(5.6)

$$2 \quad (b) \begin{bmatrix} 1 & 1 & 0 \\ 1 & \frac{3}{2} & \frac{1}{2} \\ 2 & \frac{3}{2} & \frac{1}{2} \end{bmatrix} \quad (d) \begin{bmatrix} \frac{1}{2}\lambda_0 & \frac{1}{2}\lambda_0 & \frac{1}{2}\lambda_0 \\ \frac{1}{2}\lambda_0 & \frac{1}{2}\lambda_0 & \frac{1}{2}\lambda_0 \\ \frac{1}{2}\lambda_0 & \frac{1}{2}\lambda_0 & \frac{1}{2}\lambda_0 \end{bmatrix}$$

$$(e) \begin{bmatrix} -5 & 9 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix}$$

(5.7)

$$1 \quad \begin{pmatrix} 7 & 3 \\ 6 & 8 \end{pmatrix} \quad 3 \quad (3 \ 2) \quad 4 \quad (-3 \ -1 \ 0) \quad 5 \quad (1 \ 2 \ 3)$$

$$6 \quad (7 \ 10 \ 9) \quad 7 \quad (2, 3 \ 4) \quad 8 \quad (3 \ -5 \ 1) \quad 9 \quad (1 \ -1 \ -1)$$

$$10 \quad (2 \ 3 \ 4) \quad 12 \quad -1 \ 46 \ 37 \quad 13 \quad 1 \ 2 \ 3 \quad 14 \quad 1 \ -1 \ 0$$

$$15 \quad 4 \ \frac{15}{2} \ 5 \ 6 \quad 16 \quad \frac{-17}{14} \ \frac{33}{14}$$

$$17 \quad k = -9 \quad 18 \quad k = 23 \quad 20 \quad x = -1 \ y = 1$$

(6.1)

$$1 \quad 9x^2 \quad 3 \quad \frac{5}{2}x^{-2} \quad 5 \quad \frac{1}{2}x^{-2}$$

$$7 \quad \frac{7}{3}x^{4/3} \quad 9 \quad \frac{3}{8}\sqrt{x} \quad 11 \quad 32x+15$$

$$13 \quad 12x^3+2x \quad 15 \quad b+6cx^3+3dx^2+4ex^3+5x^4$$

$$17 \quad -3x^2+14x+4 \quad 19 \quad \frac{8}{3}x(3x-7)-\frac{11}{3}$$

$$B \quad 1 \quad (6x+8) \quad 3 \quad (x^2+4x)(3x^2-4)+(x-4x)(2x+4)$$

$$5 \quad \left[\frac{(2x+3)^{3/2}}{x^2} + \frac{3\sqrt{2x+3}}{x} \right]$$

$$7 \quad \left\{ (2x^3/2-\sqrt{x})(2x+1) + (x^2+2x+1) \left(3\sqrt{x} - \frac{1}{2\sqrt{x}} \right) \right\}$$

$$9 \quad \frac{(1+\sqrt{x})(\sqrt{x}+1)}{\sqrt{x}} + \frac{x+2\sqrt{x}}{2\sqrt{x}+1}$$

$$C \quad 1 \quad \frac{15}{(x-5)^2} \quad 3 \quad \frac{32x^3+8x^2-42x+32}{(4x+3)^2}$$

- 5 $\frac{(1-x)}{2\sqrt{x}(x+1)^2}$ 7 $\frac{1}{(1-x)^2} \times \sqrt{\frac{1+x}{1-x}}$
- 9 $\frac{3}{2\sqrt{x}\sqrt{a^2-2}}$
- D 1 $8(x^2+x+1)(2x+1)$ 3 $\frac{3}{2}(3x+8)^{1/2}$
- 5 $\frac{2+x}{\sqrt{4x+x^2}}$ 8 $\frac{4x-3x^2}{2(1-x)^{3/2}}$
- E 1 $\frac{-2y}{x}$ 3 $-\left(\frac{x+1}{y+1}\right)$
- 11 $\frac{y}{x}$ 12 $\frac{\log x}{(1+\log x)^2}$ 13 $\frac{xy \log y - y^2}{xy \log x - x^2}$
- F 1 $4 + \cos x$ 3 $\frac{1}{2} \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)$
- 5 $(-1) 4 \cos 2x \sin 2x$ 7 $x^2(\cos x + \sin x) + (\sin x - \cos x)$
- 11 $\sin x(1 + \sec^2 x)$ 12 $\frac{\cos \sqrt{x}}{2\sqrt{x}}$ 13 $-\cot^2\left(\frac{x}{3}\right) / \sin^2\left(\frac{x}{3}\right)$
- G 1 $4e^{4x}$ 5 $-e^{(5-x)}$ 6 $y \sec^2 x$
- 15 $(a-b-ax)e^{-x}$ 17 $\frac{1}{2}e^{\sqrt{x}}\left(1+\frac{1}{\sqrt{x}}\right)$ 18 $\frac{2e^x}{(1-e^x)^2}$
- H 1 $\frac{2ax+b}{ax^2+bx+c}$ 3 $\left(\frac{-x}{1-x^2}\right)$ 5 $\left(\frac{4x}{1-x^4}\right)$
- 8 $2x+2x\log(x^2+1)$ 9 y 11 (a) $(10x+2)e^{1x^2+2x+2}$ (b) $16x^3-12x^2+2x$
- 10 $\frac{1}{\sqrt{a^2+x^2}}$ (c) $x^x(\log x+1)$ (f) $x^{\sin x} \left(\cos x \log x + \frac{\sin x}{x} \right)$
- I 1 $4x^3+18x^2+18x$ 2 $(x^2+5x)(2x+5)$
- J 2 $\frac{x-e^x}{\sqrt{x^2-2e^x}}$ 3 $x^{x^x} x^x \log x \left[(1+\log x) + \frac{1}{x \log x} \right]$
- 4 $\frac{-12x^2-16x+9}{(4x^2+3)^2}$
- 6 $\frac{-2x^3-6x^2-8}{(x^3+x-6)^2}$

(6.2)

- 1 (a) $f = 3x^2 - 12x + 9$, $f' = 6x - 12$, $f'' = 6$
 (b) $f = 50x^4 - 12x^2 + 5$, $f' = 200x^3 - 24x$, $f'' = 600x^2 - 24$, $f''' = 1200x$
- 3 $y_2 = \frac{-4}{(4-x^2)^{3/2}}$, value of x not defined.
- 5 $y_1 = 6$, $y_2 = -12$, $y_3 = 6$ at $x=1$ 7 yes the result holds good.
 $y_1 = -18$, $y_2 = 0$, $y_3 = 6$ at $x=3$

9. (a) $y_2 = 2a$ (c) $y_3 = \frac{a^2}{(ax-b)^2} - \frac{a^2}{(ax+b)^2}$
10. (b) $y_2 = \frac{1}{2}(e^x + e^{-x})$ (c) $y_2 = e^{-\frac{1}{2}x^2} \{x^2 - 1\}$ (d) $y_2 = e^x(x^2 + 4x + 2)$
13. (i) $\left(-\frac{1}{a}\right)^n e^{-x/a}$ (ii) $\frac{(-1)^{n-1}(n-1)!}{x^n}$ (iii) $\sin(x+n)$
- (iv) $\frac{(-1)^{n-1} 1.3.5 \dots (2n-3)}{2^n \sqrt{x^{2n-1}}}$
18. (i) $(-2)e^x \sin x$ (ii) $\frac{2}{x}$ (iii) $xa^x [x^2 \log^2 a + 6x \log a + 6]$
24. (i) $\max$ at $x = -1$, $\min$ at $x = 3$
- (ii) $\min$ at $x = -1$, $\min$ at $y = -1$
- (iii) $\max$ at $x = \frac{1}{2}$, $y \max = \frac{1}{2\sqrt{2}}$
- (iv) $\min$ at $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$
- $\max$ at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$
25. (i) Neither maxima nor minima (ii) $\max$ at $x = 0$; $\min$ at $x = -1$ and $x = +1$
- (iii) $\min$ at $x = \frac{\log 2}{2}$ (iv) $\min$ at $x = -1$, $\max$ at $x = +1$

(6.3)

1. $\begin{cases} \text{Maximum at } x=3 \text{ and} \\ \text{Maximum at } x=1 \end{cases}$ 3. Minimum at $x = \frac{1}{4}$
5. $\begin{cases} \text{Maximum at } x=2 \\ \text{Minimum at } x=1 \end{cases}$ 9. $\begin{cases} \text{Minimum at } x=2 \\ \text{Maximum at } x=0 \end{cases}$
10. Maximum at $x = -\frac{1}{2}$; Minimum at $x = \frac{1}{2}$
11. Maximum at $x = -1$; Minimum at $x = \frac{1}{3}$
20. No.
23. (i) $x < 0$ convex upwards, $x > 0$ convex downwards, inflexion $x = 0$
- (ii) Convex down (iii) Convex (iv) Inflexion at $x = 0$
- (v) inflexion at $x = b$

(7.1)

1. $-\left(\frac{1+x^2}{2x^3}\right)$ 3. $-\left[\frac{2(a-bx)}{bx}\right]$ 4. $\frac{bx^2-a}{2bx^2}$
5. $\left(-\frac{3}{2}\right)$ 6. (-3)
11. 2.10 (approx), $e_2 = -2.15$ at $p = 2.88$, $e_2 = -2.55$ at $p = 3$
13. $\frac{2}{3}$ 16. $\pi, 1$

(7.2)

- 5 $p = \frac{23}{3}$ $\pi = \frac{4}{3}$ 7 $p = \frac{10}{3}$ $\pi = \frac{5}{e} - 1$ 11 15 sets (approx)
- 13 $Q = \frac{250}{7}$ $p = \frac{115}{7}$ 21 $\lambda = 100$ $\pi = 50$
- 25 (i) $2Q = \left[1000 + \frac{1}{2} \left(\frac{Q}{50} \right)^2 \right]$ (ii) $Q = 5000$ (iii) $\pi = 4000$

(7.3)

- 1 $p = 10.4 + 0.4t$ $\pi = \frac{(4-t)^2}{20}$ Govt Rev = $\frac{4t-t^2}{10}$
- 4 $r = \frac{b-\beta+2}{2(\alpha-a)}$ 7 $r = 4$ $p = 92$ after tax $x = 3.87$ and $p = 92.51$
- 8 No maxima $t = 6$
- 9 Before tax, $x = \frac{-4 \pm \sqrt{3646}}{24}$ After tax, $x = 2.4$ (approx)

(8.1)

- 1 $\frac{\partial z}{\partial x} = 6x^2 + 10xy + y^2$, $\frac{\partial z}{\partial y \partial x} = 10x + 2y$, $\frac{\partial z}{\partial y} = 5x^2 + 2xy + 2y$
- 3 $\frac{\partial z}{\partial x} = 2e^x$, $\frac{\partial z}{\partial y \partial x} = 2e^x$, $\frac{\partial z}{\partial y} = 5e^x + (2x + 5y)e^x$
- 5 $\frac{\partial z}{\partial x} = (2x + 2y)e^x$, $\frac{\partial z}{\partial y \partial x} = 2e^x$, $\frac{\partial z}{\partial y} = (2x + 2y)e^x$
- 9 $\frac{\partial z}{\partial x} = \frac{1+2x}{1+x^2}$, $\frac{\partial^2 z}{\partial x^2} = \frac{2-2x^2}{(1+x^2)^2}$, $\frac{\partial z}{\partial y} = 2y$, -2

(8.2)

- 1 $\frac{-2p_1}{1-2p_1+p_2}$, $\frac{p_2}{1-2p_1+p_2}$ (in case of x_1)
 $\frac{-3p_2}{5-2p_1-3p_2}$, $\frac{-2p_1}{5-2p_1-3p_2}$ (in case of x_2)
- 2 -2 and -1 in case of x_1 -1 and -1 in case of x_2
- 5 Nature cannot be determined since $\frac{\partial x_1}{\partial p_1}$ is positive quantity while $\frac{\partial x_2}{\partial p_1}$ is negative one
- Own partial elasticities $(-)$ 1.5 and $(-)$ 0.4
 Cross partial elasticities 0.4 and $(-)$ 0.5

(8.3)

- 1 $MP_L = 184$, $MP_K = 33$ and $MP_{(Land)} = 97$
- 3 $\frac{\partial X}{\partial L} = 2.13L^{1.13}K^{-.34}$, $\frac{\partial X}{\partial K} = .34(1.7)^{2.13}(2)^{-.64}$

$$5 \quad (b) \frac{\partial Y}{\partial a} = 10 \times 28 \quad \frac{\partial Y}{\partial b} = 10 \times 22 \quad \frac{\partial Y}{\partial c} = 10 \times 02 \quad \frac{\partial Y}{\partial d} = 10 \times 16$$

(8.4)

$$1 \quad dz = (3x^2 + 6xy + 6y^2)dx + (3x^2 + 12xy + 6y^2)dy$$

$$3 \quad dz = (6y^2 + 8xy^2 + y)dx + (6y^2 + 8x^2y + x)dy$$

$$5 \quad dz = \left[\frac{x^2 + 2xy - y^2}{(x+y)^2} \right] dx + \left[\frac{y^2 + 2xy - x^2}{(x+y)^2} \right] dy$$

$$11 \quad dU = \frac{\partial f_1}{\partial q_1} dq_1 + \frac{\partial f_1}{\partial q} dq_2$$

$$13 \quad dP = \frac{\alpha P}{L} dL + \frac{\beta P}{K} dK$$

(8.5)

$$1 \quad dz = (4x^3 + 4y)dx + (4y^2 + 4x)dy$$

$$d^2z = 12x^2(dx)^2 + 8dxdy + 12y^2(dy)^2$$

$$3 \quad dz = 2xdx + 2ydy \quad d^2z = 2(dx)^2 + 2(dy)^2$$

$$7 \quad dU = 5e^{3x}(5x+5)$$

$$d^2U = 25e^{3x} + (5x+5).25e^{3x}$$

(8.6)

$$1 \quad \frac{y^2}{1-2xy} \quad 3 \quad \left(\frac{4y+3x^2}{4x+3y^2} \right) \quad 5 \quad \frac{y^2-6x^2}{2y-2xy} \quad 7 \quad \frac{(x)^2}{e^x}$$

(8.7)

$$1 \quad \text{Min at } \left(\frac{1}{3}, -\frac{1}{3}\right), \text{ saddle pt. at } (0, 0).$$

$$3 \quad \text{Min at } (0, 0) \quad 7 \quad \text{Min at } (3, 3)$$

$$9 \quad \text{If } a \text{ and } b \text{ are positive then minimum}$$

$$\text{If } a \text{ and } b \text{ are negative then maximum and}$$

$$\text{If } a \text{ and } b \text{ possess opposites sign then saddle point.}$$

(8.8)

$$1 \quad \text{Min at } \left(\frac{1}{4}, \frac{1}{2}\right) \quad 3 \quad \text{Min at } \left(\frac{1}{3}, -\frac{1}{3}\right) \text{ saddle at point } (0, 0)$$

$$5 \quad \text{Maximum at } (0, 0, 0)$$

(9)

$$7 \quad q_1 = 5 \text{ and } q_2 = 1 \quad 9 \quad q_1 = 4 \text{ and } q_2 = \frac{3}{2}$$

$$10 \quad x = \frac{25}{2\phi_2}$$

(10)

$$1. \quad (i) \text{ Degree} = 3 \qquad (iii) \text{ Degree} = 0 \qquad (v) \text{ Degree} = 1.03$$

$$4. \quad L = \frac{1}{2}, K = \left(\frac{1}{4}\right)^{1/2}, Q = 15.48 \text{ and } \pi = 42.20 \quad 8. Q = 10\sqrt{2}$$

$$9. \quad a = \frac{1}{2}, b = \left(\frac{1}{4}\right)^{1/2}, x = 15.48 \text{ and } \pi = 42.20$$

$$13. \quad \left. \begin{aligned} MP_a &= \frac{1}{2} A a^{-1/2} b^{1/4} c^{1/4} \\ MP_b &= \frac{1}{4} A a^{1/2} b^{-3/4} c^{1/4} \\ MP_c &= \frac{1}{4} A a^{1/2} b^{1/4} c^{-3/4} \end{aligned} \right\}$$

$$16. \quad (a) \sigma = 1 \qquad (b) \sigma = \frac{1}{1+\alpha}$$

(11.1)

$$1. \quad p_1 = 4 \text{ and } p_2 = \frac{13}{4} \qquad 5. x_1 = 20 \text{ and } x_2 = 30$$

$$6. \quad x = 2 \text{ and } y = \frac{3}{2} \qquad 7. x = 1 \text{ and } y = \frac{13}{2}$$

(11.2)

$$2. \quad x_1 = 4 \text{ and } x_2 = 4 \qquad 4. x_1 = 6 \text{ and } x_2 = 4$$

$$8. \quad Q_1 = 4 \text{ and } Q_2 = 5 \qquad 9. Q_1 = 3 \text{ and } Q_2 = 3$$

$$11. \quad (a) q_1 = q_2 = 60.33 \quad (b) \text{ Indeterminate}$$

(12.1)

$$1. \quad 2x^2 + c \qquad 3. \frac{2}{3}x^{3/2} + c \qquad 5. \frac{x^4}{4} - \frac{4x^3}{3} + \frac{x^2}{2} + c$$

$$6. \quad \frac{a}{6}x^6 + a \log x - 5x + c \qquad 7. \frac{(-3x)^4}{6} + c \qquad 8. 6 \cos x + c$$

$$9. \quad 10\sqrt{x} + c \qquad 10. \frac{x^4}{4} - 8 \log x + c \qquad 12. x \frac{x^2}{9} + \sqrt{x} + c$$

$$14. \quad 5e^x \frac{3}{x} + c \qquad 15. \frac{x^4}{2} + x^2 + c \qquad 16. 4x \frac{3}{x} \frac{2}{\sqrt{x}} + c$$

$$17. \quad -x \frac{3}{x} \qquad 19. \frac{e^{3x}}{3} + \frac{3}{2}e^{2x} + \frac{x^3}{5} + c$$

$$21. \quad \sin x - \cos x$$

$$22. \quad \frac{x^5}{5} - \frac{3}{4}x^2 - 3\sqrt{x} + 3\sqrt[3]{x} + c \qquad 23. \quad \frac{1}{2}x^4 + 3\cos x + \frac{10}{3}x\sqrt{x} + c$$

$$24. \quad \cos x - \cot x + c \qquad 25. \quad \tan x - x + c$$

$$26. \quad -\cot x - x + c \qquad 27. \quad \frac{x}{2} \frac{\sin x}{2} + c$$

28 $2 \tan^{-1} 3 \sin^{-1} x + c$

29 $\frac{a^x}{\log a} - \frac{1}{10} x^4 + c$

(12.2)

1 $\frac{2}{15}(5x+3)^{1/2} - 2 \frac{1}{7} x^{7/2}$

3 $\left[\frac{-1}{5(5x-3)} \right]$

4 $\frac{1}{2} \sec(2x+3)$

5 $\frac{1}{2}(\log x)$

6 $\frac{1}{2}(\log x)^2$

7 $-\frac{1}{2} \cos x^2$

8 $\log(\log x)$

9 $\frac{1}{b} \log(a+b \sin x)$

11 $\frac{1}{4} \tan^4 x$

12 $\frac{1}{3} \sin^3 x$

13 $\frac{1}{2} \log(1+e^{2x})$

16 $\frac{1}{6} \tan^{-1} \frac{2x}{2}$

17 $\sin^{-1} \frac{x}{4}$

18 $\log(2x^2+5x+2)$

19 $\frac{1}{3} \sin^{-1} \frac{3x}{5}$

20 $2 \cos \sqrt{x}$

21 $\frac{1}{3}(x^2+1)^4$

22 $\frac{1}{10}(x^2+3)^{10}$

23 $\frac{(1+\log x)^{1-n}}{1-n}$

24 $\frac{1}{2} \sin^2 x$

25 $-1/6(3x^2+5)$

26 $\frac{1}{a} e^{ax} + \frac{1}{a}$

27 $\log(\log \sin x)$

28 $\frac{1}{a} \tan^{-1} \frac{x}{a}$

29 $\sin^{-1} \frac{x}{a}$

36 $-e^{\cos x}$

37 $\frac{1}{3} \sqrt{x+1}$

38 $\frac{1}{2} \sqrt[3]{(1+x)^2}$

39 $2x + \frac{1}{2}(e^{2x} - e^{-2x})$

40 $2 \log |\sin x| - \cot x$

41 $\frac{\sin x - 2}{\cos x}$

42 $\frac{\tan^2 x}{2} + \log \cos x$

43 $\log \left| \tan \frac{x}{2} \right|$

44 $\frac{1}{\cos x} + \cos x$

45 $\frac{1}{4} \sin 2x - \frac{1}{16} \sin 4x$

46 $\frac{1}{2} \log |\sec x| - x$

48 $\frac{1}{2} e^{x^2}$

49 $\frac{1}{30} [x(2x+1)]^5 - 5(2x+1)^3 - 2$

50 $\frac{1}{14} (3x^2+4)(x^4-2)$

51 $2 \left[\frac{1}{2} x - \sqrt{x} + \frac{1}{2} g(\sqrt{x}+1) \right]$

52 $\frac{1}{3} (x^2+2) \sqrt{x-1}$

53 $-2 \cos \sqrt{x}$

54 $3 \sin^2 \frac{1}{2} x$

55 $\sin^{-1} e^x$

56 $\frac{2}{3} (1+\log x)^{3/2}$

57 $\frac{2}{3} \sin^3 x + c$

58 $\frac{1}{2} \tan^2 x + c$

59 $\frac{\sin^3 x}{3} + c$

60 $2\sqrt{\sec x - 1}$

61 $\log \tan^{-1} x + c$

62 $\log(x + \sqrt{x^2 + a^2}) + c$

63 $\sin^{-1} \log x + c$

64 $-2\sqrt{1+\cos x} + c$

65 $\sin^{-1} \frac{x}{2} + c$

66 $\frac{1}{2} \sin^{-1} \frac{x^2}{\sqrt{3}} + c$

67 $\tan^{-1}(x+2) + c$

68 $\frac{1}{\sqrt{2}} \sin^{-1} \frac{4x-3}{5} + c$

69 $\sin^{-1} \frac{x+2}{3}$

70 $\frac{2}{3} \tan^{-1} \left(\frac{1}{3} \tan \frac{x}{2} \right)$

71 $\frac{1}{\sqrt{2}} \tan^{-1} \sqrt{2}(x+1) + c$

74 $3 \log(x^2 - x - 1) + \frac{10}{\sqrt{5}} \log \left| \frac{(x - \frac{1}{2}) \frac{\sqrt{5}}{2}}{(x - \frac{1}{2}) + \frac{\sqrt{5}}{2}} \right|$

75 $-\frac{3}{2} \log(1 - 2x - x^2) + \frac{1}{2} \log \left| \frac{\sqrt{2} + (1+x)}{\sqrt{2} - (1+x)} \right|$

76 $\frac{2}{3} \log(x+1) + \frac{1}{6} \log(x^2 - x + 1) + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x-1}{\sqrt{2}} \right)$

77 $\frac{1}{\sqrt{5}} \sin^{-1} \left(\frac{5x-4}{6} \right)$

78 $\sin^{-1} \left(\frac{2x+1}{\sqrt{5}} \right)$

(12.3)

1 $x - 2 \log(x+2)$

2 $[-x + 2 \log(1+x)]$

3 $\frac{x^3}{2} - 9x + 81 \log(x+9)$

5 $\frac{x^3}{2} - 2x + 4 \log_b(x+2)$

6 $4 \log(x-5) - 3 \log_b(x+5)$

7 $\frac{1}{2a} \log \left(\frac{x-a}{x+a} \right)$

8 $\log \frac{(x-2)^2}{x+1}$

9 $-\frac{1}{4} \log(x^2+1) + \frac{1}{2} \log(x-1) + \frac{1}{2} \tan^{-1} x$

10 $-\frac{(2x-1)}{6(x+1)^2} + \frac{2}{9} \log \frac{x-2}{x+1}$

11 $\frac{x^3}{3} + \frac{x^2}{2} + 4x + \log \frac{x^2(x-2)^3}{(x+2)^3}$

12 $\frac{\frac{1}{6} \log \frac{(1-\cos x)(1+\cos x)^3}{(1+2\cos x)^4}}{1}$

13 $\frac{3}{x-2} + \log \frac{(x-2)^2}{x^3}$

14 $\frac{4x+3}{2(x+1)} + \log \frac{x^2}{(1+x)^2}$

15 $\frac{x-2}{2(x^2+2x+3)} - \frac{\sqrt{2}}{4} \tan^{-1} \frac{x+1}{\sqrt{2}} + \log_2(x+1)$

17 $x - 3 \log(1+x)$

24 $\frac{1}{2} \log_b \frac{(1-x)^2}{x}$

19 $-\log x + \log(1+x) - \frac{1}{x}$

30 $\frac{e^x}{(1+x)^2}$

31 $\frac{\log(x^2 - 3x + 2)(x-2)}{(x-1)}$

32 $\log \left(\frac{x}{1+x} \right)$

33 $\frac{1}{a} \log_b \left| \frac{x-a}{x} \right| + \frac{x-a}{ax}$

34 $\frac{5}{2} \log_2(x^2 + 2x + 10) + \tan^{-1} \frac{x+1}{3}$

35 $3 \log_b \left| \frac{c(x-1)}{x+2} \right| - \frac{2}{x+2}$

$$34 \quad \frac{5}{2} \log[(x^2 + 2x + 10)] \tan^{-1} \frac{x+1}{x}$$

$$35 \quad 3 \log \left| \frac{x(1-x)}{x+2} \right| - \frac{2}{x+2}$$

$$36 \quad \frac{1}{4} \log \left| \frac{x-1}{x+1} \right| - \frac{1}{2} \tan^{-1} x$$

$$37 \quad (x-4)\sqrt{x+2}$$

$$38 \quad 2 \sin \frac{1}{2} \frac{x}{2} (2-x^2) \sqrt{4-x^2}$$

$$39 \quad 4 \log \frac{x+1}{x+2} + \frac{3}{x+2}$$

$$40 \quad -\tan^{-1} x + \frac{3}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} - \frac{2}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}}$$

(12.4)

$$1 \quad x^2 e^x - 2e^x (x-1)$$

$$3 \quad \frac{e^x}{b} \left[\log x - \frac{1}{b} \right]$$

$$4 \quad -\frac{1}{x} (1 + \log x)$$

$$5 \quad \frac{1}{2} e^x (\sin x + \cos x)$$

$$6 \quad \frac{e^x}{a^2 + b^2} [a \cos(bx+cx) + b \sin(bx+cx)]$$

$$7 \quad \frac{1}{2} e^{x^2} (\sin x - \cos x)$$

$$8 \quad \frac{x^4}{4} \left(\log x - \frac{1}{4} \right)$$

$$9 \quad e^x (x^2 - 2x + 2)$$

$$10 \quad x \tan^{-1} x - \frac{1}{2} \log(1+x^2)$$

$$11 \quad \frac{1}{2} (x^2 + 1) \tan^{-1} x - \frac{1}{2} x$$

$$12 \quad -\frac{1}{4} \cos 2x + \frac{1}{8} \sin 2x$$

$$13 \quad \frac{2}{3} x^{-1/2} \left(\log x - \frac{2}{3} \right)$$

$$14 \quad \frac{e^x}{4} \left[(\log x)^2 - \frac{1}{2} \log x + \frac{1}{8} \right]$$

$$15 \quad \frac{e^x}{x+1}$$

$$19 \quad (\log x)^2 x - 3x(\log x)^2 + 6x - \log x - 6x$$

$$20 \quad \frac{1}{10} x^{-4} \sin 2x - \frac{1}{5} x^{-3} \cos 2x \quad 21 \quad \frac{1}{2} x e^x - \frac{1}{4} e^x$$

$$26 \quad e^x \left\{ \frac{1}{2} x^3 - \frac{7}{4} x^2 + \frac{3}{4} x - \frac{3}{8} \right\}$$

$$27 \quad \log \log x$$

$$28 \quad \frac{1}{2} (1 + \log x)^2$$

$$29 \quad x[(\log_b(x^2-1))^2 + 1]$$

$$30 \quad x \tan x + (\log_b x) \cos x$$

$$31 \quad \frac{1}{8} [2x^2 + 2x \sin 2x + \cos 2x]$$

$$32 \quad x \log(x^2+1) - 2x + 2 \arctan^{-1} x$$

$$33 \quad \frac{x \sec x}{(x \sin x + \cos x)} + \tan x$$

$$34 \quad \frac{\sqrt{a^2 - x^2}}{x} - \sin^{-1} \frac{x}{a}$$

$$35 \quad \frac{1}{2} x^{-1} \tan^{-1} \sqrt{x-1} - \frac{1}{2} \sqrt{x-1}$$

$$36 \quad x \log(x + \sqrt{1+x}) - \sqrt{1+x^2}$$

37 $(x \log_e x)^3 - 2x \log_e x$

39 $2\sqrt{x} \sin^{-1} \sqrt{x+2\sqrt{1-x}}$

40 $x \sin^{-1} \sqrt{\frac{x}{1+x}} - \sqrt{x} + \tan^{-1} \sqrt{x}$

41 $2\sqrt{x+1} + \log \left| \frac{x+2\sqrt{1+x}}{2} \right|$

43 $\sin^{-1} \sqrt{1+x^2}$

45 $\frac{1}{2} \tan^{-1} \frac{e^x}{2} - \frac{1}{2} x + \frac{1}{4} \log 4 + e^{2x}$

47 $3x^{1/3} - 12x^{1/6} + 24 \log(x^{1/6} + 2)$

38 $e^x \log_e x$

42 $2 \sin^{-1} \sqrt{x}$ (Hint put $x = \sin^2 t$)

44 $\frac{1}{(\tan x + 1)}$

46 $\frac{1}{2} \log \frac{x^2}{1+x} - \frac{\tan^{-1} x}{x}$

48 $-2e^{\sqrt{x}} [\sqrt{x+1}]$

(13.1)

1 $\frac{44}{3}$

4 $e^{x/2}$

7 $2 \log_e 2 - 1$

10 $\frac{1}{9} [24 \log_e 2 - 7]$

13 $\frac{\pi}{2a}$

16 $\sin^{-1} \frac{3}{4} - \sin^{-1} \frac{1}{2}$

29 (i) $\frac{1}{2}$

34 $\frac{2}{3} (2\sqrt{2\rho h}) \times h$

38 $\frac{8}{3}$

45 $-\frac{\pi}{2} \log 2$

2 $\frac{1}{2} (\log 2 - 1)$

5 $\frac{\pi}{4}$

9 $\frac{\pi}{2} - 1$

11 $\sqrt{2} - 1$

14 $\frac{10}{3}$

17 $\frac{1}{2}$

(ii) $\frac{\pi}{4}$

36 $\frac{16}{3}$

43 $\frac{\pi}{4} + \frac{1}{2}$

46 $\frac{\pi}{4}$

(13.2)

1 $q^3 - 2q^2 + 5q + 100$

2 $2\sqrt{a^2+h^2} - 2\sqrt{h}$

3 $0.3y - 0.2\sqrt{y} - 22.5$

(13.3)

1 (a) 31.25

(b) 81.75

6 $32\sqrt{6}$

7 10^3

(14.1)

A 1 Order 2 degree 1

2 Order 2 degree 4

B 1 $\log \left(\frac{y}{x} \right) - \left(\frac{x+y}{xy} \right) = C$

3 $y = \log \left(e^x + \frac{x^2}{3} + C \right)$

4 $\log xy + x y = C$

5 $\frac{1-y}{1-x} = \text{constant}$

6 $x = \frac{y+L}{1-6y}$

7 $y - 2a\sqrt{a+x} - \frac{2}{3}(a+x)^{3/2} = C$

9 $x^2 + y^2 = C^2$

10 $y = \frac{C}{x}$

11 $y = Cx$

12 $(y-a) = Ce^{1/x}$

13 $\frac{t+y}{xt} + \log \frac{x}{t} = C$

14 $2\sqrt{-\tan^{-1} s} = C$

15 $y\sqrt{1-x^2} - x\sqrt{1-y^2} = C$

16 $\sin^{-1} y \sin^{-1} x = C$

17 $t^2 + y^2 = r^2 + C$

C 9 $y = Ce^{-x} = 10e^{-x}$

11 $y = Ce^{2x} + \frac{3}{2}C = -\frac{3}{2}$

13 $x = 2t + c, c = 0$

15 $y = \frac{2}{\sqrt{3}} - \frac{2}{\sqrt{3}}e^{\sqrt{3}x}$

(14.2)

1 $x' + 2xy - C, y = -\frac{x}{2}$

2 $y = x(\log -x + C)$

3 $\frac{1}{x} - \frac{1}{y} = C, y = x$

4 $x^2 3x'y - 4y^3 = C, x^2 + 3x'y - 4y^2 + 4 = 0$

5 $y = \frac{x}{\log x + c}$

6 $x' + 2xy - y^2 = C$

7 $x^2 - y^2 - 4x^2 y^2$

9 $x^2 + y^2 - Cx^4$

12 $y - x + \log \frac{y}{x} = C$

13 $\left. \begin{aligned} \sin^{-1} \frac{y}{x} - \log x = C \text{ or} \\ 1 + 2C\sqrt{1-C^2x^2} = 0 \end{aligned} \right\}$

14 $x^2 - y^2 = Cx$

21 $-\frac{x}{2y^2} = \log(C)$

22 $(x+y)^2(2x+y)^2 = C$

23 $x^2 \sqrt{3 \log Cx}$

24 $\left\{ \begin{aligned} te^{\sqrt{s/t}} = C \text{ or} \\ a = t \log^2 \frac{C}{t} \end{aligned} \right.$

25 $(x+y-1)^2(x-y-1) = C$

26 $\log(4x+8y+5) + 8y - 4x = C$

27 $\log(2x-3) - \frac{4y+5}{2x-3} = C$

28 $\frac{(x-y)^2}{x+y-2} = C$

29 $(y-x + \log(x-y)) = C$

(14.3)

1 $x^3 + 3x^2y + y^2 = C$

2 $xy + x^2 = C$

3 $x^2 - y^2 3xy = C$

5 $\frac{1}{2}x^3 + x^2y + \frac{1}{2}y^3 = C$

7 $x^2y + 2y^2 = C$

9 $xy + y^2 = C$

11 $\log x + \frac{1}{2}y^2/x^2 = C$

12 $x^2 + y^2 - 2C + 1 = 0$

13 $y^3 + 3xy^2 = C$

15 $2x + y(x^2 + 2C) = 0$

16 $\frac{x^2}{3} + xy - y^2 = C$

17 $y^4 + 4xy = C$

18 $x^2 + y^2 = Cx^2$

19 $\frac{xy}{x-y} = C$

20 $x^2 + y^2 - 2 \sin^{-1} \frac{x}{y} = C$

(14.4)

1 $y = Ce^{-2x} + 1$

5 $y = \sin x \sin^{-1} x + C \sec x$

7 $y = \tan \frac{x}{2} [2 \sin x + 2x + C]$

8 $xy[\log x] = C$

9 $y = Ce^x - 10e^{-x}$

11 $y = -\frac{3}{2}e^{-x} + \frac{3}{2}$

13 $x = 2y + C; C = 0$

15 $y = \frac{2}{\sqrt{3}} - \frac{2}{\sqrt{3}} e^{\sqrt{3}x}$

(14.5)

4 (ii) $y = x \log x + C_1 x + C$

(iii) $\frac{x^4}{12} + C_1 x + C_2$

(iv) $C_1 e^x + C_2 e^{4x}$

(viii) $y = C_1 e^{-x} + C_2 e^{6x}$

(ix) $y = C_1 e^{2x} + C_2 e^{4x}$

(x) $y = C_1 e^{-2x} + C_2 e^{2/3x}$

(vi) $y = (C_1 + C_2 x) e^{2x}$

6 (iii) $y = C_1 e^x + C_2 e^{2x} + \frac{e^{3x}}{12}$

(iv) $y = C_1 e^{2x} + C_2 e^{-2x} - x e^{2x}$

6 (i) $y = C_1 e^{-x} + C_2 e^{-2x} + \frac{1}{3}x - \frac{4}{9}$

(ii) $y = C_1 \cos 3x + C_2 \sin 3x + C_3 \sin 3x + \left[\frac{1}{18}x^2 - \frac{1}{27} + \frac{5}{81} \right] e^{2x}$

(iii) $y = e^{-x} (C_1 \cos 2x + C_2 \sin 2x) + \frac{2}{5} \cos x + \frac{1}{5} \sin x$

(iv) $y = C_1 e^x + C_2 e^{-x} + e^{2x} \left(\frac{3}{10} \cos x + \frac{3}{5} \sin x \right)$

(v) $y = C_1 \cos 3x + C_2 \sin 3x + \frac{1}{3} C e^{3x}$

(vi) $y = C_1 e^x + C_2 e^{-x} - 5x - 2$

(viii) $y = C_1 e^x + C_2 e^{-2x} - \frac{1}{5} (6 \sin 2x + 2x + 2 \cos 2x)$

(ix) $y = C_1 e^{-ax} + C_2 e^{-bx} - \frac{x+1}{a^2}$

(15.1)

5. $P = 50 - 40e^{-0.2t}$

10 $P = 2 - x + 8e^{-x}$

11 $C = \frac{1}{2} + x - \frac{1}{2} e^{-2x}$

12 $S_1 - S_2 = e^{-2t} \lim_{t \rightarrow \infty} (e^{3t} - 1)$

14 $y = 3e^x$

16 $R = (10t^2 - x^2)^{1/2}$

18 $p_1 = \bar{p} + (\rho_0 - \bar{p}) e^{-\frac{t}{\alpha-d}}$, where $\bar{p} = \frac{b-d}{\alpha-d}$

(16.1)

1 $y = y_0 + 3t$

2 $v_t = y_0(0.85)^t$

4 $v_t = y_0 \left(\frac{a}{m} \right)^t$

6 $y_t = (0.2)^t + 10(1 - 0.2)^t$

7 $v_t = y_0(0.3)^t + \frac{1-0.3^t}{0.7}$

8 $P_t = 3 \left(\frac{1}{5} \right)^t$

9 $y_t = 10 \left(\frac{1}{L} \right)^t$

11 $y_t = \beta(\alpha)^t$

17 $y_t = \frac{7}{2} \left(\frac{3}{2} \right)^t (-1)^t$

18 $y_t = y_0 \left(\frac{1}{1-\alpha} \right)^t$

(17.1)

1 $A_2 t^2 + A_1 (-5)^t$

2 $A_1 3^t + A_2 t 3^t$

3 $A_1 (1)^t + A_2 t (-6)^t$

5 $A_1 \frac{1}{2} t + A_2 \left(-\frac{1}{2} \right)^t$

6 $\left(\frac{1}{2} \right)^{t+1} [1 + (-1)^t]$

9 $y_t = \frac{1}{8} 5^t + C_1 C_2 3^t$

10 $y_t = \frac{\pi}{2} 6^t + C_2 t e^t$

11 $y_t = \frac{1}{8} x^2 2^t + C_1 2^t + C_2 t e^t$

12 $y_t = \frac{5}{2} + \frac{3}{2} t + \frac{1}{2} t^2 + A_1 3^t + A_2 2^t$

13 $A_1 (5)^t + A_2 (-1)^t$

(17.2)

2 $y_t = \frac{2^{t+1}}{a-1} + y_0$

3 $y_t = 3^t - \frac{x}{2} - \frac{1}{4}$

4 $y_t = y_0 2t$

5 $\left(\frac{1}{2} \right)^t$

6 $y_t = \left(\frac{1}{1-\alpha} \right)^t - \frac{\beta}{\alpha} \left[1 - \left(\frac{1}{1-\alpha} \right)^t \right]$

7 Damped oscillations

11 $y_t \sim \frac{1}{2}$

16 $y_t = \frac{1}{4} \left(-\frac{1}{2} \right)^t + \frac{1}{4}$ damped oscillations

$$y_0 = \frac{6}{2}, y_1 = \frac{6}{8}, y_2 = \frac{6}{2}, y_3 = \frac{7}{12}$$

17 $y_t = 2 \left(\frac{1}{e} \right)^t + 2$ monotonic decreasing damped

$$y_1 = \frac{13}{3}, y_2 = \frac{15}{9}, y_3 = \frac{16}{27}$$

18 $y_t = \frac{2}{5} \left(\frac{1}{3} \right)^t + \frac{3}{5}$ monotonic decreasing damped

$$y_1 = \frac{13}{15}, y_2 = \frac{7}{2}, y_3 = \frac{97}{135}$$

$$19 \quad y_t = 6 - 2 \left(\frac{2}{3} \right)^t \text{ monotonic decreasing damped}$$

$$v_1 = \frac{14}{3}, v_2 = \frac{46}{9}, v_3 = \frac{146}{27}$$

$$20 \quad y_t = \frac{42}{5} (-4)^t - \frac{12}{5}, \text{ oscillator explosion}$$

(18.1)

$$1 \quad 38 \qquad 2 \quad (1) A = 4, B = 9, S_1 = 19, f = 92 \qquad 3 \quad 26 \quad (r = 2, s = 1)$$

$$4 \quad \frac{74}{3} \left(X_1 = 0, X_2 = 2, X_3 = \frac{10}{3} \right)$$

$$9 \quad 8750 (X_1 = 3750, X_2 = 500) \qquad 10 \quad \frac{45}{2} (X_1 = 1.5, X_2 = 4.5)$$

$$11 \quad Y = 2, Z = 3 \qquad 13 \quad k = 55$$

$$15 \quad (Z = 8, E = 204)$$

(18.2)

$$1 \quad \text{Minimise } f = 2705A + 221B + 455C$$

$$\text{Subject to } 10.7A + 5.4B + 7C \geq 10$$

$$5A + 10B + C \geq 15$$

$$2A + 4B \geq 20, A, B, C \geq 0$$

$$3 \quad \text{Maximise } f = 3x_1 + 4x_2 + x_3 + 7x_4$$

$$\text{Subject to } 8x_1 + 3x_2 + 4x_3 + x_4 \leq 7$$

$$2x_1 + 6x_2 + x_3 + 1x_4 \leq 3$$

$$x_1 + 4x_2 + 5x_3 + 2x_4 \leq 8$$

$$x_1, x_2, x_3 \geq 0$$

$$4 \quad \text{Maximise } f = 7A + 12B + 10C$$

$$\text{Subject to } 3A - 2B - 4C \leq 1$$

$$-A + 4B + 3C \leq -3$$

$$2A + 8C \leq 2$$

$$A \leq 0$$

$$B \leq 0$$

$$C \leq 0$$

(18.3)

$$11 \quad x_1 = 4, x_2 = 1, x_3 = 0, y = -11$$

$$25 \quad 60, 50, 380$$

$$26 \quad 5, \frac{95}{3}$$

$$27 \quad R = 5/3, S = 5, Q = T = 0$$

$$28 \quad \text{Tables } \frac{73}{8}, \text{ chairs } \frac{35}{8}, \text{ profit } = \frac{981}{8}$$

$$29 \quad x_1 = 12, x_2 = 2, \text{ all slacks} = 0 \quad \text{Dual } \left(\frac{1}{2}, \frac{1}{2} \right)$$

(19.1)

$$2 \quad X = \begin{bmatrix} 62 \\ 88.5 \\ 59 \end{bmatrix} \text{ and } \Delta X = \begin{bmatrix} 2 \\ 3/2 \\ 1 \end{bmatrix}$$

$$4 \quad V = \begin{bmatrix} 10.1 \\ 5.3 \\ 4.6 \end{bmatrix}$$

$$5 \quad X = \begin{bmatrix} 10086 \\ 4690 \end{bmatrix}$$

$$1 = \begin{bmatrix} 7684 \\ 2172 \end{bmatrix}$$

$$6 \quad 4.42 \text{ and } 6.91 \text{ per cent}$$

$$7 \quad X = \begin{bmatrix} 128 \\ 104.8 \end{bmatrix}$$

$$V = \begin{bmatrix} 89.12 \\ 10.88 \end{bmatrix}$$

$$8 \quad 42 \text{ and } 6.7 \text{ per cent}$$

$$9 \quad \Delta A = \begin{bmatrix} 10.80 \\ 21.06 \end{bmatrix}$$

$$10 \quad X = \begin{bmatrix} 247 \\ 215 \\ 230 \end{bmatrix}$$

$$L = \begin{bmatrix} 72.6 \\ 64.5 \\ 23.0 \end{bmatrix}$$

$$11 \quad V = \begin{bmatrix} 261 \\ 220 \\ 247 \end{bmatrix}$$

$$L = \begin{bmatrix} 52.2 \\ 88.0 \\ 24.7 \end{bmatrix}$$

$$12 \quad X = \begin{bmatrix} 188 \\ 149 \\ 131 \end{bmatrix}$$

$$L = \begin{bmatrix} 56.6 \\ 59.6 \\ 39.3 \end{bmatrix}$$

$$14 \quad A = \begin{bmatrix} 360 \\ 289 \\ 204 \end{bmatrix}$$

$$L = \begin{bmatrix} 72.0 \\ 66.7 \\ 61.2 \end{bmatrix}$$

$$16 \quad A = \begin{bmatrix} 422 \\ 769 \\ 373 \end{bmatrix}$$

$$L = \begin{bmatrix} 89.4 \\ 79.6 \\ 111.9 \end{bmatrix}$$

$$A = \begin{bmatrix} 5625 \\ 9250 \\ 6250 \end{bmatrix}$$

$$L = \begin{bmatrix} 2250 \\ 6475 \\ 7500 \end{bmatrix}$$

$$P = \begin{bmatrix} 14,500 \\ 14,250 \\ 19,125 \end{bmatrix}$$

Total value added = Rs. 1,62,250

LOGARITHMS

	0	1	2	3	4	5	6	7	8	9	1 2 3	4 5 6	7 8 9
10	0000	0043	0086	0128	0170						5 9 13	17 21 26	30 34 38
11	0414	0453	0492	0531	0569	0212	0253	0294	0334	0374	4 8 12	16 20 24	28 32 36
12	0792	0828	0864	0899	0934	0607	0645	0682	0719	0755	4 8 12	16 20 23	27 31 35
13	1139	1173	1206	1239	1271	0969	1004	1038	1072	1106	4 7 11	15 18 22	26 29 33
14	1461	1492	1523	1553	1584	1303	1335	1367	1399	1430	3 7 10	14 17 20	24 27 31
15	1761	790	1818	1847	1875	1514	1544	1573	1603	1632	3 6 9	12 15 19	22 25 28
16	2041	2068	2095	2122	2148	1614	1644	1673	1703	1732	3 6 9	12 14 17	20 23 26
17	2304	2330	2355	2380	2405	1903	1931	1959	1987	2014	3 6 8	11 14 17	20 23 26
18	2553	2577	2601	2625	2648	2175	2201	2227	2253	2279	3 5 8	10 13 16	18 21 24
19	2782	2810	2833	2856	2878	2430	2455	2480	2504	2529	3 5 8	10 13 15	18 20 23
20	3010	3032	3054	3075	3096	2672	2695	2718	2742	2765	2 5 7	9 12 14	17 19 21
21	3222	3243	3263	3284	3304	2900	2923	2945	2967	2989	2 4 7	9 11 13	16 18 20
22	3424	3444	3464	3483	3502	3118	3139	3160	3181	3201	2 4 6	8 11 13	15 17 19
23	3617	3636	3655	3674	3692	3324	3345	3365	3385	3404	2 4 6	8 10 12	14 16 18
24	3802	3820	3838	3856	3874	3522	3541	3560	3579	3598	2 4 6	8 10 12	14 15 17
25	3979	3997	4014	4031	4048	3711	3729	3747	3766	3784	2 4 6	7 9 11	13 15 17
26	4150	4166	4183	4200	4216	3922	3909	3927	3945	3962	2 4 5	7 9 11	12 14 16
27	4314	4330	4346	4362	4378	4055	4082	4099	4116	4133	2 3 5	7 9 10	12 14 15
28	4472	4487	4502	4518	4533	4232	4249	4265	4281	4297	2 3 5	7 8 10	11 13 15
29	4624	4639	4654	4669	4683	4393	4407	4425	4440	4456	2 3 5	6 8 9	11 13 14
30	4771	4786	4800	4814	4829	4548	4564	4579	4594	4609	2 3 5	6 8 9	11 12 14
31	4914	4928	4942	4955	4969	4713	4728	4742	4757	4771	1 3 4	6 7 9	10 12 13
32	5051	5065	5079	5092	5105	4857	4871	4886	4900	4914	1 3 4	6 7 9	10 11 13
33	5185	5198	5211	5224	5237	5011	5024	5033	5041	5050	1 3 4	6 7 8	10 11 12
34	5315	5328	5340	5353	5365	5145	5159	5172	5184	5197	1 3 4	5 7 8	9 11 12
35	5441	5453	5465	5478	5490	5276	5289	5302	5314	5327	1 3 4	5 6 8	9 10 12
36	5563	5575	5587	5599	5611	5391	5403	5416	5428	5440	1 3 4	5 6 8	9 10 11
37	5682	5694	5705	5717	5729	5527	5539	5551	5563	5575	1 2 4	5 6 7	9 10 11
38	5798	5809	5821	5832	5843	5647	5658	5670	5681	5692	1 2 4	5 6 7	8 10 11
39	5911	5922	5933	5944	5955	5742	5753	5764	5775	5786	1 2 3	5 6 7	8 9 10
40	6021	6031	6042	6053	6064	5877	5888	5899	5910	5921	1 2 3	4 5 7	8 9 10
41	6128	6138	6149	6160	6170	5977	5988	5999	6010	6021	1 2 3	4 5 6	8 9 10
42	6222	6233	6243	6253	6263	6096	6107	6117	6127	6137	1 2 3	4 5 6	7 8 9
43	6335	6345	6355	6365	6375	6284	6294	6304	6314	6324	1 2 3	4 5 6	7 8 9
44	6435	6444	6454	6464	6474	6385	6395	6405	6415	6425	1 2 3	4 5 6	7 8 9
45	6532	6542	6551	6561	6571	6493	6503	6513	6522	6532	1 2 3	4 5 6	7 8 9
46	6628	6637	6646	6656	6665	6584	6593	6603	6612	6621	1 2 3	4 5 6	7 8 9
47	6721	6730	6739	6749	6758	6684	6693	6702	6711	6720	1 2 3	4 5 6	7 8 9
48	6812	6821	6830	6839	6848	6767	6776	6785	6794	6803	1 2 3	4 5 6	7 8 9
49	6902	6911	6920	6928	6937	6866	6875	6884	6893	6902	1 2 3	4 5 6	7 8 9

LOGARITHMS

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50	6990	6998	7007	7016	7024	7033	7042	7050	7059	7067	1 2 3	3 4 5	6 7 8
51	7076	7084	7093	7101	7110	7118	7126	7135	7143	7152	1 2 3	3 4 5	6 7 8
52	7160	7168	7177	7185	7193	7202	7210	7218	7226	7235	1 2 2	3 4 5	6 7 7
53	7243	7251	7259	7267	7275	7284	7292	7300	7308	7316	1 2 2	3 4 5	6 6 7
54	7324	7332	7340	7348	7356	7364	7372	7380	7388	7396	1 2 2	3 4 5	6 6 7
55	7404	7412	7419	7427	7435	7443	7451	7459	7466	7474	1 2 2	3 4 5	5 6 7
56	7482	7490	7497	7505	7513	7520	7528	7536	7543	7551	1 2 2	3 4 5	5 6 7
57	7559	7566	7574	7582	7589	7597	7604	7612	7619	7627	1 2 2	3 4 5	5 6 7
58	7634	7642	7649	7657	7664	7672	7679	7686	7694	7701	1 1 2	3 4 4	5 6 7
59	7709	7716	7723	7731	7738	7745	7752	7760	7767	7774	1 1 2	3 4 4	5 6 7
60	7782	7789	7796	7803	7810	7818	7825	7832	7839	7846	1 1 2	3 4 4	5 6 6
61	7853	7860	7868	7875	7882	7889	7896	7903	7910	7917	1 1 2	3 4 4	5 6 6
62	7924	7931	7938	7945	7952	7959	7966	7973	7980	7987	1 1 2	3 3 4	5 6 6
63	7993	8000	8007	8014	8021	8028	8035	8041	8048	8055	1 1 2	3 3 4	5 5 6
64	8062	8069	8075	8082	8089	8096	8102	8109	8116	8122	1 1 2	3 3 4	5 5 6
65	8129	8136	8142	8149	8156	8162	8169	8176	8182	8189	1 1 2	3 3 4	5 5 6
66	8195	8202	8209	8215	8222	8228	8235	8241	8248	8254	1 1 2	3 3 4	5 5 6
67	8261	8267	8274	8280	8287	8293	8299	8305	8312	8319	1 1 2	3 3 4	5 5 6
68	8325	8331	8338	8344	8351	8357	8363	8370	8376	8382	1 1 2	3 3 4	4 5 6
69	8388	8395	8401	8407	8414	8420	8426	8432	8439	8445	1 1 2	2 3 4	4 5 6
70	8451	8457	8463	8470	8476	8482	8488	8494	8500	8505	1 1 2	2 3 4	4 5 6
71	8513	8519	8525	8531	8537	8543	8549	8555	8561	8567	1 1 2	2 3 4	4 5 5
72	8513	8579	8585	8591	8597	8603	8609	8615	8621	8627	1 1 2	2 3 4	4 5 5
73	8633	8639	8645	8651	8657	8663	8669	8675	8681	8686	1 1 2	2 3 4	4 5 5
74	8692	8698	8704	8710	8716	8722	8727	8733	8739	8745	1 1 2	2 3 4	4 5 5
75	8751	8756	8762	8768	8774	8779	8785	8791	8797	8802	1 1 2	2 3 3	4 5 5
76	8808	8814	8820	8825	8831	8837	8842	8848	8854	8859	1 1 2	2 3 3	4 5 5
77	8865	8871	8876	8882	8887	8893	8899	8904	8910	8915	1 1 2	2 3 3	4 4 5
78	8921	8927	8932	8938	8943	8949	8954	8960	8965	8971	1 1 2	2 3 3	4 4 5
79	8976	8982	8987	8993	8998	9004	9009	9015	9020	9025	1 1 2	2 3 3	4 4 5
80	9031	9036	9042	9047	9053	9058	9063	9069	9074	9079	1 1 2	2 3 3	4 4 5
81	9085	9090	9096	9101	9106	9112	9117	9122	9128	9133	1 1 2	2 3 3	4 4 5
82	9138	9143	9149	9154	9159	9165	9170	9175	9180	9186	1 1 2	2 3 3	4 4 5
83	9191	9196	9201	9206	9212	9217	9222	9227	9232	9238	1 1 2	2 3 3	4 4 5
84	9243	9248	9253	9258	9263	9269	9274	9279	9284	9289	1 1 2	2 3 3	4 4 5
85	9294	9299	9304	9309	9315	9320	9325	9330	9335	9340	1 1 2	2 3 3	4 4 5
86	9345	9350	9355	9360	9365	9370	9375	9380	9385	9390	1 1 2	2 3 3	4 4 5
87	9395	9400	9405	9410	9415	9420	9425	9430	9435	9440	0 1 1	2 2 3	3 4 4
88	9445	9450	9455	9460	9465	9469	9474	9479	9484	9489	0 1 1	2 2 3	3 4 4
89	9494	9499	9504	9509	9513	9518	9523	9528	9533	9538	0 1 1	2 2 3	3 4 4
90	9542	9547	9552	9557	9562	9566	9571	9576	9581	9586	0 1 1	2 2 3	3 4 4
91	9590	9595	9600	9605	9609	9614	9619	9624	9628	9633	0 1 1	2 2 3	3 4 4
92	9638	9643	9647	9652	9657	9661	9666	9671	9675	9680	0 1 1	2 2 3	3 4 4
93	9685	9689	9694	9699	9703	9708	9713	9717	9722	9727	0 1 1	2 2 3	3 4 4
94	9731	9736	9741	9745	9750	9754	9759	9763	9768	9773	0 1 1	2 2 3	3 4 4
95	9777	9782	9786	9791	9795	9800	9805	9809	9814	9818	0 1 1	2 2 3	3 4 4
96	9823	9827	9832	9836	9841	9845	9850	9854	9859	9863	0 1 1	2 2 3	3 4 4
97	9868	9872	9877	9881	9885	9890	9894	9899	9903	9908	0 1 1	2 2 3	3 4 4
98	9912	9917	9921	9926	9930	9934	9939	9943	9948	9952	0 1 1	2 2 3	3 4 4
99	9956	9961	9965	9969	9974	9978	9983	9987	9991	9996	0 1 1	2 2 3	3 3 4

ANTILOGARITHMS

	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
00	1000	1002	1005	1007	1009	1012	1014	1016	1019	1021	0	0	1	1	1	1	2	2	2
01	1023	1026	1028	1030	1033	1035	1038	1040	1042	1045	0	0	1	1	1	1	2	2	2
02	1047	1050	1052	1054	1057	10	1062	1064	1067	1069	0	0	1	1	1	1	2	2	3
03	1072	1074	1076	1079	1081	1084	1085	1089	1091	1094	0	0	1	1	1	2	2	2	2
04	1096	1099	1102	1104	1107	1109	1112	1114	1117	1119	0	1	1	1	1	2	2	2	2
05	1122	1125	1127	1130	1132	1135	1138	1140	1143	1146	0	1	1	1	1	2	2	2	2
06	1148	1151	1153	1156	1159	1161	1164	1167	1169	1172	0	1	1	1	1	2	2	2	2
07	1175	1173	1180	1183	1186	1189	1191	1194	1197	1199	0	1	1	1	1	2	2	2	2
08	1202	1205	1208	1211	1213	1216	1219	1222	1225	1227	0	1	1	1	1	2	2	2	3
09	1230	1233	1236	1239	1242	1245	1247	1250	1253	1256	0	1	1	1	1	2	2	2	3
10	1259	1262	1265	1268	1271	1274	1276	1279	1282	1285	0	1	1	1	1	2	2	2	3
11	1283	1291	1294	1297	1300	1303	1306	1309	1312	1315	0	1	1	1	2	2	2	2	3
12	1318	1321	1324	1327	1330	1334	1337	1340	1343	1346	0	1	1	1	2	2	2	2	3
13	1349	1352	1355	1358	1361	1365	1368	1371	1374	1377	0	1	1	1	2	2	2	3	3
14	1380	1384	1387	1390	1393	1396	1400	1403	1406	1409	0	1	1	1	2	2	2	3	3
15	1413	1416	1419	1422	1426	1429	1432	1435	1439	1442	0	1	1	1	2	2	2	3	3
16	1445	1447	1452	1455	1459	1462	1466	1469	1472	1476	0	1	1	1	2	2	2	3	3
17	1479	1483	1486	1489	1493	1496	1500	1503	1507	1510	0	1	1	1	2	2	2	3	3
18	1514	1517	1521	1524	1528	1531	1535	1538	1542	1545	0	1	1	1	2	2	2	3	3
19	1549	1552	1556	1560	1563	1567	1570	1574	1578	1581	0	1	1	1	2	2	3	3	3
20	1585	1589	1592	1596	1600	1603	1607	1611	1614	1618	0	1	1	1	2	2	3	3	3
21	1622	1626	1629	1633	1637	1641	1644	1648	1652	1656	0	1	1	1	2	2	3	3	3
22	1660	1663	1667	1671	1675	1679	1683	1687	1690	1694	0	1	1	1	2	2	3	3	3
23	1698	1702	1706	1710	1714	1718	1722	1726	1730	1734	0	1	1	1	2	2	3	3	4
24	1738	1742	1746	1750	1754	1758	1762	1766	1770	1774	0	1	1	1	2	2	3	3	4
25	1778	1782	1786	1791	1795	1799	1803	1807	1811	1816	0	1	1	1	2	2	3	3	4
26	1820	1824	1828	1832	1837	1841	1845	1849	1854	1858	0	1	1	1	2	2	3	3	4
27	1862	1866	1871	1875	1879	1884	1888	1892	1897	1901	0	1	1	1	2	2	3	3	4
28	1905	1910	1914	1919	1923	1928	1932	1936	1941	1945	0	1	1	1	2	2	3	3	4
29	1950	1954	1959	1963	1968	1972	1977	1982	1986	1991	0	1	1	1	2	2	3	3	4
30	1995	2000	2004	2009	2014	2018	2023	2028	2032	2037	0	1	1	1	2	2	3	3	4
31	2042	2046	2051	2056	2061	2065	2070	2075	2080	2084	0	1	1	1	2	2	3	3	4
32	2089	2094	2099	2104	2109	2113	2118	2123	2128	2133	0	1	1	1	2	2	3	3	4
33	2138	2143	2148	2153	2158	2163	2168	2173	2178	2183	0	1	1	1	2	2	3	3	4
34	2188	2193	2198	2203	2208	2213	2218	2223	2228	2234	1	1	2	2	3	3	4	4	5
35	2239	2244	2249	2254	2259	2265	2270	2275	2280	2286	1	1	2	2	3	3	4	4	5
36	2291	2296	2301	2307	2312	2317	2323	2328	2333	2339	1	1	2	2	3	3	4	4	5
37	2344	2350	2355	2360	2366	2371	2377	2382	2388	2393	1	1	2	2	3	3	4	4	5
38	2400	2404	2410	2415	2421	2427	2432	2438	2443	2449	1	1	2	2	3	3	4	4	5
39	2455	2460	2466	2472	2477	2483	2489	2495	2500	2506	1	1	2	2	3	3	4	4	5
40	2512	2518	2523	2529	2535	2541	2547	2553	2559	2564	1	1	2	2	3	3	4	4	5
41	2570	2576	2582	2588	2594	2600	2606	2612	2618	2624	1	1	2	2	3	3	4	4	5
42	2630	2636	2642	2649	2655	2661	2667	2673	2679	2685	1	1	2	2	3	3	4	4	5
43	2692	2693	2704	2710	2716	2723	2729	2735	2742	2748	1	1	2	2	3	3	4	4	5
44	2754	2761	2767	2773	2780	2786	2793	2799	2805	2812	1	1	2	2	3	3	4	4	5
45	2818	2825	2831	2838	2844	2852	2858	2864	2871	2877	1	1	2	2	3	3	4	4	5
46	2884	2891	2897	2904	2911	2917	2924	2931	2938	2944	1	1	2	2	3	3	4	4	5
47	2951	2958	2965	2972	2979	2985	2992	2999	3006	3013	1	1	2	2	3	3	4	4	5
48	2920	2927	3034	3041	3048	3055	3062	3069	3076	3083	1	1	2	2	3	3	4	4	5
49	3090	3097	3105	31	3119	3226	3233	3161	3245	3155	1	1	3	3	4	4	4	5	6

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51	3236	3243	3251	3258	3266	3273	3281	3289	3296	3304	1 2 2	3 4 5	5 6 7
52	3311	3319	3327	3334	3342	3350	3357	3365	3373	3381	1 2 2	3 4 5	5 6 7
53	3388	3396	3404	3412	3420	3428	3436	3443	3451	3459	1 2 2	3 4 5	6 6 7
54	3467	3475	3489	3491	3499	3508	3516	3524	3532	3540	1 2 2	3 4 5	6 6 7
55	3548	3556	3565	3573	3581	3589	3597	3606	3614	3622	1 2 2	3 4 5	6 7 7
56	3631	3639	3648	3656	3664	3673	3681	3690	3698	3707	1 2 3	3 4 5	6 7 8
57	3715	3724	3733	3741	3750	3758	3767	3776	3784	3793	1 2 3	3 4 5	6 7 8
58	3802	3811	3819	3828	3837	3846	3855	3864	3873	3882	1 2 3	4 4 5	6 7 8
59	3890	3899	3908	3917	3926	3936	3945	3954	3963	3972	1 2 3	4 5 5	6 7 8
60	3981	3990	3999	4009	4018	4027	4036	4046	4055	4064	1 2 3	4 5 6	6 7 8
61	4074	4083	4093	4102	4111	4121	4130	4140	4150	4159	1 2 3	4 5 6	7 8 9
62	4169	4178	4188	4198	4207	4217	4227	4236	4246	4256	1 2 3	4 5 6	7 8 9
63	4266	4276	4285	4295	4305	4315	4325	4335	4345	4355	1 2 3	4 5 6	7 8 9
64	4365	4375	4385	4395	4406	4416	4426	4436	4446	4457	1 2 3	4 5 6	7 8 9
65	4467	4477	4487	4498	4508	4519	4529	4539	4550	4560	1 2 3	4 5 6	7 8 9
66	4571	4581	4592	4603	4613	4624	4634	4645	4656	4667	1 2 3	4 5 6	7 9 10
67	4677	4688	4699	4710	4721	4732	4742	4753	4764	4775	1 2 3	4 5 7	8 9 10
68	4786	4797	4808	4819	4831	4842	4853	4864	4875	4887	1 2 3	4 6 7	8 9 10
69	4898	4909	4920	4932	4943	4955	4966	4977	4989	5000	1 2 3	5 6 7	8 9 10
70	5022	5023	5035	5047	5058	5070	5082	5093	5105	5117	1 2 4	5 6 7	8 9 11
71	5129	5140	5152	5164	5176	5188	5200	5212	5224	5236	1 2 4	5 6 7	8 10 11
72	5248	5250	5272	5284	5297	5309	5321	5333	5346	5358	1 2 4	5 6 7	9 10 11
73	5370	5383	5395	5408	5420	5433	5445	5458	5470	5483	1 3 4	5 6 8	9 10 11
74	5495	5508	5521	5534	5546	5559	5572	5585	5598	5610	1 3 4	5 6 8	9 10 12
75	5623	5636	5649	5662	5675	5689	5702	5715	5728	5741	1 3 4	5 7 8	9 10 12
76	5754	5768	5781	5794	5808	5821	5834	5848	5861	5875	1 3 4	5 7 8	9 11 12
77	5888	5902	5916	5929	5943	5957	5970	5984	5998	6012	1 3 4	5 7 8	10 11 12
78	6025	6039	6053	6067	6081	6095	6109	6124	6138	6152	1 3 4	6 7 8	10 11 13
79	6166	6180	6194	6209	6223	6237	6252	6266	6281	6295	1 3 4	6 7 9	10 11 13
80	6310	6324	6339	6353	6368	6383	6397	6412	6427	6442	1 3 4	6 7 9	10 12 13
81	6457	6471	6486	6501	6516	6531	6546	6561	6577	6592	2 3 5	6 8 9	11 12 14
82	6607	6622	6637	6653	6668	6683	6699	6714	6730	6745	2 3 5	6 8 9	11 12 14
83	6761	6776	6792	6808	6823	6839	6855	6871	6887	6902	2 3 5	6 8 9	11 13 14
84	6918	6934	6950	6966	6982	6998	7015	7031	7047	7063	2 3 5	6 8 10	11 13 15
85	7079	7096	7112	7129	7145	7161	7178	7194	7211	7228	2 3 5	7 8 10	12 13 15
86	7244	7261	7278	7295	7311	7328	7345	7362	7379	7396	2 3 5	7 8 10	12 13 15
87	7413	7430	7447	7464	7482	7499	7516	7534	7551	7568	2 3 5	7 9 10	12 14 16
88	7586	7603	7621	7638	7656	7674	7691	7709	7727	7745	2 4 5	7 9 11	12 14 16
89	7762	7780	7798	7816	7834	7852	7870	7889	7907	7925	2 4 5	7 9 11	13 14 16
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